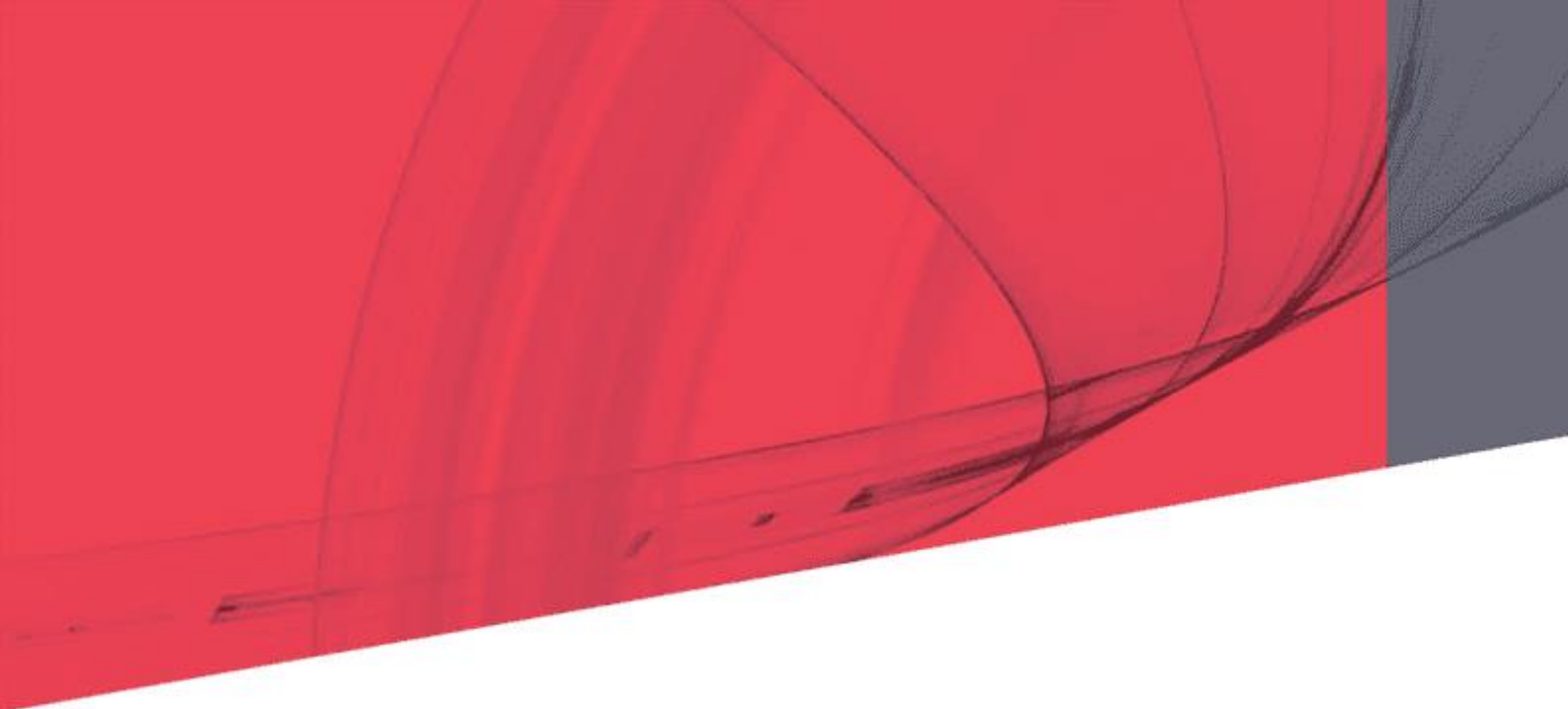




ACSPL+ Commands & Variables

Reference Guide

February 2023

Document Revision: 3.13.01

ACSPL+ Commands & Variables

Release Date: February 2023

COPYRIGHT

© ACS Motion Control Ltd., 2023. All rights reserved.

Changes are periodically made to the information in this document. Changes are published as release notes and later incorporated into revisions of this document.

No part of this document may be reproduced in any form without prior written permission from ACS Motion Control.

TRADEMARKS

Windows and Intellisense are trademarks of Microsoft Corporation.

EtherCAT® is registered trademark and patented technology, licensed by Beckhoff Automation GmbH, Germany.

Any other companies and product names mentioned herein may be the trademarks of their respective owners.

PATENTS

Israel Patent No. 235022

US Patent Application No. 14/532,023

Europe Patent application No.15187586.1

Japan Patent Application No.: 2015-193179

Chinese Patent Application No.: 201510639732.X

Taiwan(R.O.C.) Patent Application No. 104132118

Korean Patent Application No. 10-2015-0137612

www.acsmotioncontrol.com

support@acsmotioncontrol.com

sales@acsmotioncontrol.com

NOTICE

The information in this document is deemed to be correct at the time of publishing. ACS Motion Control reserves the right to change specifications without notice. ACS Motion Control is not responsible for incidental, consequential, or special damages of any kind in connection with using this document.

Revision History

Date	Revision	Description
February 2023	3.13.01	Cross-reference to Programmer's Guide Matrix section Corrections to MATRIX documentation NURBS, SPATH correction XXMxx support in GETCONF
November 2022	3.13	New Release, correct struct type variable definition
September 2022	3.12.03	Update ASSIGNPOUTS tables and links Correct BSEG parameter order
July 2022	3.12.02	Single-source XSEG Description String support
June 2022	3.12.01	Added /e switch description to PEG_I
May 2022	3.12	New Release
December 2021	3.11.01.02	Note minimum version supporting feature
December 2021	3.11.01.01	Document /q switch for XSEG, PTP, other motion commands Local Coordinates function explanations
November 2021	3.11.01	PEG & MARK Improvements, Addressing Analog I/O correction
June 2021	3.10.01	Add XXMsa table to ASSIGNPOUTS % and ++ operators
September 2020	3.02	Remove reference to obsolete SPiiPlus PCI device
July 2020	3.01	Structures and Functions
June 2020	3.00	Enhanced input shaping
July 2018	2.60	Added XRMSD, XRMSM, XRMSTD, and XRMSTM Added gantry mode commutation support Updated the following BSEG motions commands: BSEG, ARC1, ARC2, Line

Date	Revision	Description
December 2017	2.50	Added BSEG
June 2017	2.40	Updated #ETHERCAT, COEWRITE, COEREAD
October 2016	2.30.01	Changed switch to /f for end_velocity argument in XSEG
November 2022	3.13	New release
September 2022	3.12.05	Update ASSIGNPOUTS tables and links Add SWITCH explanation Explain negative LCI motion Correct BSEG parameter order
August 2022	3.12.04	Update CTIME change instructions
July 2022	3.12.03	Single-source XSEG Description Motion Duration Example Add missing ECGETREV
July 2022	3.12.02	SPTP command
June 2022	3.12.01	Added /e switch description to PEG_I, more details in DOUT
May 2022	3.12	New Release, Error mapping, FOE, Modulo, other functions
February 2022	3.11.01.06	Corrections to ARC1, ARC2, LINE, switches
February 2022	3.11.01.05	Corrections to PRATE
January 2022	3.11.01.04	Correction to HOME entry, INTERRUPTX correction
January 2022	3.11.01.03	Remove SET/GETCONF(29)
December 2021	3.11.01.02	Remove AERR peg_engine instead of axis in PEG_I, PEG_R, etc.
December 2021	3.11.01.01	Document /q switch for XSEG, PTP, other motion commands Local Coordinates function explanations XSEG example

Date	Revision	Description
November 2021	3.11.01	PEG & MARK Improvements, MFLAGSX.#SATPROT, AIN/AOUT corrections
September 2021	3.11	New and updated functions with ADK Release
June 2021	3.10.01	Formatting Corrections, ASSIGNPOUTS for XXMsa
April 2021	3.10	SLCRAT, SLVRAT corrections Note M&S State for MTIMEA/B/C Document SET/GETCONF(270) Changes for V3.10
December 2020	3.03	INSHAPEON example if CTIME<1
September 2020	3.02	Remove reference to obsolete SPiiPlus PCI device
July 2020	3.01	Fix Error Tables
June 2020	3.00	Changes supporting ADK v3.00
September 2019	2.70.10	Corrections to XCURV, XCURI and related variables
July 2019	2.70.02	SLCPA is obsolete, removed from documentation
June 2019	2.70.01	Formatting corrections, examples for functions, fixed links for system configuration variables, moved stepper loop variables to servo loop section
April 2019	2.70	GCODE Errors Many new functions, commands, and variables for new features. See version 2.70 release notes.
October 2018	2.60.10	Updated AST bits for laser and SLEC module Added FOLLOW, UNFOLLOW, EXTFAC, FOLLOWCH Added error code 5042 to the list of motion termination errors
July 2018	2.60	Updated for SPiiPlus ADK Suite v2.60
January 2018	2.50.01	Added laser control commands and functions.

Date	Revision	Description
December 2017	2.50	Updated for SPiiPlus ADK Suite v2.50
September 2017	2.40.01	Updated SETSP function.
June 2017	2.40	Updated for SPiiPlus ADK Suite v2.40
December 2016	2.30.02	Removed unsupported ServoBoost variables
October 2016	2.30.01	Added support for Absolute Encoders to SLPROUT, SLVROUT, SLCROUT Replaced references to acsc_Write and acsc_Read with acsc_Transaction
September 2016	2.30.10	Changed LINE1 to LINE, ARC2 to ARC1, ARC3 to ARC2, and ARC4 to ARC2 Updated XSEG, LINE, ARC1, and ARC2 for 6 axes support
August 2016	2.30	Updated for SPiiPlus ADK Suite v2.30
September 2014	01	First Release

Conventions Used in this Guide

Text Formats

Format	Description
Bold	Names of GUI objects or commands
BOLD + UPPERCASE	ACSPL+ variables and commands
<code>Monospace + grey background</code>	Code example
<i>Italic</i>	Names of other documents
Blue	Hyperlink
[]	In commands indicates optional item(s)
	In commands indicates either/or items

Flagged Text

	Note - includes additional information or programming tips.
	Caution - describes a condition that may result in damage to equipment.
	Warning - describes a condition that may result in serious bodily injury or death.
	Model - highlights a specification, procedure, condition, or statement that depends on the product model
	Advanced - indicates a topic for advanced users.

Related Documents

Documents listed in the following table provide additional information related to this document.

The most updated version of the documents can be downloaded by authorized users from [ACS Motion Control Resources](#) under "Downloads".

Online versions for all ACS software manuals are available to authorized users at [ACS Motion Control Knowledge Center](#).

Document	Description
SPiiPlus C Library Reference	<i>C++ and Visual Basic® libraries for host PC applications. This guide is applicable for all the SPiiPlus motion control products.</i>
<i>SPiiPlus COM Library Reference C</i>	<i>COM Methods, Properties, and Events for Communication with the Controller.</i>
<i>SPiiPlus .NET Library Reference</i>	.NET Methods, Properties, and Events for Communication with the Controller.
<i>SPiiPlusMMI Application Studio User Guide</i>	A complete guide for using the SPiiPlus MMI Application Studio and associated monitoring tools.
<i>SPiiPlus Utilities User Guide</i>	A guide for using the SPiiPlus User Mode Driver (UMD) for settingup communication with the SPiiPlus motion controller.
<i>SPiiPlus NT/DC Hardware Guide</i>	Technical description of the SPiiPlus NT/DC product line.
<i>SPiiPlus PDMnt Hardware Guide</i>	Technical description of the SPiiPlus PDMnt Network Interface.
<i>SPiiPlus SDMnt Hardware Guide</i>	Technical description of the SPiiPlus SDMnt Step Motor Drive Module.
<i>SPiiPlus UDMnt Hardware Guide</i>	Technical description of the SPiiPlus UDMnt Universal Drive Module.
<i>MC4U-CS Control Module Hardware Guide</i>	Technical description of the MC4U Control Module integrated motion control product line.
<i>HSSI Expansion Modules Guide</i>	High-Speed Synchronous Serial Interface (HSSI) for expanded I/O, distributed axes, and nonstandard devices.

Document	Description
<i>PEG and MARK Operations Application Notes</i>	Provides details on using the PEG commands in SPiiPlus systems.

Table of Contents

1. Introduction	43
2. ACSPL+ Commands	44
2.1 Axis Management Commands	48
2.1.1 BREAK	49
2.1.2 COMMUT	50
2.1.3 CONNECT	52
2.1.4 CSCREATE	54
2.1.5 CSDESTROY	58
2.1.6 DEPENDS	59
2.1.7 DISABLE/DISABLEALL	60
2.1.8 ENABLE/ENABLE ALL	61
2.1.9 ENCINIT	62
2.1.10 ENCREAD	64
2.1.11 FCLEAR	65
2.1.12 FOLLOW	66
2.1.13 GO	66
2.1.14 GROUP	67
2.1.15 HALT	68
2.1.16 HOME	69
2.1.17 IMM	71
2.1.18 KILL/KILLALL	72
2.1.19 SAFETYCONF	74
2.1.20 SAFETYGROUP	75
2.1.21 SET	76
2.1.22 SPLIT	76
2.1.23 UNFOLLOW	77
2.2 Predefined Homing Methods	78
2.2.1 Homing Method 1: Homing on the negative limit switch and index pulse	78
2.2.2 Homing Method 2: Homing on positive limit switch and index pulse	78
2.2.3 Homing Method 17: Homing on Negative Limit Switch	78
2.2.4 Homing Method 18: Homing on Positive Limit Switch	78
2.2.5 Homing Method 33 and 34: Homing on the index pulse	78

2.2.6 Homing Method 37: Homing on current position	79
2.2.7 Homing Method 50: Negative Hard Stop and index pulse (ACS Specific)	79
2.2.8 Homing Method 51: Positive Hard Stop and index pulse (ACS Specific)	79
2.2.9 Homing Method 52: Negative Hard Stop (ACS Specific)	79
2.2.10 Homing Method 53: Positive Hard Stop (ACS Specific)	79
2.3 Interactive Commands	79
2.3.1 DISP	80
2.3.2 INP	85
2.3.3 INTERRUPT	87
2.3.4 INTERRUPTX	89
2.3.5 SEND	90
2.3.6 TRIGGER	92
2.3.7 OUTP	93
2.4 PEG and MARK Commands	95
2.4.1 ASSIGNMARK	96
2.4.2 ASSIGNPEG	97
2.4.3 ASSIGNPOUTS	99
2.4.4 GETPEGCOUNT	101
2.4.5 PEG_I	101
2.4.6 PEG_R	104
2.4.7 STARTPEG	108
2.4.8 STOPPEG	108
2.4.9 SETPEGDELAY	109
2.5 Miscellaneous Commands	110
2.5.1 AXISDEF	111
2.5.2 DC	113
2.5.3 STOPDC	115
2.5.4 READ	116
2.5.5 SPDC	117
2.5.6 STOPSPDC	119
2.5.7 WRITE	120
2.5.8 SPINJECT	121
2.5.9 STOPINJECT	122
2.5.10 SPICFG	122

2.5.10.1 SPIWRITE	124
2.5.11 SPIWRITE	124
2.5.12 SPRT	125
2.5.13 SPRTSTOP	129
2.5.14 USAGESP	129
2.6 Motion Commands	129
2.6.1 ARC1	131
2.6.2 ARC1	132
2.6.3 ARC1	135
2.6.4 ARC2	136
2.6.5 ARC2	137
2.6.6 ARC2	140
2.6.7 BPTP	140
2.6.8 BPTPCalc	143
2.6.9 BSEG...ENDS	144
2.6.10 JOG	145
2.6.11 LINE	146
2.6.12 LINE	147
2.6.13 LINE	150
2.6.14 MASTER	151
2.6.15 MPOINT	152
2.6.16 MPTP...ENDS	157
2.6.17 MSEG...ENDS	160
2.6.18 PATH...ENDS	162
2.6.19 POINT	164
2.6.20 PROJECTION	167
2.6.21 PTP	169
2.6.22 PVSPLINE...ENDS	171
2.6.23 SLAVE	174
2.6.24 STOPPER	175
2.6.25 TRACK	176
2.6.26 XSEG...ENDS	177
2.6.27 NURBS	187
2.6.28 NPOINT	189

2.6.29	SPATH	191
2.6.30	SEGMENT	194
2.6.31	SMOVE	195
2.6.32	SPTP Command	195
2.6.33	Using ARC1, ARC2 and LINE Switches	196
2.7	Program Flow Commands	198
2.7.1	Assignment Command	198
2.7.2	BLOCK...END	200
2.7.3	CALL	201
2.7.4	GOTO	201
2.7.5	IF, ELSEIF, ELSE...END	202
2.7.6	INPUT	204
2.7.7	LOOP...END	205
2.7.8	ON...RET	206
2.7.9	TILL	207
2.7.10	WAIT	208
2.7.11	WHILE...END	208
2.7.12	SWITCH Statement	209
2.8	Program Management Commands	210
2.8.1	DISABLEON	210
2.8.2	ENABLEON	211
2.8.3	PAUSE	211
2.8.4	RESUME	211
2.8.5	START	212
2.8.6	STOP/STOPALL	213
2.9	Ethernet/IP ACSPL+ Support Commands	214
2.9.1	EIPGETATTR	214
2.9.2	EIPGETIND1	215
2.9.3	EIPGETIND2	216
2.9.4	EIPGETTAG	217
2.9.5	EIPSETASM	217
2.10	Laser Control Commands	218
2.10.1	LCENABLE	218
2.10.2	LCDISABLE	218

2.11 Input Shaping Commands	218
2.11.1 INSHAPEON	219
2.11.2 INSHAPEOFF	220
3. ACSPL+ Variables	221
3.1 Axis Configuration Variables	234
3.1.1 AFLAGS	236
3.1.2 ENTIME	237
3.1.3 ESTBITS	237
3.1.4 E2STBITS	238
3.1.5 EMTBITS	239
3.1.6 E2MTBITS	239
3.1.7 MFF	240
3.1.8 MFLAGS	241
3.1.9 MFLAGSX	248
3.1.10 MODULOMD	250
3.1.11 PEGQUE	254
3.1.12 SETTLE	254
3.1.13 SLPMAX	255
3.1.14 SLPMIN	256
3.1.15 STEPFB	257
3.1.16 STEPW	258
3.1.17 TARGRAD	259
3.2 Brake Variables	259
3.2.1 BOFFTIME	260
3.2.2 BONTIME	261
3.2.3 MBRKROUT	262
3.2.4 VELBRK	263
3.3 Feedback Variables	264
3.3.1 E_AOFFS	266
3.3.2 E_FREQ	267
3.3.3 E2_AOFFS	268
3.3.4 E2_FREQ	268
3.3.5 E_FLAGS	269
3.3.6 E2_FLAGS	270

3.3.7 E_PAR_A	271
3.3.8 E2_PAR_A	272
3.3.9 E_PAR_B	273
3.3.10 E2_PAR_B	273
3.3.11 E_PAR_C	274
3.3.12 E2_PAR_C	275
3.3.13 E_PAR_D	276
3.3.14 E2_PAR_D	276
3.3.15 E_PAR_E	277
3.3.16 E2_PAR_E	277
3.3.17 E_SCMUL	278
3.3.18 E2_SCMUL	279
3.3.19 E_TYPE	279
3.3.20 E2_TYPE	282
3.3.21 EFAC	282
3.3.22 E2FAC	283
3.3.23 EOFFS	284
3.3.24 E2OFFS	285
3.3.25 EPOS	286
3.3.26 FVFIL	287
3.3.27 F2ACC	288
3.3.28 HOMEDEF	288
3.3.29 HOMEVELI	289
3.3.30 HOMEVELL	290
3.3.31 RVFIL	291
3.3.32 SCSOFFS	292
3.3.33 SCCOFFS	293
3.3.34 SC2COFFS	293
3.3.35 SC2GAIN	294
3.3.36 SC2PHASE	295
3.3.37 SC2SOFFS	295
3.3.38 SLEBIASA	296
3.3.39 SLEBIASB	297
3.3.40 SLEBIASC	297

3.3.41 SLEBIASD	298
3.3.42 SLABITS	299
3.3.43 S2LABITS	300
3.3.44 SCGAIN	301
3.3.45 SCPHASE	301
3.4 Axis State Variables	302
3.4.1 AST	302
3.4.2 ASTX	305
3.4.3 IND	306
3.4.4 IST	307
3.4.5 M2ARK	308
3.4.6 MARK	309
3.4.7 MST	309
3.4.8 RMSM	311
3.4.9 RMSD	311
3.4.10 NST	312
3.4.11 AMEST	313
3.5 Safety Limits Variables	314
3.5.1 CERRA	315
3.5.2 CERRI	316
3.5.3 CERRV	317
3.5.4 DELV	318
3.5.5 DELI	319
3.5.6 E_ERR	320
3.5.7 ERRA	320
3.5.8 ERRI	321
3.5.9 ERRV	322
3.5.10 SLLIMIT	323
3.5.11 SLLROUT	323
3.5.12 SRLIMIT	324
3.5.13 XACC	325
3.5.14 XCURCDB	326
3.5.15 XCURI	327
3.5.16 XCURK	327

3.5.17 XCURV	328
3.5.18 XRMS	329
3.5.19 XRMSD	330
3.5.20 XRMSM	331
3.5.21 XRMST	332
3.5.22 XRMSTD	333
3.5.23 XRMSTM	334
3.5.24 XSACC	334
3.5.25 XVEL	335
3.6 Data Collection Variables	336
3.6.1 DCN	336
3.6.2 DCP	337
3.6.3 S_DCN	338
3.6.4 S_DCP	339
3.6.5 S_ST	339
3.7 Input and Output Variables	340
3.7.1 AIN	341
3.7.2 AINOFFS	341
3.7.3 AINSCALE	342
3.7.4 AOUT	343
3.7.5 DOUT	344
3.7.6 EXTIN	344
3.7.7 EXTOUT	345
3.7.8 IN	346
3.7.9 OUT	347
3.7.10 SPIRXN	347
3.7.11 SPIST	348
3.8 Monitoring Variables	348
3.8.1 BCODECFG	349
3.8.2 BCODEUSG	350
3.8.3 BGLOBCFG	351
3.8.4 BGLOBUSG	352
3.8.5 BSRCUSG	352
3.8.6 BSRCCFG	353

3.8.7 BVARUSG	354
3.8.8 BVARCFG	354
3.8.9 JITTER	355
3.8.10 MSSYNC	356
3.8.11 USGBUF	356
3.8.12 USGTRACE	356
3.8.13 SOFTIME	357
3.8.14 TIME	358
3.8.15 USAGE	358
3.8.16 USAGESIM	358
3.9 Motion Variables	359
3.9.1 ACC	362
3.9.2 APOS	362
3.9.3 APOSFILT	363
3.9.4 CERRK	364
3.9.5 DAPOS	365
3.9.6 DEC	365
3.9.7 DECOMP	366
3.9.8 DELK	367
3.9.9 FACC	368
3.9.10 FPOS	368
3.9.11 F2POS	369
3.9.12 FVEL	369
3.9.13 F2VEL	370
3.9.14 FEEDRF	370
3.9.15 GACC	371
3.9.16 GJERK	371
3.9.17 GMOT	372
3.9.18 GMQU	373
3.9.19 GMTYPE	373
3.9.20 GPATH	374
3.9.21 GPHASE	375
3.9.22 GRTIME	376
3.9.23 GSEG	377

3.9.24 GSFREE	377
3.9.25 GSNAP	378
3.9.26 GVEC	378
3.9.27 GVEL	379
3.9.28 JERK	379
3.9.29 KDEC	380
3.9.30 LPOS	381
3.9.31 MPOS	382
3.9.32 MTIMEA	382
3.9.33 MTIMEB	383
3.9.34 MTIMEC	384
3.9.35 NVEL	385
3.9.36 PE	386
3.9.37 PPOS	387
3.9.38 PPOSCOMP	387
3.9.39 PRFLTIME	388
3.9.40 RACC	389
3.9.41 RJERK	389
3.9.42 ROFFS	389
3.9.43 RPOS	390
3.9.44 RPOSCOMP	391
3.9.45 RPOSDEL	391
3.9.46 RSNAP	392
3.9.47 RVEL	392
3.9.48 SETTLEA	393
3.9.49 SETTLEB	393
3.9.50 SLSFF	394
3.9.51 SETTLEC	398
3.9.52 SNAP	399
3.9.53 STLTIMEA	400
3.9.54 STLTIMEB	400
3.9.55 STLTIMEC	401
3.9.56 TARGRADA	402
3.9.57 TARGRADB	403

3.9.58 TARGADC	404
3.9.59 TPOS	405
3.9.60 VEL	406
3.10 Program Execution Control Variables	407
3.10.1 ONRATE	408
3.10.2 PCHARS	408
3.10.3 PERL	409
3.10.4 PERR	409
3.10.5 PEXL	410
3.10.6 PFLAGS	410
3.10.7 PLINES	412
3.10.8 PRATE	413
3.10.9 PST	414
3.11 Safety Control Variables	414
3.11.1 E_ERR	416
3.11.2 EC2ERR	417
3.11.3 ECALERR	417
3.11.4 EC2ALERR	418
3.11.5 ECERR	418
3.11.6 ECEXTERR	419
3.11.7 ECEXTST	419
3.11.8 ECST	421
3.11.9 EC2ST	422
3.11.10 ECSYNC	423
3.11.11 EC2SYNC	424
3.11.12 ECNST	424
3.11.13 EC2NST	426
3.11.14 FAULT	427
3.11.15 FAULTSIM	430
3.11.16 FDEF	434
3.11.17 FMASK	439
3.11.18 HLLROUT	442
3.11.19 HRLROUT	445
3.11.20 MERR	447

3.11.21 SAFIN	447
3.11.22 SAFINI	449
3.11.23 S_ERR	450
3.11.24 S_FAULT	450
3.11.25 S_FDEF	454
3.11.26 S_FMASK	457
3.11.27 S_SAFIN	458
3.11.28 S_SAFINI	459
3.11.29 SS11TIME	460
3.11.30 SS12TIME	460
3.11.31 STODELAY	461
3.11.32 SYNC	462
3.12 Induction Motor Variables	462
3.12.1 SLCFIELD	462
3.12.2 SLCSLIP	464
3.13 Nanomotion Variables	465
3.13.1 SLDZMIN	466
3.13.2 SLDZMAX	467
3.13.3 SLDZTIME	468
3.13.4 SLZFF	468
3.13.5 SLFRC	469
3.13.6 SLFRCN	470
3.13.7 SLHRS	471
3.13.8 SLVKPDCF	471
3.13.9 SLPKDCF	472
3.13.10 SLVKIDCF	473
3.14 Servo-Loop Variables	473
3.14.1 DCOM	474
3.14.2 Servo-Loop Current Variables	475
3.14.2.1 SLBIASA	475
3.14.2.2 SLBIASB	476
3.14.2.3 SLBIASC	477
3.14.2.4 SLIKI	478
3.14.2.5 SLIKP	479

3.14.2.6 SLIFILT	479
3.14.2.7 SLIOFFS	480
3.14.2.8 SLILI	481
3.14.3 Servo-Loop Velocity Variables	481
3.14.3.1 SLCRAT	482
3.14.3.2 SLVKI	483
3.14.3.3 SLVKIIF	483
3.14.3.4 SLVKISF	484
3.14.3.5 SLVKITF	485
3.14.3.6 SLVKP	485
3.14.3.7 SLVKPIF	486
3.14.3.8 SLVKPSF	487
3.14.3.9 SLVKPTF	487
3.14.3.10 SLVLI	488
3.14.3.11 SLVRAT	489
3.14.4 Servo-Loop Velocity Notch Filter Variables	489
3.14.4.1 SLVNFRQ	490
3.14.4.2 SLVNWID	490
3.14.4.3 SLVNATT	491
3.14.5 Servo-Loop Velocity Low Pass Filter Variables	492
3.14.5.1 SLVSOF	492
3.14.5.2 SLVSOFD	493
3.14.6 Servo-Loop Velocity Bi-Quad Filter Variables	493
3.14.6.1 SLVB0DD	494
3.14.6.2 SLVB0DF	495
3.14.6.3 SLVB0ND	495
3.14.6.4 SLVB0NF	496
3.14.7 Servo-Loop Position Variables	497
3.14.7.1 SLDRA	497
3.14.7.2 SLDRAIF	498
3.14.7.3 SLDRX	499
3.14.7.4 SLPKI	500
3.14.7.5 SLPKIIF	501
3.14.7.6 SLPKISF	501

3.14.7.7 SLPKITF	502
3.14.7.8 SLPLI	502
3.14.7.9 SLPKP	503
3.14.7.10 SLPKPIF	504
3.14.7.11 SLPKPSF	505
3.14.7.12 SLPKPTF	505
3.14.8 Servo-Loop Compensations Variables	506
3.14.8.1 SLAFF	506
3.14.8.2 SLFRCD	507
3.14.9 Servo Loop Stepper Variables	508
3.14.9.1 MFLAGSX	508
3.14.9.2 SLSDZ	510
3.14.9.3 SLSKI	511
3.14.9.4 SLSKP	511
3.14.9.5 SLSMC	512
3.14.9.6 SLSOUT	513
3.14.9.7 SLSRL	514
3.14.10 Servo-Loop Miscellaneous Variables	514
3.14.10.1 SLCROUT	515
3.14.10.2 SLGCAXN	516
3.14.10.3 SLPROUT	517
3.14.10.4 SLP2ROUT	519
3.14.10.5 SLTFWID	521
3.14.10.6 SLVROUT	521
3.14.10.7 SLXROUT	523
3.14.11 Non-Linear Control Variables	523
3.14.11.1 SLPAP	524
3.14.11.2 SLPDP	524
3.14.11.3 SLPAI	525
3.14.11.4 SLPDI	526
3.14.11.5 SLVAP	527
3.14.11.6 SLVDP	527
3.14.11.7 SLVAI	528
3.14.11.8 SLVDI	529

3.15 Commutation Variables	529
3.15.1 SLCHALL	530
3.15.2 SLCNP	531
3.15.3 SLCOFFS	531
3.15.4 SLCORG	532
3.15.5 SLCPRD	533
3.15.6 SLHROUT	533
3.15.7 SLSTHALL	534
3.16 System Configuration Variables	535
3.16.1 CFG	536
3.16.2 CTIME	536
3.16.3 EXTFAC	537
3.16.4 FOLLOWCH	537
3.16.5 G_01WCS...G_12WCS	539
3.16.6 GPEXL	539
3.16.7 GSPEXL	540
3.16.8 GUFAC	540
3.16.9 IENA	541
3.16.10 IMASK	542
3.16.11 ISENA	543
3.16.12 S_FLAGS	544
3.16.13 S_SETUP	546
3.16.14 XSEGAMAX	548
3.16.15 XSEGAMIN	549
3.16.16 XSEGRMAX	549
3.16.17 XSEGRMIN	549
3.17 Communication Variables	550
3.17.1 BAUD	550
3.17.2 COMMCH	551
3.17.3 COMMFL	552
3.17.4 CONID	554
3.17.5 ECHO	554
3.17.6 DISPCH	556
3.17.7 GATEWAY	557

3.17.8 SUBNET	558
3.17.9 TCPIP	559
3.17.10 TCPIP2	560
3.17.11 TCPPORT	561
3.17.12 UDPPORT	562
3.18 Laser Control Variables	563
3.18.1 LCFRQ	563
3.18.2 LCDC	563
3.18.3 LCTFREQ	564
3.18.4 LCTWIDTH	564
3.19 Miscellaneous	564
3.19.1 AUNITS	565
3.19.2 AZUNITS	566
3.19.3 FK	566
3.19.4 PDESC	567
3.19.5 PN	567
3.19.6 PNAME	568
3.19.7 SN	568
3.19.8 STATIC	568
3.19.9 VR	569
3.19.10 XARRSIZE	569
4. ACSPL+ Functions	571
4.1 Arithmetical Functions	576
4.1.1 ABS	577
4.1.2 ACOS	577
4.1.3 ASIN	578
4.1.4 ATAN	578
4.1.5 ATAN2	579
4.1.6 CEIL	579
4.1.7 COS	580
4.1.8 EXP	580
4.1.9 FLOOR	581
4.1.10 HYPOT	581
4.1.11 LDEXP	582

4.1.12 LOG	582
4.1.13 LOG10	583
4.1.14 POW	583
4.1.15 SIGN	584
4.1.16 SIN	584
4.1.17 SQRT	585
4.1.18 TAN	585
4.1.19 ROUND	586
4.2 Matrix Functions	586
4.2.1 Matrix Type	587
4.2.1.1 Matrix Initialization in Compilation Time	587
4.2.2 MATRIXADD	588
4.2.3 MATRIXSUB	589
4.2.4 MATRIXMUL	590
4.2.5 MATRIXMULSC	590
4.2.6 MATRIXMULEW	591
4.2.7 MATRIXDIV	592
4.2.8 MATRIXIDENT	593
4.2.9 MATRIXTRANS	593
4.2.10 MATRIXINVERT	594
4.3 Miscellaneous Functions	594
4.3.1 GETCONF	595
4.3.2 SYSINFO	602
4.3.3 GETVAR	603
4.3.4 SETCONF	604
4.3.5 SETVAR	616
4.3.6 STR	616
4.3.7 STRTONUM	617
4.3.8 NUMTOSTR	618
4.3.9 BCOPY	619
4.3.10 SS1RESET	621
4.3.11 MDURATION	621
4.4 Array Processing Functions	624
4.4.1 AVG	625

4.4.2 COPY	625
4.4.3 DSHIFT	627
4.4.4 FILL	628
4.4.5 MAX	629
4.4.6 MAXI	630
4.4.7 MIN	630
4.4.8 MINI	631
4.4.9 SIZEOF	631
4.5 EtherCAT Functions	632
4.5.1 ECCLOSEPORT	634
4.5.2 ECCLRREG	635
4.5.3 ECEXTIN	635
4.5.4 ECEXTOUT	637
4.5.5 ECGETGRPIND	638
4.5.6 ECGETPID	639
4.5.7 ECGETMAIN	639
4.5.8 ECGETOFFSET	639
4.5.9 ECGETOPTGRP	641
4.5.10 ECGETRED	641
4.5.11 ECGETREG	641
4.5.12 ECGETREV	643
4.5.13 ECGETSLAVES	644
4.5.14 ECGETSN	644
4.5.15 ECGETSTATE	645
4.5.16 ECGETVID	645
4.5.17 ECGRPINFO	646
4.5.18 ECIN	646
4.5.19 ECOUT	648
4.5.20 ECREPAIR	650
4.5.21 ECRESCAN	651
4.5.22 ECRESCUE	652
4.5.23 ECSAVECFG	653
4.5.24 ECSAVEDCNF	653
4.5.25 ECUNMAP	654

4.5.26 ECUNMAPIN	654
4.5.27 ECUNMAPOUT	655
4.5.28 FOEDOWNLOAD	655
4.5.29 FOEUPLOAD	656
4.5.30 PDOEXT	657
4.6 CoE Functions	657
4.6.1 COEREAD	658
4.6.2 COE2READ	660
4.6.3 COEWRITE	661
4.6.4 COE2WRITE	662
4.6.5 COEGETSIZE	663
4.7 Modbus Functions	664
4.7.1 MBOPEN	665
4.7.2 MBGETHANDLE	667
4.7.3 MBCLOSE	667
4.7.4 MBREADHREG	668
4.7.5 MBREADIREG	672
4.7.6 MBWRITEHREG	676
4.7.7 MBREADCOIL	679
4.7.8 MBWRITECOIL	682
4.7.9 MBREADDIN	685
4.7.10 MBUNMAP	688
4.7.11 MBCLEAR	689
4.7.12 MBERR	690
4.7.13 #MBMAPREP	691
4.8 Servo Processor Functions	692
4.8.1 GETSP	693
4.8.2 GETSPA	694
4.8.3 GETSPV	695
4.8.4 SETSP	695
4.8.5 SETSPV	696
4.9 Signal Processing Functions	696
4.9.1 DEADZONE	697
4.9.2 DSIGN	699

4.9.3	DSTR	700
4.9.4	EDGE	701
4.9.5	INTGR	702
4.9.6	LAG	704
4.9.7	Interpolation Functions	705
4.9.7.1	Linear interpolation	705
4.9.7.2	Spline interpolation	705
4.9.7.3	MAP	711
4.9.7.4	MAPB	713
4.9.7.5	MAPN	715
4.9.7.6	MAPNB	717
4.9.7.7	MAPNS	719
4.9.7.8	MAPS	721
4.9.7.9	MAP2	723
4.9.7.10	MAP2B	725
4.9.7.11	MAP2N	727
4.9.7.12	MAP2NB	730
4.9.7.13	MAP2NS	732
4.9.7.14	MAP2S	735
4.9.7.15	MATCH	738
4.9.7.16	RAND	739
4.9.7.17	ROLL	740
4.9.7.18	SAT	741
4.10	Laser Control Functions	743
4.10.1	LCMODULATION	744
4.10.1.1	Duty cycle or frequency update	748
4.10.1.2	Duty cycle or Frequency monitoring	748
4.10.2	LCFixedDist	748
4.10.3	LCFixedInt	752
4.10.4	LCRandomDist	754
4.10.5	LCTickle	757
4.10.6	LCZone	758
4.10.6.1	LCZoneSet	760
4.10.6.2	LCZoneGet	760

4.10.7	LCStop	761
4.10.8	LCSignalSet	761
4.10.9	LCSignalGet	764
4.10.10	LCS conditioning example	764
4.10.11	Physical outputs configuration	765
4.10.11.1	LCOutputSet	765
4.10.11.2	LCOutputGet	767
4.10.12	LCDelaySet	768
4.10.13	LCDelayGet	768
4.10.14	AxListAsMask	768
4.11	Dynamic Error Compensation	769
4.11.1	ERRORMAP1D	770
4.11.2	ERRORMAPN1D	771
4.11.3	ERRORMAPA1D	772
4.11.4	ERRORMAP2D	773
4.11.5	ERRORMAPN2D	775
4.11.6	ERRORMAPA2D	777
4.11.7	ERRORMAP3DA	778
4.11.8	ERRORMAP3D2	781
4.11.9	ERRORMAP3D3	786
4.11.10	ERRORMAPN3D2	791
4.11.11	ERRORMAPN3D3	794
4.11.12	ERRORMAP3D5	798
4.11.13	ERRORMAPN3D5	801
4.11.14	ERRORMAPN3DA	804
4.11.15	ERRORMAPOFF	807
4.11.16	ERRORMAPON	807
4.11.17	#ERRORMAPREP	807
4.11.18	ERRORUNMAP	808
5.	ACSPL+ Standard Structures	809
5.1	LCI Standard Structure	809
5.1.1	LCI Functions	809
5.1.1.1	PowerPWMOut	809
5.1.1.2	PowerAnalogOut	810

5.1.1.3 PowerDigitalOut	811
5.1.1.4 FixedDistPulse	812
5.1.1.5 DistanceArrPulse	813
5.1.1.6 CoordinateArrPulse	814
5.1.1.7 Tickle	815
5.1.1.8 LaserEnable	816
5.1.1.9 LaserDisable	816
5.1.1.10 DistanceArrGate	816
5.1.1.11 CoordinateArrGate	817
5.1.1.12 AddZone	818
5.1.1.13 SetZone	819
5.1.1.14 SetCondition	819
5.1.1.15 GetCondition	822
5.1.1.16 SegmentGate	822
5.1.1.17 SegmentPulse	823
5.1.1.18 SetExtClockSync	823
5.1.1.19 PowerPWMBurst	823
5.1.1.20 SetSafetyMasks	824
5.1.1.21 Stop	824
5.1.1.22 SetMechPlatformAxes	825
5.1.1.23 SetMotionAxes	825
5.1.1.24 SetSystemDelay	826
5.1.1.25 GetSystemDelay	827
5.1.1.26 SetConfigOut	827
5.1.1.27 AssignChannels	828
5.1.1.28 SetCustomPosCalc	829
5.1.1.29 SetCustomVelCalc	829
5.1.1.30 SetCustomVelVar	830
5.1.2 LCI Structure Fields	830
5.1.2.1 MotionAxes	832
5.1.2.2 PosResolution	832
5.1.2.3 InternalPosResolution	832
5.1.2.4 PWMDutyCycle	832
5.1.2.5 PWMFrequency	832

5.1.2.6	PWMPulseWidth	833
5.1.2.7	TickleFrequency	833
5.1.2.8	TicklePulseWidth	833
5.1.2.9	PWMActive	833
5.1.2.10	TickleActive	833
5.1.2.11	InRange	833
5.1.2.12	LaserEnabled	834
5.1.2.13	OperationMode	834
5.1.2.14	Positions	834
5.1.2.15	UserPos	834
5.1.2.16	MultiAxWinSize	834
5.1.2.17	ExtraPulsesQty	835
5.1.2.18	ExtraPulsesPeriod	835
5.1.2.19	PiercePulsesNum	835
5.1.2.20	PiercePulsesWidth	835
5.1.2.21	GateOnDelay	836
5.1.2.22	GateOffDelay	836
5.1.2.23	PulseDelay	837
5.1.2.24	PowerAOutVal	837
5.1.2.25	Faults	838
5.1.2.26	PWMBurstReady	838
5.2	Diagnostics and Preventive Maintenance (DPM)	838
5.2.1	DPM_Measurement	839
5.2.1.1	DPM_Measurement Fields	839
5.2.1.2	DPM_Measurement Functions	842
5.2.2	DPM_Motion_Status	844
5.2.2.1	DPM_Motion_Status Fields	844
5.2.2.2	DPM_Motion_Status Functions	845
5.2.3	DPM Example - Adding current measurement during acceleration phase to an existing application	846
5.3	Motion Duration	847
5.3.1	MotionDuration Struct	847
5.4	Safety Zones	853
5.4.1	SafetyZone Structure and ACSPL+ Global Functions	853

5.4.1.1 SZonesEn	854
5.4.1.2 SZonesDis	854
5.4.2 SetAxes	855
5.4.3 Enable	855
5.4.4 Disable	855
5.4.5 SetZone1D	856
5.4.6 SetZone2D	856
5.4.7 SetZone3D	857
5.4.8 SetCustomZone	858
5.4.9 SetZoneDualCarriage	858
5.4.10 SetMargins	859
5.4.11 AddOffset	860
5.4.12 GetLeftLimit	861
5.4.13 GetRightLimit	862
5.4.14 GetMargin	862
5.4.15 Customized Safety Zone Definition	862
5.5 Autofocus Mode	863
5.5.1 Autofocus Signal Source Struct	863
5.5.2 Autofocus Definition Struct	866
5.5.3 ACTIVATEAF - Activation Function	869
5.5.4 DEACTIVATEAF - Deactivation Function	869
5.5.5 ASTX - Execution Status	869
5.5.6 FE	870
5.5.7 ERRF	871
5.5.8 CERRF	871
5.6 Gantry Support	873
5.6.1 CrossCouplingCompensation Struct	873
6. Terminal Commands	881
6.1 Entering Terminal Commands	881
6.2 Query Commands	881
6.2.1 Default Query Formats	883
6.2.2 Predefined Query Output Formats	883
6.2.3 User-Defined Query Output Format	885
6.3 Program Management Commands	885

6.3.1 Program Management Command Arguments	885
6.3.2 Program Buffer Commands	887
6.3.2.1 Open/Close Buffer (#)	887
6.3.2.2 D	888
6.3.2.3 F/IF	889
6.3.2.4 L	890
6.3.3 RESET	892
6.3.4 Listing Program Variables	892
6.3.4.1 VGR	892
6.3.4.2 VSD	893
6.3.4.3 VS/VSG	893
6.3.4.4 VSF/VSGF	894
6.3.4.5 VG/VGF	895
6.3.4.6 VL/VLF	895
6.3.4.7 V/VF	896
6.3.4.8 VSP	897
6.3.4.9 VST/VSGT	897
6.3.4.10 VSTF/VSGTF/VSDT	898
6.3.4.11 VGV	898
6.3.4.12 VGS/VGSF	898
6.3.5 Program Handling Commands	899
6.3.5.1 C	900
6.3.5.2 X	901
6.3.5.3 S/SR	902
6.3.5.4 P	902
6.3.6 Debug Commands	903
6.3.6.1 XS	903
6.3.6.2 XD	904
6.3.6.3 BS	904
6.3.6.4 BR	905
6.4 System Commands	906
6.4.1 SI	906
6.4.2 SIR	910
6.4.3 MEMORY	922

6.4.4 IR	922
6.4.5 U	924
6.4.6 TD	925
6.4.7 SC	925
6.4.8 ETHERCAT	927
6.4.9 #ETHERCAT2	936
6.4.10 ECMAPREP	937
6.4.11 CC	938
6.4.12 PLC	939
6.4.13 LOG	940
6.4.14 LOG HOST_TICKS	941
6.4.15 LOGP	942
7. SPiiPlus Error Codes	944
7.1 ACSPL+ Syntax Errors	944
7.2 ACSPL+ Compilation Errors	977
7.3 ACSPL+ Runtime Errors	1001
7.4 Errors	1035
7.5 Encoder Errors	1045
7.6 System Errors	1047
7.7 EtherCAT Errors	1050
7.8 EtherCAT Slave Errors	1052
7.9 MODBUS Errors	1059
8. G-Code Error Codes	1061
8.1 G-Code Syntax Errors	1061
8.2 G-Code Compilation Errors	1065
8.3 G-Code Runtime Errors	1065
Appendix A. PEG And MARK Mapping Tables	1069
A.1 ASSIGNPEG Mapping	1069
A.2 ASSIGNPOUTS Mapping	1082
A.3 ASSIGNMARK Mapping	1104

List Of Figures

Figure 2-1. CONNECT Using MAP Function	54
Figure 2-2. DISABLE and Mechanical Brake Output Process- Positive BONTIME	61
Figure 2-3. DISABLE and Mechanical Brake Output Process - Negative BONTIME	61
Figure 2-4. ENABLE and Mechanical Brake Output Process - Positive BOFFTIME	62
Figure 2-5. ARC1 Coordinate Specification	131
Figure 2-6. ARC2 Center Point and Rotation Angle Specification	136
Figure 2-7. SLAVE /pt Illustration	152
Figure 2-8. Single-Axis Motion Using MPTP	159
Figure 2-9. Two-Axis Group Motion Using MPTP/v	160
Figure 2-10. Results of Example MSEG	162
Figure 2-11. PATH...ENDS Diagram	164
Figure 2-12. PROJECTION of the XA Plane	168
Figure 2-13. FPOS - PROJECTION Example	168
Figure 2-14. PROJECTION Example - Final Result	169
Figure 2-15. PVSPLINE Motion Diagram	173
Figure 2-16. Use of STOPPER	176
Figure 2-17. Corner Processing - Exact Path Option	183
Figure 2-18. Corner Processing - Permitted Deviation, Permitted Radius and Corner Smoothing Options	184
Figure 3-1. FRF measurement of a flexible system	396
Figure 3-2. SPTP motion where SLSFF=0 (no snap feed-forward)	397
Figure 3-3. SPTP motion where SLSFF=70 (snap feed-forward is tuned)	398
Figure 4-1. Illustration of COPY Function	627
Figure 4-2. Example Mapping	672
Figure 4-3. Example Mapping	675
Figure 4-4. Example Mapping	679
Figure 4-5. Example Mapping	682
Figure 4-6. Example Mapping	685
Figure 4-7. Symmetrical Dead Zone Example	698
Figure 4-8. Asymmetrical Dead Zone Example	699
Figure 4-9. DSIGN Function Example	700
Figure 4-10. EDGE Function Example	702

Figure 4-11. INTGR Function Example	703
Figure 4-12. LAG Function Example	705
Figure 4-13. Spline Definition Range	706
Figure 4-14. Two-Dimensional Spline Definition Range	707
Figure 4-15. 5-Point Catmull-Rom Spline	708
Figure 4-16. B-Spline - Approximation of Points	709
Figure 4-17. Catmull-Ron Spline Beyond the Definition Range	710
Figure 4-18. B-Spline Map	711
Figure 4-19. MAP Example on the Scope	713
Figure 4-20. MAPB Example on the Scope	715
Figure 4-21. MAPN Example on the Scope	717
Figure 4-22. MAPNB Example on the Scope	719
Figure 4-23. MAPNS Example on the Scope	721
Figure 4-24. MAPS Example on the Scope	723
Figure 4-25. MAP2 Example on the Scope	725
Figure 4-26. MAP2B Example on the Scope	727
Figure 4-27. MAP2N Example on the Scope	730
Figure 4-28. MAP2NB Example on the Scope	732
Figure 4-29. MAP2NS Example on the Scope	735
Figure 4-30. MAP2S Example on the Scope	738
Figure 4-31. ROLL Example on the Scope	741
Figure 4-32. SAT Example on the Scope	743
Figure 4-33. Velocity	747
Figure 4-34. 2 Pulses with Pulse Width 120 msec.	836
Figure 4-35. Delays in Gating Mode	836
Figure 4-36. Illustration of one-dimensional Safety Zone	856
Figure 4-37. Illustration of two-dimensional safety zone	857
Figure 4-38. Illustration of three-dimensional Safety Zone	858
Figure 4-39. Illustration of Dual Carriage Safety Zone Gap	859
Figure 4-40. SetMargins Enlarging Safety Zone	860
Figure 4-41. Changing Safety Zone Offsets	861
Figure 5-1. Cross Coupled System Example	878
Figure 5-2. CrossCoupling System with Example Position Values	879
Figure 6-1. Communication Terminal Window	881

Figure 6-2. Interaction of Program Buffer States

899

List of Tables

Table 2-1. The ACSPL+ command set	44
Table 2-2. Homing Methods	70
Table 2-3. DISP Command Option Escape Sequences	81
Table 2-4. Type Characters	81
Table 2-5. Channel Designation for TRIGGER	93
Table 2-6. PEG Output Signal Configuration	105
Table 2-7. Commonly Monitored SPDC Variables	118
Table 2-8. Matrix Values	168
Table 2-9. IF Control Structures	202
Table 3-1. Alphabetical Listing of All ACSPL+ Variables	222
Table 3-2. AFLAG Bit Description	236
Table 3-3. MFLAGS Bit Designators	242
Table 3-4. E_FLAGS Bit Description	269
Table 3-5. Multi-Channel Feedback Support	281
Table 3-6. Homing Methods	288
Table 3-7. AST Bit Descriptions	303
Table 3-8. MST Bit Descriptions.	310
Table 3-9. NST Bit Description	312
Table 3-10. PFLAGS Bit Description 1	411
Table 3-11. PST Bit Description	414
Table 3-12. ECST Bits	421
Table 3-13. EC2ST Bits	422
Table 3-14. Axis Fault Bits	427
Table 3-15. FDEF Bit Description	435
Table 3-16. FMASK Bit Description	439
Table 3-17. SAFINI Valid Bits	449
Table 3-18. S_FAULT Fault Bits	451
Table 3-19. S_FDEF Bit Description	455
Table 3-20. S_FMASK Bit Description	457
Table 3-21. SLCROUT Values	515
Table 3-22. SLPROUT Values	518

Table 3-23. SLVROUT Values	522
Table 3-24. IENA Bit Description	541
Table 3-25. ISENA Bit Description	544
Table 3-26. S_FLAGS Bit Description	545
Table 3-27. S_SETUP Bit Designators	546
Table 3-28. COMMCH Values	551
Table 3-29. COMMFL Bit Descriptions	553
Table 3-30. ECHO Channel Numbers	555
Table 3-31. DISPCH Channel Numbers	556
Table 4-1. GETCONF Return Values	596
Table 4-2. SYSINFO Return Values	602
Table 4-3. 16-bit Binary value Template	604
Table 4-4. SETCONF Arguments	605
Table 4-5. Supported Error Counter Registers	642
Table 4-6. Modbus Error Codes	690
Table 4-7. MAP Array	712
Table 4-8. MAPB Array	714
Table 4-9. MAPN Array	716
Table 4-10. MAPNB Array	718
Table 4-11. MAPNS Array	720
Table 4-12. MAPS Array	722
Table 4-13. MAP2	725
Table 4-14. MAP2B	727
Table 4-15. MAP2B	729
Table 4-16. MAP2NB	732
Table 4-17. MAP2NS	735
Table 4-18. MAP2S	737
Table 4-19. Condition Mask for Register 0	820
Table 4-20. Condition Mask for Register 1	820
Table 4-21. Condition Mask for Register 2	821
Table 4-22. Example of Two-dimensional Safety Zone Definition	853
Table 6-1. Line Designation	886
Table 7-1. ACSPL+ Syntax Errors	944
Table 7-2. ACSPL+ Compilation Errors	978

Table 7-3. ACSPL+ Runtime Errors	1001
Table 7-4. ACSPL+ Motion Termination Errors	1035
Table 7-5. Encoder Errors	1045
Table 7-6. ACSPL+ System Errors	1047
Table 7-7. ACSPL+ EtherCAT Errors	1050
Table 7-8. EtherCAT Slave Errors	1053
Table 7-9. Modbus Errors	1059
Table A-1. Mapping PEG Engines to Encoders (Servo Processor 0) for SPiiPlusNT/DC-LT/HP/LD	1069
Table A-2. Mapping PEG Engines to Encoders (Servo Processor 1) for SPiiPlusNT/DC-LT/HP/LD	1070
Table A-3. Mapping PEG Engines to Encoders (Servo Processor 0) for SPiiPlus CMnt/CMhv/CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa/UDMhv/UDMnt/UDMpa/UDMpm/UDMpc/UDMcb	1071
Table A-4. Mapping PEG Engines to Encoders (Servo Processor 0) for UDMlc/UDlIt/UDlhp/UDMmc/PDIcl	1072
Table A-5. Mapping PEG Engines to Encoders (Servo Processor 0) for NPMpm/NPMpc-	1073
Table A-6. General Purpose Outputs Assignment for Use as PEG Pulse Outputs (Servo Processor 0) for SPiiPlusNT/DC-LT/HP/LD	1076
Table A-7. General Purpose Outputs Assignment for Use as PEG Pulse Outputs (Servo Processor 1) for SPiiPlusNT/DC-LT/HP/LD	1076
Table A-8. General Purpose Outputs Assignment for Use as PEG Pulse Outputs (Servo Processor 0) for SPiiPlus CMnt/UDMpm/CMhv/UDMhv-	1077
Table A-9. General Purpose Outputs Assignment for Use as PEG Pulse Outputs (Servo Processor 0) for UDMnt/UDMpa/UDMcb	1078
Table A-10. Engine to Encoder Assignment for IDMxx, ECMxx, and UDMxx	1079
Table A-11. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0) for SPiiPlusNT/DC-LT/HP/LD	1082
Table A-12. SPiiPlusNT/DC-LT/HP/LD Mapping of Engine Outputs to Physical Outputs (Servo Processor 1)	1083
Table A-13. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0) for CMnt/UDMpm/UDMpc/CMhv/UDMhv	1084
Table A-14. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, OUT 0-4) for CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa	1085
Table A-15. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, OUT_5-9) for CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa	1086
Table A-16. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0) for UDMnt/UDMpa/UDMcb	1087

Table A-17. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0) for UDMlc/UDMmc/UDlIt/UDlhp/PDIcl	1088
Table A-18. NPMpm/NPMpc Mapping of Engine Outputs to Physical Outputs (Servo Processor 0)	1089
Table A-19. IDMSm/ECMSm/UDMSm Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 0-5)	1091
Table A-20. IDMSm/ECMSm/UDMSm Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 6-11)	1093
Table A-21. ECMsa/IDMSa/UDMSa Mapping of Engine Outputs to Physical Outputs (Servo Processor 0)	1095
Table A-22. ECMma / IDmma / UDMma Mapping of Engine Outputs to Physical Outputs (Servo Processor 0)	1097
Table A-23. ECMdx / IDMDx / UDMdx Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 0-5)	1100
Table A-24. ECMdx / IDMDx / UDMdx Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 6-11)	1102
Table A-25. Mark-1 Inputs to Encoders Mapping for SPiiPlusNT/DC-LT/HP/LD	1104
Table A-26. Mark-2 Inputs to Encoders Mapping for SPiiPlusNT/DC-LT/HP/LD	1106
Table A-27. Mark-1 Inputs to Encoders Mapping for with SPiiPlus CMnt/UDMpm-x/UDMpc/CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa/CMhv/UDMhv	1108
Table A-28. Mark-2 Inputs to Encoders Mapping for with SPiiPlus CMnt/UDMpm/UDMpc/CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa/CMhv/UDMhv	1109
Table A-29. IDMxx/ECMxx/UDMSm/UDMSa/UDMma Encoder Mapping	1110

1. Introduction

This document details all of the elements making up the SPiiPlus ACSPL+ Programming Language as well as the command set that may be entered through the SPiiPlus MMI Application Studio Communication Terminal for use in a SPiiPlus system.

This document is intended for the use of software engineers.

2. ACSPL+ Commands

ACSPL+ comes with a complete programming command set. The commands are divided into following categories:

- > [Axis Management Commands](#)
- > [Interactive Commands](#)
- > [PEG and MARK Commands](#)
- > [Miscellaneous Commands](#)
- > [Motion Commands](#)
- > [Program Flow Commands](#)
- > [Program Management Commands](#)
- > [Laser Control Commands](#)

Table 2-1. The ACSPL+ command set

Command	Description
ARC1	Adds an arc segment to MSEG...ENDS motion.
ARC1	Adds an arc segment to XSEG...ENDS motion
ARC2	Adds an arc segment to MSEG...ENDS motion.
ARC2	Adds an arc segment to XSEG...ENDS motion
ASSIGNMARK	Marks inputs-to-encoder assignment
ASSIGNPEG	Assigns an encoder to a PEG engine and GP physical output connection.
ASSIGNPOUTS	Assignment of physical output pins.
AXISDEF	Assigns an alias to an axis
BLOCK...END	Performs a group of ACSPL+ commands in one controller cycle.
BREAK	Immediately terminates a motion and provides smooth transition to the next motion in the motion queue.
CALL	Calls subroutine.
COMMUT	Performs auto commutation for DC brushless (AC servo) motors.
CONNECT	Defines a formula for calculating reference position (RPOS).

Command	Description
"CSCREATE" on page 54	Creates a new Local Coordinate System.
"CSDESTROY" on page 58	Cancels the active Local Coordinate System.
DC	Activates data collection.
DEPENDS	Specifies a logical dependence between a motor and axes.
DISABLE/DISABLEALL	Shuts off one or more drives. DISABLE ALL provides DISABLE operation for all axes.
DISABLEON	Disables autoroutine activation in a buffer.
DISP	Builds a string and sends it to the default communication channel.
ECRESCAN	Returns the system back to the operational state if one or more slaves underwent a reset or power cycle.
ENABLE/ENABLE ALL	Activates one or more drives
ENABLEON	Enables autoroutine activation in a buffer.
FCLEAR	Clears faults.
GO	Starts a motion that was created using the /w command option.
GOTO	Transfers program execution to another point in the program.
GROUP	Defines an axis-group for coordinate multi-axis motion.
HALT	Terminates one or more motions using a third-order deceleration profile (DEC deceleration). HALTALL provides HALT operation for all axes.
IF, ELSEIF, ELSE...END	IF command structure.
IMM	Provides on-the-fly change of motion parameters.
INPUT	Suspends program execution pending user input
INTERRUPT	Causes an interrupt that can be intercepted by the host.
INTERRUPTEX	Causes an interrupt similar to that of the INTERRUPT command.
JOG	Creates a jog motion.

Command	Description
KILL/KILLALL	Terminates one or more motions using a second-order deceleration profile and the KDEC deceleration value. KILL ALL provides KILL operation for all axes.
LCDISABLE	Stops a pulse generation process, including tickle pulses.
LCENABLE	Enables a pulse generation process with current set parameters.
LINE	Adds a linear segment to MSEG...ENDS motion.
LINE	Adds a linear segment to XSEG...ENDS motion.
LOOP...END	Loop command structure.
MASTER	Defines a formula for calculating MPOS .
MPOINT	Adds a set of points to MPTP...ENDS , PATH...ENDS or PVSPLINE...ENDS motion.
MPTP...ENDS	Creates a multipoint motion.
MSEG...ENDS	Creates a segmented motion.
ON...RET	The autoroutine structure. An the autoroutine is automatically executed when a specific condition is satisfied. The routine interrupts the currently executing program, executes the commands specified in the autoroutine body, and then returns control to the interrupted program.
PATH...ENDS	Creates an arbitrary path motion with linear interpolation between the specified points.
PAUSE	Suspends program execution in a buffer.
	Defines Incremental PEG parameters. activates the PEG engine.
	Defines the Random PEG parameters. activates the PEG engine.
POINT	Adds a point to MPTP...ENDS , PATH...ENDS , or PVSPLINE...ENDS motion.
PROJECTION	An expansion command to the MSEG...ENDS set of commands, that allows the controller to perform a three dimensional segmented motion such as creating arcs and lines on a user-defined plane.

Command	Description
PTP	Creates a point-to-point motion.
PVSPLINE...ENDS	Creates an arbitrary path motion with spline interpolation between the specified points.
READ	Reads an array from a file in the flash memory.
RESUME	Resumes program execution in a buffer.
SAFETYCONF	Configures fault processing for one or more axes.
SAFETYGROUP	Activates the fault response for all axes in the <code>axis_list</code> when any axis triggers the fault, and manages the axes as a block in response to <code>KILL/KILLALL</code> and <code>DISABLE/DISABLEALL</code> .
SEND	Same as <code>DISP</code> , but also specifies the communication channel or channels.
SET	Defines the current value of either feedback (<code>FPOS</code>), reference (<code>RPOS</code>), or axis (<code>APOS</code>) position.
SLAVE	Creates a master-slave motion.
SMOVE	Define segment of movement with transition point smoothing
SPDC	Activates data collection from a Servo Processor variable.
SPRT	Activates a real-time data transfer process from the MPU to a given Servo Processor.
SPRTSTOP	Stops an active real-time data transfer process on the given SP (for cyclic command only).
SPINJECT	Initiates the transfer of MPU real-time data to the Servo Processor.
SPLIT	SPLIT breaks apart an axis group. SPLITALL breaks apart all axis groups. See <code>GROUP</code> .
SPTP Command	a multi-axis Point-to-Point motion command, using a 4th order profile
START	Activates program execution in a buffer.
	Restarts PEG at the current position if has been issued and the <i>last_point</i> has not been reached.

Command	Description
STOP/STOPALL	Terminates program execution in a buffer.
STOPDC	Terminates data collection.
STOPINJECT	Stops the transfer of MPU real-time data to the Servo Processor.
Axis Management Commands	Halts the PEG engine for the specified axis.
STOPPER	Adds a segment separator to MSEG...ENDS motion.
TILL	Delays program execution until a specified expression produces a non-zero (true) result.
TRACK	Creates tracking motion.
TRIGGER	Specifies a triggering condition. Once the condition is satisfied, the controller issues an interrupt to the host computer.
WAIT	Delays program execution for a specified number of milliseconds.
WHILE...END	While command structure.
WRITE	Writes an array to a file in the flash memory.
XSEG...ENDS	Creates extended segment motion.

2.1 Axis Management Commands

The Axis Management commands are:

Command	Description
BREAK	Immediately terminates a motion and provides smooth transition to the next motion in the motion queue.
COMMUT	Performs auto commutation for DC brushless (AC servo) motors.
CONNECT	Defines a formula for calculating reference position (RPOS).
"CSCREATE" on page 54	Creates a new Local Coordinate System.
"CSDESTROY" on page 58	Cancels the active Local Coordinate System.
DEPENDS	Specifies a logical dependence between a motor and axes.

Command	Description
DISABLE/DISABLEALL	Shuts off one or more drives. DISABLE ALL provides DISABLE operation for all axes.
ENABLE/ENABLE ALL	Activates one or more drives
ENCINIT	Used for encoder configuration
FCLEAR	Clears faults.
FOLLOW	Switches an axis into slave mode.
GO	Starts a motion that was created using the /w command option.
GROUP	Defines an axis-group for coordinate multi-axis motion.
HALT	Terminates one or more motions using a third-order deceleration profile (DEC deceleration). HALTALL provides HALT operation for all axes.
HOME	Recives parameters for a predefined set of homing methods.
IMM	Provides on the fly change of motion parameters.
KILL/KILLALL	Terminates one or more motions using a second-order deceleration profile and the KDEC deceleration value. KILLALL provides KILL operation for all axes.
SAFETYCONF	Configures fault processing for one or more axes.
SAFETYGROUP	Creates a safety axis group. When any axis in the group triggers a fault, the fault affects all axes in the group.
SET	Defines the current value of either feedback (FPOS), reference (RPOS), or axis (APOS) position.
SPLIT	SPLIT breaks apart an axis group - see GROUP . SPLITALL breaks apart all axis groups.
UNFOLLOW	Switches an axis into regular mode.

2.1.1 BREAK

Description

BREAK immediately terminates the currently executed motion of the specified axis without building a deceleration profile, and initiates the next motion in the axis motion queue, if it exists.

Syntax**BREAK** *axis_list***Arguments***axis_list*

Axis or list of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Comments**BREAK** executes differently in the following cases:

1. When the next motion waits in the motion queue, **BREAK** terminates the current motion and starts the next motion immediately.
2. When there is no next motion in the motion queue **BREAK** has no immediate effect. The current motion continues until the next motion appears in the motion queue. At that moment the controller breaks the current motion and provides a smooth velocity transition profile from motion to motion. If the current motion finishes before the next motion comes to the queue, the command has no effect.

COM Library Methods and .NET Library Methods

Break

C Library Functions

acsc_Break

Example

```

PTP 0, 1E8           !Move to point 1E8
PTP 0, -1E8          !Add another motion to the motion queue
WAIT 10000           !Wait 10 seconds
BREAK 0              !Terminate the first motion (PTP 0, 1E8)
                     !and immediately start the next motion:
                     !(PTP 0, -1E8)
STOP                 !End program

```

2.1.2 COMMUT**Description****COMMUT** performs auto commutation and may be used when the following conditions hold true:

- > The motor is DC brushless (AC servo)
- > The motor is enabled
- > The motor is idle
- > The axis is already configured and properly tuned



Versions 2.60 and higher supports **COMMUT** in GANTRY mode. Commutation of the primary axis will automatically trigger commutation of the secondary axis.

Syntax

COMMUT *axis* [,*excitation_current*][,*settle_time*][,*slope*][,*gantry_commut_delay*]

Arguments

<i>axis</i>	The affected axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>excitation_current</i>	Optional - Given as a percentage of the full current command. The default value is set to 98% of XRMS .
<i>settle_time</i>	Optional - Determines settling time for the auto commutation process initiated by COMMUT . The default value is 500 msec. The entire auto commutation process lasts approximately three times longer, since the command executes the algorithm three times for verification.
<i>slope</i>	Optional - The slope parameter is optional, and used only in special cases. If a value is assigned to this parameter then the excitation current command builds up with some slope. The parameter sets the duration of the current build-up process in milliseconds. It is usually recommended to omit this parameter, in which case the excitation current is built instantly.
<i>gantry_commut_delay</i>	Optional – can be used only in Gantry mode. It defines the delay time in milliseconds after the commutation of the primary axis is completed and before the commutation of the complimentary axis begins. The default value is 500 msec.

Comments

COMMUT is generally used in auto commutation-based startup programs.

The excitation current, settle time and slope are optional parameters for the auto commutation process initiated by **COMMUT**.

Refer to the relevant section in the Setup Guide for a complete description of the commutation process.

COM Library Methods and .NET Library Methods

Commut, WaitMotorCommutated

C Library Functions

acsc_Comm, acsc_WaitMotorCommutated

```
COMMUT 0,80,100,30      !Commut 0 axis with an excitation current
                        !of 80%. Settling time is 100msec, and a
                        !current build-up slope of 30msec.
```

2.1.3 CONNECT

Description

CONNECT defines a formula for calculating reference position (**RPOS**). This formula can include any other axes variables. **DEPENDS** must follow **CONNECT**.

Syntax

CONNECT *axis_RPOS = formula*

Comments

Care needs to be taken when using complex non-default connections. Especially with articulated robots, the non-default connections can involve inverse trigonometric functions, square roots, division, and other mathematical operations that can cause numerical errors when not properly posed. While it is recommended that **CONNECT** command be written to avoid this from occurring, it is not always possible; therefore proper handling of the numerical errors is necessary.

The following are general guidelines concerning the **CONNECT** command:

1. The default relation between an axis position (**APOS**) and its reference (motor) position (**RPOS**) is 1:1.
2. Defining a different relation can be very useful for mechanical error corrections, dynamic error compensation, backlash compensation, inverse kinematics and more.
3. If the **CONNECT** relation is based on another axis position, it creates a strict link (like a mechanical connection) between all defined axes for as long as the function is active.
4. The variable **MFLAGS**<*axis*>.17 (bit 17) disables or enables a customized (non-default) **CONNECT** formula definition. See **MFLAGS**.
5. After **CONNECT** it is recommended to initialize **ROFFS** with the first value in the correction table (as seen in the following example:

```
SET RPOS0=MAP (APOS0, ARRAY, 100, 200)
```

This forces **ROFFS** to be zero and prevents the creation of a constant offset to **RPOS**.

6. **ENABLE/ENABLE ALL**, **DISABLE/DISABLEALL** and **KILL/KILLALL** change the value of **ROFFS**. Therefore, if these commands follow **CONNECT**, then redefine the **CONNECT** formula, and **RPOS** should be initialized to nearest value.
7. To stop motion after using **CONNECT**, use **HALT** instead of **KILL**. **HALT** does not affect the **ROFFS** variable.

If a numerical error occurs when evaluating a non-default connection, the output sent to **RPOS** is undefined. As such it is recommended to toggle back to the default connection and then go back to non-default connection.

When going back to the default connection the simplest way is to set **MFLAGS**().17. When this happens **RPOS** does not change, but **APOS** will change and be set to **RPOS**. However, this sudden change of **APOS** may also cause a numerical error if **APOS** is used in a **CONNECT** function. If this happens **MFLAGS**().17 will be set, but the non-default connection will still be active.

A more robust way of handling this change is to first explicitly change the connect function of all applicable axes to **RPOS = APOS**. When this happens neither **RPOS** nor **APOS** will change

instantaneously, so no numerical error should occur. Then MFLAGS().17 can be set without causing a numerical error.

Related ACSPL+ Commands

DEPENDS

Related ACSPL+ Variables

RPOS, APOS, ROFFS

```

GLOBAL REAL ARRAY(6)           !Define ARRAY.
ARRAY(0)=60;ARRAY(1)=40;ARRAY(2)=90;ARRAY(3)=-40;ARRAY(4)=60; ARRAY(5)=10
                                !Populate Correction point ARRAY for MAP
                                !function.
MFLAGS(0).17=1                 !Set default connection between APOS and
                                !RPOS (RPOS=APOS).
ENABLE 0                        !Enable 0 axis.
MFLAGS(0).17=0                 !Set non-default connection between APOS
                                !and RPOS (RPOS is a function of APOS).
CONNECT RPOS0=APOS0+MAP(APOS0,ARRAY,100,200)
                                !CONNECT formula between RPOS0 and
                                !APOS0 using the MAP function with
                                !correction table ARRAY.
DEPENDS 0,0                     !Assign Axis 0 to Motor 0. See DEPENDS.
SET APOS0=0;SET RPOS0=0        !Initialize APOS and RPOS at 0.
PTP 0, MAP(APOS0,ARRAY,100,200)
                                !Moves axis 0 to the first point in the
                                !correction ARRAY to avoid a constant
                                !offset in ROFFS, as explained in Comment 2.
PTP 0, 1300                    !Move to 1300. Each point during the motion
                                !is modified according to the correction
                                !ARRAY in the MAP function.
STOP                           !End program

```

This illustrates the results of the example on the SPiiPlus MMI Application Studio Scope.

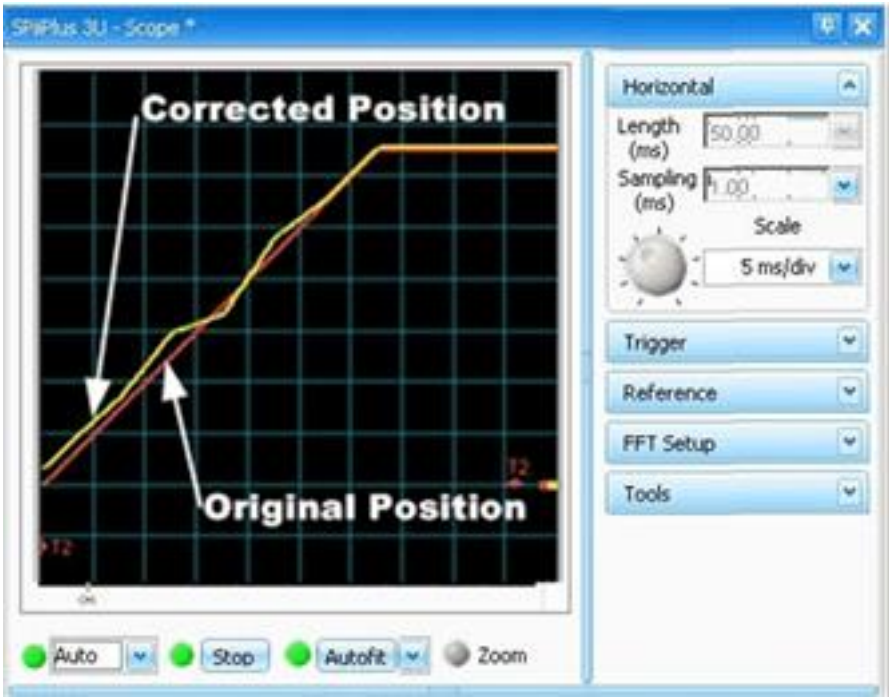


Figure 2-1. CONNECT Using MAP Function

2.1.4 CSCREATE

Description

The **CSCREATE** command creates the new Local Coordinate System (LCS) relative to the Machine Coordinate System or the previous LCS, depending on the applied switches.

Syntax

```
CSCREATE[/r] axis_list, x_trans, y_trans[, rot_axis, rot_angle]
```

or

```
CSCREATE[/r] axis_list, x_trans, y_trans, z_trans[, rot_axis, rot_angle]
```

Arguments

axis_list	The group of 2 or 3 axes. Valid values are: 0, 1, 2 ... up to the number of axes in the system minus 1.
x_trans	The new X position of the LCS in user units
y_trans	The new Y position of the LCS in user units
z_trans	The Z position of the LCS in user units. This parameter is included when axis_list includes 3 axes.

rot_axis	(Optional) The rotation axis: 0 – X, 1 – Y, 2 – Z
rot_angle	(optional) Rotation angle value: (-3.14159 : +3.14159) in radians

Switches

/r	The new LCS is relative (additive) to the existing LCS (otherwise the new LCS is relative to the Machine Coordinate System).
-----------	--

Comments

- > In function calls which include rotation parameters, the translation parameters are applied to the system first, and then the rotation parameters.
- > The enumeration of axes for the **rot_axis** parameter is a numbered list of axes in the newly created coordinate system. This enumeration relates to the virtual axes, not the physical axes of the system.
- > This command is supported in version 3.10 and higher.

Example 1

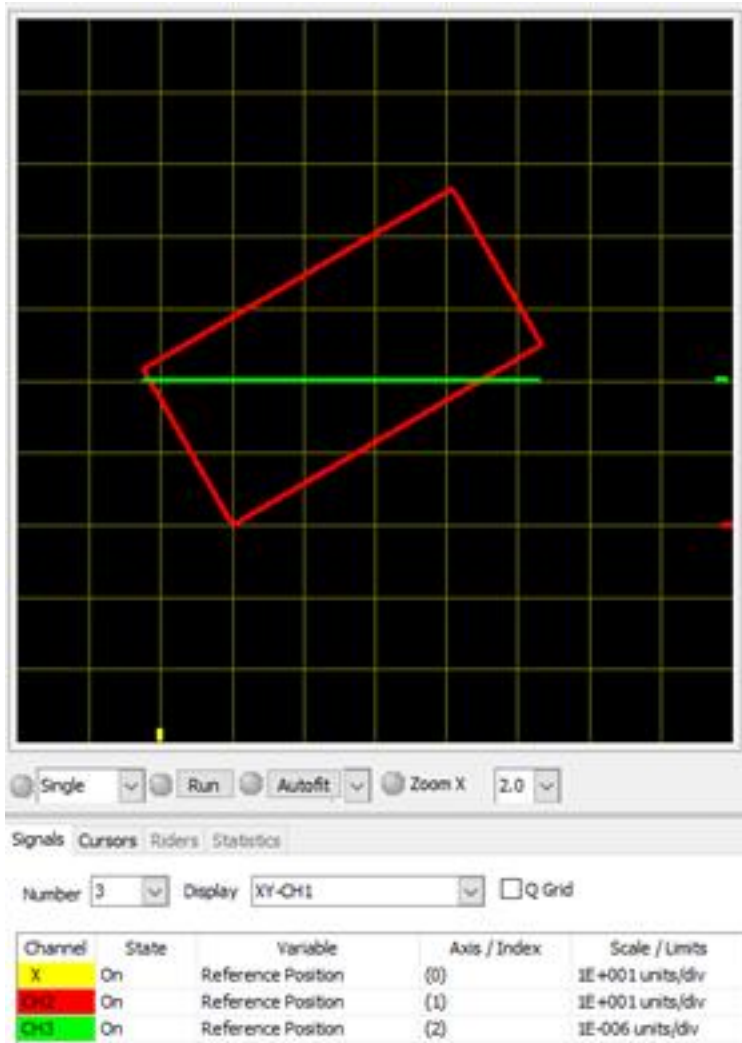
This example demonstrates rectangular motion in PTP mode.

```
!Create LCS with origin (10, 0, 0) and rotated 30 degrees around Z
REAL ang = 30* 3.141592 /180
CSCREATE (X,Y,Z), 10, 0, 0, 2, ang

PTP/ze (X,Y), 0, 0      !go to the beginning position

!Make rectangular motion
PTP/ze (X,Y),  0, 25
PTP/ze (X,Y), 50, 25
PTP/ze (X,Y), 50,  0
PTP/ze (X,Y),  0,  0

CSDESTROY (X,Y,Z)      !restore machine coordinate system
```



Example 2

This example demonstrates rectangular motion in MPTP mode.

```
!Create LCS with origin (10, 0, 0) and rotated 30 degrees around Z
REAL ang = 30* 3.141592 /180
CSCREATE (X,Y,Z), 10, 0, 0, 2, ang
PTP/ze (X,Y), 0, 0      !go to the beginning position
MPTP/z (X,Y)
    POINT (X,Y), 0, 25
    POINT (X,Y), 50, 25
    POINT (X,Y), 50, 0
    POINT (X,Y), 0, 0
ENDS (X,Y)
TILL GSEG(X) = -1      !Wait until motion complete
CSDESTROY (X,Y,Z)      !restore machine coordinate system
```

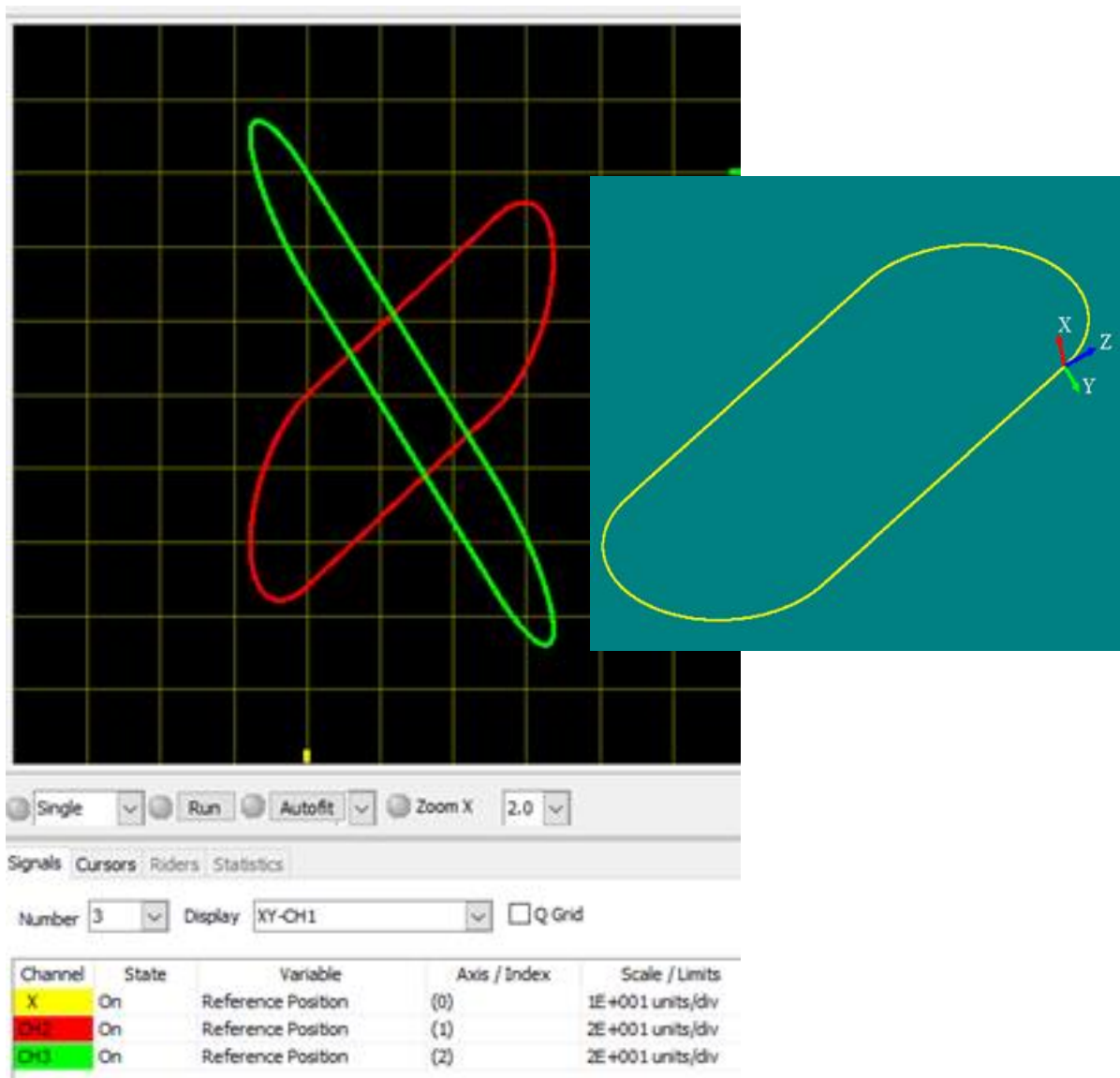
Example 3

This example demonstrates round rectangular motion in XSEG mode.

```
!Create LCS rotated 30 degrees around X
REAL ang = 30* 3.141592 /180
CSCREATE (X,Y,Z), 10, 0, 0, 0, ang
ang = 75* 3.141592 /180
CSCREATE/r (X,Y,Z), 0, 0, 0, 1, ang !Additional rotation 75 deg. Around Y

PTP/ze (X,Y), 0, 0      !go to the beginning position

XSEG/z (X,Y), 0, 0
    LINE (X,Y), 100, 0
    ARC2 (X,Y), 100, -30, -3.14159
    LINE (X,Y), 0, -60
    ARC2 (X,Y), 0, -30, -3.14159
ENDS (X,Y)
TILL GSEG(X) = -1      !Wait until motion complete
CSDESTROY (X,Y,Z)      !restore machine coordinate system
```



2.1.5 *CSDESTROY*

Description

The **CSDESTROY** command cancels the active Local Coordinate System and sets the Machine Coordinate System or previous Local Coordinate System.

Syntax

```
CSDESTROY axis_list[, restore_flag]
```

Arguments

axis_list	The group of 3 axes. Valid values are: 0, 1, 2 ... up to the number of axes in the system minus 1.
restore_flag	(Optional) Set to 1 to restore the previous LCS; 0 or omitted value restores the MCS.



Unpredictable and dangerous motion may result if **CSDESTROY** is called before the motion involving the coordinate system created by **CSCREATE** is complete.

Comments

This command is supported in version 3.10 and higher.

Examples

See [CSCREATE](#).

2.1.6 DEPENDS**Description**

DEPENDS is used only following [CONNECT](#).

DEPENDS specifies a logical dependence between a physical axis (motor) and the same or other logical axes. By default, the physical axis (motor) is assigned to its axis. **DEPENDS** is necessary because the controller is generally not capable of deriving dependence information from the **CONNECT** formula.

Syntax

DEPENDS *physical_axis*, **axis_list**

Arguments

physical_axis	Physical axis (motor).
axis_list	Axis or list of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Comments

- > Once a **CONNECT** command is executed, the controller resets the motor dependence information to the default-the motor depends only on the corresponding axis.
- > If a connection formula actually causes the motor to be dependent on another axis / axes, the **DEPENDS** command must follow to specify actual dependence.

Related ACSPL+ Commands

[CONNECT](#)

Example

See the examples from [CONNECT](#).

2.1.7 DISABLE/DISABLEALL

Description

DISABLE deactivates one, several, or using **DISABLEALL**, all drives. After **DISABLE**, **RPOS** = **FPOS** which means that no position error exists, or **PE**<*axis*> = 0.

Syntax

DISABLE *axis_list* [,*reason*]

Arguments

<i>axis_list</i>	Axis, or list of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>reason</i>	<i>reason</i> is an optional parameter. <i>reason</i> must be an integer constant or expression that equates to an integer and specifies a reason why the motor was disabled. If the parameter is specified, its value is stored in the MERR variable. If the parameter is omitted, MERR stores zero after the disable operation.

Related ACSPL+ Commands

ENABLE/ENABLE ALL, **FCLEAR**, **DISABLEALL**

Related ACSPL+ Variables

MERR, **FPOS**, **RPOS**

COM Library Methods and .NET Library Methods

Disable, DisableM, DisableAll

C Library Functions

acsc_Disable, acsc_DisableM, acsc_DisableAll

The examples illustrate the **DISABLE** process using positive and negative **BONTIME** values.

The examples are followed by screens illustrating the results of **DISABLE** for both positive and negative **BONTIME**.

Example1:

```
DISABLE (0,1)                !Disables the 0 and 1 motors. A fault
                             !notification window will be displayed.
```

Example 2:

```
DISABLE (0,1), 6100          !Disable 0 and 1 motors, store 6100 as the
                             !reason. 6100 specifies a user-defined error
                             !code which is stored in MERR.
```

Example 3:

```
DISABLE ALL                  !Disable all motors.
```

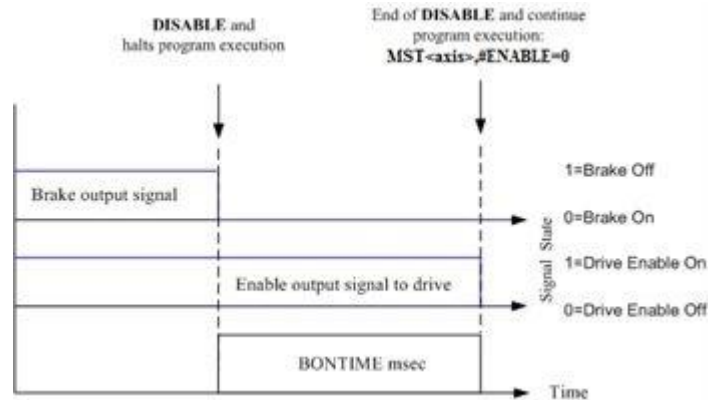



Figure 2-2. DISABLE and Mechanical Brake Output Process- Positive BONTIME

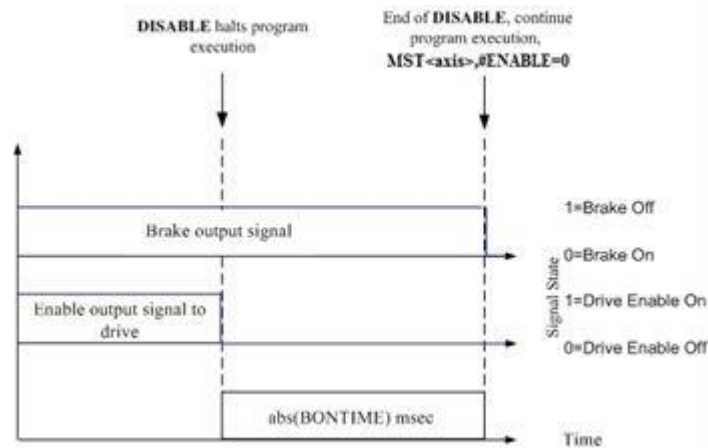


Figure 2-3. DISABLE and Mechanical Brake Output Process - Negative BONTIME

2.1.8 ENABLE/ENABLE ALL

Description

ENABLE activates one or more axes. After **ENABLE**, the motor starts following the reference position (**RPOS**) and axis faults are enabled. **ENABLE ALL** activates all axes.

Syntax

ENABLE|ENABLE ALL *axis_list*

Arguments

axis_list

Axis or list of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Comments

Motor specification is a single axis like 0 or 13, a string of axes enclosed in parentheses and separated by commas, for example: (0,1,13), or the keyword: **ALL** for all axes.

Related ACSPL+ Commands

[DISABLE/DISABLEALL](#)

Related ACSPL+ Variables

ENTIME, MFLAGS<axis>.#ENMOD

COM Library Methods and .NET Library Methods

Enable, EnableM, Wait Motor Enabled

C Library Functions

acsc_Enable, acsc_EnableM, acsc_WaitMotorEnabled

Example1:

```
ENABLE 0                                !Enable 0 axis.
```

Example 2:

```
ENABLE (0,1)                            !Enable 0 and 1 axes.
```

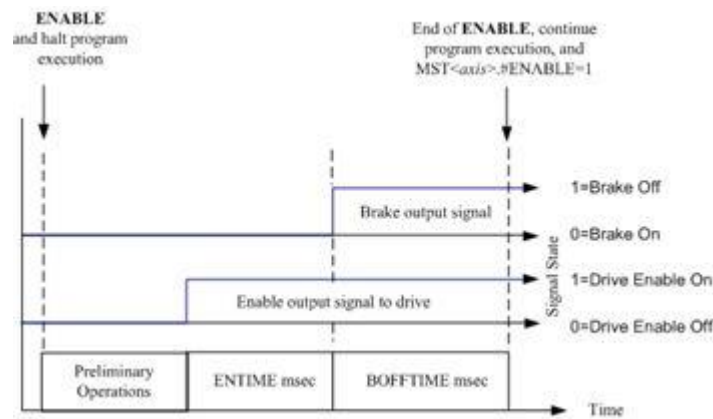


Figure 2-4. ENABLE and Mechanical Brake Output Process - Positive BOFFTIME

2.1.9 ENCINIT

Description

The **ENCINIT** function is used for encoder configuration. The **ENCINIT** function, if executed in a buffer, will wait until initialization is finished.

Syntax

```
ENCINIT (Axis, E_type, [Primary[, Slabits[, E_par_a[, E_par_b[, E_par_c [, e_freq[, e_scmul[, E_aoffs[, ESTBITS[, EMTBITS]]]]]]]]))
```

Arguments

Axis	The affected axis, valid number are: 0,1,2.. up to the number of axis in the system minus 1.
E_type	Encoder Type, according to ACSPL+ E_TYPE variable definition.

Primary	Optional. Defines the feedback – can be primary or secondary. Possible values: Primary: 1 – default Secondary: 0
Slabits	(Optional, Integer) Used for setting the total number of absolute position bits for an absolute encoder, according to ACSPL+ SLABITS variable definition.
E_par_a	(Optional, Double) Used for setting the encoder data transmission actual frequency in MHZ. According to ACSPL+ E_PAR_A definition.
E_par_b	(Optional, Integer) Used for setting the encoder data control CRC code. According to ACSPL+ E_PAR_B definition.
E_par_c	(Optional, Integer) Used for setting the interval (in microseconds) of encoder position reading. According to ACSPL+ E_PAR_C definition.
E_freq	(Optional, Integer) Used for defining the maximum encoder pulse frequency (in MHZ). According to ACSPL+ E_FREQ definition.
E_scmul	(Optional, Integer) Used for specifying the Sin-Cos multiplication factor for the encoder. According to ACSPL+ E_SCMUL definition.
E_aoffs	(Optional, double) Used for setting user-defined offset for absolute encoder. According to ACSPL+ E_AOFFS definition.
ESTBITS	(Optional, integer) Used for setting the single turn resolution (number of bits).
EMTBITS	(Optional, integer) Used for setting the multi turn resolution (number of bits).

Return Value

None

Comments

If an optional parameter is specified, the relevant ACSPL+ variable is modified as well. Otherwise, the initialization of the encoder will use the existing value of the variable.

If the Primary parameter is set to 0, secondary feedback variables are affected or used during initialization, according to the following table.

If one of the parameters is out of range, error 3041 "Assigned value is out of range" given. Not allowed E_TYPE value will trigger error 3194 "Not allowed Encoder Type".

If ESTBITS or EMTBITS are not 0 and SLABITS is not equal to ESTBITS+EMTBITS, an error is triggered.

This function is supported in version 3.00 and higher.

Primary Feedback ACSPL+ Variable	Secondary Feedback ACSPL+ Variable
SLABITS	S2LABITS
E_PAR_A	E2_PAR_A
E_PAR_B	E2_PAR_B
E_PAR_C	E2_PAR_C
E_AOFFS	E2_AOFFS
E_FREQ	E2_FREQ
E_SCMUL	E2_SCMUL
ESTBITS	E2STBITS
EMTBITS	E2MTBITS

Example

```
ENCINIT (0,10)! triggers initialization of Endat 2.2 for axis 0, primary
!feedback
```

```
STOP
```

2.1.10 ENCREAD**Description**

The **ENCREAD** function is used for reading encoder parameters. The function should be executed in a buffer, and it will wait till the execution is completed.

Syntax

```
INT ENCREAD (Axis, E_type, ParamType[, Primary])
```

Arguments

Axis	The affected axis, valid number are: 0,1,2.. up to the number of axes in the system minus 1
E_type	Encoder Type, according to ACSPL+ E_TYPE variable definition.
ParamType	Parameter Type to read. Supported values are: > 0 – Resolution (total number of bits, single turn + multi turn) > 1 – Maximum Frequency (in KHz)

Primary

Optional. Defines the feedback – can be primary or secondary.

Possible values:

- > Primary: 1 (default)
- > Secondary: 0

Return Value

Parameter value according to the requested ParamType.

Comments

Only Endat encoders are supported. For any other **E_TYPE**, the functions returns error 3196 “Requested Absolute Encoder is not supported”.

E_TYPE value is changed to 10 (Endat) after the function is called.

The function can be called only if the axis is disabled.

The function can be called only from a buffer; if called from the terminal, error 2073 is returned.

This variable is supported in version 3.10 and higher.

Example

```
IO=encread(0,10,0) !Axis 0, Endat encoder, Resolution
I1=encread(0,10,1) !Axis 0, Endat encoder, Max Frequency
```



This command is supported in the IDMSm/IDMSa/ECMSm/ECMSa/UDMSm/UDMSa products only, and only for EnDAT encoders.

2.1.11 FCLEAR

Description

FCLEAR clears the **FAULT** variable and the results of the previous fault stored in **MERR**.

Syntax

FCLEAR *axis_list*

Arguments

axis_list

Axis or list of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Comments

- > Motor specification is a single axis like 0 or 13, a string of axes enclosed in parentheses and separated by commas, for example: (0,1,13), or the keyword: ALL for all axes.
- > If the axis designation is omitted the command clears the system faults. If an axis is specified, the command clears the **FAULT** and **MERR** components for the specified axes. However, if the reason for the fault is still active, the controller will immediately set the fault again following **FCLEAR**.

- > If one of the cleared faults is an encoder error, **FCLEAR** also resets the feedback position to zero.

Related ACSPL+ Variables

MERR, FAULT

COM Library Methods and .NET Library Methods

FaultClear, FalutClearM

C Library Functions

acsc_FaultClear, acsc_FaultClearM

Example 1:

```
FCLEAR (0,1)           !Clear FAULT and MERR variables for 0 and 1 axes
```

Example 2:

```
FCLEAR ALL             !Clear FAULT and MERR variables for all axes
```



The FCLEAR.ALL command may cause an increase in USAGE.

2.1.12 FOLLOW

Description

FOLLOW switches an axis into slave mode. The specified axis will follow the profile generated by the RTC6.

Syntax

FOLLOW(axis)

Argument

axis	Axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
------	---

2.1.13 GO

Description

GO starts a motion that has been created using the /w (wait) switch.

Syntax

GO axis_list

Arguments

axis_list	Axis or list of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
-----------	---

Comments

- > Motor specification is a single axis like 0 or 13, a string of axes enclosed in parentheses and separated by commas, for example: (0,1,13), or the keyword: ALL for all axes.
- > Where **GO** specifies a single axis, that axis may not be included in any group. **GO** starts the last created motion for the same axis. If the motion was not created, or has been started before, the command has no effect.
- > Where **GO** specifies a leading axis in a group, **GO** starts the last created motion for the same axis group. If the motion was not created, or has been started before, the command has no effect.

COM Library Methods and .NET Library Methods

Go, GoM

C Library Functions

acsc_Go, acsc_GoM

Related ACSPL+ Commands

HALT , MSEG...ENDS, JOG, MPTP...ENDS, PATH...ENDS, PTP, PVSPLINE...ENDS, SLAVE, TRACK

Example

```
PTP/w (0,1), 1000, 1000      !Create PTP motion, but do not start
PTP/w 3, 8000               !Create PTP motion, but do not start
GO (0,1)                    !Start both motions at the same time
```

2.1.14 GROUP**Description**

GROUP defines an axis group for coordinate multi-axis motion. The first axis in the axes list is the leading axis. The motion parameters of the leading axis become the default motion parameters for all axes in the group. Motion on all axes in a group will start and conclude at the same time.

Syntax**GROUP** *axis_list***Arguments***axis_list*

Axis or list of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Comments

An axis can belong to only one group at a time. If the application requires restructuring the axes, it must split the existing group and only then create the new group.

Related ACSPL+ Variables

VEL, ACC, DEC, JERK, KDEC, GVEC, GVEL, GACC, GJERK

Related ACSPL+ Commands

SPLIT

COM Library Methods and .NET Library Methods

Group

C Library Functions

acsc_Group

Example

```
GROUP (0,1)           !Creates an axis group that includes axes 0 and 1.  
PTP (0,1), 1000, 10000 !PTP axis 0 to 1000, and axis 1 to 10000.
```

2.1.15 HALT

Description

In single axis motion, **HALT** terminates currently executed motion and clears all other motions waiting in the axis motion queue. The deceleration profile is defined by **DEC** (deceleration variable).
In group motion, **HALT** terminates currently executed motion of all group axes, and clears all other motions waiting in the axes motion queues. The deceleration profile is defined by the **DEC** (deceleration) variable of the leading axis.

Syntax

HALT *axis_list*[*reason*]

Arguments

<i>axis_list</i>	List of axes to be halted.
<i>reason</i>	An optional argument, which must be an integer constant or expression that equates to an integer, and specifies a cause why the axis was halted.

Switches

/e	Wait for motion termination before executing the next command.
----	--

Comments

HALT ALL terminates the motion of all axes.

Related ACSPL+ Commands and .NET Library Methods

[KILL/KILLALL](#)

COM Library Methods

Halt, HaltM

C Library Functions

acsc_Halt, acsc_HaltM

Example 1:

```
HALT 0           !Terminates 0 axis motion, the deceleration is  
                !according to DEC(0)
```


Example 2:

```
HALT (0,2)           !Terminates currently executed motion on axes
                    !0 and 2.
```

Example 3:

```
HALT ALL             !Terminates currently executed motion on all axes.
```

2.1.16 HOME**Description**

The predefined **HOME** command may receive the following parameters: Axis, HomingMethod (optional), HomingVel (optional), MaxDistance (optional), HomingOffset (optional), HomeCurrLimit (optional), HardStopThreshold (optional), SetYawToOpen (optional), SkewValue (optional), LookForTwoLS (optional), LookForTwoIndexes (optional).

If the "/e" switch is used, the optional "Timeout" parameter is also used.

Syntax

Without the "/e" switch

```
HOME Axis, [HomingMethod, HomingVel, MaxDistance, HomingOffset,
HomingCurrLimit, HardStopThreshold, SetYawToOpen, SkewValue, LookForTwoLS,
LookForTwoIndexes]
```

With the "/e" switch

```
HOME Axis, [HomingMethod, HomingVel, MaxDistance, HomingOffset,
HomingCurrLimit, HardStopThreshold, SetYawToOpen, SkewValue, LookForTwoLS,
LookForTwoIndexes, Timeout]
```

Arguments

Axis	axis index, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
HomingMethod	Optional. The number of homing method that should be used for homing. If not specified: the homing method is set according to the value of the HOMEDEF variable.
HomingVel	Optional. The velocity that will be used for the homing. If not specified: SLCPRD*EFAC/2
MaxDistance	Optional. The Maximum distance that will be used during homing. If not specified, endless motion will be used.

HomingOffset	Optional. The machine home position is found during homing. If not specified, 0 is used. If specified, after homing is completed, the zero position is offset from the home position by adding the home offset of the home position.
HomingCurrLimit	Optional. Current Limit during the homing process. If not specified: $\min(XCURV, 0.5 * XRMSP, 0.5 * XRMSP)$
HardStopThreshold	Optional. If specified, Hard Stop will be identified by: $Abs(PE) > \min(HardStopThreshold, CERRV * 0.75)$ If not specified: $Abs(PE) > CERRV * 0.75$
SetYawToOpen	For Gantry mode only. Optional. 0 – Complementary Axis keeps OPEN LOOP state during homing operation 1- Complementary Axis is set to OPEN LOOP state during homing operation
SkewValue	For Gantry mode only. Optional. Used for setting positing for the complementary axis after the homing process is finished. If not specified, the value is 0.
LookForTwoLS	For Gantry mode only. Optional. 0 (default) - only one limit is detected 1- both limits are detected
Timeout	Optional parameter. The parameter specifies the Timeout for the HOME command, in milliseconds. By default, there is no limitation.

Table 2-2. Homing Methods

Method Number	Explanation
1	Homing Method 1: Homing on the negative limit switch and index pulse
2	Homing Method 2: Homing on positive limit switch and index pulse
17	Homing Method 17: Homing on Negative Limit Switch
18	Homing Method 18: Homing on Positive Limit Switch
33/34	Homing Method 33 and 34: Homing on the index pulse
37	Homing Method 37: Homing on current position

Method Number	Explanation
50	Homing Method 50: Negative Hard Stop and index pulse (ACS Specific)
51	Homing Method 51: Positive Hard Stop and index pulse (ACS Specific)
52	Homing Method 52: Negative Hard Stop (ACS Specific)
53	Homing Method 53: Positive Hard Stop (ACS Specific)

Comments

- > The **HOME** command is non-blocking, unless the "/e" switch is used.
- > **MFLAGS.#HOME** bit will be set to 1 after the homing is completed.
- > **AST.#INHOMING** bit is 1 during the homing process • **E_TYPE** and other encoder initialization processes will reset the **#HOME** bit.
- > The predefined homing methods are defined according to the DS402 standard, except methods 50, 51, 52, 53 .



- > If the homing method is not supported, the error 3314 "Requested Homing Method is not supported" is given.
- > The axis is required to be enabled, commutated, and not in motion.
- > Disable axis during homing process will cancel the homing process.
- > Other motions cannot be executed while the axis is in home process
- > The following homing methods are supported in Gantry mode: 1, 2, 50, and 51.

Example

```

ENABLE 0
COMMUT 0
HOME 0,34          !homing on Positive Index Pulse
TILL MFLAGS0.#HOME=1

STOP

```

2.1.17 IMM

In single axis motion, **IMM** provides on-the-fly changes of the following motion parameters:

- > **VEL** (Velocity)
- > **ACC** (Acceleration)
- > **DEC** (Deceleration)
- > **JERK** (Jerk)

IMM affects the motion in progress and all motions waiting in the corresponding motion queue.

In group motion, **IMM** provides on-the-fly changes for the motion parameters of the leading axis only.

Syntax

IMM *axis_motion parameter=value or formula*

Arguments

<i>axis_motion parameter</i>	The motion parameter with the specified axis (valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1).
<i>value or formula</i>	User specified value or formula.

Related ACSPL+ Variables

[VEL](#), [ACC](#), [DEC](#), [JERK](#)

COM Library Methods and .NET Library Methods

Set Acceleration Imm, Set Deceleration Imm, Set Jerk Imm, Set Kill Deceleration Imm, Set Velocity Imm

C Library Functions

acsc_SetAccelerationImm, acsc_SetDecelerationImm, acsc_SetJerkImm, acsc_SetKillDecelerationImm, acsc_SetVelocityImm

Example

```
IMM VEL(0)=5000           !Immediately change the 0 axis velocity to 5000
```

2.1.18 KILL/KILLALL

Description

Use **KILL** after a safety event to decelerate and stop an axis faster than during normal deceleration and stop. **KILLALL** stops all axes.

In single axis motion, **KILL** terminates currently executed motion and clears all other motions waiting in the axis motion queue. The deceleration profile uses a second-order deceleration profile and the [KDEC](#) (kill deceleration) value.

In group motion, **KILL** terminates currently executed motion only for the specified axes, and clears all other motions waiting in the axis/axes motion queue. The deceleration profile uses a second-order deceleration profile and the [KDEC](#) (kill deceleration) variable of each axis.

Syntax

KILL *axis_list[,reason]*

Arguments

<i>axis_list</i>	List of axes to be killed, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1, or ALL for all axes.
<i>reason</i>	An optional argument, which must be an integer constant or expression that evaluates to an integer, and specifies a cause why the axis was killed. If the parameter is specified, its value is stored in the MERR variable. If the parameter is omitted, MERR stores zero after KILL .

Switches

<code>/e</code>	Wait for motion termination before executing the next command.
-----------------	--



For systems having more than 15 axes, the use of **KILL ALL** may cause Over Usage or Servo Processor Alarm faults.

Comments

1. **KILL ALL** terminates the motion of all axes. The deceleration profile is defined by the **KDEC** (kill deceleration) variable of each axis.
2. If several sequential **KILL** operations specify different causes for the same motor, only the first **cause** is stored in **MERR** and all subsequent causes are ignored.
3. A **cause** stored in **MERR** is cleared by **FCLEAR** or **ENABLE/ENABLE ALL**.

Related ACSPL+ Commands

HALT, **FCLEAR**, **ENABLE/ENABLE ALL**

Related ACSPL+ Variables

MERR, **KDEC**

COM Library Methods and .NET Library Methods

Kill, Kill All

C Library Functions

acsc_Kill, acsc_KillAll

Example 1:

```
KILL 1           !Kill axis 1 deceleration is according to KDEC(1)
```

Example 2:

```
KILL 0, 5011     !Kill 0 axis, store 5011 as the reason.
                  !Code 5011 corresponds to left limit error;
                  !therefore the 0 motor will be reported as
                  !disabled due to fault involving left limit.
```

Example 3:

```
KILL (0,1,2)           !Kill 0, 1 and 2 axes according to the KDEC
                        !of each specified axis.
```

Example 4:

```
KILL ALL, 6100         !Kills all axes, and stores the cause in MERR.
                        !(6100 is the code for a user-defined cause.)
```

2.1.19 SAFETYCONF

Description

SAFETYCONF configures fault processing for one or more axes, by disabling the default response to a defined axis **FAULT**, and performs one of the following responses:

- > Ignore the interrupt
- > Kill the motion
- > Disable the axis
- > Kill the motion and then disable the axis

Syntax

SAFETYCONF *axis_list*, *fault_name*, "*conf_string*"

Arguments

<i>axis_list</i>	Axis or list of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>fault_name</i>	Any axis fault name like #LL for left limit.
<i>conf_string</i>	A string enclosed in double quotation marks with one or more of the following characters determines the action of SAFETYCONF: > K (KILL/KILLALL) > D (DISABLE/DISABLEALL) > KD (KILL/KILLALL-DISABLE/DISABLEALL) > + when a fault occurs in any member of the <i>axis_list</i>, the fault response applies to all axes of the controller (each axis fault response can be unique) > - applies FMASK<<i>axis</i>>.#fault-name = 0 for all axes of the controller

Comments

If an empty string is specified, fault detection is enabled, but the controller has no response to the fault. However, an autoroutine can intercept the fault and provide a response.

In devices implementing STO, the user may not use **SAFETYCONF** to change the default response to an STO fault, which is **KILL**<axis> + **DISABLE**<axis>.

Related ACSPL+ Commands

The **#SC** Communication Terminal command shows the current fault response configuration for all axes.

Example 1:

```
SAFETYCONF 0, #PE, "K"           !The 0 motion will be killed if the 0
                                !Position Error fault occurs.
```

Example 2:

```
SAFETYCONF (3,5), #LL, "KD"      !Changes the response to the Left Limit fault
of                                of
                                !3 and 5 axes to KILL and then DISABLE.
```

Example 3:

```
SAFETYCONF ALL, #DRIVE, "D+"     !All axes will be disabled if the Drive
                                !Alarm fault occurs in any axis.
```

Example 4:

```
SAFETYCONF ALL, #VL, "-"         !Velocity Limit fault will be masked for
                                !all axes.
```

2.1.20 SAFETYGROUP

Description

SAFETYGROUP activates the fault response for all axes in the *axis_list* when any axis triggers the fault, and manages the axes as a block in response to **KILL/KILLALL** and **DISABLE/DISABLEALL**.

To cancel the defined **SAFETYGROUP**, send the command again with only the first axis as the *axis_list*.

Syntax

SAFETYGROUP *axis_list*

Arguments

axis_list

List of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

COM Library Methods and .NET Library Methods

GetSafetyInputPort

C Library Functions

acsc_GetSafetyInputPort

Example 1:

```
SAFETYGROUP (0,1,5)           !Creates safety group for axes 0, 1 and 5.
```

Example 2:

```
SAFETYGROUP 0                !Cancels the previously created safety group for
                              !axes 0, 1 and 5.
```

2.1.21 SET**Description**

SET defines the current value of either feedback (**FPOS**), reference (**RPOS**), or axis (**APOS**) position. **SET** can be initiated when the axis is disabled, or on-the-fly. **APOS** and **FPOS** are updated automatically when **SET** is specified for **RPOS**,

If a non-default **CONNECT** is used, assign different values to **APOS** and **RPOS**.

Related ACSPL+ Variables

RPOS

Syntax

SET axis_RPOS=value or formula

Arguments

axis_RPOS	The reference position of the specified axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
=	The assignment operator
value or formula	User-defined value or formula

COM Library Methods and .NET Library Methods

SetRPosition, SetFPosition

C Library Functions

acsc_SetRPosition, acsc_SetFPosition

Example

```
SET RPOS(0)=300              !Axis 0 RPOS = 300
```

2.1.22 SPLIT**Description**

SPLIT breaks down a group created using **GROUP** by designating any axis in the axis list. **SPLIT ALL** breaks down all groups.



If the **SPLIT** command specifying an axis that is currently in motion is executed within the buffer, the buffer execution is suspended until the motion is completed. However, if the **SPLIT** command is sent from the host or as a **Communication Terminal** command, it returns error 3087: "Command cannot be executed while the axis is in motion".

Syntax

SPLIT *axis_list*

Arguments

axis_list

Axis or list of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Related ACSPL+ Commands

[GROUP](#)

COM Library Methods and .NET Library Methods

Split, SplitAll

C Library Functions

acsc_Split, acsc_SplitAll

Example 1:

```
GROUP (0,1,5)      !Create an axis group including axes 0, 1 and 5.
GROUP (2,3,7)      !Create an axis group including axes 2, 3 and 7.
SPLIT 0            !Breaks down axis group 0, 1 and 5.
```

Example 2:

```
GROUP (0,1,5)      !Create an axis group including axes 0, 1 and 5.
GROUP (2,3,7)      !Create an axis group including axes 2, 3 and 7.
SPLIT ALL          !Breaks down axis group 0, 1 and 5
                  !and group 2, 3 and 7.
```

2.1.23 UNFOLLOW

Description

UNFOLLOW switches an axis into regular mode. The specified axis will follow the profile generated by the ACS controller.

Syntax

UNFOLLOW(axis)

Argument

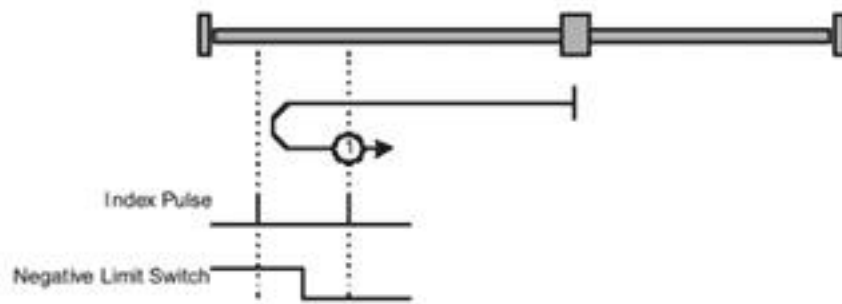
axis

Axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

2.2 Predefined Homing Methods

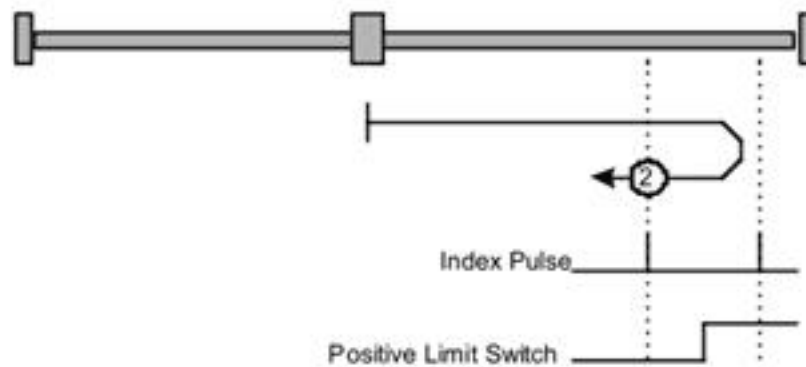
2.2.1 Homing Method 1: Homing on the negative limit switch and index pulse

With this homing method the initial direction movement is leftward if negative limit switch is inactive (shown as low in the figure above). The home position is at the first index pulse right of the position where the negative limit switch becomes active.



2.2.2 Homing Method 2: Homing on positive limit switch and index pulse

With this homing method the initial direction movement is rightward if the positive limit switch is inactive (shown as low in the figure above). The home position is at the first index pulse left of the position where the negative limit switch becomes active.



2.2.3 Homing Method 17: Homing on Negative Limit Switch

Homing method 17 is similar to method 1, except that it homes on negative limit switch only.

If negative limit switch is ON when the function is called, the homing is aborted (no homing attained).

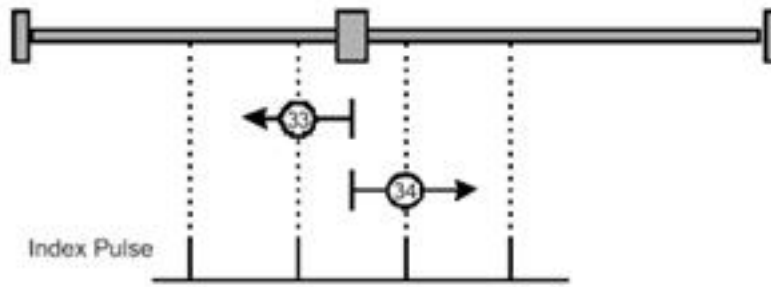
2.2.4 Homing Method 18: Homing on Positive Limit Switch

Homing method 18 is similar to method 2, except that it homes on positive limit switch only.

If positive limit switch is ON when the function is called, the homing is aborted (no homing attained).

2.2.5 Homing Method 33 and 34: Homing on the index pulse

Using methods 33 or 34, the direction of homing is negative or positive, respectively. The home position is at the index pulse found in the selected direction.



2.2.6 Homing Method 37: Homing on current position

In this method, the current position shall be taken to be the home position. FW implements the following ACSPL+ code:

```
SET FPOS(<axis>)=HomeOffset.
```

2.2.7 Homing Method 50: Negative Hard Stop and index pulse (ACS Specific)

With this homing method the initial direction movement is negative, till Hard Stop is found (by using the Position Error indication). The home position is at the first index pulse right of the position where the Hard Stop is found.

Position Error is considered as Hard Stop only if the HardStopThreshold condition is met.

2.2.8 Homing Method 51: Positive Hard Stop and index pulse (ACS Specific)

With this homing method the initial direction movement positive, till Hard Stop is found (by using the Position Error indication). The home position is at the first index pulse left of the position where the Hard Stop is found.

2.2.9 Homing Method 52: Negative Hard Stop (ACS Specific)

With this homing method the initial direction movement is negative, till Hard Stop is found (by using the Position Error indication). After the Hard Stop is found, the axis moves back to where the Hard Stop was detected, continues threshold (this is the home position), continues additional threshold multiplied by 2 and halts.

Position Error is considered as Hard Stop only if the HardStopThreshold condition is met.

2.2.10 Homing Method 53: Positive Hard Stop (ACS Specific)

With this homing method the initial direction movement is positive, till Hard Stop is found (by using the Position Error indication). After the Hard Stop is found, the axis moves back to where the Hard Stop was detected, continues threshold (this is the home position), continues additional threshold multiplied by 2 and halts.

2.3 Interactive Commands

The Interactive commands are:

Command	Description
DISP	Builds a string and sends it to the default communication channel.
INTERRUPT	Causes an interrupt that can be intercepted by the host.
INTERRUPTEX	Causes an interrupt similar to that of the INTERRUPT command.
SEND	Same as DISP , but also specifies the communication channel or channels.
TRIGGER	Specifies a triggering condition. Once the condition is satisfied, the controller issues an interrupt to the host computer.

2.3.1 DISP

Description

DISP builds an ASCII output string and sends it to a communication channel. The ASCII output can include text segments and variable values defined in various format displays. The output string is sent to the default communication channel defined by the standard variable **DISPCH**.

Syntax

DISP *argument* [, *argument. . .*]

Arguments

<i>argument</i>	An expression or string where: Expression: ACSPL+ expression as one or more variables String: "[text] [escape-sequence] [format-specifier]" String must be enclosed with double quotation marks.
[, <i>argument. . .</i>]	Optional subsequent arguments.

Command Options

An input string can include one or more of the following:

- > **Text**
- > **Escape Sequence** - appears in the output string as a non-printing character or other specified character.
- > **Formatting Specification** - determines how the results of an expression that follows the input string is formatted in the output string.

Table 2-3. DISP Command Option Escape Sequences

Escape Sequence	Added Character to Output String
<code>\r</code>	Carriage return - 0x0d
<code>\s</code>	Avoid carriage return
<code>\n</code>	New line - 0x0a
<code>\t</code>	Horizontal tab - 0x09
<code>\xHH</code>	Any ASCII character where HH represents the ASCII code of the character.



The output format specification syntax adheres to a restricted version of the C language syntax.

The format specification syntax is:

```
% [width] [.precision] type
```

where:

<i>[width]</i>	Optional specification for the minimum number of characters in the output. If <i>width</i> is smaller than the number of digits of a displayed number, the specified <i>width</i> is ignored, and the displayed number includes all digits.
<i>.[precision]</i>	Optional number that specifies the maximum number of characters printed for all or part of the output field, or the minimum number of digits printed for integer values.
<i>type</i>	Required character that determines whether the associated argument is interpreted as a character, a string, or a number, as described in Table 2-4 .

Table 2-4. Type Characters

Type	Output Format
d	Signed decimal integer
u	Unsigned decimal integer
x	Unsigned hexadecimal integer, using "abcdef"
X	Unsigned hexadecimal integer, using "ABCDEF"

Type	Output Format
e	Signed value having the form [-]d.dddd e [sign]ddd where d is a single decimal digit, dddd is one or more decimal digits, ddd is exactly three decimal digits, and sign is + or -.
E	Identical to the e format except that E rather than e introduces the exponent.
f	Signed value having the form [-]dddd.dddd, where dddd is one or more decimal digits. The number of digits before the decimal point depends on the magnitude of the number, and the number of digits after the decimal point depends on the requested precision.
g	Signed value printed in f or e format, whichever is more compact for the given value and precision. The e format is used only when the exponent of the value is less than -4 or greater than or equal to the precision argument. Trailing zeros are truncated, and the decimal point appears only if one or more digits follow it.
G	Identical to the g format, except that E , rather than e , introduces the exponent (where appropriate).

Comments

1. If an input string argument contains **n** format specifiers, the specifiers apply to the **n** subsequent expression arguments.
2. **DISP** processes arguments from left to right, as follows:
 - > **Expressions:** The expression is evaluated and the ASCII representation of the result is placed in the output string. The format of the result is determined by the formatting specifications (if any) in the input string.
 - > **Input strings:** Text is sent as-is to the output string. Escape sequences are replaced by the ASCII codes that they represent. Formatting specifications are applied to the results of any expressions that follow the string
3. **DISP** cannot be used from the SPiiPlus MMI Application Studio **Communication Terminal**, only from a program buffer.
4. **DISP** can only display the value of a single element of an array.

In order to receive unsolicited messages by a host application, perform the following:

1. Set **DISPCH** to -2.
2. Set bit 4 of **COMMFL** to 1.
3. Send **SETCONF**(306,-1,1) from the same communication channel where unsolicited messages are expected to be received.

In order to stop the receipt of unsolicited messages by a host application: send **SETCONF**(306,-1,0) from the same communication channel where there is no need any more to receive unsolicited messages.

Related ACSPL+ Commands[SEND](#), [SETCONF](#)**Related ACSPL+ Variables**[DISPCH](#), [COMMFL](#)**COM Library Methods and .NET Library Methods**

OpenMessageBuffer, GetSingleMessage, CloseMessageBuffer

C Library Functions

acsc_OpenMessageBuffer, acsc_GetMessage, acsc_CloseMessageBuffer

Example 1:

```
DISP "%15.10f",FPOS0      !Display FPOS0 in 15 digits with 10 digits
                           !following the decimal point.
                           !Output: 997.2936183303

STOP
```

Example 2:

```
DISP "0 FVEL=%15.10f", FVEL0      !Output: 0 FVEL= 997.2936183303

STOP
```

Example 3:

```
DISP "%1i",IN0.2                !Output: the current state of IN0.2 as
                                   !one digit 0 or 1.

STOP
```

Example 4:

```
DISP "IN0 as hex: %04X",IN0      !Output: IN0 as hex: 0A1D

STOP
```

Example 5:

```
DISP "IN0.0-3 as binary: %1u%1u%1u%1u", IN0.0,IN0.1,IN0.2,IN0.3
                                   !Output: IN0.0-3 as binary: 0110

STOP
```

Example 6:

```
DISP "Elapsed time is: ", TIME !Output: Elapsed time is: 4.93258E+006

STOP
```

Example 7:

```
DISP "Expression, default formatting:", FPOS0*2+FPOS1+5 , FPOS1
                                !Output: Expression, default formatting: 6.28657
                                !0.286485

STOP
```

Example 8:

```
REAL AXIS_NAME ; AXIS_NAME=0 ;
DISP "Axis",AXIS_NAME," was disabled due to error code", MERR(AXIS_NAME)
                                !Display the reason of axis disable due to a fault.
                                !Output: Axis 0  was disabled due to error code 0

STOP
```

Example 9:

```
DISP "%5i\r", FPOS0, "%5i", FPOS1
                                !Standard format, minimum 5 positions, no decimals,
                                !and a carriage return between the two values.
                                !Output:
                                !711
                                !2024

STOP
```

Example 10:

```
DISP "Hexadecimal format: %08X", MFLAGS(0), " and also %08x", MFLAGS(0)
                                !Hexadecimal format, minimum 8 positions, capital
                                !letters or lower case letters.
                                !Output: Hexadecimal format: 002A2300 and also
                                !002a2300

STOP
```

Example 11:

```
DISP "0 FPOS: %15.3e", FPOS0," and 1 FPOS: %15.3e", FPOS1
                                !Scientific format, minimum 15 positions, 3 decimals
                                !of FPOS0 and FPOS1.
                                !Output:
                                !0 FPOS: 5.000e-001 and
                                !1 FPOS: 2.865e-001

STOP
```

Example 12:

```
REAL AAA;                                !Standard or scientific format, small letters or
                                         !capital letters.
AAA=10;                                !Assigning integer value to AAA
```



```
DISP "%g", AAA;           !Output: 10
AAA=1e9                   !Assigning Hex value to AAA
DISP "%g", AAA           !Output: 1e+009
DISP "%G", AAA           !Output: 1E+009
STOP
```

2.3.2 INP

Description

INP reads data values from a specified channel and stores them to an integer array. This function is useful for creating an interface between the controller and special input devices such as a track-ball, mouse or various sensors. **INP** is also used when the controller acts as a master with the **MODBUS** protocol communication.

Before using **INP**, configure the relevant communication channel as a special communication channel using **SETCONF** function, key 302.

See also **OUTP**.

Syntax

int **INP**[/switches](channel, [array,] [start_index,] [number,] [timeout])

Arguments

<i>channel</i>	Communication channel index: > 1 - serial communication channel COM1 > 2 - serial communication channel COM2 > 6 - Ethernet network (TCP) > 7 - Ethernet network (TCP) > 8 - Ethernet network (TCP) > 9 - Ethernet network (TCP) > 10 - Ethernet Point-to-Point (UDP) > 12 - PCI bus > 16 - communication channel with Modbus slave > 36 - Ethernet network (TCP) > 37 - Ethernet network (TCP) > 38 - Ethernet network (TCP) > 39 - Ethernet network (TCP)
<i>array</i>	User-defined integer array to which the data will be stored. If array is omitted, the function purges formerly received characters in the channel.

<i>start_index</i>	The first received character is assigned to the array element with the specified index. If <i>start_index</i> is omitted, the assignment starts from the first element of the array.
<i>number</i>	The number of characters to be collected to the variable array. If <i>number</i> is omitted, the function continues receiving characters until the last element of the array is assigned, or the carriage return character is received.
<i>timeout</i>	The function waits for input not more than the specified number of milliseconds. If <i>timeout</i> is omitted, the waiting time is not limited.

Switches

Switch Name	Description
/1	Interpret the data as chars (8 bits)
/2	Interpret the data as shorts (16 bits)
/4	Interpret the data as ints (32 bits)
/f	Interpret the data as floats (32 bits)
/d	Interpret the data as doubles (64 bits)
/b	Interpret the data as big-endian (most significant byte is first in the data stream)

Comments

- > When no command options are specified, the data is interpreted as chars (8 bits).
- > Up to one type can be specified. Using more causes runtime error 3384 - The specified suffix combination is invalid.
- > When /b is not specified, the data is interpreted as little-endian (least significant byte is first in the data stream).
- > When the array's type does not match the data type, data is rounded towards zero.
- > When there is not enough data to completely interpret the requested type, the partial data is lost, and the array is not modified.

Return Value

The number of entities that have been stored into the variable.

Error Conditions

None

Example

```

GLOBAL INT MMM(10)      !Defines global user array MMM with ten elements.
SETCONF(302,2,1)        !Assigns COM2 as special input.
INP(2)                  !Purges the input buffer from old values
INP(2,MMM,0,10,1000)    !INP(2)- purge the input buffer from old values
                        !INP(2,MMM,0,10,1000) - store values from COM2 port
                        !to MMM user variable, from array index 0, total of
                        !10 values. The data collection process will end
                        !within 1000 msec or when the 10 values have been
                        !collected.

STOP                    !Ends program

```

2.3.3 INTERRUPT**Description**

INTERRUPT causes an unconditional trigger that is intercepted by the host. Once a program executes **INTERRUPT**, the interrupt signal is sent to the host application. This interrupt is detected by the COM library **EnableEvent** or C Library **acsc_SetCallback** functions which then call interrupt type **ACSC_INTR_ACSPL_PROGRAM**.

Syntax**INTERRUPT****Related ACSPL+ Commands****TRIGGER****COM Library Methods**

EnableEvent, DisableEvent, SetCallbackMask, SetCallbackPriority, GetCallbackMask

C Library Functions

acsc_InstallCallback, acsc_SetCallbackMask, acsc_SetCallbackPriority, acsc_GetCallbackMask

Example 1:

INTERRUPT used in an ACSPL+ program:

```

ENABLE 0                !Enable axis 0
SET FPOS0=0              !Set axis 0 feedback position = 0
PTP 0, 1000              !ACSPL+ executes a PTP motion.
TILL MST(0).#MOVE=0      !The motor reaches the destination point
                        !and stops.

INTERRUPT                !INTERRUPT is sent to the host application.
STOP

```

Example 2:

INTERRUPT used in a Host COM Lib application:

```

SET ch = New Channel      !Initialize ch as a COM library object
CALL ch.EnableEvent(ch.ACSC_INTR_ACSPLPROGRAM)
                        !Enable INTERRUPT as an event. The host application

```

```

!waits for an interrupt from the controller initiated
!by INTERRUPT from within the ACSPL+ program.
Private Sub ch_ACSPLPROGRAM(ByVal Param As Long) MsgBox ("Motion is
Stopped")
!When an interrupt occurs, launch a message box
!displaying "Motion is Stopped"
EndSub

```

Example 3:

INTERRUPT used in a Host C Lib application:

```

int WINAPI CallbackInput (unsigned __int64 Param,void* UserParameter);
// will be defined late
...
int CardIndex;//Some external variable, which contains card index
// set callback function to monitor digital inputs
if (!acsc_InstallCallback(Handle, // communication handle
    CallbackInput,           // pointer to the user callback function
    &CardIndex,              // pointer to the index
    ACSC_INTR_INPUT         // Type of callback
))
{
printf("callback registration error: %d\n", acsc_GetLastError());
}
...
// If callback was installed successfully, the CallbackInput function
will
// be called each time the any digital input has changed from 0 to 1.
// CallbackInput function checks the digital inputs 0 and 1
int WINAPI CallbackInput (unsigned __int64 Param,void* UserParameter)
{
if (Param & ACSC_MASK_INPUT_0 && *UserParameter == 0)
//Treat input 0 only for card with index 0
{
    // input 0 has changed from 0 to 1
    // doing something here
}
if (Param & ACSC_MASK_INPUT_1 && *UserParameter == 1)
    //Treat input 1 only for card with index 1
{
    // input 1 has changed from 0 to 1
    // doing something here
}
return 0;
}

```

2.3.4 INTERRUPTEX

Description

The **INTERRUPTEX** command operates in a manner similar to **INTERRUPT** but has the following differences:

- > It triggers the dedicated callback **ACSC_INTR_ACSPL_PROGRAM_EX** (21)
- > It accepts two mandatory integer parameters and two optional parameters. Their values will be passed along with the interrupt to the host instead of the buffer mask passed in the old **INTERRUPT** function as a 64-bit integer.
- > Adjacent ("glued") interrupts are processed differently (because parameters are not OR'ed):
 - > There is an internal queue of 256
 - > The next interrupt value will be triggered only after C Library delivers the previous one
 - > The maximum output rate is 1 interrupt per **CTIME**
 - > On queue overflow, the interrupt is lost

Syntax

INTERRUPTEX (32-bit_high_value,32-bit_low_value[,32-bit_high_value_second,32-bit_low_value_second])

Arguments

32-bit_high_value	Most significant value of a combined 64-bit integer
32-bit_low_value	Least significant value of a combined 64-bit integer
32-bit_high_value_second	Optional. High 32 bits of a combined second 64-bit integer
32-bit_low_value_second	Optional. Low 32 bits of a combined second 64-bit integer

Comments

INTERRUPTEX is supported by both by SPiiPlusNT and SPiiPlusSC products. For the SPiiPlusSC-HP products, the interrupt is passed via Shared Memory, which makes it very fast ($100 + (\text{Cycle-time})/2$ for the round trip on the average). For the SPiiPlusNT and the SPiiPlusSC-LT products, the interrupt is passed via the communication channel (Ethernet/Serial RS-232).



An application that uses C Library must make sure to empty the queue, register the callback and wait enough time until the queue is empty.

The parameters are passed to the host as a single 64-bit integer with the first parameter as the 32-bit most significant value word and the second parameter is the 32-bit least significant word.

COM Library Methods

EnableEvent, DisableEvent, SetCallbackMask, SetCallbackPriority, GetCallbackMask

C Library Functions

acsc_InstallCallback, acsc_SetCallbackMask, acsc_SetCallbackPriority, acsc_GetCallbackMask

Example 1:

```

Enable 0          ! Enable axis 0
SET FPOS0=0       ! Set axis 0 position to 0
PTP/e 0,1000      ! Move to position 1000
INTERRUPTTEX (0x12, 0x34)! interrupt with parameter, host will receive
                  ! 0x00000001200000034

Stop

```

Example 2:

```

Global integer interrupt_description (0x100)
Global integer interrupt_queue
interrupt_queue = 0
interrupt_buff = 0          ! No pending interrupts
Enable 0                   ! Enable axis 0
SET FPOS0=0                ! Set axis 0 position to 0
interrupt_description(interrupt_queue)= 0x1    ! Setting description to 1
                                              ! indicating pre-motion
                                              ! state
interrupt_queue = (interrupt_queue+1)&0xff      ! One more pending interrupt
INTERRUPTTEX (0x00, interrupt_queue)          ! Interrupt description in
                                              ! interrupt_description (0)
PTP/e 0,1000                                ! Move to position 1000
interrupt_description(interrupt_queue) = 0x2   ! Setting description to
2                                              ! indicating post-motion
                                              ! state
interrupt_queue = (interrupt_queue+1)&0xff     ! One more pending interrupt
INTERRUPTTEX (0x00, interrupt_queue)          ! Interrupt description in
                                              ! interrupt_description (1)

Stop

```

Example 3:

```

Enable 0          !Enable axis 0
SET FPOS0=0       !Set axis 0 to position 0
PTP/e 0,1000      !Move to position 1000
INTERRUPTTEX (0x12, 0x34, 0x22, 0x54)
                !interrupttex with parameter, host will receive the following;
                ! 0x00000001200000034, 0x00000002200000054

```

2.3.5 SEND**Description**

SEND is the same as **DISP**, but also specifies the communication channel for the output string.

Syntax

SEND *channel_number, argument [, argument...]*

Arguments

<i>channel-number</i>	An integer that defines the communication channel to which the message will be sent, as follows: -2: All channels -1: Last used channel 1: Serial Port 1 2: Serial Port 2 6: Ethernet network (TCP) 7: Ethernet network (TCP) 8: Ethernet network (TCP) 9: Ethernet network (TCP) 10: Ethernet point-to-point (UDP) 16: MODBUS Slave 36: Ethernet network (TCP) 37: Ethernet network (TCP) 38: Ethernet network (TCP) 39: Ethernet network (TCP)
<i>argument</i>	An expression or string where: Expression: ACSPL+ expression as one or more variables String: "[text] [escape-sequence] [format-specifier]" String must be enclosed with double quotation marks.
<i>[, argument. . .]</i>	Optional subsequent arguments.

Command Options

For a list of **Command Options**, relevant **Comments**, and **Examples**, see [DISP](#).

Related ACSPL+ Commands

[DISP](#)

Related ACSPL+ Variables

[DISPCH](#)

COM Library Methods

Send

C Library Functions

acsc_Send

2.3.6 TRIGGER

Description

TRIGGER specifies a triggering condition. Once the condition is satisfied, the controller issues an interrupt to the host computer, as follows:

1. Sets **AST<axis>.#TRIGGER** = 0
2. Examines the triggering condition every MPU cycle

Once the condition is satisfied, the controller performs the following:

1. Sets **AST<axis>.#TRIGGER** = 1
2. Produces an interrupt to the host application (software interrupt 10, enabled by **IENTA.26**).

The controller continues calculating the **TRIGGER** expression until another **TRIGGER** command is executed in the same channel. Each time the expression changes its value from zero to non-zero, the controller sets **AST<axis>.#TRIGGER** = 1 and causes an interrupt.



Full application of the **TRIGGER** command to channels greater than 7 is not currently supported.

Syntax

TRIGGER *channel*, *expression*[, *timeout*]

Arguments

<i>channel</i>	An integer number from 0 to 7 that specifies the trigger's sequential number. The number defines the AST element where the triggering bit will be set and defines the bit in the interrupt tag that is automatically sent to the host application as an interrupt.
<i>expression</i>	Specifies the triggering condition. After TRIGGER is executed, the controller checks the expression condition each MPU cycle. Triggering occurs when the expression condition is satisfied. If the argument is omitted, triggering in the specified channel is disabled.
<i>timeout</i>	Specifies triggering timeout in milliseconds. A positive number specifies the time allowed for the triggering condition to be satisfied. If the timeout has elapsed and the triggering condition has not been satisfied, the controller unconditionally raises the trigger bit. After any triggering, the controller resets timeout counting to zero. If the argument is omitted, triggering works without timeout.

Table 2-5. Channel Designation for TRIGGER

Channel	Triggering Bit	Hexadecimal Interrupt Tag (Software Interrupt 10)
0	AST0.11	0x00000001
1	AST1.11	0x00000002
2	AST2.11	0x00000004
3	AST3.11	0x00000008
4	AST4.11	0x00000010
5	AST5.11	0x00000020
6	AST6.11	0x00000040
7	AST7.11	0x00000080

Related ACSPL+ Commands**INTERRUPT****Related ACSPL+ Variables****IENA, AST****COM Library Methods**

GetCallbackMask, SetCallbackMask

C Library Functions

acsc_GetCallbackMask, acsc_SetCallbackMask

Example

```

TRIGGER 1, MST(0).#MOVE=0, 3000      !1 - once the triggering condition is satisfied,
                                     !the triggering bit AST1.#TRIGGER will be set
                                     !to "1", and the interrupt tag to the host
                                     !application is 0x00000002.
                                     !MST(0).#MOVE=0 - the triggering condition.
                                     !Actuate trigger interrupt when the motor in
                                     !the 0 axis is in position (not moving).
                                     !3000 - check the triggering condition for
                                     !3000 msec. If the triggering condition is not
                                     !satisfied after 3000 msec, then set the
                                     !triggering bit AST(1).TRIGGER to "1".

```

2.3.7 OUTP**Description**

OUTP sends data values from an integer array to a specified channel. This function is useful to create an interface between the controller and special input devices such as a track-ball, mouse and

various sensors. **OUTP** is also used when the controller acts as a master with the **MODBUS** protocol communication.

Before using **OUTP**, configure the relevant communication channel as a special communication channel using **SETCONF** function, key 302.

Each ASCII character is represented by its numerical value and is stored in a separate element of the array.

The user might have to define communication parameters for the special communication channel with **SETCONF** function keys 302, 303, 304, 309.

See also [INP](#).

Syntax

```
int OUTP[/switches] (channel, variable[, start_index[, number]])
```

Arguments

<i>channel</i>	Communication channel index: > 1 - serial communication channel COM1 > 2 - serial communication channel COM2 > 6 - Ethernet network (TCP) > 7 - Ethernet network (TCP) > 8 - Ethernet network (TCP) > 9 - Ethernet network (TCP) > 10 - Ethernet Point-to-Point (UDP) > 12 - PCI bus > 16 - communication channel with MODBUS slave > 36 - Ethernet network (TCP) > 37 - Ethernet network (TCP) > 38 - Ethernet network (TCP) > 39 - Ethernet network (TCP)
<i>variable</i>	User-defined integer array from which the data will be sent.
<i>start_index</i>	The index in the array from which to start. If start_index is omitted, number should also be omitted (in this case, all members of the array are transmitted).
<i>number</i>	The number of characters to be transmitted from the variable array. If omitted, all members of the array starting from start_index are transmitted.

Switches

Switch Name	Description
/1	Interpret the data as chars (8 bits)
/2	Interpret the data as shorts (16 bits)
/4	Interpret the data as ints (32 bits)
/f	Interpret the data as floats (32 bits)
/d	Interpret the data as doubles (64 bits)
/b	Interpret the data as big-endian (most significant byte is first in the data stream)

Comments

- > When no command options are specified, the data is interpreted as chars (8 bits).
- > Up to one type can be specified. Using more causes runtime error 3384 - The specified suffix combination is invalid.
- > When /b is not specified, the data is interpreted as little-endian (least significant byte is first in the data stream).
- > When the array's type does not match the data type, data is rounded towards zero.
- > When there is not enough data to completely interpret the requested type, the partial data is lost, and the array is not modified.

Return Value

The number of entities that have been transmitted.

Error Conditions

If the function fails, an error is generated.

2.4 PEG and MARK Commands

Command	Description
ASSIGNMARK	Assigns MARK inputs pins to encoders.
ASSIGNPEG	Assigns PEG engines to encoders.
ASSIGNPOUTS	Assigns PEG outputs.
GETPEGCOUNT	Returns the pulse counter of the required PEG engine

Command	Description
PEG_I	Activates incremental PEG, where pulses are generated at predefined, fixed position intervals.
PEG_R	Activates random, table-based PEG where predefined pulses are generated at pre-defined positions.
STARTPEG	Starts PEG motion for specified axis.
STOPPEG	Terminates PEG motion.
SETPEGDELAY	Defines the PEG signal's delay for a specific PEG engine

2.4.1 ASSIGNMARK

Description

The **ASSIGNMARK** function allows assignment of Mark inputs to encoder. It allows a mapping of encoder latching to be triggered by using different physical input pins.

Syntax

ASSIGNMARK*[/i>*i**]* *axis, mark_type, inputs_to_encoder_bit_code**



In newer products the *inputs_to_encoder_bit_code* parameter is actually a byte code. See [IDMxx/ECMxx/UDMsm/UDMsa/UDMma Encoder Mapping](#) table.

Arguments

Axis	The axis index, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1. The axis parameter determines which node unit is used. Axis parameter can be any axis number of the same unit.
mark_type	1 for Mark-1 assignment 2 for Mark-2 assignment

inputs_to_encoder_bit_code	Bit code for inputs-to-encoders mapping according to the following tables. The bit code determines which physical input pin leads to each encoder MARK latching. Mark-1 Inputs to Encoder mapping for Mark-1 Inputs to Encoders Mapping for SPiiPlusNT/DC-LT/HP/LD IDMxx/ECMxx/UDMsm/UDMsa/UDMma Encoder Mapping Mark-1 Inputs to Encoders Mapping for with SPiiPlus CMnt/UDMpm-x/UDMpc/CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa/CMhv/UDMhv Mark-2 Inputs to Encoder mapping for Mark-2 Inputs to Encoders Mapping for SPiiPlusNT/DC-LT/HP/LD Mark-2 Inputs to Encoders Mapping for with SPiiPlus CMnt/UDMpm/UDMpc/CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa/CMhv/UDMhv
-----------------------------------	---

Comments

Latching of Encoder <index> means **IST**(<index>).#MARK=1, and the **MARK**(<index>) variable value stores the feedback position of encoder <index> (**FPOS**(<index>)).

The Bit Code shown in the tables affects all of the connectors in the row.

Supported products that are not listed in the tables include only the default case.

If the switch: /i is included, the MARK input signal is inverted.

In IDMxx/ECMxx products, the latching for all encoders happens simultaneously.

2.4.2 ASSIGNPEG

Description

The **ASSIGNPEG** function is used for engine-to-encoder assignment as well as for the additional digital outputs assignment for use as PEG pulse outputs. It allows mapping of PEG engines to be triggered by the feedback of a specific encoder.

Syntax

ASSIGNPEG[/f] *axis, engines_to_encoders_code, gp_out_assign_code*

Arguments

axis	The axis index, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1. Axis parameter can be any axis number of the same unit. The axis parameter determines the Servo Processor used.
engines_to_encoder_s_	Bit code for engines-to-encoders mapping according to: SPiiPlusNT/DC-LT/HP/LD Processor 0 SPiiPlusNT/DC-LT/HP/LD Processor 1

code	SPiiPlusCMnt/UDMpm/UDMpc/CMba/CMhp/CMxa/UDMba/UDMxa/CMhv/UDMhv/UDMnt/UDMpa/UDMcb UDMlc/UDlIt/UDlhp/UDMmc/PDIcl NPMpm/NPMpc IDMsm, ECMsm, and UDMsm
gp_out_assign_code	General purpose outputs assignment to use as PEG pulse outputs according to: SPiiPlusNT/DC-LT/HP/LD SP 0 SPiiPlusNT/DC-LT/HP/LD SP 1 SPiiPlus CMnt/CMhv/UDMpm/UDMhv UDMnt/UDMpa/UDMcb IDMsm, ECMsm, UDMsm



The **axis** parameter actually serves to determine the Servo Processor used.

Comments

- > **ASSIGNPEG** is a blocking command - the ACSPL+ program moves to the next line or command only after this command has been fully executed or an error is generated.
- > The **axis** parameter can be any of the axes controlled by the same servo processor, the result will be the same.
- > If the "/f" switch is included, fast loading of Random PEG arrays is activated. This feature allows definition of state-arrays with more than 1024-members by using Random PEG. The **PEG_R** command must be called with the "/d" switch.
- > The **Bit Code** shown in the Mapping PEG Engines to Encoders tables affects all of the connectors in the row.



HSSI devices (HSSI-IO16, HSSI-ED2, etc.) cannot be used for the same Servo Processor when fast loading of Random PEG arrays is activated.

SPRT and **SPINJECT** commands cannot be used for the same Servo Processor when fast loading of Random PEG arrays is activated.

Related ACSPL+ Commands

PEG_I, ASSIGNPOUTS, PEG_R, STARTPEG, STOPPEG

COM Library Methods

None

C Library Functions

acsc_AssignPegNT

2.4.3 ASSIGNPOUTS

Description

The **ASSIGNPOUTS** function is used for assigning PEG engine output signals to physical output pins. In addition, the function allows assigning Fast General Purpose output pins and mapping between **FGP_OUT** signals to the bits of the ACSPL+ **OUT(x)** variable, where **x** is the index that has been assigned to the controller in the network during System Configuration.



The assignments can be obtained by running **#SI** in the SPiiPlus MMI Application Studio **Communication Terminal**. For example, the following is a fragment from the response to this command:

```
Axes Assignment: 8,9,10,11
Inputs/Outputs Assignment:
    Digital inputs (IN): 1.0,1.1,1.2,1.3,1.4,1.5,1.6,1.7
    Digital outputs (OUT): 1.0,1.1,1.2,1.3,1.4,1.5,1.6,1.7
```

OUT is an integer array that can be used for reading or writing the current state of the General Purpose outputs - see *SPiiPlus ACSPL+ Command & Variable Reference Guide*.

Each PEG engine has 1 PEG pulse output and 4 state outputs for a total of 5 outputs per PEG engine and a total of 30 outputs for the whole PEG generator. The controller supports 10 physical output pins that can be assigned to the PEG generator. The user defines which 10 outputs (of the 30) of the PEG generator are assigned to the 10 available physical output pins. Some of the output pins are shared between the PEG and the HSSI.

The tables in Appendix [A.2](#) define how each of the 30 outputs of the 6 PEG engines can be routed to the 10 physical output pins - 4 PEG out signals, 3 PEG state signals, and 3 HSSI signals. Note that some of the signals cannot be routed to physical pins.



Bit Code: 111 is used for switching the physical output pins to Fast General Purpose Outputs, see [ASSIGNPOUTS](#).

Syntax

```
ASSIGNPOUTS axis, peg_output, bit_code
```

Arguments

axis	The axis index, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1. For controllers with firmware version 2.15 or higher, the axis parameter can be any axis number of the unit.
peg_output	The peg output number according to the Mapping of Engine Outputs to Physical Output tables below

bit_code	Bit code for engine outputs to physical outputs mapping according to: SPiiPlusNT/DC-LT/HP/LD SP 0 SPiiPlusNT/DC-LT/HP/LD SP 1 CMnt/UDMpm/UDMpc/CMhv/UDMhv CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa (OUT0-4) CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa (OUT5-9) UDMnt/UDMpa/UDMcb UDMlc/UDMmc/UDIlt/UDIhp/PDIcl NPMpm/NPMpc IDMsm/ECMsm/UDMsm Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 0-5) IDMsm/ECMsm/UDMsm Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 6-11) ECMsa/IDMsa/UDMsa Mapping of Engine Outputs to Physical Outputs (Servo Processor 0) ECMma / IDMma / UDMma Mapping of Engine Outputs to Physical Outputs (Servo Processor 0) ECMdx / IDMdx / UDMdx Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 0-5) ECMdx / IDMdx / UDMdx Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 6-11)
-----------------	--

Comments

ASSIGNPOUTS is a blocking command in the sense that the ACSPL+ program moves to the next line or command only after this command has been fully executed or an error is generated.

A separate **ASSIGNPOUTS** command should be called for every GP output or PEG output.

Examples

The following examples illustrate using the **ASSIGNPOUTS** in order to use PEG outputs as GP outputs

Example 1:

```
ASSIGNPOUTS 0, 2, 0b111
```

This defines the **Z_PEG** output as **FGP_OUT2** and maps it to the bit 18 of the ACSPL+ **OUT** variable (see [ASSIGNPOUTS](#)).

If you run, for example:

```
OUT (x) .18=0
```

Where **x** is the index assigned to the controller during System Configuration, **FGP_OUT2** output will be activated.

Then if you run:

```
OUT (x) .18=0
```

FGP_OUT2 will be deactivated.

Example 2:

```
ASSIGNPOUTS 4, 7, 0b111
```

This defines the **X_STATE2** output as **FGP_OUT6** and maps it to the bit 22 of the ACSPL+ **OUT** variable (see [ASSIGNPOUTS](#)).

Related ACSPL+ Commands

[ASSIGNPEG](#), [PEG_I](#), [PEG_R](#), [STARTPEG](#), [STOPPEG](#)

COM Library Methods

None

C Library Functions

acsc_AssignPegOutputsNT

2.4.4 GETPEGCOUNT

Description

The function returns the pulse counter of the specified PEG engine.

Syntax

```
GetPEGCount (PEG_engine)
```

Support

- > **GETPEGCOUNT** is supported on firmware versions 3.12 and later
- > **GETPEGCOUNT** is supported on the following devices:
 - > ECMsm, ECMsa, ECMma, and ECMdx
 - > IDMsM, IDMsA, IDMma, and IDMdX
 - > UDMsm, UDMsa, UDMma, and UMDdx

2.4.5 PEG_I

Description

The **PEG_I** command is used for setting the parameters for the Incremental PEG mode.

Syntax

PEG_I[/awieczo] (peg_engine, width, first_point, interval, last_point [, err_map_static_axis1, static_axis_coord1, [err_map_static_axis2, static_axis_coord2], [err_map_max_size,], [stable_dir_dist,]] [time_based_pulses, time_based_period])

Arguments

<i>peg_engine</i>	The PEG engine number
<i>width</i>	Width of the pulse in milliseconds. Valid range is 26.6 ns to 1.745 ms.
<i>first_point</i>	A real scalar value in user units indicating the first point for the PEG generation.
<i>interval</i>	A real scalar value in user units indicating the distance between PEG events.
<i>last_point</i>	A real scalar value in user units indicating the last point for PEG generation. Optional if the /e switch is used.
<i>err_map_static_axis1</i>	The index of the first static axis when error mapping is used with PEG. Use with the /y or /yz switches.
<i>static_axis_coord1</i>	The predefined location value of the first static axis when error map correction compensation is used with PEG. Use with the /y or /yz switches.
<i>err_map_static_axis2</i>	The index of the second static axis when error mapping is used with PEG. Use with the /yz switch.
<i>static_axis_coord2</i>	The predefined location value of the second static axis when error map correction compensation is used with PEG. Use with the /yz switch.
<i>err_map_max_size</i>	Maximum array size contained error correction data. Default is 512. Range from 1 to XARRSIZE . A negative value or 0 is interpreted as default. Available only if switch /m is used in combination with /y or /yz switches.
<i>stable_dir_dist</i>	(Optional) Limit defining actual motion as opposed to jitter. If motion along the axis is greater than this value, that motion is used to determine the direction of motion.
<i>time_based_pulses</i>	Optional parameter - a real scalar value indicating the number of time-based pulses generated after each encoder-based pulse, the range is from 0 to 65,535.
<i>time_based_period</i>	Optional parameter, a real scalar in milliseconds - period of time-based pulses, the range is from 0.00005334 to 1.7476. The time-based period must be at least pulse width + 26.6667 ns (minimum distance between two pulses).



PEG is generated only after the first pre-defined start point is reached. If the current encoder position exceeds pre-defined start point no PEG pulses are fired. It is recommended to activate the PEG engine before the maximum current position for movement in the positive direction and the minimum current position for movement in the negative direction.

Comments

- > If the switch: /w is included, the execution of the command is delayed until the execution of the **STARTPEG** command.
- > If the switch: /i is included, the PEG pulse output signal is inverted.
- > If the switch: /a is included, error accumulation is prevented by taking into account the rounding of the distance between incremental PEG events. You must use this switch if **interval** does not match the whole number of encoder counts. Using this switch is recommended for any application that uses the **PEG_I** command, regardless if **interval** matches the whole number of encoder counts.
- > Valid numbers of the `peg_engine` parameter can be found in the #SI report. In case of multiple network units, the first axis number of each node indicates the first PEG engine of the node.
- > If the switch /e is included, the `last_point` parameter is optional and ignored, and the PEG process continues until the **STOPPEG** command is issued.
- > The /c switch supports 1D dynamic error compensation for Incremental PEG. 1D error compensation can be defined by the **ERRORMAP1D** or **ERRORMAP1ND** functions.
- > The /y switch supports 2D dynamic error compensation for Incremental PEG. 2D error compensation can be defined by **ERRORMAP2D**, **ERRORMAP2ND** functions. This switch requires 2 additional function arguments: `static axes index`, `static axes coordinates`.
- > The /z switch supports 3D dynamic error compensation for Incremental PEG. 3D error compensation can be defined by **ERRORMAP3DX** functions. This switch must be used in combination with switch /y (/yz) and requires 4 additional function arguments: `static axes 1 index`, `static axis 1 coordinate`, `static axes 2 index`, `static axis 2 coordinate`.
- > The /o switch is used with the `stable_dir_dist` parameter. The PEG process does not count motions smaller than this value as motion, for the purpose of determining the next PEG firing. Only motion greater than this value will be used to determine the direction of axis motion.

Product Support

The /c, /e, /y, /yz, and /o switches are supported in the following products:

- > IDM/ECM/UDMsm
- > IDM/ECM/UDMsa
- > IDM/ECM/UDMma
- > IDM/ECM/UDMdx

Example

See the Incremental PEG example in the *PEG and MARK Operations Application Note*.

Error Mapping Example

```
! PEG engine 4, Pulse width=0.1ms , Start Pos. 500,  
! Increment 200, Stop Pos. 1000, Use 1D error mapping  
PEG_I/c 4, 0.1, 500, 200, 1000  
  
! PEG engine 0, Pulse width=0.1ms , Start Pos. 500,  
! Increment 200, Stop Pos. 1000,  
! Use 2D error mapping for axis 1 coordinate = 500.  
! Maximum error mapping table size = 1024  
PEG_I/ym 0, 0.1, 500, 200, 1000, 1, 500, 1024  
! PEG engine 0, Pulse width=0.1ms , Start Pos. 500,  
! Increment 200, Stop Pos. 1000,  
! Use 3D error mapping for axis 1 coordinate = 100,  
! for axis 2 coordinate = 200  
PEG_I/yz 0, 0.1, 500, 200, 1000, 1, 100, 2, 200
```

Related ACSPL+ Commands

PEG_R, ASSIGNPEG, ASSIGNPOUTS, STARTPEG, STOPPEG

Related ACSPL+ Variables

AST

COM Library Methods

None

C Library Functions

acsc_PegIncNT, acsc_WaitPegReady

2.4.5.0.1 PEG_R

2.4.6 PEG_R

Description

The **PEG_R** command is used for setting the parameters for the Random PEG mode.

Syntax

PEG_R[/wdyzmo] (peg_engine, width, mode, first_index, last_index, POS_ARRAY, [STATE_ARRAY], [, err_map_static_axis1, static_axis_coord1, [, err_map_static_axis2, static_axis_coord2], [err_map_max_size,]] [time_based_pulses, time_based_period])

Arguments

<i>peg_engine</i>	The PEG engine number
<i>width</i>	Width of the pulse in milliseconds. Valid range is 26.6 ns to 1.745 ms.

<i>mode</i>	PEG state output signals configuration according to PEG_R. Bits 0-3 relates the PEG State 0 of the specific PEG engine Bits 4-7 relates the PEG State 1 of the specific PEG engine Bits 8-11 relates the PEG State 2 of the specific PEG engine Bits 12-15 relates the PEG State 3 of the specific PEG engine The most commonly used value is 0x4444 - PEG State Outputs 0-3 are configured with the 'State' option (bits 2, 6, 10, 14 are ON).
<i>first_index</i>	Index of first entry in the array for PEG generation
<i>last_index</i>	Index of last entry in the array for PEG generation
<i>POS_ARRAY</i>	The Random Event Positions array, maximum of 256/1024 members. If a longer array is required, use both PEG_R/d and ASSIGNPEG/f switches.
<i>STATE_ARRAY</i>	Optional parameter - the Outputs States array defining the four PEG output states, maximum of 256/1024 members. If a longer array is required, use both PEG_R/d and ASSIGNPEG/f switches.
<i>err_map_static_axis1</i>	The index of the first static axis when error mapping is used with PEG. Use with the /y or /yz switches.
<i>static_axis_coord1</i>	The predefined location value of the first static axis when error map correction compensation is used with PEG. Use with the /y or /yz switches.
<i>err_map_static_axis2</i>	The index of the second static axis when error mapping is used with PEG. Use with the /yz switch.
<i>static_axis_coord2</i>	The predefined location value of the second static axis when error map correction compensation is used with PEG. Use with the /yz switch.
<i>stable_dir_dist</i>	(Optional) Limit defining actual motion as opposed to jitter. If motion along the axis is greater than this value, that motion is used to determine the direction of motion.
<i>time-based-pulses</i>	Optional parameter - number of time-based pulses generated after each encoder-based pulse, the range is from 0 to 65,535.
<i>time-based-period</i>	Optional parameter - period of time-based pulses (milliseconds), the range is from 0.00005334 to 1.7476. Time-based period must be at least pulse width + 26.6667 nsec (minimum distance between two pulses).

Table 2-6. PEG Output Signal Configuration**PEG state output types:**

"Three state" - PEG state output is not in use

"State" - PEG state output will be changed according to the STATE_ARRAY values

"Pulse" - PEG state output will be changed according to PEG pulse value

"Pulse & State" - PEG state output will be changed according to the result of AND operation between STATE_ARRAY values AND PEG pulse value

Pulse Polarity:

If positive or negative pulse is used as PEG pulse value for the specific "PEG State Output"

State Polarity:

If positive or negative state is used as PEG pulse value for the specific "PEG State Output"

Comments

- > If the switch **/w** is included, the execution of the command is delayed until the execution of the **STARTPEG** command.
- > If the switch **/i** is specified, the PEG pulse output signal is inverted.
- > If the switch **/d** is specified, dynamic loading of positions is used. Dynamic loading can only be implemented on a PEG engine with fast loading (**ASSIGNPEG /f**). In this case, , it allows definition of a state-array which has more than 1024-members.
- > The **/y** switch supports 2D dynamic error compensation for Random PEG. 2D error compensation can be defined by **ERRORMAP2D**, **ERRORMAP2ND** functions. This switch requires 2 optional function arguments: `static axes index`, `static axes coordinates`.
- > The **/z** switch supports 3D dynamic error compensation for Random PEG. 3D error compensation can be defined by **ERRORMAP3DX** functions. This switch must be used in combination with switch **/y (/yz)** and requires 4 optional function arguments: `static axes 1 index`, `static axis 1 coordinate`, `static axes 2 index`, `static axis 2 coordinate`.
- > The parameters that can be set by the command differ from those that could be set for SPiiPlusCM/SPiiPlusSA/SPiiPlus3U controllers with the addition of the new *first_index* and *last_index* parameters.
- > When the PEG pulse is activated, the voltage between the two differential **PEG** outputs (+) and (-) drops to -5V. When the **PEG** pulse is de-activated, the voltage between the two differential **PEG** outputs is 5V.
- > In **PEG_R**, the number of position-based pulses is limited to eight pulses per controller cycle and the **minimum time interval must be >200nsec**.
- > When using a Sin-Cos encoder, **PEG** is triggered at the zero crossing of the sine-cosine waves and not at the precise interpolated position.
- > The last three arguments are optional. If **STATE_ARRAY** is omitted, the controller generates the PEG pulses at each position but does not change the state of any output. If time-based-pulses and time-based-period are omitted, the controller does not generate time based pulses.
- > The dynamic loading feature is limited by the loading frequency. If a high loading frequency is required, the loading capacity may not suffice to keep the FIFO loaded.

- > If the FIFO is emptied before all data arrays have been loaded, a memory overflow fault will be thrown.
- > Valid numbers of the `peg_engine` parameter can be found in the #SI report. In case of multiple network units, the first axis number of each node indicates the first peg engine of the node.
- > The `/m` switch supports random PEG on a modulo axis. The **POS_ARRAY** is loaded once and provides pulses every Modulo cycle. The Position values should be inside the **SLPMIN...SLPMAX** range. The PEG engine must be assigned to a Modulo axis by the **ASSIGNPEG** command.
- > The `/o` switch allows the definition of a minimum distance for which motion is considered actual motion as opposed to jitter. The name of the optional parameter is `stable_dir_dist`; motion greater than the value of this parameter is used to calculate motion direction.

Product Support

The `/m` and `/o` switches are supported in the following products:

- > IDM/ECM/UDMsm
- > IDM/ECM/UDMsa
- > IDM/ECM/UDMma

Loading Frequency Table

CTIME (ms)	Frequency (Hz)
1	200
0.5	400
0.25	800
0.2	1000

Example

See the Random PEG example in the *PEG and MARK Operations Application Note*.

Related ACSPL+ Commands

[PEG_I](#), [ASSIGNPEG](#), [ASSIGNPOUTS](#), [STARTPEG](#), [STOPPEG](#)

Related ACSPL+ Variables

AST

COM Library Methods

None

C Library Functions

`acsc_PegRandomNT`, `acsc_WaitPegReady`

2.4.7 STARTPEG

Description

The **STARTPEG** command initiates the PEG process on the specified engine. The command is used in both the Incremental and Random PEG modes by using /w switch in **PEG_I** or **PEG_R** command. If this switch is included, the execution of the **PEG_I** and **PEG_R** commands is delayed until the execution of the **STARTPEG** command.

Syntax

STARTPEG *peg_engine*

Arguments

peg_engine	The PEG engine number
-------------------	-----------------------

Comments

STARTPEG is a blocking command in the sense that the ACSPL+ program moves to the next line or command only after this command has been fully executed or an error is generated.

If **STOPPEG** has been issued before the last PEG position, you have to use **STARTPEG** to resume PEG engine firings from the current point.

Valid numbers of the *peg_engine* parameter can be found in the #SI report. In case of multiple network units, the first axis number of each node indicates the first peg engine of the node.

Example

```
PEG_I/W 0, 0.003, 1000, 1000, 3000, 2, 0.01
PTP 0, 3000      !The program initiates synchronous PEG, with PTP
!motion on axis 0.
WAIT 2          !Two milliseconds after motion starts, PEG
                !will be initiated by STARTPEG
STARTPEG 0
```

Related ACSPL+ Commands

[ASSIGNPEG](#), [ASSIGNPOUTS](#), [STOPPEG](#)

Related ACSPL+ Variables

AST

COM Library Methods

None

C Library Functions

[acsc_StartPegNT](#)

2.4.8 STOPPEG

Description

The **STOPPEG** command terminates the PEG process immediately on the specified engine. The command is used in both the Incremental and Random PEG modes.

Syntax**STOPPEG** *peg_engine***Arguments*****peg_engine***

The PEG engine number

Comments

STOPPEG is a blocking command in the sense that the ACSPL+ program moves to the next line or command only after this command has been fully executed or an error is generated.

Valid values of the *peg_engine* parameter can be found in the #SI report. In case of multiple network units, the first axis number of each node indicates the first peg engine of the node.

Example

```
PEG_I 0, 0.003, 1000, 1000, 3000, 2, 0.01
PTP 0, 3000           !The program initiates synchronous PEG, with PTP
                      !motion on axis 0.
WAIT 2               !Two milliseconds after motion starts, PEG
                      !will be terminated by STOPPEG.
STOPPEG 0
```

Related ACSPL+ Commands

[ASSIGNPEG](#), [ASSIGNPOUTS](#), [PEG_I](#), [PEG_R](#), [STARTPEG](#)

COM Library Methods

None

C Library Functions

[acsc_StopPegNT](#)

2.4.9 SETPEGDELAY**Description**

The **SETPEGDELAY** function defines the PEG signal's delay for a specific PEG engine.

The delay can be set for every available PEG signal:

- > PEG pulse
- > PEG State raising
- > PEG State falling.

Syntax

```
SETPEGDELAY (peg_engine, pulse_delay[, state_on_delay, state_off_delay])
```

Arguments

peg_engine	PEG engine number
pulse_delay	PEG pulse signal raising delay in msec
state_on_delay	(optional) Random PEG State signal raising delay in msec. The default is 0
state_off_delay	(optional) Random PEG State signal failing delay in msec. The default is 0

Comments

The **SETPEGDELAY** function should be called after PEG definition functions **PEG_I** or **PEG_R**.

The **PEG_I** and **PEG_R** functions initialize the PEG and set PEG delays to 0. The delay time is specified in msec.

Support

- > **SETPEGDELAY** is supported on firmware versions 3.13 and later
- > **SETPEGDELAY** is supported on the following devices:
 - > ECMsm, ECMsa, ECMma, and ECMdx
 - > IDMsM, IDMsA, IDMma, and IDMDx
 - > UDMsm, UDMsa, UDMma, and UMDdx

2.5 Miscellaneous Commands

The Miscellaneous commands are:

Command	Description
AXISDEF	Establishes an axis alias.
DC	Activates data collection.
SPINJECT	Returns the system back to the operational state if one or more slaves underwent a reset or power cycle.
READ	Reads an array from a file in the flash memory.
SPDC	Activates data collection from a Servo Processor variable.
SPINJECT	Initiates the transfer of MPU real-time data to the Servo Processor.
STOPINJECT	Stops the transfer of MPU real-time data to the Servo Processor.
STOPSPDC	Terminate SPDC data collection
SPICFG	Configure SPI Interface

Command	Description
SPIWRITE	Issues the SPI transaction with the number of SPI words to be sent and received in a single transaction
SPRT	Starts a real-time data transfer process from the MPU to a given Servo Processor.
SPRTSTOP	Stops an active real-time data transfer process on the given SP (for cyclic command only).
STOPDC	Terminates data collection.
WRITE	Writes an array to a file in the flash memory.

2.5.1 **AXISDEF**

Description

The **AXISDEF** command enables the user to assign an alias to one or more axes. Once assigned, the user can use the alias throughout the program in any command requiring an axis argument.

Syntax

AXISDEF *axis_alias* = *axis*

Arguments

<i>axis_alias</i>	Any string with the following restrictions: > Only one name can be defined for one axis, that is, different names cannot be used for the same axis. > The names must be unique, i.e., two axes cannot be defined with the same name. > An axis name must not be the same of any other variable name, label, keyword, etc. A compilation error occurs if one of the above restrictions is not satisfied.
<i>axis</i>	The axis number, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1. Adding (R) to the axis parameter means G-Code run on the axis will ignore the modality entry of G20.

Comments

The **AXISDEF** command can be repeated many times to define all required aliases.

The axis name must be defined in the D-Buffer. In any case, the axis definition has global scope (the definitions of the same axis in a different program must be identical as applies to all global variables).

Although postfix indexing can be used, it is recommended using explicit indexing and providing names as symbolic constants.

Related ACSPL+ Commands

None

COM Library Methods

None

C Library Functions

None

Example 1

An axis name can be used in expressions as a symbolic constant. For example, given the program includes the declaration:

```
AXISDEF Q=3
```

the following command

```
VEL(Q)=1000;
```

assigns 1000 to the required velocity of axis 3.

Example 2

As user variables, axis-related standard variables accept explicit indexing. However, axis-related standard variables also accept postfix indexing. For example, given a program that includes the declaration:

```
AXISDEF Q=3, X1=1, X2= 2
```

Example 3

Adding (R) to the axis parameter means G-Code run on the axis will ignore the modality entry of G20 in order to support axes driving a rotational motion with G-Code commands.

```
AXISDEF X=0,Y=1,Z=2,A=5 (R) ,B = 6 (R) ,C=7
```

In this example axis 5 and axis 6 will ignore the modality entry of G20.

Explicit Indexing	Postfix Indexing
VEL(3) or VEL(Q)	VEL3
ACC(1) or ACC(X1)	ACC1
SLVKI(2) or SLVKI(X2)	SLVKI2

2.5.2 DC

Description

DC (Data Collection) accumulates data of any specified standard or user-defined variable with a constant sampling rate. **DC** synchronized with motion (see Command Option **/s**) is called Axis Data Collection. **DC** not synchronized with motion is called System Data Collection.

DC terminates due to:

- > **STOPDC**
- > The defined **DC** array is completed

Syntax (except for DC/s)

DC[*/switch*] *array_name*, *number of points*, *time-interval*, *variable_1*, [*variable_2...variable_N*]

Syntax for DC/s

DC/s *axis*, *global array*, *number of points*, *time-interval*, *variable_1*, [*variable_2...variable_N*]

Arguments

global array	The name of a global array that stores samples
number of points	Define the number of samples
time-interval	Define the time interval between each sample
variable_1	Define variable/s to be sampled. The number of rows defined in array_name must match the number of variables to be sampled

Switches

/switch can be one of the following:

/s	Triggers DC with the execution of the next motion command following the call to DC/s . Motions queued before the call to DC/s will not be recorded. DC synchronized with motion is called Axis Data Collection (See in <i>ACSPL+ Programmer's Guide</i>).
/w	Create the synchronous data collection, but do not start until GO . Command option /w can only be used with the /s .
/c	Cyclic data collection

Comments

1. **DC** can include up to 24 sampled variables.
2. Only one **DC** (system data collection) process can run at the same time.
3. Up to eight **DC/s** (axis data collection) processes can be simultaneously executed where each process fills a separate array.
4. **DC/c** does not self-terminate. **STOPDC** terminates cyclic data collection.

5. **DC/c** uses the collection array as a cyclic buffer and can continue to collect data indefinitely. When the array is full, each new sample overwrites the oldest sample in the array.
6. After the cyclic data collection concludes, the controller reorganizes the sample array so that the first element represents the oldest sample and the last element represents the most recent sample.
7. Variable **S_ST.#DC** = 1 when non-synchronized **DC** is active.
8. Variable **AST<axis>.#DC** = 1 when synchronized **DC** is active.

Related ACSPL+ Commands

[STOPDC](#), [SPDC](#)

Related ACSPL+ Variables

[S_ST](#), [AST](#), [S_DCN](#), [S_DCP](#), [DCN](#), [DCP](#)

COM Library Methods and .NET Library Methods

DataCollection, WaitCollectEnd, StopCollect

C Library Functions

acsc_DataCollectionExt, acsc_WaitCollectEnd, acsc_StopCollect

Example 1:

Cyclic Data Collection

```
GLOBAL REAL ARRAY (2) (1000)

!Define a real type array for two variables

!(rows) and 1000 sampling points (columns).
DC/C ARRAY,1000,3,FPOS0,TIME

!Start cyclic data collection and store 1000

!samples in ARRAY.

!The time between each sampling point is 3 msec.

!The FPOS0 standard variable samples are stored

!in the first row of ARRAY, and the TIME

!variable values are stored in the second row.
TILL ^S_ST.#DC
!Wait until S_ST.#DC = 0 (DC collection is

!finished).
STOP
```

Example 2

Motion Synchronized Data Collection

```

GLOBAL REAL SAMPLE_ARRAY(2)(1000)
                                !Define a real type array for two variables
                                !(rows) and 1000 sampling points (columns).
DC/S 0,SAMPLE_ARRAY,1000,3,FPOS0,TIME
                                !Start cyclic data collection when motion
                                !(synchronized on axis 0) begins, and store 1000
                                !samples in SAMPLE_ARRAY. The time
                                !between each sampling point is 3 msec.
                                !The FPOS0 standard variable samples are stored in
                                !the first row of SAMPLE_ARRAY, and the TIME
                                !variable values are stored in the second row.

TILL ^AST(0).#DC
                                !Wait until AST.#DC = 0 (DC collection is
                                !finished).

STOP

```

2.5.3 STOPDC

Description

Immediately terminates **DC** and **SPDC**.

Syntax

STOPDC[/*switch*]

Switch

/*switch* can be:

/s	Terminate synchronized data collection
----	--

Comments

- > **STOPDC** with an argument delays termination of **DC**.
- > **STOPDC/s** terminates synchronous **DC** initiated by **DC/s**.
- > Multiple axis specification is not allowed.

Related ACSPL+ Commands

[DC](#), [SPDC](#)

Related ACSPL+ Variables

[S_ST](#), [AST](#), [S_DCN](#), [S_DCP](#), [DCN](#), [DCP](#)

COM Library Methods and .NET Library Methods

DataCollection, StopCollect, WaitCollectEnd

C Library Functions

acsc_DataCollectionExt, acsc_StopCollect, acsc_WaitCollectEnd

Example 1

```
STOPDC 50      !Collect an additional 50 samples and then
                !terminate DC.
```

Example 2

```
STOPDC/S 1     !Stop synchronous axis data collection for axis 1
```

2.5.4 READ

Description

Reads a file from the controller’s nonvolatile (flash) memory to a user defined array or variable. The file must exist in the nonvolatile memory by previously writing it using the **WRITE** command.

Syntax

READ *array[,filename]*

or

READ/s *user-variable[, filename]*

Switch

/s	Specifies that the user variable is a scalar and not an array
----	---

Arguments

<i>array</i>	User defined array to which the data will be imported
<i>user-variable</i>	A scalar variable for use with the /s switch, can be either REAL or INT
<i>filename</i>	Optional non-volatile memory file name.

Comments

- 1. The **filename** must not include a file name extension.
- 2. The filename maximum length is 100 chars.
- 3. The **user-array** name must be declared in the buffer where the command is executed.
- 4. The **variable** name may be declared in the buffer where the command is executed, or it may be declared in the D-buffer.
- 5. If **READ** is executed from the **Communication Terminal** as a command, **array** must specify the name of a global array.
- 6. If **READ/s** is executed from the **Communication Terminal** as a command, **variable** must specify the name of a global variable.
- 7. If the optional file name is not supplied, the variable name will be used as the file name.

The following error is supported:

- > Error 3333 "File name MAX length is 100 chars"

Related ACSPL+ Commands**WRITE****COM Library Methods**

Transaction

C Library Functions

acsc_Transaction

Examples

```
GLOBAL REAL ARRAY(5)      !Define a real global array
READ ARRAY,FILENAME       !Read the defined file to the array
```

```
GLOBAL INT VAR            !Define an integer global variable
READ/s VAR,FILENAME       !Reads from file to the variable(scalar)
STOP
```

2.5.5 SPDC**Description**

SPDC (Servo Processor Data Collection) performs fast data collection and accumulates data about the specified Servo Processor variable with a constant maximum sampling rate of 20kHz. A typical use for **SPDC** is for collecting position error (**PE**) and feedback position (**FPOS**) data at the fast Servo Processor rate.

The Servo Processor value is different from the MPU value. The Servo Processor always uses counts and not units. The Servo Processor position value is not affected by a **SET FPOS** command. An offset is added at the MPU level only. The formula (that you can find in our manuals) is:

$$FPOS = FP * EFAC + EOFFS$$

where **FPOS** is the MPU variable and **FP** is the Servo Processor calculated value.

SPDC terminates due to:

- > **STOPSPDC** SP_number
- > After accumulating the defined number of samples

Syntax

SPDC[/r] Array, number_of_samples, sampling_period, SP_number, SP_Address1, [SP_Address2]

Arguments

Array	Array name, up to XARRSIZE variable value. By default, Array is assumed to be an integer array, if the /r switch is added, it defines the array as real.
number_of_samples	The number of samples to collect, the maximum value depends on the size of the array.

sampling_period	The time, in milliseconds, that each sample is taken.
SP_number	The index of the Servo Processor to be sampled
SP_Address1	The address of the Servo Processor variable in the Servo Processor to sample.
SP_Address2	As an option, you can add another address of an other Servo Processor variable in the Servo Processor to sample. In this case, the array should be defined as (2)(N)
SP_Address3	As an option, you can add another address of an other Servo Processor variable in the Servo Processor to sample, In this case, the array should be defined as (3)(N).
SP_Address4	As an option, you can add another address of an other Servo Processor variable in the Servo Processor to sample. In this case, the array should be defined as (4)(N)



Memory addresses may vary between SPiiPlus products and revisions, so definition of a variable to represent **SP_Address** as the return value of **GETSPA** is highly recommended. **SPDC** can then use this variable in any SPiiPlus product or revision.

Related ACSPL+ Commands

STOPDC

Comments

Only one **SPDC** command per Servo Processor can run at the same time.

Table 2-7. Commonly Monitored SPDC Variables

Variable	Axis	Servo Processor Variable
Position Error	0,1,2,3	axes[0].PE
Feedback Position	0,1,2,3	axes[0].fpos
Feedback Velocity	0,1,2,3	axes[0].fvel
Sin Analog Input	0,1,2,3	axes[0].sin
Cos Analog Input	0,1,2,3	axes[0].cos
Phase A Current	0,1,2,3	axes[0].is
Phase B Current	0,1,2,3	axes[0].it
Current Command	0,1,2,3	axes[0].command

```

! ----- Declare data arrays -----
GLOBAL INT DATA(15000)
!Declare global array of integer of size 15000
REAL PAR_ADDRESS
PAR_ADDRESS=GETSPA(0,"axes[0].PE")
! ----- Define motion parameters-----
VEL(0) = 5000
ACC(0) = 100000
DEC(0) = 100000
JERK(0) =2e6
!-----Run motion and do fast data collection -----
SET FPOS(0) = 0
ENABLE 0
SPDC DATA,15000,0.05,0,PAR_ADDRESS
PTP/e 0, 1000
PTP/e 0, -1000
PTP/e 0, 1000
!Use the following if you need to convert the data to one column to
!export to Excel (otherwise you can collect 30000 points by SPDC above)
!Convert the array data from one row to one column to fit to export to
Excel.
!INT DATA1(15000)(1)
!INT J; J=0
!LOOP 15000;DATA1(J)(0)=DATA(J);J=J+1;END
STOP

```

SP_number may be set to 2 at most for most ACS products. The following products support sampling of up to 4 variables:

- > IMDsm
- > ECMsm
- > UDMsm
- > IDMSa
- > ECMsa
- > UDMsa

Related ACSPL+ Commands

STOPDC

2.5.6 STOPSPDC

Description

The **STOPSPDC** function Immediately terminates the data collection of **SPDC** (Servo processor data collection) for the specified servo processor.

Syntax

```
STOPSPDC SP_number
```

Arguments

SP_Number

The number of the Servo Processor.

Return Value

None

Comments

The following errors are supported:

> 3034 - "Illegal index value"

NST.#SPDC bit is set to off (for the relevant server processor)

This variable is supported in version 3.10 and higher.

Related ACSPL+ Commands**SPDC****Related ACSPL+ Variables****NST.#SPDC****Example**

STOPSPDC 1 !Stop Data Collection for Servo Processor 1

2.5.7 WRITE**Description**

Writes an array or scalar (any system or user-defined variable) to a file in the controller's nonvolatile (flash) memory.

Syntax**WRITE** *user-array*[, *filename*]**WRITE/s** *user-array*[, *filename*]**Switch**

/s

Specifies that the user variable is a scalar and not an array

Arguments

<i>user-array</i>	User defined array from which the data will be imported
<i>user-variable</i>	User defined scalar variable, can be either REAL or INT
<i>filename</i>	Optional non-volatile memory file name.

Comments

1. The nonvolatile memory **filename** must not include an extension.
2. The **user-array** or **user-variable** name must be declared in the buffer where the command is executed or it may be declared in a D buffer.

3. If **WRITE** is executed from the **Communication Terminal** as a command, **user-array** must specify the name of a global array.
4. If **WRITE/s** is executed from the **Communication Terminal** as a command, **user-variable** must specify the name of a global scalar variable.
5. If **filename** does not exist, it is created. If the file already exists, it is overwritten.
6. If the optional file name is not supplied, the variable name will be used as the file name.

The following error is supported:

- > Error 3333 "File name MAX length is 100 chars"

Related ACSPL+ Commands

READ

COM Library Methods

Transaction

C Library Functions

acsc_Transaction

2.5.8 SPINJECT

Description

SPINJECT initiates the transfer of MPU real-time data to the Servo Processor.

Syntax

SPINJECT([/switch] *Array*,*Nsamples*,*Node*,*Addr1*,[*Addr2*])

Arguments

Array	Data source: 1 or 2 dimensional ACSPL+ array (real or integer).
Nsamples	Number of samples from the source Array to inject. If the process is not cyclic, the injection will stop after this number of samples. The last telegram will be padded by the last element if needed.
Node	The number of the EtherCAT node as in Servo Processor data collection.
SP_Address1	Address inside the Servo Processor , it must correspond to a floating or integer variable in Servo Processor program.
SP_Address2	Address inside the Servo Processor , it must correspond to a floating or integer variable in Servo Processor program. Used only if Array is 2 dimensional.
SP_Address3	Address inside the Servo Processor , it must correspond to a floating or integer variable in Servo Processor program. Used only if Array is 3 dimensional.

SP_Address4

Address inside the **Servo Processor**, it must correspond to a floating or integer variable in **Servo Processor** program. Used only if **Array** is 4 dimensional.

Switches

/switch can be one of the following:

/c	Cyclic: For each MPU cycle the FW fills CTIME*20 values from the source Array . Once the end is reached, the process continues from the start.
/r	Designates a real source.

Examples

```
!MYDCOM is RT control of DCOM for axis 0 inside Servo Processor 0
REAL MYDCOM(20) ! 20 values are used for CTIME = 1.0
! Initialize MYDCOM here to the desired values
! ...
SPINJECT/CR MYDCOM,20,0,gets pa(0, "axes[0].direct_command")
```

2.5.9 STOPINJECT**Description**

STOPINJECT stops active injection process on the given Servo Processor.

Syntax

STOPINJECT *Servo_Processor*

Arguments

Servo_Processor	Identifies the Servo Processor upon which the injection process is operating.
------------------------	---

Example

```
STOPINJECT 1
!Stops the injection process on Servo Processor1
```

2.5.10 SPICFG**Description**

SPICFG configures and initializes the SPI interface.

Syntax

SPICFG (SlaveIndex, Mode, NumberOfWords, Polarity,Size,Frequency)

Arguments

SlaveIndex	Index of the EtherCAT slave in the EtherCAT network, or 0 in case of IDMsM/ECMsM/UDMsM
Mode	The mode of the SPI interface. The following modes are supported: > 0 - Slave > 1 - Master > 2 - SlaveListenOnly > 3 - Disable > 4 - Master Single Transaction (Used by ACSPL+ SPIWRITE)
NumberOfWords	Number of SPI Data Words used by the application (FW to SPI). Range: {0,8} Not relevant in case of Master Single Transaction Mode
Polarity	Clock Polarity. Four types are available: > Rising Edge – 0 > Rising Edge with Delay – 1 > Falling Edge – 2 > Falling Edge with Delay - 3
Size	Data size in bits, the range is {1-16}.
Frequency	An integer number which defines the frequency. The range is 200KHz-10MHz, limited values are supported (defined in the table below).

Frequency Values Supported

Value	Frequency (kHz)
2	800
3	1000
4	1500
5	2000
6	2500
7	3000
8	3500

Value	Frequency (kHz)
9	4000
10	5000

Return Value

None

Comments

When the SPI interface is not required anymore, SPICFG should be called with Mode=3 (disable) parameter.

Example*2.5.10.1 SPIWRITE**2.5.11 SPIWRITE***Description**

SPIWRITE is a function that issues the SPI transaction with the number of SPI words to be sent and received in a single transaction.

The function may be used in two modes: Slave and Single Master Transaction

Syntax

int **SPIWRITE**(SlaveIndex, NumberOfWords,SPIDataWrite,SPIDataRead,TimeOut)

Arguments

SlaveIndex	Index of the EtherCAT slave in the EtherCAT network, or 0 in case of IDmSm/ECMsm
NumberOfWords	Number of SPI Data Words to be sent/received in the single transaction. Range: {0,8}
SPIDataWrite	INT array. Size must be number of words.
SPIDataRead	INT array. Size must be number of words.
Timeout	(Optional) integer, specifies the timeout in milliseconds. Relevant only in case of slave mode. Default value is 5000.

Return Value

STATUS value, OK (0) or error.

Comments

Master mode behavior:

The data in the **SPIDataWrite** array is written to the SPI interface.

The **SPIDataRead** array contains the reply data.

The function blocks until the reply is ready (when number of received words is equal to the **NumberOfWords** parameter).

Slave mode behavior:

The data in the **SPIDataWrite** array is written to the **EXTOUT** variable (and copied to the SPI interface). The **SPIDataRead** array contains the reply data. The function will wait until one of the following conditions is true:

1. SPIRXN equals to NumberOfWords parameter
2. Timeout is reached
3. Error state is returned The function returns

Example

```

int SPIDataWrite(8)
int SPIDataRead(8)

int i=0

SPICFG(0,4,8,0,16,6) !Master Single Transaction Mode

loop 8
    SPIDataWrite(i)=i+1
    i=i+1
end
SPIWRITE(0,8,SPIDataWrite,SPIDataRead)
STOP

```

2.5.12 SPRT

Description

This function starts a real-time data transfer process from the MPU to a given Servo Processor.

Syntax

SPRT[/c] SP, Value_Array, Addr_Array

Switches

/c

Cyclic. For each MPU cycle, the firmware fills values from the source arrays.

Arguments

SP	Number of the Servo Processor to be used for real-time data transfer.
Value_ Array	Data source, 1-dimensional ACSPL+ real array > up to 20 values for $CTIME \geq 0.50$ > up to 12 values for $0.20 \leq CTIME < 0.50$
Addr_Array	Array of addresses inside that Servo Processor. It must correspond to the float or integer variable in the Servo Processor's program, as well as to the Value_Array values order and size.

Comments



The **SPRT** command cannot be used in parallel with the SPINJECT command for the same Servo Processor.

The **SPRT** command can be used for simultaneous and deterministic update of 12-20 Servo Processor variables at the controller cycle rate.

It is superior to the **SETSP** command that can only update one Servo Processor variable in each controller cycle and cannot be used for continuous update (every controller cycle).

For example, assume that the PIV gains (**SLPKP**, **SLVKP**, **SLVKI**) need to be updated simultaneously and frequently for gain scheduling. Even if the variables are set in the same program line, or with a block command, the controller still updates one Servo Processor variable every controller cycle. Each of the parameters **SLVKP**, **SLPKP**, **SLVKI** has three values according to the motion phase (0=motion, 1=idle, 2= settling) and the corresponding idle and settling gains.

If this is not needed, you could simply set the three values equal for each parameter. The update is completed within several controller cycles, that can influence the system performance.

However, using **SPRT**, the internal Servo Processor variables can be updated simultaneously within one cycle.

Note that **SPRT** affects only Servo Processor variables i.e. corresponding ACSPL+ variables don't change.

Example 1

```
GLOBAL REAL Value(9)
INT Address(9)
INT Axis
GLOBAL REAL SLPKP_value, SLVKP_value, SLVKI_value

Axis = 0
! Finding the relevant addresses can be done as one time operation
! (No need to re-use GETSPA prior to each update).

% Adress of the Servo Processor SLVKP parameter used during motion
Address(0) = getspace(0, "axes[0].params[0].SLVKP")
```

```

% Address of the Servo Processor SLVKP parameter used in idle state
Address(1) = GETSPA(0,"axes[0].params[1].SLVKP")
% Address of the Servo Processor SLVKP parameter used during settling
Address(2) = GETSPA(0,"axes[0].params[2].SLVKP")
Address(3) = GETSPA(0,"axes[0].params[0].SLPKP")
Address(4) = GETSPA(0,"axes[0].params[1].SLPKP")
Address(5) = GETSPA(0,"axes[0].params[2].SLPKP")
Address(6) = GETSPA(0,"axes[0].params[0].SLVKI")
Address(7) = GETSPA(0,"axes[0].params[1].SLVKI")
Address(8) = GETSPA(0,"axes[0].params[2].SLVKI")

BLOCK
! SLPKP=SLPK_value, SLVKP=SLVKP_Value, SLVKI=SLVKI_value must be set
! simultaneously.
! The following lines calculate the corresponding dsp variables:

! Desired ACSPL parameters:
! Desired SLVKP value
SLVKP_value = 100
! Desired SLPKP value
SLVKP_value = 50
! Desired SLVKI value
SLVKP_value = 10

! Translate the ACSPL+ variables to the low level Servo Processor
! variables and store them in an array
Value(0) = SLVKP_value /1024/32766*20000*SAT(POW(2,21)/XVEL(Axis)*EFAC
(Axis),0,1)
Value(1) = Value(0)
Value(2) = Value(0)
Value(3) = SLPKP_value /20000
Value(4)= Value(3)
Value(5) = Value(3)
Value(6) = SLVKI_value /POW(2,16)
Value(7)= Value(6)
Value(8) = Value(6)

! The following command performs the update:
SPRTSTOP 0;
SPRT 0, Value, Address
end
Stop

```

Example 2

```

! Real-Time MPU-Servo Processor Data Transfer (Cyclic)
REAL Value(2)
INT Address(2)
INT SP;
INT i

```

```

SP = 0;
i = 0;
Value(0) = 0; Value(1) = 19;
Address(0) = GETSPA(SP, "dummy_double[1]");
Address(1) = GETSPA(SP, "dummy_double[2]");

SPRTSTOP SP ! For making sure there is no previously running SPRT
commands for the same SP
SPRT/C SP, Value, Address

while 1
    BLOCK
        Value(0) = i
        Value(1) = (20 - i)
        i = i + 1
        IF (i = 20)
            i = 0
        END
    END
END
STOP

```

Example 3

```

! Real-Time MPU-Servo Processor Data Transfer
REAL Value(2)
INT Address(2)
INT SP;
INT i
SP = 0;
i = 0;
Value(0) = 0; Value(1) = 19;
Address(0) = GETSPA(SP, "dummy_double[1]");
Address(1) = GETSPA(SP, "dummy_double[2]");
SPRTSTOP SP ! For making sure there is no previously running SPRT
commands for the same SP

while 1
    BLOCK
        Value(0) = i
        Value(1) = (20 - i)
        i = i + 1
        IF (i = 20)
            i = 0
        END
        SPRT SP, Value, Address
    END
END

```

STOP

2.5.13 SPRTSTOP

Description

SPRTSTOP stops an active real-time data transfer process on the given SP (for cyclic command only).

Syntax

SPRTSTOP SP

Arguments

SP	Number of the EtherCAT node as in SP data collection.
----	---

2.5.14 USAGESP

Description

USAGESP is an array of type REAL, for each node in the EtherCAT network. The variable is used for storing the maximum servo processor usage during the preceding controller cycle.

Syntax

usage = **USAGESP**(axis)

Return Value

Maximum usage

Tag

383

Accessibility

Read-only

Comments

USAGESP stores the maximum **USAGE** value during N ticks, when N depends on the controller's cycle time. If controller's cycle is 1ms, N equals 20.

This variable is supported by the UDMxx, IDMxx and ECMxx products only.

2.6 Motion Commands

The Motion commands are:

Command	Description
ARC1	Adds an arc segment to MSEG...ENDS motion.
ARC1	Adds an arc segment to XSEG...ENDS motion.
ARC1	Adds an arc segment to BSEG...ENDS motion

Command	Description
ARC2	Adds an arc segment to MSEG...ENDS motion.
ARC2	Adds an arc segment to BSEG...ENDS motion
ARC2	Adds an arc segment to XSEG...ENDS motion
BPTP	Creates a motion boost profile
BSEG...ENDS	Creates a blended segmented motion.
JOG	Creates a jog motion.
LINE	Adds a linear segment to MSEG...ENDS motion.
LINE	Adds a linear segment to XSEG...ENDS motion.
LINE	Adds an liners segment to BSEG...ENDS motion
MASTER	Defines a formula for calculating MPOS .
MPOINT	Adds a set of points to MPTP...ENDS , PATH...ENDS or PVSPLINE...ENDS motion.
MPTP...ENDS	Creates a multipoint motion.
MSEG...ENDS	Creates a segmented motion.
PATH...ENDS	Creates an arbitrary path motion with linear interpolation between the specified points.
POINT	Adds a point to MPTP...ENDS , PATH...ENDS , or PVSPLINE...ENDS motion.
PROJECTION	An expansion command to the MSEG...ENDS set of commands, that allows the controller to perform a three dimensional segmented motion such as creating arcs and lines on a user-defined plane.
PTP	Creates a point-to-point motion.
PVSPLINE...ENDS	Creates an arbitrary path motion with spline interpolation between the specified points.
SLAVE	Creates a master-slave motion.
STOPPER	Adds a segment separator to MSEG...ENDS motion.

Command	Description
TRACK	Creates tracking motion.
XSEG...ENDS	Creates an extended segment motion.
NURBS	Creates NURBS motion
NPOINT	Creates NURBS motion segment
SEGMENT	Creates SmoothPath motion segment
SMOVE	Define segment of movement with transition point smoothing
SPATH	Creates SmoothPath motion
SPTP Command	a multi-axis Point-to-Point motion command, using a 4th order profile



For systems having more than 15 axes, avoid using motion commands to start the motion of all axes simultaneously as this may cause Over Usage or Servo Processor Alarm faults

2.6.1 ARC1

Description

ARC1 must be initialized with **MSEG...ENDS**. Use **ARC1** to specify the center point and final point coordinates of an arc and the direction of rotation. Direction is designated by a plus sign (+) or (–) for clockwise or counterclockwise rotation depending on the encoders' connections.

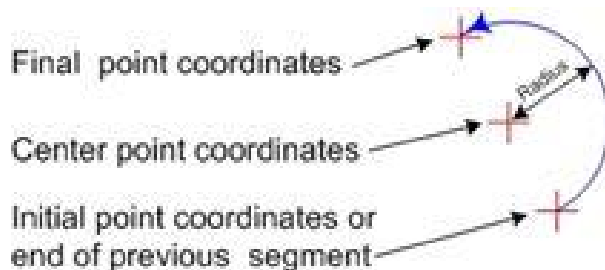


Figure 2-5. ARC1 Coordinate Specification

Syntax

ARC1 *axis_list, center-point, final-point, direction[user-specified velocity]*

Arguments

<i>axis_list</i>	List of axes involved, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1. The ARC1 axis_list can involve two or more axes, see PROJECTION.  A minimum of two axes must be specified.
<i>center-point</i>	Center point coordinates.
<i>final point</i>	Last point.
<i>direction</i>	Use + for motion in the direction of increasing encoder counts, or - for motion in the direction decreasing encoder counts.
<i>user-specified velocity</i>	If MSEG command option /v is used, the user-specified velocity must be the last parameter in the ARC1 syntax.

Comments

- > **ARC1** and **ARC2** differ only by the required arguments. **ARC1** requires the coordinates of the center point, final point, and the direction of rotation. **ARC2** requires the coordinates of the center point and the rotation angle in radians. Each command produces the same result, so selection of either **ARC1** or **ARC2** depends on the available data.
- > A single **ARC1** command can not create a complete circle because the start point and end point of the motion can not be the same. Use two **ARC1** commands, or use **ARC2**.

Related ACSPL+ Commands

[MSEG...ENDS](#), [ARC2](#), [LINE](#), [STOPPER](#), [PROJECTION](#)

COM Library Methods

Arc1, ExtArc1

C Library Functions

acsc_Arc1, acsc_ExtArc1

Example

See [MSEG...ENDS](#).

2.6.2 ARC1

Description

Use **ARC1** to specify the center point and final point coordinates of an arc and the direction of rotation. Direction is designated by a plus sign (+) or (–) for clockwise or counterclockwise rotation depending on the encoders’ connections. When **ARC1** is used for Extended Motion, it must be initialized with **XSEG...ENDS**.

Syntax

ARC1 [/switches] (axis_list), center_point_axis1, center_point_axis2, destination_point_axis1, destination_point_axis2, [destination_point_axis3, ... destination_point_axis6,] direction[, velocity] [,end_velocity][,time][,values, variables[,index [,masks]]][, lci_segment_active]

Arguments

Argument	Description
axis_list	Defines one or two axes, specified as axes numbers separated by commas or as axes names separated by commas. The axes should only be those axes specified in the corresponding XSEG command.
center_point_axis1	Center point position for the first axis
center_point_axis2	Center point position for the second axis
destination_point_axis1	Destination position of the first axis
destination_point_axis2	Destination position of the second axis
destination_point_axis3 ... destination_point_axis6	Mandatory only if AXIS_LIST contains more than 2 axes. Destination position of the rest of axes. Number of destination positions must correspond to the number of axes in the AXIS_LIST .
direction	Direction is specified as + or -. It defines clockwise or counterclockwise rotation depending on the encoder connection: "+" for motion in the direction of increasing encoder counts, or "-" for motion in the direction decreasing encoder counts.
velocity	[Mandatory with /V switch]. Defines required velocity for the current and for all subsequent segments. See Using ARC1, ARC2 and LINE Switches .
end_velocity	[Mandatory with /F switch]. Defines required velocity at the end of the current segment. See Using ARC1, ARC2 and LINE Switches .
time	[Mandatory with /T switch].

Argument	Description
	Defines segment processing time. See Using ARC1, ARC2 and LINE Switches .
values	[Mandatory with /O switch]. Defines the values to be written to variables array at the beginning of the current segment execution. values is a one-dimensional user defined array of integer or real type with maximum size of 10 elements .
variables	[Mandatory with /O switch]. Defines the user-defined array, which will be written with values data at the beginning of the current segment execution. variables is a one-dimensional user defined array of the same type and size as the values array.
index	[Optional, only used with /O switch] Defines the first element (starting from zero) of the variables array, to which values data will be written. If argument is omitted, values data is written to the variables array starting from the first element (index 0).
masks	[Optional, only used if values and variables are integer] Defines the masks that are applied to values before the values are written to variables array at the beginning of the current segment execution. masks is a one-dimensional user-defined array of integer type and the same size as the values array. The masks are only applied for integer values: $\text{variables}(n) = (\text{variables}(n) \text{ AND } (\text{NOT mask}(n))) \text{ OR } (\text{values}(n) \text{ AND mask}(n))$ If argument is omitted, all values bits are written to variables . If values is a real array, the masks argument should be omitted.
lci_segment_active	[Mandatory with /p switch] Integer value. Fire LCI State or Pulse at the beginning of current segment. The function is available if the LCI segment-based mode was previously defined by the SegmentGate or SegmentPulse functions. The value defines the state value in SegmentGate mode (1 or 0). In SegmentPulse mode the value equal 1.



For information on optional switches for this command, see **Using ARC1, ARC2 and LINE Switches**.

2.6.3 ARC1

This format of ARC1 is used for blended segment motion and in this form must be initialized with BSEG...ENDS. The command adds to the motion path an arc segment that starts in the current point and ends in the destination point with the specified center point.

Syntax

```
ARC1[/switches] (axis_list),
      center_point_axis1,center_point_axis2,
      destination_point_axis1,destination_point_axis2, direction
      [,segment_time [,acceleration_time [,jerk_time [,dwell_time]]]]
```

Arguments

Argument	Comments
<i>axis_list</i>	Defines one or two axes, specified as axes numbers separated by comma or as axes names separated by comma. The axes should only be from those axes specified in the corresponding BSEG command.
<i>center_point_axis1</i>	Center point position for the first axis
<i>center_point_axis2</i>	Center point position for the second axis
<i>destination_point_axis1</i>	Destination position of the first axis
<i>destination_point_axis2</i>	Destination position of the second axis
<i>direction</i>	Direction is specified as + or -. It defines clockwise or counterclockwise rotation depending on the encoder connection: "+" for motion in the direction of increasing encoder counts, or "-" for motion in the direction decreasing encoder counts.
<i>segment_time</i>	Only if switch/ m is specified: Segment time (T_m) in milliseconds.
<i>acceleration_time</i>	Only if switch/ a is specified: Acceleration time (T_a) in milliseconds.
<i>jerk_time</i>	Only if sufswitchfix /s is specified: Jerk time (T_j) in milliseconds
<i>dwell_time</i>	Only if sufswitchfix /d is specified: Dwell time at the final point of the segment in milliseconds. With this switch no blending will be done at the segment final point.

2.6.4 ARC2

Description

ARC2 must be initialized with **MSEG...ENDS**. Use **ARC2** to specify the center point and rotation angle in radians of an arc segment. Designate direction by positive or negative rotation angle, depending on the encoders' connections.

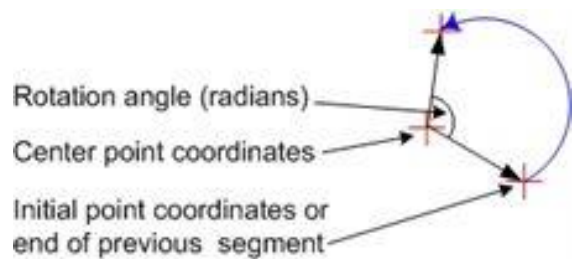


Figure 2-6. ARC2 Center Point and Rotation Angle Specification

Syntax

ARC2 *axis_list*, *center-point*, *rotation-angle and direction* [*user-specified velocity*]

Arguments

<i>axis_list</i>	List of axes involved, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1. The ARC2 <i>axis_list</i> can involve two or more axes, see PROJECTION.  A minimum of two axes must be specified.
<i>center-point</i>	center point coordinates
<i>final point</i>	last point
<i>rotation angle and direction</i>	Rotation is in radians. Use + for motion in the direction of increasing encoder counts, or - for motion in the direction decreasing encoder counts
<i>user-specified velocity</i>	If MSEG command option /v is used, the user-specified velocity must be the last parameter in the ARC2 syntax.

Comments

ARC1 and **ARC2** differ only by the required arguments. **ARC1** requires the coordinates of the center point, final point, and the direction of rotation. **ARC2** requires the coordinates of the center point and the rotation angle. Each command produces the same result, so selection of either **ARC1** or **ARC2** depends on the available data.

See **Using ARC1, ARC2 and LINE Switches** for details about function switches.

Related ACSPL+ Commands

MSEG...ENDS, ARC1, LINE, STOPPER, PROJECTION

COM Library Methods

Arc2, ExtArc2

C Library Functions

acsc_Arc2, acsc_ExtArc2

Example

See MSEG...ENDS.

2.6.5 ARC2

Description

This format of **ARC2** is used for extended segment motion and in this form must be initialized with **XSEG...ENDS**. Use **ARC2** to specify the center point and rotation angle in radians of an arc segment. Designate direction by positive or negative rotation angle, depending on the encoders' connections.

Syntax

ARC2[/switches] (axis_list), center_point_axis1,center_point_axis2, rotation_angle, [,destination_point_axis3, ...
destination_point_axis6][,velocity][,end_velocity][,time][,values, variables[,index[,masks]]]
[,external_loop_type, external_loop_type, maximum_allowed_deviation][, lci_segment_active]

Arguments

<i>axis_list</i>	List of axes numbers separated by comma or as axes names separated by comma. The axes should only be those axes specified in the corresponding XSEG command.  A minimum of two axes must be specified.
<i>center_point_axis1</i>	Center point position for the first axis
<i>center_point_axis2</i>	Center point position for the second axis
<i>destination_point_axis3</i> ... <i>destination_point_axis6</i>	Mandatory only if axis_list contains more than 2 axes. Destination position of the rest of axes. Number of destination positions must correspond to the number of axes in the axis_list.

<i>rotation_angle</i>	Defines central angle of the arc, signed according to rotation direction: plus for a counter-clock-wise arc, minus for a clock-wise arc.
<i>velocity</i>	[Optional, only used with /v switch] Defines required velocity for the current and for all subsequent segments. See Switches explanation for details.
<i>end_velocity</i>	[Optional, only used with /f switch] Defines required velocity at the end of the current segment. See Switches explanation for details.
<i>time</i>	[Mandatory with /t switch]. Defines segment processing time.
<i>values</i>	[Optional, only used with /o switch] Defines the values to be written to variables array at the beginning of the current segment execution. values is a one-dimensional user defined array of integer or real type with maximum size of 10 elements .
<i>variables</i>	[Optional, only used with /o switch] Defines the user-defined array, which will be written with values data at the beginning of the current segment execution. variables is a one-dimensional user defined array of the same type and size as the values array.
<i>index</i>	[Optional, only used with /o switch] Defines the first element (starting from zero) of the variables array, to which values data will be written. If argument is omitted, values data is written to the variables array starting from the first element (index 0).
<i>masks</i>	[Optional, only used if values and variables are integer] Defines the masks that are applied to values before the values are written to variables array at the beginning of the current segment execution. masks is a one-dimensional user-defined array of integer type and the same size as the values array. The masks are only applied for integer values: $\text{variables}(n) = (\text{variables}(n) \text{ AND } (\text{NOT mask}(n))) \text{ OR } (\text{values}(n) \text{ AND mask}(n))$ If argument is omitted, all values bits are written to variables . If values is a real array, the masks argument should be omitted.

<i>external_loop_type</i>	0 - Cancel external loop 1 - Smooth External loop (line-arc-line) 2 - Triangle External loop (line-line-line)
<i>minimum_segment_length</i>	If the lengths of both segments are more than this value, the skywriting algorithm will be applied.
<i>maximum_allowed_deviation</i>	The parameter limits the external loop deviation from the defined profile. If the value is negative – no limitation.
<i>lci_segment_active</i>	[Mandatory with /p switch] Integer value. Fire LCI State or Pulse at the beginning of current segment. The function is available if the LCI segment-based mode was previously defined by the SegmentGate or SegmentPulse functions. The value defines the state value in SegmentGate mode (1 or 0). In SegmentPulse mode the value equal 1.

Switches

The following optional ***/switches*** may be used singularly or in combination with the **ARC2** command:

<i>/v</i>	Specify required velocity. The switch requires an additional parameter that specifies the required velocity. The switch changes the required velocity for the current segment and for all subsequent segments. If the switch is not specified, the required velocity does not change.
<i>/f</i>	Decelerate to the end of segment. The switch requires an additional parameter that specifies the end velocity. The controller decelerates to the specified velocity at the end of segment. The specified value should be less than the required velocity; otherwise the parameter is ignored. The switch affects only one segment. The switch also disables corner detection and processing at the end of segment. If the switch is not specified, deceleration is not required. However, in special cases the deceleration might occur due to corner processing or other velocity control conditions.
<i>/o</i>	Synchronize user variables with segment execution. The switch requires additional two or three parameters that specify values , user variable and mask . See details in Arguments for explanation.
<i>/b</i>	Use external loops at corners. The switch requires additional parameters that specify the external loop type, the minimum segment length, and the maximum allowed deviation from profile.
<i>/p</i>	Specifies that the <i>lci_segment_active</i> parameter is required

2.6.6 ARC2

This format of ARC2 is used for blended segment motion and in this form must be initialized with BSEG...ENDS. The command adds to the motion path an arc segment that starts in the current point and specified as the center point and rotation angle.

Syntax

```
ARC2[/switches] (axis_list),
      center_point_axis1,center_point_axis2,
      rotation_angle
      [,segment_time [,acceleration_time [,jerk_time [,dwell_time]]]]
```

Arguments

Arguments	Comments
<i>axis_list</i>	Defines one or two axes, specified as axes numbers separated by comma or as axes names separated by comma. The axes should only be from those axes specified in the corresponding BSEG command.
<i>center_point_axis1</i>	Center point position for the first axis
<i>center_point_axis2</i>	Center point position for the second axis
<i>rotation_angle</i>	Defines central angle of the arc, signed according to rotation direction: plus on counter-clock-wise arc, minus on clock-wise arc.
<i>segment_time</i>	Only if switch/ m is specified: Segment time (T_m) in milliseconds.
<i>acceleration_time</i>	Only if switch/ a is specified: Acceleration time (T_a) in milliseconds.
<i>jerk_time</i>	Only if switch/ s is specified: Jerk time (T_j) in milliseconds
<i>dwell_time</i>	Only if switch/ d is specified: Dwell time at the final point of the segment in milliseconds. With this switch no blending will be done at the segment final point.

2.6.7 BPTP

Description

BPTP defines a motion profile using the MotionBoost Feature.

Syntax

```
BPTP[/switch] axis_list, destination_point, [value of Tf, value of Vf, motor_motion_delay]
```


Switches

Arguments	Comments
None	Travel time will be calculated by the MPU to achieve a the minimum possible time
/t	Minimum travel time in seconds, The calculated travel time will be at least the specified value. Incompatible with the /d switch.
/d	Travel Time – specifies the exact travel time for the motion in seconds. All other considerations are ignored, which could cause a safety fault during motion execution. Incompatible with the /t switch.
/f	User will enter final, nonzero velocity. In single axis motion the sign of the final velocity parameter has no effect, only the absolute value is considered.
/e	Wait for motion termination before executing next command.
/r	Relative motion
/v	Use velocity parameter instead of default velocity parameters.
/w	Create the motion, but to not start until the GO command is issued.
/z	Interpret entered coordinates according to the Local Coordinate System.
/m	Use the motion profile values of the axis group as a whole, rather than those of the leading axis, without exceeding any of the defined axes motion VEL , ACC , DEC , JERK values. Not compatible with /2 switch. Range is 0-25 ms.
/q	Defines actual motor movement delay in microseconds. The delay resolution is 50 microseconds. The maximum delay is 100 controller cycles: 100ms for CTIME=1ms or 20ms for CTIME=0.2ms. Allows delaying actual motor movement start (RPOS) for the specified delay, while motion profile generation (APOS) starts immediately.
/2	Use 20 kHz motion mode



Use of the /d switch to specify minimum travel time overrides all other parameters which might limit velocity and requires careful attention to safety considerations.



BPTP/2 is limited to at most 2 axes in a single function call.



The **BPTP/2** command is limited to at most 2 axes per Servo Processor and at most 4 axes per system.

Arguments

axis_list

a	Axis a will perform the motion
(axes)	Axes will perform a synchronized motion to the destination point along to a straight line connecting the start to the destination point. Axes may be any set of distinct numbers in the range 0 to 63.

destination-point

(af,bf,cf)	af is a destination point for axis a bf is a destination point for axis b cf is a destination point for axis c
------------	--

Value of Tf

Tf	Desired travel time
----	---------------------

Value of Vf

Vf	Desired final velocity
----	------------------------

Motion Delay

motor_ movement_delay	(Optional, used only with /q switch) Delay, in milliseconds, before motor motion actually starts.
------------------------------	---

GPHASE

Two options are available.

- > Four phases (For motion in positive direction; for motion in negative direction reverse the inequality signs)
 1. Acceleration buildup
 - > RJERK>0, RACC>0
 2. Acceleration finishing
 - > RJERK<0, RACC>0
 3. Deceleration buildup
 - > RJERK<0, RACC<0

- 4. Deceleration finishing
- > RJERK>0, RACC<0

Comments

This command is supported in ADK versions 2.70 and higher.

Examples

```
BPTP 0, 100
```

A simple example that moves axis 0 to position 100.
If the axis is moving when the command is issued, the controller creates the motion and inserts it into the axis motion queue. The motion waits in the queue until all motions before it finish, and only then starts.

```
BPTP/ed (0,1), 100, 200, 0.4
```

Move axes 0,1 to position 100,200.
Do not execute the next command in the program until the motion is done.
Execute the motion in exactly 0.4 seconds.

```
// execute a BPTP motion with 20 KHz resolution
// move axes 0,1 to absolute coordinates 1,1

BPTP/2 (0,1),1,1
```

2.6.8 BPTPCalc

Description

The **BPTPCALC** function calculates and allows the user to set the motion variables according to a desired motion time. When the travel time and distance are known in advance, the **BPTPCALC** should be used to generate new **VEL**, **ACC** and **JERK** values.

Syntax

```
BPTPCALC real Motion_duration, real Distance, int index
```

Arguments

Motion_duration	The desired motion time in seconds. The time will be rounded up to a whole number of controller cycles.
Distance	The travel distance in user units
Index	1 - Velocity 2 - Acceleration 3 - Jerk

Comments

This command is supported in ADK versions 2.70 and higher.

Example

```
! Calculate the motion parameter to execute a 10 unit BPTP motion in 3 ms
VEL(0) = bptpcalc(0.003, abs(10), 1)
ACC(0) = bptpcalc(0.003, abs(10), 2)
JERK(0) = bptpcalc(0.003, abs(10), 3)

enable(0)

BPTP/R(0), 10 ! Move axis 0, 10 units, in 3 ms

STOP
```

2.6.9 BSEG...ENDS

Syntax

BSEG[/switches] (*axis_list*), *initial_position_axis1*, *initial_position_axis2*, [*motor_motion_delay*,] *segment_time*, *acceleration_time*, *jerk_time*[, *dwelling_time*]

Arguments

Arguments	Comments
axis_list	Axes involved in BSEG motion, specified as axes numbers separated by comma or as axes names separated by comma
initial_position_axis1	Initial position of the first axis
initial_position_axis2	Initial position of the second axis
motor_motion_delay	(Optional, used only with /q switch) Delay, in milliseconds, before motor motion actually starts.
segment_time	Initial segment time (T_m) in milliseconds. The specified segment time will be used for all segments until <i>segment_time</i> argument is specified in segment LINE, ARC1 or ARC2 command.
acceleration_time	Initial acceleration time (T_a) in milliseconds. The specified acceleration time will be used for all segments until <i>acceleration_time</i> argument is specified in LINE, ARC1 or ARC2 command.
jerk_time	Initial jerk time (T_j) in milliseconds. The specified jerk time will be used for all segments until <i>jerk_time</i> argument is specified in LINE, ARC1 or ARC2 command.

dwel_time	Optional] Initial dwell time between segments in milliseconds. If this argument is specified, no blending will be done for all segments of the motion. That means that the motion will be stopped at the end of each segment for the specified <i>dwel_time</i> milliseconds.
------------------	---

Switches

Switch	Comments
/w	Do not start until the GO command is executed. If the switch is not specified, the motion starts immediately after the first motion segment is defined.
/r	Set the initial axis position as origin. Segment commands positions should be declared relative to the new origin.
/z	Interpret entered coordinates according to the Local Coordinate System.
/q	Defines actual motor movement delay in microseconds. The delay resolution is 50 microseconds. The maximum delay is 100 controller cycles: 100ms for CTIME=1ms or 20ms for CTIME=0.2ms. Allows delaying actual motor movement start (RPOS) for the specified delay, while motion profile generation (APOS) starts immediately.

2.6.10 JOG

Description

JOG creates a motion with constant velocity and no defined end point.

JOG motion terminates by:

- > **HALT** , **KILL/KILLALL**, **BREAK**, **DISABLE/DISABLEALL**
- > Execution of any other motion command for the same axis
- > Limit switch activation
- > Any fault activation that disables the drive or kills the motion

Syntax

JOG[*/switch*] *axis_list* [*user-specified-velocity*][*user-specified-direction*]

Arguments

<i>axis_list</i>	List of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>user-specified-velocity</i>	Optional parameter used if the /v command option is specified
<i>user-specified-direction</i>	Optional parameter. Motion direction is designated by a plus sign (+) for increasing feedback counts or (–) or decreasing feedback counts. If no operator is used, motion is in the direction of increasing encoder counts.

Switches

/switch can be:

/v	Use the specified velocity instead of the default velocity (VEL).
/w	Create the motion, but do not start until GO .

Related ACSPL+ Commands

[GO](#), [HALT](#), [KILL/KILLALL](#), [BREAK](#), [IMM](#)

Related ACSPL+ Variables

[AST](#), [MST](#), [MERR](#), [VEL](#), [ACC](#), [JERK](#), [FPOS](#), [RPOS](#), [GPHASE](#)

COM Library Methods and .NET Library Methods

Jog, JogM

C Library Functions

acsc_Jog, **acsc_JogM**

Examples

Example 1:

```
JOG/v 0, 500           !Jog 0 axis with VEL 500
```

Example 2:

```
JOG (0,1,2), -++      !Jog axes 0, 1, and 2 where axis 0 jogs in the
                        !negative direction and axes 1 and 2 jog in the
                        !positive direction.
```

2.6.11 LINE**Description**

LINE must be initialized with [MSEG...ENDS](#). **LINE** adds a linear segment to a segmented motion.

Syntax

LINE *axis_list, final-position1, final-position2[,user-specified velocity]*

Arguments

<i>axis_list</i>	List of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.  A minimum of two axes must be specified.
<i>final position1</i>	First coordinate
<i>final position2</i>	Second coordinate
<i>user specified velocity</i>	Optional user-specified velocity if LINE was initiated by MSEG/v .

Comments

1. **ENDS** informs the controller that no more segments will be specified. If **ENDS** is omitted, motion stops at the last point of the sequence and waits for the next point.
2. If the **LINE** *axis_list* involves two or more axes, see [PROJECTION](#).

Related ACSPL+ Commands

[MSEG...ENDS](#), [ARC1](#), [ARC2](#), [STOPPER](#), [PROJECTION](#)

COM Library Methods

See "Points and Segments Manipulation Methods" in *SPiiPlus COM Library Programmer's Guide*.

C Library Functions

See "Points and Segments Manipulation Methods" in *SPiiPlus C Library Reference Programmer's Guide*.

Example

See [MSEG...ENDS](#).

2.6.12 LINE

Description

This format of **LINE** is used for extended segment motion and in this form must be initialized with [XSEG...ENDS](#). Use **LINE** to add a linear segment that starts at the current point and ends in the destination point to the motion path.

Syntax

LINE [/switches] (*axis_list*), *destination_point_axis1*, *destination_point_axis2*
 [, *destination_point_axis3* ... , *destination_point_axis6*][, *velocity*][, *end_velocity*][, *time*][, *values*,
variables[, *index*[, *masks*]]] [, *external_loop_type*, *external_loop_type*, *maximum_allowed_deviation*][
 , *lc_segment_active*]

Arguments

<i>axis_list</i>	List of axes numbers separated by comma or as axes names separated by comma. The axes should only be those axes specified in the corresponding XSEG command.  A minimum of two axes must be specified.
<i>destination_point_axis1</i>	Destination position of the first axis
<i>destination_point_axis2</i>	Destination position of the second axis
<i>destination_point_axis3</i> ... <i>destination_point_axis6</i>	Mandatory if axis_list contains more than 2 axes. Destination position of the rest of axes. Number of destination positions must correspond to the number of axes in axis_list.
<i>velocity</i>	[Optional, only used with /v switch] Defines required velocity for the current and for all subsequent segments. See Switches explanation for details.
<i>end_velocity</i>	[Optional, only used with /f switch] Defines required velocity at the end of the current segment. See Switches explanation for details.
<i>time</i>	[Mandatory with /t switch]. Defines segment processing time.
<i>values</i>	[Optional, only used with /o switch] Defines the values to be written to variables array at the beginning of the current segment execution. values is a one-dimensional user defined array of integer or real type with maximum size of 10 elements .
<i>variables</i>	[Optional, only used with /o switch] Defines the user-defined array, which will be written with values data at the beginning of the current segment execution. variables is a one-dimensional user defined array of the same type and size as the values array.
<i>index</i>	[Optional, only used with /o switch]

	Defines the first element (starting from zero) of the variables array, to which values data will be written. If argument is omitted, values data is written to the variables array starting from the first element (index 0).
masks	[Optional, only used if values and variables are integer] Defines the masks that are applied to values before the values are written to variables array at the beginning of the current segment execution. masks is a one-dimensional user-defined array of integer type and the same size as the values array. The masks are only applied for integer values: $\text{variables}(n) = (\text{variables}(n) \text{ AND } (\text{NOT mask}(n))) \text{ OR } (\text{values}(n) \text{ AND mask}(n))$ If argument is omitted, all values bits are written to variables. If values is a real array, the masks argument should be omitted.
external_loop_type	0 - Cancel external loop 1 - Smooth External loop (line-arc-line) 2 - Triangle External loop (line-line-line)
minimum_segment_length	If the lengths of both segments are more than this value, the skywriting algorithm will be applied.
maximum_allowed_deviation	The parameter limits the external loop deviation from the defined profile. If the value is negative – no limitation.
lci_segment_active	[Mandatory with /p switch] Integer value. Fire LCI State or Pulse at the beginning of current segment. The function is available if the LCI segment-based mode was previously defined by the SegmentGate or SegmentPulse functions. The value defines the state value in SegmentGate mode (1 or 0). In SegmentPulse mode the value equals 1.

Switches

The following optional **/switches** may be used singularly or in combination with the **LINE** command:

/v	Specify required velocity. The switch requires an additional parameter that specifies the required velocity. The switch changes the required velocity for the current segment and for all subsequent segments. If the switch is not specified, the required velocity does not change.
/f	Decelerate to the end of segment.

	The switch requires an additional parameter that specifies the end velocity. The controller decelerates to the specified velocity at the end of segment. The specified value should be less than the required velocity; otherwise the parameter is ignored. The switch affects only one segment. The switch also disables corner detection and processing at the end of segment. If the switch is not specified, deceleration is not required. However, in special cases the deceleration might occur due to corner processing or other velocity control conditions.
/o	Synchronize user variables with segment execution. The switch requires additional two or three parameters that specify values , user variable and mask . See details in Arguments for explanation.
/b	Use external loops at corners. The switch requires additional parameters that specify the external loop type, the minimum segment length, and the maximum allowed deviation from profile.
/p	Specifies that the lci_segment_active parameter is required

2.6.13 LINE

This format of LINE is used for blended segment motion and in this form must be initialized with BSEG...ENDS. The command adds to the motion path a linear segment that starts in the current point and ends in the destination point.

Syntax

LINE[/switches] (*axis_list*),*destination_point_axis1*,*destination_point_axis2*
 [,*segment_time* [,*acceleration_time* [,*jerk_time* [,*dwel_time*]]]]

Arguments

Arguments	Comments
<i>axis_list</i>	Defines one or two axes, specified as axes numbers separated by comma or as axes names separated by comma. The axes should only be from those axes specified in the corresponding BSEG command.
<i>destination_point_axis1</i>	Destination position of the first axis
<i>destination_point_axis2</i>	Destination position of the second axis
<i>segment_time</i>	Only if the /m switch is specified: Segment time (Tm) in milliseconds.
<i>acceleration_time</i>	Only if /a switch is specified: Acceleration time (Ta) in milliseconds.
<i>jerk_time</i>	Only if the /s switch is specified: Jerk time (Tj) in millisecond

dwell_time

Only if the **/d** switch is specified: Dwell time at the final point of the segment in milliseconds.

With this switch no blending will be done at the segment final point.

2.6.14 MASTER

Description

MASTER defines master-slave motion by creating a dependency between an axis position or velocity to a variable and/or an expression. **MASTER** always follows the (axis) **MPOS** variable. **SLAVE** initiates the motion defined by **MASTER** and must follow **MASTER**.



Velocity Lock is the default state for **MASTER**. Initiate **MASTER** with **SLAVE/p** for position lock.

The following actions terminate the master-slave dependency, however, these actions do not necessarily terminate the motion:

- > **KILL** or **HALT** to the slave axis.
- > **DISABLE** to either axis or both axes.
- > Setting the logical dependence between master and slaves axes to zero. For example, **MASTER MPOS(0) = 0**.

Syntax

MASTER *axis_MPOS=formula*

Arguments

<i>axis_MPOS</i>	Define master value for axis
=	Assignment operator
<i>formula</i>	A variable and/or expression

Related ACSPL+ Commands

SLAVE, **GO**, **HALT**, **KILL/KILLALL**, **ENABLE/ENABLE ALL**, **DISABLE/DISABLEALL**, **MAP**

Related ACSPL+ Variables

AST, **XSACC**, **MFF**

COM Library Methods and .NET Library Methods

SetMaster, Slave, SlaveStalled

C Library Functions

acsc_SetMaster, acsc_Slave, acsc_SlaveStalled

Examples

Example 1:

```
MASTER MPOS (0)=5*RVEL (1)
```

```
!Creates a master-slave dependency where  
!the 0 axis velocity is slaved to five times
```

```
SLAVE 0
```

```
!the 1 axis reference velocity (RVEL)  
!Initiates the 0 axis motion
```

Example 2:

Figure 2-7 illustrates **SLAVE/pt** used in the following syntax example.

```
MASTER MPOS (0)=FPOS (1)+MARK (1)
```

```
!Create a master slave dependency, where  
!axis 0 is the slave. The master axis  
!is 1, and the FPOS (feedback position)  
!expression plus the MARK position  
!are the references for the slaved axis 0.
```

```
SLAVE/pt 0, -500, 500
```

```
!SLAVE command with position lock  
!and specified boundary switches,  
!for axis 0. -500 is the left  
!position boundary, and 500 is the  
!right position boundary.
```

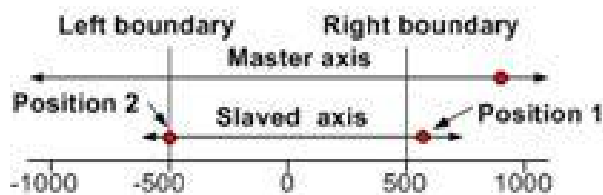


Figure 2-7. SLAVE /pt Illustration

In Figure 2-7, Position 1 is outside of the defined boundary and the master-slave dependency is stalled. Position 2 is within the defined boundary and the master-slave dependency is active.

2.6.15 MPOINT

Description

MPOINT specifies an array of destination points used by **MPTP...ENDS**, **PATH...ENDS** or **PVSPLINE...ENDS** motion commands. An **MPOINT** array must conclude with **ENDS**.

Syntax

MPOINT *axis_list, array-name (number of rows,number of points)*

Arguments

<i>axis_list</i>	List of axes involved. Axes must be in the same order as defined in MPTP , PATH or PVSPLINE .
<i>array (number of rows,number of points)</i>	The MPOINT array must be defined as Real. The number of rows is the first element and the number of columns is the second element. A three-axis, five-point array is defined as follows: real array (3)(5). Each row defines the values for the points for each axes, and depending on command preceding MPOINT, the defined velocities, and the defined times. The array must contain, at least, the same number of columns declared in the array as the number of defined points. Extra array columns are ignored. See Array Structures for a description of various array configurations.

Related ACSPL+ Commands

[MPTP...ENDS](#), [POINT](#), [PATH...ENDS](#), [PVSPLINE...ENDS](#)

COM Library Methods and .NET Library Methods

MultiPoint, MultiPointM, Spline, SplineM, AddPVPoint, AddPVPointM, AddPVTPoint, AddPVTPointM

C Library Functions

acsc_MultiPoint, acsc_MultiPointM, acsc_Spline, acsc_SplineM, acsc_AddPVPoint, acsc_AddPVPointM, acsc_AddPVTPoint, acsc_AddPVTPointM

Array Structures

There are different point array structures depending on the **MPOINT** enabling command and switches. These structures are described below.

1. A five-point, three-axis **MPOINT** array enabled by **MPTP** or **PATH** without switches appears as follows:

ARRAY (3)(5)					
	Point 1	Point 2	Point 3	Point 4	Point 5
0 Axis	1000	2000	3000	4000	5000
1 Axis	5000	4000	3000	2000	1000
5 Axis	0	2000	4000	6000	8000

See [Example 1](#), illustrating sample code based on the **ARRAY (3)(5)** structure.

2. If **MPOINT** follows **MPTP/v**, the point array must include an additional row to specify the transition velocity from the previous point to the current point and appears as follows:

ARRAY (4)(5)					
	Point 1	Point 2	Point 3	Point 4	Point 5
0 Axis	1000	2000	3000	4000	5000
1 Axis	5000	4000	3000	2000	1000
5 Axis	0	2000	4000	6000	8000
Velocity	6000	7000	5000	4000	4000

See [Example 2](#), illustrating sample code based on the **ARRAY (4)(5)** structure.

- If **MPOINT** follows **PATH/t**, the point array must include an additional row to specify the time interval between the previous point and the current point and appears as follows:

ARRAY (4)(5)					
	Point 1	Point 2	Point 3	Point 4	Point 5
0 Axis	1000	2000	3000	4000	5000
1 Axis	5000	4000	3000	2000	1000
5 Axis	0	2000	4000	6000	8000
Time	250	250	250	250	250

Time is specified in milliseconds.

- If **MPOINT** follows **PVSPLINE** without the **/t** switch, the array must include two sets of rows for each axis:
 - > The first set defines the destination points
 - > The second set defines the velocity at the destination points

For example:

ARRAY (6)(5)					
	Point 1	Point 2	Point 3	Point 4	Point 5
0 Axis	1000	2000	3000	4000	5000
1 Axis	5000	4000	3000	2000	1000
5 Axis	0	2000	4000	6000	8000
0 Axis VEL	5000	5000	5000	5000	5000

ARRAY (6)(5)					
	Point 1	Point 2	Point 3	Point 4	Point 5
1 Axis VEL	2500	2500	2500	2500	2500
5 Axis VEL	3000	3000	3000	3000	3000

5. If **MPOINT** follows **PVSPLINE/t**, the array must include an additional column that specifies the time interval between the previous point and the current point. **Time** is in milliseconds. For example:

ARRAY (7)(5)					
	Point 1	Point 2	Point 3	Point 4	Point 5
0 Axis	1000	2000	3000	4000	5000
1 Axis	5000	4000	3000	2000	1000
5 Axis	0	2000	4000	6000	8000
0 Axis VEL	5000	5000	5000	5000	5000
1 Axis VEL	2500	2500	2500	2500	2500
5 Axis VEL	3000	3000	3000	3000	3000
Time	500	500	500	500	500

See [Example 3](#), illustrating sample code based on the **ARRAY (7)(5)** structure.

6. If **MPTP/r** enables **MPOINT**, the array points are relative.

Examples

Example 1

Illustrating sample code based on the [ARRAY \(3\)\(5\)](#) structure.

```

REAL ARRAY(3)(5)           !Defines ARRAY with three rows and
                           !five columns as real.
INT NUMBER_of_POINTS       !Defines NUMBER_of_POINTS as an integer variable.
! ----- Fill the ARRAY array-----
ARRAY(0)(0)=1000;ARRAY(0)(1)=2000;ARRAY(0)(2)=3000;ARRAY(0)(3)=4000;
ARRAY(0)(4)=5000;ARRAY(1)(0)=5000;ARRAY(1)(1)=4000;ARRAY(1)(2)=3000;
ARRAY(1)(3)=2000;ARRAY(1)(4)=1000;ARRAY(2)(0)=0;ARRAY(2)(1)=2000;
ARRAY(2)(2)=4000;ARRAY(2)(3)=6000;ARRAY(2)(4)=8000
NUMBER_of_POINTS=5         !Set NUMBER_of_POINTS
ENABLE(0,1,5)              !Enable axes 0, 1 and 5
PATH(0,1,5), 1             !PATH initiates simultaneous motion for axes 0, 1,
                           !and 5. The time interval is one millisecond between

```

```

                                !each destination point.
MPOINT (0,1,5), ARRAY, NUMBER_of_POINTS
                                !Define an MPOINT array for axes 0, 1 and 5 where
                                !ARRAY and NUMBER_of_POINTS are called
ENDS (0,1,5)                    !Concludes MPOINT
STOP                            !End program

```

Example 2

Illustrating sample code based on the [ARRAY \(4\)\(5\)](#) structure.

```

REAL ARRAY(4)(5)                !Defines ARRAY with four rows and
                                !five columns as real.
INT NUMBER_of_POINTS            !Defines NUMBER_of_POINTS as an integer
                                !variable.
! ----- Fill the ARRAY array-----
ARRAY(0)(0)=1000;ARRAY(0)(1)=2000;ARRAY(0)(2)=3000;ARRAY(0)(3)=4000;
ARRAY (0)(4)=5000;ARRAY(1)(0)=5000;ARRAY(1)(1)=4000;ARRAY(1)(2)=3000;
ARRAY (1)(3)=2000;ARRAY(1)(4)=1000;ARRAY(2)(0)=0;ARRAY(2)(1)=2000;
ARRAY(2)(2)=4000;ARRAY(2)(3)=6000;ARRAY(2)(4)=8000;ARRAY(3)(0)=6000;
ARRAY(3)(1)=7000;ARRAY(3)(2)=5000;ARRAY(3)(3)=4000;ARRAY(3)(4)=4000
NUMBER_of_POINTS=5              !Set NUMBER_of_POINTS
ENABLE (0,1,5)                 !Enable axes 0, 1 and 5
PATH (0,1,5), 1
                                !PATH initiates simultaneous motion for axes
                                !0, 1 and 5. The time interval is one millisecond
                                !between each destination point.
MPOINT (0,1,5), ARRAY, NUMBER_of_POINTS
                                !Define an MPOINT array for axes 0, 1 and 5
                                !where ARRAY and NUMBER_of_POINTS are called.
ENDS (0,1,5)                    !Concludes MPOINT
STOP                            !End program

```

Example 3

Illustrating sample code based on the [ARRAY \(7\)\(5\)](#) structure.

```

REAL ARRAY(7)(5)                !Defines ARRAY with seven rows and
                                !five columns as real.
INT NUMBER_of_POINTS            !Defines NUMBER_of_POINTS as an integer variable.
! ----- Fill the ARRAY array-----
ARRAY(0)(0)=1000;ARRAY(0)(1)=2000;ARRAY(0)(2)=3000;ARRAY(0)(3)=4000;
ARRAY (0)(4)=5000;ARRAY(1)(0)=5000;ARRAY(1)(1)=4000;ARRAY(1)(2)=3000;
ARRAY (1)(3)=2000;ARRAY(1)(4)=1000;ARRAY(2)(0)=0;ARRAY(2)(1)=2000;
ARRAY(2)(2)=4000;ARRAY(2)(3)=6000;ARRAY(2)(4)=8000;ARRAY(3)(0)=5000;
ARRAY(3)(1)=5000;ARRAY(3)(2)=5000;ARRAY(3)(3)=5000;ARRAY(3)(4)=5000;
ARRAY(4)(0)=2500;ARRAY(4)(1)=2500;ARRAY(4)(2)=2500;ARRAY(4)(3)=2500;
ARRAY(4)(4)=2500;ARRAY(5)(0)=3000;ARRAY(5)(1)=3000;ARRAY(5)(2)=3000;
ARRAY(5)(3)=3000;ARRAY(5)(4)=3000;ARRAY(6)(0)=500;ARRAY(6)(1)=500;
ARRAY(6)(2)=500;ARRAY(6)(3)=500;ARRAY(6)(4)=500
NUMBER_of_POINTS=5              !Set NUMBER_of_POINTS

```



```

ENABLE (0,1,5)           !Enable axes 0, 1 and 5
PVSPLINE/t (0,1,5)       !PVSPLINE/t initiates simultaneous motion for axes
                          !0, 1 and 5 with a user-defined time interval
                          !between each point.
MPOINT (0,1,5), ARRAY, NUMBER_of_POINTS
                          !Define an MPOINT array for axes 0, 1 and 5
                          !where ARRAY and NUMBER_of_POINTS are called.
ENDS (0,1,5)             !Concludes MPOINT
STOP                     !End program

```

2.6.16 MPTP...ENDS

Description

MPTP initiates multipoint sequential positioning to a set of points. **MPTP** by itself does not specify any point, however dwell time at each point can be optionally specified. Points are specified by the **POINT** or **MPOINT** commands that follow **MPTP**.

In single axis motion, **MPTP** generates sequential motion between the defined array points, where at the end of each segment, **RVEL** = 0, as if each segment was defined by a separate **PTP** command.

In group motion, where more than one axis is declared, the first axis in the **axis_list** is the leading axis. The motion parameters of the leading axis become the default motion parameters for all axes in the group. Motion on all axes in a group start and conclude at the same time. **MPTP** generates sequential motion between the defined array points, where at the end of each set of points, **RVEL** = 0, as if each motion was defined by a separate **PTP** command.

Transition to the next motion in the motion queue, if it exists, will not occur until **ENDS** executes.

MPTP terminates with:

- > **HALT**, **KILL/KILLALL**, or **BREAK**
- > Any fault activation that disables the drive or kills the motion
- > **DISABLE/DISABLEALL** by the user

Syntax

MPTP[/switch] axis_list[,dwell][,motor_motion_delay]

Arguments

axis_list	List of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
dwell	Dwell is an optional argument, expressed in milliseconds.
motor_motion_delay	(Optional, used only with /q switch) Delay, in milliseconds, before motor motion actually starts.

Switches

/switch can be:

/w	Create the motion, but do not start until the GO command.
/v	Use the specified velocity instead of the default velocity (VEL).
/c	Use the point sequence as a cyclic array
/r	Coordinates of each point are relative to the previous point's coordinates.
/z	Interpret entered coordinates according to the Local Coordinate System.
/q	Defines actual motor movement delay in microseconds. The delay resolution is 50 microseconds. The maximum delay is 100 controller cycles: 100ms for CTIME=1ms or 20ms for CTIME=0.2ms. Allows delaying actual motor movement start (RPOS) for the specified delay, while motion profile generation (APOS) starts immediately.

Comments

1. **MPTP** motion starts only after the first point is specified
2. **MPTP** motion with all of its points is considered as one motion command in the motion queue.
3. An **MPOINT** array declaration used in **MPTP** must be defined as Real.
4. **MPTP/c** does not end automatically. Use **HALT** , **KILL/KILLALL**, or **BREAK** to stop cyclic motion.

Related ACSPL+ Commands

GO, **HALT** , **KILL/KILLALL**, **BREAK**, **IMM**, **MPOINT**, **POINT**

Related ACSPL+ Variables

ACC, **DEC**, **JERK**, **VEL**

COM Library Methods and .NET Library Methods

ToPoint, ToPointM, ExtToPoint, ExtToPointM

C Library Functions

acsc_ToPoint, acscToPointM, acsc_ExtToPoint, acsc_ExtToPointM

Examples

Example 1:

In the following example, **dwel** is not required, therefore the comma and the second argument following 1000 are omitted.

```
MPTP 0, 1000           !Create a multipoint motion of the 0 axis
                        !with a dwell of 1000 msec at each point.
```

Example 2:

```

REAL ARRAY1 (4)           !Define an array called ARRAY1 with four members
                           !as REAL
! ----- Fill the ARRAY1 array-----
ARRAY1(0)=0;ARRAY1(1)=1000;ARRAY1(2)=0;ARRAY1(3)=1000
MPTP 0, 500               !MPTP motion for 0 axis, dwell 500 msec at each point
POINT 0, 2000             !First point
MPOINT 0, ARRAY1,3        !Use three points from ARRAY1
POINT 0, 3000             !Second point
POINT 0, 5000            !Third point
ENDS 0                    !Ends the point sequence for 0 axis

```

Figure 2-8 illustrates the above code for a single-axis motion initiated by **MPTP**. Note that not all of the members defined in **ARRAY1** necessarily need to be used by **MPOINT**. In this example, only three of the defined members are called.

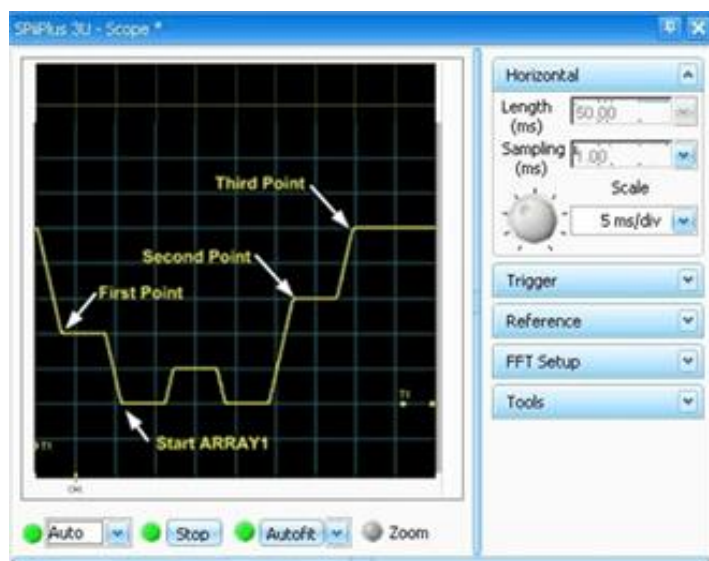


Figure 2-8. Single-Axis Motion Using MPTP

Example 3:

```

MPTP/v XY                !Create multipoint motion in group XY with
                           !user defined velocity and no dwell.
POINT XY,10000,0,5000    !Move to first point at velocity 5000
POINT XY,1000,1000,5000 !Add second point at velocity 5000
POINT XY,2000,1000,5000 !Add third point at velocity 5000
POINT XY,2000,0,100      !Add fourth point at velocity 100
ENDS XY                  !End the point sequence for XY

```

Figure 2-9 illustrates the above code for two-axis group motion initiated by **MPTP/v**.

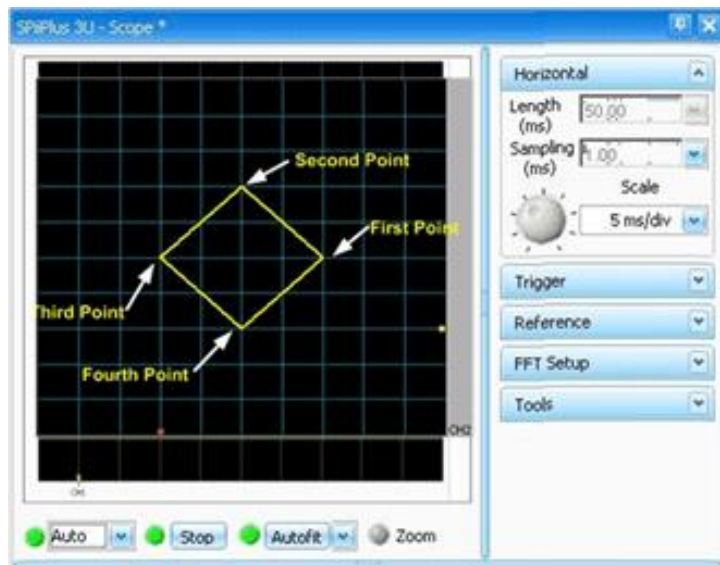


Figure 2-9. Two-Axis Group Motion Using MPTP/v

2.6.17 MSEG...ENDS

Description

MSEG initiates two-axis segmented motion. **MSEG** itself does not specify a line or arc segment. Motion starts only after the first segment is specified with a motion segment command.

Segmented motion moves axes along a continuous path where the path is defined as a sequence of line and arc segments on a plane.

Use the following commands to define segmented motion:

- > **ARC1** - adds an arc segment to a segmented motion and specifies the coordinates of center point, coordinates of the final point, and the direction of rotation
- > **ARC2** - Adds an arc segment to a segmented motion and specifies the coordinates of center point, rotation angle and direction.
- > **ENDS** - terminates the point sequence
- > **LINE** - adds a linear segment to a segmented motion.
- > **PROJECTION** - sets a projection array for a segmented motion.
- > **STOPPER** - provides a smooth transition between two segments of segmented motion.

MSEG motion terminates with:

- > **HALT**, **KILL/KILLALL**, or **BREAK**
- > Any fault activation that disables the drive or kills the motion
- > **DISABLE/DISABLEALL** by the user

Syntax

MSEG[/switch] *axis_list, initial-position-axis1,initial-position-axis2*

Arguments

<i>axis_list</i>	Axes list, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>initial-position axis1</i>	Start coordinate for the first axis
<i>initial-position-axis2</i>	Start coordinate for the second axis

Switches

/switch can be:

/w	Create the motion, but do not start until GO .
/v	Use the specified velocity instead of the default velocity (VEL).
/c	Use the MSEG segment sequence as a cyclic array.

Comments

- > Use command option /c to create cyclic motion where the final point of the last segment becomes the first point of the next motion cycle. **MSEG/c** does not automatically finish. Use **HALT**, **KILL/KILLALL**, or **BREAK** to end cyclic motion.
- > If command option /v is used, specify the user-defined velocity in each instance of **LINE**, **ARC1**, or **ARC2**.
- > The MSEG command uses a motion que buffer of up to 50 segments.

Related ACSPL+ Commands

GO, **HALT**, **KILL/KILLALL**, **BREAK**, **IMM**

Related ACSPL+ Variables

ACC, **DEC**, **JERK**, **VEL**

COM Library Methods

For **MSEG**: Segment, Line, ExtLine, Arc1, ExtArc1, Arc2, ExtArc2, Stopper, Projection

For **ENDS**: FinalPoint

C Library Functions

For **MSEG**: acsc_Segment, acsc_Line, acsc_Arc1, acsc_Arc2, acsc_ExtLine, acsc_ExtArc1, acsc_ExtArc2, acsc_Projection, acsc_Stopper

For **ENDS**: acsc_FinalPoint

Example

```

MSEG (0,1), 1000, 1000 !MSEG initiates segmented motion for the 0 and 1
axes
                                !group with initial coordinates of (1000,1000).
ARC1 (0,1), 1000, 0, 1000, -1000, -
                                !Add an arc segment with a center point located at
                                !(1000,0) and the final point located at (1000,-1000)
                                !with negative movement in terms of the encoder.
LINE (0,1), -1000, -1000
                                !Add line segment with final point (-1000,-1000).
ARC2 (0,1), -1000, 0, -3.141529
                                !Add arc segment with center (-1000,0) and a
                                !rotation angle of -p radians.
LINE (0,1), 1000, 1000
                                !Add line segment with center (1000,1000).
ENDS (0,1)                    !Ends the point sequence for the 0 and 1 axes group.
STOP                          !Ends program

```

Figure 2-10 illustrates the motion created by the example.

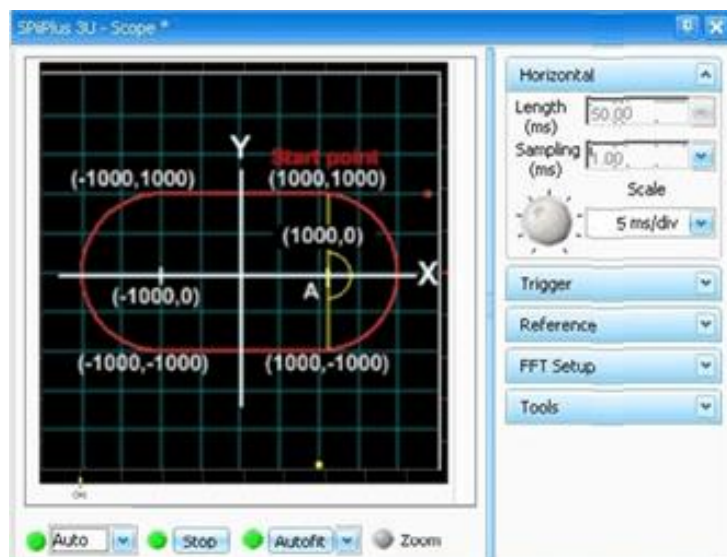


Figure 2-10. Results of Example MSEG

2.6.18 PATH...ENDS

Description

PATH initiates an arbitrary path motion with linear interpolation using **POINT**, or **MPOINT** for an array of points. The arbitrary path sequence must conclude with **ENDS**.

PATH motion terminates due to:

- > Interruption by any new motion command before the current motion concludes terminates the **PATH** motion and causes an error.
- > Any fault activation that disables the drive or kills the motion

- > User termination by **HALT** , **KILL/KILLALL**, or **DISABLE/DISABLEALL**

Syntax

PATH[/*switches*] *axis_list* [*time-interval*][, *motor_motion_delay*]

Arguments

<i>axis_list</i>	Axes list, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>time interval</i>	Optional time interval specification in milliseconds. PATH specified without \t must have the time interval specification as the last argument. The argument defines the time interval between all motion points.
<i>motor_motion_delay</i>	(Optional, used only with /q switch) Delay, in milliseconds, before motor motion actually starts.

Switches

/w	Create the motion, but do not start until the GO command.
/t	Enables a user-defined time interval in a point array
/c	Use the point sequence as a cyclic array. After arriving at the last point, continue from the first point.
/r	Coordinates of each point are relative to the previous point's coordinates.
/q	Defines actual motor movement delay in microseconds. The delay resolution is 50 microseconds. The maximum delay is 100 controller cycles: 100ms for CTIME=1ms or 20ms for CTIME=0.2ms. Allows delaying actual motor movement start (RPOS) for the specified delay, while motion profile generation (APOS) starts immediately.

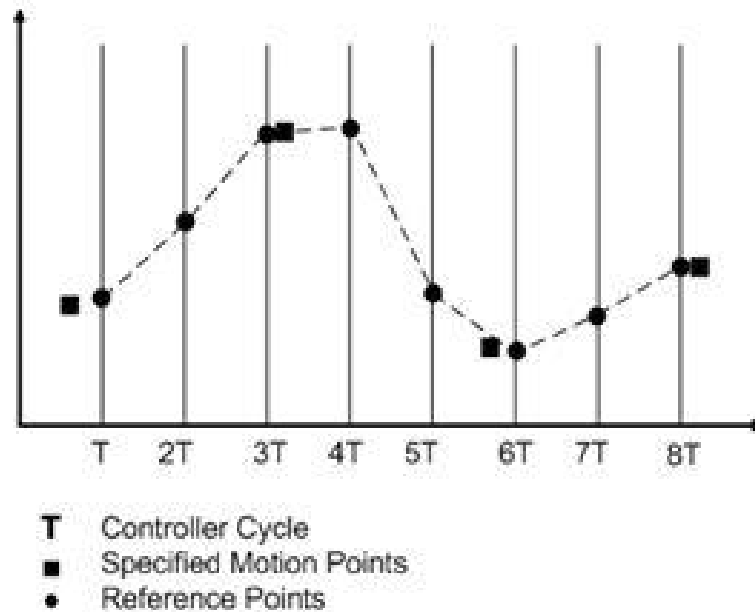


Figure 2-11. PATH...ENDS Diagram

Comments

Since a time interval and the destination point are specified, variables [VEL](#), [ACC](#), [DEC](#), [JERK](#) have no effect on this motion.

Related ACSPL+ Commands

[MPTP...ENDS](#), [POINT](#), [MPOINT](#), [PVSPLINE...ENDS](#)

COM Library Methods and .NET Library Methods

ToPoint, ToPointM, ExtToPoint, ExtToPointM

C Library Functions

acsc_ToPoint, acsc_ToPointM, acsc_ExtToPoint, acsc_ExtToPointM

Example

See [Example 2](#) of [MPOINT](#) for a three axis example of [MPTP...ENDS](#) motion.

2.6.19 POINT**Description**

POINT adds a destination point to multi-point or arbitrary motion paths. A sequence of destination points can be specified with a sequence of **POINT** commands. **POINT** must follow [MPTP...ENDS](#), [PATH...ENDS](#), or [PVSPLINE...ENDS](#). Refer to each command for a list of available command options. The sequence of specified points must conclude with **ENDS**.

Syntax

- > **POINT** syntax depends on the command options used in the initializing commands, as follows:
- > **POINT** initiated by **MPTP**:
POINT *axis_list, axis_list destination positions*

- > **POINT** initiated by **MPTP/v**:
POINT *axis_list, axis_list destination positions, vector velocity (GVEL)*
 See [Example 1](#).
- > **POINT** initialized by **PATH**:
PATH[*/command options*]*axis_list, time-interval*
POINT *axis_list, axis_list destination positions*
 See [Example 2](#).
- > **POINT** initialized by **PATH/t**:
POINT *axis_list, axis_list destination positions, time interval in milliseconds*
 See [Example 3](#).
- > **POINT** initialized by **PVSPLINE**:
PVSPLINE[*command options*]*axis_list, time-interval*
POINT *axis_list, axis_list destination positions*
 See [Example 4](#).
- > **POINT** initialized by **PVSPLINE/t**:
PVSPLINE[*command options*]*axis_list*
POINT *axis_list, axis_list destination positions, velocity at destination position, time interval in milliseconds*
 See [Example 5](#).

Comments

- > **POINT** axis specifications must follow the same order as the initializing **MPTP**, **PATH** or **PVSPLINE** commands.
- > All arguments following the *axis_list* must appear as a comma separated list, and must correspond to the number of axes.

Related ACSPL+ Commands

[MPOINT](#), [MPTP...ENDS](#), [PATH...ENDS](#), [PVSPLINE...ENDS](#)

COM Library Methods and .NET Library Methods

MultiPoint, MultiPointM, Spline, SplineM, AddPVPoint, AddPVPointM, AddPVTPoint, AddPVTPointM

C Library Functions

acsc_MultiPoint, acsc_MultiPointM, acsc_Spline, acsc_SplineM, acsc_AddPVPoint, acsc_AddPVPointM, acsc_AddPVTPoint, acsc_AddPVTPointM

Examples

Example 1

```
MPTP/v (0,1,5)           !Initiates MPTP motion for axes 0, 1 and 5.
                        !simultaneously with a user-defined velocity(GVEL)
                        !at each point.
POINT (0,1,5), 1000,4000,6000,7000
                        !Defines destination points 1000, 4000, and 6000 for
                        !axes 0, 1, and 5 with a GVEL of 7000 units.
ENDS (0,1,5)            !Ends MPTP/v point sequence for axes 0, 1, and 5.
STOP                   !Ends program
```

Example 2

```

PATH/r (0,1,5),1000      !Initiates PATH motion for axes 0, 1, and 5.
                        !simultaneously where each destination point is
                        !relative to the previous point and the time
                        !interval between each point is one second.
POINT (0,1,5), 1000,2000,3000
                        !Defines destination points 1000, 2000, and 3000 for
                        !axes 0, 1, and 5 respectively.
ENDS (0,1,5)           !Ends PATH point sequence for axes 0, 1, and 5.
STOP                   !Ends program

```

Example 3

```

PATH/t (0,1,5)          !Initiates PATH motion for axes 0, 1, and 5.
                        !simultaneously with a user-defined time interval
                        !between points.
POINT (0,1,5), 1000,2000,3000, 500
                        !Defines destination points 1000, 2000, 3000 for axes
                        !0, 1, and 5, respectively and a time interval of
                        !500 msec between points.
ENDS (0,1,5)           !Ends PATH point sequence for axes 0, 1, and 5.
STOP                   !Ends program

```

Example 4

```

PVSPLINE (0,1,5), 10    !Initiates PVSPLINE motion for axes 0, 1, and 5.
                        !Points are given at 10 msec intervals.
POINT (0,1,5), 200, 100, 300, 1000, 2000, 1500
                        !Defines points for axes 0, 1, and 5 with respective
                        !values of 200, 100, 300 and velocities at each
                        !point 1000, 2000, 1500, respectively.
ENDS (0,1,5)           !Ends PVSPLINE point sequence.
STOP                   !Ends program

```

Example 5

```

PVSPLINE/t (0,1,5)     !Initiates PVSPLINE motion for axes 0, 1, and 5, with
                        !a user-defined time interval between points.
POINT (0,1,5), 200, 100, 300, 1000, 2000, 1500, 250
                        !Defines destination points for axes 0, 1, and 5
                        !with respective values of 200, 100, 300 and
                        !velocities at each respective point of 1000, 2000,
                        !1500. The time interval between points is 250 msec.
ENDS (0,1,5)           !Ends PVSPLINE point sequence for axes 0, 1, and 5.
STOP                   !Ends program

```

2.6.20 PROJECTION

Description

PROJECTION is an expansion command to the **MSEG...ENDS** set of commands, that allows the controller to perform a three dimensional segmented motion such as creating arcs and lines on a user-defined plane. The method for this 3D segmented motion is to set a transformation matrix that defines a new plane for the segmented motion.

Syntax

PROJECTION *axes_list*, *Transformation Matrix*

Arguments

<i>axes_list</i>	List of axes, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>Transformation Matrix</i>	Defines the new plane

Comments

1. **MSEG** must precede **PROJECTION**.
2. Prior to using **PROJECTION**, the related axes must be grouped using **GROUP**.
3. The motion parameters of the segmented motion with **PROJECTION** are calculated based on the leading axis described in the **GROUP** command. The other involved axes motion parameters are not relevant. The parameters are calculated to meet a uniform travel time to all grouped axes.

Related ACSPL+ Commands

MSEG...ENDS, **LINE**, **ARC1**, **ARC2**, **GROUP**

COM Library Methods and .NET Library Methods

Projection

C Library Functions

acsc_Projection

Example

1. This example program creates a segmented motion on a new plane (in this example - a circle) on a plane rotated around Axis X by 70° relative to plane AX, as illustrated in [Figure 2-12](#).
2. **ARC2** defines the circle's center coordinates on plane AX. **PROJECTION** transforms these coordinates to the new plane, based on the values in the transformation matrix (Matrix M, in this example).
3. Populate the transformation matrix with the values from table given below. The first two rows define the relationship between the X and Y coordinates of the last **MSEG...ENDS** motion and their respective axes in the new plane. The third row defines the tangent angle of the new plane. In this example uses a 70° angle in reference to Axis A, where $\tan 70 = 2.74$.

Table 2-8. Matrix Values

Matrix Values		
Axis	X Coordinate	A Coordinate
X	1	0
A	0	1
B	0	2.74

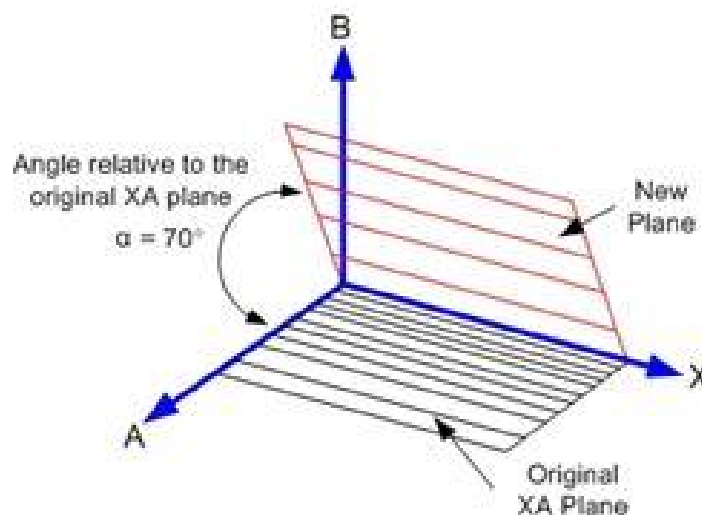


Figure 2-12. PROJECTION of the XA Plane

Figure 2-13 illustrates the reference position of axes 0, 4, and 5 (corresponding to the XAB coordinates illustrated in Figure 2-12).

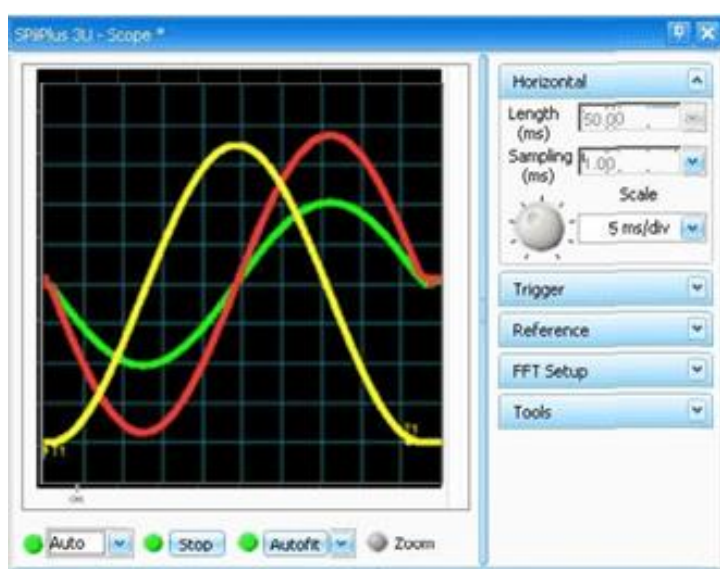


Figure 2-13. FPOS - PROJECTION Example

Figure 2-14 illustrates the circle's trajectory viewed from the XA plane.

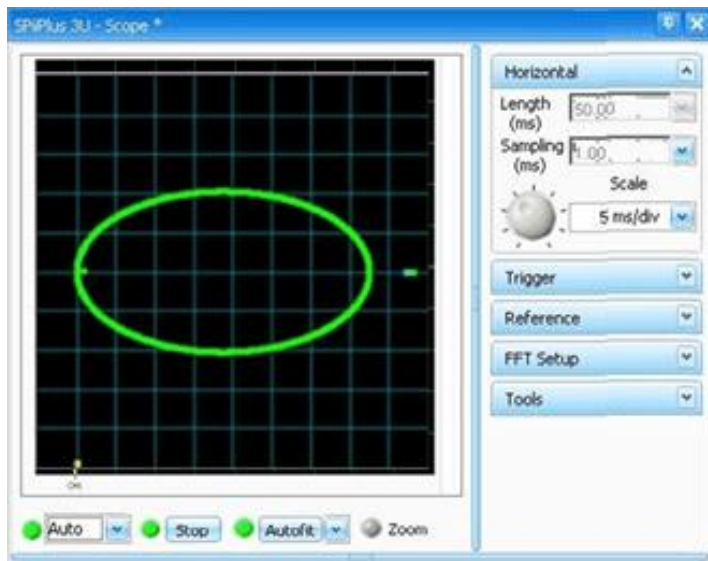


Figure 2-14. PROJECTION Example - Final Result

```

REAL M(3) (2)           !Defines Transformation Matrix M as real array
! ----- Set the transformation matrix values-----
M(0) (0)=1;M(0) (1)=0;M(1) (0)=0;M(1) (1)=1;M(2) (0)=0;M(2) (1)=2.74
VEL(0)=1000;ACC(0)=10000;DEC(0)=10000
                           !Axis motion parameters
ENABLE (0,4,5)           !Enable the 0, 4 and 5 axes (required)
                           !Corresponding to XAB coordinates shown in Figure 15.
GROUP (0,4,5)            !Group the 0, 4 and 5 axes (required)
SET FPOS(0)=0;SET FPOS(4)=0;SET FPOS(5)=0
                           !Sets the FPOS for 0, 4 and 5, respectively, to 0.
MSEG (0,4),0,0           !Define original plane.
PROJECTION (0,4,5),M     !PROJECTION of axes XAB by matrix M.
ARC2 (0,4),750,0,6,24    !ARC2 performed on new plane.
ENDS (0,4)               !Concludes MSEG.
STOP                     !End Program

```

2.6.21 PTP

Description

PTP (point-to-point) generates motion for the specified axis or axes to a specified destination point.

When **PTP** specifies a single axis, the motion profile is calculated according to [VEL](#), [ACC](#), [DEC](#), [JERK](#) values of the axis.

In group motion, when **PTP** specifies multiple axes, the group motion profile is based on the leading axis' [VEL](#), [ACC](#), [DEC](#), [JERK](#) motion values, unless **PTP/m** is used.

PTP terminates due to:

- > Interruption by any new motion command
- > Any fault activation that disables the drive or kills the motion

- > User termination by **HALT** , **KILL/KILLALL**, or **BREAK**.

Syntax

PTP[switches]*axis_list, destination-point*[*value for v, value for f, motor_movement_delay*]

Arguments

axis_list	Single axis or axis group, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
destination point	Final destination.
value for v	Optional argument for user-defined velocity.
value for f	Optional argument for user-defined velocity at a destination point
motor_movement_delay	(Optional, used only with /q switch) Delay, in milliseconds, before motor motion actually starts.

Switches

/e	Wait for motion termination before executing next command.
/f	Specify non-zero velocity at each destination point (or points) in a series of PTP motions
/m	Use the motion profile values of the axis group as a whole, rather than those of the leading axis, without exceeding any of the defined axes motion VEL , ACC , DEC , JERK values.
/r	The destination point is relative to the start point.
/v	Use the specified velocity instead of the default velocity (VEL).
/w	Create the motion, but do not start until GO.
/z	Interpret entered coordinates according to the Local Coordinate System.
/q	Defines actual motor movement delay in microseconds. The delay resolution is 50 microseconds. The maximum delay is 100 controller cycles: 100ms for CTIME=1ms or 20ms for CTIME=0.2ms. Allows delaying actual motor movement start (RPOS) for the specified delay, while motion profile generation (APOS) starts immediately.

Comments

- > Axes destination points, and relative velocity in the **PTP** command can also be an expression.

- > **PTP** can be used for executing point-to-point motion for a group of axes. For example, `PTP (0,1,2)` creates motion for axes 0, 1, and 2 as a group.
- > In single axis **PTP** motion, the final velocity (indicated by the `/f` switch) may be negative. In this case, the motion terminates at the specified coordinate with a velocity opposite to the motion direction.
- > In single axis **PTP** motion, if the motion is in the negative direction, **GVEL** can be negative.

Related ACSPL+ Commands

[MPTP...ENDS](#), [POINT](#)

COM Library Methods and .NET Library Methods

ToPoint

C Library Functions

acsc_ToPoint

Examples

Example 1:

```
PTP/v 1, 2000, 500      !PTP axis 1 to point 2000 with velocity 500
```

Example 2:

```
PTP/rw (0,1), 1000, 2000 !PTP axes 0 and 1 where the 1 target point is
1000
                           !and the 0 target point is 2000. The target points
                           !are relative to the start point. Motion will not
                           !commence until GO command is issued.
```

Example 3:

```
ENABLE 0                  !Enables axis 0
VEL(0)=10000              !Sets the default axis 0 velocity to 10000
SET RPOS(0)=0             !Sets the current axis 0 position to 0
PTP/rf 0, 2000, 1000      !Initiates a relative axis 0 motion of 2000 with end
                           !velocity of 1000 at the destination point
PTP 0, 4000               !Initiates an absolute axis 0 motion to 4000
STOP                     !Ends program
```

2.6.22 PVSPLINE...ENDS

Description

PVSPLINE (position-velocity spline) creates an arbitrary motion trajectory where the controller provides cubic spline interpolation between two points. The user specifies the end point and the end velocity for each motion segment. **ENDS** must terminate the point sequence.

PVSPLINE motion terminates due to:

- > Interruption by any new motion command
- > Any fault activation that disables the drive or kills the motion

- > User termination by [HALT](#) , [KILL/KILLALL](#), or [DISABLE/DISABLEALL](#)

Syntax

PVSPLINE[*switches*]*axis_list*[*time-interval*] [*motor_motion_delay*]

Arguments

<i>axis_list</i>	Single axis or axis group, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>time-interval</i>	Optional argument for user-defined time interval, in milliseconds, between points.
motor_motion_delay	(Optional, used only with /q switch) Delay, in milliseconds, before motor motion actually starts.

Switches

/r	The point coordinates are relative to the previous point
/w	Create the motion, but wait to start for GO
/t	Non-uniform time interval: time interval is specified for each point along with the point coordinates
/c	Perform the point sequence as a cyclic array-after the last point, start the motion again from the first point.
/q	Defines actual motor movement delay in microseconds. The delay resolution is 50 microseconds. The maximum delay is 100 controller cycles: 100ms for CTIME=1ms or 20ms for CTIME=0.2ms. Allows delaying actual motor movement start (RPOS) for the specified delay, while motion profile generation (APOS) starts immediately.

Comments

- > Since the time interval and the destination point are defined, variables [VEL](#), [ACC](#), [DEC](#), and [JERK](#) have no effect on **PVSPLINE**.
- > If **PVSPLINE** motion is ended with [HALT](#) , the controller does not follow the motion trajectory during deceleration.
- > **PVSPLINE** specified with **/t** must **NOT** have a *time-interval* specification. Instead, specify the time interval for each point as an additional argument for [POINT](#) or as an additional array row in [MPOINT](#), see [Example 5](#).

Related ACSPL+ Commands

[POINT](#), [MPOINT](#)

COM Library Methods and .NET Library Methods

Spline, SplineM, AddPVPoint, AddPVPointM, AddPVTPoint, AddPVTPointM

C Library Functions

acsc_Spline, acsc_SplineM, acsc_AddPVPoint, acsc_AddPVPointM, acsc_AddPVTPoint, acsc_AddPVTPointM

Example

```

REAL ARRAY_NAME(4) (5)      !Defines a point array of four rows and five
columns                      !and a variable that defines the number of array rows
                              !that are part of the motion.
                              !Defines NUMBER_of_POINTS as an integer variable.
INT NUMBER_of_POINTS         !Assigns 3 to the NUMBER_of_POINTS variable.
NUMBER_of_POINTS=3
PVSPLINE/r (0,1),10         !Initiates PVSPLINE motion for axis 0 and 1 with
                              !relative destination points and a time interval of
                              !10 milliseconds between each point.
POINT (0,1),200,100,1000,2000
                              !Defines values for 0 and 1 axes where:
                              !200, 100=destination point for the 0 and 1 axes,
                              !respectively; and 1000, 2000=velocity at specified
                              !points for the 0 and 1 axes, respectively.
MPOINT (0,1), ARRAY_NAME, NUMBER_of_POINTS
                              !Defines an MPOINT array for the 0 and 1 axes, where
                              !ARRAY_NAME is called, and where NUMBER_of_POINTS
                              !defines the number of array columns involved in the
                              !motion.
ENDS (0,1)                  !Ends MPOINT for 0 and 1 axes.
STOP                        !Ends program.

```

Figure 2-15 illustrates a typical **PVSPLINE** motion.

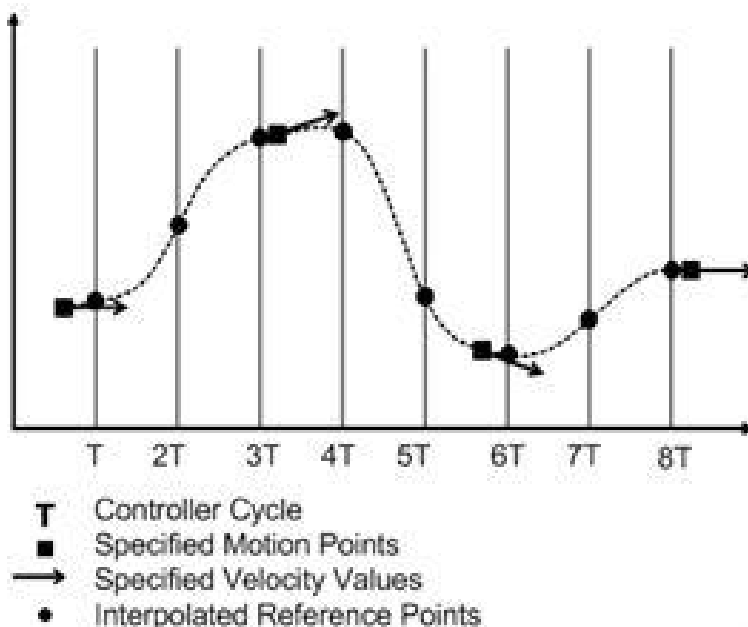


Figure 2-15. PVSPLINE Motion Diagram

2.6.23 SLAVE

Description

SLAVE initiates a motion based on position lock or velocity lock slaved to a master value or expression. Only individual axes are allowed. The initiated motion starts immediately if the axis is idle, otherwise the motion waits in the motion queue until all previously created motions finish.

SLAVE must precede **MASTER**.

MASTER - SLAVE dependency terminates with:

- > Any fault activation that disables the drive or kills the motion
- > **HALT** , **KILL/KILLALL**, **BREAK**
- > **DISABLE/DISABLEALL** to the **SLAVE** axis.
- > Setting the logical dependence between master and slave axes to zero. For example **MASTER MPOS(0)=0**

Syntax

SLAVE[/*switches*] *axis*[/*lower boundary*, *upper boundary*]

Arguments

<i>axis</i>	Axis designation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>lower boundary</i>	Optional argument used for setting the lower boundary of master-dependent SLAVE motion.
<i>upper boundary</i>	Optional argument used for setting the upper boundary of master-dependent SLAVE motion.

Switches

/w	Create the motion, but do not start until GO is executed.
/p	Master - Slave motion generated by position lock.
/t	Create slave motion within specified position boundaries.

Comments

- > When **/p** is used, the controller first initiates **velocity lock**. Only after the achieving **velocity lock** the controller will engage **position lock**.
- > If no command option is specified, the default mode is **velocity lock**.

Related ACSPL+ Commands

MASTER, **GO**, **HALT** , **KILL/KILLALL**, **DISABLE/DISABLEALL**

Related ACSPL+ Variables

XSACC, **MFF**, **JERK**, **ACC**, **VEL APOS**

COM Library Methods and .NET Library Methods**SetMaster, Slave, SlaveStalled****C Library Functions**

acsc_SetMaster, acsc_Slave, acsc_SlaveStalled

ExampleSee [MASTER](#) for an example of **MASTER - SLAVE** syntax.**2.6.24 STOPPER****Description**

STOPPER is used in conjunction with [MSEG...ENDS](#) to avoid velocity jumps at segment inflection points. When **STOPPER** is specified between two segments, the controller provides smooth deceleration to zero before **STOPPER** and a smooth acceleration to the default or specified velocity after **STOPPER**.

Syntax**STOPPER** *axis_list***Arguments***axis_list*

Single axis or axis group, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Related ACSPL+ Commands[MSEG...ENDS](#), [ARC1](#), [ARC2](#), [LINE](#), [PROJECTION](#)**COM Library Methods and .NET Library Methods**

Stopper

C Library Functions

acsc_Stopper

Example

[Figure 2-16](#) illustrates the following example. For this illustration the reference position (**RPOS**) for axes 0 and 1 was set to (0,0).

```

MSEG (0,1), 0, 0      !MSEG initiates segmented motion for the 0 and 1 axis
                       !group with the point on the plane located at (0, 0).
LINE (0,1), 1000, 2500 !Add line segment with final point at (1000,
2500).
STOPPER (0,1)         !Slow down to zero.
ARC1 (0,1), 0,2000, -1000,2500, +
                       !Add arc segment with final point (-1000, 2500) and
                       !center point (0, 2000).
STOPPER (0,1)         !Slow down to zero.
LINE (0,1), 0, 0      !Add line segment with final point (0, 0).
ENDS (0,1)            !End MSEG.
LINE (0,1), 1000, 1000 !Add line segment with final point (1000, 1000).
```

```
ENDS (0,1)           !Ends the point sequence for the 0 and 1 axis
                    !group.
```

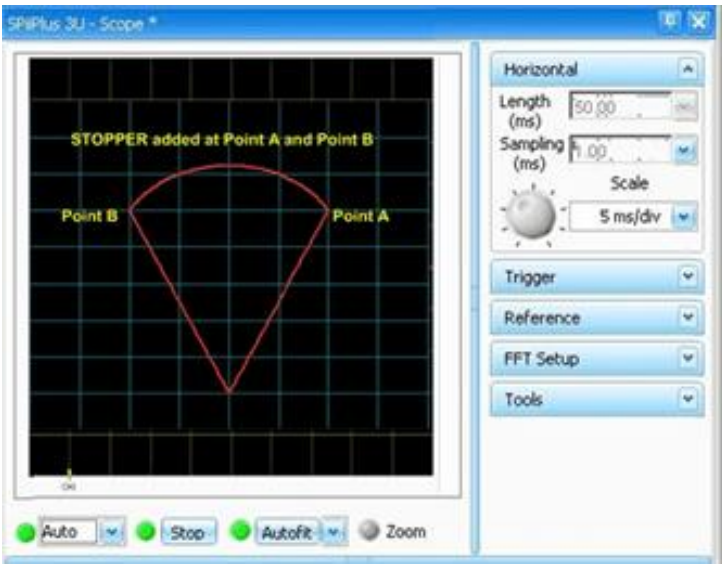


Figure 2-16. Use of STOPPER

2.6.25 TRACK

TRACK initiates track motion. In **TRACK** motion, a new point-to-point move is generated to a new target position whenever the variable **TPOS** (target position) changes. **TRACK** does not terminate automatically. If **TPOS** is not assigned a new value, motion stops at the last **TPOS** and waits. If a new **TPOS** value is assigned, motion continues.

TRACK terminates due to:

- > Any subsequent motion command (except **TRACK**) for the motion axis involved in a track motion, **except the case when the next motion is a group motion**.
- > Any fault activation that disables the drive or kills the motion.
- > User termination by **HALT** , **KILL/KILLALL**, or **DISABLE/DISABLEALL**

Syntax

TRACK [/switch] axis, [motor_motion_delay]

Arguments

axis	Axis designation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
motor_motion_delay	(Optional, used only with /q switch) Delay, in milliseconds, before motor motion actually starts.

Switch

/switch can be:

<code>/w</code>	Create the motion, but do not start until GO .
<code>/q</code>	Defines actual motor movement delay in microseconds. The delay resolution is 50 microseconds. The maximum delay is 100 controller cycles: 100ms for CTIME=1ms or 20ms for CTIME=0.2ms. Allows delaying actual motor movement start (RPOS) for the specified delay, while motion profile generation (APOS) starts immediately.

Comments

- > While **PTP** code appears shorter and simpler, there are applications where **TRACK** is preferable to **PTP**. For example, **TRACK** provides an easy way to change the destination position at any time during the motion by changing the target position (**TPOS**) variable. The controller terminates the current motion and proceeds to the next destination point, on-the-fly.
- > **TRACK** is for single axis motion only.
- > **TPOS** is updated every controller cycle.

Related ACSPL+ Commands**PTP****COM Library Methods and .NET Library Methods**

Track

C Library Functions

acsc_Track

Example

```

TRACK 0           !Initiates TRACK for 0 axis.
TPOS(0) = NewTarget !The controller generates a PTP motion to the point
                   !designated by NewTarget.

```

2.6.26 XSEG...ENDS**Description**

The **XSEG...ENDS** (Extended Segmented Motion) command block provides the following:

- > Corner detection
- > Detection of segments, where required velocity violates axis velocity/acceleration limits
- > Velocity limitation at corners and segments where required velocity violates axis velocity, acceleration and jerk limits
- > Building a velocity profile using multi-segment look-ahead algorithm
- > Corner rounding using different criteria
- > Support of up to 6 axes
- > Support for "Skywriting" - external loops at corners

Use the following commands to define the segmented motion:

- > **ARC1**- adds an arc segment to a segmented motion and specifies the coordinates of center point, coordinates of the final point, and the direction of rotation
- > **ARC2** - Adds an arc segment to a segmented motion and specifies the coordinates of center point, rotation angle and direction.
- > **LINE** - adds a linear segment to a segmented motion.
- > **ENDS**- terminates the point sequence

Syntax

XSEG

[/switches] (axis_list), initial_position_axis1,initial_position_axis2[,initial_position_axis3...,initial_position_axis6],
[,velocity][,end_velocity][,junction_velocity][,angle][,curvature_velocity][,deviation][,radius]
[,maximal_length]
[,starvation_margin [,segments_buffer]]
[,external_loop_type, minimal_segment_length, maximum_allowed_deviation]
[output_index, bit_number, polarity]

Segment commands (ARC1, ARC2, LINE)

ENDS

Arguments

<i>axis_list</i>	Axis group (specified as axes numbers separated by a comma), valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.  A minimum of two axes must be specified.
<i>initial_position_axis1</i>	Initial position of the first axis.
<i>initial_position_axis2</i>	Initial position of the second axis.
<i>initial_position_axis3...</i> <i>initial_position_axis6</i>	Mandatory only if axis_list contains more than 2 axes. Number of initial positions must correspond to the number of axes in axis_list .
<i>velocity</i>	[Optional, only used with /v switch] Defines required velocity instead of default velocity (VEL).
<i>end_velocity</i>	[Optional, only used with /f switch] Defines required velocity at the end of each segment.

<i>junction_velocity</i>	[Optional, only used with /j switch] Defines required velocity at the junction.
<i>angle</i>	[Optional, only used with /a switch] The junction will be treated as a corner if actual junction angle is more than defined.
<i>curvature_velocity</i>	[Optional, only used with switch /d] Defines required velocity at curvature discontinuity points. See Switches explanation for details.
<i>deviation</i>	[Optional, only used with /g switch] Defines maximal allowed trajectory deviation from the corner point. See Switches explanation for details.
<i>radius</i>	[Optional, only used with /u switch] Defines maximal allowed rounding radius of the additional segment. See Switches explanation for details.
<i>maximal_length</i>	[Optional, only used with /h switch] Defines the maximal length of the segment for smoothing processing. If the length of a segment that formed a corner exceeds the specified maximal length, the corner will not be smoothed.
<i>starvation_margin</i>	[Optional] Starvation margin in milliseconds. The controller sets the AST.#NEWSEGM bit starvation_margin millisecond before the starvation condition occurs. By default, if the argument is not specified, the starvation margin is 500 milliseconds.
<i>segments_buffer</i>	[Optional] One-dimensional user defined real array. The controller uses this array to store adding segments. By default, if the argument is not specified, the controller allocates internal buffer for storing 50 segments only. The argument allows the user application to reallocate the buffer for storing larger number of segments. The larger number of segments may be required if the application needs to add many very small segments in advanced.
<i>external_loop_type</i>	0 - Cancel external loop 1 – Smooth External loop (line-arc-line) 2 – Triangle External loop (line-line-line)
<i>minimum_segment_length</i>	If the lengths of both segments are more than this value, the skywriting algorithm will be applied.
<i>maximum_allowed_deviation</i>	The parameter limits the external loop deviation from the defined profile. If the value is negative – no limitation.
<i>output_index</i>	Index of digital output port to assigned to synchronization (read by OUT)

bit_number	Bit number (0 - 16) assigned to synchronization
polarity	0 or 1 - which value is considered the initial state
motor_movement_delay	(Optional, used only with /q switch) Delay, in milliseconds, before motor motion actually starts.

Switches

There are three types of optional switches:

- > General
- > Velocity look-ahead
- > Geometry look-ahead

The controller processes the specified switches in the following order:

1. The controller checks and applies geometry look-ahead options.
2. The controller checks and applies velocity look-ahead options.

Switches from different groups can be applied together. For example, it's possible to specify a velocity at curvature discontinuity points (switch **/d**) together with permitted deviation (switch **/g**). In this case, the controller first applies corner rounding for the trajectory and then calculates velocity profile for already processed trajectory.

Optional switches are for use only with the **XSEG** command:

General	
/w	Do not start until the GO command. If the switch is not specified, the motion starts immediately after the first motion segment is defined.
/v	Specify required velocity. The switch requires additional parameter that specifies required velocity. If the switch is not specified, the required velocity is derived from the leading axis parameters.
/m	Use maximum velocity under axis limits. With this switch, no required velocity should be specified. The required velocity is calculated for each segment individually on the base of segment geometry and axis velocities (VEL values) of the involved axes.

/l	Velocity limitation under axis limits. As opposed to the switch /m, with this switch the required velocity can be specified either for all motion (switch /v for xseg command) or as segment velocity (switch /v for line, arc1, arc2 commands). In both cases, the controller tries to achieve the required velocity by taking into account also velocity limits of all involved axes. If the required velocity cannot be achieved within the axis velocity limits, the maximal velocity within the axis velocity limits is used. This switch cannot be specified together with switch /m.
/z	Interpret entered coordinates according to the Local Coordinate System. With this switch, use 2 axes motion coordinate only (X,Y). Using 3 or more coordinates causes a runtime error.
/q	Defines actual motor movement delay in microseconds. The delay resolution is 50 microseconds. The maximum delay is 100 controller cycles: 100ms for CTIME=1ms or 20ms for CTIME=0.2ms. Allows delaying actual motor movement start (RPOS) for the specified delay, while motion profile generation (APOS) starts immediately.
Velocity look-ahead	
/f	Decelerate to the end of each segment. The switch requires an additional parameter that specifies end velocity. The controller decelerates to the specified velocity in the end of each segment. The specified value should be less than the required velocity; otherwise the parameter is ignored. If the switch is not specified, deceleration in each segment is not required. However, in specific segments deceleration might occur due to corner processing or other velocity control conditions.
/j	Decelerate to corner. The switch requires an additional parameter that specifies corner velocity. The controller detects corner on the path and decelerates to the specified velocity before the corner. The specified value should be less than the required velocity; otherwise the parameter is ignored. If switch j is not specified while switch a is specified, zero value of corner velocity is assumed. If switches j, a, d, and y are not specified, the controller provides automatic calculation of the corner processing.

/a	Do not treat junction as a corner, if junction angle is less than or equal to the specified value in radians. The switch requires an additional parameter that specifies negligible angle in radians. If switch a is not specified while switch j is specified, the controller accepts default value of 0.01 radians that is about 0.57 degrees. If switches j, a, d, and y are not specified, the controller provides automatic calculation of the corner processing.
/d	Decelerate to curvature discontinuity point. The switch requires an additional parameter that specifies velocity at curvature discontinuity points. Curvature discontinuity occurs in linear-to-arc or arc-to-arc smooth junctions. If the switch is not specified, the controller does not decelerate to smooth junction disregarding curvature discontinuity in the junction. If the switch is specified, the controller detects curvature discontinuity points on the path and provides deceleration to the specified velocity. The specified value should be less than the required velocity; otherwise the parameter is ignored. The switch can be specified together with switches j and/or a. If switches j, a, d, and y are not specified, the controller provides automatic calculation of the corner processing.
/y	If the switch is specified the controller provides automatic calculations as described in Enhanced automatic corner and curvature discontinuity points processing (switch /y).
/r	Set the initial axis position as origin. Segment commands positions should be declared relative to the new origin.
Geometry look-ahead	
/g	Use a corner rounding option with the specified permitted deviation The switch requires additional parameter that specifies maximal allowed deviation of motion trajectory from the corner point. The switch cannot be specified together with switches /u and /h
/u	Use a corner rounding option with the specified permitted curvature The switch requires additional parameter that specifies maximal allowed rounding radius of the additional segment The switch cannot be specified together with switches /g and /h
/h	Use a corner smoothing option. The switch requires additional parameter that specifies the maximal length of the segment for smoothing processing. If a length of one of the segments that built a corner exceeds the specified maximal length, the corner will not be smoothed. Smoothing is only applied to pair of linear segments. If one of the

	segments in a pair is arc, the smoothing is not applied for this corner. The switch cannot be specified together with switches /g and /u .
/b	Use external loops at corners. The switch requires additional parameters that specify the external loop type, the minimum segment length, and the maximum allowed deviation from profile.
/s	Defines output bit to support external loop synchronization.



XSEG without switches does not require any additional parameters except the initial point coordinates, for example, **XSEG (0,1),0,0** creates segmented motion for axes 0 and 1 with initial point (0,0) with required velocity derived from the axis 0.

Comments

For each two adjacent segments, the controller calculates the tangent vector to each segment in the junction point. If the two vectors are equal, the segments are tangent, and no special processing is required. If not, the two segments build a corner. In a corner, the controller behavior follows the corner processing option selected by the user for **XSEG** motion.

The following options are supported:

Exact path: no deviation from the specified path is permitted. The user specifies two additional parameters: threshold angle and corner velocity. The controller compares the corner angle and the threshold angle. If the corner angle is smaller, the controller ignores the corner and tries to move as if the junction is smooth (the threshold angle cannot be large, otherwise passing the junction at working velocity can produce mechanical jerk). If the corner angle is greater, the controller executes deceleration to achieve the junction point with the specified corner velocity (as shown in [Figure 2-17](#)).

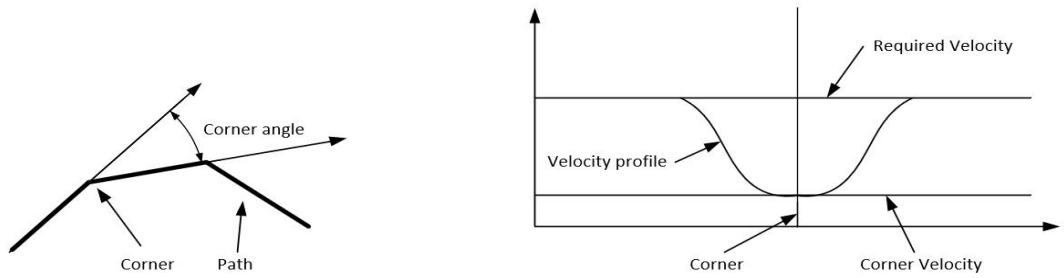


Figure 2-17. Corner Processing - Exact Path Option

Permitted deviation: the user specifies the motion trajectory maximum permitted deviation from the corner point. The controller inserts an additional segment in the corner so that the resulting path is smooth and complies with the maximum deviation.

Permitted radius: the user specifies the additional segment maximum permitted rounding radius. The controller inserts an additional segment in the corner so that the resulting path is smooth and complies with the maximum permitted radius.

Corner smoothing: the user specifies the smoothing maximum segment length. The controller applies smoothing if the length of both segments in the pair is less than the maximum segment length.

Figure 2-18 illustrates the permitted deviation, permitted radius and corner smoothing options.

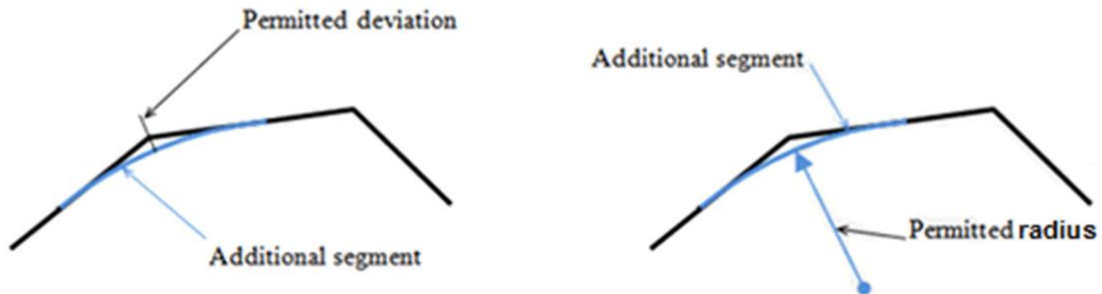


Figure 2-18. Corner Processing - Permitted Deviation, Permitted Radius and Corner Smoothing Options

XSEG updates the following motion related parameters:

- > Motion and Axis Statuses: **AST, MST**
- > Vector velocity, acceleration and jerk: **GVEL, GACC, GJERK**
- > Axis velocity: **GVEC**
- > Motion queue status and motion type: **GMQU, GMTYPE**
- > Trajectory vector distance from the beginning of the first segment: **GPATH**
- > Elapsed motion time: **GETIME**
- > Current executed segment and segments buffer status: **GSEG, GSFREE**

XSEG builds the algorithm upon the following axis motion parameters as axis constraints: **VEL, ACC,** and **JERK**.

In connection with the **segments_buffer** argument, for most applications the internal buffer size is enough and should not be enlarged.



The buffer is for the internal use of the controller only and should not be used by the user application.

The buffer size calculation rule: each segment requires about 750 bytes, so if it is necessary to allocate a buffer for 200 segments, it should be at least $750 * 200 = 150,000$ bytes. The following declaration defines a 150,000 bytes buffer:

```
real buf(18750)
```



See [XARRSIZE](#) for details of how to declare a buffer with more than 100000 elements

Examples

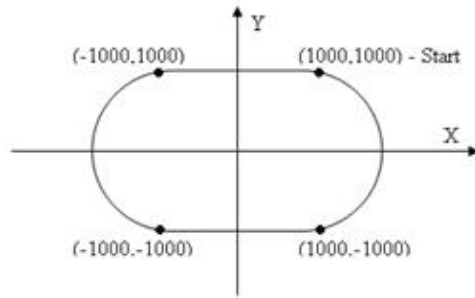
Example 1:

```
XSEG (1,0),0,0  !Segmented motion for axes 1 and 0. Required velocity
                !is derived from the axis 1, i.e., VEL(1) value. No
                !deviation from the path is permitted. If the path
                !contains a corner, and the junction angle is more
                !than default value 0.01 radians, the velocity
                !decelerates to zero in the corner point.
XSEG/vf         !Segmented motion for axes 0 and 1 with initial
(0,1),0,0,100,50 !point (0,0) with required velocity 100 units/sec;
                !at the end of each segment, the motion should
                !decelerate to 50 units/sec.
XSEG/vja       !Segmented motion for axes 1 and 2 with initial
(1,2),1000,1000, !point(1000,1000) and required velocity is 100
                !units/sec.
100,20,0.05     !If the path contains a junction, and the junction
                !angle is more than 0.05 radians, the velocity
                !decelerates to 20 unit/sec in the junction point.
```

Example 2:

```
XSEG (0,1),1000,1000  !Create segment motion in axes XY with initial point
                    !(1000,1000)
ARC1 (0,1),1000,0,1000,-1000,-!Add arc segment with center (1000,0),
final point(1000,1000), clockwise rotation
LINE (0,1),-1000,-1000!Add line segment with final point ( 1000, 1000)
ARC2 (0,1),-1000,0,-3.141529 !Add arc segment with center ( 1000,0)
                    !and rotationangle of  $-\pi$ 
LINE (0,1),1000,1000  !Add line segment with final point (1000,1000)
ENDS (0,1)            !End the segment sequence
```

The **XSEG** command creates the segment motion. The motion does not start when processing reaches the **XSEG** command. Actual motion starts once the previous motion ends and one or more segments are added. The four segment commands in example 2 specify the following path:



Example 3:

```

GLOBAL INT iXaxis, iYaxis
GLOBAL REAL buf(18750) ! For 200 segments ( ARRAY SIZE = (200[SEG]*750
[BYTES/SEG])/8 )
! Make sure that XARRSIZE value is bigger than ARRAY SIZE (if not, change
XARRSIZE on the terminal)

iXaxis = 0
iYaxis = 1

XSEG (iXaxis,iYaxis),0,0,,buf
LINE (iXaxis,iYaxis),1000,1000
LINE (iXaxis,iYaxis),1001,1001
LINE (iXaxis,iYaxis),1002,1002
!..... Add more segments .....
ENDS (iXaxis,iYaxis)

STOP

```



The **LINE** command may specify one axis that actually moves in this segment. Other axes specified in **XSEG** hold their positions while the linear segment is in progress.

The **ARC1** and **ARC2** commands always specify two axes.

The **ARC1** and **ARC2** commands differ by the required arguments. The **ARC1** command specifies the coordinates of the center point, coordinates of the final point and the direction of rotation (+ for counter-clockwise, – for clockwise rotation). The **ARC2** command specifies the coordinates of the center point and rotation angle (positive for counter clockwise, negative for clockwise rotation). The **ARC1** and **ARC2** commands may produce the same result, so the user may select the one that suits the available data. If the user knows the coordinates of the center point, coordinates of the final point and the direction of rotation, **ARC1** is preferable. If the user knows the coordinates of the center point and rotation angle, **ARC2** is preferable.

The **ENDS** command informs the controller that no more segments will be specified for the motion.

See also the section on Extended Segmented Motion in the *ACSPL+ Programmer's Guide*.

2.6.27 NURBS

Description

Creates **NURBS** motion.

Syntax

NURBS[/switches] (axis_list)[,velocity][,exc_angle][,exc_length][,segments_buffer][,motor_motion_delay]

Arguments

axis_list	Axes involved in NURBS motion, specified as axes numbers separated by comma or as axis names separated by comma. A minimum of two axes must be specified.
velocity	Optional, only used with suffix /v . Defines required feed rate. If not specified, the feed rate is derived from the leading axis parameters.
exc_angle	Optional, only used with suffix /a . The value defines exceptions from spline interpolation. If for an internal control point directions to the previous and the next control points require direction change more than the specified angle (by modulo), the control point is processed as a corner. Actually, such points divide the spline into two independent splines.
exc_length	Optional, only used with suffix /l . The value defines exceptions from spline interpolation. If a distance between two control points appears longer than the specified length, the trajectory between the points is considered straight. Actually, two independent splines are built before and after the segment.
segments_buffer	[Optional] One-dimensional user-defined real array. The controller uses this array to store added segments. By default, if the argument is not specified, the controller allocates internal buffer for storing 50 segments only. The argument allows the user application to reallocate the buffer for storing larger number of segments. The larger number of segments may be required if the spline is defined with a large number of closely specified control points; the case is typical when NURBS is used for processing minute segments prepared with a CAD system. The buffer is for the controller's internal use only and should not be used by the user application.
motor_motion_delay	(Optional, used only with /q switch) Delay, in milliseconds, before motor motion actually starts.

Switches

w	Do not start until the go command. If the switch is not specified, the motion starts once the first three spline points are specified.
v	Specify required velocity. The switch requires an additional parameter that specifies required velocity. If the switch is not specified, the feed rate is derived from the leading axis' parameters.
a	Specify exception angle. The switch requires an additional parameter that specifies the maximum angle in a control point. The value defines exceptions from spline interpolation. If defined for an internal control point, the directions to the previous and the next control points require direction change more than the specified angle (by modulo) the control point is processed as a corner. Actually, such a point divides the spline into two independent splines.
l	Specify exception length. The switch requires an additional parameter that specifies the maximum segment length. The value defines exceptions from spline interpolation. If the distance between two control points appears longer than the specified length, the trajectory between the points is considered straight. Actually, two independent splines are built before and after the segment.
q	Defines actual motor movement delay in microseconds. The delay resolution is 50 microseconds. The maximum delay is 100 controller cycles: 100ms for CTIME=1ms or 20ms for CTIME=0.2ms. Allows delaying actual motor movement start (RPOS) for the specified delay, while motion profile generation (APOS) starts immediately.



The first **NPOINT** command must define the starting point of the NURBS motion.

Return Value

None

Example 1

```

NURBS/V(0,1), 100

NPOINT    (0,1), 4.0, -6.0    ! the first point must be the motion
                                ! starting position

NPOINT    (0,1), -4.0,  1.0
NPOINT    (0,1), -1.5,  5.0
NPOINT    (0,1),  0.0,  2.0
NPOINT    (0,1),  1.5,  5.0
NPOINT    (0,1),  4.0,  1.0
NPOINT    (0,1), -4.0, -6.0

ENDS (0,1)

```

Example 2

```

! Start a NURBS motion from current position
! velocity = 1000, go to exact corner if angle > 1RAD
NURBS/AV(0,1), 1000 , 1

NPOINT    (0,1), 0, 0 ! the first point must be the motion starting
position
NPOINT    (0,1), 50, 40
NPOINT    (0,1), 100, 120
NPOINT/f (0,1), 120, 130 , 500 ! velocity for segment = 500
NPOINT/c (0,1), 180, 130 ! come to an exact stop at this point,
                            ! start a new spline after this point

NPOINT/v (0,1), 200, 120, 2000 ! change motion vel to 2000
NPOINT    (0,1), 220, 80 ! will be treated as a corner because of
                            ! angle between points

NPOINT    (0,1), 200, 60
NPOINT    (0,1), 0, 0

ENDS (0,1)
STOP

```

2.6.28 NPOINT**Description**

Add next control point and knot

Syntax

NPOINT[/switches](axis_list),coordinates,[velocity][,knot][,weight][,segment_velocity]

Arguments

axis_list	The list of axes must be the same as in the corresponding NURBS command.
coordinates	The list of coordinate values separated by commas. The list must specify one value for each axis in axis_list. The list defines coordinates of one control point of the spline.
velocity	Optional, only used with switch /v. The value changes the required velocity. The new value is valid for all spline segments after the corresponding control point.
knot	Optional, only used with switch /k. The value specifies a delta for the new knot calculation. The delta must be positive or zero. Each npoint command adds a new knot to the knot vector. The new knot is calculated as a previous knot plus delta (positive or zero). If suffix k is not specified, the delta default value is one; a new knot is calculated as the previous knot plus one. A special case occurs in the very first npoint command, if knots command was not specified, so that the previous knot does not exist. In this case, the knot parameter specifies the absolute value of the first knot; if switch /k is omitted, a zero value is assumed as the first knot. If none of the npoint commands specifies a /k switch and knots commands are omitted, the controller builds a uniform spline with evenly spread knots (except for the duplicated knots at the beginning and end).
Weight	Optional, only used with /w switch. The value specifies control point weight. Only positive weights are accepted. If a /w switch is not specified, the default value of weight is one. If none of the point commands specifies a /w switch, all control points have the same weight (one); in this case, the spline is actually a non-rational B spline.
segment_velocity	Optional, only with /f switch The required velocity for this segment only

Switches

/v	Specify new required velocity. The switch is not compatible with /f. The switch requires an additional parameter that specifies the required velocity. The value is used as the required velocity for the current and all subsequent points.
/f	Specify segment velocity.

	The switch is not compatible with /v. The switch requires an additional parameter that specifies required velocity. The value is used as the required velocity for the current point, but does not change the required velocity for subsequent points.
/k	Specify knot delta. The switch requires an additional parameter that specifies the knot delta from the previous knot. The new knot is calculated as the previous knot plus delta.
/w	Specify control point weight. The switch requires an additional parameter that specifies the weight of the control point.
/d	Mark the point specification as dummy. Dummy point specifications can either precede the first control point specification, or follow the last control point specification. Dummy points specification is required in rare cases where default calculation of starting/trailing knots is not appropriate.
/c	Mark the current point as a corner. The control point is processed as a corner. Actually, such a point divides the spline into two independent splines.

Return Value

None



The first **NPOINT** command must define the starting point of the NURBS motion.

2.6.29 SPATH**Description**

Initiate a path smoothing motion.

Syntax

SPATH [/switches] (axis_list) , starting_coordinates[,velocity][,exc_angle][,exc_length][,segments_buffer][,motor_motion_delay]

Arguments

axis_list	Axes involved in the path smoothing motion, specified as axes numbers separated by comma or as axes' names separated by comma. A minimum of two axes must be specified.
starting_coordinates	A list of coordinate values, separated by commas. The list must specify one value for each axis in axis_list . This array specifies the starting position for the SPATH motion.

velocity	Optional, only used with /v switch Defines required feed rate. If not specified, the feed rate is derived from the leading axis parameters.
exc_angle	Optional, only used with /a switch The value defines exceptions from spline interpolation. If for an internal control point directions to the previous and the next control points require direction change more than the specified angle (by modulo), the control point is processed as a corner. Actually, such point divides the spline into two independent splines.
exc_length	Optional, only used with /l switch. The value defines exceptions from spline interpolation. If a distance between two control points appears longer than the specified length, the trajectory between the points is considered straight. Actually, two independent splines are built before and after the segment.
segments_ buffer	[Optional] One-dimensional user-defined real array. The controller uses this array to store added segments. By default, if the argument is not specified, the controller allocates internal buffer for storing 50 segments only. The argument allows the user application to reallocate the buffer for storing a larger number of segments. The larger number of segments may be required if the spline is defined with a large number of closely specified control points; the case is typical when path smoothing is used to process minute segment prepared with a CAD system. The buffer is for the controller's internal use only and shouldn't be used by the user application
motor_motion_ delay	(Optional, used only with /q switch) Delay, in milliseconds, before motor motion actually starts.

Switches

w	Do not start until the go command. If the switch is not specified, the motion starts once the first three spline points are specified.
v	Specify required velocity. The switch requires an additional parameter that specifies the required velocity. If the switch is not specified, the feed rate is derived from the leading axis parameters.

a	Specify exception angle. The switch requires an additional parameter that specifies maximum angle in a control point. The value defines exceptions from spline interpolation. If for an internal control point, directions to the previous and the next control points require direction change more than the specified angle (by modulo), the control point is processed as a corner. Actually, such point divides the spline into two independent splines.
l	Specify exception length. The switch requires an additional parameter that specifies maximum segment length. The value defines exceptions from spline interpolation. If a distance between two control points appears longer than the specified length, the trajectory between the points is considered straight. Actually, two independent splines are built before and after the segment.
G	Acceleration consideration. Allow the motion generator to deviate from the specified axes acceleration parameter during velocity profile generation.
q	Defines actual motor movement delay in microseconds. The delay resolution is 50 microseconds. The maximum delay is 100 controller cycles: 100ms for CTIME=1ms or 20ms for CTIME=0.2ms. Allows delaying actual motor movement start (RPOS) for the specified delay, while motion profile generation (APOS) starts immediately.

Return Value

None

Comments

This command is supported in version 3.10 and higher.

Example

```

! Start an SPATH motion from current position (0,0),
! velocity = 1000, wait for go command before starting motion
SPATH/VW(0,1), 0, 0, 1000

SEGMENT (0,1), 50, 0
SEGMENT (0,1), 100, 0
SEGMENT/f (0,1), 100, 100 , 500 ! velocity for segment = 500
SEGMENT/c (0,1), 200, 200 ! come to an exact stop at this point,
                        ! start a new spline after this point

SEGMENT/v (0,1), 300, 200, 2000 ! change motion vel to 2000
SEGMENT (0,1), 300, 300
SEGMENT (0,1), 400, 300
SEGMENT (0,1), 400, 400

```

```
GO (0,1) !start motion

ENDS (0,1)
STOP
```

2.6.30 SEGMENT

Description

Add a new control point to the **SPATH** motion generator

Syntax

SEGMENT[/switches](axis_list),coordinates[,velocity][,segment_velocity]

Arguments

axis_list	The list of axes must be the same as in the corresponding SPATH command.
coordinates	A list of coordinate values, separated by commas/ The list must specify one value for each axis in axis_list . The list defines coordinates of the end control point of the spline.
velocity	Optional, only used with /v switch. The value changes required velocity from this point forward. The new value is valid for all spline segments after the corresponding control point.
segment_velocity	Optional, only with the /f switch The required velocity for this segment only. At the end of the segment, the velocity reverts to the value previously in use.

Switches

/v	Specify new velocity. The switch is not compatible with /f . The switch requires an additional parameter that specifies the velocity value for the current and all subsequent segments.
/f	Specify required segment velocity. The switch is not compatible with /v . The switch requires an additional parameter that specifies the required segment velocity. The value defines the required velocity in the current segment, but does not change the required velocity for subsequent segments.
/c	Mark the current point as a corner. The control point is processed as a corner. Actually, such point divides the spline into two independent splines.

Return Value

None

Comments

This command is supported in version 3.10 and higher.

2.6.31 SMOVE

The **SMOVE** command provides for positioning to specific target. **SMOVE** commands work in sequence and the next **SMOVE** command changes the previous target and provide a smooth transition from one motion direction to another, based on **ACC**, **DEC** and **JERK** values. The motion profile is optimized to pass on a rounded path near the breaking point, minimizing changes in speed and direction that would cause unwanted vibrations in the system.

Syntax

```
SMOVE[/switch] axis_list, target_point[, velocity]
```

Arguments

axis_list	Motion axes. 2 or 3 Cartesian axes can participate in the motion.
Target_point	Destination point coordinates
velocity	Optional argument for user-defined velocity

All **SMOVE** commands in the sequence must have the same axis_list parameter.

Switches

/v	Use the specified velocity instead of the default velocity (VEL).
/z	Move relative to Local Coordinate System

Example

```
SMOVE (X,Y,Z), 100, 100, 300
```

2.6.32 SPTP Command**Description**

The **SPTP** command is a multi-axis Point-to-Point motion defining motion to a specific target point, using syntax similar to the **PTP** command. Unlike **PTP**, **SPTP** uses a 4th order motion profile.

If the axis is moving when the command is issued, the controller creates the motion and inserts it into the axis motion queue. The motion waits in the queue until all motions before it finish, and only then starts.

The **SPTP** command can act on either a single axis or an axis group.

SPTP responds to **KILL** /**HALT** commands according to the same rules as other motion commands.

Syntax

```
SPTP[/switches] axis_list, target_point[, velocity]
```

Arguments

axis_list	Motion axes designation.
target_point	Target point coordinates
velocity	Optional argument for user-defined velocity

Switches

/e	Wait for motion termination before executing next command
/v	Use the specified velocity instead of the default velocity (VEL).
/w	Create the motion, but do not start until the GO command is issued
/z	Move relative to Local Coordinate System

Comments

SPTP looks like the **PTP** command. Unlike **PTP**, **SPTP** uses a 4th order motion profile.
The **SPTP** command can take either a single axis or an axis group as a parameter.

Example

```
SPTP (X,Y,Z) 100, 100, 300
```

2.6.33 Using ARC1, ARC2 and LINE Switches

The following optional switches may be used singularly or in combination with **ARC1**, **ARC2** and **LINE**:

/f	Decelerate to the end of segment. The switch requires an additional parameter that specifies the end velocity. The controller decelerates to the specified velocity at the end of segment. The specified value should be less than the required velocity; otherwise the parameter is ignored. The switch affects only one segment. The switch also disables corner detection and processing at the end of segment. If the switch is not specified, deceleration is not required. However, in special cases the deceleration might occur due to corner processing or other velocity control conditions.
/v	Specify required velocity. The switch requires an additional parameter that specifies the required velocity. The switch changes the required velocity for the current segment and for all

	subsequent segments. If the switch is not specified, the required velocity does not change.
/o	Synchronize user variables with segment execution. The switch requires additional two or three parameters that specify values , user variable and mask .
/t	Specify segment processing time The switch requires an additional parameter that specifies the segment processing time in milliseconds. Unlike the required velocity specification, the segment processing time defines velocity at the current segment only, and has no effect on the subsequent segments. The switch cannot be specified together with /V .
/b	Use external loops at corners. The switch requires additional parameters that specify the external loop type, the minimum segment length, and the maximum allowed deviation from profile. The /b switch may be defined with other corner processing options (/u, /g, etc.) . If the Skywriting algorithm is applied, other corner processing options are skipped. If Skywriting is skipped, other defined corner options will be applied. This switched is used for Extended Motion only (initialized with XSEG...ENDS)
/p	Specifies that the lci_segment_active parameter is required. This switched is used for Extended Motion only (initialized with XSEG...ENDS)

For **ARC1**, **ARC2**, and **LINE** some switches require an additional parameter to be specified. If more than one parameter is required, the parameters should be separated by a comma, and the order of parameters is fixed in the following order:

1. Required velocity (used with **/V**)
2. Final velocity (used with **/F**)
3. Segment processing time (used with **/T**)
4. **/O** requires specification of the **values**, **variables**, and **mask** parameters
5. **/B** requires specification of the External Loop Type, Minimum Segment Length, and Maximum Allowed Deviation parameters
6. **/P** requires specification of the **lci_segment_active** parameter

Examples:

LINE/v (1,0), 1000, -1000, 500	Add line segment with end point (1000, -1000) and segment velocity 500.
arc1/vf (0,1), 0, 0, 100, 100, +, 500, 100	Add arc segment with center (0,0), end point (100,100), clockwise direction, segment velocity 500 and end velocity 100

int Value(1) int Mask(1) Value(0) = 1; Mask (0) = 5 ARC2/o (0,1), 0, 0, 3.141529, Value, OUT, 2, Mask	Add arc segment with center (0,0) and 180 degree (π) angle. At the beginning of the segment execution, sets bit 0 and reset bit 2 of digital outputs OUT(2).
---	--

2.7 Program Flow Commands

Command	Description
Assignment Command	Assigns values
BLOCK...END	Executes a group of commands in one MPU cycle
CALL	Calls subroutine.
GOTO	Transfers program execution to another point in the program.
IF, ELSEIF, ELSE...END	IF command structure.
INPUT	Suspends program execution pending user input
LOOP...END	Loop command structure.
ON...RET	Defines an autoroutine
TILL	Delays program execution until a specified expression produces a non-zero (true) result.
WAIT	Delays program execution for a specified number of milliseconds.
WHILE...END	While command structure.
SWITCH Statement	SWITCH statement controlling program flow depending on value

2.7.1 Assignment Command

Description

The **Assignment** command (=) is used to assign a value to ACSPL+ standard or user-defined variables with read/write options.

Syntax

variable_name = value

Arguments

variable_name	Can be: > Name of ACSPL+ standard or user-defined variable > An element of an ACSPL+ or user array > One bit of integer variable or integer array element (applicable only to those variables having specifically named bits, for example, IMASK)
value	Can be of integer or real type. The value argument can either be: > A specific value > An expression that during runtime calculates to a value

Comments

Assigning to an ACSPL+ variable is limited by the following rules:

- > Assignment to read-only variable (for example, **FPOS**) is prohibited
- > Assignment to a protected variable (for example, **ERRI**) is allowed in only in the Configuration mode.

After assignment, the previous value of the variable is replaced by the new value.

User local and global variables must be declared before they can be used in an assignment command.

- > Explicit indexing only is allowed for user array variables.
- > If a user variable is scalar, no indexing is required.
- > If a user variable is one-dimensional array, it requires one index. Two-dimensional arrays require two indexes.

A bit can have only two possible values: 0 (false) or 1 (true), while the **value** result, which defines the bit value, can be any value. Assignments convert the value as follows:

- > If the value is zero, the bit is set to zero
- > If the value is non-zero, the bit is set to one

Although bit assignments are applicable to any integer variable or array element, they are mainly used for changing flag variables and output bits.

The controller executes assignment commands in the following order:

1. Calculate **value**
2. Convert the type of calculated value to the type of **variable_name** (if the types differ)
3. Assign the result to **variable_name**

Examples

```
VEL(0) = 1000      !Assign 1000 to axis 0 default velocity -
                   !explicit indexing.
VEL0 = 1000        !Assign 1000 to axis 0 default velocity -
```

Var1 = FPOS(0)	!postfix indexing.
Var2(0)(5) = 200	!Assign value of ACSPL+ variable to user variable.
OUT0.5 = 1	!Assign to element of user array.
	!Assign to digital output 5

2.7.2 BLOCK...END

Description

Commands specified within the **BLOCK...END** structure are executed in one MPU cycle.

Syntax

BLOCK

command-list

END

Arguments

command-list

List of commands, separated by semi-colons.

Comments

- > The structure provides an alternative to specifying *command-list* commands in one line. The commands within the structure can be specified in several lines. However, the controller executes all commands in one controller cycle, as if they were written in one line.
- > Commands and functions that may cause delay ([WAIT](#), [TILL](#), [GETSP](#), [WHILE...END](#), [LOOP...END](#) etc.) provide delay even if they are used within the **BLOCK...END** structure.

Example

INT XX,YY,ZZ	!Defines variables as integers
BLOCK	!BLOCK command
XX=TIME	!XX is assigned the TIME standard variable
YY=TIME	!(controller timer from power-up, in mSec.
	!One execution line later, YY is also
	!assigned the TIME standard variable.
	!Because both lines were written in the
	!BLOCK...END block, they are both executed
	!during the same controller cycle hence the
	!difference between the two values =0
	!(ZZ=0).
	!If both lines were NOT written within the
	!BLOCK structure, the difference would be
	!one controller cycle (ZZ=1).
END	!End the BLOCK command
ZZ=YY-XX	!The result in this example is ZZ=0.

2.7.3 CALL

Description

CALL calls a subroutine according to a specified label. All subroutines must begin with a label and conclude with **RET**.

Syntax

CALL *label*

...

STOP

label:

...

RET

Arguments

label

Unique name identifier.

Related ACSPL+ Commands

[GOTO](#)

Comments

- > A label specified by **CALL** must be defined somewhere in the same program buffer. The subroutine starts after the label and spans all commands up to **RET**.
- > **CALL** transfers program execution to the start of the subroutine and stores the return point in the stack. When **RET** executes, the return point is extracted from the stack and execution continues from the next command after **CALL**.
- > Subroutines can be nested; a subroutine can call another subroutine, and so on.
- > **CALL** can only call a sub-routine located in the same buffer

Example

```
! ----- Start subroutine labeled CHECK_PE -----
CALL CHECK_PE           !Call a subroutine named CHECK_PE.
DISABLE 0               !This line will be executed when the
                        !subroutine is terminated by the RET command.

STOP                   !Ends program.
CHECK_PE:              !Subroutine label name.
WHILE (ABS(PE(0)>20))   !Subroutine command lines.
DISP "PE is :", PE(0)
END                    !End WHILE loop.
RET                    !Terminate subroutine and return to main program.
```

2.7.4 GOTO

Description

GOTO transfers program execution to a particular point in the program specified by a unique label.



Avoid using **GOTO** to enter or exit a subroutine. Failure to do so will result in program termination due to a stack violation error. As an alternative see [CALL](#).

Syntax

GOTO *label*

Arguments

<i>label</i>	A unique label defined in the program buffer.
--------------	---

Comments

- > A label specified by **GOTO** must be defined somewhere in the same program buffer.
- > The next executed command is located after the label.

Related ACSPL+ Commands

[CALL](#)

Example

```
GOTO ARR           !Go to a label named ARR
```

2.7.5 IF, ELSEIF, ELSE...END

Description

IF, **ELSEIF**, **ELSE**, and **END** are building blocks used in the **IF** control structures. These commands specify a condition which must be met before executing a list of commands. **IF** control structures must conclude with **END**.

If the condition result is none-zero (true), the command list in the corresponding clause executes and all subsequent conditions are not validated and all other command lists are skipped. After executing the command list, the program continues from the next line after **END**.

If a condition result is zero (false), the command list is skipped. The program then checks subsequent **ELSEIF** conditions. If no condition result is true, the command list following **ELSE** executes, if it exists. The program then continues from the command following **END**.

[Table 2-9](#) describes the **IF** control structure syntax using these building blocks.

Table 2-9. IF Control Structures

Syntax

Syntax	IF Structure
IF <i>expression</i> command list END	IF-END
IF <i>expression</i> command list ELSE command list END	IF-ELSE-END
IF <i>expression</i> command list ELSEIF sequence END	IF-ELSEIF-END

Syntax	IF Structure
IF <i>expression</i> command list ELSEIF sequence ELSE <i>command list</i> END	IF-ELSEIF-ELSE-END

Comments

- > The **IF -ELSE -END** control structure has two command lists. One follows **IF** and the second follows **ELSE**. Only one command list executes, depending on the result of the expression following **IF**.
- > The **IF-ELSEIF-END**, and **IF-ELSEIF- ELSE -END** forms provide validation of a sequence of conditions.
- > **IF** control structures may contain any number of **ELSEIF** clauses. Each **ELSEIF** clause specifies its own condition. The conditions are validated in the following order:
 - > Condition after **IF**
 - > Condition after first **ELSEIF**
 - > Condition after second **ELSEIF**

Examples

Example 1:

This example program fragment activates either output 5 or 6 and shuts off the other, depending on the state of input 3.

```

IF IN0.3           !Check if the third bit of IN0 is 1.
    OUT0.5 = 1      !Set the 5th bit of OUT0 to 1.
    OUT0.6 = 0      !Set the 6th bit of OUT0 to 0.
ELSE               !Execute the next commands only if the IF
                  !condition was false
    OUT0.5 = 0      !Set the 5th bit of OUT0 to 0.
    OUT0.6 = 1      !Set the 6th bit of OUT0 to 1.
END               !End of the IF body.
STOP              !Ends program

```

Example 2:

The following program fragment implements a saturation effect by limiting variable **V0** to a range from -256 to +256.

```

IF V0 < -256       !Check if V0 is less than -256
    V0 = -256      !Assign -256 to V0 (low saturation)
ELSEIF V0 > 256    !If V0 is not less than -256, check if it
                  !is greater than 256
    V0 = 256       !Assign 256 to V0 (high saturation)
END               !End of the IF body (no saturation if
                  !V0 is equal to or more than 256)
STOP              !Ends program

```

2.7.6 INPUT

Description

INPUT suspends program execution in a specific buffer until the host or the user (via the **Terminal** option) provides a specific value or values.

Syntax

INPUT(*array*)

Arguments

array

User defined array.

Comments

- > The INPUT command accepts arguments as an array variable. The command delays program execution until the host provides an input. The input from the host is expected to consist of one or more numbers separated by spaces or commas and followed by a carriage return. The INPUT command scans the numbers in the input and stores them in the argument variable.
- > If the argument variable is scalar, INPUT takes the first number in the input and stores it in the argument variable.
- > If the argument is an array, INPUT fills the array argument with the values from the input.
- > The function execution is completed when the scalar is stored or the array is filled or a carriage return is encountered.

COM Library Methods and .NET Library Methods

SuspendBuffer

C Library Functions

acsc_SuspendBuffer

Related ACSPL+ Commands

None

Examples

Example 1:

```

INT ARR(10)      !Declares a ten-element array.
...              !Other program commands.
INPUT (ARR)      !Wait for user to enter values in ARR array.

```

Program execution halts until input is received from the host.

Assume that a user enters: 20 30 40 <ENTER>, the program stores the three values in ARR(0), ARR(1), and ARR(2) and resumes execution. Since the user has entered only three numbers, the program will store 0 (zero) as the value in the elements: ARR(3) through ARR(9).



The user may enter the value of a keyboard function key, 0 (zero) followed by an integer from 1 to 12, in which case the value is stored separately in the standard variable: **FK**.

Example 2:

```
INT ARR(2)           !Declares a two-element array.
DISP "[FILL AXIS NUMBER],[FILL DISTANCE TO GO]"
                    !Display the text in the Terminal
INPUT (ARR)          !This command stops the program until the two
                    !variables are entered into the ARR array.
SET FPOS(ARR(0))=0    !Set the feedback position to zero.
ENABLE (ARR(0))       !Enable the drive specified by the value of ARR(0).
PTP/RE (ARR(0)), (ARR(1))
                    !Move the ARR(0) axis to position ARR(1)
```

2.7.7 LOOP...END

Description

The **LOOP** command structure provides a fixed number of command list repetitions set by the definition of an exact value, or any expression. **LOOP** control structures must conclude with **END**.

Syntax

LOOP *expression*

command_list

END

Arguments

<i>expression</i>	Defines the number of loops. It can also be a specific integer number.
<i>command_list</i>	Any set of commands that are to be repeated.

Comments

- > If the *expression* result is zero or negative, the command list is not executed and the program continues from the next line after **END**.
- > If the *expression* result is not an integer the command rounds-off the number to the closest integer.

Examples

Example 1:

```
LOOP 10             !Do 10 repetitions.
J=J+1               !J is increased by 1 for each repetition.
END                 !End LOOP.
STOP                !Ends program.
```

Example 2:

```

LOOP ARR          !Do repetitions equal to value labeled by ARR.
J=J+1             !J is increased by 1 for each repetition.
END               !End LOOP.
STOP              !Ends program.

```

Example 3:

```

LOOP YY*325/TT    !Do repetitions equal to the value of the
                  !expression YY*325/TT
J=J+1             !J is increased by 1 for each repetition.
END               !End LOOP.
STOP              !Ends program.

```

2.7.8 ON...RET

Description

ON...RET defines an autoroutine. An autoroutine consists of a condition, and a body. Autoroutines must conclude with **RET**. The autoroutine condition is checked every MPU cycle. Once the autoroutine condition is met, the autoroutine interrupts, executes the lines in the body (until **RET**), and then transfers execution control back to the interrupted program line.



The controller should never directly execute **ON**. If the program execution flow comes to **ON**, the controller asserts a runtime error and aborts the program. To avoid this error, use **ON** after **STOP/STOPALL**.

Syntax

ON *condition*

```
auto_routine body
```

RET

Comments

- > Once the buffer in which an autoroutine is compiled, that autoroutine is enabled.
- > The controller implements edge-detection in autoroutine condition verification. If a condition becomes true, the controller activates the autoroutine only once. If afterwards the condition remains true, the controller does not activate the autoroutine again. The condition must become false and then become true again in order to activate the autoroutine again.
- > Only one autoroutine can be active in a buffer at a time.

Related ACSPL+ Commands

[ENABLEON](#), [DISABLEON](#)

Examples

Example 1:

Demonstrates a typical use of an autoroutine for processing controller faults. The autoroutine provides an error message when a drive alarm on 0 axis occurs.

```
STOP                                !STOP must precede the autoroutine.
ON FAULT(0).#DRIVE                 !Activate autoroutine when bit
                                   !FAULT(0).#DRIVE changes from 0 to 1
DISABLE 0                          !Disable drive X
DISP "0 Axis Drive Alarm"          !Display message
RET                                !End of autoroutine
```

Example 2:

Illustrates how all variables, not only faults, can be used in autoroutine conditions. Assuming that output OUT0.6 is connected to a LED indicator, the following autoroutine signals the motion state bit to activate the indicator, and deactivate it when the 0 axis is no longer in motion.

```
STOP                                !STOP must precede the autoroutine.
ON MST(0).#MOVE                   !When MST(0).#MOVE bit changes from 0 to
                                   !1 (signaling that the axis is moving)
OUT0.6 = 1                        !Set output 6 to 1 (turn on the LED)
RET                                !End of autoroutine
ON ^MST(0).#MOVE                  !When MST(0).#MOVE bit changes from 1 to
                                   !0 (signaling that the axis is no longer
                                   !moving)
OUT0.6 = 0                        !Set output 6 to 0 (turn off the LED)
RET                                !End of autoroutine
```

2.7.9 TILL

Description

TILL delays program execution until the result of a specified expression is met, or the return value is non-zero.

Syntax

TILL *expression* [*time_out*]

Arguments

<i>expression</i>	Defines how long to pause the program execution
<i>time_out</i>	Used to specify a time-out in milliseconds. Use <i>time_out</i> to limit the time that an expression can return zero.

Related ACSPL+ Commands

[WAIT](#)

Examples

Example 1:

```
TILL ^MST(0).#MOVE                !Wait for 0 axis termination
```

Example 2:

```
TILL IN0.0=1, 2000      !Wait until input #0 =1 or 2000 milliseconds
```

Example 3

```
TILL RPOS(0)<>0      !Wait until RPOS(0) is not equal to 0
```

2.7.10 WAIT

Description

WAIT delays program execution for the specified number of milliseconds.

Syntax

WAIT *wait_time*

Arguments

wait_time

Defines how long to delay the program execution. *wait_time* can also be defined by an expression.

Related ACSPL+ Commands

[TILL](#)

Examples

Example 1:

```
WAIT 2000      !Wait 2000 milliseconds
```

Example 2:

```
WAIT YY=65/TT      !Milliseconds defined by an expression
```

2.7.11 WHILE...END

Description

The **WHILE** command structure provides repetitive execution of commands list as long as a condition is satisfied. If the condition was not satisfied when checked for the first time, program execution continues from the command following **END**.

WHILE control structures must conclude with **END**.

Syntax

WHILE *condition*

command_list

END

Arguments

<i>condition</i>	Condition controlling the execution of the WHILE <i>command_list</i> .
<i>command_list</i>	List of commands to be executed.

Related ACSPL+ Commands

IF, ELSEIF, ELSE...END, TILL

Examples

Example 1:

```
WHILE S_ST.#DC      !Indicate data collection active.
OUT0.0=^OUT0.0      !Blink LED.
WAIT 200            !Blink period is 200msec.
END
```

Example 2:

```
WHILE (ABS(PE(0)>20)) !Do if the 0 axis Position Error is greater
                        !than 20.
DISP "PE is :", PE(0)
Display the 0 axis Position Error
END
```

Example 3:

```
WHILE 1              !Run forever
PTP/e 0, 1000
PTP/e 0, -1000
END
```

2.7.12 SWITCH Statement

The **SWITCH** statement compares the evaluated expression to several case values and starts execution of **CASE** code when an equivalent value is found.

If no **CASE** statement equaling the **SWITCH** value is found, the **DEFAULT** case will be executed, if one exists.

Execution of the selected case code will not end until an **END** command is reached. If no such command exists, execution continues with the next **CASE** or **DEFAULT** statement.

The switch expression must be of type **INT**. All **CASE** values must be **CONST** expressions of type **INT**.

Example

```
int intVar = 23;

SWITCH (intVar - 2)
```

```

CASE 20 :
    DISP "CASE 20"
    END;
CASE 21: !comment
    DISP "CASE 21";
    ! No break
DEFAULT:
    DISP "DEFAULT";
    END
END
STOP

```

2.8 Program Management Commands

The Program Management commands are:

Command	Description
DISABLEON	Disables autoroutine activation in a buffer.
ENABLEON	Enables autoroutine activation in a buffer.
PAUSE	Suspends program execution in a buffer.
RESUME	Resumes program execution in a buffer.
START	Activates program execution in a buffer.
STOP/STOPALL	Terminates program execution in a buffer. STOPALL terminates all programs.

2.8.1 DISABLEON

Description

DISABLEON disables autoroutine activation in a buffer. The command has the same functionality as **PFLAGS.#NOAUTO=1**.

Syntax

DISABLEON [(buffer-number)]

Arguments

<i>buffer-number</i>	Optional argument, if no buffer number is mentioned, the command disables autoroutines in all buffers.
----------------------	--

Related ACSPL+ Commands

ON...RET, **ENABLEON**

2.8.2 ENABLEON

Description

ENABLEON enables autoroutine activation in a buffer. The command has the same functionality as **PFLAGS.#NOAUTO=0**.

Syntax

ENABLEON [(*buffer-number*)]

Arguments

buffer-number

A number between 0 and 63 specifying the buffer in which the autoroutine is to be enabled.. If no buffer number is given, the command enables autoroutines in all buffers.

Related ACSPL+ Commands

ON...RET, **DISABLEON**

2.8.3 PAUSE

Description

PAUSE suspends program execution in a specific buffer.

Syntax

PAUSE *buffer-number*

Arguments

buffer-number

A number between 0 and 63 specifying the buffer in which the program is to be suspended.

Related ACSPL+ Commands

RESUME

Example

```
PAUSE 5           !Pauses buffer five
```

2.8.4 RESUME

Description

RESUME resumes the program execution in a specific buffer after the execution was paused.

Syntax

RESUME *buffer-number*

Arguments

buffer-number

A number between 0 and 63 specifying the buffer in which the program execution that was suspended by the **PAUSE** command is to resumed.

Related ACSPL+ Commands**PAUSE****Example**

```
RESUME 5           !Resume buffer five
```

2.8.5 START**Description**

START activates program execution at a specified line number, or a specific label in a specified buffer that is different from the buffer where **START** is enabled.

Syntax

START[/s] *buffer-number, line-number | label*

Arguments

<i>buffer-number</i>	A number between 0 and 63 specifying the buffer.
<i>line-number</i> <i>label</i>	The line number of the program within the buffer where execution is to begin, or a label identifying the line number.

Switches

Switch	Comments
/s	The "/s" switch causes the program to execute from the beginning in simulation mode up to the indicated line or label, after which execution continues normally. No motion is executed up to that point, but G-Code modality changes are registered. Execution continues normally from the indicated line or label, including motion commands.

Comments

- > Unless line #1 is used in the **START** command, it is recommended to use a label since line numbers are prone to change if programs lines are deleted or inserted.
- > **START** executes successfully if the target buffer is loaded with a program, compiled, but not running. Otherwise, **START** causes a run-time error and aborts the current program.
- > The program activated by **START** executes concurrently with the program containing the **START** command, and other active programs.
- > If the whole program needs to be run in simulation mode, for early run-time problem detection for example, the **START/s** command can be used without specifying a line number or label argument.
- > Simulation mode does not affect ACSPL+ lines, which will run normally.

Related ACSPL+ Commands**STOP/STOPALL, BREAK**

Examples

Example 1:

```
START 8,15           !Start program in buffer 8 line 15
```

Example 2:

```
START 8,Start_here   !Start program in buffer 8, at the line identified
                     !by the Start_here label.
```

Example 3 (simulation mode):

```
1  !Buffer 0
2  ENABLE(X,Y)
3  N20 G00 X0
4  N30 G01 X20 F2000
5  N40 G01 X10
6  N50 G01 X20
7  !Radius compensation
8  N10 G42 D10
9  RC:
10 N60 G01 X0
11 N30 G01 X20 F4000
12 N40 G01 X10
13 N50 G01 X20
14 N60 G01 X0
15 N70 M02
!Buffer 1
START/s 0,10 !Execution of buffer 0 in simulation mode until line 10
STOP
```

2.8.6 STOP/STOPALL

Description

STOP terminates program execution in the specified buffer. **STOPALL** terminates program execution in all buffers.

Syntax

STOP *buffer-number*

Arguments

<i>buffer-number</i>	A number between 0 and 63 specifying the buffer in which the program execution is to be halted.
----------------------	---

COM Library Methods and .NET Library Methods

StopBuffer

C Library Functions

acsc_StopBuffer

Example

```
STOP 5      !Terminate program execution in buffer five.
```

2.9 Ethernet/IP ACSPL+ Support Commands

The following ACSPL+ functions set assembly configuration and get existing assembly configuration. For more information refer to *SPiiPlus EtherNet/IP User Guide*.

2.9.1 EIPGETATTR

Description

EIPGETATTR returns value of a specific attribute. It can be a class attribute or an instance attribute.

Syntax

int eipgetattr(int class, int instance, int attr)

Arguments

Class	The following classes are supported:	
	Class Code	Class Name
	0x01	Identity
	0x02	Message Router
	0x06	Connection Manager
	0xF4	Port
	0xF5	TCP/IP Interface
	0xF6	Ethernet Link
	0x04	Assembly
	0x64	ACSPL+ Command
	0x65	ACSPL+ Variable
Instance	For the class attribute, this parameter should 0. Otherwise, the specific instance should be specified as follows	
	Class Code	Supported Instances
	0x01	1
	0x02	1

	0x06	1
	0xF4	1,2,3
	0xF5	1
	0xF6	1
	0x04	
	0x64	
	0x65	
Attr	Specifies the attribute as follows:	
	Class Code	Class Supported
		Attributes
	0x01	1
	0x02	1
	0x06	1
	0xF4	1,2,3
	0xF5	1
	0xF6	1
	0x04	1
	0x64	1,100
	0x65	1

Return Value

Returns value of a specific attribute.

-1 is returned in case of illegal parameters.

2.9.2 EIPGETIND1**Description**

EIPGETIND1 returns the first index of the requested ACSPL+ standard or user-defined variable in a one-dimensional array. The indexes start from 0.

Syntax

```
int eipgetind1(int instance, int element )
```

Arguments

<i>instance</i>	Assembly instance. The following instances are supported: <table> <tr> <th>Instance</th><th>Instance Name</th></tr> <tr> <td>> 0x64 (100)</td><td>Input Integer Assembly</td></tr> <tr> <td>> 0x65 (101)</td><td>Output Integer Assembly</td></tr> <tr> <td>> 0x66 (102)</td><td>Input Real Assembly</td></tr> <tr> <td>> 0x67 (103)</td><td>Output Real Assembly</td></tr> </table>	Instance	Instance Name	> 0x64 (100)	Input Integer Assembly	> 0x65 (101)	Output Integer Assembly	> 0x66 (102)	Input Real Assembly	> 0x67 (103)	Output Real Assembly
Instance	Instance Name										
> 0x64 (100)	Input Integer Assembly										
> 0x65 (101)	Output Integer Assembly										
> 0x66 (102)	Input Real Assembly										
> 0x67 (103)	Output Real Assembly										
<i>element</i>	The index of element in the corresponding assembly. Only indexes 0 to 123 are supported.										

Return Value

0 is returned in case of scalar variable.

-1 is returned in case of illegal index parameter.

2.9.3 EIPGETIND2**Description**

EIPGETIND2 returns the first index of the requested ACSPL+ standard or user-defined variable in a two-dimensional array. The indexes start from 0.

Syntax

```
int eipgetind2(int instance, int element)
```

Arguments

<i>instance</i>	Assembly instance. The following instances are supported: <table> <tr> <th>Instance</th><th>Instance Name</th></tr> <tr> <td>> 0x64 (100)</td><td>Input Integer Assembly</td></tr> <tr> <td>> 0x65 (101)</td><td>Output Integer Assembly</td></tr> <tr> <td>> 0x66 (102)</td><td>Input Real Assembly</td></tr> <tr> <td>> 0x67 (103)</td><td>Output Real Assembly</td></tr> </table>	Instance	Instance Name	> 0x64 (100)	Input Integer Assembly	> 0x65 (101)	Output Integer Assembly	> 0x66 (102)	Input Real Assembly	> 0x67 (103)	Output Real Assembly
Instance	Instance Name										
> 0x64 (100)	Input Integer Assembly										
> 0x65 (101)	Output Integer Assembly										
> 0x66 (102)	Input Real Assembly										
> 0x67 (103)	Output Real Assembly										
<i>element</i>	The index of element in the corresponding assembly. Only indexes 0 to 123 are supported.										

Return Value

0 is returned in case of scalar variable or one-dimensional array.

-1 is returned in case of illegal index parameter.

2.9.4 EIPGETTAG

Description

EIPGETTAG returns the tag number of requested ACSPL+ standard or user-friendly variable. User-friendly variables tags start from index 1000.

Syntax

```
int eipgettag(int instance, int element)
```

Arguments

<i>instance</i>	Assembly instance. The following instances are supported: <table> <thead> <tr> <th>Instance</th><th>Instance Name</th></tr> </thead> <tbody> <tr> <td>> 0x64 (100)</td><td>Input Integer Assembly</td></tr> <tr> <td>> 0x65 (101)</td><td>Output Integer Assembly</td></tr> <tr> <td>> 0x66 (102)</td><td>Input Real Assembly</td></tr> <tr> <td>> 0x67 (103)</td><td>Output Real Assembly</td></tr> </tbody> </table>	Instance	Instance Name	> 0x64 (100)	Input Integer Assembly	> 0x65 (101)	Output Integer Assembly	> 0x66 (102)	Input Real Assembly	> 0x67 (103)	Output Real Assembly
Instance	Instance Name										
> 0x64 (100)	Input Integer Assembly										
> 0x65 (101)	Output Integer Assembly										
> 0x66 (102)	Input Real Assembly										
> 0x67 (103)	Output Real Assembly										
<i>element</i>	The index of element in the corresponding assembly. Only indexes 0 to 123 are supported.										

Return Value

-1 is returned in case of illegal index parameter.

2.9.5 EIPSETASM

Description

EIPSETASM sets the assembly configuration.

Syntax

```
eipsetasm(int instance, int element,int tag, int first, int second )
```

Arguments

<i>instance</i>	Assembly instance. The following instances are supported: <table> <thead> <tr> <th>Instance</th><th>Instance Name</th></tr> </thead> <tbody> <tr> <td>> 0x64 (100)</td><td>Input Integer Assembly</td></tr> <tr> <td>> 0x65 (101)</td><td>Output Integer Assembly</td></tr> <tr> <td>> 0x66 (102)</td><td>Input Real Assembly</td></tr> <tr> <td>> 0x67 (103)</td><td>Output Real Assembly</td></tr> </tbody> </table>	Instance	Instance Name	> 0x64 (100)	Input Integer Assembly	> 0x65 (101)	Output Integer Assembly	> 0x66 (102)	Input Real Assembly	> 0x67 (103)	Output Real Assembly
Instance	Instance Name										
> 0x64 (100)	Input Integer Assembly										
> 0x65 (101)	Output Integer Assembly										
> 0x66 (102)	Input Real Assembly										
> 0x67 (103)	Output Real Assembly										
<i>element</i>	The index of element in the corresponding assembly. Only indexes 0 to 123 are supported.										
<i>tag</i>	The tag number of required ACSPL+ standard or user-defined variable. User-defined variables tags start from index 1000.										

<i>first</i>	The first index of required ACSPL+ standard or user-defined variable. The indexes start from 0. Used only if variable is one or two-dimension array. Should be 0 for scalar variable.
<i>second</i>	The first index of required ACSPL+ standard or user-defined variable. The indexes start from 0. Used only if variable is two-dimensional array. Should be 0 for scalar variable or one-dimensional array.

2.10 Laser Control Commands

The laser commands are:

Command	Description
LCENABLE	Enables a pulse generation process with current set parameters.
LCDISABLE	Stops a pulse generation process, including tickle pulses.

2.10.1 LCALENABLE

The **LCALENABLE** command enables a pulse generation process with current set parameters. If initially frequency or duty cycle is set to zero, than pulse generation will be pending till positive value will be set. This approach allows synchronize the pulse generation with actual motion, when, for example, frequency or duty cycle is updated as a function of motion reference velocity.

Syntax

LCALENABLE(*Index*)

Arguments

Arguments	Comment
Index	Refers to any of the axis that are allocated for the laser.

2.10.2 LCDISABLE

The **LCDISABLE** command stops a pulse generation process, including tickle pulses.

Syntax

LCDISABLE(*Index*)

Arguments

Arguments	Comment
Index	Refers to any of the axis that are allocated for the laser.

2.11 Input Shaping Commands

The input shaping commands are as follows.

Command	Description
INSHAPEON	Activate input shapping
INSHAPEOFF	Terminate input shaping

2.11.1 INSHAPEON

Description

The **INSHAPEON** function starts Input Shape algorithm for specified axis. The result is a dynamic output signal equal to the convolution of the input signal and the convolution pulses.

Syntax

INSHAPEON Axis_Index, T_array, A_array

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
T_array	One-dimensional array specifying the times of each convolution in milliseconds.
A_array	One-dimensional array specifying the amplitudes of each convolution pulse

Return Value

None

Comments

Vectors T_array and A_array define characteristics of the convolution pulses. The array sizes should be identical.

Vector T_array contains real numbers, so fractional numbers may be specified. However, the position of each pulse is rounded to a multiple of the controller cycle. If the controller cycle is one millisecond, the numbers in T_array are rounded to integers. The elements of T_array must be arranged in ascending order.

The sum of A_array entries must equal 1.

See the Using the Convolve Web Site chapter in the *ACSPL+ Programmers Guide* to get the explanation how to calculate the T_array and A_array parameters.

This function is supported in version 3.00 and higher..

Examples

```
global real CnvT(5), CnvA(5), CnvB(420)
VEL(0) = 120
ACC(0) = VEL(0)*10
JERK(0) = ACC(0)*10
```

```

CnvT(0)=0; CnvT(1)=64 ; CnvT(2)=68 ;CnvT(3)=72; CnvT(4)=139
CnvA(0)=25345/1e5; CnvA(1)=160/1e5; CnvA(2)=30987/1e5; CnvA(3)=18949/1e5;
CnvA(4)=24559/1e5

enable 0
InShapeOn 0, CnvT, CnvA
ptp/e 0,0
ptp/e 0,50
till ^MST(0).#MOVE
InShapeOff 0
stop

```

```

!INPUTSHAPE EXAMPLE: IF CTIME < 1

!In this case we need to multiply CnvT array by CTIME. Here CTIME = 0.5
global real CnvT(5), CnvA(5), CnvB(420)
CnvT(0)=0*CTIME; CnvT(1)=1*CTIME ; CnvT(2)=214*CTIME ;CnvT(3)=253*CTIME;
CnvT(4)=501*CTIME
CnvA(0)=22960/1e5; CnvA(1)=10361/1e5; CnvA(2)=3186/1e5; CnvA
(3)=45767/1e5;
CnvA(4)=17726/1e5
enable 0
InShapeOn 0, CnvT, CnvA
ptp/e 0,0
ptp/e 0,30
till ^MST(0).#MOVE
!InShapeOff 0
stop

```

2.11.2 INSHAPEOFF

Description

The **INSHAPEOFF** function stops the Input Shape algorithm for the specified axis.

Syntax

INSHAPEOFF Axis_Index

Arguments

Axis_Index

Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Comments

This variable is supported in version 3.00 and higher.

Return Value

None

3. ACSPL+ Variables

ACSPL+ supports data types of integer, real, and matrix. Matrix variables are a two-dimensional array of real values.

This chapter covers the ACSPL+ variables. ACSPL+ has a complete set of built-in variables for use in setting values that control ACSPL+ programs. The ACSPL+ variables are divided into the following categories:

- > **Axis Configuration Variables**

Along with the following subgroups:

- > Brake Variables
- > Feedback Variables
- > Safety Limits Variables
- > Axis State Variables
- > Data Collection Variables
- > Input and Output Variables
- > Monitoring Variables
- > Motion Variables
- > Program Execution Control Variables
- > Safety Control Variables
- > Induction Motor Variables
- > Nanomotion Variables
- > Servo-Loop Variables

Along with the following subgroups:

- > Servo-Loop Current Variables
- > Servo-Loop Velocity Variables
- > Servo-Loop Velocity Notch Filter Variables
- > Servo-Loop Velocity Low Pass Filter Variables
- > Servo-Loop Velocity Bi-Quad Filter Variables
- > Servo-Loop Position Variables
- > Servo-Loop Compensations Variables
- > Servo-Loop Miscellaneous Variables
- > Commutation Variables
- > System Configuration Variables
- > Communication Variables
- > Miscellaneous

The ACSPL+ programming variables are:

Table 3-1. Alphabetical Listing of All ACSPL+ Variables

Name	Description	Variable Group
ACC	Acceleration	Motion
AFLAGS	Axis Flags	Axis Configuration
AIN	Analog Inputs	Input and Output
AOUT	Analog Outputs	Input and Output
APOS	Axis Position	Motion
AST	Axis State	Axis State
ASTX	Extended Axis State	Axis State
BAUD	Serial Communication Baud Rate	Communication
BOFFTIME	Brake Deactivation Time	Axis Configuration - Brake
BONTIME	Brake Activation Time	Axis Configuration - Brake
CERRA	Critical Position Error in Accelerating	Axis Configuration - Safety Limits
CERRI	Critical Position Error in Idle	Axis Configuration - Safety Limits
CERRV	Critical Position Error in Moving	Axis Configuration - Safety Limits
CFG	Configuration Mode	System Configuration
COMMCH	Communication Channel	Communication
COMMFL	Communication Flags	Communication
CONID	Controller Identification	Communication
CTIME	Controller Cycle Time	System Configuration
DAPOS	Delayed Axis Position	Motion

Name	Description	Variable Group
DCN	Axis DC, Number of Samples	Data Collection
DCOM	Drive Command	Servo-Loop
DCP	Axis DC, Period	Data Collection
DEC	Deceleration	Motion
DELI	Delay on Transition to Idle State	Axis Configuration - Safety Limits
DELV	Delay on Transition to Velocity State	Axis Configuration - Safety Limits
DISPCH	Default Communication Channel	Communication
E_AOFFS	Sets user-defined offset for absolute encoder.	Axis Configuration - Feedback
E2_AOFFS	Sets user-defined offset for absolute encoder secondary feedback	Axis Configuration - Feedback
E_FREQ	Primary Encoder Frequency	Axis Configuration - Feedback
E_SCMUL	Primary Encoder Sin-Cos Multiplier	Axis Configuration - Feedback
E_TYPE	Primary Encoder Type	Axis Configuration - Feedback
E2FAC	Secondary Encoder Factor	Axis Configuration - Feedback
E2_FLAGS	Contains configuration bits for secondary feedback absolute encoder.	Axis Configuration - Feedback
E2_FREQ	Defines the maximum encoder pulse frequency (in MHz) for secondary feedback.	Axis Configuration - Feedback
E2OFFS	Secondary Encoder Offset	Axis Configuration - Feedback
E2_PAR_A	Sets the encoder data transmission frequency	Axis Configuration - Feedback

Name	Description	Variable Group
E2_PAR_B	Sets the encoder data control CRC code	Axis Configuration - Feedback
E2_PAR_C	Sets the interval (in microseconds) of encoder position reading	Axis Configuration - Feedback
E2_PAR_D	Defines number of status bits (LSB) in the real-time position data	Axis Configuration - Feedback
E2_PAR_E	Defines a mask for setting error bits	Axis Configuration - Feedback
E2_SCMUL	Specifies the Sin-Cos multiplication factor for the secondary feedback encoder	Axis Configuration - Feedback
ECERR	Contains EtherCAT Error Code	Safety Control
ECHO	Echo Communication Channel	Communication
ECST	Contains EtherCAT Status	Safety Control
EFAC	Primary Encoder Factor	Axis Configuration - Feedback
ENTIME	Enable Time	Axis Configuration
EOFFS	Primary Encoder Offset	Axis Configuration - Feedback
E_PAR_A		Feedback Variables
E_PAR_B		Feedback Variables
E_PAR_C		Feedback Variables
EPOS	Shows the encoder feedback	Feedback
ERRA	Position Error in Accelerating	Axis Configuration - Safety Limits
ERRI	Position Error in Idle	Axis Configuration - Safety Limits
ERRV	Position Error in Moving	Axis Configuration
EXTIN	Extended Inputs (HSSI)	Input and Output

Name	Description	Variable Group
EXTOUT	Extended Outputs (HSSI)	Input and Output
F2ACC	Defines the feedback acceleration value of the axis in secondary feedback.	Feedback
F2POS	Secondary Feedback Position	Motion
F2VEL	Secondary Feedback Velocity	Motion
FACC	Primary Feedback Acceleration	Motion
FAULT	Faults	Safety Control
FDEF	Default Response Mask	Safety Control
FK	Function Key	Miscellaneous
FMASK	Fault Mask	Safety Control
FPOS	Primary Feedback Position	Axis State
FVEL	Primary Feedback Velocity	Motion
FVFIL	Primary Feedback Velocity Filter	Axis Configuration - Feedback
G_01WCS...G_12WCS	Used for defining one of the 12 work-piece coordinate systems	System Configuration
GACC	Group Acceleration	Motion
GATEWAY	Gateway Address for 1 st Ethernet	Communication
GJERK	Group Jerk	Motion
GMOT	Motion Number	Motion
GMQU	Motion Queue	Motion
GMTYPE	Motion Type	Motion
GPATH	Group Path	Motion
GPEXL	Indicates the GSP program executed block	System Configuration

Name	Description	Variable Group
GPHASE	Motion Phase	Motion
GRTIME	Remaining Motion Time	Motion
GSEG	Motion Segment	Motion
GSFREE	Free Motion Segments	Motion
GUFAC	Holds the conversion factor from 'Common Physical Units' in [mm] to 'Controller Units'	System Configuration
GVEC	Group Vector	Motion
GVEL	Group Velocity	Motion
IENA	Interrupt Enable/Disable	System Configuration
IMASK	Interrupt Mask	System Configuration
IN	Digital Inputs	Input and Output
IND	Primary Index Position	Axis State
ISENA	Specific Interrupt Enable/Disable	System Configuration
IST	Index State	Axis State
JERK	Jerk	Motion
JITTER	Time difference between physical timer interrupt and start of SC real-time task	SPiiPlusSC
KDEC	Kill Deceleration	Motion
M2ARK	Secondary Mark Position	Axis State
MARK	Primary Mark Position	Axis State
MERR	Motor Error	Safety Control
MFF	Master Feed Forward	Axis Configuration
MFLAGS	Motor Flags	Axis Configuration

Name	Description	Variable Group
MFLAGSX	Extended Motor Flags	Axis Configuration
MPOS	Master Position	Motion
MSSYNC	Difference between master clock and bus clock	SPiiPlusSC
MST	Motor State	Axis State
NST	Reads the status of EtherCAT Sync and GPRT errors for each axis in the system.	Axis State
NVEL	Minimal Velocity	Motion
ONRATE	Autoroutine Rate	Program Execution Control
OUT	Digital Outputs	Input and Output
PCHARS	Program Size in Characters	Program Execution Control
PE	Non-Critical Position Error	Motion
PERL	Program Error Line	Program Execution Control
PERR	Program Error	Program Execution Control
PEXL	Program Executed Line	Program Execution Control
PFLAGS	Program Flags	Program Execution Control
PLINES	Program Size in Lines	Program Execution Control
PRATE	Program Rate	Program Execution Control
PST	Program State	Program Execution Control
RACC	Reference Acceleration	Motion

Name	Description	Variable Group
	RMS current of axis	Axis State
ROFFS	Reference Offset	Axis State
RPOS	Reference Position	Axis State
RVEL	Reference Velocity	Motion
RVFIL	Reference Velocity Filter	Axis Configuration - Feedback
S_DCN	System DC, Number of Samples	Data Collection
S_DCP	System DC, Period	Data Collection
S_ERR	System Error	Safety Control
S_FAULT	System Faults	Safety Control
S_FDEF	System Default Response Mask	Safety Control
S_FLAGS	System Flags	System Configuration
S_FMASK	System Fault Mask	Safety Control
S_SAFIN	System Safety Inputs	Safety Control
S_SAFINI	System Safety Inputs Inversion	Safety Control
S_SETUP	Bit mask defining various system settings	System Configuration
S_ST	State of System Data Collection	Data Collection
SAFIN	Safety Inputs	Safety Control
SAFINI	Safety Inputs Inversion	Safety Control
SC2COFFS	Defines the Sin-Cos Sine offset in secondary feedback	Axis Configuration - Feedback
SC2GAIN	Sin-Cos encoder gain compensation variable used to compensate the Cosine signal for an improper amplitude relative to the Sine signal in secondary feedback	Axis Configuration - Feedback

Name	Description	Variable Group
SC2PHASE	Sin-Cos encoder phase compensation variable and is used to compensate the Cosine signal for an improper phase difference relative to the Sine signal in secondary feedback.	Axis Configuration - Feedback
SC2SOFFS	Defines the Sin-Cos Sine offset in secondary feedback.	Axis Configuration - Feedback
SCCOFFS	Sin-Cos Offset (Cosine)	Axis Configuration - Feedback
SCGAIN	Cosine gain compensation	Axis Configuration - Feedback
SCPHASE	Cosine phase compensation	Axis Configuration - Feedback
SCSOFFS	Sin-Cos Offset (Sine)	Axis Configuration - Feedback
SETTLE	Settling Time	Axis Configuration
S2LABITS	Used for setting the total number of absolute position bits for an absolute encoder connected to a secondary feedback	Axis Configuration - Feedback
SLABITS	Absolute Position Bits	Axis Configuration - Feedback
SLAFF	Acceleration Feed Forward	Servo-Loop - Compensations
SLBIASA	Current Phase A Bias	Servo-Loop - Current
SLBIASB	Current Phase B Bias	Servo-Loop - Current
SLCFIELD	Induction Motor Excitation	Induction Motor
SLCHALL	Hall Shift	Commutation
SLCNP	Number of Motor Poles	Commutation
SLCOFFS	Commutation Offset	Commutation

Name	Description	Variable Group
SLCORG	Commutation Origin	Commutation
SLCPRD	Commutation Period	Commutation
SLCROUT	Commutation Feedback	Servo-Loop - Miscellaneous
SLCSLIP	Induction Motor Slip Factor	Induction Motor
SLDRAIF	DRA frequency	Servo-Loop - Position
SLDRAIF	Provides an Idle Factor to the SLDRA variable.	Servo-Loop - Position
SLDRX	Maximum DRA correction	Servo-Loop - Position
SLDZMAX	Maximum Dead Zone position	Servo-Loop - Nanomotion
SLDZMIN	Minimum Dead Zone position	Servo-Loop - Nanomotion
SLEBIASA	Defines encoder hardware Sine offset	Axis Configuration - Feedback
SLEBIASB	Defines encoder hardware Cosine offset	Axis Configuration - Feedback
SLEBIASC	Defines the Sin-Cos encoder's hardware compensation for the Sine offset in secondary feedback	Axis Configuration - Feedback
SLEBIASD	Defines the Sin-Cos encoder's hardware compensation for the Cosine offset in secondary feedback	Axis Configuration - Feedback
SLFRC	Static Friction	Servo-Loop - Compensations
SLFRCD	Dynamic Friction	Servo-Loop - Compensations
SLHROUT	Sets the Hall state routing	Commutation

Name	Description	Variable Group
SLIFILT	Internal Current filter	Servo-Loop - Current
SLIKI	Integrator Gain	Servo-Loop - Current
SLIKP	Integrator Proportional Gain	Servo-Loop - Current
SLIFILT	Internal Current filter	Servo-Loop - Current
SLILI	Determines output voltage	Servo-Loop - Current
SLIOFFS	Current Command Offset	Servo-Loop - Current
SLLIMIT	Soft Left Limit	Safety Limits
SLROUT	Sets the HW limits routing for the specified axis	Commutation
SLP2ROUT	Sets the feedback routing of the secondary feedback position for the specified axis	Servo Loop - Miscellaneous
SLPKITF	Increases position loop integrator coefficient	Servo-Loop - Position
SLPKP	Proportional Position Gain	Servo-Loop - Position
SLPKPIF	Provides an Idle Factor to the SLPKP variable	Servo-Loop - Position
SLPKPSF	Provides a Settling Factor to the SLPKP variable	Servo-Loop - Position
SLPKPTF	Increases position loop proportional coefficient	Servo-Loop - Position
SLPMAX	Modulo Axis Upper Limit	Axis Configuration
SLPMIN	Modulo Axis Lower Limit	Axis Configuration
SLPROUT	Position Feedback Routing	Servo-Loop - Miscellaneous

Name	Description	Variable Group
SLSTHALL	The hall state of each axis	Commutation
SLTFWID	Determines distance to target at which position and velocity loops gains increase by 50% Servo-Loop - Miscellaneous SLVBODD Bi-Quad filter	Servo-Loop - Miscellaneous
SLVBODD	Bi-Quad filter damping ratio denominator	Servo-Loop - Filter
SLVBODF	Bi-Quad filter algorithm denominator	Servo-Loop - Filter
SLVBOND	Bi-Quad filter damping ratio numerator	Servo-Loop - Filter
SLVBONF	Bi-Quad filter algorithm numerator	Servo-Loop - Filter
SLVKI	Velocity Integrator Coefficient	Servo-Loop - Velocity
SLVKIIF	Provides an Idle Factor to SLVKI variable	Servo-Loop - Velocity
SLVKISF	Provides a Settle Factor to the SLVKI variable	Servo-Loop - Velocity
SLVKITF	Increases velocity loop integrator coefficient	Servo-Loop - Velocity
SLVKP	Proportional Velocity Gain	Servo-Loop - Velocity
SLVKPIF	Provides an Idle Factor to the SLVKP variable	Servo-Loop - Velocity
SLVKPSF	Provides a Settle Factor to the SLVKP variable	Servo-Loop - Velocity
SLVKPTF	Increases the velocity loop proportional coefficient	Servo-Loop - Velocity
SLVLI	Integrator Velocity Limit	Servo-Loop - Velocity
SLVNATT	Notch Filter Attenuation	Servo-Loop - Filter
SLVNFRQ	Notch Filter Frequency	Servo-Loop - Filter
SLVNWID	Notch Filter Width	Servo-Loop - Filter

Name	Description	Variable Group
SLVRAT	Velocity Feed Forward Ratio	Servo-Loop - Velocity
SLVROUT	Velocity Feedback	Servo-Loop - Miscellaneous
SLVSOF	Low-Pass Filter Bandwidth	Servo-Loop - Filter
SLVSOFD	Low-Pass Filter Damping	Servo-Loop - Filter
SLZFF	Defines zero velocity feed forward position	Servo-Loop - Nanomotion
SRLIMIT	Soft Right Limit	Safety Limits
STEPF	Stepper Factor	Axis Configuration
STEPW	Stepper Pulse Width	Axis Configuration
STODELAY	Configures the delay time between the STO fault indication and the default response (disable) to the fault	Safety Control
SUBNET	Subnet Mask for 1 st Ethernet	Communication
SYNC	Slave Sync counter	Safety Control
TARGRAD	Target Radius	Axis Configuration
TCPIP	IP Address for 1 st Ethernet	Communication
TCPIP2	IP Address for 2 nd Ethernet	Communication
TCPPORT	TCP Port Number	Communication
TIME	Elapsed Time	Monitoring
TPOS	Target Position	Motion
UDPPORT	UDP Port Number	Communication
USAGE	MPU Usage	Monitoring
VEL	Velocity	Motion

Name	Description	Variable Group
VELBRK	Brake Velocity	Axis Configuration - Brake
XACC	Maximum Acceleration	Axis Configuration - Safety Limits
XARRSIZE	Maximum Array Size	Miscellaneous
XCURI	Maximum Current in Idle	Axis Configuration - Safety Limits
XCURCDB	Threshold of the current vector peak	Axis Configuration - Safety Limits
XCURV	Maximum Current in Moving	Axis Configuration - Safety Limits
XRMS	RMS Current Limit	Axis Configuration - Safety Limits
XR MST	RMS Current Time Constant	Axis Configuration - Safety Limits
XSACC	Maximum Slave Acceleration	Axis Configuration - Safety Limits
XVEL	Maximum Velocity	Axis Configuration - Safety Limits

3.1 Axis Configuration Variables

The Axis Configuration variables are:

Name	Description
AFLAGS	Axis Flags
ENTIME	Enable Time
ESTBITS	ESTBITS represents the single turn resolution (number of bits) of the absolute encoder.
E2STBITS	E2STBITS represents the single turn resolution (number of bits) of the absolute encoder for the secondary feedback

Name	Description
EMTBITS	EMTBITS represents the multi-turn resolution (number of bits) of the absolute encoder.
E2MTBITS	E2MTBITS represents the multi-turn resolution (number of bits) of the absolute encoder for the secondary feedback
MFF	Master Feed forward
MFLAGS	Motor Flags
MFLAGSX	Extended Motor Flags
MSTIMEA	Time elapsed from start of motion up to first entering the settled zone
MSTIMEB	Time elapsed from start of motion up to first entering the settled zone
MSTIMEC	Time elapsed from start of motion up to first entering the settled zone
PEGQUE	Count of PEG FIFO, one entry per axis
SETTLE	Axis Settling Time Parameter
SETTLE	Axis Settling Time Parameter
SLCFIELD	Defines magnetic field component of an induction motor
SLCSLIP	Defines the slip constant component of an induction motor
SLPMAX	Specifies the upper limit of modulo axis
SLPMIN	Specifies the lower limit of modulo axis
STEPF	Stepper Factor
STEPW	Stepper Pulse Width
TARGRAD	Axis settling target envelope parameter

3.1.1 AFLAGS

Description

AFLAGS is an integer array, with one element for each axis in the system, each element of which contains a set of 4 bits.

Syntax

AFLAG*axis_index*.*bit_designator* = 1|0

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>bit_designator</i>	The meanings of bit_designator are given in AFLAGS .meanings of bit_designator

Table 3-2. AFLAG Bit Description

Bit Name	No.	Description
#NOS	0	No S-Profile
#SEMIS	1	Semi-S-Profile
#AUX	2	Auxiliary Axis
#NOGROUP	3	Disable Group

Tag

3

Comments

Currently, only **AFLAGS**(axis_index).#**NOGROUP** can be set in the variable. **AFLAGS**(axis_index).#**NOGROUP** disables the axis in a group, so that any group or motion command that includes the axis in a group will fail.

Other bits are reserved for future use, and must be set to zero.

Accessibility

Read-Write



AFLAGS values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**.

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.1.2 *ENTIME*

Description

ENTIME is a real array, with one element for each axis in the system, and is used for controlling the execution of [ENABLE/ENABLE ALL](#).

Syntax

ENTIME(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 2.22507e-308 to 1.79769e+308, Default = 50.

Tag

37

Comments

Since the drive enable process is relatively long, (50 - 100msec) **ENTIME** defines a time duration in msec between [ENABLE/ENABLE ALL](#) and the moment the controller considers the drive as enabled and all faults as [FAULT\(axis_index\).#PE](#), [FAULT\(axis_index\).#CPE](#), [FAULT\(axis_index\).#DRIVE](#) are triggered.

Exact usage of the variable depends on the flag bit **MFLAGS**(axis_index).#ENMODE (bit 19). See [MFLAGS](#).

Accessibility

Read-Write



ENTIME values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Commands

[ENABLE/ENABLE ALL](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.1.3 *ESTBITS*

Description

ESTBITS is an integer array with one element for each axis. It represents the single turn resolution (number of bits) of the absolute encoder.

Syntax**ESTBITS**(axis_index)**Arguments**

index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	Value ranges from 0 to 64.

Tag

387

Comments

The **ESTBITS** variable can only be set by the **ENCINIT()** function.

Related ACSPL+ Variables**EMTBITS, SLABITS****Accessibility**

Read-only



SLABITS compatibility is ensured.

If **EMTBITS** and **ESTBITS** are 0, **SLABITS** is still a Read-Write variable.

If **ESTBITS** is not 0 or **EMTBITS** is not 0 (in other words, it was changed by a call to **ENCINIT()**), **SLABITS** cannot be assigned a new value, and the firmware will return an error if an assignment is attempted.

3.1.4 E2STBITS**Description**

E2STBITS is an integer array with one element for each axis. It represents the single turn resolution (number of bits) of the absolute encoder for the secondary feedback.

Syntax**E2STBITS**(axis_index)**Arguments**

index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	Value ranges from 0 to 64.

Tag

389

Comments

The **E2STBITS** variable can only be set by the **ENCINIT()** function.

Related ACSPL+ Variables

E2MTBITS, SLABITS

Accessibility

Read-only

3.1.5 EMTBITS

Description

EMTBITS is an integer array with one element for each axis. It represents the multi-turn resolution (number of bits) of the absolute encoder.

Syntax

EMTBITS(axis_index)

Arguments

index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	Value ranges from 0 to 64.

Tag

388

Comments

The **EMTBITS** variable can only be set by the **ENCINIT()** function.

Related ACSPL+ Variables

ESTBITS, SLABITS

Accessibility

Read-only



SLABITS compatibility is ensured.
If **EMTBITS** and **ESTBITS** are 0, **SLABITS** is still a Read-Write variable.
If **ESTBITS** is not 0 or **EMTBITS** is not 0 (in other words, it was changed by a call to **ENCINIT()**),**SLABITS** cannot be assigned a new value, and the firmware will return an error if an assignment is attempted.

3.1.6 E2MTBITS

Description

E2MTBITS is an integer array with one element for each axis. It represents the multi-turn resolution (number of bits) of the absolute encoder for the secondary feedback.

Syntax

E2MTBITS(axis_index)

Arguments

index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	Value ranges from 0 to 64.

Tag

390

Comments

The **E2MTBITS** variable can only be set by the **ENCINIT()** function.

Related ACSPL+ Variables**E2STBITS, SLABITS****Accessibility**

Read-only

3.1.7 MFF**Description**

MFF is a real array, with one element for each axis in the system, and is used for specifying the feed forward time, in milliseconds, for **MPOS** calculations.

Syntax**MFF(axis_index) = value****Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 0 to 10, Default = 2.

Tag

87

Comments

The controller calculates the **MPOS** value according to a formula supplied by the **MASTER** command. A non-zero **MFF** value provides additional extrapolation of the calculated value to the predicted value at the current time plus **MFF**. The purpose is to compensate delay introduced by the controller and the external circuits.

The default value of **MFF** depends on the controller model so that it compensates the delay introduced by the controller itself. Increase the **MFF** value if you want to compensate additional delay introduced by sensor or other circuits.

Accessibility

Read-Write



MFF values cannot be modified if protection is applied to this variable through **SPiiPlus**
MMI Application Studio → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Commands

[MASTER](#)

Related ACSPL+ Variables

[ENTIME](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.1.8 MFLAGS

Description

MFLAGS is an integer array, with one element for each axis in the system, each element of which contains a set of bits used for configuring the motor.

Syntax

MFLAGS(*axis_index*).*bit_designator* = 0/1

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>bit_designator</i>	The MFLAGS bit designators are given in MFLAGS Bit Designators .

Table 3-3. MFLAGS Bit Designators

Name	No.	Description
#DUMMY	0	0 (default) = The axis is defined as non-dummy. 1 = The axis is defined as dummy. A dummy axis is an inactive axis which is not connected to a drive and a motor. When an axis is defined as a dummy the following apply: > In SPiiPlus controllers, excluding Control Modules, the two analog outputs of the axis can be used as General Purpose outputs. > In all SPiiPlus products: all faults of the axis are disabled. > In all SPiiPlus products ENABLE/ENABLE ALL is disabled.
#OPEN	1	0 (default): Closed-loop control - for servo motors only. 1: Open-loop control. In open loop control the user defines the value of the command/s to the drive with variable DCOM.
#MICRO	2	0 (default): If MFLAGS.#PHASE2 = 1 – motor operated in full step mode. If MFLAGS.#PHASE2 = 0 – bit is ignored 1: If MFLAGS.#PHASE2 = 1 – motor operated in micro step mode. If MFLAGS.#PHASE2 = 0 – #MICRO bit should be cleared
#HOME	3	0 (default): Axis homing procedure not done. 1: Axis homing procedure done. This bit is changed by the ACSPL+ HOME command after a successful homing process. E_TYPE and other encoder initialization processes will reset the bit. The controller clears the bit during power-up.

Name	No.	Description
#STEPPER	4	#STEPPER is applicable only for PDMnt products. 0 (default): Axis is defined as a servo motor. 1: Axis is defined as a pulse-direction stepper motor in open loop.
#ENCLOOP	5	#ENCLOOP is not applicable for NT products. 0 (default): Stepper feedback loop is not active. 1: The axis provides a stepper feedback loop. In this case the FPOS(axis_index) variable will indicate the number of pulses that were sent to the pulse-direction stepper drive and not the encoder counts - if an encoder is connected and #STEPENC (bit 6 = 1). #ENCLOOP is effective only if #STEPPER (bit 4) = 1.
#STEPENC	6	0 (default): Stepper encoder feedback loop is not active. 1: The controller provides an encoder feedback to the pulse-direction stepper. In this case, the FPOS (axis_index) variable will indicate the quadrature encoder counts, and not the stepper pulse-direction pulses if #ENCLOOP (bit 5 = 1). #STEPENC is effective only if #STEPPER (bit 4) = 1. The encoder feedback is used for monitoring the axis position by a user application, and does not affect the open loop control of the stepper motor.
#NANO	7	#NANO is applicable only for UDMnt-x (new revision) and UDIhp-x products. 0 (default): Defines the axis as a servo or stepper motor. 1: Defines the axis as a Nanomotion piezo ceramic motor.
#BRUSHL	8	0 (default): The motor is a non-DC brushless type or the amplifier provides commutation itself and uses one $\pm 10V$ input from the controller. 1: The controller provides commutation for the DC brushless motor. The bit must be set only if the controller is connected to a three-phase amplifier with two-phase input commands.

Name	No.	Description
#BRUSHOK	9	#BRUSHOK is applicable only for DC Brushless motors when #BRUSHL(bit 8)=1. 0 (default): DC brushless motor is not commutated. 1: DC brushless motor is commutated. After power-up the controller clears the bit. The controller automatically sets this bit =1 only after a successful commutation or auto-commutation processes.
#PHASE2	10	0 (default): Two phase motor is not selected. 1: Two phase motor is selected.
#DBRAKE	11	0 (default): Dynamic brake is disabled 1: The controller will apply dynamic braking to stop the motor when the axis is disabled and the feedback velocity is less than the predefined value of VELBRK. See the <i>SPiiPlus Setup Guide</i>, Dynamic Brake for a complete explanation.
#INVENC	12	0 (default): Encoder counting is non-inverted. 1: The controller inverts the direction of encoder counting. This does not affect the motion direction.
#INVDOUT	13	0 (default): Drive output command/s are not inverted. 1: The controller inverts the drive output command/s. This effectively inverts the direction of the motion and the sign of the feedback.
#NOTCH	14	0 (default): Notch filter is disabled. 1: Notch filter is active
#NOFILT	15	0 (default): The control algorithm includes a second-order filter specified by bandwidth SLVSOF. 1: The control algorithm by-passes the second-order filter.

Name	No.	Description
#BI_QUAD	16	0 (default): First BiQuad Filter is disabled. 1: First BiQuad Filter is active.
#DEFCON	17	0: CONNECT is allowed. See CONNECT. 1 (default): The controller applies the default connection between APOS and RPOS (RPOS). While #DEFCON = 1, the controller does not accept CONNECT for the axis. The bit is reset to 1 every time the controller is restarted.
#FASTSC	18	0 (default): not available 1: For working with 5mHz SIN-COS encoders  Bit must be set to 1 in the middle of quadrant.
#ENMOD	19	0: Enable time is defined by ENTIME, or when the drive switches its drive alarm signal to the inactive state - whichever comes first. If the signal remains active more than ENTIME milliseconds, ENABLE/ENABLE ALL fails. 1 (default): The value of ENTIME defines the enable time.
#DUALLOOP	20	0 (default): The control algorithm implements a regular single-loop scheme. 1: The control algorithm implements a dual-loop scheme. See <i>SPIiPlus Setup Guide, Appendix B</i> for further details.
#LINEAR	21	0 (default): The axis is defined as rotary. 1: The axis is defined as linear.

Name	No.	Description
#ABSCOMM	22	0 (default): Controller doesn't automatically retrieve the commutation phase according to the absolute encoder after powerup. 1: The axis is using absolute encoder feedback for commutation and the controller automatically retrieves the commutation phase after powerup.
#BRAKE	23	0 (default): The controller does not provide mechanical brake control. 1: The brake is deactivated when the motor is enabled and activated when the motor is disabled. For detailed description, see commands ENABLE/ENABLE ALL and DISABLE/DISABLEALL, and variables BOFFTIME, BONTIME variables.
#HSSI	24	#HSSI is not currently supported by NT products. 0 (default): Direct drive connection. 1: Remote drive connection via HSSI.
#GANTRY	25	0 (defaults): standard (SISO) PIV control scheme. 1: gantry control (MIMO) scheme.
#BI_QUAD1	26	(default): Second BiQuad Filter is disabled. 1: Second BiQuad Filter is active.
#HALL	27	 #HALL applies only to DC brushless motors when #BRUSHL (bit 8) =1. 0 (default): Commutation is not based on Hall signals. 1: Commutation is based on Hall signals.

Name	No.	Description
#INVHALL	28	 #INVHALL applies only for DC brushless motors when #BRUSHL (bit 8) =1. 0 (default): Motor Hall signals counting direction is not inverted. 1: Motor Hall signals counting direction is inverted.
#MODULO	29	0 (default): Axis is defined as non-modulo. 1: Axis is defined as MODULO. In MODULO mode, physically, the motion of the axis is not limited, but each time RPOS goes out of the defined SLPMIN...SLPMAX range, the controller brings RPOS into range by changing the internal offset EOFFS. See SLPMAX and SLPMIN.
#USER1	30	The functionality of this bit can be defined by the user. 0 (default): Functionality not defined 1: User defined functionality
#USER2	31	The functionality of this bit can be defined by the user. 0 (default): Functionality not defined 1: User defined functionality

Tag

88

Comments

MFLAGS is typically configured using the **SPiiPlus MMI Application Studio → Toolbox → Setup → Adjuster** when setting the Dynamic Brake.

Use direct bit assignment for on-the-fly changes, for example, from closed-loop operation to the open-loop and vice versa.

Accessibility

Read-Write



MFLAGS values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Commands

[CONNECT](#), [ENABLE/ENABLE ALL](#), [GETCONF](#)

Related ACSPL+ Variables

[SLPROUT](#), [APOS](#), [RPOS](#)

COM Library Methods and .NET Library Methods

[ReadVariable](#), [WriteVariable](#)

C Library Functions

[acsc_ReadInteger](#), [acsc_WriteInteger](#)

3.1.9 MFLAGSX

Description

MFLAGSX is an integer array with one element for each axis in the system, each element of which contains a set of bits used for configuring the motor. It is an extension of the **MFLAGS** variable.

Syntax

```
MFLAGSX(Axis_Index).bit_designator = 0|1
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Bit_designator	An MFLAGSX bit designator as described below

MFLAGSX Bit Designators

Bit Name	No.	Description
#STCLFULL	0	0 (default): Do not use position correction for Stepper motors working under closed loop. 1: Activate Full mode of the stepper closed loop position correction mechanism . This bit is mutually exclusive to #STCLEND and #STCLSP, as such only one of them may be 1 at a time.
#STCLEND	1	0 (default): Do not use position correction for Stepper motors working under closed loop 1: Activate End mode of the stepper closed loop position correction mechanism

Bit Name	No.	Description
		This bit is mutually exclusive to #STCLFULL and #STCLSP , as such only one of them may be 1 at a time.
#STCLSP	2	0 (default): Do not use servo processor closed-loop stepper algorithm 1: Use servo processor closed-loop stepper algorithm Mutually exclusive to #STCLFULL , #STCLEND .
#VOLTMODE	3	0: Use current mode for control 1: Use voltage mode instead of current mode to compensate for low resolution
#HLIMSWAP	4	0: Left/Right limit signal swapping is disabled 1: Left/Right limit signal swapping is enabled
#SATPROT	5	1: Saturation Protection enabled (default) 0: Saturation Protection disabled

Tag

357

Comments

When saturation protection (**MFLAGSX.#SATPROT=1**) is currently available for the following products: NPMpm, UDMxx, IDMxx.

This variable is supported in ADK versions 2.70 and higher.

Related ACSPL+ Variables

SLSKI, SLSKP, SLSDZ

Accessibility

Read-Write



MFLAGSX values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio→Toolbox→Application Development→Protection

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadInteger(), acsc_WriteInteger()

3.1.10 MODULOMD

Description

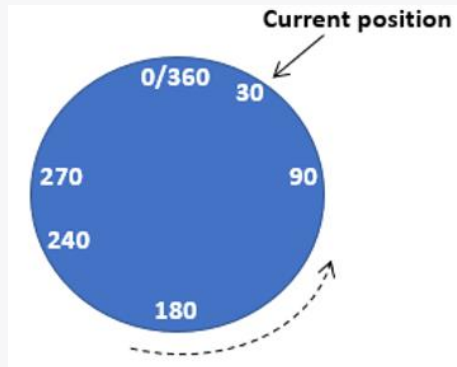
MODULOMD is an integer array, with one element for each axis in the system, the elements of which are used for storing the mode of a modulo axis.

Syntax

```
MODULOMD(axis_index) = mode
```

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
mode	Valid values are: > 0(default) – The axis feedback changes between the specified minimal (SLPMIN) and maximal(SLPMAX) positions. The reference position RPOS of the modulo axis changes in the range from SLPMIN to SLPMAX inclusively. Physically, the motion of the modulo axis is not limited, but each time when the RPOS comes out from range SLPMIN...SLPMAX, the controller brings RPOS into the range by changing the internal offset EOFFS. In the case of a default connection, the modulo operation also affects the APOS value (APOS=RPOS). > 1 (No-Pass-Through-Rollover) - Every Motion from will avoid moving through the Rollover point (where SLPMIN and SLPMAX are topologically united). This also applies to Motion Commands that exceed the Modulo range as set by SLPMAX(\$) and SLPMIN(\$). Example: Assume SLPMIN(axis_index)=0 and SLPMAX(axis_index)=360, FPOS (axis_index) is reported as 30, and a Relative point-to-point command is issued: <pre>PTP/r axis_index, 570</pre> Internal calculations will determine that the target position is (30 + 570) % 360 = 240, motion will be in positive direction passing through 180 point to 240.



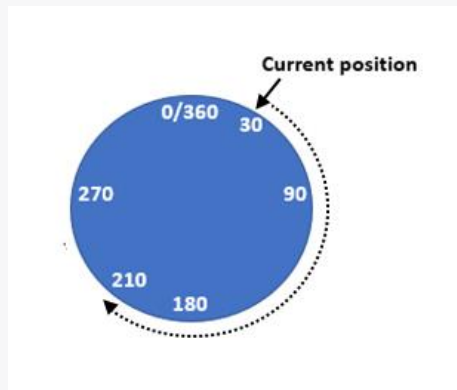
- > **2 (Positive motion only)** - Every Motion Command (excluding Jog) will cause the Axis to move in the positive Direction to the modulo of the Target Position. This also applies to motion commands that exceed the range set by **SLPMAX**(axis_index) and **SLPMIN**(axis_index).

Example 1:

Assume **SLPMIN**(axis_index)=0 and **SLPMAX**(axis_index)=360, **FPOS**(axis_index) is reported as 30, and a relative point to point command is issued:

```
PTP/r axis_index, 540
```

Internal calculations will determine that the target position is $(30 + 540) \% 360 = 210$, motion will be in positive direction passing through 180 to 210.



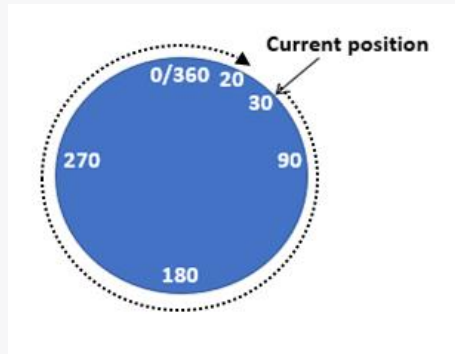
Example 2:

Assume **SLPMIN**(axis_index)=0 and **SLPMAX**(axis_index)=360, **FPOS**(axis_index) is reported as 30, and a point-to-point command is issued:

```
PTP axis_index, 20
```

Internal calculations determine that the target position is $20 \% 360 = 20$, the motion will be in positive direction passing through 180 and

360 point to 20 point.



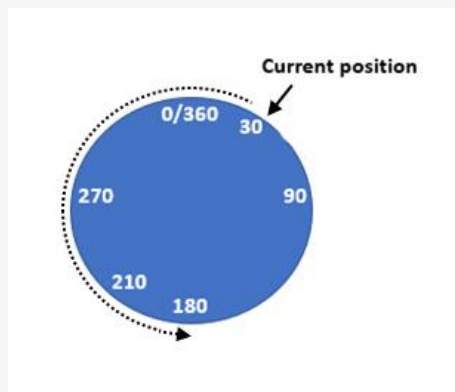
- > **3 (Negative motion only)** - Every Motion Command (excluding Jog) will cause the Axis to move in the negative direction to the modulo of the target position. This also applies to motion commands that exceed the Modulo range as set by **SLPMAX**(axis_index) and **SLPMIN**(axis_index).

Example:

Assume **SLPMIN**(axis_index)=0 and **SLPMAX**(axis_index)=360, **FPOS** (axis_index) is reported as 30, and a Relative point-to-point command is issued:

```
PTP/r axis_index, 540
```

Internal calculations will determine that the target position is $(30 + 540) \% 360 = 210$, motion will be in negative direction passing through 0 and 270 point to 210.



- > **4 (Shortest-Path)** - Each motion command will be analyzed to determine the shortest rotational distance between current position and target position, possibly moving through the Roll-Over point. This also applies to motion Commands that exceed the modulo range as set by **SLPMAX**(axis_index) and **SLPMIN**(axis_index).

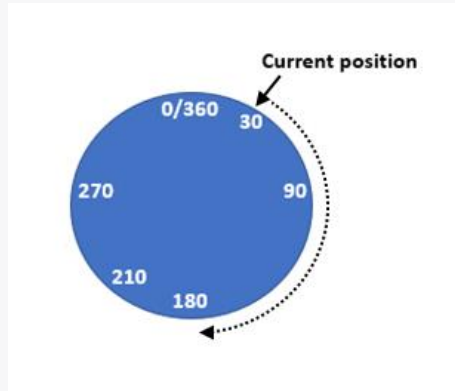
Example 1:

Assume **SLPMIN**(axis_index)=0 and **SLPMAX**(axis_index)=360, **FPOS** (axis_index) is reported as 30, and a relative point-to-point command

is issued:

```
PTP/r axis_index, 510
```

Internal calculations determine that the target position is $(30 + 510) \% 360 = 180$, motion will be in positive direction passing through 120 point to 180.

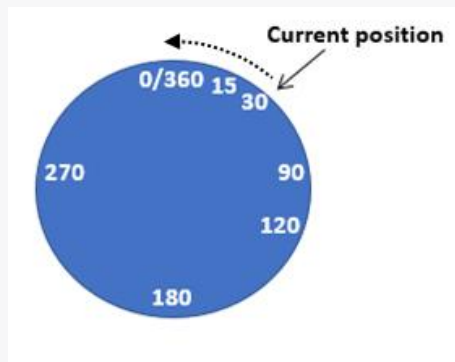


Example 2:

Assume $SLPMIN(axis_index)=0$ and $SLPMAX(axis_index)=360$, $FPOS(axis_index)$ is reported as 30, and a point-to-point command is issued:

```
PTP axis_index, 0
```

Internal calculations determine that the target position is $0 \% 360 = 0$, motion will be in negative direction passing through 15 point to 0.



Tag

417

Comments

The **MFLAGS(axis).#MODULO** bit needs to be set(1) for the axis to operate as a modulo axis.

All motion commands are calculated such that the resulting motion spans less than the Modulo range ($SLPMAX(axis) - SLPMIN(axis)$).

Axis modulo mode will not change the **EPOS** value as it does for **RPOS** and **FPOS**.

If an axis modulo mode has been turned off (**MFLAGS(axis).#MODULO=0**), then, **EPOS** is updated with the current **FPOS** value (by definition, **EPOS** = **FPOS** in a non-modulo mode).

Related ACSPL+ Variables

SLPMIN, SLPMAX, MFLAGS.#MODULO

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadInteger(), acsc_WriteInteger()

3.1.11 PEGQUE

Description

PEGQUE is an integer array with one element for each PEG engine in the system, the elements of which store the current state of PEG FIFO for that engine (the items count in the FIFO) . The parameter is updated as long as its value is different from 0.

Syntax

PEGQUE

Arguments

Tag

371

Comments

This variable is supported in version 3.00 and higher.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger

3.1.12 SETTLE

Description

SETTLE is a real array, with one element for each axis in the system, and is used for setting the Settling Time, it controls **MST(axis_index).#INPOS**.

Syntax

SETTLE(axis_index) = value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 0 to 1.79769e+308, Default = 0.

Tag

123

Comments

When the motor is not moving, the controller compares the position error (**PE** value) and the target envelope (**TARGRAD** value) every MPU cycle. **#INPOS** is raised when **PE** drops to within the range (-TARGRAD, +TARGRAD) and remains within the range for a period of time equal or greater than **SETTLE**.

If the motor starts to move or **PE** goes out of the range, **MST**(axis_index).**#INPOS** is cleared.

Accessibility

Read-Write



SETTLE values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

TARGRAD, **MST**, **PE**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.1.13 SLPMAX**Description**

SLPMAX is a real array, with one element for each axis in the system, the elements of which are used for storing the maximum range of a modulo axis.

Syntax

SLPMAX(axis_index) = value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from -1.79769e+308 to 1.79769e+308, Default = 8000.

Tag

194

Comments

SLPMAX stores the maximum range of a modulo axis, see [MFLAGS](#). **#MODULO** (bit 29). **SLPMAX** can be changed only when the motor is disabled.

Accessibility

Read-Write



SLPMAX values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

[SLPMIN](#), [EOFFS](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.1.14 *SLPMIN*

Description

SLPMIN is a real array, with one element for each axis in the system, the elements of which are used for storing the minimum range of a modulo axis.

Syntax

SLPMIN(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from -1.79769e+308 to 1.79769e+308, Default = 0.

Tag

195

Comments

SLPMIN stores the minimum range of a modulo axis, see [MFLAGS](#). **#MODULO** (bit 29). **SLPMIN** can be changed only when the motor is disabled.

Accessibility

Read-Write



SLPMIN values cannot be modified if protection is applied to this variable through
SPIiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

SLPMAX, EOFFS

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.1.15 STEPF

Description

STEPF is a real array, with one element for each axis in the system, and is used for defining the ratio between one stepper pulse and user units. See the *SPIiPlus Setup Guide*, Stepper Drive section for more information.

Syntax

STEPF(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 1e-015 to 1e+015, Default = 1.

Tag

129

Comments

STEPF = 1 (default) means that motion is performed in motor steps. For example,

PTP/R 0,320

will move the motor by 320 steps from the current position.

If another unit is required for motion programming, the user must configure an appropriate value for **STEPF**. For example, a controlled plant provides a gear ratio of 500 motor pulses per inch. If the motion programming unit must be provided in inches, the configured **STEPF** value must be 0.002 (1/500).

Accessibility

Read-Write



STEPF values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.1.16 STEPW

Description

STEPW is a real array, with one element for each axis in the system, and is used for defining the pulse width, in milliseconds, for the stepper motor. See the *SPiiPlus Setup Guide*, Stepper Drive section for more information.

Syntax

STEPW(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 12.0×10^{-6} (120 nanoseconds) to 0.050 (50 microseconds), Default = 0.001.

Tag

130

Comments

The value defines the width of the pulses generated on the pulse output for stepper control.

Accessibility

Read-Write



STEPW values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.1.17 TARGRAD

Description

TARGRAD is a real array, with one element for each axis in the system, and is used for defining the parameters for **MST**(axis_index).**#INPOS**.

Syntax

TARGRAD(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 2.22507e-308, 1.79769e+308, Default = 1.

Tag

132

Comments

When the motor is enabled but in a standstill position, the controller compares **PE** to the target envelope (**TARGRAD**) each MPU cycle. **#INPOS** = 1 when **PE** moves into the defined range (-TARGRAD, +TARGRAD) and remains within that range for a period of time equal or greater than defined by **SETTLE**.

If the motor starts to move or goes out of range, **#INPOS** is cleared.

Accessibility

Read-Write



TARGRAD values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Variables

SETTLE, **MST**(axis_index).**#INPOS**, **PE**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.2 Brake Variables

The Brake variables are:

Name	Description
BOFFTIME	Brake Deactivation Time
BONTIME	Brake Activation Time
MBRKROUT	Set the supplier for the mechanical break output signal of an axis as a specified digital output bit
VELBRK	Braking Velocity

3.2.1 [BOFFTIME](#)

Description

BOFFTIME is a real array, with one element for each axis in the system, and is used for specifying the brake release time in milliseconds.

Syntax

BOFFTIME(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from -1.79769e+308 to 1.79769e+308, Default = 50.

Tag

9

Accessibility

Read-Write



BOFFTIME values cannot be modified if protection is applied to this variable through **SPIiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Comments

See the *ACSPL+ Programmer's Guide* for information about using a mechanical brake on system startup.

Related ACSPL+ Commands

[ENABLE/ENABLE ALL](#) - The brake is deactivated automatically when the **ENABLE** command is executed.

Related ACSPL+ Variables

[MFLAGS](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.2.2 BONTIME

Description

BONTIME is a real array, with one element for each axis in the system, and is used for specifying the brake engagement time in milliseconds.

Syntax

BONTIME(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from -1.79769e+308 to 1.79769e+308, Default = 50.

Tag

10

Accessibility

Read-Write



BONTIME values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Comments

See the *ACSPL+ Programmer's Guide* for information about using a mechanical brake on system startup.

Related ACSPL+ Commands

[DISABLE/DISABLEALL](#), [FCLEAR](#), [DISABLEALL](#)



The brake is activated automatically when the **DISABLE** command is executed.

Related ACSPL+ Variables

[MERR](#), [FPOS](#), [RPOS](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.2.3 MBRKROUT

Description

MBRKROUT is an integer array, with one element for each axis in the system, and is used for setting the supplier for the mechanical brake output signal of an axis as a specified digital output bit(ACSPL+ **OUT**).

Syntax

MBRKROUT(Axis_Index) = value

Arguments

Axis	Designates the specific axis. Valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value is a 4-digit number (decimal): <NN00> where: > NN - digital output index (00-99) > 00 – specified input bit (00-31) The default value is -1, in which case the mechanical brake default assignment will be used. If Value is set to -2, the assignment for the axis is canceled and the output reverts to standard digital output behavior

Tag

381

Comments

This variable can be saved to flash memory. In other words, the mapping of a mechanical brake output using this variable will automatically occur on controller boot without the need to reconfigure it .

The following errors are supported:

- > Error 3330: "Invalid value, digital output index should range between 0-99 and bit index should range between 0-31"
- > Error 3332: "This output is already mapped as a mechanical brake to a different axis"
- > The ACSPL+ variable bit **MFLAGS**(Axis_Index). **#BRAKE** should be set to 1 in order to enable the mechanical brake for the specified axis.

This variable is supported in V3.03 and higher.

Examples

```
MBRKROUT(0) = 1000 ! output index 10 and bit 0
                  ! supplies the mechanical brake signal for axis 0.
```

```
MBRKROUT(0) = -2 ! Cancels assignment for axis 0. In other words,
                  ! the output reverts to general purpose digital output
```

Related ACSPL+ Variables

MFLAGS

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.2.4 VELBRK

Description

VELBRK is a real array, with one element for each axis in the system, and is used for defining dynamic braking.

Syntax

VELBRK(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 0 to 1.79769e+308, Default = 0.

Tag

140

Comments

If **MFLAGS(axis_index).#DBRAKE** is = 1, the controller will apply dynamic braking to stop the motor when the axis is disabled and the feedback velocity is less than the predefined value of **VELBRK**.

Accessibility

Read-Write



VELBRK values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Variables

MFLAGS(axis_index).#DBRAKE

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3 Feedback Variables

The Feedback variables are:

Name	Description
E_AOFFS	Sets user-defined offset for absolute encoder
E_FLAGS	Configuration bits for absolute encoder.
E2_AOFFS	Sets user-defined offset for absolute encoder secondary feedback
E2_FLAGS	Defines the Encoder Direction Inverse
E_FREQ	Encoder frequency
E2_FREQ	Defines the maximum encoder pulse frequency (in MHz) for secondary feedback.
E_PAR_A	Data transmission actual frequency.
E2_PAR_A	Sets the encoder data transmission frequency in MHz
E_PAR_B	Data control CRC code.
E2_PAR_B	Sets the encoder data control CRC code
E_PAR_C	The interval (in microseconds) of encoder position reading.
E2_PAR_C	Sets the interval (in microseconds) of encoder position reading.
E_PAR_D	Number of status bits (LSB) in the real-time position data (SLABITS).
E2_PAR_D	Defines the number of status bits (LSB) in the real-time position data
E_PAR_E	MASK for setting error bits.
E2_PAR_E	Defines a mask for setting error bits.
E_SCMUL	Encoder Sin-Cos Multiplier
E2_SCMUL	Specifies the Sin-Cos multiplication factor for the encoder

Name	Description
E_TYPE	Encoder Type
E2_TYPE	Defines the encoder type for the secondary feedback
EFAC	Encoder Factor
E2FAC	Secondary Encoder Factor
EOFFS	Axis Encoder Offset
E2OFFS	Axis Encoder Offset for Secondary Encoder
EPOS	Shows the encoder feedback
HOMEDF	Integer array defining homing method for each axis
HOMEVELI	Array of doubles defining the default homing velocity used by each axes for index search during ACSPL+ HOME command
HOMEVELL	Array of doubles defining the default homing velocity used for limit search during ACSPL+ HOME command
FVFIL	Primary Feedback Velocity Filter
F2ACC	Defines the feedback acceleration value of the axis
RVFIL	Reference Velocity Filter
SCSOFFS	Defines the sine offset.
SCCOFFS	Defines the cosine offset.
SC2COFFS	Defines the Sin-Cos Cosine offset
SC2GAIN	A Sin-Cos encoder gain compensation variable used to compensate the Cosine signal for an improper amplitude relative to the Sine signal

Name	Description
SC2PHASE	A Sin-Cos encoder phase compensation variable and is used to compensate the Cosine signal for an improper phase difference relative to the Sine signal
SC2SOFFS	Defines Sin-Cos Sine offset
SLEBIASA	Defines firmware sine offset.
SLEBIASB	Defines hardware cosine offset.
SLEBIASC	Defines the Sin-Cos encoder's hardware compensation for the Sine offset
SLEBIASD	Defines the Sin-Cos encoder's hardware compensation for the Cosine offset
SLABITS	Total number of absolute position bits for an absolute encoder
S2LABITS	Sets the total number of absolute position bits for an absolute encoder connected to a secondary feedback
SCGAIN	Feedback gain compensation variable
SCPHASE	Feedback phase compensation variable

3.3.1 E_AOFFS

Description

E_AOFFS is an integer array, with one element for each axis in the system, and is used for setting user-defined offset for absolute encoder.

Comments

Modifying E_AOFFS causes absolute encoder initialization.

The EOFFS variable is being modified according to E_AOFFS' value:

EOFFS=EOFFS-E_AOFFS.

Tag

300

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.2 E_FREQ

Description

E_FREQ is an integer array, with one element for each axis in the system, and is used for defining the maximum encoder pulse frequency (in MHz).

Syntax

E_FREQ(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value of each member can be one of: > 2 (shown as 2MHz but practically is 2.5MHz, default for an analog Sin-Cos encoder) > 20 (default for a digital encoder) > 60

Tag

27

Comments

The encoder is represented in the controller as a synchronous state machine that is activated by a clock, with programmable frequency in the SPiiPlus processor.

E_FREQ provides three optional clock rates that define the maximum encoder pulse frequency (in MHz) measured after an internal 4x multiplication.

In general, using a higher **E_FREQ** enables to read a higher rate of encoder input. However, the electrical noise immunity is reduced and Encoder Error **FAULT** might occur. Per case, it is recommended to use the lowest possible **E_FREQ** that does not generate an Encoder Error **FAULT**. In case of an Encoder Error, do **FCLEAR** (axis_index) and try a higher **E_FREQ** value.

For more information, see the "Encoder Input Clock" section in the *SPiiPlus Setup Guide*.

Accessibility

Read-Write



E_FREQ values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.3 E2_AOFFS

Description

E2_AOFFS is an integer array, with one element for each axis in the system, and is used for setting user-defined offset for absolute encoders for the secondary feedback.

Comments

Modifying E2_AOFFS causes absolute encoder initialization.

The E2OFFS variable is modified according to E2_AOFFS' value:

E2OFFS=E2OFFS-E2_AOFFS.

Tag

377

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.4 E2_FREQ

Description

E2_FREQ is an integer array, with one element for each axis in the system, and is used for defining the maximum encoder pulse frequency (in MHz) for secondary feedback.

Syntax

E2_FREQ(axis_index)=value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
<i>value</i>	Value of each member can be one of the following: > 2 (shown as 2MHz but practically is 2.5 MHz, default for an analog Sin-Cos encoder) > 20 (default for a digital encoder) > 60

Tag

303

Comments

The encoder is represented in the controller as a synchronous machine that is activated by a clock, with programmable frequency in the SPiiPlus processor.

E2_FREQ provides three optional clock rates that define the maximum encoder pulse frequency (in MHz) measured after an internal 4x multiplication.

In general, using a higher E2_FREQ enables to read a higher rate of encoder input. However, the electrical noise immunity is reduced and Encoder Error FAULT might occur. Per case, it is recommended to use the lowest possible E2_FREQ that does not generate an Encoder Error FAULT. In case of an Encoder Error, FCLEAR(axis_index) command should be executed and a higher E2_FREQ value should be set.

For more information, see the "Encoder Input Clock" section in the *SPiiPlus Setup Guide*.

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.5 E_FLAGS**Description**

E_FLAGS is an integer array, with one element for each axis in the system. Each element contains different configuration bits for absolute encoder.

Syntax

E_FLAGS(axis_index) = value Arguments

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
bit_designator	The meanings of bit_designator are given in Table 3-4 .

Table 3-4. E_FLAGS Bit Description

Bit Name	No.	Description
#ERRLOGIC	0	Encoder Error Logics
#UNSIGND	1	Unsigned Mode
#INVERSE	2	Encoder Direction Inverse
#HALLCORR	3	0: continuous commutation correction is disabled 1: continuous commutation is enabled

Bit Name	No.	Description
#GRAYCODE	4	0: Gray Code mode is disabled 1: Gray Code mode is enabled

Comments

#ERRLOGIC defines the status bits logic (0 or 1). The default value is 0. The Encoder 1 Error (#ENC) bit of the ACSPL+ **FAULTS** variable is triggered if the error is latched (based on the error bit logics).

If E_PAR_E is not set (0), **#ERRLOGIC**'s value has no meaning.

#UNSIGNED defines the Absolute Encoder Position mode. It can be unsigned (1) or signed (0). The default mode is signed for backward compatibility.

#INVERSE

When the bit is ON, the DSP is being notified and sends the position inverted. FW inverts the EOFFS variable as well.

The bits applies for all types of encoders.

The bit change event causes the FW to reinitialize the absolute encoder.

The bit cannot be changed if the axis is enabled.

Changing this bit may require changing the drive polarity (MFLAGS bit 13) or repeat commutation.

In case of a brushless motor, after the bit is changed the commutation will no longer be correct and the user should repeat the Adjuster commutation.

In case of a brush motor, the user should re-verify the drive polarity in the Adjuster by running Open Loop Verification.



Failure to repeat the Adjuster Commutation and Open Loop Verification may result critical position error, over current faults and in certain cases motor run away.

Tag

268

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteg

3.3.6 E2_FLAGS

Description

E2_FLAGS is an integer array, with one element for each axis in the system. Used for the secondary feedback. Each element contains different configuration bits for absolute encoder, in current version

it includes 3 bits.

E2_FLAGS Bit Description

Bit Name	Number	Description
#ERLOGIC	0	Encoder Error Logics
#UNSIGND	1	Unsigned Mode
#INVERSE	2	Encoder Direction Inverse

#ERRLOGIC defines the logics of the status bits – can be 0 or 1. If E2_PAR_E is not set (value equals to 0), ERRLOGIC value has no meaning. The default value is 0. Encoder 1 Error (#ENC) bit of the ACSPL+ FAULTS variable is triggered if error is latched (based on the error bit logics).

#UNSIGNED defines the Absolute Encoder Position mode – can be unsigned (1) or signed (0). The default mode is signed for backwards compatibility.

#INVERSE defines if the position is inverted on the DSP level. If the bit is set, the DSP sends the position inverted. FW inverts EOFFS variable.

Arguments

axis	The specific axis index. Valid numbers are: 0,1... up to the number of axes in the system, minus 1.
bit_designator	The meanings of bit_designator are defined in the table above.

Tag

309

Comments

The bits applies for all types of encoders.

The bit change event causes the FW to reinitialize the absolute encoder.

Accessibility

Read-Write

Com Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.7 E_PAR_A

Description

E_PAR_A is used for setting the encoder data transmission actual frequency in MHz. It is a double array, with one element for each axis in the system.

Syntax

E_PAR_A(axis) = value

Arguments

<i>axis</i>	The specific axis index. Valid numbers are: 0, 1... up to the number of axes in the system, minus 1.
<i>value</i>	The encoder data transmission actual frequency in MHz ranging from 1.25 to 10.



For the IDMsM/ECMsM/UDMsM products using the EnDAT encoder, only the following values are allowed for the **value** parameter: 0.1, 0.2, 1, 2, 4, 8, 16.



This variable is supported only for BiSS encoders (in all products) and EnDAT encoders (IDMsM/ECMsM/UDMsM products only).

Tag

248

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.8 E2_PAR_A

Description

E2_PAR_A is used for setting the encoder data transmission actual frequency in MHz. it is a double array, with one element for each axis in the system. Used for Secondary Feedback.

Syntax

E2_PAR_A (axis_index)=value

Arguments

<i>axis</i>	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
<i>value</i>	The encoder data transmission actual frequency in MHz ranging from 1.25 to 10.



For the IDMsM/ECMsM/UDMsM products using the EnDAT encoder, only the following values are allowed for the **value** parameter: 0.1, 0.2, 1, 2, 4, 8, 16.



This variable is supported only for BiSS encoders (in all products) and EnDAT encoders (IDMsm/ECMsm/UDMsm products only).

Tag

304

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.9 E_PAR_B

Description

E_PAR_B is used for setting the encoder data control CRC code. It is an integer array, with one element for each axis in the system.

Syntax

E_PAR_B(axis) = value

Arguments

axis	The specific axis index. Valid numbers are: 0, 1... up to the number of axes in the system, minus 1.
value	The encoder data control CRC code ranging from 0 to 255.

Tag

267

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.10 E2_PAR_B

Description

E2_PAR_B is used for setting the encoder data control CRC code. it is an integer array, with one element for each axis in the system. Used for Secondary Feedback.

Syntax**E2_PAR_B** (axis_index)=value**Arguments**

axis	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
value	The encoder data control CRC ranging from 0 to 255.

Tag

305

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.11 E_PAR_C**Description**

E_PAR_C is used for setting the interval (in microseconds) of encoder position reading. It is an integer array, with one element for each axis in the system. The default value is 0 which means 50 microseconds.

Syntax**E_PAR_C**(axis) = value**Arguments**

axis	The specific axis index. Valid numbers are: 0, 1... up to the number of axes in the system, minus 1.
value	The interval of encoder position reading in microseconds, valid range [0,..3]

Valid Values

0	50 microseconds
1	100 microseconds
2	200 microseconds
3	400 microseconds

Tag

258

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.12 E2_PAR_C

Description

E2_PAR_C is used for setting the interval (in microseconds) of encoder position reading. it is an integer array, with one element for each axis in the system. Used for Secondary Feedback. The default value is 0 which means 50 microseconds.

Syntax

E2_PAR_C (axis_index)=value

Arguments

<i>axis</i>	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
<i>value</i>	The interval of encoder position reading in microseconds valid range [0,..3]

Valid Values

0	50 microseconds (default)
1	100 microseconds
2	200 microseconds
3	400 microseconds

Tag

306

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.13 E_PAR_D

Description

E_PAR_D defines number of status bits (LSB) in the real-time position data (**SLABITS**). The status bits include warning and error bits. These bits are not part of the real-time position.

E_PAR_D is represented as an integer array with the size of maximum axes; each element for each axis in the system. The default value is 0 which means no status bits at all. Maximum value is 16.

Syntax

E_PAR_D(axis) = value

Arguments

<i>axis</i>	The specific axis index. Valid numbers are: 0, 1... up to the number of axes in the system, minus 1.
<i>value</i>	Number of status bits; the range is [0, 16].

Tag

266

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.14 E2_PAR_D

Description

E2_PAR_D defines number of status bits (LSB) in the real-time position data. The status bits include warning and error bits. These bits are not part of the real-time position. it is an integer array, with one element for each axis in the system. Used for Secondary Feedback. The default value is 0 which means no status bits at all. Maximum value is 16.

Syntax

E2_PAR_D (axis_index)=value

Arguments

<i>axis</i>	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
<i>value</i>	Number of status bits; the range is [0,16].

Tag

307

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.15 E_PAR_E**Description**

E_PAR_E defines a mask for setting error bits. It is an integer array, with one element for each axis in the system. The default value is 0, which means no error bit is set. All bits that are not defined in the mask but covered by **E_PAR_D** are considered as warning bits.

Syntax**E_PAR_E**(axis) = value**Arguments**

<i>axis</i>	The specific axis index. Valid numbers are: 0, 1... up to the number of axes in the system, minus 1.
<i>value</i>	Error bits MASK; minimum value is 0 (default – no error bits). The range is [0,2 ^{E_PAR_D} -1]

Tag

267

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.16 E2_PAR_E**Description**

E2_PAR_E defines a mask for setting error bits. It is an integer array, with one element for each axis in the system. Used for the secondary feedback. The default value is 0, which means no encoder error will be triggered. All bits that are not set in the mask but covered by **E2_PAR_D** are defined as warning bits.

Syntax**E2_PAR_E** (axis_index)=value

Arguments

<i>axis</i>	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
<i>value</i>	Error bits MASK; minimum value is 0 (default – no error bits). The range is [0,2E2_PAR_D-1]

Tag

308

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.17 E_SCMUL**Description**

E_SCMUL is an integer array, with one element for each axis in the system, and is used for specifying the Sin-Cos multiplication factor for the encoder.

Syntax**E_SCMUL(*axis_index*) = *value*****Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 2 to 16, Default = 10.

Tag

28

Comments

E_SCMUL specifies the Sin-Cos multiplication factor as a power of 2. The maximum value of 16 corresponds to a multiplication of $65536 = 2^{16}$. The minimum value of 2 corresponds to a multiplication of $4 = 2^2$.

Accessibility

Read-Write



E_SCMUL values cannot be modified if protection is applied to this variable through
SPIIPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.18 E2_SCMUL

Description

E2_SCMUL is an integer array, with one element for each axis in the system, and is used for specifying the Sin-Cos multiplication factor for the encoder. Used for secondary feedback.

Syntax

E2_SCMUL (axis_index)=value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
value	Value ranges from 2 to 16, default=10.

Tag

30

Comments

E2_SCMUL specifies the Sin-Cos multiplication factor as a power of 2. The maximum value of 16 corresponds to a multiplication of $65536 = 2^{16}$. The minimum value of 2 corresponds to a multiplication of $4 = 2^2$.

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.19 E_TYPE

Description

E_TYPE is an integer array, with one element for each axis in the system, and is used for defining the encoder type.

Syntax**E_TYPE**(*axis_index*) = *value***Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value of each element can be one of: > 0 - Up-down counter > 1 - Clock direction counter > 2 - Quadrature single-ended encoder > 3 - Quadrature encoder (Default) > 4 - 250 KHz / 500 KHz Sin-Cos encoder multiplier > 5 - HSSI encoder > 9 - Resolver > 10 - EnDat 2.2 > 11 - Smart-Abs > 12 - Panasonic > 13 - BiSS-A/B/C > 14 - Hiperface > 17 - BiSS-SSI > 19 - Sanyo Denki

Tag

29

Comments

The most common encoder type is quadrature, which corresponds to the default value 3.

A value of 2 is supported by the following products:

- > UDMlc-x-048 (where x is either 2 or 4)
- > UDIlt-x / UDlhp-x (where x is either 2 or 4)

A value of 9 is supported only by the following products:

- > SPiiPlus CMnt-x-320 (where x is either 1 or 2)
- > UDMpm-x-320 (where x is either 1 or 2)

A value of 18 is supported by the following products:

- > UDMnt-x (where x is either 1 or 2)
- > UDlhp-x (where x is either 2 or 4)

As a configuration variable, the variable can be changed only if the controller is in configuration mode.

The following table summarizes various combinations of **E_TYPE** and **E2_TYPE** values and whether they are compatible with Multi-Channel Feedback support

E_TYPE	E2_TYPE	Allowed?
3	4	✓
4	3	✓
3	3	✓
4	4	X
{10,11,12,13,17,19}	3	X
3	{10,11,12,13,17,19}	X
{10,11,12,13,17,19}	4	✓
4	{10,11,12,13,17,19}	X
{10,11,12,13,17,19}	{10,11,12,13,17,19}	X

Table 3-5. Multi-Channel Feedback Support

If the device is configured with an incompatible combination of encoders the firmware returns error code 3194, meaning "Not allowed Encoder Type".

Accessibility

Read-Write



E_TYPE values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

None

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.20 E2_TYPE

Description

E2_TYPE is an integer array, with one element for each axis in the system, and is used for defining the encoder type for the secondary feedback.

Syntax

E2_TYPE(axis_index)=value

Arguments

<i>axis</i>	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
<i>value</i>	value of each member can be one of the following: > 0 – not defined > 1 - Clock direction counter > 2 - Quadrature single-ended encoder > 3 - Quadrature encoder (Default) > 4 - 250 KHz / 500 KHz Sin-Cos encoder multiplier > 5 - HSSI encoder > 9 - Resolver > 10 - EnDat 2.2 > 11 - Smart-Abs > 12 - Panasonic > 13 - BiSS-C > 14 – Hiperface > 17 - SSI > 18 - 10 MHz Sin-Cos encoder multiplier > 19 – Sanyo Denki

Tag

31

Comments

See [E_TYPE](#) for Multi-channel feedback support details.

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.21 EFAC

Description

EFAC is a real array, with one element for each axis in the system, and is used for defining a factor between the raw feedback in encoder counts and the **FPOS** value calculated by the controller.

Syntax

EFAC(*axis_index*) = *value*

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value of each member ranges between 1e-15 to 1e+15, Default = 1.

Tag

36

Comments

When reading the feedback position from the SP, the controller executes feedback transform according to the formula:

$$FPOS = FP * EFAC + EOFFS$$

where **FPOS** is the controller feedback position in user units, **FP** is an SP-calculated feedback position in encoder counts, **EFAC** is a user-defined value of the corresponding **EFAC** factor, and **EOFFS** represents an offset.

As a configuration variable, the **EFAC** value is normally defined by **SPiiPlus MMI Application Studio** → **Toolbox** → **Setup** → **Adjuster** during the setup procedure of the system.

Accessibility

Read-Write



EFAC values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Variables

All standard variables that are based on position units.

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.22 E2FAC

Description

E2FAC is a real array, with one element for each axis in the system, and is used for defining a factor between the secondary raw feedback in encoder counts and the **F2POS** value calculated by the controller.

Syntax

E2FAC(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value of each member ranges between 1e-15 to 1e+15, Default = 1.

Tag

32

Comments

When reading the secondary feedback position from the SP, the controller executes feedback transform according to the formula:

$$F2POS = FP2 * E2FAC + E2OFFS$$

where F2POS is the secondary controller feedback position in user units, FP2 is an SP-calculated secondary feedback position in encoder counts, E2FAC is a user-defined value of the corresponding E2FAC factor, and E2OFFS represents an offset.

As a configuration variable, the **E2FAC** value is normally defined by **SPiiPlus MMI Application Studio** → **Toolbox** → **Setup** → **Adjuster** during the setup procedure of the system.

Accessibility

Read-Write



E2FAC values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Variables

[F2POS](#), [EOFFS](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.23 *EOFFS*

Description

EOFFS is a real array, with one element for each axis in the system, and is used for defining the offset between the raw feedback from the encoder counts and the [FPOS](#) value calculated by the controller.

Syntax

EOFFS(*axis_index*) = *value*

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value can be any integer.

Tag

38

Comments

EOFFS provides the offset between the raw feedback in encoder counts and the **FPOS** value calculated by the controller. The value of **EOFFS** changes when the set command defines a new origin for an axis.

When reading the feedback position from the SP, the controller executes feedback transform according to the formula:

$$FPOS = FP * EFAC + EOFFS$$

where FPOS is the controller feedback position in user units, **FP** is an SP-calculated feedback position in encoder counts, **EFAC** is a user-defined value of the corresponding EFAC factor, and EOFFS represents an offset.

Accessibility

Read-Only

Related ACSPL+ Variables[EFAC](#), [FPOS](#)**COM Library Methods and .NET Library Methods**

ReadVariable

C Library Functions

acsc_ReadReal

3.3.24 E2OFFS**Description**

E2OFFS is a real array, with one element for each axis in the system, and is used for defining the offset between the raw feedback from the secondary encoder counts and the **FPOS** value calculated by the controller.

Syntax**E2OFFS(axis_index) = value****Arguments**

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value can be any integer.

Tag

34

Comments

E2OFFS provides the offset between the raw feedback from the secondary encoder (in encoder counts) and the **F2POS** value calculated by the controller. The value of **E2OFFS** changes when the set command defines a new origin for the axis's secondary feedback.

When reading the secondary feedback position from the SP, the controller executes feedback transform according to the formula:

$$F2POS = FP2 * E2FAC + E2OFFS$$

where **F2POS** is the secondary controller feedback position in user units, **FP2** is an SP-calculated secondary feedback position in encoder counts, **E2FAC** is a user-defined value of the corresponding **E2FAC** factor, and **E2OFFS** represents an offset.

Accessibility

Read-Only

Related ACSPL+ Variables**E2FAC**, **F2POS****COM Library Methods and .NET Library Methods**

ReadVariable

C Library Functions

acsc_ReadReal

3.3.25 EPOS**Description**

EPOS is a real array, one element for each feedback, and is used for showing the encoder feedback. Not affected by Gantry mode.

Comments

EPOS value is affected by ACSPL+ **SET** command only when applied to **FPOS** variable with the same index (a relevant offset is being added to EPOS value).

EPOS variable can be useful for displaying the encoder position in Gantry mode.

Tag

299

Accessibility

Read only

Com Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.3.26 FVFIL

Description

FVFIL is a real array, with one element for each axis in the system, and is used for setting the intensity (in %) of the filter that the controller uses when calculating **FVEL**.

Syntax

FVFIL(*axis_index*) = *value*

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 0 to 100, Default = 30.

Tag

54

Comments

FVFIL = 0 corresponds to no filtering. In this case the controller calculates **FVEL** as the derivative of the **FPOS** variable:

$$\Delta = (FPOS_n - FPOS_{n-1}) * K$$

$$FVEL_n = \Delta$$

where K is a scaling factor that reduces **FVEL** to user units per second.

As the **FPOS** value is supplied by a discrete physical sensor, the **FVEL** value calculated without filtering contains a considerable amount of noise.

Non-zero value of **FVFIL** provides additional filtering in the **FVEL** calculation according to the formula:

$$FVEL_n = \Delta * ((1 - FVFIL)/100) + FVEL_{n-1} * (FVFIL/100)$$

Accessibility

Read-Write



FVFIL values cannot be modified if protection is applied to this variable through **SPiiPlus** MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

FVEL

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.27 F2ACC

Description

F2ACC is a real array, with one element for each axis in the system, and is used for defining the feedback acceleration value of the axis. Used for secondary feedback.

Tag

317

Accessibility

Read-Only

Com Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.3.28 HOMEDEF

Description

HOMEDEF is an integer array with one element for each axis in the system, and defines the default homing method for an axis used by the ACSPL+ **HOME** command.

Default value is 0, and should be set to a valid homing method number.

Syntax

```
HOMEDEF(Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
------------	---

Table 3-6. Homing Methods

Method Number	Explanation
1	Homing Method 1: Homing on the negative limit switch and index pulse
2	Homing Method 2: Homing on positive limit switch and index pulse
17	Homing Method 17: Homing on Negative Limit Switch
18	Homing Method 18: Homing on Positive Limit Switch
33/34	Homing Method 33 and 34: Homing on the index pulse
37	Homing Method 37: Homing on current position

Method Number	Explanation
50	Homing Method 50: Negative Hard Stop and index pulse (ACS Specific)
51	Homing Method 51: Positive Hard Stop and index pulse (ACS Specific)
52	Homing Method 52: Negative Hard Stop (ACS Specific)
53	Homing Method 53: Positive Hard Stop (ACS Specific)

TAG

358

Comments

HOMEDEF should be used with the **HOME** command. **HOME**(<axis>) will use the homing method based on HOMEDEF variable.\

Related ACSPL+ Commands**HOME****Accessibility**

Read-Write

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadInteger(), acsc_WriteInteger()

3.3.29 HOMEVELI**Description**

HOMEVELI is a double array with one element for each axis in the system, and defines the default homing velocity used for index search during ACPSL+ HOME command.

Default value is 0. If the value is 0, the velocity is calculated based on the **HomingVel** parameter.

Syntax

```
HOMEVELI(Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 0 to 1e100.

Tag

360

Comments

HOMEVELL should be used with the **HOME** command. The following homing methods are affected by this parameter: 17,18,33,34,50,51

This variable is supported in version 3.00 and higher.

Related ACSPL+ Commands and Variables

HOME, HOMEDF, HOMEVELL

Accessibility

Read-Write

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadInteger(), acsc_WriteInteger()

3.3.30 HOMEVELL**Description**

HOMEVELL is a double array with one element for each axis in the system, and defines the default homing velocity used for limit search during ACSPL+ **HOME** command. Default value is 0.

If the value is 0, the velocity is calculated based on the HomingVel parameter.

Syntax

```
HOMEVELL(Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 0 to 1e100.

Tag

366

Comments

HOMEVELL should be used with the **HOME** command. The following homing methods are affected by this parameter: 1,2,17,18

This variable is supported in version 3.00 and higher.

Related ACSPL+ Commands and Variables

HOME, HOMEDF, HOMEVELL

Accessibility

Read-Write

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadInteger(), acsc_WriteInteger()

3.3.31 RVFIL

Description

RVFIL is a real array, with one element for each axis in the system, and is used for specifying the power of the filter that the controller uses to calculate **RVEL**.

Syntax

RVFIL(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value is a percent ranging from 0 to 100, Default = 0.

Tag

110

Comments

The value is specified as a percent where 0 means no filtering. In this case the controller calculates **RVEL** as the first difference of **RPOS** as follows:

$$\Delta = (RPOS - RPOS_{n-1}) * K$$

$$RVEL_n = \Delta$$

where K is a scaling factor that translates **RVEL** to position units per second.

A non-zero value for **RVFIL** provides additional filtering in the **RVEL** calculation according to the formula:

$$RVEL_n = \Delta * (1 - (RVFIL/100)) + (RVEL_{n-1} * (RVFIL/100))$$

The default value of **RVFIL** is zero. No filtering is required as long as the axis motion is not a **MASTER-SLAVE** motion. In this case **RPOS** is a calculated value and the first difference provides a smooth approximation of velocity.

If an axis is involved in a **MASTER-SLAVE** motion, **RPOS** usually contains a signal from a discrete physical sensor that causes a certain amount of noise in the first difference. Increase the value of **RVFIL** if a smoother approximation is required.

Accessibility

Read-Write



RVFIL values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Commands

MASTER

Related ACSPL+ Variables

RVEL

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.32 SCSOFFS

Description

SCSOFFS is a real array, with one element for each axis in the system, and is used for defining a Sin-Cos encoder's software compensation for the Sine offset.

Syntax

SCSOFFS(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value designates the offset ranging from -32766 to 32766.

Tag

202

Comments

The digital range corresponds to ± 0.5 Volts. This number is used to modify the offset of the Sin-Cos encoder signal related to the axis. The ratio between this number and the offset is 9.6.

Accessibility

Read-Write



SCSOFFS values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.33 SCCOFFS

Description

SCCSOFFS is a real array, with one element for each axis in the system, and is used for defining a Sin-Cos encoder's software compensation for the Cosine offset.

Syntax

SCCOFFS(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value designates the offset ranging from -32766 to 32766.

Tag

203

Comments

The digital range corresponds to ± 0.5 Volts. This number is used to modify the offset of the Sin-Cos encoder signal related to the axis. The ratio between this number and the offset is 9.6.

Accessibility

Read-Write



SCCOFFS values cannot be modified if protection is applied to this variable through **SPIiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.34 SC2COFFS

Description

SC2COFFS is a real array, with one element for each axis in the system, and is used for defining the Sin-Cos Cosine offset. Used for secondary feedback.

Syntax

SC2COFFS (axis_index)=value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
<i>value</i>	value designates the offset ranging from -32766 to 32766.

Tag

311

Comments

The digital range corresponds to ± 0.5 Volts. This number is used to modify the offset of the Sin-Cos encoder signal related to the axis. The ratio between this number and the offset is 9.6.

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.35 SC2GAIN**Description**

SC2GAIN is a real array, with one element for each axis in the system, and is a Sin-Cos encoder gain compensation variable used to compensate the Cosine signal for an improper amplitude relative to the Sine signal. Used for secondary feedback.

Syntax**SC2GAIN** (axis_index)=value**Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0.5 and 1.5; Default: 1.

Tag

312

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.36 SC2PHASE

Description

SC2PHASE is a real array, with one element for each axis in the system, and is a Sin-Cos encoder phase compensation variable and is used to compensate the Cosine signal for an improper phase difference relative to the Sine signal. Used for Secondary Feedback.

Syntax

SC2PHASE (axis_index)=value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
value	value in degrees, the value ranges between -15 and 15; default is 0.

Tag

313

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.37 SC2SOFFS

Description

SC2SOFFS is a real array, with one element for each axis in the system, and is used for defining the Sin-Cos Sine offset. Used for secondary feedback.

Syntax

SC2SOFFS (axis_index)=value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
value	value designates the offset ranging from -32766 to 32766.

Tag

314

Comments

The digital range corresponds to ± 0.5 Volts. This number is used to modify the offset of the Sin-Cos encoder signal related to the axis. The ratio between this number and the offset is 9.6.

Accessibility

Read-Write

Com Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.38 SLEBIASA

SLEBIASA is a real array, with one element for each axis in the system, and is used for defining a Sin-Cos encoder's hardware compensation for the Sine offset.

Syntax

SLEBIASA(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value designates the offset ranging from -50 to 50; Default: 0.

Tag

164

Comments



SLEBIASA performs the same function as **SCSOFFS** with the difference being that it corresponds to hardware offset compensation of encoder signals. Only certain SPiiPlus products: SPiiPlusNT-HP, CMnt, and UDMpc have an option for hardware offset compensation. Hardware compensation has some advantages over software compensation, such as, the possibility to get analog signals out of saturation, and making hardware based features like PEG more accurate.

SLEBIASA is normally set as part of the SPiiPlus MMI Application Studio **Sin-Cos Encoder Analyzer** tool routine. The tool first calculates the software compensation variable (**SCSOFFS**), and then writes the final value to the hardware variable **SLEBIASA** and resets the software one. Then during verification phase new value for **SCSOFFS** is found and stored along with previously found **SLEBIASA**.

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.39 SLEBIASB

SLEBIASB is a real array, with one element for each axis in the system, and is used for defining a Sin-Cos encoder's hardware compensation for the Cosine offset.

Syntax

SLEBIASB(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value designates the offset ranging from -50 to 50; Default: 0.

Tag

165

Comments



SLEBIASB performs the same function as **SCCOFFS** with the difference being that it corresponds to hardware offset compensation of encoder signals. Only certain SPiiPlus products: SPiiPlusNT-HP, CMnt, and UDMpc have an option for hardware offset compensation. Hardware compensation has some advantages over software compensation, such as, the possibility to get analog signals out of saturation, and making hardware based features like PEG more accurate.

SLEBIASB is normally set as part of the SPiiPlus MMI Application Studio **Sin-Cos encoder analyzer** tool routine. The tool first calculates the software compensation variable (**SCCOFFS**), and then writes the final value to the hardware variable **SLEBIASB** and resets the software one. Then during verification phase new value for **SCCOFFS** is found and stored along with previously found **SLEBIASB**.

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.40 SLEBIASC

Description

SLEBIASC is a real array, with one element for each axis in the system, and is used for defining the Sin-Cos encoder's hardware compensation for the Sine offset. Used for secondary feedback.

Syntax

SLEBIASC (*axis_index*)=value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
<i>value</i>	value designates the offset ranging from -50 to 50; default: 0.

Tag

315

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.41 SLEBIASD**Description**

SLEBIASD is a real array, with one element for each axis in the system, and is used for defining the Sin-Cos encoder's hardware compensation for the Cosine offset. Used for secondary feedback.

Syntax**SLEBIASD** (axis_index)=value**Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0,1,2,... up to the number of axes in the system minus 1.
<i>value</i>	value designates the offset ranging from -50 to 50; default: 0.

Tag

316

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.3.42 SLABITS

Description

SLABITS is an integer array, with one element for each axis in the system, and is used for setting the total number of absolute position bits for an absolute encoder.

Syntax

SLABITS(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 16 to 50, Default = 49.

Tag

220

Comments

SLABITS is used for setting the total number of absolute position bits for an absolute encoder. This is the sum of the multi-turn resolution bits, the turn resolution bits, and the number of status bits. Absolute encoders typically include two status bits (Warning and Error), which should be taken into account.

Examples:

Rotary encoder with 17 bits turn resolution and 16 bits multi-turn resolution:

SLABITS = 17+16 = 33

Rotary encoder with 17 bits turn resolution, 16 bits multi-turn resolution, and 2 status bits:

SLABITS = 17+16+2 = 35

Linear Encoder with 32 bits absolute resolution and 2 status bits:

SLABITS = 32+2 = 34

The number of status bits may vary for different encoders, so the user should take care to add the correct number of status bits for the specific encoder in use.

Accessibility

Read-Write



SLABITS values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

[E_TYPE](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.43 S2LABITS

Description

S2LABITS is an integer array, with one element for each axis in the system, and is used for setting the total number of absolute position bits for an absolute encoder connected to a secondary feedback.

Syntax

S2LABITS(axis_index)=value

<i>axis_index</i>	Specific axis, range: 0 up to number of axis in the system minus 1.
<i>value</i>	Value ranges between 16 to 50, default is 49.

Comments

S2LABITS is used for setting the total number of absolute position bits for an absolute encoder. This is the sum of the multi-turn resolution bits, the turn resolution bits, and the number of status bits. Absolute encoders typically include two status bits (Warning and Error), which should be taken into account.

Examples:

Rotary encoder with 17 bits turn resolution and 16 bits multi-turn resolution:

S2LABITS = 17+16 = 33

Rotary encoder with 17 bits turn resolution, 16 bits multi-turn resolution, and 2 status bits:

S2LABITS = 17+16+2 = 35

Linear Encoder with 32 bits absolute resolution and 2 status bits:

S2LABITS = 32+2 = 34

The number of status bits may vary for different encoders, so the user should take care to add the correct number of status bits for the specific encoder in use.

Tag

310

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, Write Variable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.44 SCGAIN

SCGAIN is a real array, with one element for each axis in the system, and is a Sin-Cos encoder gain compensation variable used to compensate the Cosine signal for an improper amplitude relative to the Sine signal.

Syntax

SCGAIN(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges between 0.5 and 1.5; Default: 1.

Tag

204

Comments

The value of **SCGAIN** is normally set by the SPiiPlus MMI Application Studio **Sin Cos Encoder Analyzer** tool routine when calculating the optimum encoder compensation.

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.3.45 SCPHASE

SCPHASE is a real array, with one element for each axis in the system, and is a Sin-Cos encoder phase compensation variable and is used to compensate the Cosine signal for an improper phase difference relative to the Sine signal.

Syntax

SCPHASE(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value in degrees, the value of which ranges between -15 and 15; Default: 0.

Tag

205

Comments

The value of **SCPHASE** is normally set by the SPiiPlus MMI Application Studio **Sin Cos Encoder Analyzer** tool routine when calculating the optimum encoder compensation.

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.4 Axis State Variables

The Axis State variables are:

Name	Description
AST	Axis State
ASTX	Extended Axis State
IND	Index Position
IST	Index State
AFLAGS	Mark Position
MARK	Secondary Mark Position
MST	Motor State
NST	Status of EtherCAT Sync and GPRT errors for each axis in the system.
	RMS current
ROFFS	Reads the offset calculated by the controller in the connect formula.

3.4.1 AST

Description

AST is an integer array, with one element for each axis in the system, the elements of which contain a set of bits used for displaying the current axis state.

Syntax

[command] AST(axis_index).bit_designator

Arguments

<i>command</i>	Typical commands are DISP and the like.
<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1. In the case of bit 2 (#PEGREADY) this parameter designates the PEG engine, not the axis.
<i>bit_designator</i>	A description of the AST bit designators is given in Table 3-7 .

Table 3-7. AST Bit Descriptions

Bit Name	No.	Description
#LEAD	0	1 = axis is leading in a group
#DC	3	1 = Axis data collection is in progress
#PEGREADY	4	1 = all values are loaded and the Incremental/Random PEG is ready to respond to movement When referring to this bit, <i>axis_index</i> designates the PEG engine rather than the axis.
#MOVE	5	1 = Axis is involved in a motion
#ACC	6	1 = Axis in accelerating motion state
#BUILDUP	7	1 = Segments build-up
#VELLOCK	8	1 = Slave is synchronized to master in velocity lock mode - slave velocity strictly follows the master velocity.
#POSLOCK	9	1 = Slave is synchronized to master in position lock mode - slave position strictly follows the master position.
#TRIGGER	11	1 = Produces an interrupt to the host application, enabled by IENA.26
#NEWSEGM	16	The controller sets the bit to inform that a new segment is required to be provided by the application. The bit is set starvation_margin ms before the starvation condition occurs. The starvation condition is indicated by #STARV bit.

Bit Name	No.	Description
#STARV	17	The controller sets the bit to indicate starvation condition. The starvation condition means that there are not enough further segments to continue the motion with required velocity. In this case, the controller starts decelerating the motion with $\frac{1}{2}$ JERK in order to prevent motion discontinuity and avoid mechanical jerks. Once the application begins supplying segments at a sufficient rate, the controller returns the motion back to normal condition. Note, that often the starvation condition causes inefficient velocity generation and increases the time required for completing the required motion path.
#ENCWARN	18	Indicates if there is an encoder warning. Cleared by the ACSPL+ FCLEAR command.
#ENC2WARN	19	Indicates if there is an secondary encoder warning. Cleared by the ACSPL+ FCLEAR command.
#INRANGE	20	Laser In Range
#LCTICKLE	21	0: tickle mode is off 1: tickle mode is active
#LCMODUL	22	0: modulation is off 1: modulation is active
#FOLLOWED	23	0: Axis in regular mode 1: Axis in slave mode and follows the profile generated by RTC6
#HOLD	24	0: hold is off 1: hold is in progress
#INHOMING	25	0: homing is not in process 1: homing is in process
#DECOMPON	26	0: dynamic error compensation is switched off 1: dynamic error compensation is switched on The bit is set to 0 in the following cases: > Calling ERRORMAPOFF function > Calling ERRORUNMAP function (in case there are no other active dynamic error compensation

Bit Name	No.	Description
		zones of the specified axis) > During homing process initiated by HOME command (switched ON back at the end of the process) > Encoder Error / Encoder Not Connected faults > Changing one of the encoder-related parameters (E_TYPE, E_FREQ, E_SCMUL, SLPBITS, E_AOFSS) > Changing encoder routing (SLPROUT)
#INSHAPE	27	0: Input Shaping not Active 1: Input Shaping Active
#ENCPROC	29	0: Encoder Initialization not in process 1: Encoder Initialization in process

Tag

7

Accessibility

Read-Only

Related ACSPL+ Commands[MASTER](#), [SLAVE](#)**Related ACSPL+ Variables**[MST](#)**COM Library Methods and .NET Library Methods**

ReadVariable, GetAxisState

C Library Functions

acsc_ReadInteger, acsc_GetAxisState

3.4.2 ASTX**Description**

ASTX is an extension of the **AST** variable. It is an integer array, with one element for each axis in the system, the elements of which contain a set of bits used for displaying the current axis state.

Syntax

```
[command]ASTX(axis_index).bit_designator
```

ASTX Bit Descriptions

Bit Name	No	Description
#GCACOMP	0	Indicates if Gantry Cross Axis Coupling Compensation is in process for the specified axis
#AFACTIVE	1	Autofocus is activated for the axis
#AF_RANGE	2	Autofocus feature is in_range – position control is now executed by the DSP
#AF_FOCUS	3	Autofocus feature is in focus – focus has been achieved (Control signal < Threshold)

Tag

437

Accessibility

Read-only

3.4.3 IND**Description**

IND is a real array, with one element for each axis in the system, the elements of which store the position of the last encountered encoder index in user-defined units. The variable operates in connection with [IST\(axis_index\).#IND](#).

Tag

72

Comments

After power-up, [IST\(axis_index\).#IND](#) is reset and the value of **IND** is undefined because an index capture has not yet occurred. When the motor encounters an encoder index, [IST\(axis_index\).#IND](#) is raised and the current [FPOS](#) position is latched to **IND**.

Subsequent index values are ignored as long as **#IND** remains raised.

To resume the latching logic [IST\(axis_index\).#IND](#) must be explicitly cleared by the command [IST\(axis_index\).#IND=0](#).

Accessibility

Read-Only



IND values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables[IST](#), [FPOS](#)

COM Library Methods and .NET Library Methods

ReadVariable, GetIndexState, ResetIndexState

C Library Functions

acsc_ReadReal, acsc_GetIndexState, acsc_ResetIndexState

3.4.4 IST**Description**

IST is an integer array, with one element for each axis in the system, the elements of which contain a set of bits that indicate the state of the **IND** and the **MARK** variables for the given axis.

Syntax**IST(axis_index).bit_designator = value****Arguments**

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
bit_designator	The IST has four bit designators: > #IND (bit 0) - Primary encoder index > #IND2 (bit 1) - Secondary encoder index > #MARK (bit 2) - Mark 1 > #MARK2 (bit 3) - Mark 2
value	value can be zero or non-zero.

Tag

79

Comments

The controller processes Index/Mark signals as follows: when an Index/Mark signal is encountered for the first time, the controller latches **FPOS** or **F2POS** to one of the variables **IND**, **MARK**, **M2ARK** and **sets the corresponding IST bit = 1**.

When finding an Index for the first time, the correct procedure is:

1. Start by setting the index flag to 1: **IST(axis).#IND=1**.

Then reset the flag to 0: **IST(axis).#IND=0**.

This puts the system in the correct mode for finding the Index.

As long as an **IST** bit is raised, the controller does not latch another value to the corresponding variable. To resume the latching logic, the user application must explicitly reset the corresponding **IST** bit to 0.

Accessibility

Read-Write

Related ACSPL+ Variables

FPOS, F2POS, IND, MARK, M2ARK

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetIndexState, ResetIndexState

C Library Functions

acsc_ReadInteger, acsc_WriteInteger, acsc_GetIndexState, acsc_ResetIndexState

3.4.5 M2ARK**Description**

M2ARK is a real array, with one element for each axis in the system, the elements of which store the position of the last encountered MARK2 signal in **IST(axis_index).#MARK2**.

Tag

84

Comments

After power-up **IST(axis_index).#MARK2** is reset and the value of **M2ARK** is undefined. When the motor encounters a MARK2 signal, the bit is raised and the current **FPOS** position is latched to **IST(axis_index).#MARK2**.

If the motion continues and the motor encounters another M2ARK signal, the new value is ignored as long as **IST(axis_index).#MARK2 = 1**.

To resume the latching logic, **IST(axis_index).#MARK2** must be explicitly cleared with the command **IST(axis_index).#MARK2=0**.

Example

```
ON IST(Axis).#MARK2           ! When axis MARK is tripped
DISP "MARK Position:",M2ARK(Axis) ! Display the MARK position
IST(Axis).#MARK2=0           ! Reset MARK flag
RET
```

Accessibility

Read-Only

Related ACSPL+ Variables

IST, FPOS, F2POS, MARK

IST, FPOS, F2POS, MARK

COM Library Methods and .NET Library Methods

ReadVariable, GetIndexState,

C Library Functions

acsc_ReadReal, acsc_GetIndexState

3.4.6 MARK

Description

MARK is a real array, with one element for each axis in the system, the elements of which store the position of the last encountered MARK1 signal in **IST(axis_index).#MARK**.

Tag

85

Comments

After power-up, **IST(axis_index).#MARK** is reset and the value of **MARK** is undefined. When the motor encounters a MARK1 signal, the bit is raised and the current **FPOS** position is latched to **IST(axis_index).#MARK**.

If the motion continues and the motor encounters another MARK1 signal, the new value is ignored as long as **IST(axis_index).#MARK=1**.

To resume the latching logic, **IST(axis_index).#MARK** must be explicitly cleared with the command **IST(axis_index).#MARK=0**.



MARK values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection.

Example

```
ON IST(Axis).#MARK           ! When axis MARK is tripped
DISP "MARK Position:", MARK(Axis) ! Display the MARK position
IST(Axis).#MARK=0           ! Reset MARK flag
RET
```

Accessibility

Read-Only

Related ACSPL+ Variables

[IST](#), [FPOS](#), [F2POS](#), [IND](#), [M2ARK](#), [IENA](#)

COM Library Methods and .NET Library Methods

[ReadVariable](#), [GetIndexState](#)

C Library Functions

[acsc_ReadReal](#), [acsc_GetIndexState](#)

3.4.7 MST

Description

MST is an integer array, with one element for each axis in the system. The elements of which contain a set of bits that display the current motor state, as given in [Table 3-8](#), for the given axis.

Syntax

MST(node_index).bit_designator = 1/0

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>bit_designator</i>	The meanings of bit_designator are given in 3.4.7 .

Table 3-8. MST Bit Descriptions.

Bit Name	No	Description
#ENABLED	0	0: motor is disabled 1: motor is enabled.
#OPEN	1	0: motor is operating with closed loop control 1: motor is operating with open loop control.
#INPOS	4	0: Motor is moving or is out of range 1: Motor is not moving and has reached the target position (see variables TARGRAD and SETTLE)
#MOVE	5	0 = Axis is not involved in a motion 1 = Axis is involved in a motion
#ACC	6	0 = Motor is not accelerating 1 = Motor is accelerating.
#INTARGA	25	0 = No motion has settled in Target Radius A 1 = Motion has settled in Target Radius A
#INTARGB	26	0 = No motion has settled in Target Radius B 1 = Motion has settled in Target Radius B
#INTARGC	27	0 = No motion has settled in Target Radius C 1 = Motion has settled in Target Radius C

Tag

90

Accessibility

Read-Only

Related ACSPL+ Commands

All motion commands.

Related ACSPL+ Variables

[FPOS](#), [F2POS](#), [APOS](#), [RPOS](#)

COM Library Methods and .NET Library Methods

ReadVariable, GetMotorState

C Library Functions

acsc_ReadInteger, acsc_GetMotorState

Example

```
WHILE MST(0).#MOVE
WAIT 300
```

3.4.8 RMSM

Description

RMSM is a real array with one element for each axis in the system, the elements of which store the motor RMS current for an axis (in % of drive peak). The value ranges between 0 and 100.

Tag

363

Comments

This variable is supported in ADK versions 2.70 and higher.

Accessibility

Read-Only

ACSPL+ Variables

XRMSM, XRMSTM

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.4.9 RMSD

Description

RMSD is a real array, with one element for each axis in the system, the elements of which store the drive RMS current for an axis (in % of drive peak). The value ranges between 0 and 100.

Tag

362

Comments

This variable is supported in ADK versions 2.70 and higher.

Accessibility

Read-Only

Related ACSPL+ Variables

XRMSD, XRMSTD

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.4.10 NST**Description**

NST is an integer array, with one element for each EtherCAT node in the system, each element of which contains a set of 2 bits. The variable enables users to differentiate between different causes of servo processor alarm faults.



An axis not associated to a physical drive will have a high Servo Processor Alarm fault.

Syntax

NST(*node_index*).*bit_designator* = 1/0

Arguments

node_index	Designates the specific node, valid numbers are: 0, 1, 2, ... up to the number of nodes in the system minus 1.
bit_designator	The meanings of bit_designator are given in Table 3-9 .

Table 3-9. NST Bit Description

Bit Name	Bit No.	Description
#SYNC	0	Sync error
#GPRT	1	GPRT error
#MISSING	2	EtherCAT Node Missing
#SPDC	3	0: SPDC data collection not active 1: SPDC data collection active
#FSYNC	4	FPGA Sync Fault
#SPRT	5	0: Real-time data transfer process from the MPU to a corresponding Servo Processor is not active 1: Real-time data transfer process from the MPU to a corresponding Servo Processor is active

Bit Name	Bit No.	Description
#LCI	10	LCI Fault

Tag

229

Comments

If the #SYNC or #GPRT error bit is set in **NST**, there will be a network error in all axes, and servo processor alarm in all the axes related to the node related to the **NST**. These faults indicate a problem in the interface between the firmware and the node.

The setting of the #SYNC error bit means that one or more slaves are out of synchronization with the master.

The setting of the #GPRT error bit means that the queue (the size, of which, is 400) for the GPRT commands (commands that are sent by request) was full and some commands to be sent were lost. For example, for the SPiiPlusDC-LT-4 controller 8 such commands can be sent every cycle, these commands can be found in a reserved place in the EtherCAT telegram for the command1,...,command8.

The following commands will cause the NST.#SPRT bit to be 1:

- > **SPINJECT**
- > **SPRT**
- > **ASSIGNPEG/f**
- > **BPTP/2** (20 kHz motion profile)
- > **FOLLOW** (in case of customized servo algorithm for 20 kHz motion profile)

SPINJECT, **SPRT**, **ASSIGNPEG/f**, **BPTP/2** and **FOLLOW** are mutually exclusive, meaning only one of the features can be active at the given time. So the **NST.#SPRT** bit should be checked before using any of these commands.

FCLEAR for any axis associated with the node axis will reset all bits of the **NST** variable of that node.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadInteger

C Library Functions

acsc_ReadInteger

3.4.11 AMEST**Description**

AMEST is an integer array with an element for each axis in the system. It is updated after each READ or WRITE to an AME device, and contains the device status word.

If the READ/WRITE operation is successful, the status word should contain the value 0x2 (POWER bit on).

Status Word

Bit 0	Bit 1	Bit 2	Bits 8-15
Err	PWR	CONFIG MODE	Error Code

Tag

386

Accessibility

Read-Only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadInteger()

3.5 Safety Limits Variables

The Safety Limits variables are:

Name	Description
CERRA	Critical Position Error (Accelerating)
CERRI	Critical Position Error (Idle)
CERRV	Critical Position Error (Velocity)
DELI	Delay on Transition to Idle State
DELV	Delay on Transition to Velocity State
ERRA	Tolerable Error (Accelerating)
ERRI	Tolerable Error (Idle)
ERRV	Tolerable Error (Velocity)
SLLIMIT	Software Left Limit-feedback count down limit
SLLROUT	Sets the HW limits routing for the specified axis
SRLIMIT	Software Right Limit-feedback count up limit
XACC	Over Acceleration fault parameter
XCURCDB	Threshold of the current vector peak

Name	Description
XCURI	Maximum idle motor current
XCURK	Current limit during kill operation
XCURV	Maximum drive current during motion
XRMS	Drive RMS over current fault parameter
XRMSD	Drive RMS over current fault parameter
XRMSM	Motor RMS over current fault parameter
XRMST	Drive RMS Current Time Constant
XRMSTD	Drive RMS Current Time Constant
XRMSTM	Motor RMS Current Time Constant
XSACC	Maximum slave axis acceleration
XVEL	Over Velocity fault parameter

3.5.1 CERRA

Description

CERRA is a real array, with one element for each axis in the system, and is used for defining the Position Error criterion for acceleration/deceleration states.

Syntax

CERRA(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value of each member ranges between 2.22507e-308 and 1.79769e+308, Default = 1000.

Tag

11

Comments

CERRA defines critical position error fault ([FAULT](#)(*axis_index*).#**CPE**) criterion when the motor is in acceleration or deceleration motion states.

As a configuration variable, the **CERRA** value is normally defined by **SPiiPlus MMI Application Studio** → **Toolbox** → **Setup** → **Adjuster** during the setup procedure of the system.

Accessibility

Read-Write



CERRA values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Variables

FAULT(axis_index).#CPE

CERRI, **CERRV**, **DELI**, **DELV**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.5.2 CERRI

Description

CERRI is a real array, with one element for each axis in the system, and is used for defining the critical Position Error when the motor is idle.

Syntax

CERRI(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value of each member ranges between 2.22507e-308 and 1.79769e+308, Default = 1000.

Tag

12

Comments

As a configuration variable, the **CERRI** value is normally defined by **SPiiPlus MMI Application Studio → Toolbox → Setup → Adjuster** during the setup procedure of the system.

Accessibility

Read-Write



CERRI values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Variables

[FAULT\(axis_index\).#CPE](#)

[CERRA](#), [CERRV](#), [DELI](#), [DELV](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.5.3 CERRV

Description

CERRV is a real array, with one element for each axis in the system, and is used for defining the critical Position Error when the motor is moving with constant velocity.

Syntax

CERRV(axis_index) = value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value of each member ranges between 2.22507e-308 and 1.79769e+308, Default = 1000.

Tag

13

Comments

As a configuration variable, the **CERRV** value is normally defined by **SPiiPlus MMI Application Studio** → **Toolbox** → **Setup** → **Adjuster** during the setup procedure of the system.

Accessibility

Read-Write



CERRV values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Variables

[FAULT\(axis_index\).#CPE](#)

[CERRA](#), [CERRI](#), [DELI](#), [DELV](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.5.4 DELV

Description

DELV is a real array, with one element for each axis in the system, and is used for defining the delay of transition to the Constant Velocity state.

Syntax

DELV(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value of each member ranges between 2.22507e-308 and 1.79769e+308, Default = 50.

Tag

24

Comments

DELV is defined in msec and applies a delay when the motion state changes to a constant velocity state (**GPHASE**(*axis_index*) = 4).

DELV affects the following faults:

- > **FAULT**(*axis_index*).#PE
- > **FAULT**(*axis_index*).#CPE

Accessibility

Read-Write



DELV values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Variables

ERRA, **ERRI**, **ERRV**, **DELI**, **FAULT**.#CPE, **FAULT**.#PE

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.5.5 DELI

Description

DELI is a real array, with one element for each axis in the system, and is used for defining the delay of transition to the Idle state.

Syntax

DELI(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value of each member ranges between 2.22507e-308 and 1.79769e+308, Default = 50.

Tag

23

Comments

DELI is defined in milliseconds and applies a delay when the motion state changes from any motion state to idle (**GPHASE**(*axis_index*) = 0 or 12).

DELI affects the following faults and current limits:

- > **FAULT**(*axis_index*).#**PE**
- > **FAULT**(*axis_index*).#**CPE**
- > **XCURI**(*axis_index*)
- > **XCURV**(*axis_index*)

Accessibility

Read-Write



DELI values cannot be modified if protection is applied to this variable through **SPiiPlus** MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

ERRA, **ERRI**, **ERRV**, **DELV**, **FAULT**(*axis_index*).#**CPE**, **FAULT**(*axis_index*).#**PE**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.5.6 E_ERR

Description

E_ERR is an integer array for each axis. It contains the encoder error code that was identified during the encoder initialization process.

The encoder errors range from 5121 to 5128 and are latched in the **E_ERR** variable. The error codes are specified in [Table 7-5](#) in the Error Codes section.

Comments

This variable is supported in version 3.00 and higher.

Accessibility

Read-Only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadInteger()

3.5.7 ERRA

Description

ERRA is a real array, with one element for each axis in the system, and is used for defining the Position Error criterion for Acceleration/Deceleration states.

Syntax

ERRA(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 2.22507e-308 to 1.79769e+308, Default = 100.

Tag

39

Comments

ERRA defines the maximum tolerable position error (**FAULT**(axis_index).#PE) when the motor is moving with acceleration.

As a configuration variable, the **ERRA** value is normally defined by **SPiiPlus MMI Application Studio** → **Toolbox** → **Setup** → **Adjuster** during the setup procedure of the system.

Accessibility

Read-Write



ERRA values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Variables

[FAULT\(axis_index\).#PE](#), [FDEF](#)

[ERRI](#), [ERRV](#), [DELI](#), [DELV](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable,

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.5.8 *ERRI*

Description

ERRI is a real array, with one element for each axis in the system, and is used for defining the maximum tolerable Position Error ([FAULT\(axis_index\).#PE](#)) when the motor is idle.

Syntax

ERRI(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 2.22507e-308 to 1.79769e+308, Default = 100.

Tag

40

Comments

As a configuration variable, the **ERRI** value is normally defined by **SPiiPlus MMI Application Studio → Toolbox → Setup → Adjuster** during the setup procedure of the system.

Accessibility

Read-Write



ERRI values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Variables

[FAULT\(axis_index\).#PE](#)

[ERRA](#), [ERRV](#), [DELI](#), [DELV](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.5.9 ERRV

Description

ERRV is a real array, with one element for each axis in the system, and is used for defining the maximum tolerable Position Error (**FAULT**(axis_index).#PE) when the axis is moving with constant velocity.

Syntax

ERRV(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 2.22507e-308 to 1.79769e+308, Default = 100.

Tag

41

Comments

As a configuration variable, the **ERRV** value is normally defined by **SPiiPlus MMI Application Studio** → **Toolbox** → **Setup** → **Adjuster** during the setup procedure of the system.

Accessibility

Read-Write



ERRV values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Variables

FAULT(axis_index).#PE

ERRA, **ERRI**, **DELI**, **DELV**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.5.10 SLLIMIT

Description

SLLIMIT is a real array, with one element for each axis in the system, and is used for defining the minimum allowed Left position for the motor.

Syntax

SLLIMIT(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from -1.79769e+308 to 1.79769e+308, Default = 2e+014.

Tag

124

Comments

If reference position **RPOS** is less than this value, a software Left Limit fault results and bit **FAULT** (axis_index).#**SLL** is = 1.

Accessibility

Read-Write



SLLIMIT values cannot be modified if protection is applied to this variable through **SPIiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Variables

RPOS, **FAULT**(axis_index).#**SLL**, **SRLIMIT**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetFault

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_GetFault

3.5.11 SLLROUT

HW Limits Routing is available using ACSPL+ variable: **SLLROUT**

Description

SLLROUT is an integer array, with one element for each axis in the system, and is used for setting the HW limits routing for the specified axis.

Syntax

SLLROUT(<axis>)=value

Arguments

Value	HW Limits
0	According to SLPROUT
001	From channel 0
101	From channel 1
201	From channel 2
301	From channel 3

Comments

If **SLLROUT(<axis>)=0**, the routing is being done according to SLPROUT. In particular, if SLPROUT (<axis>)=0, the HW limits are being taken from the axis itself.

Tag

318

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.5.12 SRLIMIT

Description

SRLIMIT is a real array, with one element for each axis in the system, and is used for defining the minimum allowed Right position for the motor.

Syntax

SRLIMIT(*axis_index*) = value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from -1.79769e+308 to 1.79769e+308, Default = 2e+014.

Tag

128

Comments

If reference position **RPOS** is greater than this value, a software Right Limit fault results and **FAULT** (axis_index).#**SRL** is = 1.

Accessibility

Read-Write



SRLIMIT values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection.

Related ACSPL+ Variables

RPOS, **FAULT**(axis_index).#**SRL**, **SLLIMIT**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetFault

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_GetFault

3.5.13 XACC

Description

XACC is a real array, with one element for each axis in the system, and is used for defining the maximum allowed acceleration for the motor.

Syntax

XACC(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 2.22507e-308 to 1.79769e+308, Default = 1e+007.

Tag

141

Comments

If the reference acceleration **RACC** exceeds this value, the Acceleration Limit fault is activated and bit **#AL** is set in variable **FAULT**.

Accessibility

Read-Write



XACC values cannot be modified if protection is applied to this variable through **SPiiPlus**
MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

FAULT(axis_index).#AL, **RACC**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetFault

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_GetFault

3.5.14 XCURCDB

Description

XCURCDB is a real array, the size of which is determined by the total number of axes in the system. It is used for defining the threshold of the current vector peak.

Syntax

```
XCURCDB(index) = value
```

Arguments

index	A number between 0 up to a maximum number of axes in the system minus 1.
value	value range is [0,100] (as a percentage of the maximum peak output) Default value: 0. If the value is 0, the actual value will be the minimum of ACSPL+ variables XRMSM and XRMSD .

Comments

The parameter is relevant only if the Controlled Current Dynamic Brake Mode is active.

This variable is supported in version 3.10 and higher.

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.5.15 XCURI

Description

XCURI is a real array, with one element for each axis in the system, and is used for limiting the drive output when the motor is enabled but in standstill position. **XCURI** is defined as a percentage of the maximum peak output. For products that incorporate the Drive Power Electronics (UDMnt, etc...), this value directly limits the Output Current. For products that output Voltage Signals to external Amplifiers (universal analog drive controllers such as the UDI), this value limits the Output Signal to percentage of $\pm 10V$.

Syntax

XCURI(*axis_index*) = *value*

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 0 to 100, Default = 50.

Tag

142

Comments

XCURI is defined as a percentage of the maximum peak output. For drive command products, the command scales the voltage output range. For example, in the UDMnt-10/20, the maximum output voltage is $\pm 20V$. Setting **XCURI** to 50 will limit the drive output to $\pm 10V$ when the motor is idle. For other products setting **XCURI** will scale the peak current output.

Accessibility

Read-Write



XCURI values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Variables

FAULT(*axis_index*).#**CL**, **XRMS**, **XCURV**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetFault

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_GetFault

3.5.16 XCURK

Description

XCURK is a double array with one element for each axis in the system. It sets the current limit (in percentage) for the axis to be applied during a KILL MOTION operation.

Syntax

```
XCURK(Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 0 to 100. Default value is 10.

Tag

375

Comments

The value transferred to the DSP is:

$$\frac{\mathbf{XCURV} * \mathbf{XCURK}}{10000}$$

This variable is supported in version 3.00 and higher.

Related ACSPL+ Commands and Variables

XCURV

Accessibility

Read-Write

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadDouble(), acsc_WriteDouble()

3.5.17 XCURV

Description

XCURV is a real array, with one element for each axis in the system, and is used for limiting the drive output when the motor is moving. **XCURV** is defined as a percentage of the maximum peak output. For products that incorporate the Drive Power Electronics (UDMnt, etc...), this value directly limits the Output Current. For products that output Voltage Signals to external Amplifiers (universal analog drive controllers such as the UDI), this value limits the Output Signal to percentage of ±10V.

Syntax

```
XCURV(axis_index) = value
```

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 0 to 100, Default = 50.

Tag

143

Comments

XCURV is defined as a percentage of the maximum peak output. For drive command products, the command scales the voltage output range. For example, in the UDMnt-10/20, the maximum output voltage is $\pm 20V$. Setting **XCURV** to 50 will limit the drive output to $\pm 10V$ when the motor is idle. For other products setting **XCURV** will scale the peak current output.

Accessibility

Read-Write



XCURV values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection.

Related ACSPL+ Variables

FAULT(axis_index).#**CL**, **XRMS**, **XCURI**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetFault

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_GetFault

3.5.18 XRMS**Description**

XRMS is a real array, with one element for each axis in the system, and is used for setting the maximum allowable rms current for the motor. **XRMS** is defined as a percentage of the maximum peak output. For products that incorporate the Drive Power Electronics (UDMnt, etc...), this value directly limits the Output Current. For products that output Voltage Signals to external Amplifiers (universal analog drive controllers such as the UDI), this value limits the Output Signal to percentage of $\pm 10V$.

Syntax

XRMS(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 0 to 100, Default = 50.

Tag

144

Comments

The SP program calculates **RMS** of the corresponding motor. If the calculated value exceeds the **XRMS** value, an overcurrent fault occurs. **XRMS** is defined as a percentage of the maximum output voltage.

Accessibility

Read-Write



XRMS values cannot be modified if protection is applied to this variable through **SPiiPlus**
MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

FAULT(axis_index).#**CL**, **XCURI**, **XCURV**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetFault

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_GetFault

3.5.19 XRMSD

Description

XRMSD is a real array, with one element for each axis in the system, and is used for setting the maximum allowable RMS current on the drive, as opposed to **XRMSM** which sets the maximum allowed RMS current for the motor. **XRMSD** is defined as a percentage of the maximum peak output. For products that incorporate the Drive Power Electronics (UDMnt, etc...), this value directly limits the Output Current. For products that output Voltage Signals to external Amplifiers (universal analog drive controllers such as the UDI), this value limits the Output Signal to percentage of $\pm 10V$.

Syntax

```
XRMSD(Axis_Index) = Value
```

Arguments

Axes_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 0 to 100, Default = 50.

Tag

345

Comments

XRMSD is used to define the desired maximum current for the drive.

The SP program calculates RMS of the corresponding controller drive output. If the calculated value exceeds the **XRMSD** value, an overcurrent fault occurs. **XRMSD** is defined as a percentage of the maximum output current.

Related ACSPL+ Variables

XRMSD, **FAULT**(Axis_Index).#**CL**, **XCURI**, **XCURV**

Accessibility

Read-Write



XRMSD values cannot be modified if protection is applied to this variable through SPiPlus MMI Application Studio > Toolbox > Application Development > Protection.

.NET Library Method

ReadVariable(), WriteVariable(), GetFault

C Library Function

acsc_ReadReal(), acsc_WriteReal(), acsc_GetFault

3.5.20 XRMSM

Description

XRMSM is a real array, with one element for each axis in the system, and is used for setting the maximum allowable RMS current on the motor, as opposed to **XRMSD** which sets the maximum allowed RMS current for the drive. **XRMSM** is defined as a percentage of the maximum peak output. For products that incorporate the Drive Power Electronics (UDMnt, etc...), this value directly limits the Output Current. For products that output Voltage Signals to external Amplifiers (universal analog drive controllers such as the UDI), this value limits the Output Signal to percentage of $\pm 10V$.

Syntax

```
XRMSM(Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 0 to 100, Default = 100.

Tag

351

Comments

The SP program calculates RMS of the corresponding controller drive output. If the calculated value exceeds the **XRMSM** value, an overcurrent fault occurs. **XRMSM** is defined as a percentage of the maximum output current.

XRMSM is used to define the desired maximum current for the motor.

Related ACSPL+ Variables

XRMSTM, **FAULT**(Axis_Index).#**CL**, **XCURI**, **XCURV**

Accessibility

Read-Write



XRMSTM values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio→Toolbox→Application Development→ Protection.

.NET Library Method

ReadVariable(), WriteVariable(), GetFault

C Library Function

acsc_ReadReal(), acsc_WriteReal(), acsc_GetFault

3.5.21 XRMST

Description

XRMST is a real array, with one element for each axis in the system, and is used for setting the time constant in milliseconds for the **XRMS** to activate the overcurrent protection. For calculation of **XRMS** activation time, see *SPiiPlus Setup Guide* .

Syntax

XRMST(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 200 to 3260, Default = 3230.

Tag

145

Accessibility

Read-Write



XRMST values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

FAULT(axis_index).#**CL**, **XCURI**, **XCURV**, **XRMS**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetFault

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_GetFault

3.5.22 XRMSTD

Description

XRMSTD is a real array, with one element for each axis in the system, and is used for setting the time constant in milliseconds for **XRMSD** to activate the overcurrent protection for the drive. For calculation of **XRMSD** activation time, see *SPiiPlus Setup Guide*.

Syntax

```
XRMSTD (Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 200 to 60,000, Default = 3230.

Tag

346

Related ACSPL+ Variables

XRMSD, FAULT(Axis_Index).#CL, XCURI, XCURV

Accessibility

Read-Write



XRMSTM values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio >Toolbox > Application Development > Protection.

.NET Library Method

ReadVariable(), WriteVariable(), GetFault

C Library Function

acsc_ReadReal(), acsc_WriteReal(), acsc_GetFault



If Drive X RMS protection is triggered, the error 5049 "Drive Overcurrent" is given.

3.5.23 XRMSTM

Description

XRMSTM is a real array, with one element for each axis in the system, and is used for setting the time constant in milliseconds for **XRMSM** to activate the overcurrent protection for the motor. For calculation of **XRMSM** activation time, see *SPiiPlus Setup Guide*.

Syntax

```
XRMSTM(Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 200 to 600,000, Default = 20,000.

Tag

352

Related ACSPL+ Variables

XRMSM, FAULT(Axis_Index).#CL, XCURI, XCURV

Accessibility

Read-Write



XRMSTM values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio > Toolbox > Application Development > Protection.

.NET Library Method

ReadVariable(), WriteVariable(), GetFault

C Library Function

acsc_ReadReal(), acsc_WriteReal(), acsc_GetFault



If Motor XRMS protection is triggered, the error 5048 "Motor Overcurrent" is given.

3.5.24 XSACC

Description

XSACC is a real array, with one element for each axis in the system, and is used for defining the maximum allowed slave acceleration in **MASTER** - **SLAVE** motion.

Syntax

XSACC(axis_index) = value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 2.22507e-308 to 1.79769e+308, Default = 1e+007.

Tag

146

Comments

When a slave is synchronized to a master, the controller verifies the slave acceleration against the **XSACC** value each MPU cycle. If the slave acceleration exceeds **XSACC**, the motion falls out of synchronization. The controller tries to regain synchronism by having the slave pursue the master with the maximum allowed motion parameters.

Accessibility

Read-Write



XSACC values cannot be modified if protection is applied to this variable through
SPIIPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Commands[MASTER](#), [SLAVE](#)**COM Library Methods**

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.5.25 XVEL**Description**

XVEL is a real array, with one element for each axis in the system, and is used for defining the maximum allowed velocity for the axis.

Syntax***XVEL(axis_index) = value*****Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 2.22507e-308 to 1.79769e+308, Default = 2e+006.

Tag

147

Comments

If **RVEL** reference velocity exceeds **XVEL**, **FAULT**(axis_index).#VL = 1.



Trying to adjust the position and velocity loops when **XVEL** is not correctly set will produce poor results. Verify that **XVEL** is correctly defined to fit the application and other requirements before adjusting the loops.

Accessibility

Read-Write



XVEL values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Variables

FAULT(axis_index).#VL, **RVEL**, **FVEL**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.6 Data Collection Variables

The Data Collection variables are:

Name	Description
DCN	Axis Data Collection, Number of Samples
DCP	Axis Data Collection, Period
S_DCN	System Data Collection, Number of Samples
S_DCP	System Data Collection, Period
S_ST	State of System Data Collection

3.6.1 DCN

Description

DCN is an integer array, with one element for each axis in the system, the elements of which store the number of data collection samples per given axis.

Tag

19

Comments

DCN stores a defined number of axis data collection samples, as follows:

1. While an axis data collection is in progress **DCN** displays the index of the array element that stores the next sample.
2. When an axis data collection terminates for the corresponding axis, **DCN** stores the number of actually collected samples. If the data collection terminates automatically, the variable is always equal to the requested number of samples specified in **DC**. If **STOPDC** terminates data collection, **DCN** may contain less than the specified number of samples.

If an axis data collection is in progress, **DCN** increments each time the next sample is stored. When the data collection terminates, the **DCN** holds the last value, until the next data collection starts for the same axis.

Accessibility

Read-Only

Related ACSPL+ Commands

[DC](#), [STOPDC](#)

Related ACSPL+ Variables

[AST](#), [DCP](#)

COM Library Methods and .NET Library Methods

[ReadVariable](#), [DataCollection](#), [StopCollect](#), [WaitCollectEnd](#)

C Library Functions

[acsc_ReadInteger](#), [acsc_DataCollectionExt](#), [acsc_StopCollect](#), [acsc_WaitCollectEnd](#)

3.6.2 DCP

Description

DCP is a real array, with one element for each axis in the system, the elements of which store the axis data collection samples based on a specified sampling *period*. When an axis data collection terminates, **DCP** stores the sampling period.

Tag

21

Comments

DCP is generally equal to the specified *period*, however because *period* is rounded to an integer number of controller cycles, the actual *period* may differ from the *period* specified in the [DC](#) command.

If **DC/t** (temporal data collection) was executed, **DCP** may be greater than the requested minimal period.

When a system data collection starts, **DCP** is assigned a real data collection period.

Accessibility

Read-Only

Related ACSPL+ Commands[DC](#), [STOPDC](#)**Related ACSPL+ Variables**[AST](#), [DCN](#)**COM Library Methods and .NET Library Methods**

ReadVariable, DataCollection, StopCollect, WaitCollectEnd

C Library Functions

acsc_ReadReal, acsc_DataCollectionExt, acsc_StopCollect, acsc_WaitCollectEnd

3.6.3 *S_DCN***Description****S_DCN** is a scalar integer that stores a defined number of system data collection samples.**Tag**

111

Comments**S_DCN** stores a defined number of system data collection samples, as follows:

1. While a system data collection is in progress **S_DCN** displays the index of the array element that stores the *next* sample.
2. When a system data collection terminates, the variable stores the number of actually collected samples. If the data collection terminates automatically, the variable is always equal to the requested number of samples specified in the dc command. If the data collection terminates due to the [STOPDC](#) command, the variable may be less than the requested number of samples.

For *cyclic* data collection **S_DCN** displays the current number of collected samples and changes as follows:

1. At the start of data collection, **S_DCN** is assigned with zero.
2. With each sampling, **S_DCN** is incremented until it reaches the specified size of the sample array
3. **S_DCN** remains unchanged - the newest sample overwrites the oldest, so the total number of samples remains the same.

As long as cyclic data collection is in progress, the application cannot use the sample array. After the cyclic data collection finishes, the controller repacks the sample array so that the first element represents the oldest sample and the last element represents the most recent sample.

Accessibility

Read-Only

Related ACSPL+ Commands[DC](#), [STOPDC](#)

Related ACSPL+ Variables[S_ST](#), [S_DCP](#)**COM Library Methods and .NET Library Methods**

ReadVariable, DataCollection, StopCollect, WaitCollectEnd

C Library Functions

acsc_ReadInteger, acsc_DataCollectionExt, acsc_StopCollect, acsc_WaitCollectEnd

3.6.4 S_DCP**Description****S_DCP** is real variable that stores the period of system data collection sampling.**Tag**

112

Comments

When a system data collection terminates, the **S_DCP** stores the sampling *period*. Unless a temporal data collection was executed, the variable is always equal to the requested period specified in the [DC](#) command.



S_DCP is generally equal to the specified period, however because the period is rounded to an integer number of controller cycles, the actual period may differ from the period specified in the **DC** command.

Accessibility

Read-Only

Related ACSPL+ Commands[DC](#), [STOPDC](#)**Related ACSPL+ Variables**[S_ST](#), [S_DCN](#)**COM Library Methods and .NET Library Methods**

ReadVariable, DataCollection, StopCollect, WaitCollectEnd

C Library Functions

acsc_ReadReal, acsc_DataCollectionExt, acsc_StopCollect, acsc_WaitCollectEnd

3.6.5 S_ST**Description****S_ST** is a scalar integer variable that provides the state of System Data Collection.**Tag**

120

Comments

S_ST provides a bit that indicates if system data collection is currently in progress.

Bit 3:

0 - System data collection off

1 - System data in progress.

Accessibility

Read-Only

Related ACSPL+ Commands

[DC](#), [STOPDC](#)

Related ACSPL+ Variables

[S_DCN](#), [S_DCP](#)

COM Library Methods and .NET Library Methods

ReadVariable, DataCollection, StopCollect, WaitCollectEnd

C Library Functions

acsc_ReadInteger, acsc_DataCollectionExt, acsc_StopCollect

3.7 Input and Output Variables

The Input and Output variables are:

Name	Description
AIN	Analog Inputs
AINOFFS	Percent offset of analog signal from external source
AINSCALE	Define scale of analog input signal
AOUT	Analog Outputs
COMMCH	Returns the last activated communication channel.
DCOM	Drive command - in open loop
DOUT	Drive output.
EXTIN	Extended digital inputs (HSSI)
EXTOUT	Extended digital outputs (HSSI)
IN	General Purpose Digital Inputs
OUT	General Purpose Digital Outputs

Name	Description
SPIRXN	Number of words transmitted from SPI interface
SPIST	Integer array with one element for each EtherCAT node in the system. It shows the current state of the SPI communication channel.

3.7.1 AIN

Description

AIN is a real array, the size of which is determined by the total number of analog input signals in the system, and is used for defining the level of an analog signal from an external source such as a sensor or a potentiometer.

Syntax

```
value = AIN(index)
```

Arguments

index	A number between 0 up to the maximum number of analog input signals minus one.
value	value is the scaling, by percent, of the signal and ranges from -100 to +100, Default = 0.

Tag

4

Comments

None

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable, GetAnalogInput

C Library Functions

acsc_ReadReal, acsc_GetAnalogInput

3.7.2 AINOFFS

Description

AINOFFS is a real array, the size of which is determined by the total number of analog input signals in the system and is used for defining the percent offset of an analog signal from an external source such as a sensor or a potentiometer.

Syntax

```
AINOFFS(index) = value
```

index	A number between 0 up to the maximum number of analog inputs signals minus one.
value	Value is the offset, by percent, ranges from -100 to +100, Default = 0.

Tag

378

Comments

- > If used in combination with the analog input scaling feature(**AINSCALE**) then, the order of operations will be first offset and then scaling.
- > The **AIN** variable range is now between -200 to +200.

This variable is supported in version 3.10 and higher.

Related ACSPL+ Variables

AIN, AINSCALE

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable(), WriteVariable()

C Library Functions

acsc_ReadReal(), acsc_WriteReal()

3.7.3 AINSCALE

Description

AINSCALE is a real array, the size of which is determined by the total number of analog input signals in the system and is used for defining the scaling of an analog signal from an external source such as a sensor or a potentiometer.

Syntax

```
AINSCALE(index) = value
```

Arguments

index	A number between 0 up to the maximum number of analog inputs signals minus one.
value	Value is the scaling factor, ranges from -2 to +2, Default = 1.

Tag

391

Comments

If combined with the analog input offset feature(**AINOFFS**) then the order of operations will be offset and then scaling.

The **AIN** variable range is between -200 to +200.

This variable is supported in version 3.10 and higher.

Related ACSPL+ Variables**AIN, AINOFFS****Accessibility**

Read-Write

3.7.4 AOUT**Description**

AOUT is a real array, the size of which is determined by the total number of analog output signals in the system, and is used for defining the level of a general purpose analog signal that is sent to an external device.

Syntax

```
AOUT(index) = value
```

Arguments

<i>index</i>	A number between 0 up to the maximum number of analog output signals minus one.
<i>value</i>	value is the scaling, by percent, of the signal and ranges from -100 to +100, Default = 0.

Tag

5

Comments

1. Some aspects of **AOUT** are model-dependent, including the number of analog outputs and type of analog inputs (differential or single-ended).
2. In SPiiPlus controllers (not CM) **AOUT** can be used only when the axis is defined as Dummy - see [MFLAGS](#).

To define the analog output command to a drive (connected to a motor) in open loop, refer to [DCOM](#).

Accessibility

Read-Write

Related ACSPL+ Variables**MFLAGS****COM Library Methods and .NET Library Methods**

ReadVariable, WriteVariable, GetAnalogOutput, SetAnalogOutput

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_GetAnalogOutput, acsc_SetAnalogOutput

3.7.5 DOUT**Description****DOUT** is an integer array that stores the drive command (velocity loop output) for each axis.**Tag**

26

Comments**DOUT** values range from -32767 to 32767, which corresponds to the range of +/-100% command.In open loop mode (**MFLAGS.1=1**), **DOUT** is determined by the **DCOM** variable.In closed loop mode (**MFLAGS.1=0**), **DOUT** is determined by **DCOM** plus the velocity loop output.In gantry mode (**MFLAGS.25=1**) **DOUT** of the primary axis indicates the longitudinal (force) command and **DOUT** of the secondary axis indicates the rotational (torque) command.**DOUT** is updated at the MPU rate (**CTIME**) in most products, see note below.*The following ACS products only support lower rate update (once per 100msec) of DOUT:*

- > UDMpc
- > CMnt
- > UDMpm
- > MC4U with SPiiPlus NT-HP/LT/LD
- > MC4U with SPiiPlus DC-HP/LT/LD
- > SPiiPus SAnt

Accessibility

Read-Only, ReadVariable

COM Library Methods and .NET Library Methods

ReadVariable, C Library Functions, acsc_ReadInteger

3.7.6 EXTIN**Description****EXTIN** is an integer array, the size of which is determined by the total number of SPI and HSSI input signals in the system, and reads the current state of the inputs. The number of inputs depends on the number of SPI and HSSI modules in the system.

For details about the HSSI, see the *HSSI Modules Hardware Guide*.

Comments

The **SPIRXN** variable is updated every cycle with the number of active elements.

Tag

42

Accessibility

Read-Only

Related ACSPL+ Variables

EXTOUT, **IN**

COM Library Methods and .NET Library Methods

ReadVariable, GetExtInput, GetExtInputPort

C Library Functions

acsc_ReadInteger, acsc_GetExtInput, acsc_GetExtInputPort

3.7.7 EXTOUT

Description

EXTOUT is an integer array, the size of which is determined by the total number of SPI and HSSI output signals in the system, which can be used for reading or setting the current state of the outputs. The number of outputs depends on the number of HSSI modules and SPI inputs in the system.

For details about the HSSI, see the *HSSI Modules Hardware Guide*.

Syntax

EXTOUT(index) = value

Arguments

<i>index</i>	A number between 0 and 511.
<i>value</i>	value ranges from -2147483648, 2147483647, Default = 0.

Comments

When used with an SPI interface in master mode, the **EXOUT** function should be used.

The **SPICFG** function sets the number of elements which contain data.

Tag

43

Accessibility

Read-Write

Related ACSPL+ Variables

EXTIN, **OUT**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetExtOutput, SetExtOutput, GetExtOutputPort, SetExtOutputPort

C Library Functions

acsc_ReadInteger, acsc_WriteInteger, acsc_GetExtOutput, acsc_SetExtOutput, acsc_GetExtOutputPort, acsc_SetExtOutputPort

3.7.8 IN

Description

IN is an integer array, the size of which is determined by the total number of digital input signals in the system, and stores the current state of the General Purpose digital inputs.

Syntax

IN(*port*).*bit*

Arguments

<i>port</i>	A number between 0 and the total number of ports in the system minus one.
<i>bit</i>	<i>bit</i> can be 0-31.

Tag

71

Comments

General Purpose inputs are represented by bits 0..31 of **IN(port)**. Each bit reports the state of one General Purpose input.

For example, entering the query command **?IN(0).0** in the SPiiPlus MMI Application Studio **Communication Terminal** will return "0" if inputs #0 is non-active or "1" when active.

 In some SPiiPlus controllers, the digital input pins can also be used as **MARK**.

Accessibility

Read-Only

Related ACSPL+ Variables

OUT, **EXTOUT**

COM Library Methods and .NET Library Methods

ReadVariable, GetInput, GetInputPort

C Library Functions

acsc_ReadInteger, acsc_GetInput, acsc_GetInputPort

3.7.9 OUT

Description

OUT is an integer array, the size of which is determined by the total number of digital output signals in the system, and can be used for reading or writing the current state of the General Purpose digital outputs.

Syntax

OUT(port).bit

Arguments

<i>port</i>	A number between 0 the total number of ports in the system minus one.
<i>bit</i>	<i>bit</i> can be 0-31.

Tag

94

Comments

General purpose outputs are represented by bits 0..31 of **OUT(port)**. Each bit reports the state of one general purpose output for the given port.

For example, the query command **?OUT(23).0 = 1** through the SPiiPlus MMI Application Studio **Communication Terminal** will activate the outputs #0 of port 23.



In some SPiiPlus controllers the digital output pins can also be used as **PEG** - see .

Accessibility

Read-Write

Related ACSPL+ Variables

[IN](#), [EXTIN](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, SetOutput, GetOutput, SetOutputPort, GetOutputPort

C Library Functions

acsc_ReadInteger, acsc_WriteInteger, acsc_SetOutput, acsc_GetOutput, acsc_SetOutputPort, acsc_GetOutputPort

3.7.10 SPIRXN

Description

SPIRXN is a variable that shows the number of actual words that contain data transmitted from the SPI external interface. The range is from 0 to 8.

Accessibility

Read-only

Tag

384

Comments

SPIRXN value is 0 if the SPI is disabled. This variable is supported by the UDMxx, IDMxx and ECMxx products only.

3.7.11 SPIST**Description**

SPIST is an integer array with one element for each EtherCAT node in the system. It shows the current state of the SPI communication channel.

Bit Field

The contents of each entry are interpreted according to the following table

Bit	Name	Description
0	SPI RX Overflow	1 - more than 16 SPI words received 0 - No Event
1	SPI TX Full	1 - Attempt to add another SPI word for transmission while the SPI TX FIFO is full. The FIFO depth is 16. 0 - No Event
2-15		Reserved

Comments

If a specific element has the value 0, then there are no errors in that node

TAG

385

Accessibility

Read-only

3.8 Monitoring Variables

The Monitoring variables are:

Name	Description
BCODEUSG	An integer array, with one element for each buffer in the system. It shows the actual memory allocated by a program code in KB, for each buffer.
BCODECFG	An integer array, with one element for each buffer in the system. It is used for configuration of the amount of memory in KB that is pre-allocated for program code for each buffer.

Name	Description
BGLOBCFG	Sets the amount of memory pre-allocated for global variables in the D-Buffer.
BGLOBUSG	A scalar that shows the amount of memory used by the global variables defined in the D-Buffer.
BSRCCFG	An integer array with one element for each buffer in the system. It is used for configuration of the amount of memory in KB that is pre-allocated for a program source for each buffer.
BSRCUSG	An integer array, with one element for each buffer in the system. It shows, in KB, the actual memory allocated by a program source for each buffer.
BVARUSG	An integer array with one element for each buffer in the system. It shows, in KB, the actual memory allocated by a program local variables for each buffer.
BVARCFG	An integer array with one element for each buffer in the system. It is used for configuration of the amount of memory that is pre-allocated for local variables in KB, for each buffer.
JITTER	Elapsed time between the physical timer interrupt and the SC real-time task starts working.
MSSYNC	Time difference between the master clock and the bus clock.
USGBUF	Stores the amount of MPU usage as a percentage of the specific ACSPL+ buffer in the controller cycle during the execution of real-time tasks.
USGTRACE	Stores the amount of MPU usage as a percentage of the specific real-time task in the controller cycle during the execution of real-time tasks.
SOFTIME	Parameter which specifies the EtherCAT frame delivery time in microseconds.
TIME	Elapsed Time
USAGE	MPU Usage

3.8.1 BCODECFG

Description

BCODECFG is an integer array, with one element for each buffer in the system. It is used for configuration of the amount of memory in KB that is pre-allocated for program code for each buffer.

Syntax

```
BCODECFG(Buffer_Index) = value
```

Arguments

Buffer_Index	Buffer index.
--------------	---------------

Tag

409

Comments

The values are in KB.

The default value is 128KB.

BCODEUSG can be useful for setting the value of the **BCODECFG** variable.

A new value takes affect only after controller reboot.

Related ACSPL+ Variables

BCODEUSG

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.8.2 BCODEUSG

Description

BCODEUSG is an integer array, with one element for each buffer in the system. It shows the actual memory allocated by a program code in KB, for each buffer.

Syntax

```
[command]BCODEUSG(Buffer_Index)
```

Arguments

Buffer_Index	Buffer index.
--------------	---------------

Tag

410

Comments

- > The values are in KB.
- > For empty buffer, the value is 0.

- > BCODEUSG can be useful for setting the value of the **BCODECFG** variable.

Related ACSPL+ Variables

BCODECFG

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.8.3 *BGLOBCFG*

Description

BGLOBCFG is a scalar. It sets the amount of memory pre-allocated for global variables in the D-Buffer.

Syntax

```
BGLOBCFG = value
```

Arguments

NONE

Tag

413

Comments

- > The value is in KB.
- > The default value is 1024KB.
- > **BGLOBUSG** can be useful for setting the value of the **BGLOBCFG** variable.
- > Setting the **BGLOBCFG** variable changes the value of **BGLOBCFG** only if the **S_SETUP.#VRMEMVAR** bit is ON
- > The **BGLOBCFG** variable applies only for global variables defined in the D-Buffer.
- > A new value takes effect only after controller reboot.

Related ACSPL+ Variables

BGLOBUSG

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, Write Variable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.8.4 BGLOBUSG

Description

BGLOBUSG is a scalar that shows the amount of memory used by the global variables defined in the D-Buffer.

Syntax

```
value = BGLOBUSG
```

Arguments

NONE

Tag

414

Comments

- > The value is in KB.
- > **BGLOBUSG** can be useful for setting the value of the **BGLOBCFG** variable.
- > The **BGLOBUSG** variable has a valid value only if S_SETUP.#VRMEMVAR bit is ON
- > The **BGLOBUSG** variable applies only for global variables defined in the D-Buffer

Related ACSPL+ Variables

BGLOBCFG

Accessibility

Read-only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.8.5 BSRCUSG

Description

BSRCUSG is an integer array, with one element for each buffer in the system. It shows, in KB, the actual memory allocated by a program source for each buffer.

Syntax

```
value = BSRCUSG(Buffer_Index)
```

Arguments

Buffer_Index

Buffer index.

Tag

412

Comments

- > The values are in KB
- > For empty buffers the value is 0
- > **BSRCUSG** can be useful for setting the value of the **BSRCCFG** variable
- > The value should equal the size of the source code

Related ACSPL+ Variables

BSRCCFG

Accessibility

Read-only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.8.6 BSRCCFG**Description**

BSRCCFG is an integer array with one element for each buffer in the system. It is used for configuration of the amount of memory in KB that is pre-allocated for a program source for each buffer.

Syntax

```
BSRCCFG(Buffer_Index) = value
```

Arguments

Buffer_Index

Buffer index.

Tag

411

Comments

- > The values are in KB.
- > The default value is 64KB.
- > **BSRCUSG** can be useful for setting the value of the **BSRCCFG** variable.
- > A new value takes effect only after controller reboot.

Related ACSPL+ Variables

BSRCUSG

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.8.7 BVARUSG**Description**

BVARUSG is an integer array with one element for each buffer in the system. It shows, in KB, the actual memory allocated by a program local variables for each buffer.

Syntax

```
value = BVARUSG(Buffer_Index)
```

Arguments

Buffer_Index

Buffer index.

Tag

416

Comments

- > The values are in KB.
- > For an empty buffer, the value is 0.
- > **BVARUSG** can be useful for setting the value of the **BVARCFG** variable.
- > The **BVARUSG** variable has a valid value only if **S_SETUP.#VRMEMVAR** bit is ON

Related ACSPL+ Variables

BVARCFG

Accessibility

Read-only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.8.8 BVARCFG**Description**

BVARCFG is an integer array with one element for each buffer in the system. It is used for configuration of the amount of memory that is pre-allocated for local variables in KB, for each buffer.

Syntax

```
BVARCFG(Buffer_Index) = value
```

Arguments

Buffer_Index	Buffer index.
--------------	---------------

Tag

415

Comments

- > The values are in KB.
- > The default value is 10KB.
- > **BVARUSG** can be useful for setting the value of the **BVARCFG** variable.
- > The **BVARCFG** variable takes effect only if **S_SETUP**.#VRMEMVAR bit is ON.
- > A new value takes effect only after controller rebooted.

Related ACSPL+ Variables

BVARCFG

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.8.9 JITTER

Description

JITTER is a real variable that contains the time, in microseconds, that elapsed from the physical timer interrupt until the SC real-time task starts working. This parameter shows the influence of overall hosting PC load on real-time endurance of SC.

Tag

224

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.8.10 MSSYNC

Description

MSSYNC is a real variable that contains the difference, in microseconds, between clocks of the master and the bus. This parameter shows how close the synchronization is between the two clocks.

Tag

225

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.8.11 USGBUF

Description

USGBUF array stores the amount of MPU usage as a percentage of the specific ACSPL+ buffer in the controller cycle during the execution of real-time tasks.

USGBUF is a real array with one element for each ACSPL+ buffer as follows:

- > USGBUF(0) - Buffer 0
- > USGBUF(63) - Buffer 63
- > USGBUF(64) - D-Buffer
- > USGBUF(65) - Buffer for immediate commands execution (C / COM libraries, communication terminal, etc.)
- > USGBUF(66) - Buffer for MACRO execution

Tag

246

Comments

The real-time tasks always have the greatest priority. If the usage reaches 80% or more, the response time of the controller deteriorates. In addition, it is dangerous and may cause jerks in the motion profile.

The **USGBUF** variable should be used for debugging purposes only and should not be used in real production applications. In order to use a tracing mechanism, bit 1 of **S_SETUP** variable should be set to 1.

3.8.12 USGTRACE

Description

USGTRACE array is used for storing the amount of MPU usage as a percentage of the specific real-time task in the controller cycle during the execution of real-time tasks.

USGTRACE is a real array with one element for each real-time task according to the following list:

- > USGTRACE(0) - EtherCAT communication (including communication jitter)
- > USGTRACE(1) - reading inputs, prerequisite operations for motion generator
- > USGTRACE(2) - motion generator and real-time objects
- > USGTRACE(3) - operations on axes and Servo Processor interfaces
- > USGTRACE(4) - execution of ACSPL+ buffers
- > USGTRACE(5) - writing outputs, house keeping operations



Values 6...9 are not used and reserved for future needs

Tag

245

Comments

The real-time tasks always have the greatest priority. If the usage reaches 80% or more, the response time of the controller deteriorates. In addition, it is dangerous and may cause jerks in the motion profile.

The **USGTRACE** variable should be used for debugging purposes only and should not be used in real production applications. In order to use a tracing mechanism, bit 1 of **S_SETUP** variable should be set to 1.

3.8.13 SOFTIME

Description

SOFTIME is a read-only real array, with one element for each EtherCAT node in the system, which specifies the EtherCAT frame delivery time in microseconds.

Tag

255

Accessibility

Read Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

Comments

A Bit 2 (#SOFTIME) setting enables measuring the EtherCAT frame delivery time. Currently this feature is supported by the following products only:

- > "IOMnt (rev.B2 and higher)
- > "PDMnt (rev. B3 and higher)
- > "SDMnt (rev. B2 and higher)

3.8.14 TIME

Description

TIME is a real variable that provides the defines the elapsed time (in milliseconds) from the controller power-up.

Tag

134

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.8.15 USAGE

Description

USAGE is a real variable used for storing the amount of MPU usage as a percentage of the real-time tasks in the controller cycle during the execution of real-time tasks.

Tag

137

Comments

The real-time tasks always have the greatest priority. If the usage reaches 80% or more, the response time of the controller deteriorates, in addition, it is dangerous and may cause jerks in the motion profile.

The **USAGE** variable value can be used in autoroutines for halting the application should usage exceed a certain value.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.8.16 USAGESIM

Description

USAGESIM is a real variable used for storing MPU usage as a percentage of the real-time tasks in the controller cycle during the execution of real-time tasks in the simulator.

Tag

376

Comments

The **USAGESIM** variable should be used for debugging purposes only and should not be used in real production applications.

Since the simulator runs in a non-real-time OS, USAGESIM can frequently be above 100% .

This variable is supported in version 3.00 and higher.

Accessibility

Read-only

Related ACSPL+ Variables

[USAGE](#)

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9 Motion Variables

The Motion variables are:

Name	Description
ACC	Default Acceleration
APOS	Axis Position
APOSFILT	Integer array storing the current desired motor position, including the filtering operation result
CERRK	Critical Position Error criterion for the KILL state
DEC	Default Deceleration
DECOMP	Error Correction
DELK	Defines the time delay after a kill process when CERRK is used to indicate a critical position error.
DAPOS	Delayed Axis Position
FACC	Feedback Acceleration
FEEDRF	Feedrate Override
FPOS	Feedback Position
F2POS	Secondary Feedback Position

Name	Description
FVEL	Feedback Velocity
F2VEL	Secondary Feedback Velocity
GACC	Group Acceleration
GJERK	Group Jerk
GMOT	Motion Number
GMQU	Motion Queue
GMTYPE	Motion Type
GPATH	Group Path
GPHASE	Motion Phase
GRTIME	Remaining Motion Time
GSEG	Motion Segment
GSFREE	Free Motion Segments
GSNAP	A real array, with one element for each axis in the system, and is used for defining the current calculated snap vector snap of group motion.
GVEC	Group Vector
GVEL	Group Velocity
JERK	Default jerk
KDEC	Default kill deceleration
LPOS	Axis position in LCS
NVEL	Sets a non-zero axis velocity in stepper motor applications
SETTLEA	Sets the time to wait inside the target radius before triggering
SETTLEB	Sets the time to wait inside the target radius before triggering
SETTLEC	Sets the time to wait inside the target radius before triggering
SLSFF	A SNAP feed forward parameter.

Name	Description
SNAP	A real array with one element for each axis in the system, and is used for defining the snap (jerk derivative) of the motion profile in user-defined units.
STLTIMEA	Last settling time of axis, according to SETTLEA and TARGRADA criteria
STLTIMEB	Last settling time of axis, according to SETTLEB and TARGRADB criteria
STLTIMEC	Last settling time of axis, according to SETTLEC and TARGRADC criteria
SETTLEC	Sets the time to wait inside the target radius before triggering
TARGRADA	Sets the target radius around which you wish the motion to settle.
TARGRADB	Sets the target radius around which you wish the motion to settle.
TARGRADC	Sets the target radius around which you wish the motion to settle.
TPOS	Target position for track motion
PE	Position Error
RACC	Reference Acceleration
PPOS	Reference Position, 20kHz
PRFLTIME	Holds the time(In milliseconds) passed from the moment that the move starts until the motion profile ends
PPOSCOMP	Reference Jerk
RJERK	Reference Position
RPOSCOMP	Reference Position
RPOSDEL	Actual motor motion delay
RSNAP	A real array, with one element for each axis in the system, and is used for defining the current calculated reference snap (jerk derivation) in user-defined units.
RVEL	Reference Velocity
VEL	Velocity

3.9.1 ACC

Description

ACC is a real array, with one element for each axis in the system, and is used for defining the motion profile acceleration.

Syntax

ACC(*axis_index*) = *value*

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 2.22507e-308 to 1.79769e+308, Default = 100000.

Tag

1

Comments

For single-axis motion, **ACC** defines the axis acceleration. If the axis is a leading axis in a group, **ACC** defines the vector acceleration of the common motion.

If **ACC** is changed when a motion is in progress, the change does not affect currently executing motions, or motions that were created before the change.

Accessibility

Read-Write



ACC values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Commands

IMM, **SLAVE**, and all motion commands where the profile is generated by the controller.

Related ACSPL+ Variables

DEC, **JERK**, **KDEC**, **VEL**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, SetAcceleration. GetAcceleration, SetAccelerationImm

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_SetAcceleration. acsc_GetAcceleration, acsc_SetAccelerationImm

3.9.2 APOS

Description

APOS is a real array, with one element for each axis in the system, and is used for defining the current reference value for the axis in user-defined units.

Syntax

See [SET](#)

Tag

6

Comments

APOS is updated each MPU cycle if a motion that involves the axis is in progress.

If the corresponding motor has a default connection ([MFLAGS](#)(axis_index).#DEFCON =1), **APOS** = [RPOS](#).

Accessibility

Read-Only - Can be changed using [SET](#).

Related ACSPL+ Commands

[MASTER](#), [SLAVE](#)

Related ACSPL+ Variables

[MPOS](#)

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.3 APOSFILT**Description**

APOSFILT is real array with one element for each axis in the system. The array elements store the current desired motor position, including the filtering operation result, such as input shaping.



APOSFILT updates on every controller cycle according to the filtering algorithm. When the filtering algorithm is not configured, **APOSFILT** = **APOS**.

TAG

368

Comments

This variable is supported in version 3.00 and higher.

ACCESSIBILITY

Read-Only

RELATED ACSPL+ COMMANDS

All motion commands

RELATED ACSPL+ VARIABLES

FPOS, RPOS, APOS,

.NET LIBRARY METHODS

ReadVariable(), WriteVariable()

C LIBRARY FUNCTIONS

acsc_ReadReal(), acsc_WriteReal()

3.9.4 CERRK

Description

CERRK is a real array, with one element for each axis in the system, and is used for defining the Critical Position Error criterion for the **KILL** state.

Syntax

```
CERRK(Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 2.22507e-308 to 1.79769e+308. Default: 1000

Tag

347

Comments

CERRK defines the maximum tolerable critical position error (**FAULT(axis_index).#PE**) during the kill operation.

The value of **CERRK** is always equal or higher than that of **CERRA**.

If **CERRK** is assigned a new value which is lower than **CERRA**, **CERRK** is set to **CERRA** automatically.

If **CERRA** is assigned a new value which is higher than **CERRK**, **CERRK** is set to **CERRA** automatically.

Related ACSPL+ Variables

FAULT(Axis_Index).#PE, FDEF

Accessibility

Read-Write



CERRK values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio > Toolbox > Application Development > Protection.

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadReal(), acsc_WriteReal()

3.9.5 DAPOS

Description

DAPOS is a real array, with one element for each axis in the system. **DAPOS** reads the delayed Axis Position value which is synchronized with **RPOS** and **FPOS**.

Syntax

DAPOS is activated as part of the SPiiPlus MMI Application Studio **Scope**.

Tag

18

Comments

Use **DAPOS** only to view the axis position in the **Scope** when comparing the axis position to the **RPOS**.

In the SPiiPlus, **APOS** (axis position) is not synchronized with **RPOS** and **FPOS** and are characterized by a few msec delay.

When implementing a non-default **CONNECT**, it may be necessary to monitor **APOS** versus **RPOS** with the Scope.

Accessibility

Read-Only

Related ACSPL+ Commands

CONNECT

Related ACSPL+ Variables

APOS, **RPOS**

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.6 DEC

Description

DEC is a real array, with one element for each axis in the system and is used for specifying the motion profile deceleration in milliseconds.

Syntax

DEC(axis_index) = value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 2.22507e-308 to 1.79769e+308, Default = 100000.

Tag

22

Comments

For single-axis motion, **DEC** defines axis deceleration. If the axis is a leading axis in a group, **DEC** defines the vector deceleration of the common motion.

If **DEC** is changed when a motion is in progress, the change does not affect currently executing motions or motions that were created before the change.

Accessibility

Read-Write



DEC values cannot be modified if protection is applied to this variable through **SPiiPlus**
MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Commands

[HALT](#) , [IMM](#), [SLAVE](#)

Related ACSPL+ Variables

[ACC](#), [JERK](#), [KDEC](#), [VEL](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, SetDeceleration. GetDeceleration, SetDecelerationImm

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_SetDeceleration. acsc_GetDeceleration, acsc_SetDecelerationImm

3.9.7 *DECOMP*

Description

DECOMP is a real array, with one element for each axis in the system, and is used for displaying the error correction for the mechanical error compensation that was applied to the axis. **DECOMP** displays the difference between **RPOS** and **RPOSCOMP**.

Tag

344

Accessibility

Read-Only

Related ACSPL+ Commands

All motion commands

Related ACSPL+ Variables

FPOS, RPOS, RPOSCOMP, APOS, PE .

NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadReal(), acsc_WriteReal()

3.9.8 DELK**Description**

DELK is a real array, with one element for each axis in the system, and is used for defining the time delay after a kill process in which we still use **CERRK** to indicate a critical position error.

Syntax

```
DELK(Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 0 to 1000.

Tag

350

Comments

CERRK defines the maximum tolerable critical position error (**FAULT(axis_index).#CPE**) during the kill operation. After the kill operation in order to ensure a smooth transition, we still use **CERRK** to define the maximal tolerable position error, for **DELK** time.

Related ACSPL+ Variables

FAULT(Axis_Index).#CPE, CERRK

Accessibility

Read-Write



DELK values cannot be modified if protection is applied to this variable through SPiPlus MMI Application Studio > Toolbox > Application Development > Protection.

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadReal(), acsc_WriteReal()

3.9.9 *FACC*

Description

FACC is a real array, with one element for each axis in the system, and is used for defining the feedback acceleration value of the axis.

Tag

46

Accessibility

Read-Only

Related ACSPL+ Variables

[FVEL](#)

COM Library Methods and .NET Library Methods

ReadVariable, GetAcceleration

C Library Functions

acsc_ReadReal, acsc_GetAcceleration

3.9.10 *FPOS*

Description

FPOS is a real array, with one element for each axis in the system, and is used for defining the current feedback position for the motor.

Tag

52

Comments

The user can shift the origin of feedback position using [SET](#).

The user can select the units of feedback position by setting the [EFAC](#) variable.

Accessibility

Read-Only

Related ACSPL+ Commands

[SET](#)

Related ACSPL+ Variables

[APOS](#), [RPOS](#), [SLPROUT](#)

COM Library Methods and .NET Library Methods

ReadVariable, GetFPosition, SetFPosition

C Library Functions

acsc_ReadReal, acsc_GetFPosition, acsc_SetFPosition

3.9.11 F2POS

Description

F2POS is a real array, with one element for each axis in the system, and is used for defining the current secondary feedback value for the motor in user-defined units.

Tag

44

Comments

The user can shift the origin of secondary feedback position using [SET](#).

The user can select the units of secondary feedback position by setting the [E2FAC](#) variable.

The application needs to explicitly clear [IST\(axis_index\).](#)**#IND2** in order to resume the latching logic.

Accessibility

Read-Only - Can be changed by [SET](#).

Related ACSPL+ Commands

[SET](#), [SLP2ROUT](#)

Related ACSPL+ Variables

[IST](#)

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.12 FVEL

Description

FVEL is a real array, with one element for each axis in the system, the elements of which store the measured velocity.

Tag

53

Accessibility

Read-Only

Related ACSPL+ Variables

[FVFIL](#), [RVEL](#), [XVEL](#)

COM Library Methods and .NET Library Methods

ReadVariable, GetFVelocity

C Library Functions

acsc_ReadReal, acsc_GetFVelocity

3.9.13 F2VEL

Description

F2VEL is a real array, with one element for each axis in the system, the elements of which store the measured secondary velocity.

Tag

45

Accessibility

Read-Only

Related ACSPL+ Variables

[FVFIL](#), [RVEL](#), [XVEL](#), [FVEL](#)

COM Library Methods and .NET Library Methods

ReadVariable, GetFVelocity

C Library Functions

acsc_ReadReal, acsc_GetFVelocity

3.9.14 FEEDRF

Description

FEEDRF is real array, with one element for each axis in the system, the elements of which store the feedrate factor. The feedrate factor modifies the calculation of motion velocity for all relevant motion profiles.

Examples

```
FEEDRF(2) = 1.23
```

```
IMM FEEDRF(0) = 0.5
```

Tag

362

Comments

This variable may be updated immediately using the "IMM" qualifier.

The allowed range is 0.1 to 2.0.

Acceleration and Jerk are NOT affected, which actually changes the trajectory characteristics (may result in Triangular instead of Trapezoidal Trajectory).

It takes effect on the next Trajectory calculation, according to the specified velocity of that trajectory; situation varies according to PTP Switches.

In case of group motion, the FEEDRF of the leading Axis will be in effect.

Motion Modes in which FEEDRF is supported: PTP, JOG, TRACK, MPTP, XSEG

Accessibility

Read-Write

Related ACSPL+ Commands

IMM, PTP, JOG, TRACK, MPTP, XSEG

Related ACSPL+ Variables

VEL

3.9.15 GACC**Description**

GACC is a real array, with one element for each axis in the system, and is used for deriving the vector acceleration of a group motion.

For example when the three axes 0, 1 and 2 are moving as a group, the **GACC** is calculated by:

$$GACC = \sqrt{(Acceleration_0)^2 + (Acceleration_1)^2 + (Acceleration_2)^2}$$



GPATH, **GVEL**, **GACC**, **GPHASE**, **GJERK**, and **GRTIME** Variables are updated while the motion is in progress.

Tag

55

Accessibility

Read-Only

Related ACSPL+ Commands**GROUP****Related ACSPL+ Variables****GVEL**, **GJERK**, **GPATH**, **GPHASE**, **GRTIME****COM Library Methods and .NET Library Methods**

ReadVariable, GetAcceleration

C Library Functions

acsc_ReadReal, acsc_GetAcceleration

3.9.16 GJERK**Description**

GJERK is a real array, with one element for each axis in the system, and is used for deriving the vector acceleration of a group motion.

For example when the three axes 0, 1 and 2 are moving as a group, the **GJERK** is calculated by:

$$GJERK = \sqrt{(Jerk_0)^2 + (Jerk_1)^2 + (Jerk_2)^2}$$



GPATH, **G**VEL, **G**ACC, **G**JERK, **G**PHASE, and **G**RTIME variables are updated while the motion is in progress.

Tag

57

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.17 **GMOT**

Description

GMOT is an integer array, with one element for each axis in the system, and defines the ordinal number of the current motion.

Tag

58

Comments

The **GMOT** value is valid only if one of the following is true:

- > Single-axis motion in progress
- > The axis is a leading axis in a group and motion in the group is in progress

After power-up, **GMOT** is zero and increments each time a motion of the corresponding axis/axis group terminates.

GMOT resets to zero each time the axis group is created or split.

Accessibility

Read-Only.

Related ACSPL+ Commands

GROUP, **S**PLIT

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.9.18 GMQU

Description

GMQU is an integer array, with one element for each axis in the system, and defines the total number of motions in the motion queue including the currently executing motion. The maximum motion queue per axis is 5.

Tag

59

Comments

GMQU is valid only if one of the following is true:

- > Single-axis motion in progress
- > The axis is a leading axis in a group and motion in the group is in progress

After power-up **GMQU** is zero. The variable is incremented by one each time a new motion of the corresponding axis/axis group is issued. It is decremented by one each time a motion of the corresponding axis/axis group terminates.

GMQU resets to zero each time an axis is regrouped, i.e., a group that contains the axis is created or split-up.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.9.19 GMTYPE

Description

GMTYPE is an integer array, with one element for each axis in the system. The MPU updates **GMTYPE** each time a motion involving the corresponding axis or axis group starts or terminates.

Tag

60

Comments

GMTYPE is updated according to the type of the motion as follows:

- 0 - no motion
- 1 - **PTP** motion
- 2 - **MPTP...ENDS** motion
- 3 - **TRACK** motion
- 4 - **MSEG...ENDS** motion
- 5 - **JOG** motion

6 - **SLAVE** motion

7 - **PATH...ENDS** motion

8 - **PVSPLINE...ENDS** motion

10 - **XSEG...ENDS** motion

11 - **BPTP** motion

12 - **BSEG...ENDS** motion

43 - **BPTP/2** motion using 20 kHz control

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.9.20 **GPATH**

Description

GPATH is a real array, with one element for each axis in the system. **GPATH** defines the current path value, defined as the distance from the motion origin to the current motion point, or in the case of Extended Segmented Motion, the distance from the beginning of the first segment.

Tag

61

Comments

GPATH updates each MPU cycle if one of the following is true:

- > Single-axis motion in progress
- > The axis is a leading axis in a group and motion in the group is in progress

If either of these conditions is not true, **GPATH** retains its previous value.

For single-axis motion, **GPATH** defines a positive distance from the initial point of the motion.

If the axis is a leading axis, **GPATH** defines a vector distance along the trajectory from the motion origin.



GPATH, **GVEL**, **GACC**, **GJERK**, **GPHASE**, and **GRTIME** variables are updated while the motion is in progress.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.21 GPHASE

Description

GPHASE is an integer array, with one element for each axis in the system. **GPHASE** defines the current phase of a motion.

Tag

62

Comments

GPHASE can have the following values:

0 - no motion

1 - acceleration buildup

2 - constant acceleration

3 - acceleration finishing

4 - constant velocity

5 - deceleration buildup

6 - constant deceleration

7 - deceleration finishing

8 - kill deceleration

9 - asynchronous phase of master-slave motion

10 - synchronous phase of master-slave motion

11 - stalled phase of master-slave motion.

13 - dwell phase in [JOG](#) or [MPTP...ENDS](#) motions, or no defined target point in [PATH...ENDS](#), [PVSPLINE...ENDS](#) or [MPTP...ENDS](#) motions.

31 – Jerk buildup when acceleration buildup

32 – Jerk finishing when acceleration buildup

33 – Jerk buildup when acceleration finishing

34 – Jerk finishing when acceleration finishing

35 – Jerk buildup when deceleration buildup

36 – Jerk finishing when deceleration buildup

37 – Jerk buildup when deceleration finishing

38 – Jerk finishing when deceleration finishing



GPATH, **GVEL**, **GACC**, **GJERK**, **GPHASE**, and **GRTIME** variables are updated while the motion is in progress.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.9.22 GRTIME

Description

GRTIME is a real array, with one element for each axis in the system. **GRTIME** defines an estimated value of time (in milliseconds) remaining until the end of the current motion.

Tag

63

Comments

GRTIME updates each MPU cycle if one of the following is true:

- > Single-axis motion in progress
- > The axis is a leading axis in a group and motion in the group is in progress

GRTIME does not update if the motion is **JOG** or **MASTER SLAVE**. If **GRTIME** does not update, it retains its previous value.

Normally, 1-2 msec after motion starts, **GRTIME** accepts the correct value. In rare cases, the **GRTIME** value remains high during motion phases 1 and 2, and accepts correct value at the beginning of phase 3.



GPATH, **GVEL**, **GACC**, **GJERK**, **GPHASE**, and **GRTIME** variables are updated while the motion is in progress.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.23 GSEG

Description

GSEG is an integer array, with one element for each axis in the system. **GSEG** defines the ordinal number of the currently executing segment.

Tag

64

Comments

GSEG updates only under one of the following conditions:

- > Single-axis motion in progress, or
- > The axis is a leading axis in a group and motion in the group is in progress

If either of these conditions is not true, **GSEG** retains its previous value.

GSEG updates as follows:

- > If the current motion in the axis/axis group is not **MSEG...ENDS**, the **GSEG** value is -1.
- > The value resets to zero when a multi segment motion starts
- > The value increments each time when the motion passes from one segment to the next.
- > The value decrements each time the motion passes from one segment to the previous (possible only in master-slave motion).
- > Because the motion returns to the start point in cyclic motion, **GSEG** may appear greater than the number of a segment in the motion, if the motion overruns the segment sequence in positive direction.
- > For master-slave cyclic motion, **GSEG** may appear negative, if the motion overruns the segment sequence in a negative direction.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.9.24 GSFREE

Description

GSFREE is an integer array, with one element for each axis in the system. **GSFREE** is updated for the leading axis with the number of free cells in the segment queue.

Tag

65

Comments

If **GSFREE** is zero, the segment queue is full and the next coming **POINT** or **MPOINT** command will be delayed until the required number of cells are freed.

Accessibility

Read-Only

Related ACSPL+ Variable

GSEG

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.9.25 GSNAP**Description**

GSNAP is a real array, with one element for each axis in the system, and is used for defining the current calculated snap vector snap of group motion.

Tag

407

Accessibility

Read-write

3.9.26 GVEC**Description**

GVEC is a real array, with one element for each axis in the system. **GVEC** is updated each MPU cycle, if a motion involving the axis is in progress. If the motion is not in progress, **GVEC** retains its previous value.

Tag

66

Comments

In single-axis motion, **GVEC** = 1 or -1, depending on the motion direction.

In multi-axis group motion, **GVEC** values for all axes in the group are updated each MPU cycle and together build up a tangent vector for the motion trajectory.

GVEC can also be used for retrieving a tangent vector.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.27 GVEL

Description

GVEL is a real array, with one element for each axis in the system, and is used for deriving the vector velocity of a group motion.

For example when the three axes 0, 1 and 2 are moving as a group, the **GVEL** is calculated by:

$$GVEL = \sqrt{(Velocity_0)^2 + (Velocity_1)^2 + (Velocity_2)^2}$$



GPATH, **GVEL**, **GACC**, **GJERK**, **GPHASE**, and **GRTIME** variables are updated while the motion is in progress

Tag

67

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Lisbrary Functions

acsc_ReadReal

3.9.28 JERK

Description

JERK is a real array, with one element for each axis in the system, and is used for defining the jerk of the motion profile.

Syntax

JERK(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 2.22507e-308 to 1.79769e+308, Default = 2e+007.

Tag

80

Comments

For single-axis motion, **JERK** defines the axis jerk. If the axis is a leading axis in a group, **JERK** defines vector jerk of the common motion.

If **JERK** is changed when a motion is in progress, the change does not affect currently executing motions, or motions that were created before the change.

Accessibility

Read-Write



JERK values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Commands

IMM, and all motion commands where the profile is generated by the controller.

Related ACSPL+ Variables

ACC, **DEC**, **KDEC**, **VEL**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetJerk, SetJerk, SetJerkImm

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_GetJerk, SetJerk, acsc_SetJerkImm

3.9.29 KDEC

Description

KDEC is a real array, with one element for each axis in the system, and is used for defining deceleration when a motion is killed by the user or fails due to a fault.

Syntax

KDEC(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 2.22507e-308 to 1.79769e+308, Default = 100000.

Tag

81

Comments

For single-axis motion, the value defines axis deceleration. If the axis is a leading axis in a group, **KDEC** defines the vector deceleration when the common motion is killed or fails.

If **KDEC** is changed when a motion is in progress, the change does not affect currently executing or motions that were created before the change.

Accessibility

Read-Write



KDEC values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Variables

[ACC](#), [DEC](#), [JERK](#), [VEL](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, SetKillDeceleration, GetKillDeceleration, SetKillDecelerationImm

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_SetKillDeceleration, acsc_GetKillDeceleration, acsc_SetKillDecelerationImm

3.9.30 LPOS

Description

LPOS is a real array, with one element for each axis in the system, the elements of which store the axis position in the Local Coordinate System.

Syntax

LPOS(*axis_index*) = value

Arguments

<i>axis_index</i>	Designates the specific axis, valid values are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 2.22507e-308 to 1.79769e+308; default = 50.

Tag

372

Accessibility

Read-Only

Related ACSPL+ Variables

"FPOS" on page 368, "RPOS" on page 390, "APOS" on page 362

.NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.9.31 MPOS

Description

MPOS is a real array, with one element for each axis in the system, and defines the current master position value for the axis in user units.

Tag

89

Comments

MASTER must precede **MPOS** for the specified axis. **MPOS** updates each controller cycle according to the formula specified in **MASTER**.

Accessibility

Read-Only

Related ACSPL+ Commands

MASTER, **SLAVE**

Related ACSPL+ Variables

FPOS, **F2POS**, **APOS**

COM Library Methods and .NET Library Methods

ReadVariable, SetMaster, Slave

C Library Functions

acsc_ReadReal, acsc_SetMaster, acsc_Slave

3.9.32 MTIMEA

Description

MTIMEA returns the time elapsed from start of motion up to first entering the settled zone, using **SETTLEA** and **TARGRADA** to determine the time and radius required for settling.

Syntax

```
Value = MTIMEA(Axis_Index)
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
------------	---

Tag

341

Comments

We compare the distance between current position and target position to the given target radii, represented by **TARGRADA**. This indicates if the motor is in the target zone, if it stays in the target zone for at least **SETTLEA** time - bit **MST.#INTARGA** is raised, respectively, and depending on the operating mode – further inspection will be stopped or continued.

The time from the beginning of motion until first entering the settled zone is represented by **MSTIMEA** and is only valid when the **#INTARGA** bit is on.

Related ACSPL+ Variables

TARGRADA, SETTLEA, MST(axis_index).#INTARGA

Accessibility

Read-only



MSTIMEA values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio > Toolbox > Application Development > Protection.



MSTIMEA is only updated if the Move & Settle feature is enabled by using **SETCONF(318)** to enable either single mode or auto mode.

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadReal()

3.9.33 *MSTIMEB*

Description

MSTIMEB returns the time elapsed from start of motion up to first entering the settled zone, using **SETTLEB** and **TARGRADB** to determine the time and radius required for settling.

Syntax

```
Value = MSTIMEB (Axis_Index)
```

Arguments

Axis_
Index

Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Tag

342

Comments

We compare the distance between current position and target position to the given target radii, represented by **TARGRADB**. This indicates if the motor is in the target zone, if it stays in the target zone for at least **SETTLEB** time - bit **MST.#INTARGB** is raised, respectively, and depending on the operating mode – further inspection will be stopped or continued.

The time from the beginning of motion until first entering the settled zone is represented by **MSTIME_B** and is only valid when the **#INTARGB** bit is on.

Related ACSPL+ Variables

TARGRADB, SETTLEB, MST(axis_index).#INTARGB

Accessibility

Read-only



MSTIMEB values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio > Toolbox > Application Development > Protection.



MSTIMEB is only updated if the Move & Settle feature is enabled by using **SETCONF(318)** to enable either single mode or auto mode.

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadReal()

3.9.34 *MSTIMEC*

Description

MSTIMEC returns the time elapsed from start of motion up to first entering the settled zone, using **SETTLEC** and **TARGRADC** to determine the time and radius required for settling.

Syntax

```
Value = MSTIMEC(Axis_Index)
```

Arguments

Axis_
Index

Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Tag

343

Comments

We compare the distance between current position and target position to the given target radii, represented by **TARGRADC**. This indicates if we the motor is in the target zone, if it stays in the target zone for at least **SETTLEC** time - bit **MST.#INTARGC** is raised, respectively, and depending on the operating mode – further inspection will be stopped or continued.

The time from the beginning of motion until first entering the settled zone is represented by **MSTIMEC** and is only valid when the **#INTARGC** bit is on.

Related ACSPL+ Variables

TARGRAD, SETTLEC, MST(axis_index).#INTARGC

Accessibility

Read-only



MSTIMEC values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio > Toolbox > Application Development > Protection.



MSTIMEC is only updated if the Move & Settle feature is enabled by using **SETCONF**(318) to enable either single mode or auto mode.

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadReal()

3.9.35 NVEL

Description

NVEL is a real array, with one element for each axis in the system, and is used for specifying the start and the end velocities for an axis in stepper motor applications.

Syntax

NVEL(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 0 to 1.79769e+308, Default = 0.

Tag

91

Comments

1. An **NVEL** element affects the motion of the corresponding axis and the multi-axis motions if the axis is a leading axis in the group.
2. If an element is zero, the normal motion profile starts from zero velocity and finishes at zero velocity.
If an element is non-zero, at the beginning of motion the velocity immediately jumps to the **NVEL** value and then continues the regular motion profile. At the end of the motion, the

motion approaches the final point at the velocity specified by **NVEL**, and then immediately drops to zero. For example, **KILL/KILLALL** and **HALT** slow the velocity to the value specified in **NVEL**, and then the velocity drops to zero.

Accessibility

Read-Write



NVEL values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection.

Related ACSPL+ Commands

IMM, and all motion commands where the profile is generated by the controller.

Related ACSPL+ Variables

VEL

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.9.36 PE

Description

PE is a real array, with one element for each axis in the system, and is used for displaying the difference between **RPOS** and **FPOS** (the current position error) denoting a noncritical position error.



The **PE** value is valid only if the motor is enabled.

Tag

98

Accessibility

Read-Only

Related ACSPL+ Commands

All motion commands.

Related ACSPL+ Variables

FPOS, **RPOS**, **FAULT**

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.37 PPOS

Description

PPOS is real array, with one element for each axis in the system, the elements of which store the current desired motor reference position. Unlike **RPOS**, this holds the current value, rather than a value taking into account the delay for reading the actual current position from the encoder.

Tag

364

Comments

When the motor is disabled, **RPOS** = **FPOS**.

This variable is supported in ADK versions 2.70 and higher.

Accessibility

Read-Only

Related ACSPL+ Commands

SET, CONNECT, and all motion commands.

Related ACSPL+ Variables

FPOS, RVEL, RACC

COM Library Methods and .NET Library Methods

ReadVariable, GetRPosition, SetRPosition

C Library Functions

acsc_ReadReal, acsc_GetRPosition, acsc_SetRPosition

3.9.38 PPOSCOMP

Description

PPOSCOMP is real array, with one element for each axis in the system, the elements of which store the current desired motor reference position including dynamic error compensation. Unlike **RPOSCOMP**, this holds the current value, rather than a value taking into account the delay for reading the actual current position from the encoder.

Tag

365

Comments

This variable is supported in ADK versions 2.70 and higher.

Accessibility

Read-Only

Related ACSPL+ Commands

All motion commands.

Related ACSPL+ Variables

PPOS, FPOS, RPOS, APOS, PE, DECOMP

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadReal(), acsc_WriteReal()

3.9.39 PRFLTIME**Description**

PRFLTIME is a real array, the size of which is determined by the total number of axes in the system. It holds the time(In milliseconds) passed from the moment that the move starts until the motion profile ends.

Syntax

```
value = PRFLTIME(index)
```

Arguments

index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
-------	---

Tag

405

Comments

- > The **PRFLTIME** variable is updated with the latest profile time for the relevant axes.
- > If we have multi-axes move, the profile time for all the involved axes is the same and will be updated once the profile has ended for all the axes.
- > A kill/error event is regarded as profile end.
- > If the profile starts and ends in the same cycle (for example, move from the current axis position to the same position), the profile time will be 0.

Related ACSPL+ Variables**Accessibility**

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable()

C Library Functions

acsc_ReadReal()

3.9.40 RACC

Description

RACC is a real array, with one element for each axis in the system, and defines the current reference acceleration value for the motor in user-defined units.

Tag

106

Comments

RACC updates each controller cycle, and is calculated by digital differentiation of [RVEL](#).

Accessibility

Read-Only

Related ACSPL+ Commands

All motion related commands.

Related ACSPL+ Variables

[RVEL](#), [RPOS](#)

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

Real

3.9.41 RJERK

Description

RJERK is a real array, with one element for each axis in the system, and defines the current calculated reference jerk value for the motor in user-defined units.

RJERK is updated each controller cycle and is calculated as digital differentiation of [RACC](#).

Tag

259

Accessibility

Read-Only

3.9.42 ROFFS

Description

ROFFS is a real array, with one element for each axis in the system, the elements of which store the Reference Offset.

Tag

107

Comments

As long as the motor is in the default connection (**MFLAGS**(*axis*).#**DEFCON** = 1), offset **ROFFS** is zero. Once a user specifies connect formula such as:

CONNECT **RPOS**(0) = F(...)

the controller calculates offset **ROFFS**(0) to prevent a sudden change in **RPOS**(0) that may cause the motor to jump. The controller then calculates:

RPOS(0) = F(...) + **ROFFS**(0)

each controller cycle.

The controller recalculates **ROFFS** to prevent motor jump when the commands **CONNECT**, **SET**, **ENABLE/ENABLE ALL**, **DISABLE/DISABLEALL**, **KILL** are executed. **ROFFS** reads the current value of the offset.



Watching the **ROFFS** value facilitates development and debugging of applications with complex kinematics.

Accessibility

Read-Only

Related ACSPL+ Commands

CONNECT, **SET**, **ENABLE/ENABLE ALL**, **DISABLE/DISABLEALL**, **KILL/KILLALL**

Related ACSPL+ Variables

MFLAGS(axis_index).#**DEFCON** (bit 17 = Default Connection)

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.43 RPOS**Description**

RPOS is real array, with one element for each axis in the system, the elements of which store the current desired motor reference position.

Tag

108

Comments

RPOS updates each MPU cycle according to the connection specified for the motor, see **CONNECT**.

When the motor is disabled, **RPOS** = **FPOS**.

Accessibility

Read-Only

Related ACSPL+ Commands

[SET](#), [CONNECT](#), and all motion commands.

Related ACSPL+ Variables

[FPOS](#), [RVEL](#), [RACC](#)

COM Library Methods and .NET Library Methods

ReadVariable, GetRPosition, SetRPosition

C Library Functions

acsc_ReadReal, acsc_GetRPosition, acsc_SetRPosition

3.9.44 RPOSCOMP

Description

RPOSCOMP is real array, with one element for each axis in the system, the elements of which store the current desired motor reference position including dynamic error compensation.



RPOSCOMP updates every controller cycle according to the configured dynamic error compensation, see **ERRORMAP1D**, **ERRORMAPN1D**, **ERRORMAP2D**, **ERRORMAPN2D**.

When the dynamic error compensation is not configured, **RPOSCOMP** = **RPOS**.

Tag

348

Accessibility

Read-Only

Related ACSPL+ Commands

All motion commands.

Related ACSPL+ Variables

[FPOS](#), [RPOS](#), [APOS](#), [PE](#), [DECOMP](#)

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadReal(), acsc_WriteReal()

3.9.45 RPOSDEL

Description

RPOSDEL shows the actual delay time that is currently set. The delay value is rounded (ceiling function) to 50 µsec. At the beginning of the motion, which is delayed, the parameter indicates the specified delay. At the end of the motion, the delay is gradually reduced to zero.

Syntax

```
INT val = RPOSDEL
```

Tag

329

Accessibility

Read-Only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadInteger()

3.9.46 *RSNAP*

Description

RSNAP is a real array, with one element for each axis in the system, and is used for defining the current calculated reference snap (jerk derivation) in user-defined units.

Tag

408

3.9.47 *RVEL*

Description

RVEL is a real array, with one element for each axis in the system, the elements of which store the current motor reference velocity in user-defined units.

Tag

109

Accessibility

Read-Only

Related ACSPL+ Commands

All motion commands.

Related ACSPL+ Variables

[RPOS](#), [RACC](#), [FVEL](#)

COM Library Methods and .NET Library Methods

ReadVariable, GetRVelocity

C Library Functions

acsc_ReadReal, acsc_GetRVelocity

3.9.48 SETTLEA

Description

SETTLEA allows you to set the time to wait inside the TARGRADA before triggering MST.#INTARGA.

Syntax

```
SETTLEA(Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 2.22507e-308, 1.79769e+308, Default = 0.

Tag

338

Comments

We compare the distance between current position and target position to the given target radii, represented by **TARGRADA**. This indicates if we the motor is in the target zone, if it stays in the target zone for at least **SETTLEA** time - bit **MST.#INTARGA** is raised, respectively, and depending on the operating mode – further inspection will be stopped or continued.

The time from the beginning of motion until first entering the settled zone is represented by **MSTIMEA** and is only valid when the **#INTARGA** bit is on.

Related ACSPL+ Variables

TARGRADA, MST(axis_index).#INTARGA, MSTIMEA

Accessibility

Read-Write



SETTLEA values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio > Toolbox > Application Development > Protection .

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadReal(), acsc_WriteReal()

3.9.49 SETTLEB

Description

SETTLEB allows you to set the time to wait inside the TARGRADB before triggering MST.#INTARGB.

Syntax

```
SETTLEB(Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 2.22507e-308, 1.79769e+308, Default = 0.

Tag

339

Comments

We compare the distance between current position and target position to the given target radii, represented by **TARGRADB**. This indicates if the motor is in the target zone, if it stays in the target zone for at least **SETTLEB** time - bit **MST.#INTARGB** is raised, respectively, and depending on the operating mode – further inspection will be stopped or continued.

The time from the beginning of motion until first entering the settled zone is represented by **MSTIMEB** and is only valid when the **#INTARGB** bit is on.

Related ACSPL+ Variables

TARGRADB, MST(axis_index).#INTARGB, MSTIMEB

Accessibility

Read-Write



SETTLEB values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio > Toolbox > Application Development > Protection .

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadReal(), acsc_WriteReal()

3.9.50 SLSFF

Description

SLSFF is a snap feed forward parameter.

Syntax

```
SLSFF(Axis_index) = Value
```

Arguments

Axis_ index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Resonance frequency (Hz) of a non-collocated system. Value ranges from 0 to 20000; Default: 0.

Tag

418

Comments

Flexible systems consist of a non-collocated feedback sensor, which is mounted away from the motor or the load. Those systems can suffer from vibration and dynamic performance due to mechanical resonance, which is caused by compliance between two or more components in the mechanical transmission, mostly between motor and load. The snap feed-forward reduces the resonance frequency oscillations and improves the high-speed and large acceleration of jerk profiles.

The **SLSFF** value is the resonance frequency in Hz.

The **SLAFF** (acceleration feedforward) must be tuned before activating the **SLSFF**.

SLSFF can be used with **SPTP** or **BPTP** motions.

SLSFF is not supported in the 20KHz motion mode (**BPTP/2**).

Example

Figure 3-1 shows an FRF measurement of the velocity and position loop of a flexible system, including a motor and encoder on opposite sides. The resonance frequency of the system is 70 Hz.

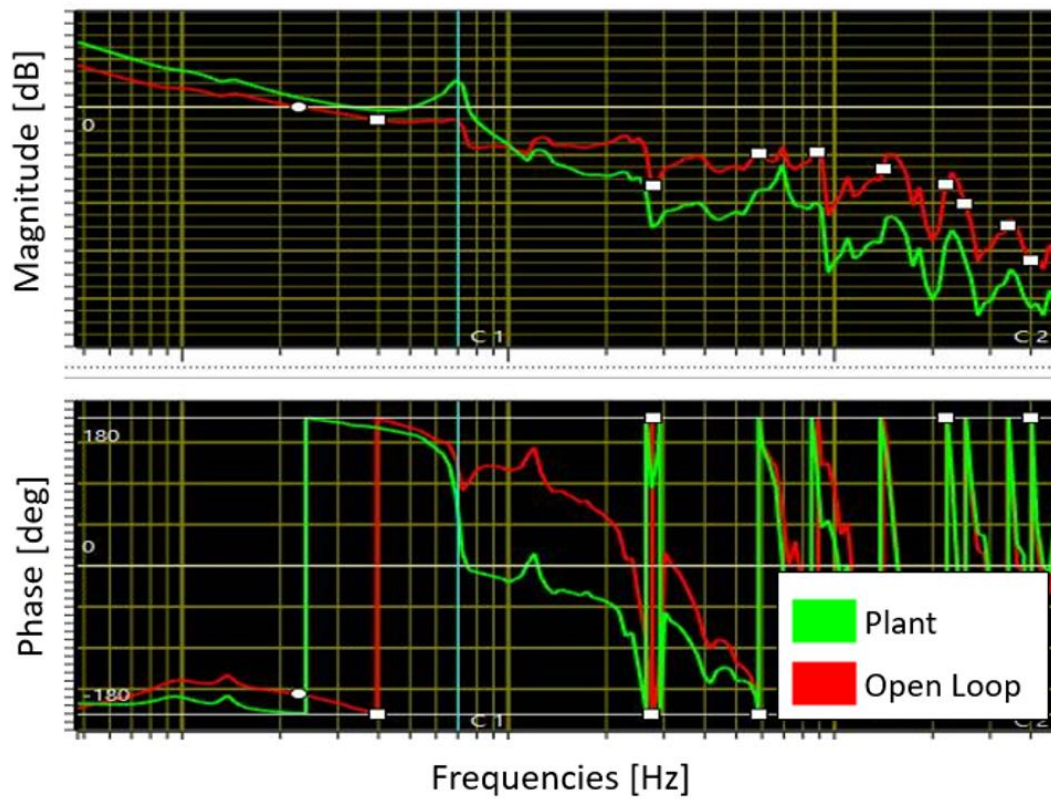


Figure 3-1. FRF measurement of a flexible system

The resonance frequency is excited during a short and aggressive movement. [Figure 3-2](#) describes a short SPTP motion without snap feed-forward (**SLSFF=0**).

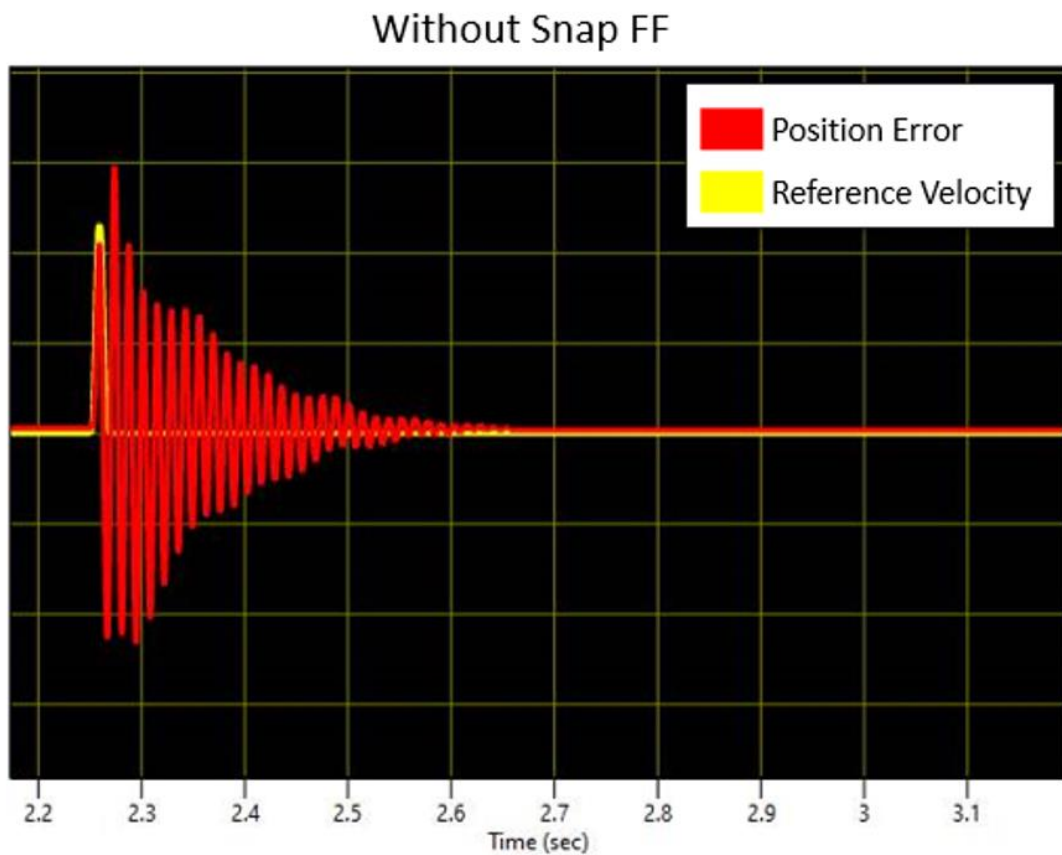


Figure 3-2. SPTP motion where SLSFF=0 (no snap feed-forward)

Snap feedforward is a form of control that accounts for the flexibility of a system and significantly improves performance and reduces resonance vibrations.

Figure 3-3 shows the same SPTP motion where setting the SLSFF motion as the resonance frequency of the system (**SLSFF=70**):

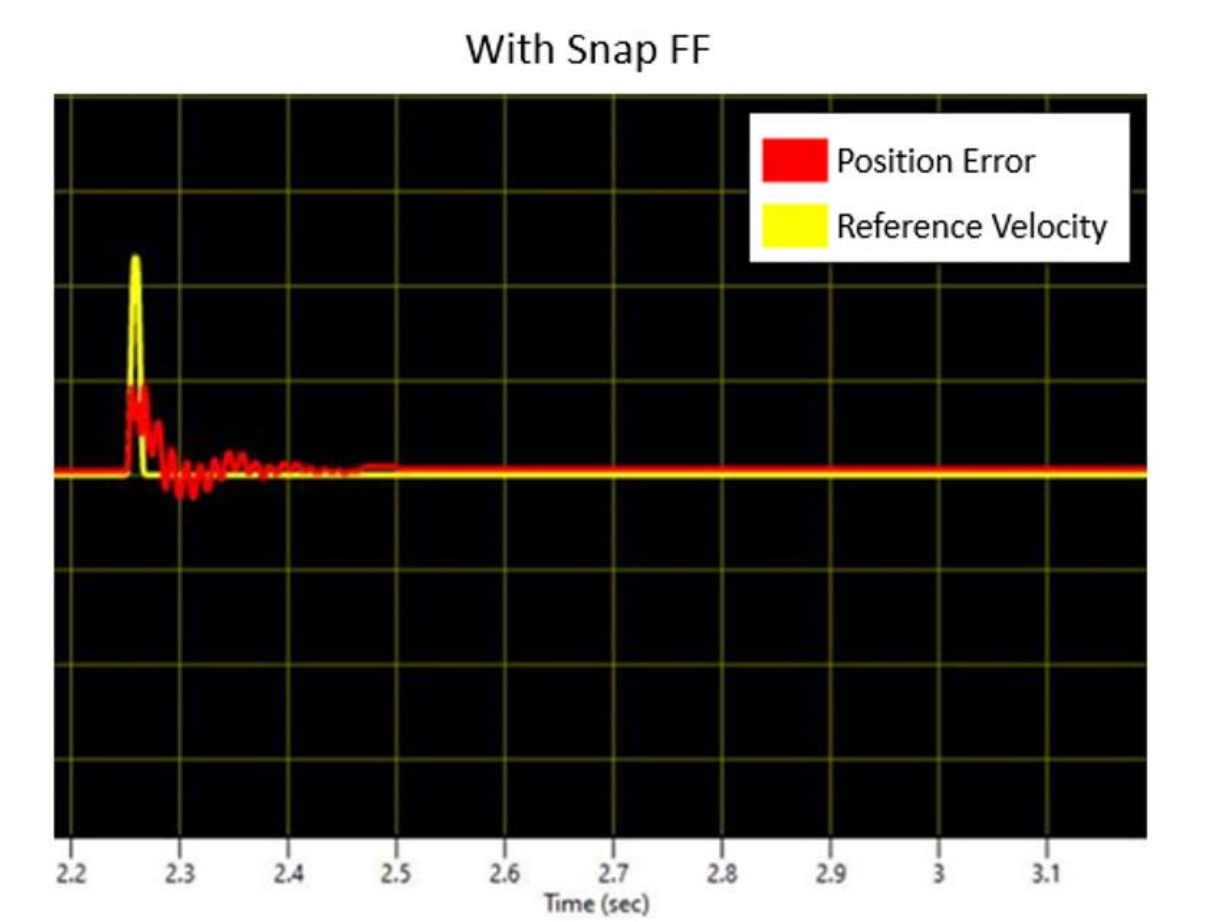


Figure 3-3. SPTP motion where SLSFF=70 (snap feed-forward is tuned)

3.9.51 *SETTLEC*

Description

SETTLEC allows you to set the time to wait inside the TARGRADC before triggering MST.#INTARGC.

Syntax

```
SETTLEC(Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 2.22507e-308, 1.79769e+308, Default = 0.

Tag

340

Comments

We compare the distance between current position and target position to the given target radii, represented by **TARGRADC**. This indicates if the motor is in the target zone, if it stays in the target zone for at least **SETTLEC** time - bit **MST.#INTARGC** is raised, respectively, and depending on the operating mode – further inspection will be stopped or continued.

The time from the beginning of motion until first entering the settled zone is represented by **MSTIMEC** and is only valid when the **#INTARGC** bit is on.

Related ACSPL+ Variables

TARGRADC, MST(axis_index).#INTARGC, MSTIMEC

Accessibility

Read-Write



SETTLEC values cannot be modified if protection is applied to this variable through SPiPlus MMI Application Studio > Toolbox > Application Development > Protection .

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadReal(), acsc_WriteReal()

3.9.52 SNAP

Description

SNAP is a real array with one element for each axis in the system, and is used for defining the snap (jerk derivative) of the motion profile in user-defined units.

Syntax

```
SNAP(axis_index) = value
```

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value value ranges from 2.22507e-308 to 1.79769e+308, Default = 1e+009

Tag

406

Accessibility

Read-write

3.9.53 STLTIMEA

Description

STLTIMEA is a real array, with one element for each axis in the system. Each element holds the last settling time of the axis, i.e., the time passed since the profile generation has been completed up to first entering the settled zone, using **SETTLEA** and **TARGRADA** to determine the time and radius required for settling.

Syntax

```
value = STLTIMEA(axis_index)
```

Arguments

axis_index

Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Tag

392

Comments

- > In cases where the motion settled within the specified radius before the profile has finished, **STLTIME** = 0. Otherwise, **STLTIME** [axis_index] = **MSTIME**[axis_index] - **PRFLTIME** [axis_index].
- > This feature has two operating modes, which are controlled through **SETCONF** with key 318, where index indicates the axis number and the value the mode, value of 0 is the default and deactivates the feature. See **SETCONF** documentation for more details.
- > Only motions which support **TPOS** may be used to measure the settle time. These motions are: **PTP**, **MPTP**, and **TRACK**.
- > Once the **MST**(axis_index).#INTARGA bit is set(1), the **STLTIMEA** variable holds an updated value of the axis settling time.

Related ACSPL+ Variables

MSTIMEA, **TARGRADA**, **SETTLEA**, **MST**(axis_index).#INTARGA

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.54 STLTIMEB

Description

STLTIMEB is a real array, with one element for each axis in the system. Each element holds the last settling time of the axis, i.e., the time passed since the profile generation has been completed up to

first entering the settled zone, using **SETTLEB** and **TARGRADB** to determine the time and radius required for settling.

Syntax

```
value = STLTIMEB(axis_index)
```

Arguments

axis_index

Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Tag

393

Comments

- > In cases where the motion settled within the specified radius before the profile has finished, **STLTIME** = 0. Otherwise, **STLTIME** [axis_index] = **MSTIME**[axis_index] - **PRFLTIME** [axis_index].
- > This feature has two operating modes, which are controlled through **SETCONF** with key 318, where index indicates the axis number and the value the mode, value of 0 is the default and deactivates the feature. See **SETCONF** documentation for more details.
- > Only motions which support **TPOS** may be used to measure the settle time. These motions are: **PTP**, **MPTP**, and **TRACK**.
- > Once the **MST**(axis_index).#INTARGB bit is set(1), the **STLTIMEB** variable holds an updated value of the axis settling time.

Related ACSPL+ Variables

MSTIMEB, **TARGRADB**, **SETTLEB**, **MST**(axis_index).#INTARGB

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.55 STLTIMEC

Description

STLTIMEC is a real array, with one element for each axis in the system. Each element holds the last settling time of the axis, i.e., the time passed since the profile generation has been completed up to first entering the settled zone, using **SETTLEC** and **TARGRADC** to determine the time and radius required for settling.

Syntax

```
value = STLTIMEC(axis_index)
```

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
------------	---

Tag

392

Comments

- > In cases where the motion settled within the specified radius before the profile has finished, **STLTIME** = 0. Otherwise, **STLTIME** [axis_index] = **MSTIME**[axis_index] - **PRFLTIME** [axis_index].
- > This feature has two operating modes, which are controlled through **SETCONF** with key 318, where index indicates the axis number and the value the mode, value of 0 is the default and deactivates the feature. See **SETCONF** documentation for more details.
- > Only motions which support **TPOS** may be used to measure the settle time. These motions are: **PTP**, **MPTP**, and **TRACK**.
- > Once the **MST**(axis_index).#INTARGC bit is set(1), the **STLTIMEC** variable holds an updated value of the axis settling time.

Related ACSPL+ Variables

MSTIMEC, TARGRAD, SETTLE, MST(axis_index).#INTARGC

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.9.56 TARGRADA

Description

TARGRADA is a variable designed to define the target radius around which you wish the motion to settle.

Syntax

```
TARGRADA(Axis_Index) = Value
```

Arguments

Axis_ Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 2.22507e-308, 1.79769e+308, Default = 1.

Tag

335

Comments

We compare the distance between current position and target position to the given target radii, represented by **TARGRADA**. This indicates if the motor is in the target zone, if it stays in the target zone for at least **SETTLEA** time - bit **MST.#INTARGA** is raised, respectively, and depending on the operating mode – further inspection will be stopped or continued.

The time from the beginning of motion until first entering the settled zone is represented by **MSTIMEA** and is only valid when the **#INTARGA** bit is on.

Related ACSPL+ Variables

SETTLEA, MST(axis_index).#INTARGA, MSTIMEA

Accessibility

Read-Write



TARGRADA values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio > Toolbox > Application Development > Protection.

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadReal(), acsc_WriteReal()

3.9.57 TARGRADB**Description**

TARGRADB is a variable designed to define the target radius around which you wish the motion to settle.

Syntax

```
TARGRADB(Axis_Index) = Value
```

Arguments

Axis_ Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 2.22507e-308, 1.79769e+308, Default = 1.

Tag

336

Comments

We compare the distance between current position and target position to the given target radii, represented by **TARGRADB**. This indicates if we the motor is in the target zone, if it stays in the target zone for at least **SETTLEB** time - bit **MST.#INTARGB** is raised, respectively, and depending on the operating mode – further inspection will be stopped or continued.

The time from the beginning of motion until first entering the settled zone is represented by **MSTIMEB** and is only valid when the **#INTARGB** bit is on.

Related ACSPL+ Variables

SETTLEB, MST(axis_index).#INTARGB, MSTIMEB

Accessibility

Read-Write



TARGRADB values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio > Toolbox > Application Development > Protection.

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadReal(), acsc_WriteReal()

3.9.58 TARGRADC**Description**

TARGRADC is a variable designed to define the target radius around which you wish the motion to settle.

Syntax

```
TARGRADC(Axis_Index) = Value
```


Arguments

Axis_ Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value ranges from 2.22507e-308, 1.79769e+308, Default = 1.

Tag

337

Comments

We compare the distance between current position and target position to the given target radii, represented by **TARGRADC**. This indicates if we the motor is in the target zone, if it stays in the target zone for at least **SETTLEC** time - bit **MST.#INTARGC** is raised, respectively, and depending on the operating mode – further inspection will be stopped or continued.

The time from the beginning of motion until first entering the settled zone is represented by **MSTIMEC** and is only valid when the **#INTARGC** bit is on.

Related ACSPL+ Variables

SETTLE_C, MST(axis_index).#INTARGC, MSTIMEC

Accessibility

Read-Write



TARGRADC values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio > Toolbox > Application Development > Protection.

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadReal(), acsc_WriteReal()

3.9.59 TPOS**Description**

TPOS is a real array, with one element for each axis in the system, and is used for defining or updating the target position in TRACK motion.

Syntax

TPOS(axis_index) = value

Arguments

axis_ index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from -1.79769e+308 to 1.79769e+308, Default = 0.

Tag

135

Comments

The controller update occurs as follows:

- > When the controller executes **PTP** motion, the axes' target coordinates are stored in the **TPOS** elements.
- > During **MPTP...ENDS** motion, the controller updates the target coordinates each time motion to the next point starts.
- > When the controller executes **TRACK** motion, the axes' target coordinates are stored in the **TPOS** elements.

Accessibility

Read-Write



TPOS values cannot be modified if protection is applied to this variable through SPiPlus
MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Commands

TRACK

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, Track

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_Track

3.9.60 VEL

Description

VEL is a real array, with one element for each axis in the system, and is used for defining the default velocity of the motion profile. If a motion command does not specify a specific velocity, the default value is used.

Syntax

VEL(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from -1.79769e+308 to 1.79769e+308, Default = 10000.

Tag

139

Comments

For single-axis motion, the value defines axis velocity. If the axis is a leading axis in a group, its value defines a vector velocity of common motion.

If **VEL** is changed when a motion is in progress, the change does not affect currently executing motions or motions that were created before the change.

Accessibility

Read-Write



VEL values cannot be modified if protection is applied to this variable through **SPiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Commands

IMM, and all motion commands where the profile is generated by the controller.

Related ACSPL+ Variables

ACC, **DEC**, **JERK**, **KDEC**,

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, SetVelocity, GetVelocity, SetVelocityImm

C Library Functions

acsc_ReadReal, acsc_WriteReal, acsc_SetVelocity, acsc_GetVelocity, acsc_SetVelocityImm

3.10 Program Execution Control Variables

The Program Execution Control variables are:

Name	Description
ONRATE	Autoroutine Rate
PCHARS	Program Size in Characters
PERL	Program Error Line
PERR	Program Error
PEXL	Executed Line
PFLAGS	Program Flags
PLINES	Number of Lines
PRATE	Program Rate
PST	Program State

3.10.1 ONRATE

Description

ONRATE is an integer array with one element for each program buffer plus one for the D-Buffer and is used for controlling the autoroutine execution rate.

Syntax

ONRATE(*buffer_index*) = *value*

Arguments

<i>buffer_index</i>	buffer index - a number between 0 and the total number of buffers minus one (the highest number is that of the D-Buffer).
<i>value</i>	value ranges from 1 to 10, Default = 1.

Tag

93

Comments

ONRATE is set through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Program Manager → Program Buffer Parameters**.

When an autoroutine executes in the program buffer, the execution rate is **ONRATE** lines per each MPU cycle. The normal rate of program execution (when no autoroutine is activated) is defined by **PRATE**.

Accessibility

Read-Write



ONRATE values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.10.2 PCHARS

Description

PCHARS is an integer array with one element for each program buffer plus one for the D-Buffer that stores the total number of characters stored in the buffer.

Tag

95

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.10.3 PERL**Description**

PERL is an integer array with one element for each program buffer plus one for the D-Buffer that stores the line number where the error occurred.

Tag

99

Comments

If an error occurs during ACSPL+ program execution, the controller stores the line number where the error occurred in the corresponding element of the **PERL** array.

Accessibility

Read-Only

Related ACSPL+ Variables[PERR](#)**COM Library Methods and .NET Library Methods**

ReadVariable, GetProgramError

C Library Functions

acsc_ReadInteger, acsc_GetProgramError

3.10.4 PERR**Description**

PERR is an integer array with one element for each program buffer plus one for the D-Buffer that stores an error code.

Error codes are found in [Table 7-2](#) and [Table 7-3](#).

Tag

100

Comments

If an error occurs during ACSPL+ program execution, the controller stores the error code in the corresponding element of the **PERR** array.

Accessibility

Read-Only

Related ACSPL+ Variables[PERL](#)**COM Library Methods and .NET Library Methods**

ReadVariable, GetProgramError

C Library Functions

acsc_ReadInteger, acsc_GetProgramError

3.10.5 PEXL**Description**

PEXL is an integer array with one element for each program buffer plus one for the D-Buffer that stores the number of the currently executed line.

Tag

101

Comments

PEXL stores the number of the currently executed line in the buffer. If the program has not executed, the variable reads zero.

Accessibility

Read-Only

Related ACSPL+ Variables[PERL](#), [PERR](#)**COM Library Methods and .NET Library Methods**

ReadVariable

C Library Functions

acsc_ReadInteger



PEXL does not support D-Buffer.

3.10.6 PFLAGS**Description**

PFLAGS is an integer array with one element for each program buffer plus one for the D-Buffer, each element of which contains a set of bits that defines the behavior of the program buffer.

When the #JIT(Just In Time) bit is ON the controller waits for a file loading operation (via MMI or host application program) and can start executing the commands in the buffer immediately after the loading process is completed.

The Just in Time buffer acts as a FIFO for the ACSPL+ commands which are read from the file. After the file is loaded, the buffer can be executed any number of times.



The **#JIT** bit can be set to ON only if the buffer is empty. Error 3204 "JIT and Dynamic modes require the buffer to be empty" is returned if the buffer is not empty.

Syntax

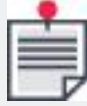
PFLAGS(buffer_index).(bit) = 0|1

Arguments

<i>buffer_index</i>	buffer index - a number between 0 and 64 (64 being the D-Buffer).
<i>bit</i>	Table 3-10

Table 3-10. PFLAGS Bit Description 1

Bit Name	No.	Description
#NOAUTO	0	0 (default): Autoroutines (if exist in the buffer) are enabled. 1: Autoroutines (if exist in the buffer) are disabled.
#NOEDIT	1	0 (default): User program can be viewed and edited. 1: User program can be viewed but cannot be edited.  Supported after applying protection, see Protection Wizard in the <i>MMI Application Studio User Guide</i>.
#DYNAMIC	2	0 (default): Buffer works in normal order. 1: Buffer works in Dynamic mode.
#JIT	3	0 (default): Buffers works in normal order. 1: Buffer works in JIT mode.
#PRIVLG	4	0 (default): Buffer works in normal order. 1: Sets the buffer as privileged which means that the program in the buffer can change the values of protected variables, start and stop other ACSPL+ programs, and execute any other action that in a regular buffer would cause a protection violation.
#DEBUG	5	0 (default): Buffer works in normal order. 1: Not applicable (obsolete option)

Bit Name	No.	Description
#NOVIEW	6	0 (default): The program is visible in the buffer. 1: The program in the buffer hidden from being viewed.  Supported after applying protection, see Protection Wizard in the <i>MMI Application Studio User Guide</i>.

Comments

The bit cannot be applied for the D-Buffer. Attempt to setting the bit for the D-Buffer will result in error 3200 "JIT is not allowed for D-Buffer".

Tag

102

Accessibility

Read-Write



PFLAGS values cannot be modified if protection is applied to this variable through **SPIIPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

Example

```
PFLAGS (0) #JIT=1
```

3.10.7 PLINES**Description**

PLINES is an integer array with one element for each program buffer plus one for the D-Buffer each element of which contains the total number of lines stored in the associated buffer.

Tag

103

Accessibility

Read-Only

Related ACSPL+ Variables

[PCHARS](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger

3.10.8 PRATE

Description

PRATE is an integer array with one element for each program buffer, each element of which is used for defining the program execution rate for that buffer.

Syntax

PRATE(*buffer_index*) = *value*

Arguments

<i>buffer_index</i>	buffer index - a number between 0 and N, N being the number of buffers in the product or system.
<i>value</i>	value ranges from 1 to 10, Default = 1.

Tag

104

Comments

PRATE is set through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Program Manager → Program Buffer Parameters**.

PRATE defines the program execution rate. The execution rate is **PRATE** lines per each MPU cycle.

PRATE is used only if no autoroutine is activated in the buffer. While an autoroutine is executed, **ONRATE** defines execution rate.

For example, if the controller is configured so that **PRATE(2)** is 1, but **ONRATE(2)** is 4, the program in Buffer 2 will be executed one line per one controller cycle, and any autoroutine specified in Buffer 2 that interrupts the program will be executed four lines per one controller cycle. When the **RET** command that terminates the autoroutine is executed, the controller switches back to the rate of one line per one cycle.

Accessibility

Read-Write



PRATE values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Variables

ONRATE

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.10.9 PST

Description

PST is an integer array with one element for each program buffer plus one for the D-Buffer each element of which contains a set of bits that display the current state of the given program buffer.

The **PST** bits are detailed in [Table 3-11](#).

Table 3-11. PST Bit Description

Bit Name	No.	Description
#COMPILED	0	0: Program in buffer is not compiled 1: Program in buffer is compiled
#RUN	1	0: Program is not running. 1: Program is running.
#SUSPEND	2	0: Program in buffer is not suspended. 1: Program is suspended after STEP or due to a breakpoint in debug mode.
#DEBUG	5	0: Buffer works in normal order (default). 1: Not applicable (obsolete option)
#AUTO	7	0: Autoroutine is not running 1: Autoroutine is running.

Tag

105

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable, GetProgramState

C Library Functions

acsc_ReadInteger, acsc_GetProgramState

3.11 Safety Control Variables

The Safety Control variables are:

Name	Description
CERRA	Critical Position Error (Accelerating)
CERRI	Critical Position Error (Idle)
CERRV	Critical Position Error (Velocity)
DELI	Delay on Transition to Idle State
DELV	Delay on Transition to Velocity State
E_ERR	An integer array for each axis containing the encoder error code identified during the encoder initialization process
EEXTST	EtherCAT state of the SPiiPlusES slave
EEXTERR	SPiiPlusES EtherCAT error code (based on Application Level Error Code).
FAULT	Faults
FAULTSIM	Fault Simulation
ECST	Contains EtherCAT status
EC2ST	EtherCAT status in dual EC configuration
ECERR	Contains EtherCAT error code
ECALError	Contains EtherCAT slave error code
EC2ALERR	Contains EtherCAT slave error code for second network in dual EC configuration
EC2ERR	EC error code for second network in dual EC configuration
ECSYNC	Indicates synchronization between controller and node
EC2SYNC	Indicates synchronization between controller and node in second EC network
ECNST	Provides information EC process information
EC2NST	Provides information EC process information for the second EC network
FDEF	Default Response Mask
FMASK	Fault Mask

Name	Description
HLLROUT	Integer array used for mapping between an axis hardware left limit to a specified digital input bit
HRLROUT	Integer array used for mapping between an axis hardware righthlimit to a specified digital input bit
MERR	Motor Error
SAFIN	Safety Inputs
SAFINI	Safety Inputs Inversion
S_ERR	System Error
S_FAULT	System Faults
S_FDEF	System Default Response Mask
S_FMASK	System Fault Mask
S_SAFIN	System Safety Inputs
S_SAFINI	System Safety Inputs Inversion
SS1TIME	Store SS1-t channel A time between emergency stop request and drive switching to torque off mode
SS12TIME	Store SS1-t channel B time between emergency stop request and drive switching to torque off mode
STODELAY	Configures the delay time between the STO fault indication and the default response (disable) to the fault
SYNC	Slave synchronization indicator

3.11.1 **E_ERR**

Description

E_ERR is an integer array for each axis. It contains the encoder error code that was identified during the encoder initialization process.

The encoder errors range from 5121 to 5128 and are latched in the **E_ERR** variable. The error codes are specified in [Table 7-5](#) in the Error Codes section.

Comments

This variable is supported in version 3.00 and higher.

Accessibility

Read-Only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadInteger()

3.11.2 EC2ERR**Description**

EC2ERR is a scalar variable containing an EtherCAT error code referring to the second EtherCAT network in a Dual EtherCAT system. The EtherCAT error codes are given in [Table 7-7](#).

Syntax**EC2ERR****Arguments**

None

Tag

420

Comments

Any EtherCAT error sets **EC2ST.#OP** to false and the error code is latched in **EC2ERR**.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.11.3 ECAERR**Description**

ECAERR is an integer array for each EtherCAT slave in the configuration (ENI file). It contains the AL Status Code error of the slave, when the value "0" indicates no error.

Tag

328

Accessibility

Read-Only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadInteger()

The error codes are defined according to AL Status Code (ETG 1020).

The error codes are listed in [Table 7-8](#).

3.11.4 EC2ALERR**Description**

EC2ALERR is an integer array for each EtherCAT slave in the configuration (ENI file) of the second EtherCAT network in a Dual EtherCAT configuration. It contains the AL Status Code error of the slave, when the value "0" indicates no error.

Tag

421

Accessibility

Read-Only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadInteger()

The error codes are defined according to AL Status Code (ETG 1020).

The error codes are listed in [Table 7-8](#).

3.11.5 ECERR**Description**

ECERR is a scalar variable containing an EtherCAT error code. The EtherCAT error codes are given in [Table 7-7](#).

Syntax**ECERR****Arguments**

None

Tag

239

Comments

Any EtherCAT error sets **ECST.#OP** to false and the error code is latched in **ECERR**.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.11.6 ECEXTERR

Description

ECEXTERR is a scalar (INT) variable representing the EtherCAT error code of the SPiiPlusES (based on Application Level Error Code). The error code range is 7000-7999. The error codes are specified in [Table 7-8](#).



Can be used only by SPiiPlusES.

Syntax

ECEXTERR

Arguments

None

Comments

If the controller is not SPiiPlusES, the value is always 0.

Tag

323

Accessibility

Read-Only

Com Library Methods and .NET Library Methods

ReadVariable

C Library Functions

Acsc_ReadInteger

3.11.7 ECEXTST

Description

ECEXTST is a scalar (INT) variable representing the EtherCAT state of the SPiiPlusES slave. The state is reflected in the relevant bits.



Can be used only by SPiiPlusES.

Bit	Designator	Description
0	#INIT	SPiiPlusES is in INIT state
1	#BOOT	SPiiPlusES is in BOOT state (currently not supported)
2	#PREOP	SPiiPlusES is in PREOP state
3	#SAFEOP	SPiiPlusES is in SAFEOP state
4	#ES_OP	SPiiPlusES is in OP state
5	#ES_DC	Distributed Clocks are ON
6	#WATCHDOG	PDI Watchdog status. 0: expired, 1: reloaded
7	#LNKPORTA	Physical Link Port A
8	#LNKPORTB	Physical Link Port B
9	#EXTSYNC	Synchronized to External Clock

Syntax**ECEXTST****Arguments**

None

Comments

Comments

If the controller is not SPiiPlusES, , MPU3U, or IDMSm, all bits are 0. If external EtherCAT master is not connected, the slave is in the INIT state.

Bit 9 #EXTSYNC is relevant for IDMSm only.

Tag

322

Accessibility

Read-Only

Com Library Methods and .NET Library Methods

ReadVariable

C Library Functions

Acsc_ReadInteger

3.11.8 ECST

Description

ECST is a scalar variable affecting the EtherCAT state. The EtherCAT state is reflected in the first six bits, as shown in [Table 3-12](#).

Table 3-12. ECST Bits

Bit	Designator	Description
0	#SCAN	The scan process was performed successfully, that is, the Master was able to detect what devices are connected to it.
1	#CONFIG	There is no deviation between XML and actual setup. The Master succeeded to initialize the network by steps described in configuration file.
2	#INITOK	All bus devices are successfully set to INIT state. The Master started all devices to the initial state.
3	#CONNECTED	Indicates valid Ethernet cable connection to the master. The physical link of EtherCAT cable is OK on the Master side.
4	#INSYNC	If DCM is used, indicates synchronization between the Master and the bus.
5	#OP	The EtherCAT bus is operational. The Master successfully turned each Slave into full operational mode and the bus is ready for full operation.
6	#DCSYNC	Distributed clocks are synchronized.
7	#RINGMODE	Ring topology mode is selected.
8	#RINGCOMM	Ring Communication is active.
9	#EXTCONN	External clock is connected
10	#DCXSYNC	External clock/slaves are synchronized

Syntax

ECST.bit_designator = 1|0

Arguments

None

Tag

238

Comments

All bits (except #INSYNC in some cases) should be true for proper bus functioning.

For monitoring the bus state, checking bit #OP is sufficient. Any bus error will reset the #OP bit.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.11.9 EC2ST**Description**

EC2ST is a scalar variable reflecting the EtherCAT state of the second EtherCAT network in a Dual EtherCAT configuration. The EtherCAT state is reflected in the first six bits, as shown in [Table 3-13](#).

Table 3-13. EC2ST Bits

Bit	Designator	Description
0	#SCAN	The scan process was performed successfully, that is, the Master was able to detect what devices are connected to it.
1	#CONFIG	There is no deviation between XML and actual setup. The Master succeeded to initialize the network by steps described in configuration file.
2	#INITOK	All bus devices are successfully set to INIT state. The Master started all devices to the initial state.
3	#CONNECTED	Indicates valid Ethernet cable connection to the master. The physical link of EtherCAT cable is OK on the Master side.
4	#INSYNC	If DCM is used, indicates synchronization between the Master and the bus.
5	#OP	The EtherCAT bus is operational. The Master successfully turned each Slave into full operational mode and the bus is ready for full operation.
6	#DCSYNC	Distributed clocks are synchronized.
7	#RINGMODE	Ring topology mode is selected.
8	#RINGCOMM	Ring Communication is active.

Bit	Designator	Description
9	#EXTCONN	External clock is connected
10	#DCXSYNC	External clock/slaves are synchronized

Syntax

EC2ST.*bit_designator* = 1|0

Arguments

None

Tag

419

Comments

EC2ST reflects the EtherCAT state of the second EtherCAT network, if it's not supported all bits are OFF.

All bits (except #INSYNC in some cases) should be true for proper bus functioning.

For monitoring the bus state, checking bit #OP is sufficient. Any bus error will reset the #OP bit.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.11.10 ECSYNC**Description**

ECSYNC is an integer array with an element for each ACS EtherCAT node in the first EtherCAT network. The value represents synchronization between the controller and the node.

Syntax

```
ECSYNC(index)
```

Arguments

EtherCAT node index

Tag

429

Comments

ECSYNC's values depend on CTIME and on the product type.

Accessibility

Read-only

.NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.11.11 EC2SYNC

EC2SYNC is an integer array with an element for each ACS EtherCAT node in the second EtherCAT network in a Dual EtherCAT setup. The value represents synchronization between the controller and the node.

Syntax

```
EC2SYNC (index)
```

Arguments

EtherCAT node index

Tag

422

Comments

EC2SYNC values depend on CTIME and on the product type.

Accessibility

Read-only

.NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.11.12 ECNST**Description**

ECNST is an integer array, each element for ACS EtherCAT node in the first EtherCAT network. The variable provides information regarding different processes such as GPRT, Data collection and fault indication.

Syntax

```
ECNST (index) .bit_designator=1|0
```

Arguments

EtherCAT node index

Bit Name	Bit No.	Description
#SYNC	0	Sync Error
#GPRT	1	GPRT error
#MISSING	2	EtherCAT Node is missing
#SPDC	3	0: SPDC data collection not active 1: SPDC data collection active
#FSYNC	4	FPGA Sync Fault
#SPRT	5	0: Real-time data transfer process from the MPU to a corresponding Servo Processor is not active 1: Real-time data transfer process from the MPU to a corresponding Servo Processor is active
#LCI	10	LCI Fault

Tag

430

Comments

If the #SYNC or #GPRT error bit is set in ECNST, there will be a network error in all axes in the first EtherCAT network, and servo processor alarm in all the axes controlled by the node. These faults indicate a problem in the interface between the firmware and the node.

The setting of the #SYNC error bit means that the node is out of synchronization with the master. The setting of the #GPRT error bit means the queue (which has the size of 400) for the GPRT commands (which are being sent by request) was full and some commands were lost.

The following commands will set the **ECNST**.#SPRT bit to 1:

- > **SPINJECT**
- > **SPRT**
- > **ASSIGNPEG/f**
- > **BPTP/2** (20 kHz motion profile)
- > **FOLLOW** (in case of customized servo algorithm for 20 kHz motion profile)

SPINJECT, **SPRT**, **ASSIGNPEG/f**, **BPTP/2** and **FOLLOW** are mutually exclusive, meaning only one of those features can be active at a given time. So the **NST**.#SPRT bit should be checked before using any of these commands. **FCLEAR** for any axis associated with the node will reset all bits of the **NST** variable of that node.

Accessibility

Read-only

.NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.11.13 EC2NST**Description**

EC2NST is an integer array, with an element for each ACS EtherCAT node in the second EtherCAT network.

The variable provides information regarding different processes such as **GPRT**, data collection and fault indication.

Syntax

```
EC2NST(index).bit_designator=1|0
```

Arguments

EtherCAT node index

Bit Name	Bit No.	Description
#SYNC	0	Sync Error
#GPRT	1	GPRT error
#MISSING	2	EtherCAT Node is missing
#SPDC	3	0: SPDC data collection not active 1:SPDC data collection active
#FSYNC	4	FPGA Sync Fault
#SPRT	5	0: Real-time data transfer process from the MPU to a corresponding Servo Processor is not active 1:Real-time data transfer process from the MPU to a corresponding Servo Processor is active
#LCI	10	LCI Fault

Tag

423

Comments

If the #SYNC or #GPRT error bit is set in **EC2NST**, there will be a network error in all axes in the second EtherCAT network, and servo processor alarm in all the axes controlled by the node. These faults indicate a problem in the interface between the firmware and the node.

The setting of the #SYNC error bit means that the node is out of synchronization with the master. The setting of the #GPRT error bit means the queue (which has the size of 400) for the **GPRT** commands (which are being sent by request) was full and some commands were lost.

The following commands will set the **EC2NST**.#SPRT bit to 1:

- > **SPINJECT**
- > **SPRT**
- > **ASSIGNPEG/f**
- > **BPTP/2** (20 kHz motion profile)
- > **FOLLOW** (in case of customized servo algorithm for 20 kHz motion profile)

SPINJECT, **SPRT**, **ASSIGNPEG/f**, **BPTP/2** and **FOLLOW** are mutually exclusive, meaning only one of the features can be active at the given time. So the NST/#SPRT bit should be checked before using any of these commands.

FCLEAR for any axis associated with the node will reset all bits of the NST variable of that node.

Accessibility

Read-only

.NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.11.14 FAULT

Description

FAULT is an integer array, with one element for each axis in the systems, the elements of which contain a set of bits that stores axis-related fault bits.

The fault bits are detailed in [Table 3-14](#).

Table 3-14. Axis Fault Bits

Bit	Fault	Fault Description
0	#RL	Hardware Right Limit. 1 = Right limit switch is activated.
1	#LL	Hardware Left Limit. 1 = Left limit switch is activated.
2	#NT	Network Error. 1 = EtherCAT network error is activated.
4	#HOT	Motor Overheat. 1 = Motor's temperature sensor indicates overheat.

Bit	Fault	Fault Description
5	#SRL	Software Right Limit. 1 = Axis reference position (RPOS) is greater than the software right limit margin (SRLIMIT).
6	#SLL	Software Left Limit. 1 = Axis reference position (RPOS) is less than the software left limit margin (SLLIMIT).
7	#ENCNC	Encoder Not Connected. 1 = Primary encoder (for digital encoder type only) is not connected.
8	#ENC2NC	Encoder 2 Not Connected. 1 = Secondary encoder (for digital encoder type only) is not connected.
9	#DRIVE	Drive Alarm. 1 = Signal from the drive reports a failure.
10	#ENC	Encoder Error. 1 = Primary encoder miscounts.
11	#ENC2	Encoder 2 Error. 1 = Secondary encoder miscounts.
12	#PE	Position Error. 1 = Position error (PE) has occurred. PE is defined by the following variables: > ERRI - Maximum position error while the axis is idle > ERRV - Maximum position error while the axis is moving with constant velocity > ERRA - Maximum position error while the axis is accelerating or decelerating > DELI - Delay on transition from ERRA to ERRI > DELV - Delay on transition from ERRA to ERRV

Bit	Fault	Fault Description
13	#CPE	>Critical Position Error. 1 = Position error (#PE) exceeds the value of the critical limit. #CPE errors occur outside normal range of operation and #CPE > #PE. The critical limit depends on the axis state and is defined by the following variables: > CERRI if the axis is idle (not moving) > CERRV if the axis is moving with constant velocity > CERRA if the axis is accelerating or decelerating > DELI - Delay on transition from ERRA to CERRI > DELV - Delay on transition from ERRA to CERRV
14	#VL	Velocity Limit. 1 = Absolute value of the reference velocity (RVEL) exceeds the limit defined by the XVEL parameter.
15	#AL	Acceleration Limit. 1 = Absolute value of the reference acceleration (RACC) exceeds the limit defined by the XACC parameter.
16	#CL	Current Limit. 1 = RMS current calculated in the Servo Processor exceeds the limit value defined by the XRMS parameter.
17	#SP	Servo Processor Alarm. 1 = Axis Servo Processor loses its synchronization with the MPU. The fault indicates a fatal problem in the controller.
18	#STO	Safe Torque Off 1 = STO is active
20	#HSSINC	Hssi Not Connected. 1 = HSSI module is not connected.

Tag

47

Comments

FAULT indicates axis related fault bits as detected by the safety mechanism. When each of the faults is active (such as Left Limit), the corresponding fault bit becomes = 1 while the fault is active, and automatically reverts to 0 when the fault is no longer active.

- > Each fault can be masked by **FMASK**.
- > The logic of some faults can be inverted by **SAFINI**.
- > The default response of each fault can be disabled by **FDEF**. In this case, any customized default response can be implemented by autoroutines - see **ON...RET**.

For a list of **S_FAULT** related system fault bits see [S_FAULT Fault Bits](#)

Accessibility

Read-Only



FAULT values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Variables

[S_FAULT](#), [FDEF](#), [S_FDEF](#), [FMASK](#), [S_FMASK](#), [SAFIN](#), [S_SAFIN](#), [SAFINI](#), [S_SAFINI](#)

COM Library Methods and .NET Library Methods

ReadVariable, GetFault

C Library Functions

acsc_ReadInteger, acsc_GetFault

3.11.15 FAULTSIM

Description

FAULTSIM is an integer array, with one element for each axis in the system. Each such element consists of bits representing the errors in the axis or the system itself. This variable is used to simulate these faults and raising a certain bit will trigger the fault the axis/system thereby raising **FAULT** and **S_FAULT**.

The fault bits are indicated in the following table:

Bit	Fault	Fault Description
Axis Faults		
0	#RL	Hardware Right Limit 1 = Right limit switch is activated
1	#LL	Hardware Left Limit 1 = Left limit switch is activated.

Bit	Fault	Fault Description
Axis Faults		
2	#NT	Network Error 1 = EtherCAT network error is activated
4	#HOT	Motor Overheat 1 = Motor's temperature sensore indicates overheating
5	#SRL	Software Left Limit 1 = Axis reference postion (RPOS) is greater than the software right limit margin (SRLIMIT).
6	#SLL	Software Left Limit 1 = Axis reference position (RPOS) is less than the software left limit margin (SLLIMIT).
7	#ENCNC	Encoder Not Connected 1 = Primary encoder (for digital encoder type only) is not connected.
8	#ENC2NC	Encoder 2 Not Connected 1 = Secondary encoder (for digital encoder type only) is not connected.
9	#DRIVE	Drive Fault 1 = Signal from the drive reports a failure
10	#ENC	Encoder Error 1 = Primary encoder miscounts.
11	#ENC2	Encoder 2 Error 1 = Secondary encoder miscounts.
12	#PE	1 = Position error (PE) has occurred. PE is defined by the following variables: > ERRI - Maximum position error while the axis is idle > ERRV - Maximum position error while the axis is moving with constant velocity > ERRA - Maximum position error while the axis is accelerating or decelerating > DELI - Delay on transition from ERRA to ERRI > DELV - Delay on transition from ERRA to ERRV

Bit	Fault	Fault Description
Axis Faults		
13	#CPE	Critical Position Error 1 = Position error (#PE) exceeds the value of the critical limit. #CPE errors occur outside normal range of operation and #CPE > #PE. The critical limit depends on the axis state and is defined by the following variables: > CERRI if the axis is idle (not moving) > CERRV if the axis is moving with constant velocity > CERRA if the axis is accelerating or decelerating > DELI- Delay on transition from ERRA to CERRI > DELV - Delay on transition from ERRA to CERRV
14	#VL	Velocity Limit 1 = Absolute value of the reference velocity (RVEL) exceeds the limit defined by the XVEL parameter.
15	#AL	Acceleration Limit 1 = Absolute value of the reference acceleration (RACC) exceeds the limit defined by the XACC parameter.
16	#CL	Current Limit 1 = RMS current calculated in the Servo Processor exceeds the limit value defined by the XRMS parameter.
17	#SP	Servo Processor 1 = Axis Servo Processor loses its synchronization with the MPU. The fault indicates a fatal problem in the controller.
20	#HSSINC	HSSI Not Connected 1 = HSSI module is not connected.
System Faults		
23	#EXTNT	External Network Error
24	#TEMP	MPU Overheat Fault Activated at CPU temperature > 90°C or System temperature > 70°C Default response - none

System Faults		
25	#PROG	Program Fault 1 = Run time error occurs in one of the executing ACSPL+ programs.
26	#MEM	Memory Overflow 1 = User application requires too much memory.
27	#TIME	MPU Overuse 1 = User application consumes too much time in the controller cycle.
28	#ES	Hardware Emergency Stop 1 = ES signal is activated.
29	#INT	Servo Interrupt 1 = The servo interrupt that defines the controller cycle is not generated. The fault indicates a fatal controller problem.
30	#INTGR	File Integrity 1 = The integrity of the user application in controller RAM is checked by the controller at power-up and whenever an #IR Terminal command is issued.
31	#FAILURE	Component Failure 1 = An MC4U hardware component other than the drive, such as the Power Supply, I/O card, or encoder card, has failed.
Axis Faults		
		When the bus voltage is not supplied to the MC4U, a component failure fault is reported. The fault is system wide and prevents all axes from operating unless the fault is masked or bus voltage is supplied to the power supply. When a component failure is reported, the affected power supply is identified by its address. To determine the faulty unit, use the MMI System Viewer and Diagnostics

Tag

332

Comments

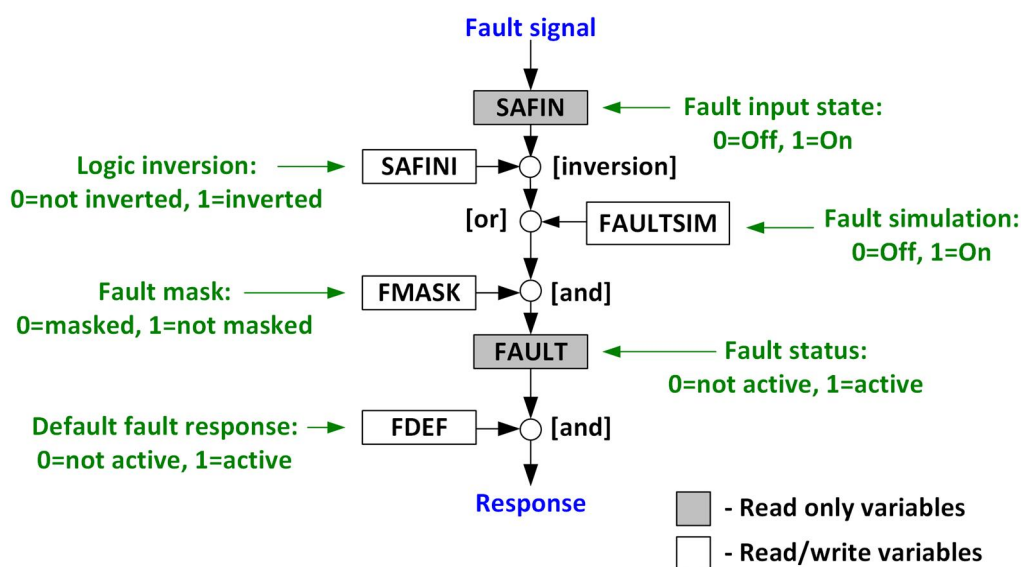
This variable allows you to simulate Axis & System faults, by raising a certain bit on a given axis – you effectively set the fault and the appropriate response will be triggered.



In order to reset some faults you must use **FCLEAR** command, setting the **FAULTSIM** bit to 0 might not be enough.

FAULTSIM variable is not saved to flash and will be reset upon controller restart.

FAULTSIM variable does not interact with **SAFINI**/**SAFIN** variables and therefore does not affect the variables.



Accessibility

Read/Write

Related ACSPL+ Variables

FAULT, **S_FAULT**, **SAFINI**, **S_SAFINI**, **FMASK**, **S_FMASK**

3.11.16 FDEF

Description

FDEF is an integer array, with one element for each axis in the system, the elements of which contain a set of bits used for setting a default response to an axis fault.

Syntax

FDEF(axis_index)[.bit_designator] = value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>bit_designator</i>	The FDEF bit designators are given in FDEF Bit Description .
<i>value</i>	value ranges from -2147483648 to 2147483647, Default = -1.

Table 3-15. FDEF Bit Description

Bit	Fault	Fault Description	Default Response (FDEF)
0	#RL	Hardware Right Limit 1 = Right limit switch is activated.	The controller kills the violating axis. As long as the fault is active, the controller kills any motion that tries to move the axis in the direction of the limit; however, motion within the permissible range is allowed.
1	#LL	Hardware Left Limit 1 = Left limit switch is activated.	Same as for #RL.
2	#NT	Network Error 1 = EtherCAT network error detected.	Halts all program buffers and waits for receipt of network Sync signal.
4	#HOT	>Motor Overheat 1 = Motor's temperature sensor indicates overheat.	None.
5	#SRL	Software Right Limit 1 = Axis reference position (RPOS) is greater than the software right limit margin (SRLIMIT).	The controller kills the violating axis. As long as the fault is active, the controller kills any motion that tries to move the axis in the direction of the limit. Motion in the direction out of the limit is allowed.
6	#SLL	Software Left Limit	Same as #SRL.

Bit	Fault	Fault Description	Default Response (FDEF)
		1 = Axis reference position (RPOS) is less than the software left limit margin (SLLIMIT).	
7	#ENCNC	Encoder Not Connected 1 = Primary encoder (for digital encoder type only) is not connected.	The controller disables the violating axis.
8	#ENC2NC	Encoder 2 Not Connected 1 = Secondary encoder (for digital encoder type only) is not connected.	No default response.
9	#DRIVE	Drive Fault 1 = Signal from the drive reports a failure.	The controller disables the violating axis. This fault is only detected when the axis is enabled. To catch this fault in an ACSPL+ program, write an autoroutine.
10	#ENC	Encoder Error 1 = Primary encoder miscounts.	The controller disables the violating axis. The faults remain active until the user resolves the problems and enables the axis again or executes FCLEAR .
11	#ENC2	Encoder 2 Error 1 = Secondary encoder miscounts.	Same as #ENC.
12	#PE	Position Error. 1 = Position error (PE) has occurred. PE is defined by the following variables: > ERRI - Maximum position error while the axis is idle	None.

Bit	Fault	Fault Description	Default Response (FDEF)
		> ERRV - Maximum position error while the axis is moving with constant velocity > DELI - Delay on transition from ERRA to ERRI > DELV - Delay on transition from ERRA to ERRV	
13	#CPE	Critical Position Error 1 = Position error (#PE) exceeds the value of the critical limit. #CPE errors occur outside normal range of operation and #CPE > #PE. The critical limit depends on the axis state and is defined by the following variables: > CERRI if the axis is idle (not moving) > CERRV if the axis is moving with constant velocity > CERRA if the axis is accelerating or decelerating > DELI - Delay on transition from ERRA to CERRI > DELV - Delay on transition from CERRA to ERRV	The controller disables the violating axis.

Bit	Fault	Fault Description	Default Response (FDEF)
14	#VL	>Velocity Limit 1 = Absolute value of the reference velocity (RVEL) exceeds the limit defined by the XVEL parameter.	The controller kills the violating axis.
15	#AL	>Acceleration Limit 1 = Absolute value of the reference acceleration (RACC) exceeds the limit defined by the XACC parameter.	The controller kills the violating axis.
16	#CL	Current Limit 1 = RMS current calculated in the Servo Processor exceeds the limit value defined by the XRMS XpParameter.	The controller disables the violating axis.
17	#SP	Servo Processor Alarm 1 = Axis Servo Processor loses its synchronization with the MPU. The fault indicates a fatal problem in the controller.	The controller disables the violating axis and kills the motion that involves the axis.
18	#STO	Safe Torque Off 1 = STO is activated	Blocks the PWM signals to the power stage of the drive
20	#HSSINC	Hssi Not Connected 1 = HSSI module is not connected.	None.

Tag

48

Comments

When an **FDEF** bit = 1, the controller executes the default response when the corresponding fault occurs. If the **FDEF** bit = 0, the default response is disabled.

Not every fault has a default response. For a fault that has no default response, the corresponding **FDEF** bit is inoperative.

Accessibility

Read-Write



FDEF values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Variables

[FAULT](#), [S_FAULT](#), [S_FDEF](#), [FMASK](#), [S_FMASK](#), [SAFIN](#), [S_SAFIN](#), [SAFINI](#), [S_SAFINI](#)

COM Library Methods and .NET Library Methods

[ReadVariable](#), [WriteVariable](#), [GetResponseMask](#), [SetResponseMask](#), [GetFaultMask](#), [SetFaultMask](#)

C Library Functions

[acsc_ReadInteger](#), [acsc_WriteInteger](#), [acsc_GetResponseMask](#), [acsc_SetResponseMask](#), [acsc_GetFaultMask](#), [acsc_SetFaultMask](#)

3.11.17 FMASK

Description

FMASK is an integer array, with one element for each axis in the system, the elements of which contain a set of bits used for enabling or disabling each axis fault bit.

Syntax

FMASK(*axis_index*)[*.bit_designator*] = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>bit_designator</i>	The FDEF bit designators are given in FMASK Bit Description .
<i>value</i>	value ranges from -2147483648 to 2147483647, Default=1040414435.

Table 3-16. FMASK Bit Description

Bit	Fault	Fault Description
0	#RL	Hardware Right Limit 1 = Right limit switch is activated.
1	#LL	Hardware Left Limit 1 = Left limit switch is activated.
2	#NT	Network Error 1 = EtherCAT network error detected.
4	#HOT	Motor Overheat

Bit	Fault	Fault Description
		1 = Motor's temperature sensor indicates overheat.
5	#SRL	Software Right Limit 1 = Axis reference position (RPOS) is greater than the software right limit margin (SRLIMIT).
6	#SLL	Software Left Limit 1 = Axis reference position (RPOS) is less than the software left limit margin (SLLIMIT).
7	#ENCNC	Encoder Not Connected 1 = Primary encoder (for digital encoder type only) is not connected.
8	#ENC2NC	Encoder 2 Not Connected 1 = Secondary encoder (for digital encoder type only) is not connected.
9	#DRIVE	Drive Fault 1 = Signal from the drive reports a failure.
10	#ENC	Encoder Error 1 = Primary encoder miscounts.
11	#ENC2	Encoder 2 Error 1 = Secondary encoder miscounts.
12	#PE	Position Error 1 = Position error (PE) has occurred. PE is defined by the following variables: > ERRI - Maximum position error while the axis is idle > ERRV - Maximum position error while the axis is moving with constant velocity > ERRA - Maximum position error while the axis is accelerating or decelerating > DELI - Delay on transition from ERRA to ERRI > DELVDELV - Delay on transition from ERRAERRA to ERRV

Bit	Fault	Fault Description
13	#CPE	Critical Position Error 1 = Position error (#PE) exceeds the value of the critical limit. #CPE errors occur outside normal range of operation and #CPE > #PE. The critical limit depends on the axis state and is defined by the following variables: > CERRI if the axis is idle (not moving) > CERRV if the axis is moving with constant velocity > CERRA if the axis is accelerating or decelerating > DELI - Delay on transition from ERRA to CERRI > DELV - Delay on transition from CERRA to ERRV
14	#VL	Velocity Limit 1 = Absolute value of the reference velocity (RVEL) exceeds the limit defined by the XVEL parameter.
15	#AL	Acceleration Limit 1 = Absolute value of the reference acceleration (RACC) exceeds the limit defined by the XACC parameter.
16	#CL	Current Limit 1 = RMS current calculated in the Servo Processor exceeds the limit value defined by the XRMS parameter.
17	#SP	Servo Processor Alarm 1 = Axis Servo Processor loses its synchronization with the MPU. The fault indicates a fatal problem in the controller.
18	#STO	Safe Torque Off 1 = STO is active
20	#HSSINC	Hssi Not Connected 1 = HSSI module is not connected.

Tag

51

Comments

The default value = 1 and causes the controller to check for the fault associated with that bit, as follows:

0 = the corresponding **FAULT** bit is disabled.

1 = the corresponding **FAULT** is enabled and examined each MPU cycle.

Accessibility

Read-Write



FMASK values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

FAULT, **S_FAULT**, **S_FDEF**, **S_FMASK**, **SAFIN**, **S_SAFIN**, **SAFINI**, **S_SAFINI**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetResponseMask, SetResponseMask, GetFaultMask, SetFaultMask

C Library Functions

acsc_ReadInteger, acsc_WriteInteger, acsc_GetResponseMask, acsc_SetResponseMask, acsc_GetFaultMask, acsc_SetFaultMask

3.11.18 HLLROUT

Description

HLLROUT is an integer array with one element for each axis in the system, and it is used for mapping the hardware left limit of an axis to a specified digital input bit (ACSPL+ **IN**).

Syntax

HLLROUT(Axis_Index) = value

Arguments

Axis	Designates the specific axis. Valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value is a 4-digit number (decimal): <NN00> where: > NN - digital input index (00-99) > 00 – specified input bit (00-31) The default value is -1, in which case mapping will not occur.

Tag

379

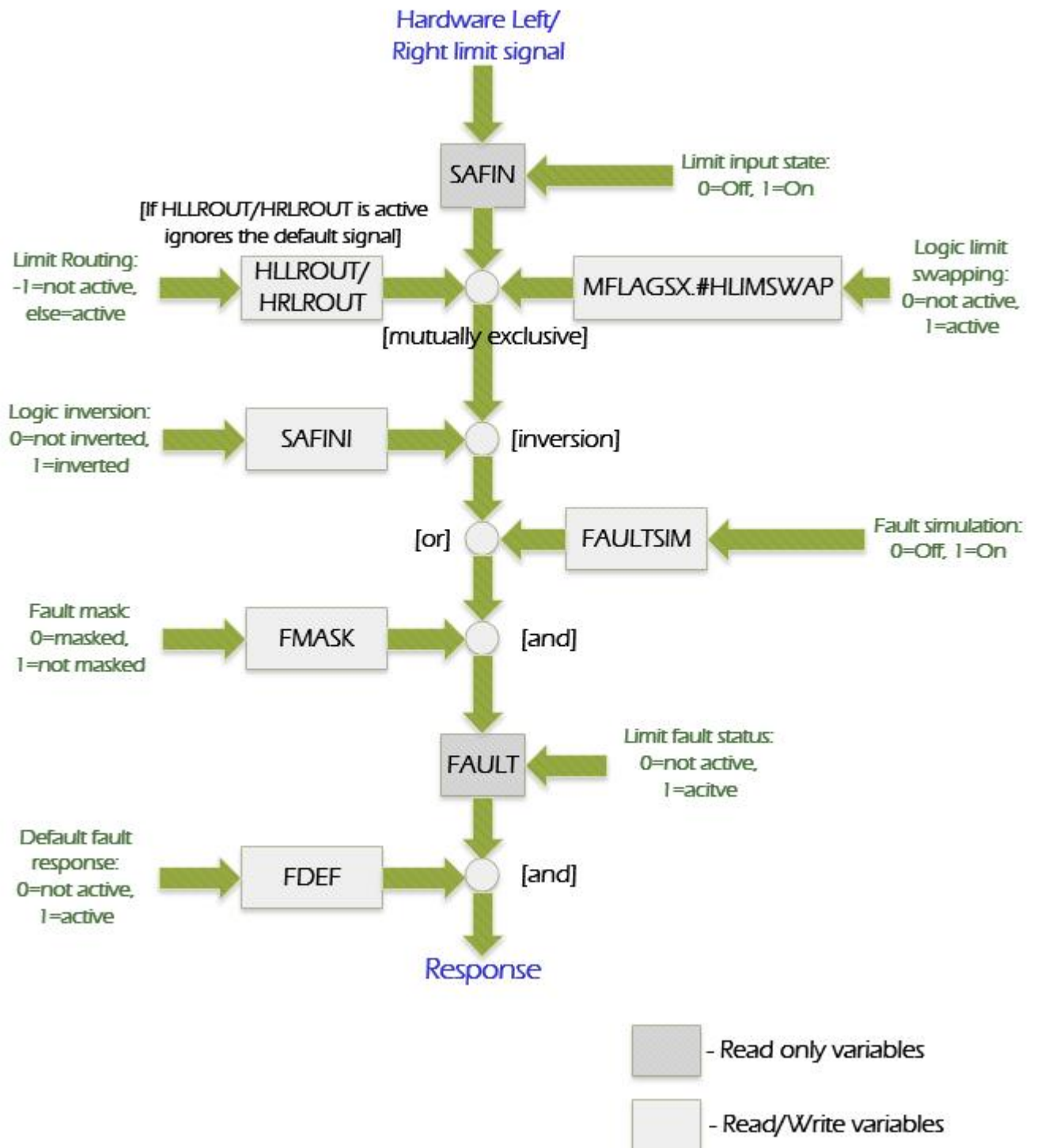
Comments

A value of -1 disables the mapping of the digital input to the hardware limit and restores the default behavior.

The following errors are supported:

- > Error 3329: "Invalid value, digital input index should range between 0-99 and bit index should range between 0-31"
- > Error 3332: "Hardware limit swapping(**MFLAGSX.#HLIMSWAP**) and limit routing are mutually exclusive"

The following diagram illustrates the behavior of the firmware when the left limit signal is set:



Related ACSPL+ Variables

HLLROUT, IN

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.11.19 HRLROUT

Description

HRLROUT is an integer array with one element for each axis in the system, and it is used for mapping the hardware right limit of an axis to a specified digital input bit (ACSPL+ **IN**).

Syntax

HRLROUT(Axis_Index) = value

Arguments

Axis	Designates the specific axis. Valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Value	Value is a 4-digit number (decimal): <NN00> where: > NN - digital input index (00-99) > 00 – specified input bit (00-31) The default value is -1, in which case mapping will not occur.

Tag

379

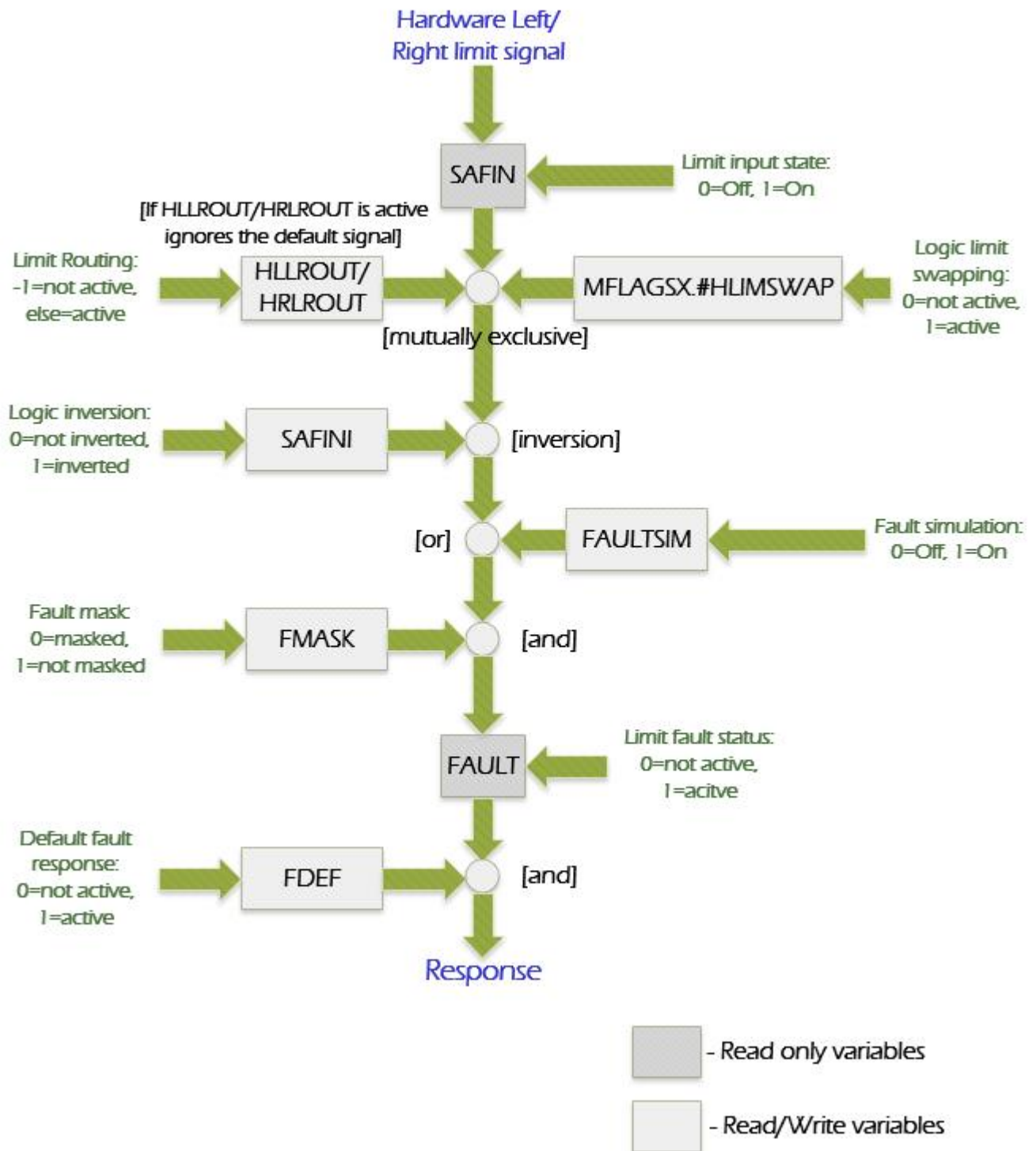
Comments

A value of -1 disables the mapping of the digital input to the hardware limit and restores the default behavior.

The following errors are supported:

- > Error 3329: "Invalid value, digital input index should range between 0-99 and bit index should range between 0-31"
- > Error 3332: "Hardware limit swapping(**MFLAGSX.#HLIMSWAP**) and limit routing are mutually exclusive"

The following diagram illustrates the behavior of the firmware when the left limit signal is set:



Related ACSPL+ Variables

HLLROUT, IN

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.11.20 MERR

Description

MERR is an integer array, with one element for each axis in the system, the elements of which store a code indicating the termination cause of the last motion of an axis.



An error code= 5027, "Motor Failed: Servo Processor Alarm" fault is activated when an axes does not have a physical drive associated to it.

The **MERR** return values are listed in [Table 7-4](#)

Tag

86

Comments

MERR is updated every time the axis motion is terminated. MERR stores the last termination code until either [FCLEAR](#) or [ENABLE/ENABLE ALL](#) is executed.

Accessibility

Read-Only

Related ACSPL+ Commands

[FCLEAR](#)

Related ACSPL+ Variables

[PERR](#)

COM Library Methods and .NET Library Methods

ReadVariable, GetMotorError

C Library Functions

acsc_ReadInteger, acsc_GetMotorError

3.11.21 SAFIN

Description

SAFIN is an integer array, with one element for each axis in the system, the elements of which contain a set of bits that indicates the raw state, before processing, of the axis safety inputs.

SAFIN(<axis>).17 (#STO1) and **SAFIN(<axis>).18 (#STO2)** present the status of the emergency stop request (24V switched off).

For the products that supports SS1-t:

SAFIN(<axis>).19 (#SS11) and **SAFIN(<axis>).20 (#SS12)** present the status of switching in the torque off mode (5V switched off).

SAFIN(<axis>).16 (#SS1TERR) presents the status of SS1-t timing error.

The value of each element in the array ranges from -2147483648 to 2147483647, Default=0.

Tag

121

Comments

1. The **SAFIN** uses the same bit numbers as in **S_SAFIN** and as the corresponding faults in **FAULT** and **S_FAULT**.
2. **SAFIN** is normally read-only. However, when working with the Simulator, read/write is permitted to simulate safety inputs.
3. Only the **SAFIN** bits below are valid.

Bit Name	No.	Description
#RL	0	Hardware Right Limit
#LL	1	Hardware Left Limit
#HOT	4	Motor Overheat
#DRIVE	9	Drive Fault
#SS1TERR	16	Status of SS1-t timing error
#STO1	17	Status of the emergency stop request (24V switched off)
#STO2	18	Status of the emergency stop request (24V switched off)
#SS11	19	Status of switching in the torque off mode (5V switched off)
#SS12	20	Status of switching in the torque off mode (5V switched off)
#ES	28	Hardware Emergency Stop 1 = ES signal is activated
#COMP	31	Component Failure

Accessibility

Read-Only



SAFIN can be written to when working with the SPiPlus Simulator.

Related ACSPL+ Variables

FAULT, S_FAULT, FDEF, S_FDEF, FMASK, S_FMASK, S_SAFIN, SAFINI, S_SAFINI

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.11.22 SAFINI

Description

SAFINI is an integer array, with one element for each axis in the system, the elements of which contain a set of bits defining the active state of the **axis** safety input variable (**SAFIN**) specifying inversion of the signal input logic, if required.

Syntax

SAFINI(axis_index)[.bit_designator] = value

Arguments

axis_index	axis_index designates the specific axis: 0, 1, 2, .. up to the number of axes in the system minus 1.
bit_designator	The valid SAFINI bits are given in Table 3-17 .
value	value ranges from -2147483648 to 2147483647, Default=0.

Table 3-17. SAFINI Valid Bits

Bit Name	No.	Description
#RL	0	Hardware Right Limit
#LL	1	Hardware Left Limit
#HOT	4	Motor Overheat
#DRIVE	9	Drive Fault
#ES	28	Hardware Emergency Stop 1 = ES signal is activated

Tag

122

Comments

1. When a **SAFINI** bit=0, the corresponding signal is not inverted and the high voltage state is considered active.
2. When a **SAFINI** bit=1, the bit is inverted and the low voltage state is considered active.

Accessibility

Read-Write



SAFINI values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection.

Related ACSPL+ Variables

[FAULT](#), [S_FAULT](#), [FDEF](#), [S_FDEF](#), [FMASK](#), [S_FMASK](#), [SAFIN](#), [S_SAFIN](#), [S_SAFINI](#).

COM Library Methods and .NET Library Methods

[ReadVariable](#), [WriteVariable](#), [SetSafetyInputPortInv](#), [GetSafetyInputPortInv](#)

C Library Functions

[acsc_ReadInteger](#), [acsc_WriteInteger](#), [acsc_SetSafetyInputPortInv](#), [acsc_GetSafetyInputPortInv](#)

3.11.23 S_ERR**Description**

S_ERR is a scalar integer that contains the code of the initialization error set during powerup.

The error codes are specified in [Table 7-6](#).

Tag

113

Accessibility

Read-Only

Related ACSPL+ Variables

None

COM Library Methods and .NET Library Methods

[ReadVariable](#)

C Library Functions

[acsc_ReadInteger](#)

3.11.24 S_FAULT**Description**

S_FAULT is a scalar integer variable consisting of a set of bits equating to the occurrence of faults. **S_FAULT** has two categories of bits, Axis Faults and System Faults (faults that are not related to any specific axis).

The **S_FAULT** bits are described in [Table 3-18](#).

Table 3-18. S_FAULT Fault Bits

Bit	Fault	Fault Description
Axis Faults		
0	#RL	Hardware Right Limit 1 = Right limit switch is activated.
1	#LL	Hardware Left Limit 1 = Left limit switch is activated.
2	#NT	Network Error. 1 = EtherCAT network error is activated.
4	#HOT	Motor Overheat 1 = Motor's temperature sensor indicates overheat.
5	#SRL	Software Right Limit 1 = Axis reference position (RPOS) is greater than the software right limit margin (SRLIMIT).
6	#SLL	Software Left Limit 1 = Axis reference position (RPOS) is less than the software left limit margin (SLLIMIT).
7	#ENCNC	Encoder Not Connected 1 = Primary encoder (for digital or SinCos encoder types) is not connected.
8	#ENC2NC	Encoder 2 Not Connected 1 = Secondary encoder (for digital or SinCos encoder types) is not connected.
9	#DRIVE	Drive Fault 1 = Signal from the drive reports a failure.
10	#ENC	Encoder Error 1 = Primary encoder miscounts.
11	#ENC2	Encoder 2 Error 1 = Secondary encoder miscounts.

Bit	Fault	Fault Description
12	#PE	Position Error 1 = Position error (PE) has occurred. PE is defined by the following variables: > ERRI - Maximum position error while the axis is idle > ERRV - Maximum position error while the axis is moving with constant velocity > ERRA - Maximum position error while the axis is accelerating or decelerating > DELI - Delay on transition from ERRA to ERRI > DELV - Delay on transition from ERRA to ERRV
13	#CPE	Critical Position Error 1 = Position error (#PE) exceeds the value of the critical limit. #CPE errors occur outside normal range of operation and #CPE > #PE. The critical limit depends on the axis state and is defined by the following variables: > CERRI if the axis is idle (not moving) > CERRV if the axis is moving with constant velocity > CERRA if the axis is accelerating or decelerating > DELI - Delay on transition from ERRA to CERRI > DELV - Delay on transition from ERRA to CERRV
14	#VL	Velocity Limit. 1 = Absolute value of the reference velocity (RVEL) exceeds the limit defined by the XVEL parameter.
15	#AL	Acceleration Limit 1 = Absolute value of the reference acceleration (RACC) exceeds the limit defined by the XACC parameter.
16	#CL	Current Limit 1 = RMS current calculated in the Servo Processor exceeds the limit value defined by the XRMS parameter.
17	#SP	Servo Processor Alarm

Bit	Fault	Fault Description
		1 = Axis Servo Processor loses its synchronization with the MPU. The fault indicates a fatal problem in the controller.
18	#STO	Safe Torque Off 1 = STO is active
20	#HSSINC	HSSI Not Connected 1 = HSSI module is not connected.
System Faults		
23	#EXTNT	External Network Error
24	#TEMP	MPU Overheat Fault Activated at CPU temperature > 90°C or System temperature > 70°C Default response - none.
25	#PROG	Program Fault 1 = Run time error occurs in one of the executing ACSPL+ programs.
26	#MEM	Memory Overflow 1 = User application requires too much memory.
27	#TIME	MPU Overuse 1 = User application consumes too much time in the controller cycle.
28	#ES	Hardware Emergency Stop 1 = ES signal is activated.
29	#INT	Servo Interrupt 1 = The servo interrupt that defines the controller cycle is not generated. The fault indicates a fatal controller problem.
30	#INTGR	File Integrity 1 = The integrity of the user application in controller RAM is checked by the controller at power-up and whenever an #IR Terminal command is issued.

Bit	Fault	Fault Description
31	#FAILURE	Component Failure 1 = An MC4U hardware component other than the drive, such as the Power Supply, I/O card, or encoder card, has failed. When the bus voltage is not supplied to the MC4U, a component failure fault is reported. The fault is system wide and prevents all axes from operating unless the fault is masked or bus voltage is supplied to the power supply. When a component failure is reported, the affected power supply is identified by its address. To determine the faulty unit, use the MMI System Viewer and Diagnostics

Tag

114

Comments

An **S_FAULT** bit, such as Left Limit, will be = 1 whenever one or more Left Limit fault bits are = 1. In this manner, **S_FAULT** provides an indication of the aggregate state of each **FAULT** bit.

Accessibility

Read-Only



S_FAULT values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

FAULT, **FDEF**, **S_FDEF**, **FMASK**, **S_FMASK**, **SAFIN**, **S_SAFIN**, **SAFINI**, **S_SAFINI**

COM Library Methods and .NET Library Methods

ReadVariable, GetFault

C Library Functions

acsc_ReadInteger, acsc_GetFault

3.11.25 S_FDEF

Description

S_FDEF is a scalar integer variable consisting of a set of bits for defining the default response for the system faults contained in **S_FAULT**. **S_FDEF** is connected to **S_FAULT** in the same way that **FDEF** is connected with **FAULT**.

Syntax

S_FDEF[*bit_designator*] = *value*

Arguments

bit_designator	The S_FDEF bits and associated responses are given in Table 3-19 .
value	value ranges from -2147483648 to 2147483647, Default = 1.

Table 3-19. S_FDEF Bit Description

Bit	Fault	Fault Description	Default Response (S_FDEF)
23	#EXTNT	External Network Error	The controller disables all axes.
24	#TEMP	MPU Overheat	S_SETUP.#USGTEMP = 0: No response S_SETUP.#USGTEMP=1: Default response is to disable all axes
25	#PROG	Program Fault 1 = Run time error occurs in one of the executing ACSPL+ programs.	The controller kills all axes.
26	#MEM	Memory Overflow 1 = User application requires too much memory.	The controller kills all axes.
27	#TIME	MPU Overuse 1 = User application consumes too much time in the controller cycle.	No default response.
28	#ES	Hardware Emergency Stop 1 = ES signal is activated.	The controller disables all axes, and sets the offset of each axis to 0.  It does not stop the program buffers.

Bit	Fault	Fault Description	Default Response (S_FDEF)
29	#INT	Servo Interrupt 1 = The servo interrupt that defines the controller cycle is not generated. The fault indicates a fatal controller problem.	The controller disables all axes.
30	#INTGR	File Integrity 1 = The integrity of the user application in controller RAM is checked by the controller at power-up and whenever an #IR Terminal command is issued.	No default response
31	#FAILURE	Component Failure 1 = An MC4U hardware component other than the drive, such as the Power Supply, I/O card, or encoder card, has failed.	No default response The user has to supply a user-defined fault response.

Tag

115

Comments

The default value for all **S_FDEF** bits is 1, which enables the default response. If an **S_FDEF** bit = 0, the default response is disabled.

Accessibility

Read-Write



S_FDEF values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

[FAULT](#), [S_FAULT](#), [FDEF](#), [FMASK](#), [S_FMASK](#), [SAFIN](#), [S_SAFIN](#), [SAFINI](#), [S_SAFINI](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetResponseMask, SetResponseMask, GetFaultMask, SetFaultMask

C Library Functions

acsc_ReadInteger, acsc_WriteInteger, acsc_GetResponseMask, acsc_SetResponseMask, acsc_GetFaultMask, acsc_SetFaultMask

3.11.26 S_FMASK

Description

S_FMASK is scalar integer variable consisting of a set of bits for enabling or disabling the system faults contained in **S_FAULT**. **S_FMASK** is connected to **S_FAULT** in the same way that **FMASK** is connected with **FAULT**.

Syntax

S_FMASK[*bit_designator*] = *value*

Arguments

bit_designator	The S_FMASK bits and associated responses are given in Table 3-20 .
value	value ranges from -2147483648 to 2147483647, Default=0.

Table 3-20. S_FMASK Bit Description

Bit	Fault	Fault Description
23	#EXTNT	External Network Error
24	#TEMP	MPU Overheat
25	#PROG	Program Fault 1 = Run time error occurs in one of the executing ACSPL+ programs.
26	#MEM	Memory Overflow 1 = User application requires too much memory.
27	#TIME	MPU Overuse 1 = User application consumes too much time in the controller cycle.
28	#ES	Hardware Emergency Stop 1 = ES signal is activated.
29	#INT	Servo Interrupt 1 = The servo interrupt that defines the controller cycle is not generated. The fault indicates a fatal controller problem.
30	#INTGR	File Integrity

Bit	Fault	Fault Description
		1 = The integrity of the user application in controller RAM is checked by the controller at power-up and whenever a #IR immediate command is issued.
31	#FAILURE	Component Failure 1 = An MC4U hardware component other than the drive, such as the Power Supply, I/O card, or encoder card, has failed.

Tag

117

Comments

The **S_FMASK** default value = 1 and causes the controller to check for the fault associated with that bit, as follows:

0: The corresponding **FAULT** bit is disabled

1: The corresponding **FAULT** is enabled and examined each MPU cycle.

Accessibility

Read-Write



S_FMASK values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Variables

FAULT, **S_FAULT**, **FDEF**, **S_FDEF**, **FMASK**, **SAFIN**, **S_SAFIN**, **SAFINI**, **S_SAFINI**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetFaultMask, SetFaultMask

C Library Functions

acsc_ReadInteger, acsc_WriteInteger, acsc_GetFaultMask, acsc_SetFaultMask

3.11.27 S_SAFIN**Description**

S_SAFIN is a scalar integer variable that indicates the raw state of the **#ES** bit (Emergency Stop) input stored in the **SAFIN** variable and indicates the **#FAILURE** bit (system fault) stored in the **S_FAULT** variable.

The value ranges from -2147483648 to 2147483647, Default=0.

Tag

118

Comments

S_SAFIN uses the same bit numbers as in **SAFIN** and as the corresponding faults in **FAULT** and **S_FAULT**, but only the **#ES** bit is meaningful.



S_SAFIN can be written to when working with the SPiiPlus Simulator.

Accessibility

Read-Only

Related ACSPL+ Variables

FAULT, **S_FAULT**, **FDEF**, **S_FDEF**, **FMASK**, **S_FMASK**, **SAFIN**, **SAFINI**, **S_SAFINI**

COM Library Methods and .NET Library Methods

ReadVariable, GetSafetyInputPort

C Library Functions

acsc_ReadInteger, acsc_GetSafetyInputPort

3.11.28 S_SAFINI**Description**

S_SAFINI is a scalar integer variable used for defining the active state of the **system** safety input variable (**S_SAFIN**) specifying inversion of the signal input logic, if required.

Tag

119

Comments

1. When a **S_SAFINI** bit=0, the corresponding signal is not inverted and the high voltage state is considered active.
2. When a **S_SAFINI** bit=1, the bit is inverted and the low voltage state is considered active.

Accessibility

Read-Write



S_SAFINI values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables

FAULT, **S_FAULT**, **FDEF**, **S_FDEF**, **FMASK**, **S_FMASK**, **SAFIN**, **S_SAFIN**, **SAFINI**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, SetSafetyInputPortInv, GetSafetyInputPortInv

C Library Functions

acsc_ReadInteger, acsc_WriteInteger, acsc_SetSafetyInputPortInv, acsc_GetSafetyInputPortInv

3.11.29 SS1TIME

Description

SS1TIME is a integer array with one element for each EtherCAT node in the system, the elements of which store the last SS1-t channel A time between the emergency stop request (24V switched off) and the point in time when the drive in fact switched the torque off mode (5V switched off). The value ranges between 0 and 500. The user can read this value in order to determine whether the system stops motion within the time required by the system functional safety requirements.



SS1TIME should be used for SS1-t diagnostics only

Tag

373

Comments

This variable is supported in version 3.00 and higher

Accessibility

Read-Only

Related ACSPL+ Variables

SS12TIME

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.11.30 SS12TIME

Description

SS12TIME is a integer array with one element for each EtherCAT node in the system, the elements of which store the last SS1-t channel B time between the emergency stop request (24V switched off) and the point in time when the drive in fact switched the torque off mode (5V switched off). The value ranges between 0 and 500. The user can read this value in order to determine whether the system stops motion within the time required by the system functional safety requirements.



SS12TIME should be used for SS1-t diagnostics only

Tag

374

Comments

This variable is supported in version 3.00 and higher

Accessibility

Read-Only

Related ACSPL+ Variables

SS11TIME

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.11.31 STODELAY

Description

STODELAY is a real array, with one element for each axis in the system. It is used to configure the delay time between the STO fault indication and the default response (disable) to the fault. During this time the user can activate his own response (auto-routine) to kill the motion.

Syntax

STODELAY(axis) = value

Arguments

<i>axis</i>	The specific axis index. Valid numbers are: 0,1... up to the number of axes in the system, minus 1.
<i>value</i>	The value is the delay time between the STO fault indication and the default response (disable) to the fault. value can range between 0 (minimum) to 200 (maximum). The default is 50.

Tag

319

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.11.32 SYNC

Description

SYNC is an integer array (one element per each slave node) the elements of which contain a slave synchronization indicator for the node.

Tag

222

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.12 Induction Motor Variables

The Induction Motor variables are:

Name	Description
SLCFIELD	Induction Motor Excitation
SLCSLIP	Induction Motor Slip Factor

3.12.1 SLCFIELD

Description

SLCFIELD is a real array with one element for each axis in the system. It is used along with **SLCSLIP** for controlling permanent magnet (PM) synchronous motors (so called "DC Brushless motors"). **SLCFIELD** defines the magnetic field component.

Syntax

SLCFIELD (*axis_index*)>= *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value provides the percentage of the peak current of the amplifier ranging from 0 to 25.

Tag

217

Comments

Vector control, also known as Field Oriented Control, is a control technique that imitates the DC motor operation and applies it to AC motors: PM synchronous motors (DC brushless motors) and induction motors. This technique decomposes the motor current into two independent components:

- > The magnetizing (direct) component that influences the total magnetic flux.
- > The torque-producing (quadrature) component that influences the generated torque. This component is perpendicular to the magnetizing component.

In PM synchronous motors, the magnetizing component is in phase with the permanent magnet field. The goal is usually to keep this component zero, so all the current is dedicated to torque production. This maximizes the torque/current ratio and improves the efficiency and dynamic performance.

The magnetic field variable, **SLCFIELD**, is usually set equal to the nominal magnetizing current of the motor. It should be around the "knee" of the magnetizing curve of the motor - it should be high enough to maximize the torque constant of the motor, but must not be too high to prevent magnetic saturation. The exact value is provided by the motor manufacturer, but it can also be estimated based on 10-40% of the nominal current.

The value of **SLCFIELD** is expressed as percentage of the peak current of the amplifier and is calculated according to the following formula:

$$Field = \frac{\sqrt{2}I_{mag}}{I_{peak}} 100$$

where:

I_{mag}	The nominal excitation current of the motor (rms value).	Provided by the manufacturer, of 10-40% of the nominal current
I_{peak}	The amplitude of the maximum current of the amplifier.	



SLCFIELD = 0 identifies an induction motor.

Accessibility

Read-Write



SLCFIELD values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

Related ACSPL+ Variables

[SLCSLIP](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.12.2 SLCSLIP

Description

SLCSLIP is a real array with one element for each axis in the system. It is used along with [SLCFIELD](#) for controlling permanent magnet (PM) synchronous motors (so called "DC Brushless motors"). **SLCSLIP** defines the slip constant.

Syntax

SLCSLIP(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value designates the slip frequency ranging from 0 to 5000.

Tag

218

Comments

The range of **SLCSLIP** is typically 200-1000. A reference value is calculated as follows:

$$SLCSLIP = 131 \times f_n [Hz] \times s_n [\%] \times \frac{I_{peak}}{\sqrt{2}I_n}$$

where:

f_n - The nominal supply frequency

s_n - The nominal slip, given by:

where:

n_s - The synchronous velocity

n_n - The nominal velocity

I_{peak} - The peak current of the amplifier

I_n - The nominal motor current (rms)

For example, the nominal data of a 1.05kW motor is:

$n = 2870\text{rpm}$, $n = 3000\text{rpm}$, $I_n = 8.1\text{A}$, and $f_n = 50\text{Hz}$

So the nominal slip of the motor is:

$$s = \frac{3000-2870}{3000} = 0.043$$

For $I_{peak} = 10\text{A}$, the value of **SLCSLIP** will be:

$$SLCSLIP = 131 \times 50 \times 0.043 \frac{10}{\sqrt{2} \times 8.1} = 248$$

Accessibility

Read-Write



SLCSLIP values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection.

Related ACSPL+ Variables

[SLCFIELD](#)

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.13 Nanomotion Variables

The UDIhp-x / UDMnt-x (new revision) products' control algorithm supports Nanomotion motors, based on Nanomotion servo algorithm. Each UDIhp-x / UDMnt-x (new revision) products can support up to four Nanomotion axes using AB1 amplifier or two Nanomotion axes using AB2 amplifier (with automatic DC/AC mode switching).



AB2 amplifiers are not supported by SPiiPlus ADK Suite 2.40 or later. If AB2 amplifiers are used and there is not a need to upgrade the FW, it is recommended to continue using FW 2.30.03.

If an upgrade is needed, consult ACS Applications and the relevant DSP will be provided.

To activate the Nanomotion algorithm set the seventh bit of **MFLAGS** variable to 1

> **MFLAGS(<axis>).7 = 1**, or alternatively **MFLAGS(<axis>).#NANO = 1**

The following variables should be used in support of Nanomotion piezo-ceramic motor motion:

Name	Description
SLDZMIN	Parameter which specifies the minimum position of the Dead Zone (when the servo is turned off)
SLDZMAX	Parameter which specifies the maximum position of the Dead Zone (when the servo is turned on).
SLDZTIME	Parameter which specifies the duration (in msec) required for settling after entering the SLDZMIN.
SLZFF	Parameter which specifies the distance from target to stop the velocity Feed Forward.
SLFRC	Parameter which specifies the initial non-zero command to overcome the static friction in positive direction.
SLFRCN	Parameter which specifies the initial non-zero command to overcome the static friction in negative direction.
SLHRS	Parameter which specifies the modulation ratio of the drive command.
SLVKPDCF	Parameter which specifies the multiplication factor of the velocity loop gain (SLVKP) in DC mode.
SLPKPDCF	Parameter which specifies the multiplication factor of the position loop gain (SLPKP) in DC mode.
SLVKIDCF	Parameter which specifies the multiplication factor of the velocity loop integrator (SLVKI) in DC mode.

3.13.1 SLDZMIN

Description

SLDZMIN is a real array, with one element for each axis in the system, and is used for defining the minimum position of the Dead Zone (when Servo is turned off).

Syntax

SLDZMIN(*axis_index*) = *value*

Arguments

<i>axis_index</i>	<i>axis_index</i> designates the specific axis: 0 - X, 1 - Y, etc.
<i>value</i>	<i>value</i> ranges between 0 to 1.79769e+308, Default = 1.

Tag

162

Comments

The Dead Zone mechanism stops the motor when the position approaches the target within the value of **SLDZMIN**. The value depends on the system specifications; usually **SLDZMIN** is between 1.0 to 2.0 counts (with an equivalent value in user units).

Accessibility

Read-Write



SLDZMIN values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.13.2 SLDZMAX

Description

SLDZMAX is a real array, with one element for each axis in the system, and is used for defining the maximum position of the Dead Zone.

Syntax

SLDZMIN(axis_index) = value

Arguments

axis_index	axis_index designates the specific axis: 0 - X, 1 - Y, etc.
value	value ranges between 0 to 1.79769e+308, Default = 10.

Tag

163

Comments

The Dead Zone mechanism starts the motor again when the error radius increases above the value **SLDZMAX**. The value depends on the system specifications; usually **SLDZMAX** is between 4.0 to 10.0 counts (with an equivalent value in user units).

Accessibility

Read-Write



SLDZMAX values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.13.3 SLDZTIME

Description

SLDZTIME is a real array, with one element for each axis in the system, which defines the duration (in msec) required for settling after entering the **SLDZMIN**. Only after this duration the controller starts monitoring the position error and returns the servo if |PE| exceeds **SLDZMAX**.

Syntax

SLDZTIME(axis_index) = value

Arguments

axis_index	axis_index designates the specific axis: 0 - X, 1 - Y, etc.
value	value ranges between 0.1 to 1.79769e+308, Default = 20.

Tag

251

Accessibility

Read-Write



SLDZMAX values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.13.4 SLZFF

Description

SLZFF is a real array, with one element for each axis in the system, and is used for defining the distance from target to stop the velocity Feed Forward.

Syntax

SLZFF(axis_index) = value

Arguments

axis_index	axis_index designates the specific axis: 0 - X, 1 - Y, etc.
value	value ranges between 0 to 1.79769e+308, Default = 300.

Tag

189

Comments

Using **SLZFF** improves settling time by stopping the feed forward velocity when the axis is getting close to the target position. The distance from the target is defined by **SLZFF** (in user units). The proper value of **SLZFF** depends on the total moving mass and the resolution of the encoder. Usually:

- > For an HR1 motor with encoder resolution of 0.1 μ M, set **SLZFF** to 100 - 300 counts (with an equivalent value in user units).
- > For an HR8 motor with encoder resolution of 0.1 μ M, set **SLZFF** to 300 - 400 counts (with an equivalent value in user units).

Accessibility

Read-Write



SLZFF values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.13.5 SLFRC**Description**

SLFRC is a real array, with one element for each axis in the system, which defines initial non-zero command to overcome the static friction in a positive direction.

Syntax**SLFRC**(axis_index) = value**Arguments**

axis_index	axis_index designates the specific axis: 0 - X, 1 - Y, etc.
value	value ranges between 0 to 50, Default = 0.

Tag

167

Comments

SLFRC is expressed as a percentage of the maximum output.

Accessibility

Read-Write



SLFRC values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.13.6 SLFRCN

Description

SLFRCN is a real array, with one element for each axis in the system, which defines initial non-zero command to overcome the static friction in a negative direction.

Syntax

SLFRCN(*axis_index*) = *value*

Arguments

axis_index	axis_index designates the specific axis: 0 - X, 1 - Y, etc.
value	value ranges between 0 to 50, Default = 0.

Tag

250

Comments

SLFRCN is expressed as a percentage of the maximum output.

Accessibility

Read-Write



SLFRCN values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.13.7 SLHRS

Description:

SLHRS is a real array, with one element for each axis in the system, which defines the modulation ratio of the drive command.

Syntax:

SLHRS(*axis_index*) = *value*

Arguments

<i>axis_index</i>	axis_index designates the specific axis: 0 - X, 1 - Y, etc.
<i>value</i>	value ranges from 0 to 100, Default = 100.

Tag:

169

Accessibility

Read-Write



DCOM values cannot be modified if protection is applied to this variable through **SPiiPlus**
MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, Write Variable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.13.8 SLVKPDCF

Description

SLVKPDCF is a real array, with one element for each axis in the system, which defines multiplication factor of the velocity loop gain (**SLVKP**) in DC mode. The normal gain is increased by setting a **SLVKPDCF** value larger than 1.

Syntax

SLPKPDCF *axis_index* = *value*

Arguments

<i>axis_index</i>	axis_index designates the specific axis: 0 - X, 1 - Y, etc.
<i>value</i>	value ranges between -1.79769e+308 to 1.79769e+308, Default = 1.

Tag

252

Accessibility

Read-Write



SLVKPDCF values cannot be modified if protection is applied to this variable through
SPIiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.13.9 SLPKPDCF

Description

SLPKPDCF is a real array, with one element for each axis in the system, which defines multiplication factor of the position loop gain (**SLPKP**) in DC mode. The normal gain is increased by setting the **SLPKPDCF** to a value larger than 1.

Syntax

SLPKPDCF *axis_index* = *value*

Arguments

<i>axis_index</i>	axis_index designates the specific axis: 0 - X, 1 - Y, etc.
<i>value</i>	value ranges between 0 to 1.79769e+308, Default = 1.

Tag

253

Accessibility

Read-Write



SLPKPDCF values cannot be modified if protection is applied to this variable through
SPIiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.13.10 SLVKIDCF

Description

SLVKIDCF is a real array, with one element for each axis in the system, which defines multiplication factor of the velocity loop integrator (**SLVKI**) in DC mode. The normal gain is increased by setting the **SLVKIDCF** to a value larger than 1.

Syntax

SLVKIDCF *axis_index* = *value*

Arguments

<i>axis_index</i>	axis_index designates the specific axis: 0 - X, 1 - Y, etc.
<i>value</i>	value ranges between 0 to 1.79769e+308, Default = 1.

Tag

254

Accessibility

Read-Write



SLVKIDCF values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14 Servo-Loop Variables



Servo-Loop variables are fully accessible at the ACSPL+ level. While ACSPL+ programs generally do not refer to servo-loop variables at run time, an advanced program could change a servo-loop variable on-the-fly to provide adaptive control

The servo-loop variables are used for configuration and adjustment, and are set through **SPiiPlus MMI Application Studio → Setup → Adjuster**.

The Servo-Loop variable is:

Name	Description
DCOM	Drive Command

Additional servo-loop variables are grouped as follows:

- > Servo-Loop Current Variables
- > Servo-Loop Velocity Variables
- > Servo-Loop Velocity Notch Filter Variables
- > Servo-Loop Velocity Low Pass Filter Variables
- > Servo-Loop Velocity Bi-Quad Filter Variables
- > Servo-Loop Position Variables
- > Servo-Loop Compensations Variables
- > Servo Loop Stepper Variables
- > Servo-Loop Miscellaneous Variables
- > Non-Linear Control Variables

3.14.1 DCOM

Description

DCOM is a real array, with one element for each axis in the system. It is used for defining the commanded current as a percentage of the maximum drive command..

Syntax

DCOM(*axis_index*) = *value*

Arguments

<i>axis_index</i>	<i>axis_index</i> designates the specific axis: 0 - X, 1 - Y, etc.
<i>value</i>	value ranges from -100 to 100, Default = 0.

Tag

20

Comments

DCOM defines a percentage of the maximum current command **XCURV**. When operating in the open loop mode (**MFLAGS.1=1**), **DCOM** supplies this current directly to the motor windings.

DCOM defines a percentage of the maximum drive command, for example, the UDI provides differential drive output from -10 to +10V. Therefore, assigning 100 to **DCOM** provides +10V on the drive output, -100 corresponds to -10V and 0 to 0V.

When operating in the closed loop mode (**MFLAGS.1=0**), **DCOM** offsets the normal commanded output current from the drive..

Accessibility

Read-Write



DCOM values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

Related ACSPL+ Commands

MFLAGS

COM Library Methods and .NET Library Methods

ReadVariable, Write Variable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.2 Servo-Loop Current Variables

The Servo-Loop Current variables are:

Name	Description
SLBIASA	Current phase A bias
SLBIASB	Current phase B bias
SLBIASC	Current phase C bias
SLIKI	Integrator gain
SLIKP	Integrator proportional gain
SLIFILT	Internal current filter
SLIOFFS	Offset to be added to the result of the current loop control
SLILI	Used to limit the drive's output voltage

3.14.2.1 SLBIASA

Description

SLBIASA is a real array, with one element for each axis in the system, and is used for defining offset of the current in phase "S" or offset in command of phase "S".

Syntax

SLBIASA(axis_index) = value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between -10 to 10, Default = 0.

Tag

149

Comments

SLBIASA is expressed as a percentage of the maximum controller voltage output.

1. For **integrated models**: **SLBIASA** is read-only and displays the measured value of the current input bias.
2. For **non-integrated models**: **SLBIASA** is read-write and specifies the bias of the drive output. The controller uses the value only for brushless motors commutated by the controller.

Accessibility

Read-Only (integrated models)

Read-Write (nonintegrated models)



SLBIASA values cannot be modified if protection is applied to this variable through **SPIIPlus MMI Application Studio → Toolbox → Application Development → Protection Wizard**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.2.2 SLBIASB

Description

SLBIASB is a real array, with one element for each axis in the system, and is used for defining offset of the current in phase "T" or offset in command of phase "T".

Syntax

SLBIASB(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between -10 to 10, Default = 0.

Tag

150

Comments

SLBIASB is expressed as a percentage.

1. For **integrated models**: **SLBIASB** is read-only and displays the measured value of the current input bias.
2. For **nonintegrated models**: **SLBIASB** is read-write and specifies the bias of the drive output. The controller uses the value only for brushless motors commutated by the controller.

Accessibility

Read-Only (integrated models)

Read-Write (nonintegrated models)



SLBIASB values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.2.3 SLBIASC**Description**

SLBIASC is a real array, with one element for each axis in the system, and is used for defining offset of the current in phase "R" or offset in command of phase "R".

Syntax

```
SLBIASC(axis_index)=value
```

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges between -10 to 10, Default = 0.

Tag

404

Comments

SLBIASC is expressed as a percentage of the maximum controller voltage output.

1. For integrated models: **SLBIASC** is read-only and displays the measured value of the current input bias.
2. For non-integrated models: **SLBIASC** is read-write and specifies the bias of the drive output. The controller uses the value only for brushless motors commutated by the controller.

Related ACSPL+ Variables**SLBIASA, SLBIASB****Accessibility**

Read-Only (integrated models)

Read-Write (nonintegrated models)



SLBIASC values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio -> Toolbox -> Application Development -> Protection Wizard

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.2.4 SLIKI

Description

SLIKI is a real array, with one element for each axis in the system, and is used for specifying the current loop.

Syntax

SLIKI(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0 - 65000.

Tag

170

Comments

SLIKI is active only in integrated models.

Accessibility

Read-Write



SLIKI values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.2.5 SLIKP

Description

SLIKP is a real array, with one element for each axis in the system, and is used for specifying the current loop proportional coefficient.

Syntax

SLIKP(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0 - 256000, Default = 1000.

Tag

171

Comments

SLIKP is active only in integrated models.

Accessibility

Read-Write



SLIKP values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.2.6 SLIFILT

Description

SLIFILT is a real array, with one element for each axis in the system, and is used for defining the UDM current filter frequency.

Syntax

SLIFILT(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0-5000.

Tag

226

Accessibility

Read-Write



SLIFILT values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Variables and .NET Library Methods

[SLIKI](#), [SLIKP](#)

COM Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.2.7 SLIOFFS

Description

SLIOFFS is a real array, with one element for each axis in the system, and is used for offset to be added to the result of the current loop control.

Syntax

SLIOFFS(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between -50 to 50, Default = 0.

Tag

172

Comments

The variable contains value in percents of maximal drive output.

The primary goal of the variable is to compensate for an active component of the motor load. For example, in a vertical axis the weight of the carriage can be compensated.

The variable is valid for DC brush and brushless motors.

Normally, the variable is changed in the process of adjustment (use **SPiiPlus MMI Application Studio → Toolbox → Setup → Adjuster Wizard**).

Accessibility

Read-Write



SLIOFFS values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.2.8 SLULI

Description

SLULI is a real array, with one element for each axis in the system, and is used to limit the drive's output voltage. If raised, a higher speed can be achieved for the given drive (higher output voltage).

Syntax

SLULI(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value is a percentage of the maximal drive output, and consequently ranges between 0 to 100; Default: 88.

Tag

221

Comments



It is recommended to set value **SLULI(axis)=97** to get a higher speed only for SPiiPlus CMnt, SPiiPlus UDMpm and SPiiPlus UDMpc

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.3 Servo-Loop Velocity Variables

The Servo-Loop Velocity variables are:

Name	Description
SLCRAT	Defines gear ratio between velocity feedback resolution and commutation feedback resolution
SLVKI	Sets velocity integrator coefficient
SLVKIIF	Provides an Idle Factor to the SLVKI variable
SLVKISF	Provides a Settle Factor to the SLVKI variable
SLVKP	Sets the proportional velocity gain.
SLVKPIF	Provides an Idle Factor to the SLVKP variable
SLVKPSF	Provides a Settle Factor to the SLVKP variable
SLVLI	Integrator velocity limit
SLVRAT	Velocity feed forward ratio

3.14.3.1 SLCRAT

Description

SLCRAT is a real array, with one element for each axis in the system defining the ratio between the velocity feedback resolution and the commutation feedback resolution. It is used during the commutation phase.

Syntax

SLCRAT(*axis_index*)

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between -8.38861e+006 to 8.38861e+006, Default = 1.

Tag

154

Accessibility

Read-Write



SLCRAT values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.3.2 SLVKI

Description

SLVKI is a real array, with one element for each axis in the system, and is used for specifying the velocity loop integrator coefficient.

Syntax

SLVKI(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0 to 20000; Default = 200.

Tag

179

Accessibility

Read-Write



SLVKI values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.3.3 SLVKIIF

Description

SLVKIIF is a real array with one element for each axis in the system. It is used for providing an Idle Factor to the **SLVKI** (Integrator Gain - Velocity) variable.

Syntax

SLVKIIF *axis_index* = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0.0 and 100.0.

Tag

233

Accessibility

Read-Write



SLVKIIF values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.3.4 SLVKISF

Description

SLVKISF is a real array with one element for each axis in the system. It is used for providing a Settle Factor to the **SLVKI** variable.

Syntax

SLVKISF *axis_index* = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0.0 and 100.0.

Tag

234

Accessibility

Read-Write



SLVKISF values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.3.5 SLVKITF

Description

SLVKITF increases the velocity loop integrator coefficient when the axis is close to the target position. It is a real array with one element for each axis in the system.

Syntax

SLVKITF *axis_index* = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 1.0 and 100.0. Default = 1.

Tag

271

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.3.6 SLVKP

Description

SLVKP is a real array, with one element for each axis in the system, and is used for specifying the velocity loop proportional coefficient.

Syntax

SLVKP(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0 to 16777215, Default = 100.

Tag

180

Accessibility

Read-Write



SLVKP values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.3.7 *SLVKPIF*

Description

SLVKPIF is a real array with one element for each axis in the system. It is used for providing an Idle Factor to the SLVKP (Proportional Gain - Velocity) variable.

Syntax

SLVKPIF *axis_index* = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0.0 and 100.0.

Tag

235

Accessibility

Read-Write



SLVKPIF values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.3.8 SLVKPSF

Description

SLVKPSF is a real array with one element for each axis in the system. It is used for providing a Settle Factor to the SLVKP variable.

Syntax

SLVKPSF *axis_index* = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0.0 and 100.0.

Tag

236

Accessibility

Read-Write



SLVKPSF values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.3.9 SLVKPTF

Description

SLVKPTF increases the velocity loop proportional coefficient when the axis is close to the target position. It is a real array with one element for each axis in the system.



SLVKPTF is supported for the NanoPWM drives only.

Syntax

SLVKPTF *axis_index* = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 1.0 and 100.0. Default = 1.

Tag

272

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.3.10 SLVLI

Description

SLVLI is a real array, with one element for each axis in the system, and is used for providing an integrator limit for the velocity of the specified axis.

Syntax

SLVLI(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0 to 100, Default = 50.

Tag

181

Comments

SLVLI is expressed as a percentage of the maximum value.

Accessibility

Read-Write



SLVLI values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.3.11 SLVRAT

Description

SLVRAT is a real array, with one element for each axis in the system. It defines the velocity feed forward ratio during the commutation phase..

Syntax

SLVRAT(*axis_index*)

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges between -8.38861e+006 to 8.38861e+006, Default = 1.

Tag

187

Comments

Velocity feed forward compensates for the velocity feedback, achieving zero position error at constant velocity.

Accessibility

Read-Write



SLVRAT values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.4 Servo-Loop Velocity Notch Filter Variables

Notch Filter variables serve for notching the velocity servo-loop. The Servo-Loop Velocity Notch Filter variables are:

Name	Description
SLVNFRQ	Notch filter frequency.
SLVNWID	Notch filter width.

Name	Description
SLVNATT	Notch filter attenuation.

3.14.4.1 SLVNFRQ

Description

SLVNFRQ is a real array, with one element for each axis in the system, and is used for providing a notch filter frequency.

Syntax

SLVNFRQ(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0.1 to 4000, Default = 300.

Tag

182

Comments

SLVNFRQ is expressed in Hz.

Accessibility

Read-Write



SLVNFRQ values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.4.2 SLVNWID

Description

SLVNWID is a real array, with one element for each axis in the system, and is used for providing a notch filter width.

Syntax

SLVNWID(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0.1 to 4000, Default = 30.

Tag

183

Comments**SLVNWID** is expressed in Hz.**Accessibility**

Read-Write



SLVNWID values cannot be modified if protection is applied to this variable through
SPIiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.4.3 SLVNATT**Description**

SLVNATT is a real array, with one element for each axis in the system, and is used for providing the attenuation of the notch frequency at frequency specified by **SLVNFRQ**.

Syntax

SLVNATT(*axis_index*) = *value***Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0.05 to 20; Default = 5.

Tag

184

Accessibility

Read-Write



SLVNATT values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.5 Servo-Loop Velocity Low Pass Filter Variables

Low Pass Filter variables serve for setting the velocity low pass filtering parameters. The Servo-Loop Velocity Low Pass Filter variables are:

Name	Description
SLVSOF	Sets filter bandwidth
SLVSOFD	Sets filter damping

3.14.5.1 SLVSOF

Description

SLVSOF is a real array, with one element for each axis in the system, and is used for providing a second order filter bandwidth for the velocity of the specified axis.

Syntax

SLVSOF(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges between 0.1 to 4000, Default = 700.

Tag

185

Comments

SLVSOF is expressed in Hz.

Accessibility

Read-Write



SLVSOF values cannot be modified if protection is applied to this variable through
SPIIPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.5.2 SLVSOFD

Description

SLVSOFD is a real array, with one element for each axis in the system, and is used for providing a second order filter damping factor for the velocity of the specified axis.

Syntax

SLVSOFD(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0.3 to 1, Default = 0.707.

Tag

186

Accessibility

Read-Write



SLVSOFD values cannot be modified if protection is applied to this variable through
SPIIPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.6 Servo-Loop Velocity Bi-Quad Filter Variables

Bi-Quad Filter variables serve for setting the velocity bi-quad filtering parameters. The Servo-Loop Velocity Bi-Quad Filter variables are:

Name	Description
SLVBODD	Sets the damping ratio denominator for a Bi-Quad filter.
SLVBODF	Sets the denominator value of the Bi-Quad filter.
SLVBOND	Sets the damping ratio numerator for a Bi-Quad filter.
SLVBONF	Sets the numerator value of the Bi-Quad filter.

3.14.6.1 SLVBODD

Description

SLVBODD is a real array, with one element for each axis in the system, and is used for setting the damping ratio denominator for a Bi-Quad filter to the velocity loop control in addition to the existing 2nd order Low-pass and Notch filters for the given axis.

Syntax

SLVBODD(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value designates denominator value of the Bi-Quad damping ratio ranging from 0.1 to 1. See <i>SPiiPlus ACSPL+ Programmer Guide</i> .

Tag

208

Comments

The Bi-Quad filter is the most general 2nd order filter. It has two poles and two zeros. It can be thought of as a high-pass filter in series with a low-pass filter.

Accessibility

Read-Write



SLVBODD values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.6.2 SLVBODF

Description

SLVBODF is a real array, with one element for each axis in the system, and is used for setting the denominator natural frequency value of the Bi-Quad filter algorithm applied to the velocity loop control in addition to the existing 2nd order Low-pass and Notch filters for the given axis.

Syntax

SLVBODF(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value designates denominator value of the Bi-Quad filter algorithm ranging from 0.1 to 4000 [Hz]. See <i>SPiiPlus ACSPL+ Programmer Guide</i> .

Tag

209

Comments

The Bi-Quad filter is the most general 2nd order filter. It has two poles and two zeros. It can be thought of as a high-pass filter in series with a low-pass filter.

Accessibility

Read-Write



SLVBODF values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functionss

acsc_ReadReal, acsc_WriteReal

3.14.6.3 SLVBOND

Description

SLVBOND is a real array, with one element for each axis in the system, and is used for setting the damping ratio numerator for a Bi-Quad filter to the velocity loop control in addition to the existing 2nd order Low-pass and Notch filters for the given axis.

Syntax

SLVBOND(axis_index) = value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value designates numerator value of the Bi-Quad damping ratio ranging from 0.1 to 1. See <i>SPiiPlus ACSPL+ Programmer Guide</i> .

Tag

210

Comments

The Bi-Quad filter is the most general 2nd order filter. It has two poles and two zeros. It can be thought of as a high-pass filter in series with a low-pass filter.

Accessibility

Read-Write



SLVBOND values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.6.4 SLVBONF**Description**

SLVBONF is a real array, with one element for each axis in the system, and is used for setting the numerator natural frequency value of the Bi-Quad filter algorithm applied to the velocity loop control in addition to the existing 2nd order Low-pass and Notch filters for the given axis.

Syntax**SLVBONF(*axis_index*) = *value*****Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value designates numerator value of the Bi-Quad filter algorithm ranging from 0.1 to 4000 [Hz]. See <i>SPiiPlus ACSPL+ Programmer Guide</i> .

Tag

211

Comments

The Bi-Quad filter is the most general 2nd order filter. It has two poles and two zeros. It can be thought of as a high-pass filter in series with a low-pass filter.

Accessibility

Read-Write



SLVBONF values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.7 Servo-Loop Position Variables

The Servo-Loop Position variables are:

Name	Description
SLDRA	Defines disturbance rejection.
SLDRAIF	Provides an Idle Factor to the SLDRA.
SLDRX	Defines the maximum DRA correction for a given axis.
SLPKI	Specifies the position loop integrator coefficient
SLPKIIF	Provides the Idle Factor to the SLPKI (Integrator Gain - Position) variable
SLPKISF	Provides the Settle Factor to the SLPKI (Integrator Gain - Position) variable
SLPLI	Defines the limit for the position of the specified axis
SLPKP	Sets the proportional coefficient of the position for the specified axis.
SLPKPIF	Provides an Idle Factor to the SLPKP variable.
SLPKPSF	Provides an Settle Factor to the SLPKP variable.

3.14.7.1 SLDRA**Description**

SLDRA is a real array, with one element for each axis in the system, and is used for defining the DRA frequency for the given axis.

The ACS proprietary Disturbance Rejection Algorithm (DRA) is used to improve the disturbance rejection response of the servo, and helps to minimize the position error during the settling phase and shorten the settling time.

Syntax

SLDRA(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value designates the DRA frequency ranging from 0 to 1500 [Hz].

Tag

206

Comments

The most common use of DRA is to improve the settling of systems mounted on passive isolation platforms. Passive isolation is typically used to isolate systems from disturbances transmitted from the floor. They employ a seismic mass supported on a soft spring made of rubber, metal, or air. The spring's damping action absorbs vibrations above the spring's resonance. For this reason, passive isolation manufacturers usually try to lower spring resonant frequency to increase the effective isolation range. When a servo force is applied to generate motion, it also acts on the isolated stationary base, causing it to vibrate. Because the frequency is low (usually below 1 Hz, to 10 Hz) and damping is very light, the isolation system continues vibrating long after the motion profile has ended. This vibration acts as disturbance to the servo system, introduces position error, and extends the settling time.

The DRA is used to minimize the latter effect and improve the position error during settling.

Accessibility

Read-Write



SLDRA values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.7.2 SLDRAIF

Description

SLDRAIF is a real array with one element for each axis in the system. It is used for providing an Idle Factor to the **SLDRA** variable.

Syntax**SLDRAIF** *axis_index* = *value***Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0.0 and 100.0.

Tag

230

Accessibility

Read-Write



SLDRAIF values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.7.3 SLDRX**Description**

SLDRX is a real array, with one element for each axis in the system, and is used for defining the maximum DRA correction for the given axis.

The ACS proprietary Disturbance Rejection Algorithm (DRA) is used to improve the disturbance rejection response of the servo, and helps to minimize the position error during the settling phase and shorten the settling time.

Syntax**SLDRX** (*axis_index*) = *value***Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value designates the DRA correction ranging from 0 to 2 ²³ .

Tag

207

Comments

The most common use of DRA is to improve the settling of systems mounted on passive isolation platforms. Passive isolation is typically used to isolate systems from disturbances transmitted from the floor. They employ a seismic mass supported on a soft spring made of rubber, metal, or air. The spring's damping action absorbs vibrations above the spring's resonance. For this reason, passive isolation manufacturers usually try to lower spring resonant frequency to increase the effective isolation range. When a servo force is applied to generate motion, it also acts on the isolated stationary base, causing it to vibrate. Because the frequency is low (usually below 1 Hz, to 10 Hz) and damping is very light, the isolation system continues vibrating long after the motion profile has ended. This vibration acts as disturbance to the servo system, introduces position error, and extends the settling time.

The DRA is used to minimize the latter effect and improve the position error during settling.

Accessibility

Read-Write



SLDRX values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.7.4 SLPKI

Description

SLPKI is a real array, with one element for each axis in the system, and is used for specifying the position loop integrator coefficient.

Syntax

SLPKI(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0 to 20000; Default = 0.

Tag

174

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.7.5 SLPKIIF

Description

SLPKIIF is a real array with one element for each axis in the system. It is used for providing an Idle Factor to the [SLPKI](#) (Integrator Gain - Position) variable.

Syntax

SLPKIIF(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0.0 to 100.0; Default = 1.

Tag

260

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

acsc_ReadReal, acsc_WriteReal

3.14.7.6 SLPKISF

Description

SLPKISF is a real array with one element for each axis in the system. It is used for providing a Settle Factor to the [SLPKI](#) (Integrator Gain - Position) variable.

Syntax

SLPKISF(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0.0 to 100.0; Default = 1.

Tag

261

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.7.7 SLPKITF

Description

SLPKITF increases the velocity loop proportional coefficient when the axis is close to the target position. It is a real array with one element for each axis in the system.



SLPKITF is supported for the NanoPWM drives only.

Syntax

SLPKITF(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0.0 to 100.0; Default = 1.

Tag

269

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.7.8 SLPLI

Description

SLPLI is a real array, with one element for each axis in the system, and is used for providing an integrator limit for the position of the specified axis.

Syntax**SLPLI**(*axis_index*) = *value***Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0 to 100; Default = 0.

Tag

176

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.7.9 SLPKP**Description**

SLPKP is a real array, with one element for each axis in the system, and is used for setting the proportional coefficient of the position for the specified axis.

Syntax**SLPKP**(*axis_index*) = *value***Arguments**

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges between 0 to 16777215, Default = 0.

Tag

175

Comments

Motor movement during commutation largely depends on the servo-loop parameters.

COMMUT will not operate properly if **SLPKP** is set to zero, or the integrator is very low.

Accessibility

Read-Write



SLPKP values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

Related ACSPL+ Commands

COMMUT

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.7.10 SLPKPIF

Description

SLPKPIF is a real array with one element for each axis in the system. It is used for providing an Idle Factor to the **SLPKP** variable.

Syntax

SLPKPIF *axis_index* = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0.0 and 100.0.

Tag

231

Accessibility

Read-Write



SLPKPIF values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.7.11 SLPKPSF

Description

SLPKPSF is a real array with one element for each axis in the system. It is used for providing a Settle Factor to the **SLPKP** variable.

Syntax

SLPKPSF *axis_index* = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0.0 and 100.0.

Tag

232

Accessibility

Read-Write



SLPKPSF values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.7.12 SLPKPTF

Description

SLPKPTF increases the position loop proportional coefficient when the axis is close to the target position. It is a real array with one element for each axis in the system.



SLPKPTF is supported for the NanoPWM drives only.

Syntax

SLPKPTF *axis_index* = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 1.0 and 100.0. Default = 1.

Tag

270

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.8 Servo-Loop Compensations Variables

Servo-Loop Compensations variables are used to set various parameters that provide compensation to overcome certain motion problems. The Servo-Loop Compensation variables are:

Name	Description
SLAFF	Defines acceleration feed forward for a given axis
SLFRCD	Compensation for dynamic friction

3.14.8.1 SLAFF**Description**

SLAFF is a real array, with one element for each axis in the system, and is used for specifying the acceleration feed forward of the specified axis.

Syntax

SLAFF(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 0 to 16777215; Default: 0.

Tag

148

Accessibility

Read-Write



SLAFF values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.8.2 SLFRCD

Description

SLFRCD is a real array, with one element for each axis in the system, and is used for providing dynamic friction compensation at the maximum velocity.

Syntax

SLFRCD(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges between 0% to 5% of maximum command.

Tag

168

Comments

SLFRCD provides dynamic compensation at the maximum velocity **XVEL**. For lower velocities, the compensation is reduced proportionally with the velocity. The value of **SLFRCD** is given as a percentage (range is 0 to 50%) of the maximum command.

Accessibility

Read-Write



SLFRCD values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.9 Servo Loop Stepper Variables

This section contains a collection of variables employed for calculation of the position correction for stepper motors.

Name	Description
MFLAGSX	Extended Motor Flags
SLSDZ	Array of dead zone values for position correction
SLSKI	Array of integral gain values for position correction
SLSKP	Array of proportional gain values for position correction
SLSMC	Array of maximum allowed stepper correction values
SLSOUT	Returns calculated stepper correction for given axis
SLSRL	Array of rate limiter values for position correction

3.14.9.1 MFLAGSX

Description

MFLAGSX is an integer array with one element for each axis in the system, each element of which contains a set of bits used for configuring the motor. It is an extension of the **MFLAGS** variable.

Syntax

```
MFLAGSX(Axis_Index).bit_designator = 0|1
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
Bit_designator	An MFLAGSX bit designator as described below

MFLAGSX Bit Designators

Bit Name	No.	Description
#STCLFULL	0	0 (default): Do not use position correction for Stepper motors working under closed loop. 1: Activate Full mode of the stepper closed loop position correction mechanism . This bit is mutually exclusive to #STCLEND and #STCLSP, as such only one of them may be 1 at a time.
#STCLEND	1	0 (default): Do not use position correction for Stepper motors working under closed loop 1: Activate End mode of the stepper closed loop position correction mechanism This bit is mutually exclusive to #STCLFULL and #STCLSP, as such only one of them may be 1 at a time.
#STCLSP	2	0 (default): Do not use servo processor closed-loop stepper algorithm 1: Use servo processor closed-loop stepper algorithm Mutually exclusive to #STCLFULL,#STCLEND.
#VOLTMODE	3	0: Use current mode for control 1: Use voltage mode instead of current mode to compensate for low resolution
#HLIMSWAP	4	0: Left/Right limit signal swapping is disabled 1: Left/Right limit signal swapping is enabled
#SATPROT	5	1: Saturation Protection enabled (default) 0: Saturation Protection disabled

Tag

357

Comments

When saturation protection (**MFLAGSX.#SATPROT=1**) is currently available for the following products: NPMpm, UDMxx, IDMxx.

This variable is supported in ADK versions 2.70 and higher.

Related ACSPL+ Variables

SLSKI, SLSKP, SLSDZ

Accessibility

Read-Write



MFLAGSX values cannot be modified if protection is applied to this variable through SPiiPlus MMI Application Studio→Toolbox→Application Development→ Protection

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadInteger(), acsc_WriteInteger()

3.14.9.2 SLSDZ

Description

SLSDZ is a real array with one element for each axis in the system, it is used for the closed loop operation of steppers and represents the dead zone of the position correction.

Syntax

```
SLSDZ (Axis_Index) = Value
```

Arguments

Axis_ Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
----------------	---

Tag

356

Comments

Dead zone [user units]. Inside this zone ($|PE| < \mathbf{SLSDZ}$) algorithm is inactive.

This variable is supported in ADK versions 2.70 and higher.

Related ACSPL+ Variables

SLSKI, SLSKP, SLSRL, MFLAGSX

Accessibility

Read-Write



SLSDZ values cannot be modified if protection is applied to this variable through **SPiiPlus** MMI Application Studio→Toolbox→Application Development→ Protection

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadInteger(), acsc_WriteInteger()

3.14.9.3 *SLSKI***Description**

SLSKI is a real array with one element for each axis in the system, it is used for the closed loop operation of steppers and represents the integral gain[rad/sec] of the position correction.

Syntax

```
SLSKI (Axis_Index) = Value
```

Arguments

Axis_Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
------------	---

Tag

353

Comments

This variable is supported in ADK versions 2.70 and higher.

Related ACSPL+ Variables

SLSKP, SLSRL, SLSDZ

Accessibility

Read-Write



SLSKI values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio→Toolbox→Application Development→Protection**

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadInteger(), acsc_WriteInteger()

3.14.9.4 *SLSKP***Description**

SLSKP is a real array with one element for each axis in the system, it is used for the closed loop operation of steppers and represents proportional gain of the position correction.

Syntax

```
SLSKP (Axis_Index) = Value
```

Arguments

Axis_ Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
----------------	---

Tag

354

Comments

Proportional gain is unitless.

This variable is supported in ADK versions 2.70 and higher.

Related ACSPL+ Variables

SLSKI, SLSRL, SLSDZ, MFLAGSX

Accessibility

Read-Write



SLSKP values cannot be modified if protection is applied to this variable **through SPiiPlus MMI Application Studio→Toolbox→Application Development→Protection**

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadInteger(), acsc_WriteInteger()

3.14.9.5 SLSMC**Description**

SLSMC is a real array with one element for each axis in the system, it is used for the closed loop operation of steppers and represents the maximum allowed stepper correction.

Syntax

```
SLSMC(Axis_Index) = Value
```

Arguments

Axis_ Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
----------------	---

Tag

360

Comments

Maximum correction value [user units]. If this value is reached FAULT.#PE is set.

This variable is supported in ADK versions 2.70 and higher.

Related ACSPL+ Variables

SLSKI, SLSKP, SLSRL, SLSOUT, MFLAGSX

Accessibility

Read-Write



If the accumulated correction of a closed-loop algorithm reaches the maximum limit as defined by **SLSMC**, a non-critical Position Fault error is raised (error 5022).

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadInteger(), acsc_WriteInteger()

3.14.9.6 SLSOUT

Description

SLSOUT is a real array with one element for each axis in the system, it is used for the closed loop operation of steppers and represents the calculated stepper correction

Syntax

```
SLSOUT(Axis_Index)
```

Arguments

Axis_ Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
----------------	---

Tag

359

Comments

Output of the PI loop [user units].

This variable is supported in ADK versions 2.70 and higher.

SLSOUT is reset upon disabling of an axis, or upon setting **RPOS** using the set command.

Related ACSPL+ Variables

SLSKI, SLSKP, SLSRL, MFLAGSX

Accessibility

Read only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadInteger()

3.14.9.7 SLSRL

Description

SLSRL is a real array with one element for each axis in the system, it is used for the closed loop operation of steppers and represents the rate limiter of the position correction.

Syntax

```
SLSRL(Axis_Index) = Value
```

Arguments

Axis_ Index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
----------------	---

Tag

355

Comments

Rate limiter of the PI output [user unit/sec].

This variable is supported in ADK versions 2.70 and higher.

Related ACSPL+ Variables

SLSKI, SLSKP, SLSDZ

Accessibility

Read-Write



SLSRL values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio→Toolbox→Application Development→Protection**

.NET Library Method

ReadVariable(), WriteVariable()

C Library Function

acsc_ReadInteger(), acsc_WriteInteger()

3.14.10 Servo-Loop Miscellaneous Variables

This section contains a collection of miscellaneous variables employed for specific servo-loop purposes. The Servo-Loop Miscellaneous variables are:

Name	Description
SLCROUT	Commutation feedback routing , see Commutation Variables
SLGCAXN	Specifies the complementary gantry axis
SLPROUT	Position feedback routing
SLP2ROUT	Sets the feedback routing of the secondary feedback position
SLVROUT	Velocity feedback routing

3.14.10.1 SLCROUT

Description

SLCROUT is an integer array, with one element for each axis in the system, and is used for setting the feedback routing of the velocity commutation for the specified axis.

Syntax

SLCROUT(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	The value values and the feedback sources associated with them are given in Table 3-21 . Default = 0.

Table 3-21. SLCROUT Values

SLCROUT	FACC (SPonly)
0	According to E_TYPE velocity
001	From channel 0 quadrature velocity or Absolute Encoder velocity
002	From channel 0 SINCOS velocity
003	From channel 0 HSSI velocity
004	From analog input 0
005	From channel 0 resolver velocity
101	From channel 1 quadrature velocity Absolute Encoder velocity
102	From channel 1 SINCOS velocity

SLCROUT	FACC (SPonly)
103	From channel 1 HSSI velocity
104	From analog input 1
105	From channel 1 resolver velocity
201	From channel 2 quadrature velocity Absolute Encoder velocity
202	From channel 2 SINCOS velocity
203	From channel 2 HSSI velocity
204	From analog input 2
205	From channel 2 resolver velocity
301	From channel 3 quadrature velocity Absolute Encoder velocity
302	From channel 3 SINCOS velocity
303	From channel 3 HSSI velocity
304	From analog input 3
305	From channel 3 resolver velocity

Tag

159

Accessibility

Read-Write



SLCROUT values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.14.10.2 SLGCAXN**Description**

SLGCAXN is a read-only integer array, with one element for each axis in the system, which specifies the complementary gantry axis. The value can be viewed SPiiPlus MMI Application Studio

Communication Terminal window or its value can be assigned to another variable, for example:
Var = **SLGCAXN**(axis_index).



In order to change gantry axis allocation, the **SETCONF**(267) command should be used.

Syntax

SLGCAXN(axis_index)

Arguments

axis_index	axis_index designates the specific axis: 0 - X, 1 - Y, etc.
-------------------	---

Tag

256

Accessibility

Read Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.14.10.3 *SLPROUT*

Description

SLPROUT is a real array, with one element for each axis in the system, and is used for setting the feedback routing of the position for the specified axis.

Syntax

SLPROUT(axis_index) = value

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	The value values and the feedback sources associated with them are given in Table 3-22 . Default = 0.

Table 3-22. SLPROUT Values

SLPROUT	FPOS
0	According to E_TYPE position
001	From channel 0 quadrature position Absolute Encoder position
002	From channel 0 SINCOS position
003	From channel 0 HSSI position
004	From analog input 0
005	From channel 0 resolver position
101	From channel 1 quadrature position Absolute Encoder position
102	From channel 1 SINCOS position
103	From channel 1 HSSI position
104	From analog input 1
105	From channel 1 resolver position
201	From channel 2 quadrature position Absolute Encoder position
202	From channel 2 SINCOS position
203	From channel 2 HSSI position
204	From analog input 2
205	From channel 2 resolver position
301	From channel 3 quadrature position Absolute Encoder position
302	From channel 3 SINCOS position
303	From channel 3 HSSI position
304	From analog input 3
305	From channel 3 resolver position

Tag

177

Comments

The controller supports a standard control loop configuration where 0 feedback position (**FPOS**) is obtained from the 0 encoder, **FPOS(1)** from the 1 encoder, etc.

SLPROUT¹ 0 indicates **FPOS** is from an alternative sensor, for example, if **SLPROUT(0)** is 0104, **FPOS** is obtained from an analog input 0 rather than from the encoder. In this case, the feedback source could be a potentiometer or any other device that produces analog voltage proportional to the motor position.

The meaning of the routing value depends on the axis and the controller model. For example, a value of 1 specified for the 0 or 2 axis selects the 0 encoder, the same value for the 1 or 2 axis selects the 1 encoder.

Accessibility

Read-Write



SLPROUT values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.10.4 SLP2ROUT

SLP2ROUT is variable for setting the feedback routing of the secondary feedback position.

Description

SLP2ROUT is an integer array, one element for each axis in the system, and is used for setting the feedback routing of the secondary feedback position for the specified axis.

Syntax

SLP2ROUT(<axis>)=value

Arguments

The value values and the feedback sources associated with them are given below. The default value is 0.

Value	F2POS
0	Secondary Feedback is not used
001	From channel 0 quadrature position or Absolute Encoder position
002	From channel 0 SINCOS position
003	From channel 0 HSSI position101

Value	F2POS
004	From analog input 0
005	From channel 0 resolver position
101	From channel 1 quadrature position or Absolute Encoder position
102	From channel 1 SINCOS position
103	From channel 1 HSSI position
104	From analog input 1
105	From channel 1 resolver position
201	From channel 2 quadrature position or Absolute Encoder position
202	From channel 2 SINCOS position
203	From channel 2 HSSI position
204	From analog input 2
205	From channel 2 resolver position
301	From channel 3 quadrature position or Absolute Encoder position
302	From channel 3 SINCOS position
303	From channel 3 HSSI position
304	From analog input 3
305	From channel 3 resolver position

Comments

SLP2ROUT variable can be used for routing when applied on secondary feedback only.

In a system configured for Multi-Channel Feedback, **SLP2ROUT** is used in a different way.

Example

```

E_TYPE(0)=3      ! AqB
E2_TYPE(0) = 4    ! SinCos
SLP2ROUT(0) = 002 ! in other products the value is 102

```

Tag

301

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.14.10.5 SLTFWID

Description

SLTFWID determines the distance to the target at which the position and velocity loops gains will be increased by 50%. It is a real array with one element for each axis in the system.



SLTFWID is supported for the NanoPWM drives only.

Syntax

SLTFWID(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	A real value ranging between 0.0 and 4294967295. Default = 0.

Tag

273

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.10.6 SLVROUT

Description

SLVROUT is a real array, with one element for each axis in the system, and is used for setting the feedback routing of the velocity for the specified axis.

Syntax

SLVROUT(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	The value values and the feedback sources associated with them are given in Table 3-23 . Default = 0.

Table 3-23. SLVROUT Values

SLVROUT	FVEL (SP only)
0	According to E_TYPE velocity
001	From channel 0 quadrature velocity Absolute Encoder velocity
002	From channel 0 SINCOS velocity
003	From channel 0 HSSI velocity
004	From analog input 0
005	From channel 0 resolver velocity
101	From channel 1 quadrature velocity Absolute Encoder velocity
102	From channel 1 SINCOS velocity
103	From channel 1 HSSI velocity
104	From analog input 1
105	From channel 1 resolver velocity
201	From channel 2 quadrature velocity Absolute Encoder velocity
202	From channel 2 SINCOS velocity
203	From channel 2 HSSI velocity
204	From analog input 2
205	From channel 2 resolver velocity
301	From channel 3 quadrature velocity Absolute Encoder velocity
302	From channel 3 SINCOS velocity
303	From channel 3 HSSI velocity

SLVROUT	FVEL (SP only)
304	From analog input 3
305	From channel 3 resolver velocity

Tag

188

Accessibility

Read-Write



SLVROUT values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.14.10.7 SLXROUT

Description

SLXROUT is a real array

Syntax

Tag

166

Accessibility

Read-Only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadReal()

3.14.11 Non-Linear Control Variables

See *Non-Linear Control Application Note* for more information about use of non-linear control variables.

The following variables are used to implement non-linear control.

Name	Description
SLPAP	Holds the exponent (α) of the Position-Loop Proportional Non-Linear Control
SLPDP	The Linear Range (δ) of the Position-Loop Proportional Non-Linear Control
SLPAI	The exponent (α) of the Position-Loop Integral Non-Linear Control
SLPDI	The Linear Range (δ) of the Position-Loop Integral Non-Linear Control
SLVAP	The exponent (α) of the Velocity-Loop Proportional Non-Linear Control
SLVDP	The Linear Range (δ) of the Velocity-Loop Proportional Non-Linear Control
SLVAI	The exponent (α) of the Velocity-Loop Integral Non-Linear Control
SLVDI	he Linear Range (δ) of the Velocity-Loop Integral Non-Linear Control

3.14.11.1 SLPAP

Description

SLPAP is a real array, the size of which is determined by the total number of axes in the system.

SLPAP holds the exponent (α) of the Position-Loop Proportional Non-Linear Control.

Syntax

SLPAP(index) = *value*

Tag

395

Comments

This variable is supported in version 3.10 and higher.

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable(), WriteVariable()

C Library Functions

acsc_ReadReal(), acsc_WriteReal()

3.14.11.2 SLPDP

Description

SLPDP is a real array, the size of which is determined by the total number of axes in the system. This is the Linear Range (δ) of the Position-Loop Proportional Non-Linear Control.

Syntax

SLPDP(index) = value

Arguments

index	Axis index, from 0 to number of axes in the system-1
value	Default value is 1. The range is [0,1]

Tag**Comments**

A value of 1 sets the Linear range of the Non-Linear Gain curve to the Position-Error Limit.

If the value is not default and the required license for this feature is missing, error 3300 "Non Linear Control License is required" is given.

This variable is supported in version 3.10 and higher.

Related ACSPL+ Variables

SLPAP, SLPAI, SLPDI, SLVAP, SLVDP, SLVAI, SLVDI

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable(), WriteVariable()

C Library Functions

acsc_ReadReal(), acsc_WriteReal()

3.14.11.3 SLPAI**Description**

SLPAI is a real array, the size of which is determined by the total number of axes in the system. This is the exponent (α) of the Position-Loop Integral Non-Linear Control.

Syntax

SLPAI(index) = value

Arguments

index	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1
value	Default value is 1. The range is [0,100]

Tag**Comments**

A value of 1 sets the Linear Range of the Non-Linear Gain curve to the Position-Error Limit.

If the value is not default and the required license for this feature is missing, error 3300 "Non Linear Control License is required" is given.

This variable is supported in version 3.10 and higher.

Related ACSPL+ Variables**SLPAP, SLPDP, SLPAI, SLVAP, SLVDP, SLVAI, SLVDI****Accessibility**

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable(), WriteVariable()

C Library Functions

acsc_ReadReal(), acsc_WriteReal()

3.14.11.4 SLPDI**Description**

SLPDI is a real array, the size of which is determined by the total number of axes in the system. This is the Linear Range (δ) of the Position-Loop Integral Non-Linear Control.

Syntax**SLPDI**(index) = value**Arguments**

index	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1
value	Default value is 1. The range is [0,1]

Tag

398

Comments

A value of 1 sets the Linear Range of the Non-Linear Gain curve to the Position-Error Limit.

If the value is not default and the required license for this feature is missing, error 3300 "Non Linear Control License is required" is given.

This variable is supported in version 3.10 and higher.

Related ACSPL+ Variables**SLPAP, SLPDP, SLPAI, SLVAP, SLVDP, SLVAI, SLVDI****Accessibility**

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable(), WriteVariable()

C Library Functions

acsc_ReadReal(), acsc_WriteReal()

3.14.11.5 *SLVAP***Description**

SLVAP is a real array, the size of which is determined by the total number of axes in the system. This is the exponent (α) of the Velocity-Loop Proportional Non-Linear Control.

Syntax

SLVAP(index) = value

Arguments

index	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1
value	Default value is 1. The range is [0,1]

Tag**Comments**

A value of 1 sets the Velocity-Loop Proportional Gain to Linear.

If the value is not default and the required license for this feature is missing, error 3300 "Non Linear Control License is required" is given.

This variable is supported in version 3.10 and higher.

Related ACSPL+ Variables

SLPAP, SLPDP, SLPAI, SLPDI, SLVDP, SLVAI, SLVDI

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable(), WriteVariable()

C Library Functions

acsc_ReadReal(), acsc_WriteReal()

3.14.11.6 *SLVDP***Description**

SLVDP is a real array, the size of which is determined by the total number of axes in the system. This is the Linear Range (δ) of the Velocity-Loop Proportional Non-Linear Control.

Syntax

SLVDP(index) = value

Arguments

index	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1
value	

Switches**Return Value**

None

Tag

400

Comments

This variable is supported in version 3.10 and higher.

Example

3.14.11.7 SLVAI

Description

SLVAI is a real array, the size of which is determined by the total number of axes in the system. This is the exponent (α) of the Velocity-Loop Integral Non-Linear Control.

Syntax

SLVAI(index) = value

Arguments

index	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1
value	Default value is 1. The range is [0,1]

Tag

401

Comments

A value of 1 sets the Velocity-Loop Integral Gain to Linear.

If the value is not default and the required license for this feature is missing, error 3300 "Non Linear Control License is required" is given.

This variable is supported in version 3.10 and higher.

Related ACSPL+ Variables

SLPAP, SLPDP, SLPAI, SLPDI, SLVDP, SLVAI, SLVDI

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable(), WriteVariable()

C Library Functions

acsc_ReadReal(), acsc_WriteReal()

3.14.11.8 SLVDI

Description

SLVDI is a real array, the size of which is determined by the total number of axes in the system. This is the Linear Range (δ) of the Velocity-Loop Integral Non-Linear Control.

Syntax

```
SLVDI(index) = value
```

Arguments

index	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1
value	Default value is 1. The range is [0,1]

Tag

402

Comments

A value of 1 sets the Linear Range of the Non-Linear Gain curve to the Maximum Current Command. If the value is not default and the required license for this feature is missing, error 3300 "Non Linear Control License is required" is given.

This variable is supported in version 3.10 and higher.

Related ACSPL+ Variables

SLPAP, SLPDP, SLPAI, SLPDI, SLVDP, SLVAI, SLVDI

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable(), WriteVariable()

C Library Functions

acsc_ReadReal(), acsc_WriteReal()

3.15 Commutation Variables

The Servo-Loop Commutation variables are:

Name	Description
SLCHALL	Hall Shift
SLCNP	Number of Poles
SLCOFFS	Commutation Offset

Name	Description
SLCORG	Commutation Origin
SLCPRD	Commutation Period
SLHROUT	Setting Hall State Routing
SLSTHALL	Getting Hall State



The low-level variables in this section are normally not used by the user. Generally, these variables are defined during the axis adjustment using **SPiiPlus MMI Application Studio** → **Toolbox** → **Setup** → **Adjuster** or the [COMMUT](#) command.

3.15.1 SLCHALL

Description

SLCHALL is an integer array, with one element for each axis in the system, and serves for storing the Hall shift.

Tag

192

Comments

The **Adjuster** commutation program calculates this parameter and saves it.



Do not change this parameter manually.

Accessibility

Read-Write



SLCHALL values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.15.2 SLCNP

Description

SLCNP is an integer array, with one element for each axis in the system, and defines the number of poles for a rotary motor.

Syntax

SLCNP(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from 2 to 1000, Default = 4.

Tag

152

Comments

For linear motors, set **SLCNP**=2.

Accessibility

Read-Write



SLCNP values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.15.3 SLCOFFS

Description

SLCOFFS is a real array, with one element for each axis in the system, and defines a commutation offset in electrical degrees to be added to the commutation phase.

Syntax

SLCOFFS(*axis_index*) = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from -60 to 60, Default=0.

Tag

153

Comments

SLCOFFS defines **SLCOFFS** is valid only if a brushless motor is specified (**MFLAGS**(axis_index).#**BRUSHL** = 1).

Assignment to **SLCOFFS** immediately changes the commutation phase. Use **SLCOFFS** to introduce a small correction to the commutation phase.

Accessibility

Read-Write



SLCOFFS values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.15.4 *SLCORG*

Description

SLCORG is a real array, with one element for each axis in the system, that defines the commutation phase in electrical degrees at the point of origin which is usually at the index point.

Syntax

SLCORG(axis_index) = value

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>value</i>	value ranges from: > -60 to 60, Default=0 > 0 to 360, Default=0

Tag

156

Comments

SLCORG is valid only if a brushless motor has been specified, i.e., **MFLAGS**(axis_index).#**BRUSHL** = 1.

Accessibility

Read-Write



SLCORG values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.15.5 SLCPRD

Description

SLCPRD is a real array, with one element for each axis in the system, that defines the servo-loop commutation period.

Syntax

SLCPRD(*axis_index*) = *value*

Arguments

axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value ranges from 256 to 16777215, Default=8000.

Tag

158

Comments

SLCPRD defines the feedback counts per revolution for rotary motors and the feedback counts per two magnetic pitches for linear motors.

Accessibility

Read-Write



SLCPRD values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.15.6 SLHROUT

Hall Routing is available using ACSPL+ variable: **SLHROUT**

Description

SLHROUT is an integer array, with one element for each axis in the system, and is used for setting the Hall state routing for the specified axis.

Syntax

SLHROUT(<axis>)=value

Value	SLSTHALL
0	Default
001	From channel 0
101	From channel 1
201	From channel 2
301	From channel 3

Comments

Hall state is an integer number within the range [0,5]. `Getconf(262,<index>)` returns Hall state of axis with index <index>. If `SLHROUT(<index>)` is not 0, `Getconf(262,<index>)` returns the hall state of the channel based on the table defined above.

Tag

302

Accessibility

Read-Write

Com Library Methods and .NET Library Methods

`ReadVariable`, `WriteVariable`

C Library Functions

`acsc_ReadInteger`, `acsc_WriteInteger`

3.15.7 SLSTHALL

Description

SLSTHALL is an integer array, with one element for each axis in the system, and is used for getting the Hall state of each axis.

The value is an integer number, in range of {-1,5}. -1 means invalid Hall state.

Comments

The **SLSTHALL** variable's value is being updated every cycle.

Tag

196

Accessibility

Read only

Com Library Methods and .NET Library Methods

ReadVariable

3.16 System Configuration Variables

The System Configuration variables are:

Name	Description
CFG	Configuration Mode
CTIME	Controller Cycle Time
EXTFAC	Conversion factor between the SL2-100 protocol transferred units (microns [μm]) and ACS user units.
FOLLOWCH	Maps an axis to a data channel of a SLEC module
G_01WCS...G_12WCS	Defines one of the 12 work-piece coordinate systems a user can set as part of the CNC setup/configuration
GPEXL	Indicates the GSP program executed block
GSPEXL	Indicates the line in a running buffer that called a subroutine
GUFAC	Holds the value a conversion factor, from 'Common Physical Units' in [mm] to 'Controller Units'
IENA	Interrupt Enable/Disable
IMASK	Interrupt Mask
ISENA	Interrupt-Specific Enable/Disable
S_FLAGS	System Flags

Name	Description
<code>S_SETUP</code>	System Settings
<code>XSEGAMAX</code>	Maximal processing angles
<code>XSEGAMIN</code>	Minimal processing angles
<code>XSEGRMAX</code>	Maximal arc radius difference
<code>XSEGRMIN</code>	Minimal arc radius

3.16.1 *CFG*

Description

CFG is an integer variable that indicates the application protection configuration mode.

Syntax

CFG = *value*

Arguments

<i>value</i>	value can be one of the following: 0: Controller is in protected mode 1: Controller is in normal mode
--------------	--

Tag

14

Comments

An attempt to assign a value to a protected variable when **CFG** = 0 causes an error.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Commands

acsc_ReadInteger

3.16.2 *CTIME*

CTIME is a real variable that defines the controller cycle time.

Syntax

CTIME = *value*

Arguments**value****value** can be 0.2, 0.25, 0.5, or 1.0 milliseconds (depending on the controller model).**Tag**

17

Comments

Many operations in the controller are synchronized to the controller cycle. For example, profile generation is executed each controller cycle.



After setting **CTIME** to a new value, the configuration should be saved to flash memory. Afterwards run the MMI System Setup procedure.

For information about valid **CTIME** values see the EtherCAT Cycle Rate section in the Installation Guide for the relevant controller.

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

3.16.3 EXTFAC**Description**

EXTFAC is a real array, with one element for each axis in the system. It is used as a conversion factor between the SL2-100 protocol transferred units (microns [μm]) and ACS user units.

Syntax**EXTFAC**(axis)**Argument****axis**

Axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

Tag

326

3.16.4 FOLLOWCH**Description**

FOLLOWCH is an integer array, with one element for each axis in the system, the elements of which maps an axis to a data channel of a SLEC EtherCAT node.

Syntax

FOLLOWCH(axis)xbit

Arguement

axis	Axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
bit	A description of the AST bit designators is:
bit	Description
0-15	Data channel of SLEC
16-31	EtherCAT slave unit ID of the SLEC EtherCAT node

Tag

237

Example 1

For an EtherCAT system that includes the following devices:

- > SPiiPlusEC motion controller and EtherCAT master
- > SPiiPlusCMHV EtherCAT control module
- > SLEC. EtherCAT node

SLEC node number 1:

- > Axis 0 will follow channel 0
- > Axis 2 will follow channel 1

```
FOLLOWCH (0) = 0x00010000
FOLLOWCH (2) = 0x00010001
```

Example 2

For an EtherCAT system that includes the following devices:

- > SpiiPlusEC motion controller and EtherCAT master
- > SLEC EtherCAT node
- > MC4U(8 axes)
- > MC4U(8 axes)
- > SLEC.EtherCAT node

 Each MC4U includes 2 drives with 4 axes each for a total of 16 total axes..

SLEC node number 0:

- > Axes 0 will follow channel 0
- > Axes 1 will follow channel 1.

SLEC node number 3:

- > Axes 8 will follow channel 0
- > Axes 9 will follow channel 1.

```
FOLLOWCH (0) = 0x00000000
FOLLOWCH (1) = 0x00000001
FOLLOWCH (8) = 0x00050000
FOLLOWCH (9) = 0x00050001
```

3.16.5 G_01WCS...G_12WCS

Description

G_01WCS to **G_012WCS** are each a real array, with one element for each axis in the system (up to 9 axes). Each is used for defining one of the 12 work-piece coordinate systems a user can set as part of the CNC setup/configuration. During the execution of a GSP program, a user can select one of them to work. If user choses going back to work in the Machine Coordinate System, he can then chose to clear it.

Syntax

N/A

Arguments

N/A

Tag

284...295

Accessibility

Read Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.16.6 GPEXL

Description

GPEXL is an integer array, with one element for each program buffer in the system. It indicates the GSP program executed block. If block executes motion then GPEXL will indicate this line till this motion completion.

If a buffer is not running, GPEXL equals 0.

Syntax

N/A

Arguments

N/A

Tag

296

Accessibility

Read Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.16.7 GSPEXL

Description

GSPEXL is an integer array, with one element for each program buffer in the system. It indicates the line in a running buffer that called a subroutine.

If a buffer is not running, GSPEXL equals 0.

Syntax

N/A

Arguments

N/A

Tag

347

Accessibility

Read Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.16.8 GUFAC

Description

GUFAC is a real array, with one element for each program buffer in the system. Each entry in the array holds the value a conversion factor, from 'Common Physical Units' in [mm] to 'Controller Units' (In the world of ACSPL+, those 'Controller Units' are sometimes related to as 'User Units'). As part of GSP modality data, default value of GUFAC is 1.0 for all buffers.

Syntax

N/A

Arguments

N/A

Tag

298

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

3.16.9 IENA

Description

IENA is a 23-bit mask variable used for enabling or disabling software and hardware interrupts from a specific source.

Syntax

IENA.*bit_designator* = 1|0

Arguments

<i>bit_designator</i>	The meanings of bit_designator are given in Table 3-24 .
-----------------------	---

Table 3-24. IENA Bit Description

Bit	Interrupt
7	Enable MARK1 0 interrupt
8	Enable MARK2 0 interrupt
9	Enable MARK1 2 interrupt
10	Enable M2ARK2 2 interrupt
11	Enable MARK1 4 interrupt
12	Enable M2ARK2 4 interrupt
13	Enable MARK1 5 interrupt
14	Enable M2ARK2 5 interrupt
15	Enable Emergency Stop interrupt
16	Enable Physical motion end interrupt

Bit	Interrupt
17	Enable Logical motion end interrupt
18	Enable Motion failure (Motion interruption due to a fault) interrupt
19	Enable Motor failure (Motor disable due to a fault) interrupt
20	Enable Program termination interrupt
21	Enable Dynamic buffer interrupt
22	Enable ACSPL+ interrupt by INTERRUPT command
23	Enable Digital input interrupt
24	Enable Motion start interrupt
25	Enable Motion phase interrupt
26	Enable ACSPL+ interrupt by TRIGGER command

Tag

69

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.16.10 IMASK

Description

IMASK is an integer array, with one element for each axis in the system, the elements of which contain a set of bits that define which motor index and mark signals are processed.

Syntax

IMASK(*axis_index*).*bit_designator* = *value*

Arguments

<i>axis_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>bit_designator</i>	IMASK has four bit designators:

	> #IND (bit 0) - Primary encoder index > #IND2 (bit 1) - Secondary encoder index > #MARK (bit 2) - Mark1 > #MARK2 (bit 3) - Mark2
value	value ranges from -2147483648 to 2147483647, Default = 13

Tag

70

Comments

If a bit is zero, the controller neither analyzes or latches the corresponding **INDEX** or **MARK** signal.

Every axis does not provide all **INDEX** and **MARK** signals. The secondary encoder index is available only if a secondary encoder is used. **MARK** signals are only available for axes 0, 1, 4, and 5.

Accessibility

Read-Write



IMASK values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, GetIndexState, ResetIndexState

C Library Functions

acsc_ReadInteger, acsc_WriteInteger, acsc_GetIndexState, acsc_ResetIndexState

3.16.11 ISENA

Description

ISENA is an integer array, with one element for each axis in the system, the elements of which contain a set of 8 bits used for enabling or disabling software interrupts within a specific interrupt status bit for a specific axis or buffer. Each element corresponds to one software interrupt status bit and specifies which axis, buffers or inputs are enabled to cause interrupt.

Syntax

ISENA*array_index.bit_designator* = 1|0

Arguments

<i>array_index</i>	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
<i>bit_designator</i>	The meanings of bit_designator are given in Table 3-25 .

Table 3-25. ISENA Bit Description

Bit	Interrupt
0	Controls Physical Motion End interrupt.
1	Controls Logical Motion End interrupt.
2	Controls Motion Failure interrupt.
3	Controls Motor Failure interrupt.
4	Controls Program Termination interrupt.
5	Controls Command Execution interrupt (dynamic buffer only).
6	Controls ACSPL+ interrupt (by INTERRUPT command).
7	Controls Digital Input interrupt.



If the Physical Motion is PEG-related, then bit 0 is not supported in NT 1.0.

Tag

78

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.16.12 *S_FLAGS*

Description

S_FLAGS is an integer variable containing a set of bits that define different settings for the controller.

Syntax

S_FLAGS.*bit_designator* = 1|0

Arguments

<i>bit_designator</i>	The meanings of bit_designator are given in Table 3-26 .
-----------------------	---

Table 3-26. S_FLAGS Bit Description

Bit Name	No.	Description	Remarks
#S_FLAGS	1	S_FLAGS.1 controls whether the controller allots a controller cycle to processing non-executable lines such as comments, empty lines and labels. 0: Comments, empty lines, labels are skipped during execution and not allotted a controller cycle. (Default) 1: These lines are each allotted a controller cycle.	Changes to S_FLAGS.1 take effect only after a program is recompiled.
#FCLEAR	2	0: Sets the controller to regular mode. (Default) 1: Sets the controller to strict mode	In the regular mode the next motion command simply clears the reason for the previous KILL/KILLALL. In the strict mode, the next motion command cannot activate the motion and fails. Motion cannot be activated as long as the reason for the previous KILL is non-zero for any involved motor. Motion can continue only after clearing the MERR variable with ENABLE/ENABLE ALL or FCLEAR.

Tag

116

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.16.13 S_SETUP

Description

An integer variable containing a bit mask for defining various settings for the system.

Syntax

S_SETUP.bit_designator = 1|0

Arguments

bit_designator	The S_SETUP bit designators are given in Table 3-27 .
----------------	---

Table 3-27. S_SETUP Bit Designators

Name	No.	Description
#USGTRACE	1	0 (default) - Usage tracing is disabled 1 - Enables usage tracing (for debugging purposes only)
#SOFTIME	2	0 (default) - EtherCAT frame delivery time measurement is disabled 1 - Enables EtherCAT frame delivery time measurement (for debugging purposes only)
#FRMLOSS	3	0 (default) - single EtherCAT frame loss is not allowed. Every frame loss causes an EtherCAT error. 1 - single EtherCAT frame loss is allowed without causing an EtherCAT error.  A single EtherCAT frame loss mode can affect the performance of ACS units, if this mode is in use, then the ACS units MUST be connected before non-ACS units in the network.
#ENHPROT	4	0 - backward compatible application protection. 1 (default) - allow enhanced application protection.
#CONFPROT	5	0 (default) - prevent system reconfiguration In Protected Mode. 1 - allow system reconfiguration In Protected Mode.

Name	No.	Description
#GMODE	6	0 - backward compatible gantry mode for PEG, MARK, INDEX. 1 (default) - enable enhanced gantry mode for PEG, MARK, INDEX.
#POSSYNC	7	0 - backward compatibility for FPOS/RPOS synchronization 1 (default) - enable FPOS/RPOS synchronization
#ESDBMODE	8	SPiPlusES Debug Mode
#IOMNTMAP	9	0 - IOMNT units are not mapped to ACSPL+ IN and OUT variables 1 (default) - IOMNT units are mapped to ACSPL+ IN and OUT variables
#IRMSLEG*	10	0 (default) - RMS Protection is separated between Motor and Drive 1 - Legacy RMS protection only, using X RMS+X RMST
#USGTEMP	11	0: Do not use default response for Motor Overheat, MPU Overheat or MPU Overuse 1: Exception default response for Motor Overheat, MPU Overheat or MPU Overuse is 'DISABLE'.
#HOME402	13	CiA402 Homing Methods Enabled
#COMM402	14	CiA402 Drive uses ACSPL+ COMMUT Command
#FASTPWUP	15	Optimized power-up time
#CTDBRAKE	19	When the bit is ON, products that support Controlled Current Dynamic Brake will operate in the new mode. If the bit is OFF, Dynamic Brake is applied. Default value is ON.
#SINCOSREP	20	When the bit is ON, the mechanism compensation for any quadrature misalignment between the Analog counter and the quadrature counter during initialization is applied. The bit is ON by default.

Name	No.	Description
#FRM2LOSS	22	0 (default) single EtherCAT frame loss in the second EtherCAT network is not allowed. Every frame loss causes an EtherCAT error. 1 – single EtherCAT frame loss in the second EtherCAT network is allowed without causing an EtherCAT error. A single EtherCAT frame loss mode can affect the performance of ACS units. If this mode is in use, then the ACS units MUST be connected in the first EtherCAT network.



*Starting from version 2.60, after changing the RMSLEG bit, the system should be reconfigured using the System Setup component

Tag

240

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.16.14 XSEGAMAX

Description

A real variable that defines the maximal angle required when configuring look-ahead processing angles.

Syntax

XSEGAMAX= value

Tag

262

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.16.15 XSEGAMIN

Description

A real variable that defines the minimal angle required when configuring look-ahead processing angles.

Syntax

XSEGAMIN= value

Tag

263

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.16.16 XSEGRMAX

Description

A real variable that defines the maximal radius difference required when configuring arcs.

Syntax

XSEGRMAX= value

Tag

264

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.16.17 XSEGRMIN

Description

A real variable that defines the minimal arc radius required when configuring arcs.

Syntax

XSEGRMIN= value

Tag

265

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.17 Communication Variables

Communication variables are used for establishing various communication parameters.

The Communication variables are:

Name	Description
BAUD	Serial Communication Baud Rate
COMMCH	Communication Channel
COMMFL	Communication Flags
CONID	Controller Identification
DISPCH	Default Communication Channel
ECHO	Echo Communication Channel
GATEWAY	Contains the address of a network router that serves accessing another network segments.
SUBNET	Used to determine to what subnet an IP address belongs.
TCPIP	IP Address for 1 st Ethernet
TCPIP2	IP Address for 2 nd Ethernet
TCPPORT	TCP port identifier
UDPPORT	UDP port identifier

3.17.1 BAUD

Description

BAUD is an integer variable that defines the serial communication rate, given in bits per second.

Syntax

BAUD = *value*

Arguments

value	value can be one of the following:	
	>	300
	>	1200
	>	4800
	>	9600
	>	19200
	>	57600
	>	115200

Tag

8

Comments

Changes to **BAUD** take effect only after controller restart.

Accessibility

Read-Write



BAUD values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable, OpenComSerial

C Library Commands

acsc_ReadInteger, acsc_WriteInteger, acsc_OpenComSerial

3.17.2 COMMCH

Description

COMMCH is an integer that stores a number representing the last activated communication channel.

Table 3-28. COMMCH Values

Value	Description
1	Serial port 1
2	Serial port 2
6	Ethernet network (TCP)
7	Ethernet network (TCP)

Value	Description
8	Ethernet network (TCP)
9	Ethernet network (TCP)
10	Ethernet Point-to-Point (UDP)
12	PCI bus
16	MODBUS Slave
36	Ethernet network (TCP)
37	Ethernet network (TCP)
38	Ethernet network (TCP)
39	Ethernet network (TCP)

Tag

15

Comments

When queried through a communication channel, **COMMCH** reads the number of the current communication channel.

COMMCH can be used in [SEND](#), or assigned to [DISPCH](#).

Accessibility

Read-Only.

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.17.3 COMMFL

This variable is for advanced users. Changing the default values of these bits is not recommended!

Description

COMMFL is a scalar variable containing a set of bits that affect controller communication.

Syntax

COMMFL.*bit_designator* = 1|0

Arguments

<i>bit_designator</i>	The COMMFL bits and the meanings of their values are given in Table 3-29 .
-----------------------	---

Table 3-29. COMMFL Bit Descriptions

Bit Name	Bit No.	Description
#VERBOSE	0	Controls error message form. 0 = The controller provides the error number only to the C Library function or COM method, when an error occurs. 1 = The controller provides an extended message
	1	1 = Enable motor messages
	2	1 = Enable Axis messages
	3	1 = Enable Program messages
#SAFEMSG	4	1 = Controller sends unsolicited messages in Safe communication format
#CSUMMSG	6	1 = A checksum is included in unsolicited messages. Normally the user does not need to change this bit.
#NOCOMM	7	Controls the communication in protected mode. 1 = The controller ignores any command received via communication channels except the queries that start from '?' character. The bit is not effective if the controller is in configuration mode. The default value is 0.
#NOQUERY	8	Controls the communication in protected mode. 1 = The controller ignores any query received via communication channels. The bit is not effective if the controller is in configuration mode. The default value is 0.

Tag

16

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Commands

acsc_ReadInteger, acsc_WriteInteger

3.17.4 CONID

Description

CONID is an integer variable that contains the controller identification.

Syntax

CONID = *value*

Arguments

value

An integer ranging between 0 and 65536.

Tag

193

Comments

The controller identification can be used for many different purposes like Modbus Slave ID, CAN Slave ID or user-defined unique ID within the user network.



According to the Modbus specification, the controller must have an individual address from 1 to 247.

In order to specify the Modbus Slave address, **CONID** should be initialized to the Modbus Slave address value.

By default, the variable has zero value.

Accessibility

Read-Write



CONID values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Commands

acsc_ReadInteger, acsc_WriteInteger

3.17.5 ECHO

Description

ECHO is a 10-member mask integer variable that defines an echo communication channel.

Syntax

ECHO = *channel_number*

Arguments

<i>channel_number</i>	The values of channel_number and their meanings are given in Table 3-30 .
-----------------------	--

Table 3-30. ECHO Channel Numbers

Channel Number	Channel Name
-1	Echo NOT active
-2	All channels
1	Serial port 1
2	Serial port 2
6	Ethernet network (TCP)
7	Ethernet network (TCP)
8	Ethernet network (TCP)
9	Ethernet network (TCP)
10	Ethernet Point-to-Point (UDP)
12	PCI bus
16	MODBUS Slave
36	Ethernet network (TCP)
37	Ethernet network (TCP)
38	Ethernet network (TCP)
39	Ethernet network (TCP)

Tag

35

Comments

If **ECHO** specifies a valid communication channel, the controller sends an echo of each command received from any communication channel to the specified channel. Address each channel as follows:

The default value for **ECHO** is -1, in order to select an echo channel, the user needs to select a channel number.

ECHO cannot be saved to flash. After power-up the value is set to -1.

Use [DISPCH](#) to configure the communication channels related to the controller. Use [COMMCH](#) to retrieve the current controller channel’s physical connection (only the channel that is connected to the terminal on which the query is sent).

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.17.6 DISPCH

Description

DISPCH is a scalar integer variable that defines a communication channel between the controller and a host application, SPiiPlus MMI Application Studio or any device connected to the controller's communication ports.

Syntax

DISPCH = *channel_number*

Arguments

<i>channel_number</i>	The values of channel_number and their meanings are given in Table 3-31 .
-----------------------	--

Table 3-31. DISPCH Channel Numbers

Channel Number	Description
-2	All channels
-1	No default channel is specified, the command uses the last channel activated by the host.
1	Serial port 1
2	Serial port 2
6	Ethernet network (TCP)
7	Ethernet network (TCP)
8	Ethernet network (TCP)
9	Ethernet network (TCP)
10	Ethernet Point-to-Point (UDP)

Channel Number	Description
12	PCI bus
16	MODBUS Slave
36	Ethernet network (TCP)
37	Ethernet network (TCP)
38	Ethernet network (TCP)
39	Ethernet network (TCP)

Tag

25

Comments

DISPCH is relevant only to messages sent with **DISP** and **SEND** (described also as “Unsolicited Messages”).

In order to view unsolicited messages in the SPiiPlus MMI Application Studio **Communication Terminal** window, select the check box in the lower right corner of the Terminal window to enable **Show Unsolicited Messages**.

If **DISPCH** specifies a valid communication channel, all unsolicited messages (messages that are sent with **DISP** and **SEND** from the program buffers) are sent to this channel irrespective of the channel used for immediate commands.

Accessibility

Read-Write

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.17.7 GATEWAY**Description**

GATEWAY is an integer variable used for setting the address of a network router that serves for accessing another network segment.



The configuration is only available for the first Ethernet port.

Syntax**GATEWAY** = *value***Arguments****value****value** an hexadecimal number - format: 0xAABBCCDD.**Tag**

227

Comments

The **GATEWAY** address **value** consists of 4 individual bytes, each byte containing a decimal number ranging from 0 to 255. The bytes, when read, include a dot between each byte, with the least significant byte of the value representing the first decimal number. For example, the value 0x0100000A is the address: 10.0.0.1.

If controller is configured to obtain network settings from a DHCP server, **GATEWAY** contains the gateway address received from DHCP server

Accessibility

Read-Write



GATEWAY values cannot be modified if protection is applied to this variable through SPIiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.17.8 SUBNET**Description**

SUBNET is an integer variable used to determine to what subnet an IP address belongs.



The configuration is only available for the first Ethernet port.

Syntax**SUBNET** = *value***Arguments****value****value** an hexadecimal number - format: 0xAABBCCDD.**Tag**

228

Comments

The **SUBNET value** consists of 4 individual bytes, each byte containing a decimal number ranging from 0 to 255. The bytes, when read, include a dot between each byte, with the least significant byte of the value representing the first decimal number. For example, the value 0x00FFFFFF represents mask 255.255.255.0.

If controller is configured to obtain network settings from a DHCP server, **SUBNET** contains the address received from DHCP server.

Accessibility

Read-Write



SUBNET values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.17.9 TCPIP

Description

TCPIP is an integer variable used for setting the TCP/IP for the Ethernet port N1.

Syntax

TCPIP = *value*

Arguments

value

value an hexadecimal number - format: 0xAABBCCDD.

Tag

133

Comments

If **TCPIP** has a non-zero value, the controller uses the value as its TCP/IP address. In this case, other configuration parameters receive the following default values:

- > Subnet mask - 255.255.255.0
- > Gateway address - no gateway, i.e., no routing is supported

If **TCPIP** is zero, the controller uses the DHCP protocol to receive the network configuration from the DHCP server. The network configuration received from the DHCP server includes the following parameters:

- > Controller's TCP/IP address
- > Subnet mask
- > Gateway address

The **TCPIP** variable value has to be in hex, for example:

```
TCPIP=0x6400000a
```

assigns a TCP/IP address of: 10.0.0.100. Note that the address is calculated starting from the least significant byte of the value.

To retrieve the assigned address in an ACSPL+ program, use the [GETCONF](#) function with key 310.

Accessibility

Read-Write



TCPIP values cannot be modified if protection is applied to this variable through **SPiiPlus** **MMI Application Studio** → **Toolbox** → **Application Development** → **Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Commands

acsc_ReadInteger, acsc_WriteInteger, acsc_GetEthernetCards (to find all SPiiPlus controllers in the network segment)

3.17.10 TCPIP2

Description

TCPIP2 is an integer variable used for setting the TCP/IP for a second Ethernet port: N2.

Syntax

TCPIP2 = *value*

Arguments

value

value an hexadecimal number - format: 0xAABBCCDD.

Tag

198

Comments

If **TCPIP2** is zero, the address will be automatically obtained at the controller start-up through DHCP protocol. The default address for second Ethernet port is 192.168.0.100.

The **TCPIP2** variable value has to be in hex: 0xAABBCCDD, for example:

```
TCPIP=0x6400000a
```

assigns a TCP/IP address of: 10.0.0.100. Note that the address is calculated starting from the least significant byte of the value.

To retrieve the assigned address in an ACSPL+ program, use the [GETCONF](#) function with key 310.

Accessibility

Read-Write



TCPIP2 values cannot be modified if protection is applied to this variable through **SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection**

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Commands

acsc_ReadInteger, acsc_WriteInteger, acsc_GetEthernetCards (to find all SPiiPlus controllers in the network segment)

3.17.11 TCPPORT

Description

TCPPORT is a scalar integer that stores a number representing a TCP port.

Syntax

TCPPORT = *Port_number*

Arguments

Port_number

An integer ranging between 0 and 65536.

Tag

200

Comments

TCPPORT defines Ethernet ports in the controller for TCP. By default, this variable is set to 701. In order to establish communication with the controller through a port different from default port numbers, the following should be done:

1. Set **TCPPORT** to a value other than 701.



Some of the ports are used by the controller firmware and cannot be used. It is recommended to use ports starting from 1024.

2. Save system parameters to the flash.
3. Restart the controller.
4. Try to establish communication using new ports by providing them in client user application. If communication isn't established, try to set other values.



This new port value is used by the client user application only. The SPiiPlus Tools and SPiiPlus C/COM Library continue to use the default ports.

Accessibility

Read-Write



TCPPORT values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.17.12 UDPPORT

Description

UDPPORT is a scalar integer that stores a number representing a UDP port.

Syntax

UDPPORT = *Port_number*

Arguments

Port_number

An integer ranging between 0 and 65536.

Tag

201

Comments

UDPPORT defines Ethernet ports in the controller for UDP. By default, it is set to 700. In order to establish communication with the controller through different from default port numbers, the following should be done:

1. Set **UDPPORT** to a value other than 700.



Some of the ports are used by the controller firmware and cannot be used. It is recommended to use ports starting from 1024.

2. Save system parameters to the flash.
3. Restart the controller.
4. Try to establish communication using new ports by providing them in client user application. If communication isn't established, try to set other values.



This new port value is used by the client user application only. The SPiiPlus Tools and SPiiPlus C/COM Library continue to use the default ports.

Accessibility

Read-Write



UDPPORT values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

3.18 Laser Control Variables

3.18.1 LCFRQ

Description

LCFRQ is a real type array of 64 elements. Each element corresponds to the specific axis that belongs to an LCM unit. The same index that is defined in **LCMODULATION** should be used as an index for **LCFRQ**.

LCFRQ indicates the actual frequency in [Hz] for modes 2 and 3, or a configured frequency value for mode 1.

Tag

321

Accessibility

Read-Only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadReal()

3.18.2 LCDC

Description

LCDC is real type array of 64 elements. Each element corresponds to the specific axis that belongs to an LCM unit. The same index that is defined in **LCMODULATION** should be used as the index for **LCDC**.

LCDC indicates the actual duty cycle in [%] for mode 1, or the configured duty cycle value for modes 2 and 3.

Tag

320

Accessibility

Read-Only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadInteger()

3.18.3 LCTFREQ

Description

LCTFREQ is a real type array of 64 elements. Each element corresponds to the specific axis that belongs to the LCM unit. The same index that is defined in **LCTICKLE** should be used as the index for **LCTFREQ**.

LCTFREQ indicates the actual tickle pulse frequency in Herz.

Tag

333

Accessibility

Read-Only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadReal()

3.18.4 LCTWIDTH

Description

LCTWIDTH is a real type array of 64 elements. Each element corresponds to the specific axis that belongs to LCM unit. The same index that is defined in **LCTICKLE** should be used as the index for **LCTWIDTH**.

LCTWIDTH indicates the actual tickle pulse width in milliseconds.

Tag

334

Accessibility

Read-Only

.NET Library Method

ReadVariable()

C Library Function

acsc_ReadReal()

3.19 Miscellaneous

The Miscellaneous variables are:

Name	Description
	Secondary user unit name
AUNITS	User unit name
FK	Function Key
PDESC	Buffer description
PN	Controller part number
PNAME	Buffer name
SN	Controller serial number
STATIC	Tag used to define a global variable as STATIC
VR	Controller FW version
XARRSIZE	Maximum Array Size

3.19.1 AUNITS

Description

AUNITS is an array of strings, with one string for each axis in the system. Each string may be at most 19 characters in length.

AUNITS is read-only, but the value can be set during the adjusting process in the MMI.

The value of the string will be saved as part of the Par<Axis>.\$\$\$ file, and may be restored from that file.

Syntax

```
DISP AUNITS (1)
```

Arguments

None

Tag

431

Comments

Units' names can only be assigned during the adjusting process in the MMI.

Related ACSPL+ Variables

EFAC

Accessibility

Read-Only

3.19.2 A2UNITS

Description

A2UNITS is an array of strings, with one string for each axis in the system. Each string may be at most 19 characters in length.

A2UNITS is read-only, but the value can be set during the adjusting process in the MMI.

The value of the string will be saved as part of the Par<Axis>.\$\$\$ file, and may be restored from that file.

Syntax

```
DISP
A2UNITS (1)
```

Arguments

None

Tag

432

Comments

Units' names can only be assigned during the adjusting process in the MMI.

Related ACSPL+ Variables

E2FAC

Accessibility

Read-Only

3.19.3 FK

Description

FK is a scalar integer and is used by the controller to store function keys in an input string processed with the **INPUT** command.

Tag

50

Comments

If the controller encounters a zero character in the input string, it stores the value of the next character in **FK**.

This is the usual way for loading function key codes (F1 - F12).



An autoroutine can be used to respond to changes in **FK**.

Accessibility

Read-Only

COM Library Methods and .NET Library Methods

ReadVariable

C Library Functions

acsc_ReadInteger

3.19.4 PDESC**Description**

PDESC is a string type array holding the description given to buffer programs by the user. The maximum length of each string is 249. The variables will be part of the system information variables and should be saved to flash (and later retrieved) upon command.

Syntax

```
PDESC(1) = "A buffer used to calibrate axes 0-4"
```

Arguments

None

Tag

425

Related ACSPL+ Variables

PNAME

Accessibility

Read-Write

3.19.5 PN**Description**

PN is a string variable containing the controller part number.

Syntax

```
DISP PN
```

Arguments

None

Tag

427

Accessibility

Read-only

3.19.6 PNAME

Description

PNAME is a string type array that holds the names given to buffer programs by the user. The maximum length of each string is 49 characters. The variables will be part of the system information variables and should be saved to flash (and later retrieved) upon command.

Syntax

```
PNAME(1) = "Calibration buffer"
```

Arguments

None

Tag

424

Related ACSPL+ Variables

PDESC

Accessibility

Read-Write

3.19.7 SN

Description

SN is a string type variable that contains the controller serial number.

Example

```
DISP SN
```

Tag

426

Accessibility

Read-only

3.19.8 STATIC

Description

A new tag can be used to define a global variable as **STATIC**.

STATIC variables can only be defined in the D-buffer and once defined can only be freed using the **#VGV** command, that is: removing the definition from the D-buffer does not remove the variable.

Use the **#VGS/#VGSF** command to list all static variables in the system.

Syntax

```
GLOBAL INT/REAL STATIC Var
```

3.19.9 VR

Description

VR String type variable that contains the controller FW version.

Syntax

```
DISP VR
```

Tag

428

Accessibility

Read-only

3.19.10 XARRSIZE

Description

XARRSIZE is a scalar integer that stores maximum size of an array.

Syntax

XARRSIZE = *value*

Arguments

value

value ranges from 10,000 to 10,000,000 (elements); Default = 100,000

Tag

219

Comments

By default the maximum size for a user array is 100,000 elements; if, however, an application requires larger arrays, the user may change this value to accommodate a larger array size. However the following should be taken into consideration:

1. Defining large arrays may use too much memory and may cause an out of memory fault. To avoid this, the RAM size available for user data in the specific controller model should be checked. One element in an array requires 8 bytes of RAM, 131,072 elements require 1 MB.
2. The processing time required for operations on large arrays in ACSPL+ may cause an over usage fault. Therefore, arrays should be defined only with the size actually required for the application.



It is strongly recommended that users change the **XARRSIZE** variable only if necessary, and that such changes be tested under safe conditions.

Accessibility

Read-Write



XARRSIZE values cannot be modified if protection is applied to this variable through
SPiiPlus MMI Application Studio → Toolbox → Application Development → Protection

COM Library Methods and .NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadInteger, acsc_WriteInteger

4. ACSPL+ Functions

ACSPL+ functions are divided into the following categories:

- > [Arithmetical Functions](#)
- > [Matrix Functions](#)
- > [Miscellaneous Functions](#)
- > [EtherCAT Functions](#)
- > [CoE Functions](#)
- > [Modbus Functions](#)
- > [Servo Processor Functions](#)
- > [Signal Processing Functions](#)
- > [Laser Control Functions](#)
- > [Dynamic Error Compensation](#)

This chapter covers the ACSPL+ functions.

The ACSPL+ Functions, in alphabetical order, are:

Function	Description
ABS	Calculates the absolute value.
ACOS	Calculates the arc cosine.
AINOFFS	Percent offset of analog signal from external source
AINSCALE	Define scale of analog input signal
ASIN	Calculates the arc sine.
ATAN	Calculates the arctangent.
ATAN2	Calculates the arctangent of X/Y.
AxListAsMask	Mask for defining axes.
AVG	Finds the average of all values in an array.
BCOPY	Copies a given number of bytes from a source array into a target array.
CEIL	Calculates the ceiling of a value.
COPY	The function copies one array to another.

Function	Description
COEREAD	Read CoE slave Object Directory entry.
COS	Calculates the cosine.
COEWRITE	Write into CoE slave Object Directory.
DEADZONE	Implements dead-zone routine.
DSIGN	Implements a dynamic version of the standard SIGN function.
DSTR	Converts a string to an integer array.
ECIN	Copy EtherCAT offset data into ACSPL+ variable
ECGETGRPIND	Returns an array that contains optional groups' indexes that are part of the current configuration
ECGETOPTGRP	Returns number of actually connected optional groups, not including the mandatory group.
ECGETREV	Returns EtherCAT Revision ID of the node
ECGETSLAVES	Returns the number of EtherCAT slaves in the network
ECGETSN	Returns Serial Number of the node.
ECGRPINFO	> Fills array with nodes' indexes which are members of a given optional group > Returns the number of members in the group.
ECOUT	Copy data of ACSPL+ variable into EtherCAT offset
ECRESCUE	Triggers execution of a rescue scan of the EtherCAT network
ECSAVECFG	Saves to flash an array of optional groups based on current configuration, based on <code>ecgetgrpind()</code> function.
ECSAVEDCNF	Returns array that contains optional groups' indexes that are part of the last saved configuration
ECUNMAP	Undoes any results from running ECIN and ECOUT
ECUNMAPIN	Undoes any results from running ECIN to a specific offset.
ECUNMAPOUT	Undoes any results from running ECOUT to a specific offset.

Function	Description
EDGE	Returns 1 on positive edge of x.
ENCREAD	Read encoder parameters
ERRORMAP1D	Configures and activates 1D error correction for the mechanical error compensation for the specified zone,
ERRORMAP2D	Configures and activates 2D error correction for the mechanical error compensation of the 'axis0' command for the specified zone,
ERRORMAPN1D	Configures and activates 1D error correction for the mechanical error compensation for the specified zone
ERRORMAPN2D	Configures and activates 2D error correction for the mechanical error compensation of the 'axis0' command for the specified zone,
ERRORMAPOFF	Deactivates error mapping correction for the mechanical error compensation for the specified zone.
ERRORMAPON	Activates error correction for the mechanical error compensation for the specified zone.
#ERRORMAPREP	Generates a report of all activated zones of error mapping for all axes in the system.
ERRORUNMAP	Deactivates error correction for the mechanical error compensation for the specified zone.
EXP	Calculates the exponent.
FLOOR	Calculates the floor of a value.
GETCONF	Reads hardware and firmware parameters.
GETSP	Reads a value from the specified SP address.
GETSPA	Retrieves address of the SP variable specified by name.
GETSPV	Reads a value from the specified SP variable name.
HYPOT	Calculates the hypotenuse.
INP	Reads data characters from the specified channel and stores them into integer array.
INTGR	Implements an integrator with DEADZONE and SAT .

Function	Description
LAG	Provides delayed switching on argument change (anti-bouncing effect).
LCDelayGet	Returns the actual currently configured delay in microseconds.
LCDelaySet	Sets the pulse generation delay in microseconds.
LCFixedDist	Initializes the fixed distance pulse firing mode.
LCFixedInt	Initializes the fixed distance pulse firing mode.
LCMODULATION	Initializes Pulse modulation mode and sets initial values for the unit internal registers.
LCOutputGet	Returns the laser outputs configuration.
LCOutputSet	Configures the laser physical outputs.
LCRandomDist	Initializes either array based pulse firing mode or gating mode.
LCSignalGet	Returns the laser control signal (LCS) output conditioning state.
LCSignalSet	Configures the laser control signal (LCS) output conditioning state.
LCStop	Stops any previously initialized laser mode and resets the previously defined mode parameters.
LCTickle	Initializes Tickle mode.
LCZone	Sets a laser operation zone area.
LCZoneGet	Returns the limits of the laser operation zone that was previously defined
LCZoneSet	Changes the minimal and/or maximal laser operationzone limit.
LDEXP	Calculates a value of $x*2^{\text{exp}}$.
LOG	Calculates the natural logarithm.
LOG10	Calculates the base-10 logarithm.
MAP	Implements a table-defined function with constant step.
MAPB	One-dimensional uniform b-spline.

Function	Description
MAPN	One-dimensional non-uniform linear interpolation (replaces obsolete MAPBY1 and MAPBY2).
MAPNB	One-dimensional non-uniform B-spline.
MAPNS	One-dimensional non-uniform Catmull-Rom spline.
MAPS	One-dimensional uniform Catmull-Rom spline.
MAP2	Implements a table-defined function with two arguments and constant step along each argument.
MAP2B	Two-dimensional uniform B-spline.
MAP2N	Two-dimensional non-uniform linear interpolation (replaces obsolete MAP2FREE).
MAP2NB	Two-dimensional non-uniform B-spline.
MAP2NS	Two-dimensional non-uniform Catmull-Rom spline.
MAP2S	Two-dimensional uniform Catmull-Rom spline.
MATCH	Calculates axis position that matches current reference position of the same axis with zero offset.
MAX	Finds the maximum value in an array.
MAXI	Finds the element with maximum value in an array or in a section of an array and returns its index.
MIN	Finds the minimum value in an array.
MINI	Finds the element with minimum value in an array or in a section of an array and returns its index.
NUMTOSTR	Converts a number to an ASCII string.
POW	Calculates x raised to the power of y.
RAND	Implements a random number generator.
ROLL	Calculates a result rolled-over to within one pitch.
SAT	Implements a saturation characteristic.

Function	Description
SETCONF	Writes hardware and firmware parameters.
SETSP	Sets a value for the specified SP address.
SETSPV	Sets a value for the specified SP variable name.
SETVAR	Writes a value to a system or user variable, scalar or array, that was declared as a Tag number.
SIGN	Returns -1, 0 or 1 depending on the sign of x.
SIN	Calculates the sine.
SQRT	Calculates the square root.
STR	Converts an integer array to a string.
STRTONUM	Converts an ASCII string representing a number to the number it represents.
SYSINFO	Returns certain system information based on the argument that is specified.
TAN	Calculates the tangent.

4.1 *Arithmetical Functions*

The Arithmetical functions are:

Function	Description
ABS	Calculates the absolute value
ACOS	Calculates the arc cosine
ASIN	Calculates the arc sine
ATAN	Calculates the arctangent
ATAN2	Calculates the arctangent of Y/X
CEIL	Calculates the ceiling of a value
COS	Calculates the cosine
EXP	Calculates the exponent
FLOOR	Calculates the floor of a value

Function	Description
HYPOT	Calculates the hypotenuse
LDEXP	Calculates a value of $x*2^{\text{exp}}$
LOG	Calculates the natural logarithm
LOG10	Calculates the base 10 logarithm
POW	Calculates x raised to the power of y
SIGN	Returns -1, 0 or 1 depending on the sign of x
SIN	Calculates the sine
SQRT	Calculates the square root
TAN	Calculates the tangent
ROUND	Rounds REAL to INTEGER

4.1.1 ABS

Description

ABS calculates the absolute value

Syntax

ABS(*input*)

Arguments

input

input can be a real number or an expression.

Return Value

Real number - returns the absolute value of the input.

Error Conditions

None

Example

```
XX = ABS (-3.14)
DISP XX !Output = 3.14
```

4.1.2 ACOS

Description

ACOS calculates arc cosine.

Syntax**ACOS**(*input*)**Arguments***input**input* can be a real number or an expression.**Return Value**Real number - returns the arc cosine of the argument in the range from 0 to π radians.**Error Conditions**The value of **input** must be between -1 to 1, otherwise the function returns Error 3045, Numerical Error in Standard Function.**Example**

```
XX = ACOS (-1)
DISP XX                               !Output = 3.141592654
```

4.1.3 ASIN**Description****ASIN** calculates the arc sine.**Syntax****ASIN**(*input*)**Arguments***input**input* can be a real number or an expression.**Return Value**Real number - returns the arc sine of XX in the range $-\pi/2$ to $\pi/2$.**Error Conditions**The value of **input** must be between -1 to 1, otherwise the function returns Error 3045, Numerical Error in Standard Function.**Example**

```
XX = ASIN (-1)
DISP XX                               !Output = -1.570796327
```

4.1.4 ATAN**Description****ATAN** calculates the arctangent.**Syntax****ATAN**(*input*)

Arguments*input***input** can be a real number or an expression.**Return Value**Real number - returns the arctangent of the input in the range $-\pi/2$ to $\pi/2$ radians.**Error Conditions**

None

Example

```
XX = ATAN(-1)
DISP XX                !Output = -0.7853981634
```

4.1.5 ATAN2**Description****ATAN2** calculates the arctangent of **X/Y**.**Syntax****ATAN2**(*X_input*,*Y_input*).**Arguments***X_input***X_input** can be a real number or an expression.*Y_input***Y_input** can be a real number or an expression.**Return Value**Real number - returns the arctangent value of **X_input**, **Y_input**. **ATAN2** calculates a value in the range of $-\pi$ to π radians using the signs of both parameters to determine the quadrant of the return value. If both parameters are 0, the function returns 0. **ATAN2** is well defined even if Y equals 0.**Error Conditions**

None

Example

```
X_input = -1; Y_input = 0
XX=ATAN2(X_input,Y_input)
DISP XX                !Output = -1.5708
```

4.1.6 CEIL**Description****CEIL** calculates the ceiling of a value.**Syntax****CEIL**(*input*)

Arguments*input***input** can be a real number or an expression.**Return Value**Integer number - returns a value that represents the smallest integer that is ³ **input**.**Error Conditions**

None

Example

```

XX=CEIL(3)
YY=CEIL(-3)
ZZ=CEIL(2.1)
TT=CEIL(-2.1)
DISP XX, YY, ZZ, TT !Output = 3 -3 3 -2

```

4.1.7 COS**Description****COS** calculates the cosine.**Syntax****COS**(*input*)**Arguments***input***input** can be a real number or an expression (which is treated as radians by the function).**Return Value**

Real number - returns the cosine of X in the range of -1 to 1.

Error Conditions

None

Example

```

XX = COS(-3.141592654)
DISP XX !Output = -1

```

4.1.8 EXP**Description****EXP** calculates the e^ **input****Syntax****EXP**(*input*)

Arguments***input*****input** can be a real number or an expression.**Return Value**

Real number - returns the exponential value of **input**. On overflow, the function returns the largest real number.

Error Conditions

None

Example

```
XX = EXP(1)
DISP XX !Output = 2.718281828
```

4.1.9 FLOOR**Description**

FLOOR calculates the floor of a value.

Syntax

FLOOR(*input*)

Arguments***input*****input** can be a real number or an expression.**Return Value**

Integer number - returns a value representing the largest integer that is $\leq$ to **input**.

Error Conditions

None

Example

```
XX=FLOOR(3)
YY=FLOOR(-3)
ZZ=FLOOR(2.1)
TT=FLOOR(-2.1)
DISP XX,YY,ZZ,TT !Output = 3 -3 2 -3
```

4.1.10 HYPOT**Description**

HYPOT calculates the hypotenuse of a right triangle

Syntax

HYPOT(*X_input*, *Y_input*)

Arguments

<i>X_input</i>	X_input can be a real number or an expression.
<i>Y_input</i>	Y_input can be a real number or an expression.

Return Value

Real number - calculates the length of the hypotenuse of a right triangle, given the length of the two sides **X_input** and **Y_input**. **HYPOT** is equivalent to the square root of $X^2 + Y^2$.

Error Conditions

None

Example

```
XX=HYPOT(3,4)
DISP XX           !Output = 5
```

4.1.11 LDEXP

Description

LDEXP calculates the value of $x*2^{exp}$.

Syntax

LDEXP(*X_input*, *Y_input*)

Arguments

<i>X_input</i>	X_input can be a real number or an expression.
<i>Y_input</i>	Y_input can be a real number or an expression.

Return Value

Real number - returns the value of $X_input*2^{Y_input}$. On overflow **LDEXP** returns the largest real number with a sign, depending on the sign of **X_input**

Error Conditions

None

Example

```
XX= LDEXP(1,2)
DISP XX           !Output = 4
```

4.1.12 LOG

Description

LOG calculates the natural logarithm.

Syntax

LOG(*input*)

Arguments

<i>input</i>	input can be a real number or an expression.
--------------	---

Return Value

Real number - returns the natural logarithm of **input**.

Error Conditions

input must be > 0, otherwise the function returns Error 3045, Numerical Error in Standard Function.

Example

```
XX=LOG(2.718281829)
DISP XX                !Output = 1
```

4.1.13 LOG10

Description

LOG10 calculates the base 10 logarithm.

Syntax

LOG10(*input*)

Arguments

<i>input</i>	input can be a real number or an expression.
--------------	---

Return Value

Real number - returns the base 10 logarithm of **input**.

Error Conditions

input must be >0, otherwise the function returns Error 3045, Numerical Error in Standard Function.

Example

```
XX=LOG10(10)
DISP XX                !Output = 1
```

4.1.14 POW

Description

POW calculates X raised to the power of Y.

Syntax

POW(*X_input*, *Y_input*)

Arguments

<i>X_input</i>	X_input can be a real number or an expression.
<i>Y_input</i>	Y_input can be a real number or an expression.

Return Value

Real number - returns the value of $(X_input)^{Y_input}$.

Error Conditions

None

Example

```
XX=POW(2,3)
DISP XX           !Output = 8
```

4.1.15 SIGN**Description**

SIGN returns -1, 0 or 1 depending if the input is negative, zero or positive.

Syntax

SIGN(*input*)

Arguments

<i>input</i>	<i>input</i> can be a real number or an expression.
--------------	---

Return Value

Real number - returns:

-1 if *input* < 0;

0 if *input* = 0;

1 if *input* > 0

Error Conditions

None

Example

```
XX=SIGN(-5), SIGN(0), SIGN(5)
DISP XX           !Output = -1 0 1
```

4.1.16 SIN**Description**

SIN calculates the sine.

Syntax

SIN(*input*)

Arguments

<i>input</i>	<i>input</i> can be a real number or an expression (which is treated as radians by the function).
--------------	---

Return Value

Real number - returns the sine value of **input** in the range of -1 to 1.

Error Conditions

None

Example

```
XX=SIN(1.570796327)
DISP XX                !Output = 1
```

4.1.17 SQRT**Description**

SQRT calculates the square root.

Syntax

SQRT(*input*)

Arguments

input	input can be a real number or an expression.
--------------	---

Return Value

Real number - returns the square root of **input**.

Error Conditions

input must be ≥ 0 , otherwise the function returns Error 3045, Numerical Error in Standard Function.

Example

```
XX=SQRT(4)
DISP XX                !Output = 2
```

4.1.18 TAN**Description**

TAN calculates the tangent.

Syntax

TAN(*input*)

Arguments

input	input can be a real number or an expression (which treated as radians by the function).
--------------	--

Return Value

Real number - returns the tangent value of **input**.

Error Conditions

None

Example

```
REAL PI
PI = 3.141592654
DISP TAN(PI/4)                !Output = 1
```

4.1.19 ROUND**Description**

Round a REAL number to closest integer value

Syntax

ROUND(input)

Arguments

Input - Arbitrary real number

Return Value

Closest integer value

Comments

ROUND calculates the closest integer number according to arithmetical rules.

This command is supported in ADK versions 2.70 and higher.

Examples:

```
DISP (ROUND (1.5) )           !Output = 2
DISP (ROUND (1.4) )           !Output = 1
DISP (ROUND (-1.5) )          !Output = -2
DISP (ROUND (-1.4) )          !Output = -1
```

4.2 Matrix FunctionsACSPL+ supports a **MATRIX** type starting with version 3.10.A **MATRIX** is defined as 2-dimensional **REAL** array.**Syntax**

- > MATRIX A(N)(N) – A is square matrix of size NxN where N is a positive constant.
- > MATRIX B(N)(M) – B is NxM matrix having N rows and M columns, where N and M are positive constants.

The matrix functions are:

Function	Overloaded Operator	Description
<code>MATRIXADD</code>	<code>+</code>	Add matrices
<code>MATRIXSUB</code>	<code>-</code>	Subtract matrices
<code>MATRIXMUL</code>	<code>*</code>	Multiply matrices
<code>MATRIXMULSC</code>	<code>*</code>	Multiply matrix by scalar
<code>MATRIXMULEW</code>	<code>.*</code>	Element-wise matrix multiplication
<code>MATRIXDIV</code>	<code>/</code>	Matrix division
<code>MATRIXIDENT</code>		Identity matrix
<code>MATRIXTRANS</code>		Transpose matrix
<code>MATRIXINVERT</code>		Invert matrix

For details about programming considerations in ACSPL+ using MATRIX type variables, see the MATRIX Type chapter in the *ACSPL+ Programmer's Guide*.

4.2.1 Matrix Type

ACSPL+ supports a **MATRIX** type starting with version 3.10.

A **MATRIX** is defined as 2-dimensional **REAL** array.

Syntax

- > `MATRIX A(N)(N)` – A is square matrix of size NxN where N is a positive constant.
- > `MATRIX B(N)(M)` – B is NxM matrix having N rows and M columns, where N and M are positive constants.

Example

```
MATRIX A(2)(2) !Defines a square matrix of 2x2 size
```

4.2.1.1 Matrix Initialization in Compilation Time

1. Regular Initialization (similar to 2D arrays initialization):
`MATRIX A(M)(N)` can be initialized by a 2D array (array of M rows, each of N size)
 - a. Preconditions:
 - i. Number of rows of initialization values = matrix first dimension
 - ii. Number of columns of initialization values = matrix second dimension
 - b. Error conditions:
 - i. If preconditions are violated, then a compilation error will occur.
 - c. Example:

```
!///Compilation-time initializations///!  
MATRIX A(2)(2)=( (1,2) , (3,4) ) !Regular initialization of a 2x2 matrix
```

2. Default Initialization:

A matrix which not initialized by the user on definition is automatically initialized with zeros.

Example:

```
!///Compile-time initializations///!  
MATRIX C(2)(2) !Default initialization, filled by zeros
```

For details about MATRIX type functions and operators see Matrix Functions in the *ACSPL+ Commands & Variables Reference Guide*.

4.2.2 MATRIXADD

Description

Add two matrices.

Syntax

```
MATRIXADD([in] MATRIX A, [in] MATRIX B, [out] MATRIX C)
```

Arguments

MATRIX A	First operand matrix
MATRIX B	Second operand matrix
MATRIX C	Result Matrix

Return Value

None

Comments

1. All argument matrices must be of the same size.
2. If preconditions are violated, then a compilation error will occur.
3. This function is supported in versions 3.11 and higher

Example

```
MATRIX A(2)(2)=( (1,2) , (3,4) )  
MATRIX B(2)(2)=( (5,6) , (7,8) )  
MATRIX C(2)(2)  
MATRIXADD(A,B,C)
```


Example With Overloaded Operator

```
MATRIX A (2) (2) = ( (1,2) , (3,4) )
MATRIX B (2) (2) = ( (5,6) , (7,8) )
MATRIX C (2) (2)
C=A+B
STOP
```

4.2.3 *MATRIXSUB*

Description

Subtract two matrices.

Syntax

MATRIXSUB([in] MATRIX A, [in] MATRIX B, [out] MATRIX C)

Arguments

MATRIX A	First operand matrix
MATRIX B	Second operand matrix
MATRIX C	Result Matrix

Return Value

None

Comments

1. All argument matrices must be of the same size.
2. If preconditions are violated, then a compilation error will occur.
3. This function is supported in versions 3.11 and higher

Example

```
MATRIX A (2) (2) = ( (1,2) , (3,4) )
MATRIX B (2) (2) = ( (5,6) , (7,8) )
MATRIX C (2) (2)
MATRIXSUB (B,A,C)
```

Example With Overloaded Operator

```
MATRIX A (2) (2) = ( (1,2) , (3,4) )
MATRIX B (2) (2) = ( (5,6) , (7,8) )
MATRIX C (2) (2)
C=B-A
STOP
```

4.2.4 MATRIXMUL

Description

Multiply matrices

Syntax

MATRIXMUL([in] MATRIX A, [in] MATRIX B, [out] MATRIX C)

Arguments

MATRIX A	First operand matrix
MATRIX B	Second operand matrix
MATRIX C	Result Matrix

Return Value

None

Comments

1. Matrix A's column dimension must equal Matrix B's row dimension.
2. The dimension of Matrix C's rows are equal to the row dimension of Matrix A
3. The dimension of Matrix C's columns are equal to the column dimension of Matrix B
4. If preconditions are violated, then a compilation error will occur.
5. This function is supported in versions 3.11 and higher

Example

```
MATRIX A (3) (2) = ( (1, 2) , (3, 4) , (5, 6) )
MATRIX B (2) (2) = ( (7, 8) , (9, 10) )
MATRIX C (3) (2)
MATRIXMUL (A, B, C)
```

Example With Overloaded Operator

```
MATRIX A (3) (2) = ( (1, 2) , (3, 4) , (5, 6) )
MATRIX B (2) (2) = ( (7, 8) , (9, 10) )
MATRIX C (3) (2)
C=A*B
```

4.2.5 MATRIXMULSC

Description

Multiply matrix by scalar value.

Syntax

MATRIXMULSC([in] MATRIX A, [in] REAL S, [out] MATRIX C)

Arguments

MATRIX A	First operand matrix
REAL S	Scalar operand
MATRIX C	Result Matrix

Return Value

None

Comments

1. The result matrix is of the same size as operand matrix.
2. If preconditions are violated, then a compilation error will occur.
3. This function is supported in versions 3.11 and higher

Example

```
MATRIX A(3)(2) = ((1,2), (3,4), (5,6))  
REAL r=7  
MATRIX C(3)(2)  
MATRIXMULSC(A,r,C)  
MATRIXMULSC(A,5.5,C)  
STOP
```

4.2.6 MATRIXMULEW**Description**

Element-wise matrix multiplication

Syntax

MATRIXMULEW([in] MATRIX A, [in] REAL S, [out] MATRIX C)

Arguments

MATRIX A	First operand matrix
REAL S	Scalar operand
MATRIX C	Result Matrix

Return Value

None

Comments

1. All parameter matrices must be of the same size.
2. If preconditions are violated, then a compilation error will occur.
3. This function is supported in versions 3.11 and higher

Example

```
MATRIX A (2) (2) = ( (1, 2) , (3, 4) )  
MATRIX B (2) (2) = ( (5, 6) , (7, 8) )  
MATRIX C (2) (2)  
MATRIXMULEW (A, B, C)
```

Example With Overloaded Operator

```
MATRIX A (2) (2) = ( (1, 2) , (3, 4) )  
MATRIX B (2) (2) = ( (5, 6) , (7, 8) )  
MATRIX C (2) (2)  
C=A.*B  
STOP
```

4.2.7 MATRIXDIV

Description

Matrix division

Syntax

MATRIXDIV([in] MATRIX A, [in] MATRIX B, [out] MATRIX C)

Arguments

MATRIX A	Dividend matrix
MATRIX B	Divisor matrix
MATRIX C	Result Matrix

Return Value

None

Comments

- 1. Matrix B is a square matrix.
- 2. The dimension of Matrix A's columns are equal to the order of Matrix A
- 3. C dimension = (A's rows, B's order) = A dimension
- 4. If preconditions are violated, then a compilation error will occur.
- 5. If Matrix B is not convertible (determinant(B) == 0) a run-time error will occur.
- 6. This function is supported in versions 3.11 and higher

Example

```
MATRIX A (2) (3) = ( (1, 2, 3) , (4, 5, 6) )  
MATRIX B (3) (3) = ( (1, 2, 3) , (4, 5, 6) , (7, 8, 1) )  
MATRIX C (2) (3)  
MATRIXDIV (A, B, C)
```

Example With Overloaded Operator

```
MATRIX A (2) (3) = ( (1, 2, 3) , (4, 5, 6) )
MATRIX B (3) (3) = ( (1, 2, 3) , (4, 5, 6) , (7, 8, 1) )
MATRIX C (2) (3)
C=A/B
```

4.2.8 MATRIXIDENT

Description

Generate identity matrix, a matrix filled with values of 1 on the main diagonal and all other elements are 0.

Syntax

MATRIXIDENT([in out] MATRIX A)

Arguments

MATRIX A	Square Matrix
----------	---------------

Return Value

Identity matrix of dimension matching the input matrix.

Comments

- 1. The result matrix must be square.
- 2. If preconditions are violated, then a compilation error will occur.
- 3. This function is supported in versions 3.11 and higher

Example

```
MATRIX A (3) (3)
MATRIXIDENT (A)
```

4.2.9 MATRIXTRANS

Description

Transpose a matrix.

Syntax

MATRIXTRANS([in]MATRIX A, [out] MATRIX B)

Arguments

MATRIX A	Input matrix to be transposed
MATRIX B	Output, transposed version of MATRIX A

Return Value

None

Comments

1. The target matrix should be defined so that its row number is equal to the target matrix column number, and its column number is equal to target matrix row number.
2. If preconditions are violated, then a compilation error will occur.
3. This function is supported in versions 3.11 and higher

Example

```
MATRIX A (2) (3) = ( (1, 2, 3) , (4, 5, 6) )  
MATRIX B (3) (2)  
MATRIXTRANS (A, B)
```

4.2.10 MATRIXINVERT

Description

Invert a matrix.

Syntax

MATRIXINVERT([in] MATRIX A, [out] MATRIX B)

Arguments

MATRIX A	Input matrix to be inverted
MATRIX B	Output, inverted version of MATRIX A

Return Value

None

Comments

1. The argument matrix must be square.
2. The result matrix should be of the same size as the argument matrix.
3. If preconditions are violated, then a compilation error will occur.
4. This function is supported in versions 3.11 and higher

Example

```
MATRIX A (3) (3) = ( (1, 2, 3) , (4, 5, 6) , (7, 8, 1) )  
MATRIX B (3) (3)  
MATRIXINV (A, B)
```

4.3 Miscellaneous Functions

The Miscellaneous functions are:

Function	Description
GETCONF	Reads hardware and firmware parameters.
SYSINFO	Returns certain system information based on the argument that is specified.
GETVAR	Reads the current value of the variable and returns it as a real number.
SETCONF	Writes hardware and firmware parameters.
SETVAR	Provide write access to all ACSPL+ variables and to user variables declared with tag .
STR	Converts an integer array to a string.
STRTONUM	Converts an ASCII string representation of a number to the number it represents.
NUMTOSTR	Converts a number to an ASCII string.
BCOPY	Copies bytes from a source array to a target array.
SS1RESET	Resets the values of the last SS1-t channel A and channel B times
ENCREAD	Read encoder parameters

4.3.1 [GETCONF](#)



GETCONF should be used only by knowledgeable users.

Description

GETCONF retrieves system configuration data that was configured by [SETCONF](#).



Some keys relate to data that is set by the system and not by **SETCONF**, for keys set by **SETCONF** see **SETCONF** Arguments.

Syntax

GETCONF(key,index)

Arguments

key	Specifies the configured feature.
index	Specifies the axis, buffer, or type of information requested.

Return Value

GETCONF return values are described in [Table 4-1](#) according to key.

Table 4-1. GETCONF Return Values

Key	Value - Bit and Explanation
26	Returns the mask that determines, for each digital input, whether the leading or trailing signal edge triggers an interrupt. The mask contains a bit for each available input signal. The location of bits in the mask corresponds to the location of bits in variable INO. For each bit: 1: The controller generates an interrupt on the falling edge of the corresponding input signal. 0: Controller generates an interrupt on the rising edge of the corresponding input signal. After power-up, all bits in the mask = 0.
37	Returns the mask that determine whether a digital input triggers on a single edge, or on both edges. If value = 0, the trigger edge is determined by key 26. The location of bits in the mask corresponds to the location of bits in variable INO. 1: The controller generates an interrupt on both edges. 0: The controller generates an interrupt on one edge. After power-up the mask contains 0 in all bits.
71	Used to view the actual assignment of digital outputs to PEG states and PEG pulses outputs. Returns the bit code according to or for SPiiPlusNT/DC-LT/HP/LD-x, or for SPiiPlus CMnt-x-320/UDMpm-x-320, depending on the axis.
72	Used to view the actual encoder PEG engine assignment. Returns the bit code according to or for SPiiPlusNT/DC-LT/HP/LD-x, or for SPiiPlus CMnt-x-320/UDMpm-x-320, depending on the axis.
73	Used to view the actual output pins assignment for PEG engines. Returns the bit code according to or ,for SPiiPlusNT/DC-LT/HP/LD-x, or or for SPiiPlus CMnt-x-320/UDMpm-x-320, depending on the axis.
74	Returns the saved EtherCAT topology configuration Index = 1 : EtherCAT topology mode that is saved on the controller's non-volatile memory: 0: line topology mode 1: ring topology mode 2: two lines topology mode

Key	Value - Bit and Explanation
	Index =2 : Number of nodes connected to the EtherCAT master's main line that is saved on the controller's non-volatile memory Index =3 : Number of nodes connected to the EtherCAT master's redundant line that is saved on the controller's non-volatile memory.
76	Returns the the System or MPU temperature (in degrees °C): 0: Current System temperature 1: Current temperature in controller's CPU/MPU 2: Current DSP temperature (IDMsm/sa, ECMsm/sa, UDMsa)
78	Returns the status if fast loading of Random PEG arrays for the relevant Servo Processor is activated or deactivated: 0: fast loading of Random PEG arrays is deactivated 1: fast loading of Random PEG arrays is activated.  Index is the axis of the relevant Servo Processor: 0, 1, 2, ..., up to total number of axes in system minus 1.
79	Returns the maximum USAGE value since power-up or since last call to the SETCONF(79) command.
80	Returns the UnitID of the network unit that the specified digital input is assigned to. Index = 0, 1, 2... up to total number of digital inputs in the system minus 1.
81	Returns the UnitID of the network unit that the specified digital output is assigned to. Index = 0, 1, 2... up to total number of digital outputs in the system minus 1.
82	Returns the UnitID of the network unit that the specified analog input is assigned to. Index = 0, 1, 2... up to total number of analog inputs in the system minus 1.
83	Returns the UnitID of the network unit that the specified analog output is assigned to. Index = 0, 1, 2... up to total number of analog outputs in the system minus 1.
86	Returns the number of allowed single EtherCAT frames that were actually lost.

Key	Value - Bit and Explanation
99	Index = 0: Returns number of regular ACSPL+ program buffers. Index = 7: Returns total number of axes to which the controller is configured. Index = 8:Key Returns the maximum number of data bytes in the SAFE format message.
203	Returns the value of MFLAGS.1 (Open Loop) 1: Open loop 0: Not open loop
204	Returns the value of MFLAGS.9 (Commutation OK). 1: Commutation OK) 0: Commutation not OK)
214	Only valid for brushless motors (MFLAGS.8 = 1). Returns the commutation phase (degrees) at the current point.
216	Only valid for brushless motors (MFLAGS.8 = 1). Returns the commutation state (MFLAGS.9): 0: Commutation is not OK (not initialized) 1: Commutation is OK.
229	Returns the mechanical brake output: 1: Mechanical brake is inactive 0: Mechanical brake is active

Key	Value - Bit and Explanation
246	Upon receipt of a Drive Alarm signal, the controller stores a general Drive Alarm code (5019) in the MERR variable. The extended Drive Fault status code can be obtained by executing GETCONF(246, Axis). The following extended Drive Fault statuses are supported (the MERR code appears in brackets) by the DDM3U Motor Drive: > Drive Alarm (5060) > Drive Alarm: Short circuit (5061) > Drive Alarm: External Protection Activated (5062) > Drive Alarm: Power Supply Too Low (5063) > Drive Alarm: Power supply too high (5064) > Drive Alarm: Temperature too high (5065) > Drive Alarm: Power Supply 24VF1 (5066) > Drive Alarm: Power Supply 24VF2 (5067) > Drive Alarm: Emergency Stop (5068) > Drive Alarm: Power down (5069) > Drive Alarm: Phase lost. (5070) > Drive Alarm: Drive not ready (5071) > Drive Alarm: Over current (5072) > Not in use (reserved) (5073) > Drive Alarm: Damper is not ok (5074) > Drive Alarm: Digital Drive Interface not Connected (5075) Using the GETCONF function, the faults 5064, 5065, 5069, 5071 can be read before the ENABLE command is executed.  The GETCONF function provides a delay until the extended Drive Fault is received. This behavior differs from the implementation in the SPiiPlus CM, where the extended Drive Fault status is stored in MERR immediately upon receipt of the Drive Alarm signal.
253	Returns the state of the STO signals for the axis selected by index bit 0: ST01 bit 1: ST02

Key	Value - Bit and Explanation
262	Returns the current Hall state, which can be 0, 1, 2, 3, 4, or 5, of the axis given by axis_def (a number: 0, 1, 2, ... up to the number of axes in the system minus 1). It returns -1 for invalid states.
265	When a SIN-COS encoder is used, there are rare cases in which a homing repeatability error of 1 quadrant (quarter of a sine-period) may occur. This key is used for supporting SIN-COS repeatability. For example: <pre> !!!Original homing procedure here ... TILL IST(AXIS).#IND; !Index was found ! Move to a middle of a quadrant, close to the index location PTP(AXIS), IND(AXIS) + POW(2, (E_SCMUL(AXIS)-3)) * EFAC(AXIS) TILL ^MST(AXIS).#MOVE WAIT 1000 ! Repeatability correction SET FPOS(AXIS) = FPOS(AXIS) - IND(AXIS) - GETCONF(265,AXIS) PTP(AXIS), 0 ! 0 = index location </pre>
270	If an axis is enabled while moving, the motor back-EMF may generate high currents during the ENABLE process which can potentially damage the drive or the motor. To avoid such damage the controller should check the motor velocity during the ENABLE process and triggers a fault (error 5104 – “motor is moving”) if it is above a threshold. The threshold is proportional to SLCPRD (commutation period) for a brushless motor (MFLAGS().8=1). It is proportional to XVEL(maximal velocity) for a DC-brush motor (MFLAGS().8=0) Usually the user should not modify the factor, but in special specific cases it may need to be increased. A typical example where modification might be needed is a dual loop system with high resolution encoder on the load and low resolution encoder at the motor (used for commutation). The threshold may be multiplied by a factor using a special SETCONF command. > SETCONF(270, <axis>, <value>) The value is – 1.0 by default. > GETCONF(270, <axis>) returns the current value of the factor. The SETCONF command should be executed after each controller powerup.  Because of the potential damage to the drive and the motor, the user is advised to set this value only after consulting the factory. Contact support@acsmotioncontrol.com.
301	Returns the size of the segment queue in segmented motions (MPTP...ENDS , MSEG...ENDS , PATH...ENDS , PVSPLINE...ENDS).

Key	Value - Bit and Explanation
310	Returns an integer value that contains the TCP/IP address currently assigned to the controller. The index argument has to be zero, for example, GETCONF(310, 0) If a TCP/IP protocol is not configured, or not supported, the return value is zero.
312	Returns the RAM load in percentage, the amount of total physical memory, or the amount of free physical memory as: 0 – memory load in percentage 1 – amount of total physical memory 2 – amount of free physical memory
318	Returns the current mode of operation of move & settle: 0 – Feature is off, no sampling of settling times around radius 1 – Single mode, measure settling time up to first successful settle 2 – Auto mode, keep measuring even after first successful settle
322	Returns 1 if the channel is connected, 0 otherwise

Comments

The IDMxx and EDMxx devices do not support **GETCONF(76,0)** and **GETCONF(76,1)**.

COM Library Methods

GetConf

C Library Functions

acsc_GetConf

Examples

Example 1:

```
?B/GETCONF (229, 0)
```

Returns the following mask:

```
00000000,00000000,00000000,00000001
```

Reports the actual state of the mechanical brake for the given axis. The output is presented in binary base (/B).

Example 2:

```
?X/GETCONF (229, 0)
```

Output:

```
00000001
```

Reports the actual state of the output pins of the mechanical brake for the given axis in hexadecimal format.

Example 3:

```
?X/GETCONF (310,0)
```

Output:

6e00000a

The address of a controller whose TCP/IP address is 10.0.0.110

4.3.2 *SYSINFO*

Description

SYSINFO retrieves a value related to the SPiiPlus controller system based on the argument that is specified.

Syntax

SYSINFO(*Int*)

Arguments

<i>Int</i>	A positive integer ranging between 1 to 16.
------------	---

Return Value

Returns a value based on the specified *Int*.

The possible values of *Int* and the associated return values are given in [Table 4-2](#).

Table 4-2. **SYSINFO** Return Values

Int	Value Returned
1	The SPiiPlus model number. For all EtherCAT products, returns 60. A returned value of ≤ 0 means that the connection is to the Simulator. Note: To get the product ID, use the function ECGETPID.
2	The SPiiPlus version number.
10	Number of regular ACSPL+ program buffers.
11	D-Buffer index.
13	Total number of axes in the current configuration, whether a single SPiiPlus controller or an EtherCAT network.
14	Number of EtherCAT nodes

Int	Value Returned
15	Number of data collection channels per DSP.
16	EtherCAT support: 1 - Yes 0 - No

COM Library Methods

None

C Library Functions

acsc_SysInfo

4.3.3 GETVAR**Description**

GETVAR retrieves a value from a variable (ACSPL+ or user-defined variable, scalar or array) that was declared as a Tag number.

Syntax**GETVAR(Tag, [Index1, Index2])****Arguments**

Tag	The variable Tag number (positive integer)
Index1, Index2	If the variable is an array, the indexes point to the specific location in the array. If the variable is scalar, omit the indexes.

Return Value

Returns the value of the variable specified by the **Tag**.

Example

```

GLOBAL REAL TAG 1001 EE(2) (2)           !Defines user variable array EE as Tag 1001.
SETVAR (15,1001,1,1)                     !Sets value 15 to user variable array

DISP GETVAR (1001,1,1)                   !described in Tag 1001 cell (1)(1).
                                           !Display the value in user variable array
                                           !designated by Tag 1001 cell (1)(1).
STOP                                     !Ends program

```

The controller displays the return value of the **GETVAR** function which is = 15.

4.3.4 SETCONF



SETCONF should be used only by knowledgeable users.

Description

SETCONF defines system configuration data.

All the keys that can be set by **SETCONF** listed in [SETCONF Arguments](#) can also be retrieved by [GETCONF](#).

Syntax

SETCONF(*key,index,value*)

Arguments

<i>key</i>	Specifies the configured feature.
<i>index</i>	Specifies axis or buffer number.
<i>value</i>	The set of bit states for the defined key. <i>value</i> is set up according to a 16-bit binary template, illustrated in 16-bit Binary value Template. The controller strips all leading zeros. The controller understands <i>value</i> in binary, hexadecimal, or decimal format. The following prefixes determine <i>value</i> format: OB - binary OH - hexadecimal A decimal value does not require any prefix.

Table 4-3. 16-bit Binary *value* Template

Bit	3 1	3 0	2 9	2 8	2 7	.. .	6	5	4	3	2	1	0
Pre fix [O B] [O H] <i>val ue</i>	0	0	0	0	0	.. .	0	0	0	0	0	0	0

SETCONF arguments are detailed in [Table 4-4](#).

Table 4-4. SETCONF Arguments

Key	Index	Value - Bit and Explanation
26	Don't Care	Sets the mask that determines, for each digital input, whether the leading or trailing signal edge triggers an interrupt. The mask contains a bit for each available input signal. The location of bits in the mask corresponds to the location of bits in variable INO. For each bit - 1: The controller generates an interrupt on the falling edge of the corresponding input signal. 0: Controller generates an interrupt on the rising edge of the corresponding input signal. After power-up, all bits in the mask = 0.
37		Sets the mask that determine whether a digital input triggers on a single edge, or on both edges. If value = 0, the trigger edge is determined by key 26. The location of bits in the mask corresponds to the location of bits in variable INO. 1: The controller generates an interrupt on both edges. 0: The controller generates an interrupt on one edge. After power-up the mask contains 0 in all bits.
79		Clears the stored maximum usage value. This value is the maximum system usage since power-up or a previous call to SETCONF(79). See GETCONF(79).
86		Resets the counter of allowed single EtherCAT frames that were actually lost
203	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1	Sets the value of MFLAGS.1 (Open Loop) 1: Open loop 0: Not open loop
204	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1	Sets the value of MFLAGS.9 (Commutation OK). 1: Commutation OK

Key	Index	Value - Bit and Explanation
		0: Commutation not OK
214	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1	Valid only for brushless motors (MFLAGS.8 = 1). SETCONF with the 214 key sets the commutation offset at the current position to the specified value (in degrees). There are, however, two cases that have to be considered when using SETCONF: > Motor Not Commutated (MFLAGS.9=0) If the motor has not yet been commutated, SETCONF sets the commutation offset at the current position to the value specified by the value argument. > Motor Commutated (MFLAGS.9=1) The behavior of SETCONF for commutated motors in SPiiPlus NT controllers is different from that in non-NT SPiiPlus controllers. For NT controllers once a motor is commutated, the commutation phase has an additional 90°. SETCONF modifies the commutation phase prior to this 90° addition. Therefore, to change the commutation phase of a commutated motor, it is recommended that the user enter ?GETCONF(214,axis) in the Communication Terminal and subtract 90 from the returned value. From this the user can calculate what the value of the value argument should be to get the proper phase.
216	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1	Valid only for brushless motors (MFLAGS.8 = 1). If value = 1: > MFLAGS.9=0 (Commutation is not OK) > MFLAGS bits 1, 4, 5, 6 are set to zero > MFLAGS.8 is set to 1 > DCOM is reset to zero > Sets RPOS = FPOS

Key	Index	Value - Bit and Explanation
		If value = 0: > MFLAGS.9=0 (Commutation is not OK) > MFLAGS bits 1, 4, 5, 6 are set to zero > MFLAGS.8 is set to 10 > Variable DCOM is reset to zero.
217	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1	Valid only for brushless motors (MFLAGS.8 = 1) where the encoder has encountered the index and IND contains a valid value. value: Not Relevant. > Adjusts the commutation offset so that the commutation phase at the last index point is equal to the specified value in degrees. > MFLAGS.9=0 (Commutation is not OK) > MFLAGS bits 1, 4, 5, 6 are set to zero > MFLAGS.8 is set to 1 > DCOM is reset to zero > Sets RPOS = FPOS
229	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1	Mechanical brake output: 1: Deactivates mechanical brake output. 0: Activates mechanical brake output. The motor must be disabled to execute this setting.
246	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1	SETCONF(246, Axis, 0) is used to clear the fault status on all axes that relate to the DDM3U Motor Drive that handles the specified axis.  The FCLEAR command does not clear the fault status in the MC4U, SETCONF(246, Axis, 0) has to be used instead.

Key	Index	Value - Bit and Explanation
249	Axis: Axis: 0, 1, 2, ..., up to total number of axes in system minus 1	There are situations where automatic current bias measurement (in ACS control modules) can lead to problems. For example, when running an air bearing stage, the automatic current offset measurement may be inconsistent with every enable due to air bearing stage movement or drift during enable. Thus any small offset can create a relatively large oscillation during constant velocity. SETCONF(249, Axis, 0) disables the automatic current bias measurement for the given axis.

Key	Index	Value - Bit and Explanation
267	0, 1, 2, ..., up to total number of axes in system minus 1	Changes the gantry pair's allocation to all axes within the Servo Processor that the specified axis belongs to. Gantry pair's allocation is done according to the following: 0: for pairs (0,1) and (2,3) 1: for pairs (0,2) and (1,3) For example: 4 axes MC4U system Command for setting pairs (0,1) and (2,3) - SETCONF(267,0,0) Command for setting pairs (0,2) and (1,3) - SETCONF(267,0,1) 8-axes MC4U system Command for setting pairs (0,1) and (2,3) - SETCONF(267,0,0) Command for setting pairs (0,2) and (1,3) - SETCONF(267,0,1) Command for setting pairs (4,5) and (6,7) - SETCONF(267,4,0) Command for setting pairs (4,6) and (5,7) - SETCONF(267,4,1) Two 8-axes MC4U systems connected in the network Command for setting pairs (0,1) and (2,3) - SETCONF(267,0,0) Command for setting pairs (0,2) and (1,3) - SETCONF(267,0,1) Command for setting pairs (4,5) and (6,7) - SETCONF(267,4,0) Command for setting pairs (4,6) and (5,7) - SETCONF(267,4,1) Command for setting pairs (8,9) and (10,11) - SETCONF(267,8,0)

Key	Index	Value - Bit and Explanation
		Command for setting pairs (8,10) and (9,11) - SETCONF(267,8,1) Command for setting pairs (12,13) and (14,15) - SETCONF(267,12,0) Command for setting pairs (12,14) and (13,15) - SETCONF(267,12,1)
270	Axis: 0, 1, 2, ..., up to total number of axes in system minus 1	If an axis is enabled while moving, the motor back-EMF may generate high currents during the ENABLE process which can potentially damage the drive or the motor. To avoid such damage the controller should check the motor velocity during the ENABLE process and triggers a fault (error 5104 – “motor is moving”) if it is above a threshold. The threshold is proportional to SLCPRD (commutation period) for a brushless motor (MFLAGS().8=1). It is proportional to XVEL (maximal velocity) for a DC-brush motor (MFLAGS().8=0) Usually the user should not modify the factor, but in special specific cases it may need to be increased. A typical example where modification might be needed is a dual loop system with high resolution encoder on the load and low resolution encoder at the motor (used for commutation). The threshold may be multiplied by a factor using a special SETCONF command. > SETCONF(270, <axis>, <value>) The value is – 1.0 by default. > GETCONF(270, <axis>) returns the current value of the factor. The SETCONF command should be executed after each controller powerup.

Key	Index	Value - Bit and Explanation
		 Because of the potential damage to the drive and the motor, the user is advised to set this value only after consulting the factory. Contact support@acsmotioncontrol.com.
302	2 1	The following decimal values specify a communication channel for special input: 3: Set the channel to Modbus Master mode. 2: Set the channel to Modbus Slave mode 1: Assigns the channel for special input. 0: Set the channel to regular command processing mode (default channel mode). If a channel is assigned for special input, the controller does not process commands from this channel. Output to the channel is provided by regular DISP and SEND commands. index specifies the channel.
303	2 1	The value sets the baud rate for the specified serial channel, where the baud rate is the decimal value. 115200 (default), 57600, 19200, 9600, 4800, 2400, 1200, 600, 300. index specifies the channel.

Key	Index	Value - Bit and Explanation
304	2 1	Sets communication options: Bit 2 - 1: extended stop bit 0: normal stop bit Bit 3 - 1: check parity 0: no parity Bit 4 - 1: even parity 0: odd parity index specifies the channel.
306	-1	In order to be able to receive messages, use specific channels (SEND and DISP commands) 1 - Enable receiving messages on a specific channel 0 - Stop receiving messages from a host application
308	19 18 17	When the controller acts as a Modbus Master, the function establishes or closes the Modbus TCP connection with the Slave device using the specified slaveIP. Up to three slaves can be connected to the master controller (17, 18 or 19). Value=SlaveIP. The slaveIP value specifies the Modbus Slave device IP address. The slaveIP address is calculated as follows: If the Slave has the following address: 192.168.1.10, the slaveIP parameter should be $10 \times 2^{24} + 1 \times 2^{16} + 168 \times 2^8 + 192 = 167880896$ (0x0A01A8C0). If the specified channel is already open, the function closes the opened channel and then opens new one. If slaveIP is zero, the function closes the TCP connection.
309	Don't Care	Defines the sequence for the two 16-bit Modbus interface registers -

Key	Index	Value - Bit and Explanation
		1: Low word will be first, then high word 0: High word will be first, then low word (default configuration)

Key	Index	Value - Bit and Explanation
310	0	Used for providing access to the controller TCP/IP address. Where: Value = TCP/IP Address, which is a 32-bit (four bytes) integer, each byte of which contains one part of the TCP/IP address, for example, 0x6400000a assigns address 10.0.0.100 (the number is read from right to left and fills in the address left to right). If value is zero, SETCONF activates a new execution of the DHCP protocol and obtains a new TCP/IP address from the host (the host may configure the same address as before). SETCONF does not change the TCPIP variable. After power-up, the controller is initialized with the TCP/IP address set in the TCPIP variable. There are several limitations when using SETCONF(310): > If the TCPIP variable stored in the flash is zero, SETCONF(310) must be used only with zero address argument. In other words, if the controller is configured for dynamic addressing, assigning static addresses is not allowed. > If the TCPIP variable stored in the flash is not zero, SETCONF(310) must be used only with non-zero address arguments. In other words, if the controller is configured for static addressing, switching to a dynamic address is not allowed. > SETCONF(310) has a long execution time. During this time, communication with the controller is impossible using any communication channel. Use SETCONF(310) only within the controller initialization sequence. Avoid attempts to communicate with the controller and the motor ENABLE command or motion commands while SETCONF(310) is in progress.

Key	Index	Value - Bit and Explanation
318	Axis: 0, 1, 2, ..., up to total number of axis in system minus 1	Move & Settle: 0 – Feature is off, no sampling of settling times around radius 1 – Single mode, measure settling time up to first successful settle 2 – Auto mode, keep measuring even after first successful settle.
322	> TCP Server Communication Port > Communication Channel (channels 26-29 are reserved for this task, no other channels may be used) Index value calculated as follows: Index = Port*100+Channel	The value is the TCP Server IP Address The IP address is calculated as follows: If the TCP Server has the following address: 192.168.1.10, the IP parameter should be $10*2^{24} + 1*2^{16} + 168*2^8 + 192 = 167880896$ (0x0A01A8C0). If the specified channel is already open, the function closes the opened channel and then opens new one. If an IP is zero, the function closes the TCP connection.
325	0	Range of Values: 0-400 > 0 – Matrix related operations will not be performed in real-time. > 400 - Operations on matrices with up to 400 elements (e.g., 20x20) will be performed in real-time. Operations on matrices larger than 400 are not allowed to run in real-time due to critical impact on MPU usage.

Return Value

None

COM Library Methods

SetConf

C Library Functions

acsc_SetConf

Example

The following example illustrates setting the value of MFLAGS.1 to 1, configuring axis 2 to open loop control:

```
SETCONF(203, 1, 1)
```

4.3.5 SETVAR

Description

SETVAR writes a value to an ACSPL+ or user variable, scalar or array, that was declared as a Tag number.

Syntax

SETVAR(*value*, *Tag*, [*Index1*, *Index2*])

Arguments

<i>value</i>	A real or integer value assigned to the variable.
<i>Tag</i>	The variable Tag number (positive integer).
<i>Index1</i> , <i>Index2</i>	If the variable is an array, the indexes point to the specific location in the array. If the variable is scalar , omit the indexes.

Return Value

None

Error Conditions

None

Example

```
GLOBAL REAL TAG 1001 EE(2)(2)    !Defines user variable array EE as Tag 1001
SETVAR (15,1001,1,1)             !Sets value 15 to user variable array
                                  !described in Tag 1001 cell (1)(1).
DISP GETVAR (1001,1,1)           !Retrieves the value in user variable array
                                  !described in Tag 1001 cell (1)(1).
STOP                             !Ends program
```

The controller displays the **GETVAR** return value which is = 15.

4.3.6 STR

Description

STR converts an integer array to a string. Each element of the array is interpreted as an ASCII character.

Syntax

string **STR**(*array_name*, [*start_index*,] [*number*])

Arguments

<i>array_name</i>	Name of user-defined integer array.
<i>start_index</i>	Index in the array from which to start converting.
<i>number</i>	Number of characters to convert.

Comments

If an element value is in the range from 0 to 255, it is directly converted to the corresponding ASCII character. Otherwise, the value will be cyclic, based on 256.

If **start_index** is omitted, the assignment starts from the first element of the array.

If neither **start_index** nor **number** is specified, the conversion takes all elements of the array. If only **start_index** is specified, the conversion takes all characters from the specified index until the end of the array. **number** limits the number of characters in the resulting string.

Return Value

String composed of the array elements interpreted as characters.

Example

```
GLOBAL INT BIBI(3)
BIBI(0)=65
BIBI(1)=67
BIBI(2)=83
DISP STR(BIBI)
STOP
```

The function transforms each of the array members of BIBI to ASCII code characters. When **DISP** is applied, the displayed value will be: "ACS"

4.3.7 STRTONUM**Description**

This function converts ASCII encoded element string to a number.

Syntax

double **STRTONUM**(*Source, Type, From, N*)

Arguments

<i>Source</i>	An integer array
<i>Type</i>	Designates the format of the converted number, it can be: > 0 - decimal > 1- hex > 2 - floating point

From	Index of the first element in the array that may contain the number
N	The number of elements in source that may be used

Return Value

The number converted from the ASCII string

Example

```
real number
int string(4);
number = 0
string(0) = 49; !1
string(1) = 50; !2
string(2) = 46; !.
string(3) = 50; !2
number = strtonum(string, 0, 0, 4);
disp"number = %f", number;
number = strtonum(string, 1, 0, 4);
disp"number = %f", number;
number = strtonum(string, 2, 0, 4);
disp"number = %f", number;
stop
! output:
! number = 12.000000
! number = 18.000000
! number = 12.200000
```

4.3.8 NUMTOSTR

Description

This function converts a number to a string of elements where each element is an ASCII encoded value.

Syntax

int **NUMTOSTR**(*Number, Target, Type, From, N*)

Arguments

Number	The number to be converted
Target	An integer array used in which to put the results
Type	Designates the format of the number to be converted, it can be: > 0 - decimal > 1- hex > 2 - floating point

From	Index of the first element in the array that may contain the number
N	The number of elements in target that may be used

Return Value

The number of elements used.

Example

```

real number
int target(12);
int i;
int j;
number = 12.2
i=0; loop 12 target(i) = 0; i = i+1; end;
j = numtostr(number,target,0,0,12)
disp "target = %X,%X,%X,%X,%X num elements = %d", target(0),target
(1),target(2),target(3),target(4), j
i=0; loop 12 target(i) = 0; i = i+1; end;
j = numtostr(number,target,1,0,12)
disp "target = %X,%X,%X,%X,%X num elements = %d", target(0),target
(1),target(2),target(3),target(4), j
i=0; loop 12 target(i) = 0; i = i+1; end;
j = numtostr(number,target,2,0,12)
disp "target = %X,%X,%X,%X,%X num elements = %d", target(0),target
(1),target(2),target(3),target(4), j
stop
! output:
! target = 31,32,0,0,0 num elements = 2
! target = 63,0,0,0,0 num elements = 1
! target = 31,32,2E,32,30 num elements = 9

```

4.3.9 BCOPY**Description**

This function can be used to copy bytes from the source array to the target array.

Syntax

int BCOPY(Source_array, Target_array, CopyBytes, S_SkipBytes, T_SkipBytes, From, N)

Arguments

Source_array	Array containing input data
Target_array	Array containing output data
CopyBytes	Number specifying how many bytes will each copy operation
S_SkipBytes	Number specifying how many bytes will be skipped in the source array following each copy operation

<i>T_SkipBytes</i>	Number specifying how many bytes will be skipped in the target array following each copy operation
<i>From</i>	Index of the first element in the source array that may be used
<i>N</i>	The number of elements in the source that may be used

Return Value

The number of elements (in the source) used.

Comments

The function copies **CopyBytes** from the source array to the target array, skip **S_SkipBytes** in the source and skip **T_SkipBytes** in the target, and then copy **CopyBytes** again. The operation starts from the **From** element in source and is applied to a maximum of **N** elements. This means no bytes from any element other than the specified **N** elements will be affected. Elements less than **N** can be affected if the target array is too short to complete the whole operation.

The difference between elements and bytes throughout this function should be noted.

Examples

In the examples that follow use is made of:

bcopy(source,target,1,3,0) - Copy every 4 byte element from source to a 1 byte element in target

bcopy(source,target,2,2,0) - Copy every 4 byte element from source to a 2 byte element in target

bcopy(source,target,1,0,3) - Copy every 1 byte element from source to a 4 byte element in target

bcopy(source,target,2,0,2) - Copy every 2 byte element from source to a 4 byte element in target

1. Unpacking a single element in source to 1, 2, or 4 elements in target

```
int source(1)
int target(4)
int j
source(0) = 0x01020304;
j = bcopy(source,target,4,0,0,0,1) => target = 1020304,0,0,0 num elements
= 1
j = bcopy(source,target,2,0,2,0,1) => target = 304,102,0,0 num elements =
1
j = bcopy(source,target,1,0,3,0,1) => target = 4,3,2,1 num elements = 1

stop
```

2. Packing multiple elements in source to 1 element in target.

```
int source(4)
int target(1)
int i
int j
target(0) = 0;
i = 0
loop 4
```



```

source(i) = 1;
i=i+1;
end
j = bcopy(source,target,4,0,0,0,4) => target(0) = 1 j = 1
j = bcopy(source,target,2,2,0,0,4) => target(0) = 10001 j = 2
j = bcopy(source,target,1,3,0,0,4) => target(0) = 1010101 j = 4
stop

```

4.3.10 SS1RESET

Description

Resets the values of the last SS1-t channel A and channel B times.



This function should be used for SS1-t diagnostics only.

Syntax

SS1RESET [/f] *axis*

Arguments

axis	The axis index, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1. Axis parameter can be any axis number of the same unit.
------	---

Comments

This variable is supported in version 3.00 and higher

Return Value

None

4.3.11 MDURATION

Description

The **MDURATION** function calculates time of certain motion profile phase. The function can be executed in buffer or terminal, and it will wait till the execution is completed.

Syntax

```

REAL MDURATION(MotionType, Distance, Vel, Acc, Dec, Jerk, phaseNumber
(optional), EncoderFactor(optional))

```

Arguments

MotionType	An integer number representing the motion type 0: PTP motion Currently only PTP motion is supported
Distance	A real number representing the distance of motion. The velocity in [user units]
Vel	A real number representing the desired velocity. The velocity in . Velocity ranges from -1.79769e+308 to 1.79769e+308.
Acc	A real number representing the desired acceleration. The acceleration in . Acceleration ranges from 2.22507e-308 to 1.79769e+308.
Dec	A real number representing the desired deceleration. The acceleration in . Deceleration ranges from 2.22507e-308 to 1.79769e+308.
Jerk	A real number representing the desired jerk. The jerk in . Jerk ranges from 2.22507e-308 to 1.79769e+308.
phaseNumber	(Optional, Integer) Determines which motion profile component will be returned. 0: t – Motion overall time [sec] – default 1: t1 – Acceleration buildup [sec] 2: t2 – Constant Acceleration [sec] 3: t3 – Acceleration finishing [sec] 4: t4 – Constant velocity [sec] 5: t5 – Deceleration buildup [sec] 6: t6 – Constant deceleration [sec] 7: t7 – Deceleration finishing [sec]
EncoderFactor	A real representing the factor between the raw feedback in encoder counts and the FPOS value calculated by the controller. EncoderFactor ranges between 1e-15 to 1e+15, Default = 1.

Return Value

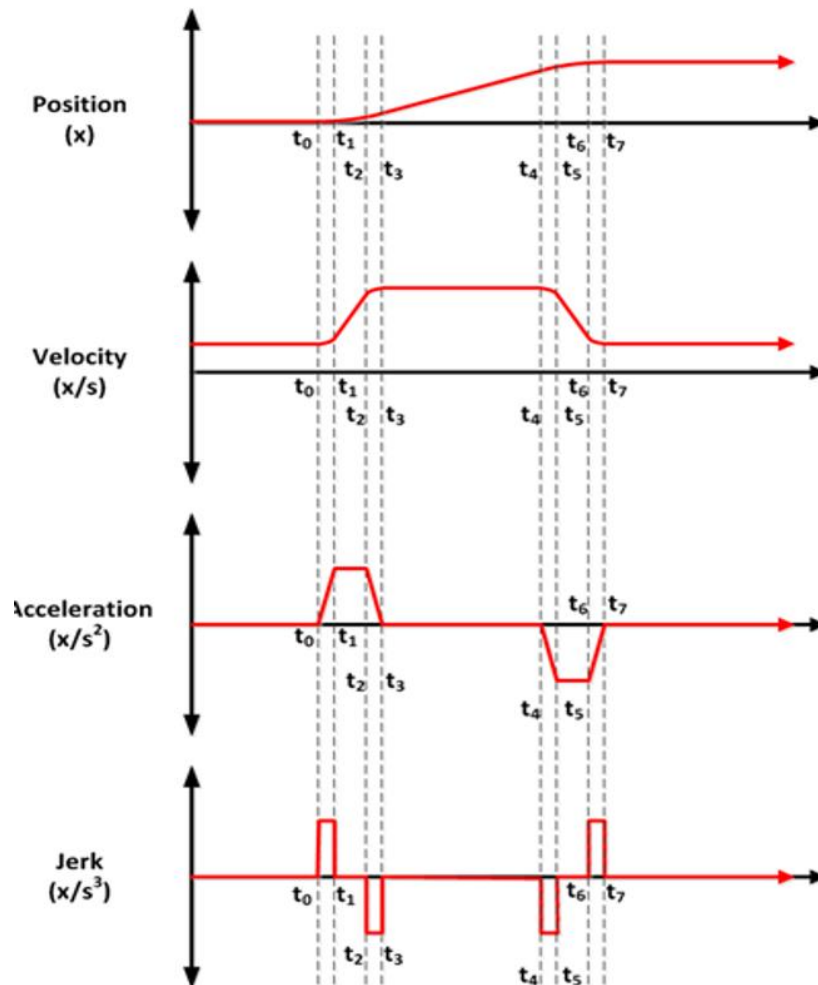
Returns motion profile time according to `phaseNumber`.

Comments

Assume that the motion will start and complete when velocity and acceleration are 0.

Input Shaping is not supported

`phaseNumber` options corresponds to a third order point-to-point profile:



Assume that the motion will start and complete when velocity and acceleration are 0.

Input Shaping is not supported.

Error Conditions

The function detects the following error conditions.

- > **3207** - phaseNumber parameter is not between 0 to 7
- > **3412** - The MotionType parameter is intended to be one of supported motion types. Currently only PTP motion is supported.
- > **3045** - At least one of struct members is infinite.
- > **3041** - Vel, Acc, Dec, Jerk, Or EncoderFactor is out of range.

Examples

```
! Example 1
! MotionType = 0 - PTP
! Distance = 20000, Vel= 10000, Acc = 100000, Dec = 100000, Jerk =
20000000
! phaseNumber = default = 0 - Motion overall time in seconds
```

```
! EncoderFactor = default = 1
V0 = MDURATION(0, 20000, 10000, 100000, 100000, 20000000)
! Return value: Motion overall time V0 = 2.105[sec]
STOP
```

```
! Example 2
! MotionType = 0 - PTP
! Distance = 20000, Vel= 10000, Acc = 100000, Dec = 100000, Jerk =
20000000
! phaseNumber = 4 - Constant velocity
! EncoderFactor = default = 1
V0 = MDURATION(0, 20000, 10000, 100000, 100000, 20000000, 4)
! Return value: Constant velocity V0 = 1.895[sec]
STOP
```

```
! Example 3
! MotionType = 0 - PTP
! Distance = 20000, Vel= 10000, Acc = 100000, Dec = 100000, Jerk =
20000000
! phaseNumber = 4 - Constant velocity
! EncoderFactor = 2
V0 = MDURATION(0, 20000, 10000, 100000, 100000, 20000000, 4, 2)
! Return value: Constant velocity V0 = 1.895[sec]
STOP
```

4.4 Array Processing Functions

The Array Processing functions are:

Function	Description
AVG	Finds the average of all values in an array.
COPY	Copies data from one user array to another.
DSHIFT	Shifts all of the elements of the array to one position left.
FILL	Fills an array or a section of array with the specified value.
MAX	Finds the maximum value in an array
MAXI	Finds the maximum value in an array or in a section of array and returns its index.
MIN	Finds the minimum value in an array.
MINI	Finds the element with minimum value in an array or in a section of an array and returns its index.

Function	Description
SIZEOF	Returns number of columns or number of rows of the given array

4.4.1 AVG

Description

AVG finds the average of all values in an array.

Syntax

AVG(array_name)

Arguments

array_name	The name of an array that has been declared in the program.
-------------------	---

Return Value

Real number - returns the average of all elements in **array_name**.

Example

```
REAL Ar(3)
Ar(0) = 1; Ar(1)=0.5; Ar(2)=3;
DISP AVG(Ar)
!Output = 1.5
```

4.4.2 COPY

Description

COPY copies data from one user array to another.

Syntax

COPY(source, destination, from_source_row, to_source_row, from_source_col, to_source_col, from_destination_row, to_destination_row, from_destination_col, to_destination_col)

Arguments

source	The name of an array that has been declared in the program from which the data is to be copied.
destination	The name of an array that has been declared in the program to which the data is to be copied.
from_source_row	The index of the first row of the source to begin copying.
to_source_row	The index of the last row of the source to end copying.

<i>from_source_col</i>	The index of the first column of the source to begin copying. Used only for matrix type arrays, otherwise it can be omitted.
<i>to_source_col</i>	The index of the last column of the source to end copying. Used only for matrix type arrays, otherwise it can be omitted.
<i>from_destination_row</i>	The index of the first row of the destination to begin copying into.
<i>to_destination_row</i>	The index of the last row of the destination to end copying into.
<i>from_destination_col</i>	The index of the first column of the destination to begin copying into. Used only for matrix type arrays, otherwise it can be omitted.
<i>to_destination_col</i>	The index of the last column of the destination to begin copying into. Used only for matrix type arrays, otherwise it can be omitted.

Comments

If the matrix indexes are omitted, the entire source matrix will be copied to the destination matrix.

If the destination matrix has different dimensions than the source matrix then the destination matrix will use the source matrix values to fill each row completely and move to the next row.

Return Value

None

Error Conditions

Error 3034, Illegal index value - the destination matrix is smaller than the source matrix.

Example

```

INT GLOBAL SOURCE(3)(3), DESTINATION(3)(3)      !Define source and destination
                                                !matrices
!----- Assign values to SOURCE matrix -----
-
SOURCE(0)(0)=1;SOURCE(0)(1)=2;SOURCE(0)(2)=3;SOURCE(1)(0)=4;SOURCE(1)
(1)=5;
SOURCE(1)(2)=6;SOURCE(2)(0)=7;SOURCE(2)(1)=8;SOURCE(2)(2)=9
!----- Assign values to DESTINATION matrix -----
----
COPY(SOURCE,DESTINATION,0,1,0,2,1,2,0,2)      !COPY command -
                                                !The program copies the first
                                                !two rows from the source
                                                !matrix to the destination
                                                !matrix second two rows. The
                                                !matrices are as follows:
                                                !Source:
                                                !1 2 3

```

STOP

```
!4 5 6
!7 8 9
!Destination:
!0 0 0
!1 2 3
!4 5 6
!Ends program
```

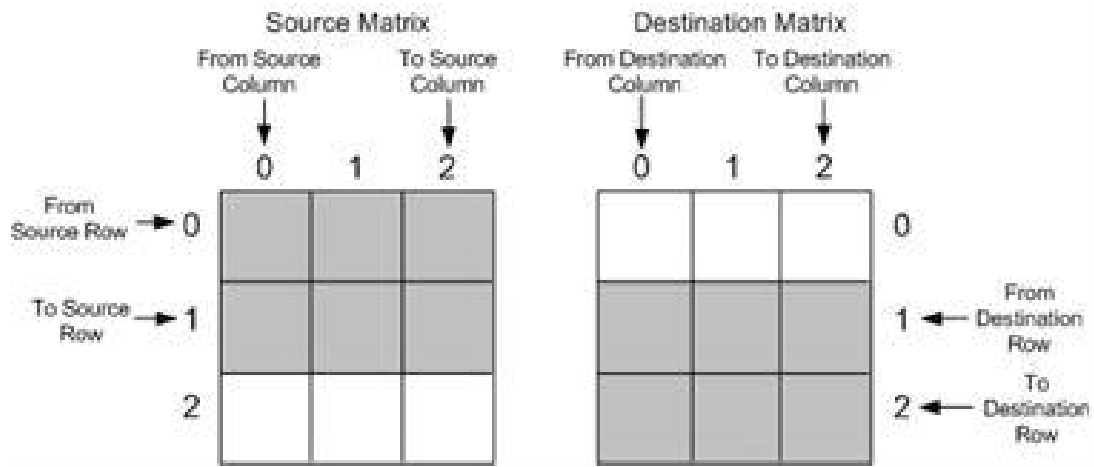


Figure 4-1. Illustration of COPY Function

4.4.3 DSHIFT

Description

DSHIFT shifts all elements of the array to one position left.

Syntax

real **DSHIFT**(*array, value, index*)

Arguments

<i>array</i>	An array of real with the maximal size of 10,000 elements
<i>value</i>	The real value to be inserted to the array. The Value is inserted to position of the element with the index Index.
<i>index</i>	The position of the element the Value is inserted to. The Index should be equal or less than declared size of the array

Return Value

The first element (element with index 0) of the array.

Comments

Each time the function is called the first element of the array (element with the index 0) is returned. All of the other elements of the array shift one position to the left (element with index 1 to 0, element with index 2 to 1, etc.). The Value parameter is inserted to the element with the index Index.

This function only works with arrays with up to 10,000 elements.

The function is useful when it is necessary to delay data for a number of the controller cycles.

Example 1

Example 1 shows how to delay the result of the motion trajectory generator.

```
global real Array(20)    ! Array for delay
global int Axis          ! Axis index
Axis=0\
fill(APOS(Axis), Array) ! FILL array with initial APOS value
MFLAGS(Axis).#DEFCON=0  ! Motion trajectory delay is implemented
                        ! using CONNECT function
! Motion trajectory is delayed for 10 controller cycles
CONNECT RPOS(Axis) = dshift(Array, APOS(Axis),8)
DEPENDS Axis, Axis
stop
```

Example 2

The example 2 shows how to delay the pulses of laser generator. In this example it's assumed that the laser control module generates pulses as function of vector velocity of the axis. The PFGPAR parameter is used to transfer a vector velocity value to the laser control module.

```
global real Array(20)    ! Array for delay
global int Axis          ! Axis index
Axis=0
fill(0, Array)          ! FILL array with zero values
! Laser control module pulses are delayed for 10 controller cycles
while (1); PFGPAR(Axis) = dshift(Array, GVEL(Axis), 9); end;
stop
```

4.4.4 FILL

Description

FILL fills an array or a section of array with a specified value.

Syntax

FILL (*real value, destination, from_array_row, to_array_row, from_array_column, to_array_column*)

Arguments

<i>real value</i>	Any real number.
<i>destination</i>	The name of an array that has been declared in the program to which the value is to be copied.
<i>from_array_row</i>	The index of the first row of the destination array to begin filling.
<i>to_array_row</i>	The index of the last row of the destination array to end filling.

<i>from_array_ column</i>	The index of the first column of the destination array to begin filling. Used only for matrix type arrays, otherwise it can be omitted.
<i>to_array_ column</i>	The index of the last column of the destination array to end filling. Used only for matrix type arrays, otherwise it can be omitted.

Comments

If the matrix indexes are omitted, the entire matrix will be filled with the requested value.

Example

```
GLOBAL ARR(6)(3)
FILL(3,ARR,0,4,1,2)
STOP

!The program fills ARR with the value 3, from row 0 to row 4,
!and from column 1 to column 2. After executing the program,
!the resulting ARR values are:
!0      3      3
!0      3      3
!0      3      3
!0      3      3
!0      3      3
!0      0      0
```

4.4.5 MAX

Description

MAX finds the maximum value in an array or in a section of an array.

Syntax

MAX(array_name, From1, To1, From2, To2)

Arguments

<i>array_name</i>	The name of an array that has been declared in the program.
From1	The initial element
To1	The final element
From2	The initial element
To2	The final element

Return Value

Real number - returns the maximum element in the array.

Example

```
REAL AR(3)
AR(0) = 1; AR(1)=0.5; AR(2)=3;
DISP MAX(AR) !Output = 3
```

4.4.6 MAXI

Description

MAXI finds the maximum value in an array or in a section of array and returns its index.

Syntax

MAXI(array_name, From1, To1, From2, To2)

Arguments

<i>array_name</i>	The name of an array that has been declared in the program.
From1	The initial element
To1	The final element
From2	The initial element
To2	The final element

Return Value

Integer - **MAXI** returns the index of maximum element in the array, or in the specified section of the array. In case of a two-dimensional array only the column index is returned.

Error Conditions

None

Example

```
REAL AR(3)
AR(0)= 1; AR(1)= 0.5; AR(2)= 3
DISP MAXI(AR) !Output = 2
```

4.4.7 MIN

Description

MIN finds the minimum value in an array or in a section of an array.

Syntax

MIN(array_name, From1, To1, From2, To2)

Arguments

<i>array_name</i>	The name of an array that has been declared in the program.
-------------------	---

Return Value

Real number - returns the minimum element in the array.

Error Conditions

None

Example

```
REAL Ar(3)
Ar(0) = 1; Ar(1)=0.5; Ar(2)=3;
DISP MIN(Ar)                !Output = 0.5
```

4.4.8 MINI

Description

MINI finds the element with the minimum value in an array or in a section of an array and returns its index.

Syntax

MINI(array_name, From1, To1, From2, To2)

Arguments

<i>array_name</i>	The name of an array that has been declared in the program.
From1	The initial element
To1	The final element
From2	The initial element
To2	The final element

Return Value

Integer - **MINI** returns the index of the minimum element in array X, or in the specified section of array x. In case of a two-dimensional array, only the column index is returned.

Example

```
REAL AR(3)
AR(0) = 1; AR(1)= 0.5; AR(2)= 3
DISP MINI(AR) !Output = 1
```

4.4.9 SIZEOF

Description

The function returns number of columns or number or rows of the given array (can be user-defined or ACSPL+). If the array is one-dimension, number of rows is always one and the number of columns represents the number of elements.

Syntax

```
sizeof(Array, index(optional) )
```

Arguments

Array	The array name, can be local or global
Index	Optional parameter. If specified, can be 1 or 2

Return Value

- > No index is specified: total number of elements in the array
- > Index parameter equals to 1: number of columns for two-dimensional array, or number of elements for one-dimensional array
- > Index parameter equals to 2: number of rows for two-dimensional array, or 1 for one-dimensional array.

Comments

In case of wrong parameters, the corresponding runtime error will be generated. The function is intended to be used for arrays only, meaning that an error is generated if a scalar is given as a parameter.

Example

```
IO=sizeof(NST)
```

4.5 EtherCAT Functions

EtherCAT functions include:

COGETSIZE	Returns the size, in bits, of a specific entry in the object dictionary of a specific slave.
ECCLOSEPORT	The function closes the specified port of specified EtherCAT node.
ECCLRREG	Clears the contents of error counters registers.
ECEXTIN	Used for mapping input variables (TxPDO) to non-ACS EtherCAT network.
ECEXTOUT	Used for mapping output variables (RxPDO) from non-ACS EtherCAT network to the SPiiPlusES.
ECGETGRPIND	The function returns an array that contains optional groups' indexes that are part of the current configuration (including mandatory group, which is 0).

ECGETPID	Returns product ID of the node.
ECGETMAIN	The function returns the number of EtherCAT slaves connected to the main EtherCAT line.
ECGETOFFSET	The function returns offset of the specified variable of the specific EtherCAT node.
ECGETOPTGRP	The function returns number of actually connected optional groups, not including the mandatory group.
ECGETRED	The function returns the number of EtherCAT slaves connected to the redundant EtherCAT line.
ECGETREV	Returns EtherCAT Revision ID of the node
ECGETREG	Gets the contents of ESCs error counters registers.
ECGETSLAVES	Returns the number of slaves in an EtherCAT network.
ECGETSN	Returns Serial Number of the node.
ECGETSTATE	Returns the state of the node.
ECGETVID	Returns vendor ID of the node.
ECGRPINFO	The function fills array with nodes' indexes which are members of a given optional group. In addition, it returns the number of members in the group.
ECIN	Copies the EtherCAT network input variable to an ACSPL+ variable.
ECOUT	Copies ACSPL+ variable to EtherCAT network output variable.
ECRESCAN	Triggers the system to rescan the EtherCAT network after a slave has been removed or been added in order to refresh the network composition data.
ECRESCUE	Triggers execution of a rescue scan of the EtherCAT network
ECREPAIR	Returns the system back to the operational state if one or more slaves underwent a reset or power cycle.
ECSAVECFG	Saves the current network topology configuration into the non-volatile memory.

ECSAVEDCNF	The function returns array that contains optional groups' indexes that are part of the last saved configuration (including mandatory group, which is 0).
ECUNMAP	Resets mapping of ECIN and ECOUT .
ECUNMAPIN	Resets mapping of ECIN of a specified offset.
ECUNMAPOUT	Resets mapping of ECOUT of a specified offset.
FOEDOWNLOAD	Downloads a file over EtherCAT from the controller's flash to slave
FOEUPLOAD	Upload file over EtherCAT and saves it to the controller's flash memory
PDOEXT	Prints the PDO configuration defined by master (SPiiPlusES only)

4.5.1 ECCLOSEPORT

Description

The function closes the specified port of specified EtherCAT node.

Syntax

Int **ECCLOSEPORT**(*name*, *index*)

Arguments

Index	EtherCAT node index.
Port	EtherCAT port index (0 - EtherCAT IN port, 1 - EtherCAT OUT port)

Return Value

None

Comments

This function can only be used if ring topology is configured and ring communication is active. In case there is a cable / port failure on a specific EtherCAT node, it is recommended (as long as the erroneous situation exists) that the machine should start up directly in the line topology mode. This is very useful, because it prevent a manual stop of the machine with a suspicious communication link / device).

Example

```
ECCLOSEPORT(<slave index>, <port index>)
FCLEAR ALL
ECSAVECFG
STOP
```

In this example of the AUTOEXEC program that should run on power-up in this case:

The first parameter is the slave index in the network and the second parameter is the port index of the slave that should be closed.

After "fclear All", two Lines Are Stored Into The Controller's Non-volatile Memory.

4.5.2 ECCLRREG

Description

ESC Error Counters Registers Clear. The **ECCLRREG** function clears the contents of the error counters registers.

Syntax

```
void ECCLRREG[0 /1] (index,offset)
```

Arguments

Index	EtherCAT slave index.
Offset	Register offset in the Beckhoff memory.

Switches

/0	First EtherCAT network. Equivalent to no switch
/1	Second EtherCAT network

Return Value

None

Comments

When the Offset value is -1, all error counters in all slaves are cleared. Otherwise, only the specific register at the specified Offset is cleared.

After executing the **ECCLRREG** function, we recommend to execute the **FCLEAR** function without parameters before running **ECGETREG**.

Example

Run the following code example in a Program Buffer.

```
ECCLRREG (0,0x310)  
FCLEAR  
STOP
```

You can also enter this code in the SPiiPlus MMI Application Studio Connection Terminal: **ECCLRREG (0,-1)** .

4.5.3 ECEXTIN

Description

The **ECEXTIN** function is used for mapping input variables (TxPDO) to non-ACS EtherCAT network.



Can be used only by SPiiPlusES.

Syntax

```
ECEXTIN (int PDOIndex, int Index, int Subindex, string Varname)
```

Arguments

PDOIndex	One of the TxPDO indexes defined by external master’s configuration.
Index	Index of the variable mapped to the TxPDO
Subindex	Sub-Index of the variable mapped to the TxPDO
Varname	Valid name of a global ACSPL+ variable.

Return Value

None

Comments

Once the function is called successfully, the Firmware copies the value of the network input variable into the ACSPL+ variable every controller cycle. The input is from the external EtherCAT master (e.g. TwinCAT) point of view (a value provided by SPiiPlusES to the external master).

There is no restriction on number of the mapped network variables.

The mapping is allowed only when the SPiiPlusES is in OP state.

All types supported by the SPiiPlusES are also supported by the ECEXTIN function.



Error 3309 “Function is supported only by SPiiPlusES” is returned when called on a controller which is not SPiiPlusES.

If SPiiPlusES is not in OP state, error 3310 “SPiiPlusES is not in OP state. PDO is not enabled” error is given.

The mapping is rested by the ECUNMAP command.

COM Library Methods

None

C Library Functions

None

Example

```
D-Buffer:
GLOBAL INT ActualPositionTwinCAT
```



```
Regular Buffer:  
ecextin (0x1A01,0x6064,0, ActualPositionTwinCAT)  
  
STOP
```

4.5.4 ECEXTOUT

Description

The **ECEXTOUT** function is used for mapping output variables (RxPDO) from non-ACS EtherCAT network to the SPiiPlusES.



Can be used only by SPiiPlusES.

Syntax

```
ECEXTOUT (int PDOIndex, int Index, int Subindex, string Varname)
```

Arguments

PDOIndex	One of the RxPDO indexes defined by external master's configuration.
Index	Index of the variable mapped to the TxPDO
Subindex	Sub-Index of the variable mapped to the TxPDO
Varname	Valid name of a global ACSPL+ variable.

Return Value

None

Comments

Once the function is called successfully, the Firmware copies the value of the network output variable into the ACSPL+ variable every controller cycle. The output is from the external EtherCAT master (e.g. TwinCAT) point of view (a value provided to SPiiPlusES by the external master).

There is no restriction on number of the mapped network variables.

Error 3309 "Function is supported only by SPiiPlusES" is returned when called on a controller which is not SPiiPlusES.

If SPiiPlusES is not in OP state, error 3310 "SPiiPlusES is not in OP state. PDO is not enabled" error is given.



The mapping is allowed only when the SPiiPlusES is in OP state.
All types supported by the SPiiPlusES are also supported by the ECEXTOUT function.

The mapping is rested by the ECUNMAP command.

COM Library Methods

None

C Library Functions

None

Example

```
D-Buffer:
GLOBAL INT TargetPositionTwinCAT

Regular Buffer:
ecextout (0x1601,0x607A,0, TargetPositionTwinCAT)

STOP
```

4.5.5 ECGETGRPIND

Description

The function returns an array that contains optional groups' indexes that are part of the current configuration (including mandatory group, which is 0). The array ends with {-1}.

Syntax

ECGETGRPIND[/0]/[1](groups_array)

Arguments

groups_array	Array of type INT, filled with groups' indexes. {-1} marks the end.
--------------	---

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

None.

Example:

```
int groups(5)
ecgetgrpind(groups) ! groups array is filled with: {0,1,-1,0,0},
```

!meaning that there is one optional group (with index 1) defined in the actual system.

4.5.6 ECGETPID

Description

ECGETPID returns product ID of the node.

Syntax

```
ECGETPID[/0 /1] ( index)
```

Arguments

Index	EtherCAT slave index, starting from 0.
-------	--

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

Product ID of the node.

4.5.7 ECGETMAIN

Description

The function returns the number of EtherCAT slaves connected to the main EtherCAT line.

Syntax

```
ECGETMAIN()
```

Return Value

The number of EtherCAT slaves connected to the main EtherCAT line

4.5.8 ECGETOFFSET

Description

The function returns offset of the specified variable of the specific EtherCAT node. The "/b" switch is used to retrieve the offset on the variable in bits.

Syntax

```
Int ECGETOFFSET[/b /0 /1](name, index, InOut(optional), occurrence(optional))
```

Arguments

Name	Name of variable as shown in System Setup of #ETHERCAT command.
Index	EtherCAT node index.

InOut (optional)	Can be Output (=0) or Input (=1). By default, the variable is being searched in Inputs and, if not found, in Outputs.
Occurrence (optional)	By default the value is 0.

Switches

/b	Returns the bit offset of the variable
/0	First EtherCAT network, equivalent to no switch
/1	Second EtherCAT network

Comments

The offset in bits can be calculated by following method:

Offset in Bytes as provided by **#ETHERCAT** report.

The return value of the **ECGETOFFSET/B** will be equal to:

<Offset in Bytes>*8+<Offset in Bits>

Return Value

Offset of the specified variable.

Example

```
GLOBAL INT IN0_OFFSET
! Assuming that IOMnt is the first EtherCAT node in the network
IN0_OFFSET = ECGETOFFSET("Digital Inputs 0", 0)
! Assuming that WAGO is the first EtherCAT node in the network.
! Returns offset of network variable (PDO) "Channel 1 Data", slave 0,
Output, first occurrence
I0=ECGETOFFSET("Channel 1 Data",0)
!Returns offset of network variable (PDO) "Channel 1 Data", slave 0,
Input, first occurrence
I1=ECGETOFFSET("Channel 1 Data",0,1,1)
!Returns offset of network variable (PDO) "Channel 1 Data", slave 0,
Input, second occurrence
```

Example

```
WAGO_Offset_Bit, WAGO_Offset_Byte
WAGO_Offset_Bit =ECGETOFFSET/b ("Input(s).Channel 2, Word 1",0)
WAGO_Offset_Byte = ECGETOFFSET ("Input(s).Channel 2, Word 1",0)
```

Example

```
I0=ecgetoffset/1 ("Digital Inputs 0",1) !Second EtherCAT network, offset  
in !bytes  
I1=ecgetoffset/1b ("Digital Inputs 0",1) !Second EtherCAT network, offset  
!in bits  
  
STOP
```

4.5.9 ECGETOPTGRP**Description**

The function returns number of actually connected optional groups, not including the mandatory group.

Syntax

ECGETOPTGRP [/0][/1] ()

Arguments

None.

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

Returns number of actually connected optional groups, not including the mandatory group.

Example:

```
?ecgetoptgrp()  
2
```

4.5.10 ECGETRED**Description**

The function returns the number of EtherCAT slaves connected to the redundant EtherCAT line.

Syntax

ECGETRED()

Return Value

The number of EtherCAT slaves connected to the redundant EtherCAT line.

4.5.11 ECGETREG**Description**

ESC Error Counters Registers (Beckhoff Memory). The ESCs have numerous error counters that help you detect and locate errors. The ECGETREG function enables you to view the contents of these

registers.

Syntax

```
int ECGETREG[/0/1](index,offset)
```

Arguments

Index	EtherCAT slave index.
Offset	Register offset in the Beckhoff memory.

Switches

/0	First EtherCAT network. Equivalent to no switch
/1	Second EtherCAT network

Return Value

None

Comments

The following table lists supported error counter registers.

Table 4-5. Supported Error Counter Registers

Offset	Name	Description
0x300	Port Error Counter (CRC A)	Error Counted at the Auto-Forwarded (per port). Each register contains two counters: > Invalid Frame Counter: 0x300/2/4/6 > RX Error Counter: 0x301/3/5/7
0x302	Port Error Counter (CRC B)	
0x304	Port Error Counter (CRC C)	
0x306	Port Error Counter (CRC D)	
0x308	Forwarded RX Error Counter (CRC A/B)	Invalid frame with marking from previous ESC detected (per port).
0x309	Forwarded RX Error Counter	

Offset	Name	Description
0x30A	Forwarded RX Error Counter (CRC C/D)	
0x30B	Forwarded RX Error Counter	
0x30C	ECAT Processing Unit Error Counter	Invalid frame passing the EtherCAT Processing Unit (additional checks by processing unit).
0x30D	PDI Error Counter	Physical Errors detected by the PDI.
0x310	Lost Link Counter, Port A (IN)	Link Lost events (per port).
0x311	Lost Link Counter, Port B (OUT)	
0x312	Lost Link Counter, Port C	
0x313	Lost Link Counter, Port D	



If a cable is unplugged, we recommend using the **FCLEAR** command before using **ECGETREG**.

The mapping is allowed only when stack is operational.

Example

Run the following code example in a Program Buffer.

```
I0=ECGETREG (0, 0x310)
STOP
```

You can also enter this code in the SPiiPlus MMI Application Studio Connection Terminal:

```
?ECGETREG (0, 0x310)
```

4.5.12 ECGETREV

Description

The function returns EtherCAT Revision ID of the node.

Syntax

```
ECGETREV ( )
```

Arguments

Index	EtherCAT node index, starting from 0.
-------	---------------------------------------

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

The EtherCAT Revision ID of the node

4.5.13 *ECGETSLAVES*

Description

This function is used to retrieve the number of slaves in an EtherCAT network.

Syntax

```
ECGETSLAVES [/0] [/1] ()
```

Arguments

None

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

Number of EtherCAT slaves in the network.

Comments

If a slave was added or removed, the [ECRESCAN](#) command should be used before using **ECGETSLAVES** again.

4.5.14 *ECGETSN*

Description

ECGETSN returns the EtherCAT Serial Number of the node.

Syntax

```
ECGETSN[/0/1] (index)
```

Arguments

Index	EtherCAT node index, starting from 0.
-------	---------------------------------------

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

Serial Number of the node.

4.5.15 ECGETSTATE

Description

ECGETSTATE returns the state of the node.

Syntax

```
ECGETSTATE [/0/1] (index)
```

Arguments

<i>Index</i>	EtherCAT slave index, starting from 0.
--------------	--

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

The State of the node.

State	Returned Value
INIT	1
PREOP	2
SAFEOP	4
OP	8

4.5.16 ECGETVID

Description

ECGETVID returns the vendor ID of the node.

Syntax

```
ECGETVID [/0] [/1] (index)
```

Arguments

<i>Index</i>	EtherCAT slave index, starting from 0.
--------------	--

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

The ACS vendor ID of the node.

4.5.17 ECGRPINFO**Description**

The function fills array with nodes' indexes which are members of a given optional group. In addition, it returns the number of members in the group.

Syntax

ECGRPINFO[/0[/1](group_index, nodes_array)

Arguments

<i>group_index</i>	Optional Group Index, starting from 0 (0 is the mandatory group)
<i>nodes_array</i>	Array of type INT, filled with nodes' indexes. {-1} marks the end.

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

Returns the number of the members in the specified optional group.

Example:

```
int nodes (5)
I0=ecgrpinfo (1,nodes) ! returns number of nodes in optional group 1
! nodes array is filled with: {0,-1,0,0,0}
```

4.5.18 ECIN**Description**

This function is used to copy the EtherCAT network input variable at the corresponding EtherCAT offset into the specified ACSPL+ variable.

Syntax

ECIN[/b /0 /1](int offset, Varname)

Arguments

offset	Internal EtherCAT offset (in bytes or in bits) of network variable derived from the SPiiPlus MMI Application Studio Communication Terminal <u>ETHERCAT</u> command. > No switch: EtherCAT offset in bytes > "/b" switch: the offset is in bits
Varname	Valid name of ACSPL+ variable, global or standard.

Switches

/b	Offset is in bits, not bytes
/0	First EtherCAT network. Equivalent to no switch
/1	Second EtherCAT network

Return Value

None

Comments

Once the function is called successfully, the Firmware copies the value of the network input variable at the corresponding EtherCAT offset into the specified ACSPL+ variable, every controller cycle.

There is no restriction on number of mapped network variables.



The mapping is allowed only when stack is operational.

In the event of wrong parameters or stack state, the function will produce a corresponding runtime error.



It is recommended to use the **ECGETOFFSET** function to retrieve the offset, in bytes or in bits.

COM Library Methods

None

C Library Functions

acsc_MapEtherCATInput

Example

```
D-Buffer:
GLOBAL INT IOMNT_IN(4)
GLOBAL INT IOMNT_OUT(4)
```

```

Regular Buffer:
AUTOEXEC:

INT IN0_OFFSET
INT IN1_OFFSET
INT IN2_OFFSET
INT IN3_OFFSET

INT OUT0_OFFSET
INT OUT1_OFFSET
INT OUT2_OFFSET
INT OUT3_OFFSET

! Assuming that IOMnt is the first EtherCAT node in the network
IN0_OFFSET = ECGETOFFSET("Digital Inputs 0", 0)
IN1_OFFSET = ECGETOFFSET("Digital Inputs 1", 0)
IN2_OFFSET = ECGETOFFSET("Digital Inputs 2", 0)
IN3_OFFSET = ECGETOFFSET("Digital Inputs 3", 0)

OUT0_OFFSET = ECGETOFFSET("Digital Outputs 0", 0)
OUT1_OFFSET = ECGETOFFSET("Digital Outputs 1", 0)
OUT2_OFFSET = ECGETOFFSET("Digital Outputs 2", 0)
OUT3_OFFSET = ECGETOFFSET("Digital Outputs 3", 0)

ECIN(IN0_OFFSET, IOMNT_IN(0))
ECIN(IN1_OFFSET, IOMNT_IN(1))
ECIN(IN2_OFFSET, IOMNT_IN(2))
ECIN(IN3_OFFSET, IOMNT_IN(3))

ECOUT(OUT0_OFFSET, IOMNT_OUT(0))
ECOUT(OUT1_OFFSET, IOMNT_OUT(1))
ECOUT(OUT2_OFFSET, IOMNT_OUT(2))
ECOUT(OUT3_OFFSET, IOMNT_OUT(3))

STOP

```

4.5.19 ECOUT

Description

This function is used to copy the value of ACSPL+ variable into the network output variable at the corresponding EtherCAT offset.

Syntax

ECOUT[/b /0 /1]/[r] (*int offset, Varname*)

Arguments

<i>offset</i>	Internal EtherCAT offset (in bytes or in bits) of network variable derived from the SPiiPlus MMI Application Studio Communication Terminal <u>ETHERCAT</u> command. > No switch: EtherCAT offset in bytes > "/b" switch: the offset is in bits
<i>Varname</i>	Valid name of ACSPL+ variable, global or standard.

Switches

<i>/r</i>	Copies the EtherCAT network output (RxPDO) variable at the corresponding EtherCAT offset into the specified ACSPL+ variable.
<i>/b</i>	Offset is in bits, not bytes
<i>/0</i>	First EtherCAT network. Equivalent to no switch
<i>/1</i>	Second EtherCAT network

Return Value

None

Comments

Once the function is called successfully, the Firmware copies the value of the specified ACSPL+ variable network input variable into the given EtherCAT offset, every controller cycle.

There is no restriction on number of mapped network variables.



Mapping is allowed only when stack is operational. ACS recommends retrieving the offset using ACSPL+ **ECGETOFFSET()**.

In the event of wrong parameters or stack state, the function will produce corresponding runtime error.

COM Library Methods

None

C Library Functions

acsc_MapEtherCATOutput

Example 1

```
D-Buffer:
GLOBAL INT IOMNT_IN(4)
GLOBAL INT IOMNT_OUT(4)
Regular Buffer:
AUTOEXEC:
```

```

INT IN0_OFFSET
INT IN1_OFFSET
INT IN2_OFFSET
INT IN3_OFFSET
INT OUT0_OFFSET
INT OUT1_OFFSET
INT OUT2_OFFSET
INT OUT3_OFFSET
! Assuming that IOMnt is the first EtherCAT node in the network
IN0_OFFSET = ECGETOFFSET("Digital Inputs 0", 0)
IN1_OFFSET = ECGETOFFSET("Digital Inputs 1", 0)
IN2_OFFSET = ECGETOFFSET("Digital Inputs 2", 0)
IN3_OFFSET = ECGETOFFSET("Digital Inputs 3", 0)
OUT0_OFFSET = ECGETOFFSET("Digital Outputs 0", 0)
OUT1_OFFSET = ECGETOFFSET("Digital Outputs 1", 0)
OUT2_OFFSET = ECGETOFFSET("Digital Outputs 2", 0)
OUT3_OFFSET = ECGETOFFSET("Digital Outputs 3", 0)
ECIN(IN0_OFFSET, IOMNT_IN(0))
ECIN(IN1_OFFSET, IOMNT_IN(1))
ECIN(IN2_OFFSET, IOMNT_IN(2))
ECIN(IN3_OFFSET, IOMNT_IN(3))
ECOUT(OUT0_OFFSET, IOMNT_OUT(0))
ECOUT(OUT1_OFFSET, IOMNT_OUT(1))
ECOUT(OUT2_OFFSET, IOMNT_OUT(2))
ECOUT(OUT3_OFFSET, IOMNT_OUT(3))
STOP

```

Example 2

```

D-Buffer:
GLOBAL INT TargetPosition
Regular Buffer:
!Assuming the first EtherCAT node in the network has "Target Position"
network variable
ecout/r ( ECGETOFFSET("Target Position",0),TargetPosition)
STOP

```

4.5.20 ECREPAIR

Description

ECREPAIR serves to return the system back to the operational state if one or more slaves underwent a reset or power cycle. It provides an ability to recover EtherCAT network when there is a need to replace unit for maintenance without the need to perform commutation, homing, etc., to all other units within the EtherCAT network.

Syntax

ECREPAIR[/0][/1]

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Comments

ECREPAIR performs the following steps:

1. Detects which nodes are not communicating
2. Brings all slaves to the EtherCAT OP state
3. Establishes inter-slaves and master-slaves synchronization
4. Downloads Servo Processor programs to repaired nodes only
5. Restores all Servo Processor variable values accordingly

ECREPAIR can take a long time to complete; therefore it is recommended calling **ECREPAIR** from a Program Buffer. It is possible to execute the **ECREPAIR** command from the SPiiPlus MMI Application Studio Communication Terminal; however, in this case the communication timeout should be configured to be longer.



It is strongly recommended not to take any other actions until **ECREPAIR** completes.

Once the process is complete, the system state can be evaluated through:

- > ECST
- > SYNC values
- > Servo Processor Alarm indication on all axes
- > The **View System Configuration** task of the System Configuration Wizard

If all of these indicators show normal operation status, the **ECREPAIR** operation was successful.

EtherCAT slaves that were operational before **ECREPAIR** activation will keep their feedback and commutation valid.

4.5.21 ECRESCAN

Description

ECRESCAN triggers the system to rescan the EtherCAT network after a slave has been removed or been added in order to refresh the network composition data.

Syntax

```
ECRESCAN[/0] [/1]
```

Arguments

None

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Comments

The command can be entered either through a Program Buffer or via the SPiiPlus MMI Application Studio **Communication Terminal**.

During controller power-up, the controller automatically detects the EtherCAT network change and informs the user, making a call to **ERESCAN** unnecessary.

4.5.22 ECRESCUE**Description**

ECRESCUE triggers execution of a rescue scan of the EtherCAT network. The scan will detect the failure location preventing return of EtherCAT frames to the EtherCAT master.

Syntax

```
ECRESCUE[/0 /1]
```

Arguments

None

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Comments

The command can be entered through a Program Buffer in the SPiiPlus MMI Application Studio.

Procedure when using an application developed by the user:

1. Issue the **ECRESCUE** command to execute EtherCAT network rescue scan
2. Check **ECERR** value
3. In case of **ECERR** = 6024, Issue the **?ECGETMAIN()** and **?ECGETRED()** commands to identify the location of the failure
4. The controller must be rebooted after running the command **ECREPAIR**

In most cases the **ECRESCUE** command shouldn't be explicitly executed by the user. During the controller power up, the controller automatically detects the EtherCAT network failure and informs the user.

4.5.23 ECSAVECFG

Description

The command saves to flash an array of optional groups based on current configuration, based on `ecgetgrpind()` function. For example, actual configuration that contains 1 optional group will be represented as: {0,1,-1}. This array will be read upon power-up of the controller. Actual configuration will be checked against the approved configuration. If not identical, error 6016, "The actual network configuration doesn't match the last approved configuration.", is displayed.

Syntax

ECSAVECFG

Arguments

None

Return Value

None.

Example:

```
ecsavecfg
```

4.5.24 ECSAVEDCNF

Description

The function returns array that contains optional groups' indexes that are part of the last saved configuration (including mandatory group, which is 0). The array ends with {-1}.

Syntax

ECSAVEDCNF[/0]/[1](groups_array)

Arguments

groups_array	Array of type INT, filled with groups' indexes. {-1} marks the end.
---------------------	---

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

None.

Example:

```
int groups(5)
ecsavedcnf(groups) ! groups array is filled with: {0,1,-1,0,0},
>!meaning that there is one optional group (with index 1) in the last
saved configuration.
```

4.5.25 ECUNMAP

Description

This function is used to reset all previous mapping defined by [ECIN](#) and [ECOUT](#).

Syntax

ECUNMAP [/0 /1]

Arguments

None

Switches

/0	First EtherCAT network. Equivalent to no switch
/1	Second EtherCAT network

Return Value

None

Comments

The mapping is allowed only when stack is operational.

COM Library Methods

None

C Library Functions

acsc_UnmapEtherCATInputsOutputs

4.5.26 ECUNMAPIN

Description

This function is used to reset all previous mapping defined by [ECIN](#) to a specific offset.

Syntax

ECUNMAPIN [/0 /1](*EOffset*)

Arguments

<i>EOffset</i>	An integer providing the offset to which a variable was mapped using ECIN .
----------------	---

Return Value

None

Switches

/0	First EtherCAT network. Equivalent to no switch
/1	Second EtherCAT network

Comments

The mapping is allowed only when stack is operational.

Example

Given the previous execution of `ECIN(48,I0)`, `ECUNMAPIN(48)` will unmap only I0.

4.5.27 ECUNMAPOUT

Description

This function is used to reset all previous mapping defined by `ECOUT` to a specific offset.

Syntax

ECUNMAPOUT [/0 /1] (EOffset)

Arguments

<i>EOffset</i>	An integer providing the offset to which a variable has been mapped by <code>ECOUT</code> .
----------------	---

Return Value

None

Switches

/0	First EtherCAT network. Equivalent to no switch
/1	Second EtherCAT network

Example

Assuming previous execution of `ECOUT(48,I0)` and `ECOUT(50,I1)`, executing `ECUNMAPOUT(48)` will unmap only I0.

4.5.28 FOEDOWNLOAD

Description

The function downloads a file over EtherCAT from the controller's flash memory to a slave device.

Syntax

```
FOEDOWNLOAD [/0][/1] [/b]( string LocalPath, string TargetFileName, int SlaveIndex)
```

Arguments

LocalPath	The path of the file in ACS controller's flash
TargetFileName	The name of the file to be transferred to the slave
SlaveIndex	Index of the slave in the EtherCAT network

Switches

/0	First EtherCAT network. Equivalent to no switch.
/1	Second EtherCAT network
/b	With switch "b", the FW will attempt to set the slave to BOOTSTRAP mode before the FoE transfer operation.

Comments

Switch **/b**: even if the slave doesn't support BOOTSTRAP, the firmware will attempt to set it back to OP state without executing the FoE operation.

The function blocks the ACSPL+ buffer until completion.

It cannot be called from the terminal.

The following errors are supported in case of a failure:

- > Error 3317 "FoE Error: Access Denied"
- > Error 3308 "The disk is full or the file is too big"
- > Error 3307 "FoE Protocol is not support by Slave"
- > Error 3040 "Unable to open file"

This function is supported in version 3.00 and higher.



This function is intended for use with EtherCAT slaves that support the FoE protocol.

Example

```
FOEDOWNLOAD/b ("C:\Test.txt","Test",0)
```

4.5.29 FOEUPLOAD

Description

The function reads a file over EtherCAT from a slave and saves it to the controller's flash memory.

Syntax

```
FOEUPLOAD [/0][/1] ( string LocalPath, string TargetFileName, int  
MaxFileSize, int SlaveIndex)
```

Arguments

LocalPath	The path of the file as it will be saved to the ACS controller's flash memory
TargetFileName	The name of the file on the slave

MaxFileSize	The maximum size of the file (in bytes)
SlaveIndex	Index of the slave in the EtherCAT network

Switches

/0	First EtherCAT network. Equivalent to no switch.
/1	Second EtherCAT network

Comments

- > The function blocks the ACSPL+ buffer processing until completion. It cannot be called from the terminal. For complete list of errors, see the FoE Errors table.
- > The function is intended for use with EtherCAT slaves that support the FoE protocol.

Example

```
! Reads file "FileToRead" from slave 0, saves it as "Text.xml"
! Maximum size is 10MB
FOEUPLOAD("Text.xml","FileToRead",10000000,0)
STOP
```

4.5.30 PDOEXT

The PDOEXT command may be used with ACS DS402 products. The command prints the PDO configuration defined by the master that controls the device.

Example

In/Out	ObjectIndex	SubIndex	Size	PdoIndex
In	0x6041	0x00	16	0x1a02
In	0x606c	0x00	32	0x1a02
In	0x6841	0x00	16	0x1a11
In	0x6864	0x00	32	0x1a11
Out	0x6040	0x00	16	0x1602
Out	0x60ff	0x00	32	0x1602
Out	0x6840	0x00	16	0x1611
Out	0x687a	0x00	32	0x1611

4.6 CoE Functions

CoE functions are required for SDO transfers in CoE. SDO are part of the cyclic EtherCAT data transfer. It is impossible to define a generic function for any kind of mailbox transfer, as protocols like EoE, FoE and VoE have their own definitions. So CoE is supported first.



The SPiiPlus MMI Application Studio **Communication Terminal** **ETHERCAT** command reports for every slave if it has Mailbox support.



CoE functions cannot be used with ACS EtherCAT slaves since the CoE protocol is not supported.

The CoE functions are:

Function	Description
COEREAD	Read CoE slave Object Directory entry.
COE2READ	Read CoE slave Object Directory entry in second EtherCAT network.
COEWRITE	Write into CoE slave Object Directory.
COE2WRITE	Write into CoE slave Object Directory in second EtherCAT network.
COEGETSIZE	Returns the size, in bits, of a specific entry in the object dictionary of a specific slave.

4.6.1 COEREAD

Description

This function is used to read Object Dictionary entry from CoE slave.

This function with the “/d” switch provides the capability of reading a double type (64 bit).

In addition, parameter slave with the “-1” value means SPiiPlusES’ Object Dictionary. The following objects can be read from the SPiiPlusES’ Object Dictionary:

- > CiA402 objects, range: 0x6000-0x9FFF

This function with the “/l” (non-capital L) switch is an extension to the existing COEREAD function, and enables reading of objects bigger than 64 bits, e.g. strings.

Syntax

COEREAD/d[/size] (int slave,int Index,int Subindex)

COEREAD/l (int Slave, int Index, int Subindex, int Len, int Array[])

Arguments

size	1/2/4 number of bytes in the Object Dictionary /f for floating (32 bit) or /d for double (64 bit)
slave	Slave number (which can be obtained by running the SPiiPlus MMI Application Studio Communication Terminal ETHERCAT command). -1 means "SPiiPlusES"
Index	Index in the Object Dictionary.
Subindex	Sub-index in the Object Dictionary.
Len	Number of bytes to read if "l" switch is used. The maximum value is 100.
Array	A user-defined one-dimensional integer ACSPL+ array that will store the result value. Each element is treated as a single byte, so size of Array should be at least "Len".

Return Value

COEREAD/d returns the value stored in the variable in the specified **Index** of the Object Dictionary.

COEREAD/l has no return value, the result is found in the **Array** parameter

Comments

If the object doesn't exist, error 3303 "Error SDO: Object doesn't exist in the Object Dictionary" is returned. In case of wrong parameters, the corresponding runtime error will be generated. The function cannot be used in the **Communication Terminal**. The function delays the buffer execution on its line until it is successful or fails the whole buffer with timeout or other error.

COM Library Methods

None

C Library Functions

None

Example 1

```
COEREAD/4 (0,0x6040,0)
```

This reads 4 bytes from **slave** 0, at **Index** 0x6040, **Subindex** 0 and returns the value that is stored in the variable at this **Index**.

Example 2

```
V0=Coeread/d (0,0x2801,1)
```

```
STOP
```

In the example above, 8 bytes are read from slave 0, index 0x2801, subindex 1. The returned value is being stored in the V0 global REAL variable.

Example 3

4.6.2 COE2READ

Description

This function is used to read an Object Dictionary entry from a slave via the CoE protocol, in the second EtherCAT network of a Dual EtherCAT configuration. It is equivalent in meaning and syntax to the **COERead** command, which reads data from the first network.

Syntax

```
COE2READ[/[size][f][d] (int Slave, int Index, int Subindex))  
COE2READ/l (int Slave, int Index, int SubIndex, int Len, int Array[])
```

Arguments

Size suffix	1/2/4 number of bytes in the Object Dictionary, /f for floating (32 bit), /d for double (64 bit) and /l for an array
Slave	Slave number in the second EtherCAT network, can be obtained by running the SPiiPlus MMI Application Studio Communication Terminal ETHERCAT2 command).
Index	Index in the Object Dictionary
Subindex	Sub-index in the Object Dictionary
Len	Number of bytes to read if "l" suffix is used. The maximum value is 100.
Array	A user-defined one-dimensional integer ACSPL+ array that will store the result value. Each element is treated as a single byte, so size of Array should be at least "Len". Used only if "l" suffix is used.

Return Value

COE2READ[/size] returns the value stored in the specified entry in the Object Dictionary of the slave.

COE2READ/l has no return value, the result is stored in the Array parameter.

Comments

If the object doesn't exist, error 3303 "Error SDO: Object doesn't exist in the Object Dictionary" is returned. In case of wrong parameters, the corresponding runtime error will be generated. The function should not be used in the Communication Terminal. The function delays the buffer

execution on its line until it's successful or fails. In case of failure, buffer execution will stop with a timeout or other error.

Com Library Methods

None

C Library Functions

None

Example

```
int arr(20)
coe2read/l (0,0x1008,0,20,arr)      !object 0x1008 is the Device Name
STOP
```

4.6.3 COEWRITE

Description

This function is used to write a value into the CoE slave Object Dictionary.

This function with the "/d" switch is an extension to the existing coeread function, and provides the capability of reading a double type (64 bit).

This function with the "/l" (non-capital L) switch is an extension to the existing COEREAD function, and enables reading of objects bigger than 64 bits, e.g. strings.

Syntax

COEWRITE/d[/size] (int slave,int Index,int Subindex,double Value)
COEREAD/l (int Slave, int Index, int Subindex, int Len, int Array[])

Arguments

<i>size</i>	1, 2 or 4 - the number of bytes in the Object Dictionary /f for floating.(32 bit) or /d for double (64 bit)
<i>slave</i>	Slave number (which can be obtained by running the SPiiPlus MMI Application Studio Communication Terminal ETHERCAT command)
<i>Index</i>	Index in the Object Dictionary.
<i>Subindex</i>	Sub-index in the Object Dictionary.
<i>Value</i>	The value to be written.
<i>Len</i>	Number of bytes to read if "l" switch is used. The maximum value is 100.
<i>Array</i>	A user-defined one-dimensional integer ACSPL+ array that will store the result value. Each element is treated as a single byte, so size of Array should be at least "Len".

Return Value

None

Comments

If the object doesn't exist, error 3303 "Error SDO: Object doesn't exist in the Object Dictionary" is returned. In case of wrong parameters, the corresponding runtime error will be generated. The function cannot be used in the **Communication Terminal**. The function delays the buffer execution on its line until it is successful or fails the whole buffer with timeout or other error.

COM Library Methods

None

C Library Functions

None

Example 1

```
COEWRITE/4 (0,0x6041,0,0x0)
```

This writes the **Value** 0 (4 bytes) into **slave** 0, at **Index** 0x6041, **Subindex** 0.

Example 2

```
Coewrite/d (0,0x2801,1,555.666)  
  
STOP
```

In the example above, the value "555.666" is being written to slave 0, index 0x2801, subindex 1.

Example 3**4.6.4 COE2WRITE****Description**

The **COE2WRITE** function writes a value to an Object Dictionary entry of a slave via the CoE protocol, in the second EtherCAT network of a Dual EtherCAT configuration. It is equivalent in meaning and syntax to the **COEWRITE** command, which writes data to the first network.

Syntax

```
COE2WRITE[/size](int Slave, int Index, int SubIndex, double Value)  
COE2WRITE/1 (int Slave, int Index, int SubIndex, int Len, int Array[])
```

Arguments

Size suffix	1/2/4 number of bytes in the Object Dictionary, /f for floating (32 bit), /d for double (64 bit) and /l for an array
Slave	Slave number (which can be obtained by running the SPiiPlus MMI Application Studio Communication Terminal ETHERCAT2 command).
Index	Index in the Object Dictionary
Subindex	Sub-index in the Object Dictionary
Len	Number of bytes to write if "l" suffix is used. The maximum value is 100.
Array	A user-defined one-dimensional integer ACSPL+ array that stores the data to be written. Each element is treated as a single byte, so size of Array should be at least "Len". Used only if "l" suffix is used.

Return Value

None.

Comments

If the object doesn't exist, error 3303 "Error SDO: Object doesn't exist in the Object Dictionary" is returned. In case of wrong parameters, the corresponding runtime error will be generated. The function should not be used in the Communication Terminal. The function delays the buffer execution on its line until it's successful or fails. In case of failure, buffer execution will stop with a timeout or other error.

Com Library Methods

None

C Library Functions

None

Example

```
int arr(5)
arr(0)=0; arr(1)=1; arr(2)=2; arr(3)=3;arr(4)=4;
coe2write/l (0,0x2022,0,5,arr) !writing to object 0x2022:0
STOP
```

4.6.5 COEGETSIZE**Description**

COEGETSIZE returns the size, in bits, of a specific entry in the object dictionary of a specific slave.

Syntax

int **COEGETSIZE** [/0 /1](*Slave, Index, Subindex*)

Arguments

<i>Slave</i>	An integer representing slave number, starting from 0
<i>Index</i>	An integer representing index in the object dictionary
<i>Subindex</i>	An integer representing sub-index in the object dictionary

Return Value

Size in bits of the entry in the object dictionary.

Switches

<i>/0</i>	First EtherCAT network. Equivalent to no switch.
<i>/1</i>	Second EtherCAT network

Example

```
I0=coegetsize(0,0x1000,0)

!Return value: 32 bits. Object 0x1000 usually means the device type.
```

Comments:

The function returns the received value or fails with runtime error. The function cannot be used in the SPiiPlus MMI Application Studio Communication Terminal. The function delays the buffer execution on its line until it's successful or fails the whole buffer with timeout or other error.

4.7 Modbus Functions

Function	Description
MBOPEN	Establishes a Modbus TCP connection with a Modbus server device using the server's IP address
MBGETHANDLE	Returns the server's communication handle if a connection has been established using the specified IP address
MBCLOSE	Closes an open Modbus TCP connection using the server's communication handle or IP address
MBREADHREG	Maps a Modbus server's holding register to a specified ACSPL+ variable. The value in the register is then read to the variable, either once or at a specified interval.
MBREADIREG	Used to map a Modbus Server's input register to a specified ACSPL+ variable. The value in the register is then read to the variable, either once or at a specified interval.

Function	Description
MBWRITEHREG	The function is used to map a client's ACSPL+ variable to a Modbus server's register. The value in the client variable is then written to the server's register, either once or at a defined interval.
MBREADCOIL	The function is used to map a Modbus server coil to a specified ACSPL+ variable and read the state of the coil
MBWRITECOIL	The function is used to map a Modbus server coil to a specified ACSPL+ variable and write the variable's value to the coil.
MBREADDIN	The function maps a Modbus server discrete input to an ACSPL+ variable
MBUNMAP	Unmaps a specific request using the request ID
MBCLEAR	Clears the error codes from the Modbus requests error array (MBERR) and reactivates the requests that experienced the error condition
MBERR	Holds the most recent Modbus error code that occurred for the request during the communication process
#MBMAPREP	Displays a report of all the active Modbus connections and mapped variables

4.7.1 MBOPEN

Description

MBOPEN establishes a Modbus TCP connection with a Modbus server device using the server's IP address.

Syntax

int **MBOPEN**[/switches] (server_ip[, server_id, word_order])

Arguments

server_ip	The server IP address as a string. For example, "192.168.1.10".
server_id	(Optional) Modbus server ID (1 to 247), the default value is 1

word_order	(Optional) Specifies the order or sequence in which words (registers) are received or sent. To transfer 32-bit (4 byte) or 64-bit (8 byte) values using the Modbus protocol, you must define the order in which the registers are received. (Most vendors choose to map the least significant word onto the lower address of the register pair). > big-endian (0) > little-endian (1) (default)
------------	--

Return Value

On success: A positive integer value that is used as a communication handle for the server

On failure: A runtime error will occur

Comments

- > Up to three servers can be connected to the client controller at any given time.
- > A runtime error will occur if the client fails to open a connection with the server.
- > If a connection with the specified IP address is already open, the function will return the communication handle for the server.
- > To communicate with an ACS server device, a valid **server_id** must be specified (the ACS server device has a **CONID** variable that specifies the server's ID).
- > Some I/O devices might have a Modbus connection timeout; after a period of inactivity in the channel, the connection is closed (this behavior and the timeout period are usually user-defined).

See [Section 7.3, ACSPL+ Runtime Errors](#) for supported error codes

Example 1

This example demonstrates how to open a connection to a Modbus server on controller power-up by using the AUTOEXEC label.

```
AUTOEXEC:
  GLOBAL INT server_handle
  !Opens a Modbus TCP connection with a server device
  server_handle = MBOPEN("10.0.0.100")
STOP
```

Example 2

This example demonstrates how to open a connection to a Modbus server by specifying the server_id and word_order optional arguments.

```
GLOBAL INT server_handle, server_id, word_order
server_id= 10
word_order = 0 !big-endian
!Opens a Modbus TCP connection with a server device
server_handle = MBOPEN("10.0.0.100", server_id, word_order)
STOP
```

Related ACSPL+ Commands

MBCLOSE, MBGETHANDLE

4.7.2 MBGETHANDLE

Description

MBGETHANDLE returns the server's communication handle if a connection has been established using the specified IP address.

Syntax

int **MBGETHANDLE**(IP_address)

Arguments

IP_address	IP address, as a string
------------	-------------------------

Return Value

On success: Server's communication handle

On failure: -1

Example

This example shows how to verify the existence of a connection to a Modbus server device.

```
GLOBAL INT server_handle
!Checks if a connection with the specified IP has been established
server_handle = MBGETHANDLE("10.0.0.100")
IF server_handle <> -1
    DISP "There is an open connection with the specified IP"
END
STOP
```

Related ACSPL+ Commands

MBOPEN, MBCLOSE

4.7.3 MBCLOSE

Description

Closes an open Modbus TCP connection using the server's communication handle or IP address

Syntax

int **MBCLOSE**[/switches](server_communication_handle|server_ip_address)

Arguments

server_communication_ handle	The server's communication handle, as received from the MBOPEN function or The server's IP address (when used with the /s switch)
---------------------------------	--

Switches

/s

Using the function with the /s switch specifies that the server IP address is provided, rather than the handle

Return Value

On success: 0

On failure: A runtime error will occur

Comments



Closing a connection will stop all the active Modbus mapping requests.

See [Section 7.3, ACSPL+ Runtime Errors](#) for supported error codes

Example 1

This example demonstrates how to open and close a connection to a Modbus server device using the device handle.

```
GLOBAL INT server_handle
!Opens a Modbus TCP connection with a server device
server_handle = MBOPEN("10.0.0.100")
MBCLOSE(server_handle) !Closes the connection
STOP
```

Example 2

This example demonstrates how to open and close a connection to a Modbus server device using the IP address.

```
GLOBAL INT server_handle
!Opens a Modbus TCP connection with a server device
server_handle = MBOPEN("10.0.0.100")
MBCLOSE/s("10.0.0.100") !Closes the connection
STOP
```

Related ACSPL+ Commands

MBOPEN, MBGETHANDLE

4.7.4 MBREADHREG

Description

The **MBREADHREG** function maps a Modbus server's holding register to a specified ACSPL+ variable. The value in the register is then read to the variable, either once or at a specified interval.

Syntax

For Scalar variables

MBREADHREG[/switches] server_handle, mapped_scalar, starting_address[, request_frequency]

For Arrays

MBREADHREG/a server_handle, mapped_array, array_index, starting_address[, number_of_elements, request_frequency]

Arguments

server_handle	The server's communication handle (received from the MBOPEN function)
mapped_scalar/mapped_array	A valid ACSPL+ variable name (can be either static or standard, and scalar or array)
array_index (if /a switch is used)	Starting index for the mapping
starting_address	The starting address of the register to be mapped to the specified ACSPL+ variable.
number_of_elements	(Optional) The number of elements that will be mapped (sequentially from the starting address). Element type depends on the switch that has been used. For example, the /f switch specifies that each element is regarded as a single-precision floating-point number (32 bits, 2 registers).
request_frequency	(Optional) Specifies the Modbus request frequency (in milliseconds). The request is sent every "request_frequency" period. Default value: 10 milliseconds Minimum value: 5 milliseconds

Switches

Switch Name	Description
/s	Indicates a one-time read (as opposed to reading the value from the client at the requested frequency). The function copies the value from the server's register to the mapped variable.
/2	The data read should be regarded as integer type (16 bit).
/f	The data read should be regarded as float type (32 bit).
/d	The data read should be regarded as double type (64 bit).
/a	The mapped variable is an array.

Return Value

On success: A unique request ID

On failure: A runtime error will occur

Comments

- > If the mapping request tries to read a location that does not exist on the server, a runtime error will occur (the user will know if they tried to access an invalid address), and the request will not be created.
- > If an error occurs during the Modbus mapping process, the mapping will stop, and the **MBERR**(request_id) will hold the error code. Also, the request will become inactive. To remove or restore a Modbus request, see the **MBUNMAP** and **MBCLEAR** functions.
- > The request ID can be used to check if an error has occurred for the request using the **MBERR**(request_id). Also, it can be used to unmap a specific request. (The request ID is unique per request).
- > If no switch is specified, the data to be mapped is regarded as 32-bit (2 registers) integer value.
- > Each element is converted to the format of the user's mapped variable type. For example, if the user mapped an Integer variable using the /f suffix (each element from the server device is regarded as a single-precision floating-point value), the value of the element is converted into a two's-complement integer representation before it is stored.
- > Loss of precision can occur when converting different data types. For example, reading a real (floating point) value to an integer element will result in truncation, which may cause a loss of precision.
- > A runtime error will occur if the specified mapped_variable length is incompatible with the specified number_of_elements.
- > The maximum number of registers that can be mapped is 32. (32 shorts, 16 integers, 16 floats, or 8 doubles).
- > The maximum number of mapping requests is 32 per server device.
- > If repeated timeout errors occur during the mapping process, it is recommended to use a single-time mapping. Such errors may indicate that the server device is having problems keeping up with the request rate. (This might be required for Festo AG & Co. KG and Beckhoff I/O devices.)

See [Section 7.3, ACSPL+ Runtime Errors](#) for supported error codes

Example 1

This example demonstrates the mapping of 2 holding registers to the client's ACSPL+ global scalar variable, where the value read is treated as an integer number.

```
D-Buffer:
  GLOBAL STATIC INT mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  starting_address = 0           !First holding register address
```

```

server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
request_id = MBREADHREG server_handle, mapped_scalar, starting_address
STOP

```

Example 2

This example demonstrates the mapping of 2 holding registers to the client's ACSPL+ global scalar variable, where the value read is treated as a single-precision floating-point number.

```

D-Buffer:
  GLOBAL STATIC REAL mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  starting_address = 0 !First holding register address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADHREG/f server_handle, mapped_scalar, starting_
address
STOP

```

Example 3

This example demonstrates the mapping of 10 holding registers to the client's ACSPL+ global array, where each element, is treated as a single-precision floating-point number (2 registers).

```

D-Buffer:
  GLOBAL STATIC REAL mapped_array(20)
Buffer 1:
  INT server_handle, starting_address, starting_index, number_of_
elements, request_id
  starting_index = 10
  starting_address = 0 !First holding register address
  number_of_elements = 5
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADHREG/fa server_handle, mapped_array, starting_
index, starting_address, number_of_elements
STOP

```

Example 4

This example demonstrates a one-time mapping of 2 holding registers to the client's ACSPL+ global scalar variable, where the value read is treated as a single-precision floating-point number.

```

D-Buffer:
  GLOBAL STATIC REAL mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  starting_address = 0 !First holding register address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADHREG/fs server_handle, mapped_scalar, starting_
address
STOP

```

Figure 4-2 illustrates the mapping of two Modbus device registers to a 32-bit little-endian variable.

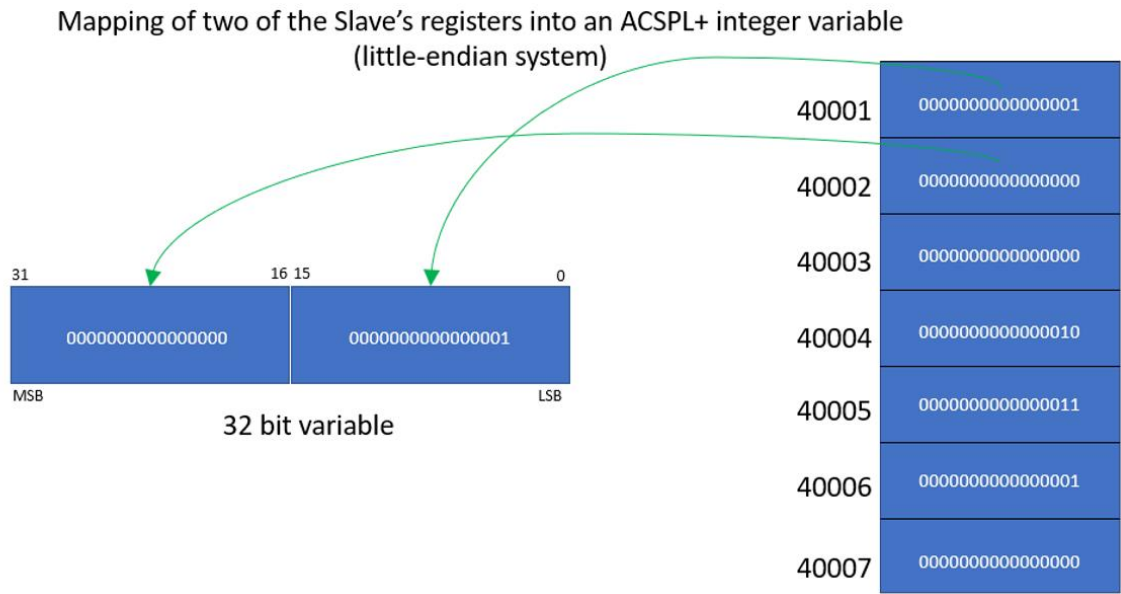


Figure 4-2. Example Mapping

Related ACSPL+ Commands

MBOPEN, MBCLOSE, MBWRITEREG, MBWRITECOIL

4.7.5 MBREADIREG

Description

The **MBREADIREG** function is used to map a Modbus Server's input register to a specified ACSPL+ variable. The value in the register is then read to the variable, either once or at a specified interval.

Syntax

MBREADIREG[/switches] server_handle, mapped_scalar, starting_address[, request_frequency]

MBREADIREG/a server_handle, mapped_array, array_index, starting_address[, number_of_elements, request_frequency]

Arguments

server_handle	The server's communication handle (received from the MBOPEN function)
mapped_scalar / mapped_array	A valid ACSPL+ variable name (can be either static or standard, and scalar or array)
array_index (if the /a switch is used)	Starting index for the mapping
starting_address	The starting address of the register to be mapped to the specified ACSPL+ variable.

number_of_elements	(Optional) The number of elements that will be mapped (sequentially from the starting address). The element type depends on the switch that is used. For example, the /f switch specifies that each element is regarded as a single-precision floating-point number (32 bits, 2 registers).
request_frequency	(Optional) The Modbus request frequency (in milliseconds). The request is sent every "request_frequency" period. Default value: 10 milliseconds Minimum value: 5 milliseconds

Switches

Name	Description
/s	Indicates a one-time read. The function copies the value from the server's register to the mapped variable.
/2	The data read should be interpreted as integer type (16 bit).
/f	The data read should be interpreted as float type (32 bit).
/d	The data read should be interpreted as double type (64 bit).
/a	The mapped variable is an array.

Return Value

On success: A unique request ID

On failure: A runtime error will occur

Comments

- > If the mapping request tries to read a location that does not exist on the server, a runtime error will occur (the user will know if they tried to access an invalid address), and the request will not be created.
- > If an error occurs during the Modbus mapping process, the mapping will stop, and the **MBERR**(request_id) will hold the error code. Also, the request will become inactive. To remove or restore a Modbus request, see the **MBUNMAP** and **MBCLEAR** functions.
- > The request ID can be used to check if an error has occurred for the request using the **MBERR**(request_id). Also, it can be used to unmap a specific request. (The request ID is unique per request).
- > If no switch is specified, the data to be mapped is regarded as 32-bit (2 registers) integer value.
- > Each element is converted to the format of the user's mapped variable type. For example, if the user mapped an Integer variable using the /f suffix (each element from the server

device is regarded as a single-precision floating-point value), the value of the element is converted into a two's-complement integer representation before it is stored.

- > Loss of precision can occur when converting different data types. For example, reading a real (floating point) value to an integer element will result in truncation, which may cause a loss of precision.
- > A runtime error will occur if the specified mapped_variable length is incompatible with the specified number_of_elements.
- > The maximum number of registers that can be mapped is 32. (32 shorts, 16 integers, 16 floats, or 8 doubles).
- > The maximum number of mapping requests is 32 per server device.
- > If repeated timeout errors occur during the mapping process, it is recommended to use a single-time mapping. Such errors may indicate that the server device is having problems keeping up with the request rate. (This might be required for Festo AG & Co. KG and Beckhoff I/O devices.)

See [Section 7.3, ACSPL+ Runtime Errors](#) for supported error codes

Example 1

This example demonstrates the mapping of 2 input registers to the client's ACSPL+ global scalar variable, where the value read is interpreted as an integer number.

```
D-Buffer:
  GLOBAL STATIC INT mapped_scalar
Buffer 1:
  INT server_handle, starting_address
  starting_address = 0           !First input register address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADIREG server_handle, mapped_scalar, starting_address
STOP
```

Example 2

This example demonstrates the mapping of 2 input registers to the client's ACSPL+ global scalar variable, where the value read is interpreted as a single-precision floating-point number.

```
D-Buffer:
  GLOBAL STATIC REAL mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  starting_address = 0           !First input register address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADIREG/f server_handle, mapped_scalar, starting_
address
STOP
```

Example 3

This example demonstrates the mapping of 10 input registers to the client's ACSPL+ global array, where each element is interpreted as a single-precision floating-point number (2 registers).

```
D-Buffer:
  GLOBAL STATIC REAL mapped_array(20)
Buffer 1:
  INT server_handle, starting_address, starting_index, number_of_
elements, request_id
  starting_index = 10
  starting_address = 0           !First input register address
  number_of_elements = 5
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADIREG/fa server_handle, mapped_array, starting_
index, starting_address, number_of_elements
STOP
```

Example 4

This example demonstrates a one-time mapping of 2 input registers to the client's ACSPL+ global scalar variable, where the value read is interpreted as a single-precision floating-point number.

```
D-Buffer:
  GLOBAL STATIC REAL mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  starting_address = 0 !First input register address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADIREG/fs server_handle, mapped_scalar, starting_
address
STOP
```

Figure 4-3 illustrates the mapping of two Modbus device registers to a 32-bit little-endian variable.

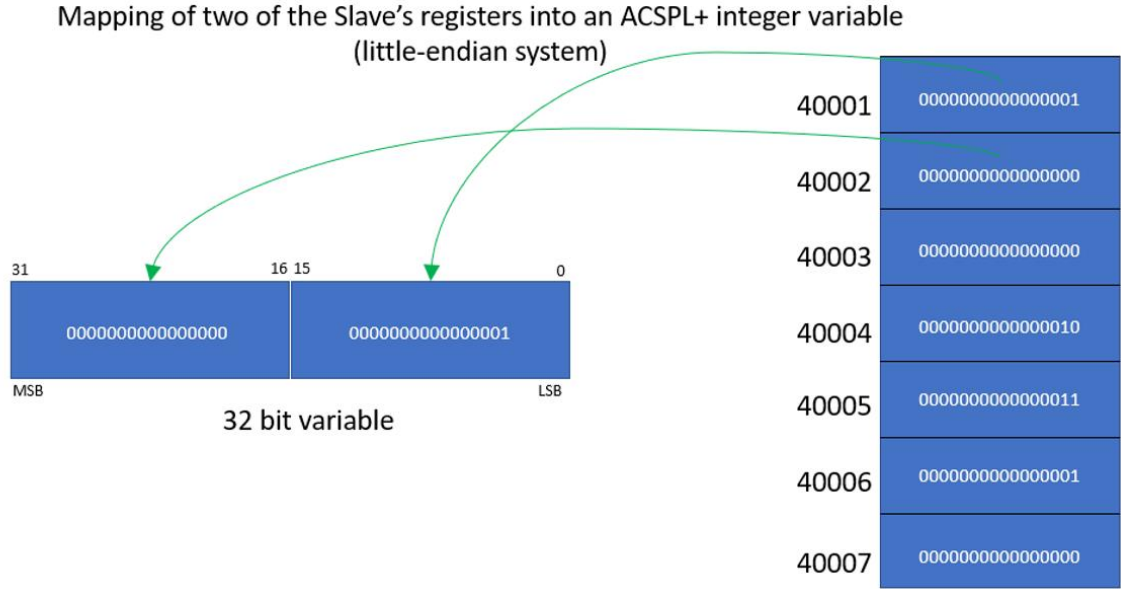


Figure 4-3. Example Mapping

4.7.6 MBWRITEHREG

Description

The **MBREADIREG** function is used to map a client's ACSPL+ variable to a Modbus server's register. The value in the client variable is then written to the server's register, either once or at a defined interval.

Syntax

MBWRITEHREG[/switches] server_handle, mapped_scalar, starting_address[, request_frequency]

MBWRITEHREG/a server_handle, mapped_array, array_index, starting_address[, number_of_elements, request_frequency]

Arguments

server_handle	The server's communication handle (received from the MBOPEN function).
mapped_scalar / mapped_array	A valid ACSPL+ variable name (may be either static or standard, and scalar or array).
array_index (if the /a switch is used)	The array index from which mapping will begin.
starting_address	The starting address of the register to be mapped to the specified ACSPL+ variable.
number_of_elements	(Optional) The number of elements to be mapped (sequentially from the starting address). The element type depends on the switch used. For example, the /f suffix specifies that each element is interpreted as a single-precision floating-point number (32 bits, 2 registers).
request_frequency	The Modbus request frequency (in milliseconds). The request is sent every "request_frequency" period. Default value: 10 milliseconds Minimum value: 5 milliseconds

Switches

/s	Generates a single write request
/2	The data should be written as a 16-bit integer type
/f	The data should be written as a 32-bit float type
/d	The data should be written as a 64-bit double type
/a	The mapped variable is an array

Return Value

On success: A unique request ID

On failure: A runtime error will occur

Comments

- > If the mapping request tries to write a location that does not exist on the server, a runtime error will occur (the user will know if they tried to access an invalid address), and the request will not be created.
- > If an error occurs during the Modbus mapping process, the mapping will stop, and the **MBERR**(request_id) will hold the error code. Also, the request will become inactive. To remove or restore a Modbus request, see the **MBUNMAP** and **MBCLEAR** functions.
- > The request ID can be used to check if an error has occurred for the request using the **MBERR**(request_id). Also, it can be used to unmap a specific request. (The request ID is unique per request).
- > If no switch is specified, the data to be mapped is regarded as 32-bit (2 registers) integer value.
- > Each element is converted to the format of the user's mapped variable type. For example, if the user mapped an Integer variable using the /f suffix (each element from the server device is regarded as a single-precision floating-point value), the value of the element is converted into a two's-complement integer representation before it is stored.
- > Loss of precision can occur when converting different data types. For example, reading a real (floating point) value to an integer element will result in truncation, which may cause a loss of precision.
- > A runtime error will occur if the specified mapped_variable length is incompatible with the specified number_of_elements.
- > The maximum number of registers that can be mapped is 32. (32 shorts, 16 integers, 16 floats, or 8 doubles).
- > The maximum number of mapping requests is 32 per server device.
- > If repeated timeout errors occur during the mapping process, it is recommended to use a single-time mapping. Such errors may indicate that the server device is having problems keeping up with the request rate. (This might be required for Festo AG & Co. KG and Beckhoff I/O devices.)

See [Section 7.3, ACSPL+ Runtime Errors](#) for supported error codes

Example 1

This example demonstrates the mapping of 2 input registers to the client's ACSPL+ global scalar variable, where the mapped variable value is interpreted as an integer value.

```
D-Buffer:
    GLOBAL STATIC INT mapped_scalar
Buffer 1:
    INT server_handle, starting_address, request_id
    mapped_scalar = 100
```

```

starting_address = 0 !First input register address
server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
request_id = MBWRITEHREG server_handle, mapped_scalar, starting_
address
STOP

```

Example 2

This example demonstrates the mapping of 2 input registers to the client's ACSPL+ global scalar variable, where the mapped variable value is interpreted as a single-precision floating-point number.

```

D-Buffer:
  GLOBAL STATIC REAL mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  mapped_scalar = 100.25
  starting_address = 0 !First input register address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBWRITEHREG/f server_handle, mapped_scalar, starting_
address
STOP

```

Example 3

This example demonstrates the mapping of 10 input registers to the client's ACSPL+ global array, where each element is interpreted as a single-precision floating-point number (2 registers).

```

D-Buffer:
  GLOBAL STATIC REAL mapped_array(20)
Buffer 1:
  INT server_handle, starting_address, starting_index, number_of_
elements, request_id
  starting_index = 10
  starting_address = 0 !First input register address
  number_of_elements = 5
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBWRITEHREG/fa server_handle, mapped_array, starting_
index, starting_address, number_of_elements
STOP

```

Example 4

This example demonstrates a one-time mapping of 2 input registers to the client's ACSPL+ global scalar variable, where the mapped variable value is interpreted as a single-precision floating-point number.

```

D-Buffer:
  GLOBAL STATIC REAL mapped_scalar
Buffer 1:

```

```
INT server_handle, starting_address, request_id
mapped_scalar = 1.1
starting_address = 0 !First input register address
server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
request_id = MBWRITEHREG/fs server_handle, mapped_scalar, starting_
address
STOP
```

Figure 4-4 illustrates the mapping of a 32-bit little-endian variable to two Modbus device registers.

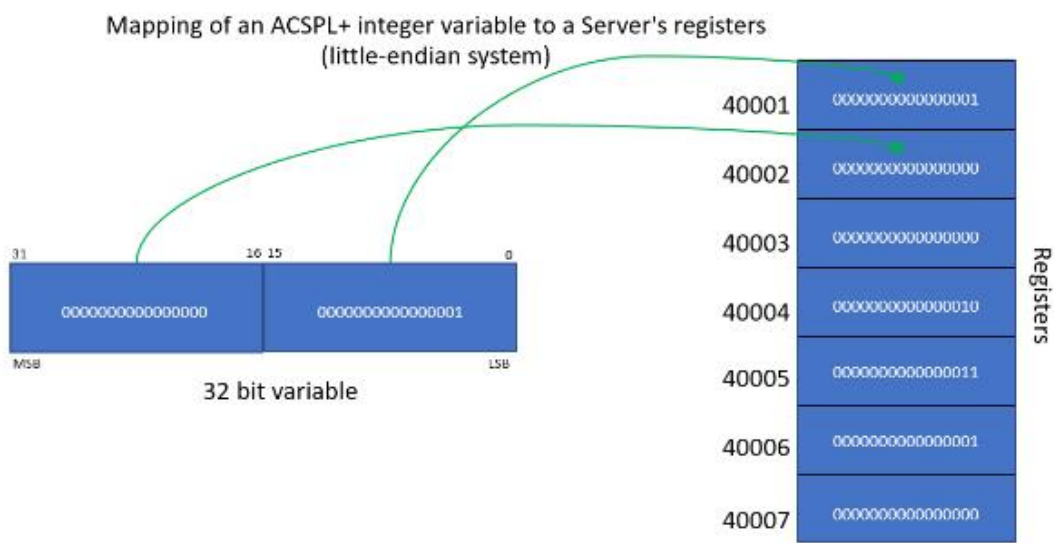


Figure 4-4. Example Mapping

4.7.7 MBREADCOIL

Description

The **MBREADCOIL** function is used to map a Modbus server coil to a specified ACSPL+ variable and read the state of the coil.

Syntax

MBREADCOIL[/switches] server_handle, mapped_scalar, starting_address[, number_of_coils, request_frequency]

MBREADCOIL/a server_handle, mapped_array, array_index, starting_address[, number_of_coils, request_frequency]

Arguments

server_handle	The server’s communication handle (received from the MBOPEN function)
mapped_scalar / mapped_array	A valid ACSPL+ variable name (may be either static or standard, and scalar or array)

array_index (if the /a switch is used)	The array index from which mapping will begin.
starting_address	The starting address of the coils to be mapped to the specified ACSPL+ variable.
number_of_coils	(Optional) The number of coils to map (sequentially from starting_address)
request_frequency	(Optional) The Modbus request frequency (in milliseconds). The request is sent every "request_frequency" period. Default value: 10 milliseconds Minimum value: 5 milliseconds

Switches

/s	A one-time read. Copies the value from specified coils to mapped_scalar .
/a	The mapped variable is an array
/b	Bitwise mapping. Coils are mapped to bits of the specified variable rather than to successive elements of an array.

Return Value

On success: A unique request ID

On failure: A runtime error will occur

Comments

- > If the mapping request tries to read to or write from a location that does not exist on the server, a runtime error will occur (the user will know if they tried to access an invalid address), and the request will not be created.
- > If an error occurs during the Modbus mapping process, the mapping will stop and **MBERR** (request_id) will hold the error code. Also, the request will become inactive. To remove or restore a Modbus request, see the **MBUNMAP** and **MBCLEAR** functions.
- > The request ID can be used to check if an error has occurred for the request using **MBERR** (request_id). The request ID can also be used for unmapping a specific request. (The request ID is unique per request.)
- > A runtime error will occur if the specified mapped_scalar length is incompatible with the specified number_of_coils.
- > The maximum number of coils that can be mapped is 32.
- > The maximum number of mapping requests is 32 per server device.
- > If repeated timeout errors occur during the mapping process, it is recommended to use a single-time mapping. Such errors may indicate that the Server device is having problems

keeping up with the request rate. (This might be required for Festo AG & Co. KG and Beckhoff I/O devices.)

See [Section 7.3, ACSPL+ Runtime Errors](#) for supported error codes

Example 1

This example demonstrates mapping a server coil to the client's ACSPL+ global variable.

```
D-Buffer:
  GLOBAL STATIC INT mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  starting_address = 1           !First Coil address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADCOIL server_handle, mapped_scalar, starting_address
STOP
```

Example 2

This example demonstrates mapping three server coils to the client's ACSPL+ global array.

```
D-Buffer:
  GLOBAL STATIC INT mapped_array(5)
Buffer 1:
  INT server_handle, starting_address, number_of_coils
  INT array_index, request_id
  array_index = 2
  number_of_coils = 3
  starting_address = 1           !First Coil address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADCOIL/a server_handle, mapped_array, array_index,
starting_address, number_of_coils
STOP
```

Example 3

This example demonstrates bitwise mapping of 20 server coils to the client's ACSPL+ global variable.

```
D-Buffer:
  GLOBAL STATIC INT mapped_scalar
Buffer 1:
  INT server_handle, starting_address, number_of_coils, request_id
  number_of_coils = 20
  starting_address = 1           !First Coil address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADCOIL/b server_handle, mapped_scalar, starting_
address, number_of_coils
STOP
```

Example 4

This example demonstrates a one-time mapping from a server coil to the client's ACSPL+ global variable.

```
D-Buffer:
  GLOBAL STATIC INT mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  starting_address = 1           !First Coil address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADCOIL/s server_handle, mapped_scalar, starting_
address
STOP
```

Figure 4-5 illustrates the mapping of a server coil to an integer variable.

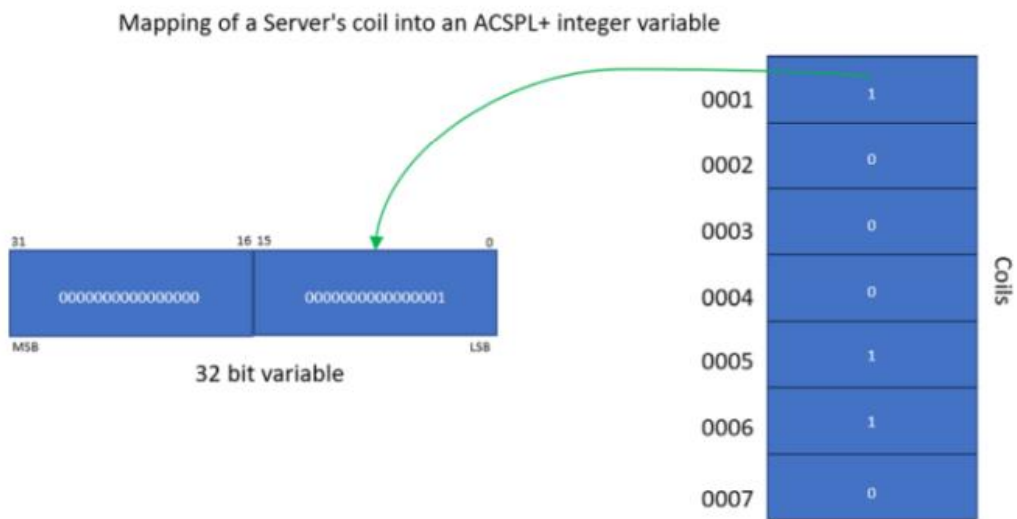


Figure 4-5. Example Mapping

4.7.8 MBWRITECOIL

Description

The **MBWRITECOIL** function is used to map a Modbus server coil to a specified ACSPL+ variable and write the variable's value to the coil.

Syntax

MBWRITECOIL[/switches] server_handle, mapped_scalar, starting_address[, number_of_coils, request_frequency]

MBWRITECOIL/a server_handle, mapped_array, array_index, starting_address[, number_of_coils, request_frequency]

Arguments

server_handle	The server's communication handle (received from the MBOPEN function)
mapped_scalar / mapped_array	A valid ACSPL+ variable name (can be either static or standard, and scalar or array)
array_index (if the /a switch is used)	The array index from which mapping will begin.
starting_address	The starting address of the coils to be mapped to the specified ACSPL+ variable.
number_of_coils	(Optional) The number of coils to map (sequentially from starting_address)
request_frequency	(Optional) The Modbus request frequency (in milliseconds). The request is sent every "request_frequency" period. Default value: 10 milliseconds Minimum value: 5 milliseconds

Switches

/s	A one-time write. Copies the value from specified coils to mapped_scalar .
/a	The mapped variable is an array
/b	Bitwise mapping. Coils are mapped to bits of the specified variable rather than to successive elements of an array.

Return Value

On success: A unique request ID

On failure: A runtime error will occur

Comments

- > If the mapping request tries to read from or write to a location that does not exist on the server, a runtime error will occur (the user will know if they tried to access an invalid address), and the request will not be created.
- > If an error has occurred during the mapping process, the mapping for the request will stop and MBERR(request_id) will hold the error code that occurred. Also, the request will become inactive. To remove or restore a Modbus request, see the **MBUNMAP** and **MBCLEAR** functions.
- > The request ID can be used to check if an error has occurred for the request using MBERR(request_id). Also, it can be used for unmapping a specific request. (The request ID is unique per request.)

- > A runtime error will occur if the specified mapped_scalar length is incompatible with the specified number_of_coils.
- > The maximum number of coils that can be mapped is 32.
- > The maximum number of mapping requests is 32 per server device.
- > If repeated timeout errors occur during the mapping process, it is recommended to use a single-time mapping. Such errors may indicate that the Server device is having problems keeping up with the request rate. (This might be required for Festo AG & Co. KG and Beckhoff I/O devices.)

See [Section 7.3, ACSPL+ Runtime Errors](#) for supported error codes

Example 1

This example demonstrates mapping a server coil to the client's ACSPL+ global variable.

```
D-Buffer:
  GLOBAL STATIC INT mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  mapped_scalar = 1
  starting_address = 1           !First Coil address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBWRITECOIL server_handle, mapped_scalar, starting_
address
STOP
```

Example 2

This example demonstrates mapping three server coils to the client's ACSPL+ global array.

```
D-Buffer:
  GLOBAL STATIC INT mapped_array(5)
Buffer 1:
  INT server_handle, starting_address, number_of_coils
  INT array_index, request_id
  number_of_coils = 3
  array_index = 2
  starting_address = 1           !First Coil address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBWRITECOIL/a server_handle, mapped_array, array_index,
starting_address, number_of_coils
STOP
```

Example 3

This example demonstrates bitwise mapping of 20 server coils to the client's ACSPL+ global variable.

```
D-Buffer:
  GLOBAL STATIC INT mapped_scalar
Buffer 1:
```



```

INT server_handle, starting_address, number_of_coils, request_id
number_of_coils = 20
starting_address = 1           !First Coil address
server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
request_id = MBWRITECOIL/b server_handle, mapped_scalar, starting_
address, number_of_coils
STOP

```

Example 4

This example demonstrates a one-time mapping from a server coil to the client's ACSPL+ global variable.

```

D-Buffer:
  GLOBAL STATIC INT mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  starting_address = 1           !First Coil address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBWRITECOIL/s server_handle, mapped_scalar, starting_
address
STOP

```

Figure 4-6 illustrates the mapping of an integer variable to a server coil.

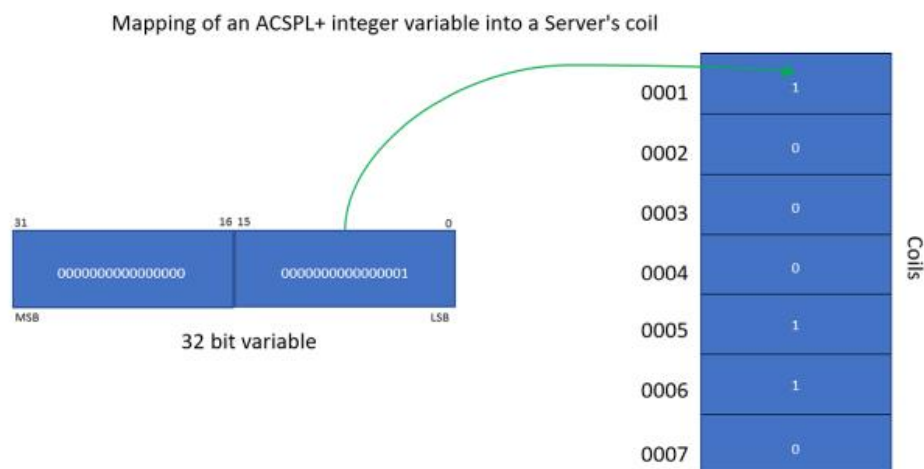


Figure 4-6. Example Mapping

4.7.9 MBREADDIN

Description

The **MBREADDIN** function maps a Modbus server discrete input to an ACSPL+ variable.

Syntax

MBREADDIN[/switches] server_handle, mapped_scalar, starting_address[, number_of_discrete_inputs, request_frequency]

MBREADDIN/a server_handle, mapped_array, array_index, starting_address[, number_of_discrete_inputs, request_frequency]

Arguments

server_handle	The server's communication handle (received from the MBOPEN function)
mapped_scalar / mapped_array	A valid ACSPL+ variable name (can be either static or standard, and scalar or array)
array_index (if the /a switch is used)	The array index from which mapping will begin.
starting_address	The starting address of the discrete inputs to be mapped to the specified ACSPL+ variable.
number_of_discrete_inputs	(Optional) The number of discrete inputs to map (sequentially from starting_address)
request_frequency	(Optional) The Modbus request frequency (in milliseconds). The request is sent every "request_frequency" period. Default value: 10 milliseconds Minimum value: 5 milliseconds

Switches

/s	A one-time read. Copies the value from the specified discrete input to mapped_scalar .
/a	The mapped variable is an array
/b	Bitwise mapping. Discrete inputs are mapped to bits of the specified variable rather than to successive elements of an array.

Return Value

On success: A unique request ID

On failure: A runtime error will occur

Comments

- > If the mapping request tries to read to or write from a location that does not exist on the server, a runtime error will occur (the user will know if they tried to access an invalid address), and the request will not be created.

- > If an error has occurred during the mapping process, the mapping for the request will stop, and **MBERR**(request_id) will hold the error code that occurred. Also, the request will become inactive. To remove or restore a Modbus request, see the **MBUNMAP** and **MBCLEAR** functions.
- > The request ID can be used to check if an error has occurred for the request using **MBERR**(request_id). Also, it can be used for unmapping a specific request. (The request ID is unique per request.)
- > A runtime error will occur if the specified mapped_scalar length is incompatible with the specified number_of_discrete_inputs.
- > The maximum number of discrete inputs that can be mapped is 32.
- > The maximum number of mapping requests is 32 per server device.
- > If repeated timeout errors occur during the mapping process, it is recommended to use a single-time mapping. Such errors may indicate that the Server device is having problems keeping up with the request rate. (This might be required for Festo AG & Co. KG and Beckhoff I/O devices.)

See [Section 7.3, ACSPL+ Runtime Errors](#) for supported error codes

Example 1

This example demonstrates mapping a server's discrete input to the client's ACSPL+ global variable.

```
D-Buffer:
  GLOBAL STATIC INT mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  starting_address = 1           !First discrete input address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADDIN server_handle, mapped_scalar, starting_address
STOP
```

Example 2

This example demonstrates mapping three of the server's discrete inputs to the client's ACSPL+ global array.

```
D-Buffer:
  GLOBAL STATIC INT mapped_array(5)
Buffer 1:
  INT server_handle, starting_address, number_of_coils
  INT array_index, request_id
  array_index = 2
  number_of_discrete_inputs = 3
  starting_address = 1           !First discrete input address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADDIN/a server_handle, mapped_array, array_index,
starting_address, number_of_discrete_inputs
STOP
```

Example 3:

This example demonstrates bitwise mapping of 20 of the server's discrete inputs to the client's ACSPL+ global variable.

```
D-Buffer:
  GLOBAL STATIC INT mapped_scalar
Buffer 1:
  INT server_handle, starting_address, number_of_discrete_inputs,
  request_id
  number_of_discrete_inputs = 20
  starting_address = 1           !First discrete input address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADDIN/b server_handle, mapped_scalar, starting_
  address, number_of_discrete_inputs
  STOP
```

Example 4

This example demonstrates a one-time mapping from a server's discrete input to the client's ACSPL+ global variable.

```
D-Buffer:
  GLOBAL STATIC INT mapped_scalar
Buffer 1:
  INT server_handle, starting_address, request_id
  starting_address = 1           !First discrete input address
  server_handle = MBOPEN("10.0.0.100") !Opens a Modbus connection
  request_id = MBREADDIN/s server_handle, mapped_scalar, starting_
  address
  STOP
```

4.7.10 MBUNMAP

Description

The **MBUNMAP** function unmaps a specific request using the request ID.

Syntax

MBUNMAP [request_id]

Arguments

request_id	(Optional) A request ID received from the MBREAD or MBWRITE function
-------------------	---

Return Value

The number of requests that have been unmapped (0 or 1).

Comments

- > Unmapping a request will also clear its error code(if such exists) from the MBERR array.
- > If no request ID is specified, all active Modbus requests will be removed.

See [Section 7.3, ACSPL+ Runtime Errors](#) for supported error codes

Example 1

This example demonstrates the unmapping of a specific Modbus request using the request ID.

```
MBUNMAP request_id
STOP
```

Example 2

This example demonstrates the unmapping of all active Modbus requests.

```
MBUNMAP
STOP
```

4.7.11 MBCLEAR

Description

The **MBCLEAR** function clears the error codes from the Modbus requests error array (**MBERR**) and reactivates the requests that experienced the error condition.

Syntax

MBCLEAR [request_id]

Arguments

request_id	(Optional) A request ID received from the MBREAD or MBWRITE function.
-------------------	--

Return Value

None

Comments

If **request_id** is not specified, the entire Modbus requests error array(MBERR) will be cleared (set to 0) and all request that experienced errors will be reactivated.

See [Section 7.3, ACSPL+ Runtime Errors](#) for supported error codes

Example 1

This example demonstrates how to clear a Modbus error and reactivate the relevant request.

```
MBCLEAR request_id
STOP
```

Example 2

This example demonstrates how to clear all Modbus errors and reactivate all Modbus requests.

```
MBCLEAR
STOP
```

4.7.12 MBERR

Description

MBERR is an integer array with one element for each possible Modbus request ID (between 0 - 95). It holds the most recent Modbus error code that occurred for the request during the communication process.

Table 4-6. Modbus Error Codes

Error Code	Description
0	No Error
8001	Illegal Function - Function code received in the query is not recognized or allowed by the server.
8002	Illegal Data Address - Data addresses of some or all the required entities are not allowed or do not exist in the server.
8003	Illegal Data Value - A value contained in the query data field is not an allowable value for the server.
8004	Server Device Failure - An unrecoverable error occurred while the server was attempting to perform the requested action.
8005	Acknowledge - Server has accepted the request and is processing it, but a long duration of time is required. This response is returned to prevent a timeout error from occurring in the client.
8006	Server Device Busy - Server is engaged in processing a long-duration command. The client should retry later.
8012	Transaction Identifier Mismatch – The received response transaction identifier did not match the expected value.
8013	Response Length Mismatch - The received response length did not match the expected length.
8014	Response Out of Bounds – The received response value is invalid. This may be due to the value being out of the allowed range, an attempt to write to a protected variable, etc.
8015	Timeout – No response has been received from the server device for the timeout period(5 seconds). The connection with the server device has been terminated.
8016	Connection Closed - The connection with the server device has been closed.

Comments

MBERR can be cleared by unmapping the erroneous request using the **MBUNMAP** function or by activating the erroneous request using the **MBCLEAR** function.

Example

This example demonstrates monitoring the error state of a Modbus mapping request.

```
D-buffer:
    GLOBAL INT request_id
Buffer 1:
    request_id = MBWRITEHREG server_handle, mapped_scalar, starting_
address

ON MBERR(request_id)
DISP "An error has occurred for the request, request ID=",request_id
!Here we decide how we would like to handle this error, un-mapping
!the request, closing the communication channel, or clearing the
!fault and restore the request to an active state
.
.
.
MBCLEAR request_id !For example, clearing the error of the request,
!and restoring it to an active state
RET
STOP
```

Tag

403

Accessibility

Read-Only

4.7.13 #MBMAPREP

Description

The **#MBMAPREP** command displays a report of all the active Modbus connections and mapped variables.

Syntax

```
#MBMAPREP
```

Example

```
#MBMAPREP
      Modbus Server Address:10.0.0.103 (channel:18)
      =====
Server ID
=====
1

Endianness
=====
Little-Endian

Mapped Variables
=====
None

      Modbus Server Address:10.0.0.31 (channel:19)
      =====
Server ID
=====
1

Endianness
=====
Little-Endian

Mapped Variables
=====
1)Variable Name:bit_status | Type:Integer
   Mapping Type:Write Coils
   Starting Address:0
   Number of Coils:1
   Bit Mapping:False
   Request Frequency:5.000000[milliseconds]
   Request State:Sleeping
   Request ID:64
   Request Error Code:0
```

4.8 Servo Processor Functions

SPiiPlus™, a proprietary ACS Motion Control Servo Processor (SP), executes the real-time tasks such as implementation of the real time control algorithms. Each SPiiPlus can control from two to eight axes (depending on the product). The SPiiPlus includes all the necessary peripherals that are needed

for a high performance axis control, such as encoder counters, Digital-to-Analog interface, smart inputs and outputs.

Servo Processor functions are used to read and monitor SP values.



The number of Servo Processors is model-dependent. Check the controller Hardware Guide for the specific SPiiPlus model.

The Servo Processor functions are:

Function	Description
GETSP	Reads a value from the specified SP address
GETSPA	Retrieves address of the SP variable specified by name.
GETSPV	Reads a value from the specified SP variable name
SETSP	Sets a value for the specified SP address
SETSPV	Sets a value for the specified SP variable name

4.8.1 [GETSP](#)

Description

GETSP reads a value from the specified SP address.

The read value is treated as a signed integer number.

The minimal delay for the function response is more than three MPU cycles.

Syntax

GETSP [/0]/[1] (*int SP, int Address*)

Arguments

<i>SP</i>	The SP number in the system. Use <code>getconf(260, axis)</code> , where <code>axis</code> is axis in the system to get the SP number.
<i>Address</i>	Address within the SP. Get the address by using <code>getspa(SPnumber, varname)</code> command.



For **Address**, use the memory address as obtained by [GETSPA](#).

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

Value read from the specified SP address.

Error Conditions

The function causes an error if an SP number is specified other than 0–3, or if an illegal address is specified.

Example

```
REAL PAR_ADDRESS
PAR_ADDRESS=GETSPA(0,"PE(0)")    ! Get the memory address of PE(0) in SP#0
GETSP(0, PAR_ADDRESS)           ! The return value is the PE(0) in SP#0
```

4.8.2 GETSPA

Description

GETSP reads a value from the specified SP address.

The read value is treated as a signed integer number.

The minimal delay for the function response is for or more MPU cycles, depending on the product.

Syntax

```
GETSPA [/0][/1](int SP, int Address)
```

Arguments

<i>SP_number</i>	The SP number in the system.
<i>SP_variable</i>	String representing the name of an SP variable. In order to avoid unexpected results, place SP_variable in quotes.

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Return Value

Address of the variable in the SP memory, or -1 if the variable does not exist. The return value can be used in **GETSP** and **SETSP**.

Example

```
XX= GETSPA(0,"axes[0].PE")      ! Retrieve the address Position error
                                ! variable of axis 0 in SP 0.
```

4.8.3 GETSPV



This function is obsolete and is replaced by [GETSP](#).

4.8.4 SETSP



The **SETSP** function is for advanced users only. Misuse of this function may damage the drive.

Description

SETSP writes a value to the specified SP address.

Syntax

SETSP [/0][/1] (*int SP_number, int Address, value*)

Arguments

<i>SP_number</i>	The SP number in the system.
<i>Address</i>	Address within the SP. Look up addresses by using getspa function
<i>value</i>	The value to write to the address.

Switches

/0	First EtherCAT network (equivalent to no switch)
/1	Second EtherCAT network

Error Conditions

Error 3126, Illegal SP number.

Example

```
REAL PAR_ADDRESS
PAR_ADDRESS=GETSPA(0,"axes[0].params[0].SLVKP")      !Gets the memory
address of axes[0].params[0].SLVKP
SETSP(0, PAR_ADDRESS, 1000)                          !Changes the value of the axes
[0].params[0].SLVKP in SP 0 to 1000.
```

4.8.5 SETSPV



This function is obsolete and is replaced by [SETSP](#).

4.9 Signal Processing Functions

The Signal Processing functions are:

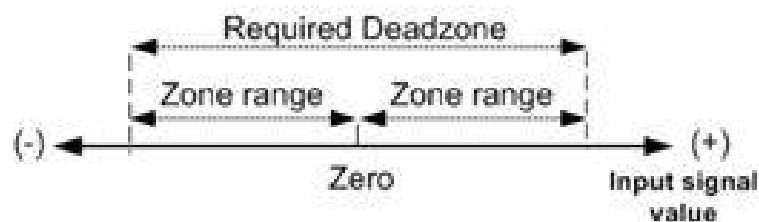
Function	Description
COPY	The function copies one array to another.
DEADZONE	Implements dead-zone routine
DSIGN	Implements a dynamic version of the standard SIGN function.
DSTR	Converts a string to an integer array.
EDGE	Returns 1 on positive edge of x
INP	Reads data characters from the specified channel and stores them into integer array
INTGR	Implements an integrator with DEADZONE and SAT .
LAG	Provides delayed switching on argument change (anti-bouncing effect)
MAP	Implements a table-defined function with constant step
MAPB	Implements a one-dimensional uniform B-spline interpolation
MAPN	One-dimensional non-uniform linear interpolation (replaces the obsolete MAPBY1 and MAPBY2 functions)
MAPS	One-dimensional non-uniform linear interpolation (replaces the obsolete MAPBY1 and MAPBY2 functions)
MAPNB	One-dimensional non-uniform b-spline
MAPNS	One-dimensional non-uniform Catmull-Rom spline
MAPS	One-dimensional uniform Catmull-Rom spline
MAP2	Implements a table-defined function with two arguments and constant step along each argument.

Function	Description
MAP2B	Two-dimensional uniform b-spline
MAP2N	Two-dimensional non-uniform linear interpolation (replaces the obsolete MAP2FREE function)
MAP2NB	Two-dimensional non-uniform b-spline
MAP2NS	Two-dimensional non-uniform Catmull-Rom spline
MAP2S	Two-dimensional uniform Catmull-Rom spline
MATCH	Calculates axis position that matches current reference position of the same axis with zero offset.
RAND	Implements a random number generator.
ROLL	Calculates a result rolled-over to within one pitch.
SAT	Implements a saturation characteristic

4.9.1 DEADZONE

Description

DEADZONE returns values based on a defined analog or other input signal with a defined symmetrical or asymmetrical dead zone around zero.



DEADZONE is useful for anti-bouncing effect.

Syntax

DEADZONE (*input_signal*, *zone1*[*zone2*])

Arguments

<i>input_signal</i>	Any real number or integer expression
<i>zone1</i>	Any real number or integer expression for a symmetrical deadzone. See Example 1 .
<i>zone2</i>	Any real number or integer expression, required for an asymmetrical deadzone requires two values. See Example 2 .

Return Value

DEADZONE returns a **real number** as follows:

If $-zone < input_signal < zone$, return value = 0

If $input_signal < -zone$, return value = $input_signal + zone$

If $input_signal > zone$, return value = $input_signal - zone$

Error Conditions

A negative zone range returns Error 3045, Numerical Error in Standard Function.

Examples

Example 1

Symmetrical Dead Zone

In this example, illustrated in [Figure 4-7](#), input_signal ranges from -20: +20, and creates a symmetrical dead zone from -10: +10.

```
return_value = DEADZONE(input_signal,10)
```

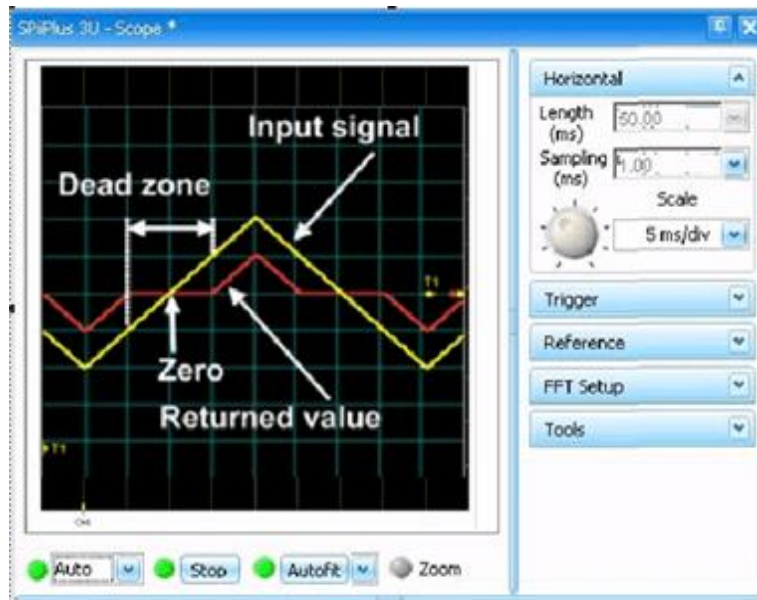


Figure 4-7. Symmetrical Dead Zone Example

Example 2

Asymmetrical Dead Zone

In this example, illustrated in [Figure 4-8](#), input_signal ranges from -20: +20, and creates an asymmetrical dead zone from -5 – +15.

```
return_value = DEADZONE(input_signal,-5,10)
```

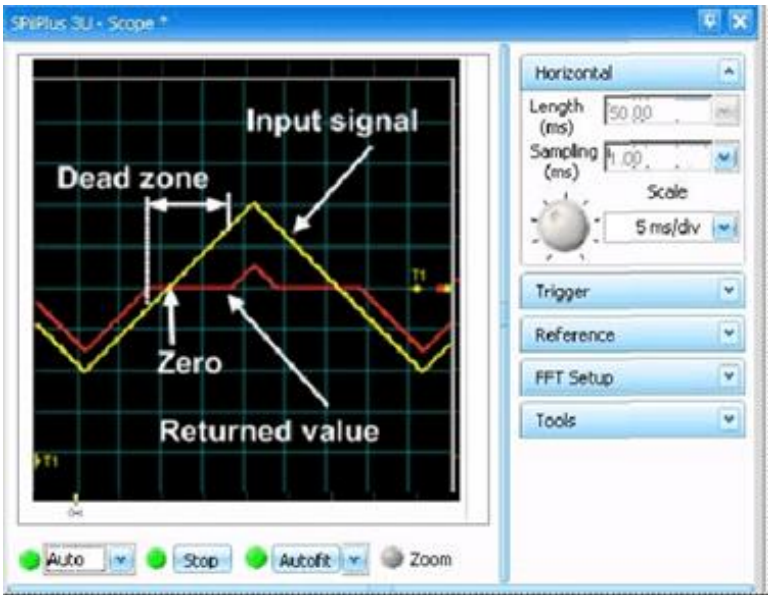


Figure 4-8. Asymmetrical Dead Zone Example

4.9.2 *DSIGN*

Description

DSIGN returns values between -1 to 1 based on a defined input variable with defined delay time and ramp time.

Syntax

real **DSIGN**(*X*, *Delay_Time*, *Ramp_Time*)

Arguments

<i>X</i>	Input variable name declared as real or real expression.
<i>Delay_Time</i>	Real number that determines on zero crossing of <i>X</i> , continues the previous return value for defined delay time in msec.
<i>Ramp_Time</i>	Real number that controls the rate of the returned value change between -1 to 1 (and vice versa), defined in msec.

Return Values

0: when the input variable is < zero while **Ramp Time** = 0.

1: when the input variable is > zero while **Ramp Time**= 0.

-1 to 1: when the input variable changes between positive to negative (or vice versa), while **Ramp Time** is > 0.

Error Conditions

Delay time and ramp time should be positive. If any of them becomes negative, Error 3045, Numerical Error in Standard Function appears.

Example

```
REAL XX, YY
YY = DSIGN(XX, 50, 100)
STOP
```

Figure 4-9 illustrates the **DSIGN** function.

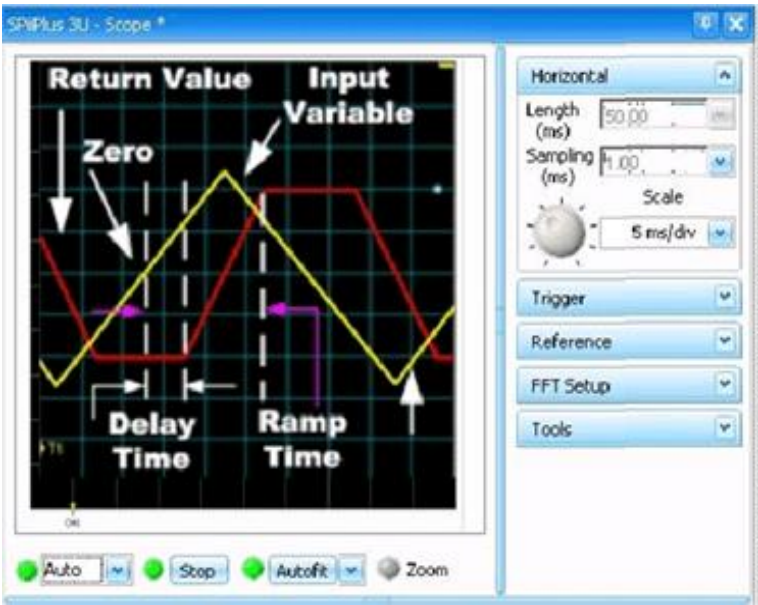


Figure 4-9. DSIGN Function Example

4.9.3 DSTR

Description

DSTR converts a string to an integer array based on an ASCII transformation code. The function decomposes a string to characters and assigns the characters to the sequential elements of the variable array.

Each ASCII character is represented as its numerical value and stored in a separate element of the array.

Syntax

int **DSTR**(*string*, *array_name*, [*start_index*,] [*number*])

Arguments

<i>string</i>	String of characters enclosed in double quotation marks.
<i>array_name</i>	User-defined integer array.

start_index	Index in the array from where the transformed numbers will be placed. If start_index is omitted, the assignment starts from the first element of the array.
number	Number of characters from the string to be transformed. If number is omitted, the function assigns all characters of the string. If number is specified, the function assigns the specified number of characters. In both cases the assignment stops when the last array element is reached.

Return Value

The number of actually transformed characters.

Example

In the example below **DSTR** transforms each of "ACS-Motion_Control" characters to its corresponding numeric value and assigns then to a user defined array "BIBI" (each character to a separate array member. BIBI content is as follows:

65 67 83 45 84 69 67 72 56 48

```
GLOBAL INT BIBI(10)
DSTR("ACS-Motion_Control",BIBI)
STOP
```

4.9.4 *EDGE*

Description

EDGE returns 0 or 1, based on a defined input variable. **EDGE** is mainly useful in PLC implementation when an action must be taken once a condition becomes true.

Syntax

real **EDGE**(X)

Arguments

X	Input real variable name or real expression.
---	--

Return Value

EDGE returns 1 when the input variable changes from 0 to (+1) or from 0 to (-1) until the input variable changes to other values. The function returns 0 in all other cases.

Error Conditions

None

Example

```
GLOBAL REAL XX,XI,YY           !Declares three local variables
XX=-5 ; XI=1                   !Assigns values to the variables XX and XI.
WHILE 1                         !Run the following routine forever
  XX=XX+XI
  YY = EDGE(XX)
  IF (XX>5) | (XX<-5) XI=-XI    !Keeps XX between -5 to 5.
END !Ends IF END               !Ends WHILE
STOP                           !Ends program
```

Figure 4-10 illustrates the **EDGE** function.

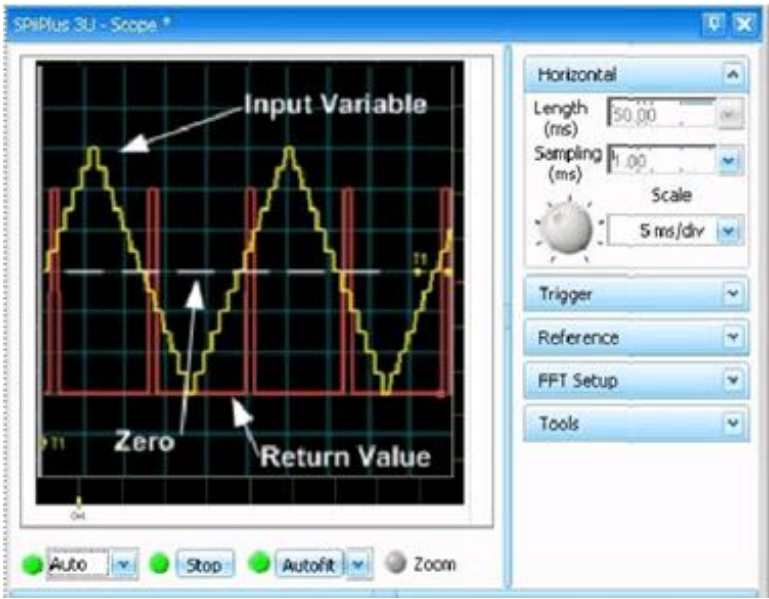


Figure 4-10. EDGE Function Example

4.9.5 INTGR

Description

INTGR returns an integrator with optional deadzone and saturation limits.

Syntax

real **INTGR**(*X*, *Deadzone*, *Min*, *Max*[, *Initial_Value*])

Arguments

<i>X</i>	Real variable name or real expression.
<i>Deadzone</i>	A real number. When the Deadzone value falls into the range of - Deadzone to + Deadzone , INTGR retains its previous value as if <i>X</i> = 0.
<i>Min</i>	A real number representing the low integrator saturation limit.

Max	A real number representing the high integrator saturation limit.
Initial_Value	A real number setting the initial value of the return value. (optional)

Return Value

INTGR returns a value in the range from *Min* to *Max*.

Error Conditions

None

Example

Input variable XX is changing between -20 to +20.

```
GLOBAL REAL YY, XX      !Defines global real variables.
REAL ZZ                 !Defines increment variable as real
MARK:                   !Set GOTO line
ZZ=0.1; XX=-20          !Set variable values
WHILE XX<=20            !Set upper limit for XX
    XX=XX+ZZ            !Set increment for XX
    YY=INTGR (XX,5,-10,10,20)
END                     !End WHILE
WHILE XX>=-20           !Set lower limit for XX
    XX=XX-ZZ            !Set increment for XX
    YY=INTGR (XX,5,-10,10,20)
END                     !End WHILE
GOTO MARK               !GOTO MARK to create a program loop
STOP                   !End program
```

Figure 4-11 illustrates the **INTGR** function.

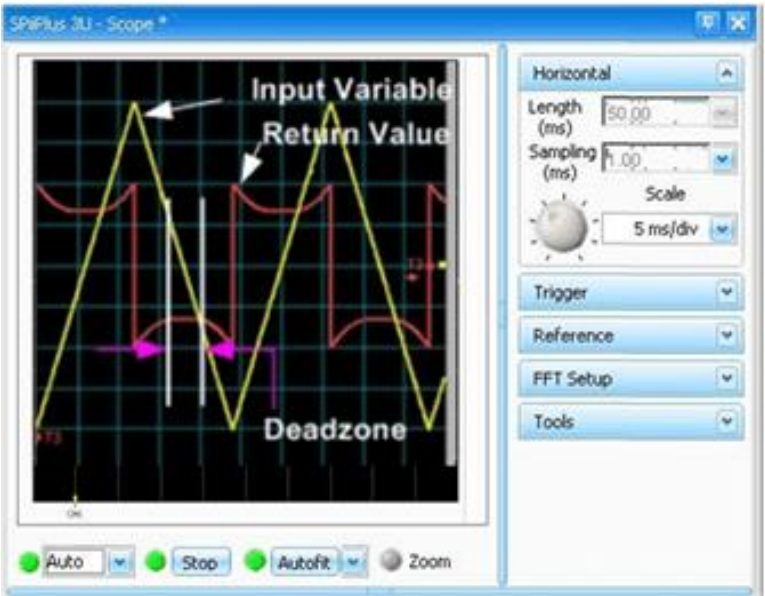


Figure 4-11. INTGR Function Example

4.9.6 LAG

Description

LAG returns values based on a defined input signal with defined up delay or down delay. This function is useful for an anti-bouncing effect.

Syntax

LAG(*X*, *up_delay*, *down_delay*)

Arguments

X	Real variable name or real expression. Although LAG accepts any real or integer argument for X , the typical use implies a logical X argument, for example, an integer variable or expression that supplies only 0 or 1. If X supplies values other than 0 or 1, LAG treats any non-zero value as 1.
p_delay	A real number representing the delay on the positive edge in msec.
down_delay	A real number representing the delay on the negative edge in msec.

Return Value

1: when **X** remains non-zero for at least **up_delay** msec.

0: when **X** remains zero for at least **down_delay** msec.

Error Conditions

None

Example

The example here demonstrates **LAG** by generating a reference signal **XX** that changes from 0 to 1 every 100 msec, and based on **XX** and **YY**, implements a lag of 50msec and 20msec.

INT XX,YY,XT	!Define three integers
XT=TIME; XX=0	!XT equals the time elapsed from
	!startup. XX equal zero.
WHILE 1	!Run an infinite routine
IF TIME-XT>100; XX=^XX; XT=TIME; END	!Switch the value of XX - from ON
to	
	!OFF
YY=LAG (XX, 20, 50)	!YY is a returned value that is delayed
	!20msec after the transfer of XX from
	!0 to 1 and delayed 50msec after the
	!transfer of XX from 1 to 0
END	!Ends WHILE
STOP	!Ends program

Figure 4-12 illustrates the **LAG** function.

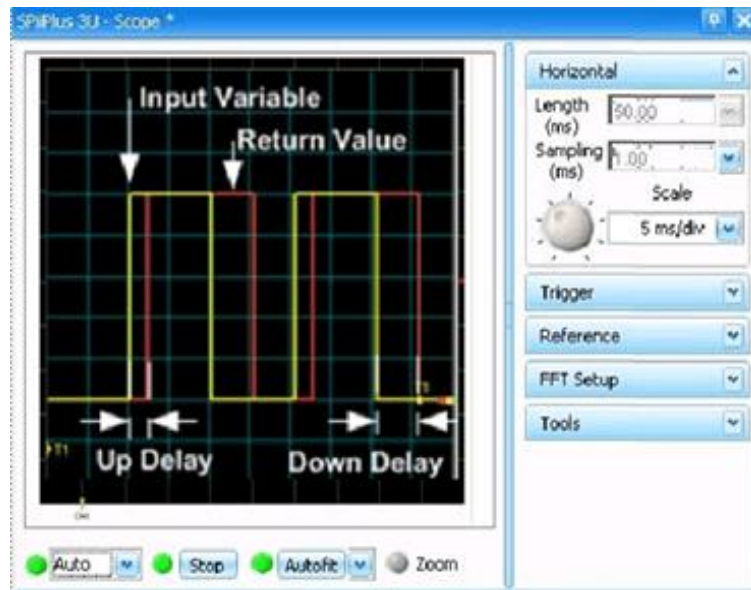


Figure 4-12. LAG Function Example

4.9.7 Interpolation Functions

There are a number of interpolation methods between points among them are Linear and Spline.

4.9.7.1 Linear interpolation

Linear interpolation of function $y(x)$ between distant points a and b are calculated as follows:

$$y(x_i) = \frac{y(b) - y(a)}{b - a} x_i + y(a) - a \frac{y(b) - y(a)}{b - a}, x_i \in [a, b]$$

4.9.7.2 Spline interpolation

A spline is a special function defined piecewise by polynomials. The spline is a piecewise polynomial function spread over an interval $[a, b]$ consists of polynomial pieces, such that:

$$a = t_0 < t_1 < t_2 < \dots < t_{k-2} < t_{k-1} = b$$

The points t_i are called **knots**. The vector is called a **knot vector** for the spline. If the knots are equidistantly distributed in the interval $[a, b]$, we say the spline is **uniform**, otherwise we say it is **non-uniform**.

In many cases functional dependence between two or more values cannot be expressed as an analytic formula. The most common presentation of those functions is a table of function values in specific points.

For example, a machine axis was graduated with an external laser interferometer. The result of graduation is a table like the following:

Commanded position (x)	100	200	300	400	500	600	700	...
Actual position (p)	103	199	294	402	500	598	705	...

The table defines a functional dependence $p=f(x)$ that cannot be expressed analytically.

The argument values for x in the definition table are knots, and the function values for p are control points.

A 3rd order polynomial spline provides an approximation of the table-driven function that can provide the function value not only in the knots, but at any point. Between each two knots the spline is expressed as:

$$p = a_0 + a_1x + a_2x^2 + a_3x^3 = \sum_{i=0}^3 a_i x^i$$

where coefficients a_0, a_1, a_2, a_3 have different values at different intervals.

The SPiiPlus controller also supports two-dimensional splines. In this case, the definition table is a two-dimensional matrix. Knot points are defined for two arguments x and y , and the matrix contains corresponding p values. Knot values divide the XY plane into rectangular cells. The matrix defines the function values in the cell vertices. Within each cell, the interpolating spline is expressed as:

$$p = \sum_{i,j=0}^3 a_{ij} x^i y^j$$

Many different spline approximations can be provided for one definition table. The SPiiPlus controller supports two kinds of splines: Catmull-Rom and B-Splines (see description below).

If the distance between the knots in the table is constant, the spline is called uniform. On the contrary, a non-uniform spline corresponds to a table that contains function values in arbitrary points. However, the definition table always arranges the knot values in ascending order, so that $x_i \leq x_{i+1}$.

All knot points constitute the **definition range** of the spline. Figure 4-13 illustrates definition range of a function defined with six non-uniform knots:

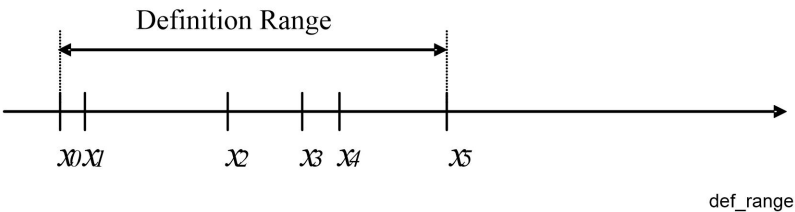


Figure 4-13. Spline Definition Range

In a two-dimensional case, definition range is a rectangular area, as illustrated in [Figure 4-14](#):

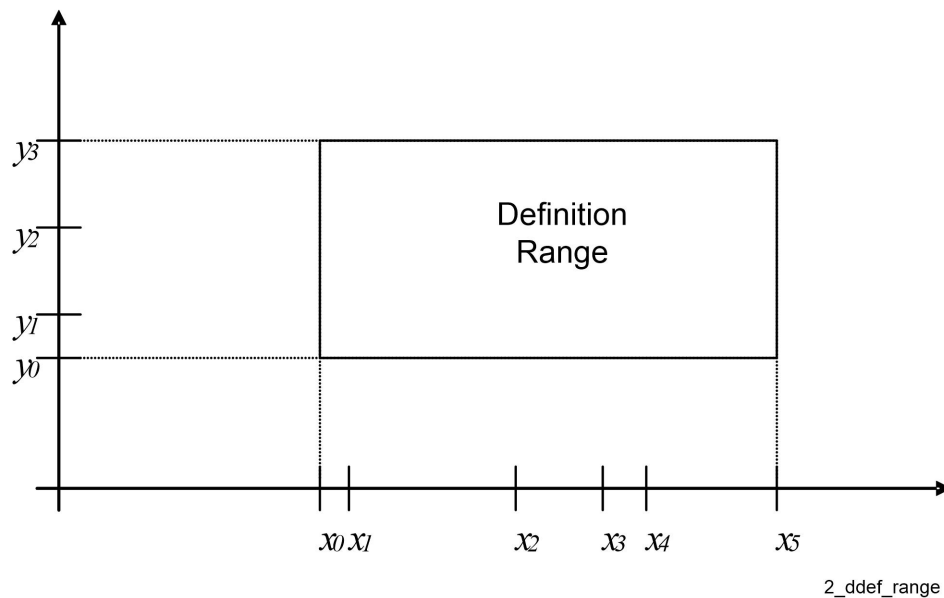


Figure 4-14. Two-Dimensional Spline Definition Range

One-Dimensional Catmull-Rom Spline

Assume a definition table provides N control points $p_0, p_1, p_2, \dots, p_{N-1}$ in knots $x_0, x_1, x_2, \dots, x_{N-1}$.

The Catmull-Rom construction process is described as follows:

- > At each internal knot x_i ($1 \leq i \leq N-2$) calculate the derivative:

$$v_i = (p_{i+1} - p_{i-1}) / (x_{i+1} - x_{i-1})$$
- > At the first and last knots assume zero derivative: $v_0 = v_{N-1} = 0$.
- > In interval i ($1 \leq i \leq N-2$), build a 3rd order polynomial $p=f(x)$ that satisfies four bound conditions $p_i, v_i, p_{i+1}, v_{i+1}$.
- > Beyond the definition range the spline is defined as follows:

$$p = p_0 \text{ if } x < x_0$$

$$p = p_{N-1} \text{ if } x > x_{N-1}$$

[Figure 4-15](#) illustrates a Catmull-Rom spline that interpolates a 5-component definition table.

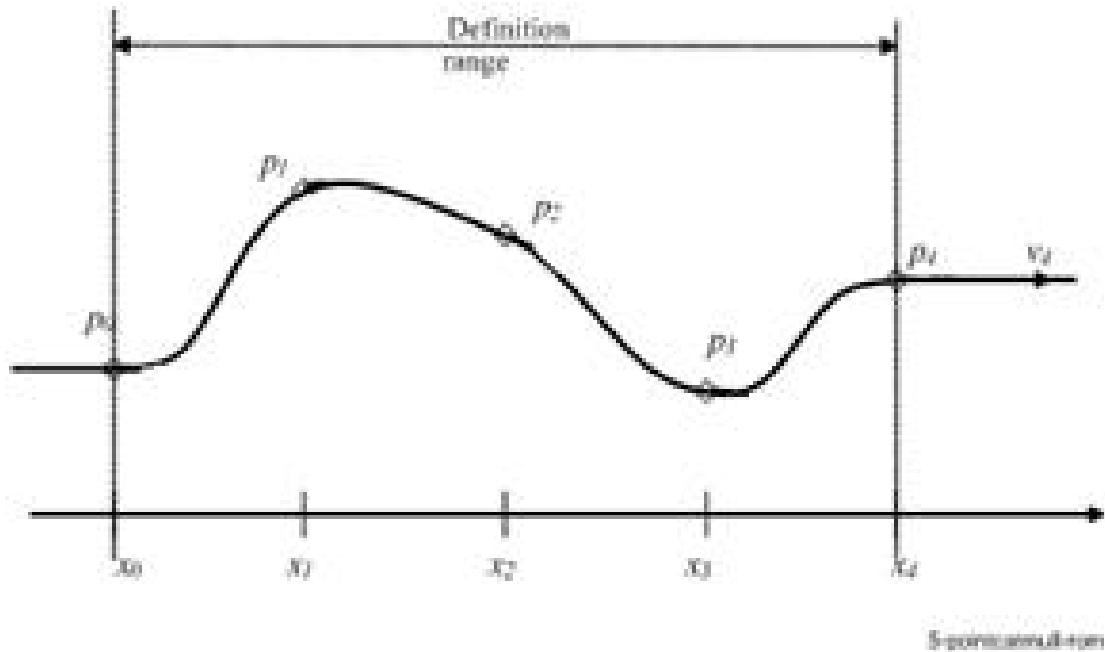


Figure 4-15. 5-Point Catmull-Rom Spline

Features of the Catmull-Rom spline:

- > The spline is C^1 -continuous, meaning the curve and its first derivative are continuous functions. The second derivative has discontinuities in the knot points.
- > The curve interpolates all control points; meaning that the curve goes through each control point.
- > At internal control point number i , the first derivative vector is parallel to the line connecting control points $i-1$ and $i+1$.
- > At the first and the last control points, the first derivative is zero.
- > The spline yields a constant value equal to p_0 on the interval from $-\infty$ to x_0 .
- > The spline yields constant value equal to p_{N-1} on the interval from x_{N-1} to $+\infty$.

One-Dimensional B-spline

Assume a definition table provides N control points $p_0, p_1, p_2, \dots, p_{N-1}$ in knots $x_0, x_1, x_2, \dots, x_{N-1}$.

Unlike the Catmull-Rom spline, a B-Spline does not go through the control points. Actually, it approximates the control points as illustrated in [Figure 4-16](#).

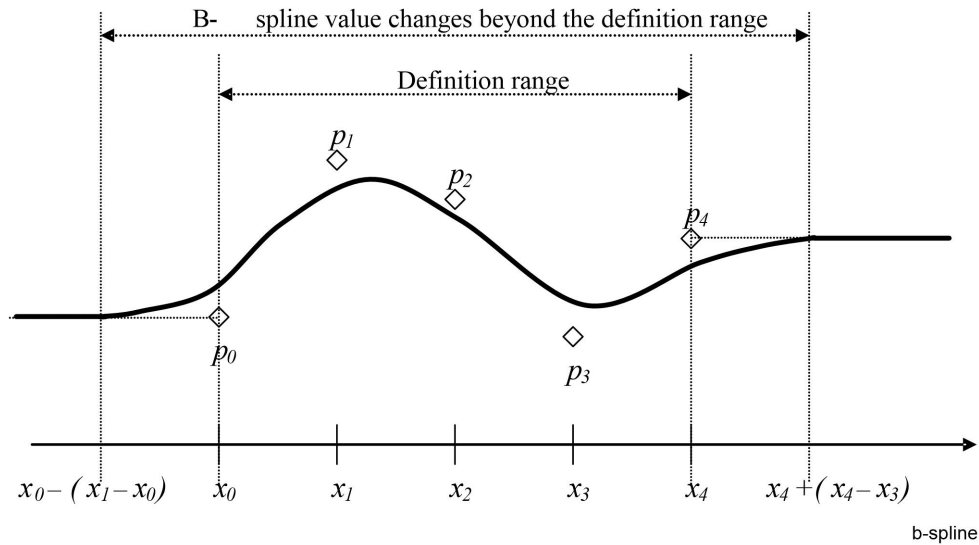


Figure 4-16. B-Spline - Approximation of Points

Compared to a Catmull-Rom spline, a B-Spline generates a smoother curve. Used in the controller, a B-spline provides continuous velocity and acceleration; where Catmull-Rom spline provides continuous velocity only, and acceleration may change by jumping at the control points.

Features of the B-Spline:

- > The spline is C^2 -continuous, meaning the curve, its first and second derivatives are all continuous functions.
- > The curve approximates the control points; and does not go through each control point.
- > The spline yields changing value in the interval from $x_0 - (x_1 - x_0)$ to $x_{N-1} + (x_{N-1} - x_{N-2})$.
- > The spline yields constant value equal to p_0 in the interval from $-\infty$ to $x_0 - (x_1 - x_0)$.
- > The spline yields constant value equal to p_{N-1} in the interval from $x_{N-1} + (x_{N-1} - x_{N-2})$ to $+\infty$.

Not-passing the control points is not always a drawback. If values $p_0, p_1, p_2 \dots p_{N-1}$ are obtained from some measuring process, the values include measuring error that has a stochastic component. B-Spline tends to filter out the stochastic error, thereby improving overall accuracy.

Two-Dimensional Splines

The SPiiPlus NT controllers support two-dimensional Catmull-Rom and B-Splines.

A two-dimensional spline approximates a definition table that provides $N \times M$ control points $p_{00}, p_{01}, \dots, p_{0,M-1}, p_{10}, p_{11}, \dots, p_{N-1,M-1}$ on the grid defined by knots $x_0, x_1, x_2 \dots x_{M-1}$ and $y_0, y_1, y_2 \dots y_{N-1}$.

A two-dimensional spline is defined as tensor product of two one-dimensional splines. Two-dimensional splines share many features with the corresponding one-dimensional splines, for example:

Catmull-Rom Spline Surface	B-Spline Surface
C^1 -continuous	C^2 -continuous
Interpolates control points	Approximates control points

A section of a two-dimensional spline surface along any direction provides a curve, which is a corresponding one-dimensional file. For example, a two-dimensional Catmull-Rom spline is cut on the grid line that corresponds to knot y_2 . The section is a one-dimensional Catmull-Rom spline built upon control points $p_{20}, p_{21}, p_{22}, \dots, p_{2,M-1}$.

The behavior of a two-dimensional spline beyond the definition range is more complex than in the case of a one-dimensional spline. [Figure 4-17](#) illustrates the Catmull-Rom spline beyond the definition range:

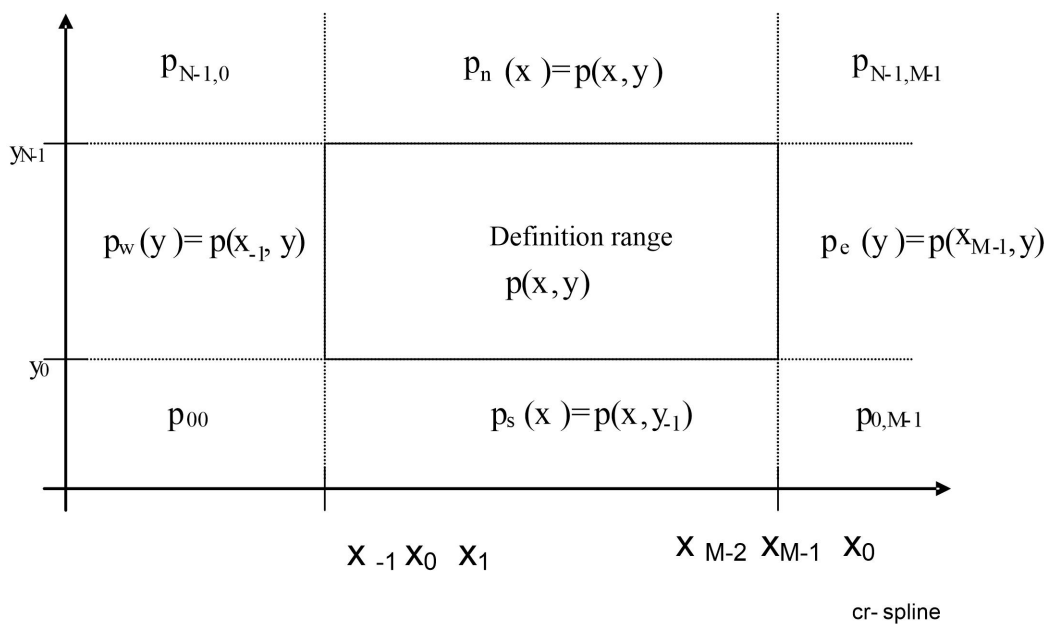


Figure 4-17. Catmull-Ron Spline Beyond the Definition Range

The behavior of Catmull-Rom depends on the area as follows:

- > Within definition range: function of two arguments $p(x, y)$.
- > Southeast to definition range: constant value equal to control point p_{00} .
- > Northeast to definition range: constant value equal to control point $p_{N-1,0}$.
- > Northwest to definition range: constant value equal to control point $p_{N-1,M-1}$.
- > Southwest to definition range: constant value equal to control point $p_{0,M-1}$.
- > South to definition range: function of x , $p_s(x) = p(x, y_0)$
- > North to definition range: function of x , $p_n(x) = p(x, y_{N-1})$
- > West to definition range: function of y , $p_w(y) = p(x_0, y)$
- > East to definition range: function of y , $p_e(y) = p(x_{M-1}, y)$

Similar to one-dimensional B-spline, two-dimensional B-spline has an extended range of result change. To specify the extended range, let us define four artificial knot points:

$$\begin{aligned}x_{-1} &= x_0 - (x_1 - x_0) \\ x_M &= x_{M-1} + (x_{M-1} - x_{M-2}) \\ y_{-1} &= y_0 - (y_1 - y_0) \\ x_N &= x_{N-1} + (x_{N-1} - x_{N-2})\end{aligned}$$

Then, the extended range spans from x_{-1} to x_M and from y_{-1} to y_N .

Area map of a B-spline is similar to the Catmull-Rom spline, but the extended range takes place of the definition range:

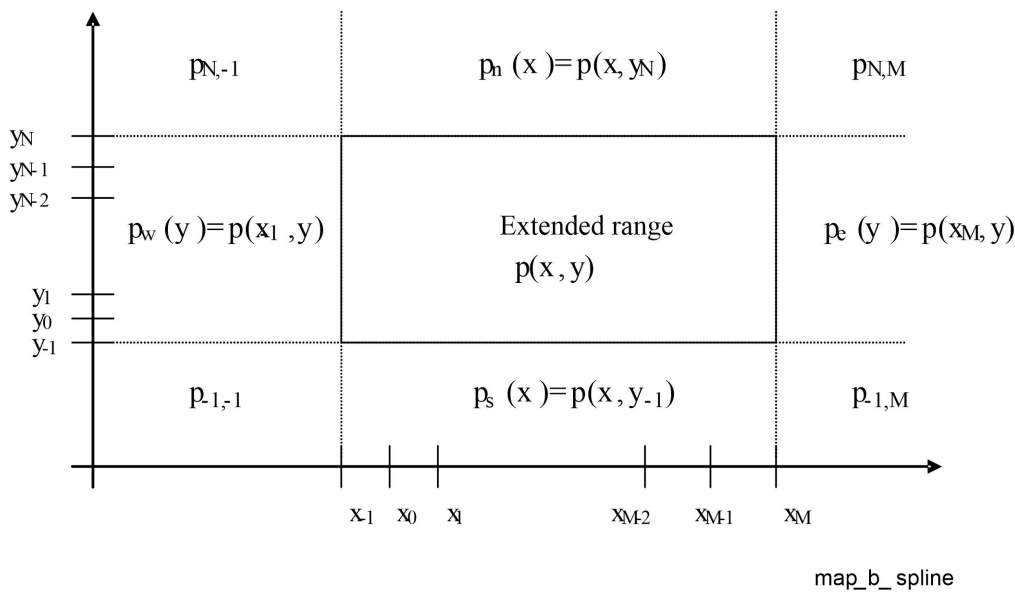


Figure 4-18. B-Spline Map

4.9.7.3 MAP

Description

MAP returns a value from an array of points per input variable value, base value of the variable and fixed defined intervals of the variable. The return values between the array points are linearly interpolated. **MAP** is useful for creating a correction table for mechanical error compensation.

Syntax

MAP(*X*, *array*, *base*, *step*)

Arguments

<i>X</i>	Real variable name or real expression.
<i>array</i>	The name of a real one-dimensional array that specifies points
<i>base</i>	A real number representing the value of XX that corresponds to the first point in array.

step	A real number representing the value of XX that defines fixed intervals between the array points.
-------------	--

Return Value

MAP returns a linearly interpolated value from the **array** for each value of **XX**.

Error Conditions

The function detects the following error conditions:

- > Error 2044, Index is out of range, when the defined **array** size is less than the defined number of **array** points.
- > Error 3113, The step in the table is zero or negative, when the **step** argument is zero or negative.

Example

```
REAL ARRAY(5);REAL XX; REAL YY; REAL XI
!Defines array ARRAY, and variables XX, YY, and XI.
!----- Assign values to ARRAY array -----
ARRAY(0)= -10; ARRAY(1)=3; ARRAY(2)=5; ARRAY(3)=14; ARRAY(4)=12
!Assign values to ARRAY
XX=-20; XI=0.1 !Assign values to XX and XI
WHILE XX<20 !Set conditions
XX=XX+XI
YY=MAP(XX,ARRAY,-10,5) !MAP command where:
!XX is a variable name
!ARRAY is a one-dimensional point array
!-10 is the value of XX that corresponds to base
!point!5 is the value of XX that defines fixed intervals
!between the ARRAY points.
END !Ends mapping
STOP !Ends program
```

Table 4-7 shows the XX and YY values.

Table 4-7. MAP Array

XX	-20	-10	-7	-5	0	5	10	15	20
ARRAY	N/A	-10	N/A	3	5	14	12	N/A	N/A
YY	-10	-10	-2.2	3	5	14	12	12	12

Figure 4-19 illustratesIllustration of this **MAP** example on the Scope.

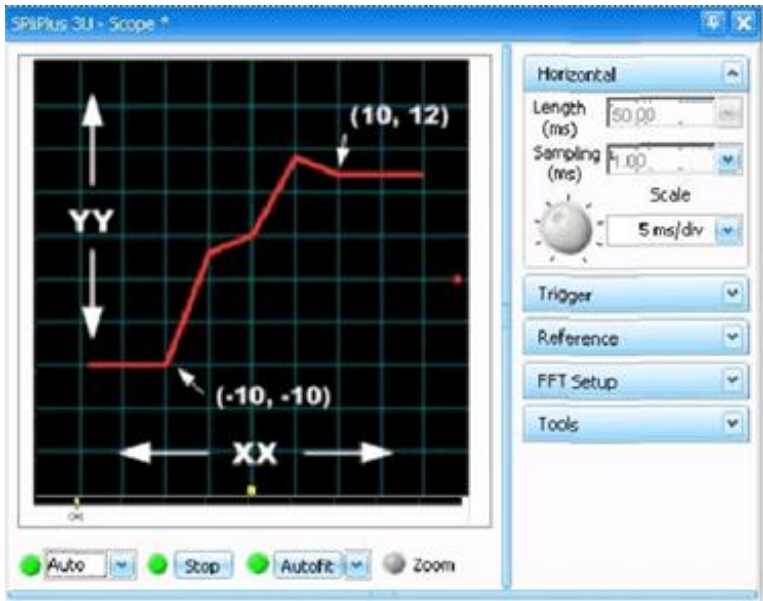


Figure 4-19. MAP Example on the Scope

4.9.7.4 MAPB

Description

MAPB returns a value from an array of points according to an input variable value, base value of the variable and fixed defined intervals of the variable. The return values between the array points are interpolated by a third order B-spline. **MAPB** is useful for creating a correction table for mechanical error compensation.

Syntax

MAPB(*XX*, *array*, *base*, *step*)

Arguments

<i>XX</i>	Real variable name or real expression.
<i>array</i>	The name of a real one-dimensional array that specifies points
<i>base</i>	A real number representing the value of <i>XX</i> that corresponds to the first point in <i>array</i> .
<i>step</i>	A real number representing the value of <i>XX</i> that defines fixed intervals between the <i>array</i> points.

Return Value

MAPB returns the third order B spline interpolated value from the *array* according to the value of variable *XX*.

Error Conditions

The function detects the following error conditions:

- > Error 2044, Index is out of range, when the defined **array** size is less than the defined number of **array** points.
- > Error 3113, The step in the table is zero or negative, when the **step** argument is zero or negative.

Example

```
REAL ARRAY(5);REAL XX; REAL YY; REAL XI
!Defines array ARRAY, and variables XX,YY, and XI
!----- Assign point values to ARRAY array -----
---
ARRAY(0)= -10; ARRAY(1)=3; ARRAY(2)=5; ARRAY(3)=14; ARRAY(4)=12
XX=-20; XI=0.1                                !Assign values to XX and XI
WHILE XX<20                                     !Set conditions
XX=XX+XIYY=MAPB(XX,ARRAY,-10,5) !MAPB command where:!XX is a variable
name                                           !ARRAY is a one-dimensional point array
                                              !-10 is the value of XX that corresponds to base
                                              !point
                                              !5 is the value of XX that defines fixed
                                              intervals
                                              !between the ARRAY points.
END                                           !Ends mapping
STOP                                         !Ends program
```

Table 4-8 shows the XX and YY values.

Table 4-8. MAPB Array

XX	-20	-10	-7	-5	0	5	10	15	20
ARRA Y	N/A	-10	N/A	3	5	14	12	N/A	N/A
YY	-10	- 7.83	- 2.45	1.16	6.16	12.33	12.33	12	12

Figure 4-20 illustrates this **MAPB** example on the Scope.

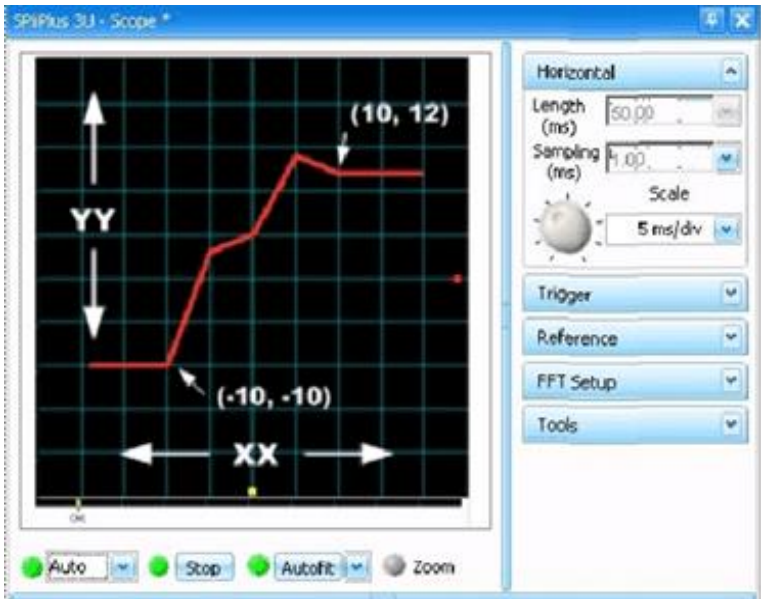


Figure 4-20. MAPB Example on the Scope

4.9.7.5 MAPN

Description

MAPN returns a value from an array of points according to the value of an input variable, and a look-up array. The return values between the array points are linearly interpolated. **MAPN** is useful for creating a correction table for mechanical error compensation.

Syntax

MAPN(*XX*, *X_table*, *Y_table*)

Arguments

<i>XX</i>	Real variable name or real expression.
<i>X_table</i>	The name of a real one-dimensional array that specifies the values of <i>XX</i> that correspond to the point array.
<i>Y_table</i>	The name of a real one-dimensional array that specifies points.

Return Value

MAPN returns a linearly interpolated value from the array according to the value variable **XX**.

- > To obtain a valid return value from the MAPN function after the Y_table array is changed, the buffer must be recompiled before calling the MAPN function.
- > If a MAPN function is called without recompiling the buffer, then the return value is calculated according to the values from the Y_table array that was provided to the function after the last compilation.

Error Conditions

The function detects the following error conditions:

- > The **Xtable** or **Ytable** argument has the wrong dimension
- > The arguments in **Xtable** are not arranged in ascending order.

Example

```
REAL Xtable(5);REAL Ytable(5);REAL XX; REAL YY; REAL XI!Defines arrays
Xtable, Ytable, and variables XX,
!YY, and XI.
!----- Assign values to Xtable array -----
Xtable(0)= -10;Xtable(1)=-3;Xtable(2)=6;Xtable(3)=12;Xtable(4)=14
!----- Assign values to Ytable array -----
Ytable(0)= -10;Ytable(1)=3;Ytable(2)=5;Ytable(3)=14;Ytable(4)=4
XX=-20; XI=0.1          !Assigns values to XX and XI.
WHILE XX<20             !Set conditions.
XX=XX+XI
YY=MAPN(XX,Xtable,Ytable)!MAPN where:      !XX is the defined variable.
                                !Xtable is a one-dimensional point array that
                                !specifies the values of XX corresponding to the
                                !point array.
                                !Ytable is a one-dimensional array that specifies
                                !points.
END                        !Ends mapping
STOP                       !Ends program
```

Table 4-9 shows the XX and YY values.

Table 4-9. MAPN Array

XX	-20	-10	-7	-3	6	12	14	15	20
ARRAY	N/A	-10	N/A	-3	6	12	12	N/A	N/A
YY	-10	-10	-4.42	-3	6	12	14	14	14

Figure 4-21 illustrates this **MAPN** example on the Scope.

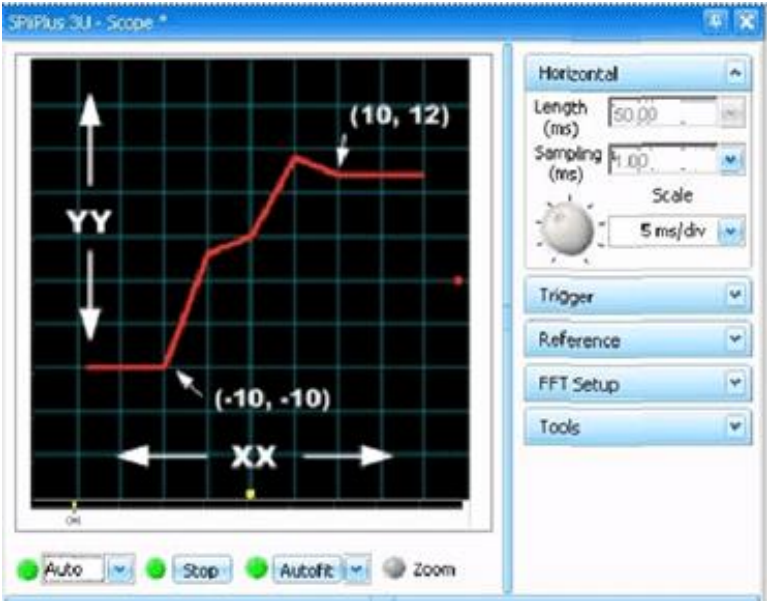


Figure 4-21. MAPN Example on the Scope

4.9.7.6 MAPNB

Description

MAPNB returns a value from a pre-defined control point array based on a defined variable and a look-up table. The return values between the table intervals are interpolated according to a third order B-spline. **MAPNB** is useful for creating a correction table for mechanical error compensation.

Syntax

MAPNB(XX, X_table, Y_table)

Arguments

XX	Real variable name or real expression.
X_table	The name of a real one-dimensional array that specifies the values of XX that correspond to the point array.
Y_table	The name of a real one-dimensional array that specifies points.

Return Value

MAPNB returns the third order B spline interpolated value from the Y_table per variable XX value.



- > To obtain a valid return value from the MAPNB function after the Y_table array is changed, the buffer must be recompiled before calling the MAPNB function.
- > If a MAPNB function is called without recompiling the buffer, then the return value is calculated according to the values from the Y_table array that was provided to the function after the last compilation.

Error Conditions

The function detects the following error conditions:

- > The **X_table** or **Y_table** argument has the wrong dimension
- > The arguments in **X_table** are not arranged in ascending order sequence.

Example

```
REAL Xtable(5);REAL Ytable(5);REAL XX; REAL YY; REAL XI
!Defines arrays Xtable, Ytable, and variables XX,
!YY, and XI.
!----- Assign values to Xtable array -----
Xtable(0)= -10; Xtable(1)=-3; Xtable(2)=6; Xtable(3)=12; Xtable(4)=14
!----- Assign values to Ytable array -----
Ytable(0)= -10; Ytable(1)=3; Ytable(2)=5; Ytable(3)=14; Ytable(4)=4
XX=-20; XI=0.1          !Assigns values to XX and XI.
WHILE XX<20             !Set conditions.
XX=XX+XI
YY=MAPNB(XX,Xtable,Ytable)!MAPNB where:
                        !XX is the defined variable.
                        !Xtable is a point array that specifies the values
                        !of XX corresponding to the point array.
                        !Ytable is an array that specifies Y points.
END                      !Ends mapping
STOP                     !Ends program
```

Table 4-10 shows the XX and YY values.

Table 4-10. MAPNB Array

XX	-20	-10	-7	-3	6	12	14	15	20
ARRA Y	N/A	-10	N/A	-3	6	12	12	N/A	N/A
YY	-10	- 8.02	- 4.65	- 0.41	7.64	9.23	5	4.124	4

Figure 4-22 illustrates this **MAPNB** example on the Scope.

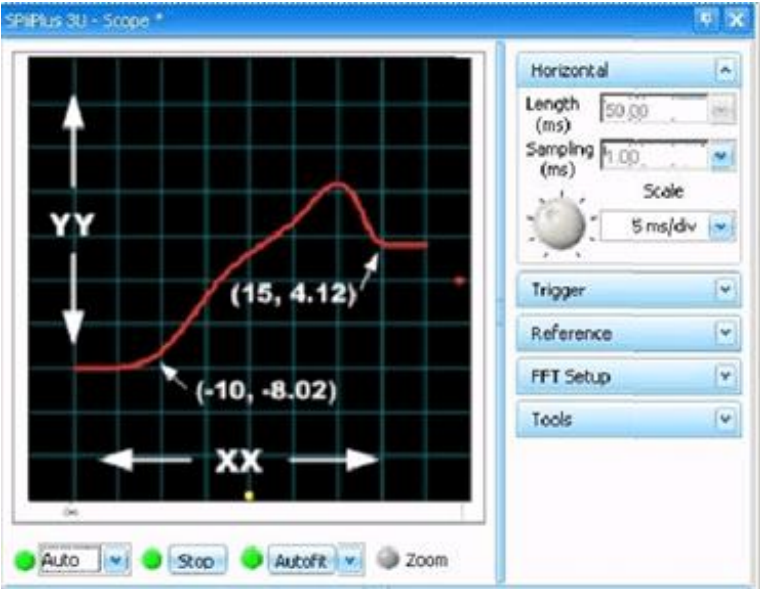


Figure 4-22. MAPNB Example on the Scope

4.9.7.7 MAPNS

Description

MAPNS returns a value from an array of points according to the value of an input variable, and a look-up array. The return values between the array points are interpolated according to a third order Catmull-Rom spline. **MAPNS** is useful for creating a correction table for mechanical error compensation.

Syntax

MAPNS(*XX*, *X_table*, *Y_table*)

Arguments

<i>XX</i>	Real variable name or real expression.
<i>X_table</i>	The name of a real one-dimensional array that specifies the values of <i>XX</i> that correspond to the point array.
<i>Y_table</i>	The name of a real one-dimensional array that specifies points.

Return Value

MAPNS returns the third order Catmull-Rom spline interpolated value from the *X_table* per variable *XX* value.



- > To obtain a valid return value from the MAPNS function after the Y_table array is changed, the buffer must be recompiled before calling the MAPNS function.
- > If a MAPNS function is called without recompiling the buffer, then the return value is calculated according to the values from the Y_table array that was provided to the function after the last compilation.

Error Conditions

The function detects the following error conditions:

- > The **X_table** or **Y_table** argument has the wrong dimension
- > The arguments in **X_table** are not arranged in ascending order sequence.

Example

```
REAL Xtable(5);REAL Ytable(5);REAL XX; REAL YY; REAL XI
!Defines arrays Xtable, Ytable, and variables XX,
!YY, and XI
!----- Assign values to Xtable array -----
Xtable(0)= -10; Xtable(1)=-3; Xtable(2)=6; Xtable(3)=12; Xtable(4)=14
!----- Assign values to Ytable array -----
Ytable(0)= -10; Ytable(1)=3; Ytable(2)=5; Ytable(3)=14; Ytable(4)=4
XX=-20; XI=0.1                !Assigns values to XX and XI.
WHILE XX<20                    !Set conditions.
XX=XX+XI
YY=MAPNS (XX,Xtable,Ytable)    !MAPNS command where:
                                !XX is the defined variable.
                                !Xtable is a one-dimensional point array that
                                !specifies the values of XX corresponding to the
                                !point array.
                                !Ytable is a one-dimensional array that specifies
                                !points.
END                             !Ends mapping
STOP                           !Ends program
```

Table 4-11 shows the XX and YY values.

Table 4-11. MAPNS Array

XX	-20	-10	-7	-3	6	12	14	15	20
ARRAY	N/A	-10	N/A	-3	6	12	12	N/A	N/A
YY	-10	-10	-4.42	-3	6	12	14	14	14

Figure 4-23 illustrates this **MAPNS** example on the Scope.

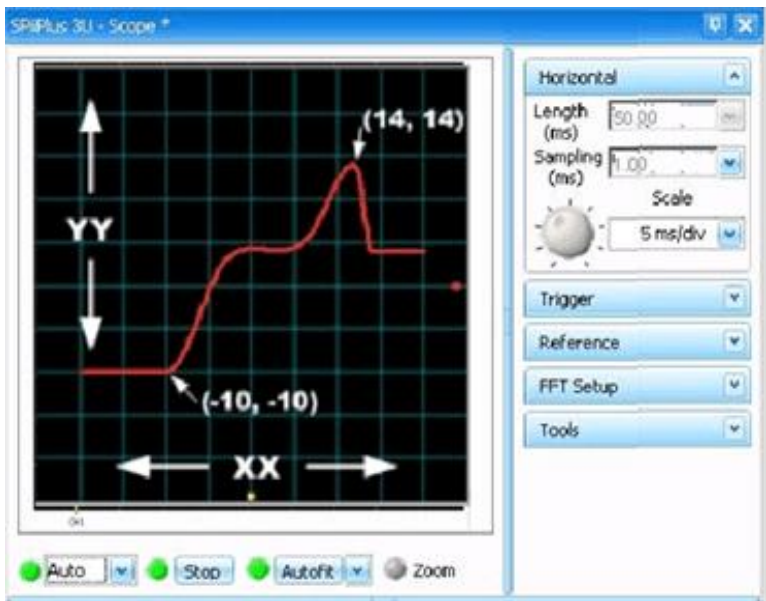


Figure 4-23. MAPNS Example on the Scope

4.9.7.8 MAPS

Description

MAPS returns a value from an array of points according to the value of an input variable, the base value of the variable, and the fixed defined intervals. The return values between the array-points are interpolated according to a third order Catmull-Rom spline. **MAPS** is useful for creating a correction table for mechanical error compensation.

Syntax

MAPS(*XX*, *array*, *base*, *step*)

Arguments

<i>XX</i>	Real variable name or real expression.
<i>array</i>	The name of a real one-dimensional array that specifies points
<i>base</i>	A real number representing the value of <i>XX</i> that corresponds to the first point in array.
<i>step</i>	A real number representing the value of <i>XX</i> that defines fixed intervals between the array points.

Return Value

MAPS returns the third order Catmull-Rom spline interpolated value from the array according to the value of *XX*.

Error Conditions

The function detects the following error conditions:

- > Error 2044, Index is out of range, when the defined **array** size is less than the defined number of **array** points.
- > Error 3113, The step in the table is zero or negative, when the **step** argument is zero or negative.

Example

```
REAL ARRAY(5);REAL XX; REAL YY; REAL XI
!----- Assign values to ARRAY array -----
ARRAY(0)= -10; ARRAY(1)=3; ARRAY(2)=5; ARRAY(3)=14; ARRAY(4)=12
XX=-20; XI=0.1
WHILE XX<20
XX=XX+XI
YY=MAPS(XX,ARRAY,-10,5)
END
STOP
```

Table 4-12 shows the XX and YY values.

Table 4-12. MAPS Array

XX	-20	-10	-7	-5	0	5	10	15	20
ARRAY	N/A	-10	N/A	3	5	14	12	N/A	N/A
YY	-10	-10	-2.65	3	5	14	12	12	12

Figure 4-24 illustrates this **MAPS** example on the Scope.

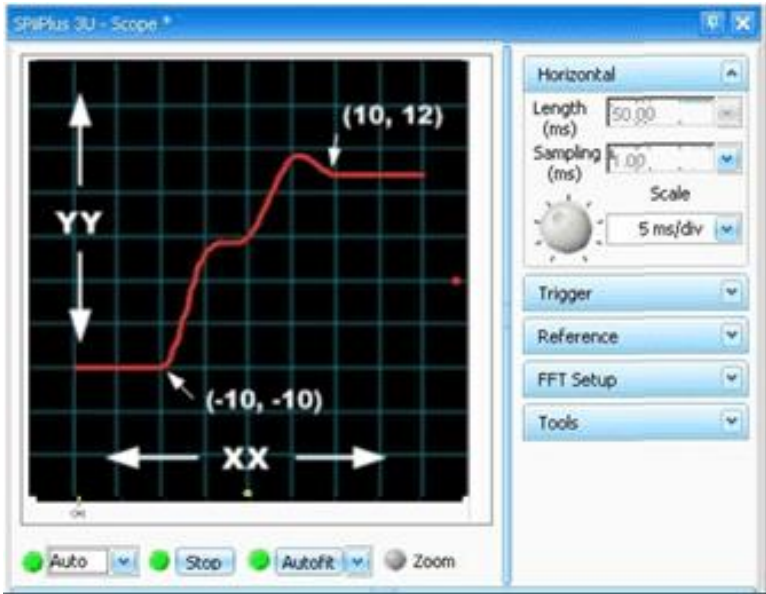


Figure 4-24. MAPS Example on the Scope

4.9.7.9 MAP2

Description

MAP2 returns a value from a two dimensional point array for each set of two input variable values, the base values of the input variables, and a fixed, defined interval for the variables. The return values between the array-points are linearly interpolated. **MAP2** is useful for dynamic error compensation.

Syntax

MAP2(*XX*, *ZZ*, *table*, *baseX*, *stepX*, *baseY*, *stepY*)

Arguments

<i>XX</i>	Real variable name or real expression.
<i>ZZ</i>	Real variable name or real expression.
<i>table</i>	The name of a real two-dimensional array that specifies points
<i>baseX</i>	A real number representing the value of XX that corresponds to the first point in the array.
<i>stepX</i>	A real number representing the value of XX that defines fixed intervals between the array points.
<i>baseY</i>	A real number representing the value of ZZ that corresponds to the first point in the array.
<i>stepY</i>	A real number representing the value of ZZ that defines fixed intervals between the array points.

Return Value

MAP2 returns a linearly interpolated value from the array according to the value of variables **XX** and **ZZ**.

Error Conditions

The function detects the following error conditions:

- > Error 3072, Wrong array size, if **table** has only one dimension.
- > Error 3113, The step in the table is zero or negative, when the **stepX** or **stepY** arguments are zero or negative.

Example

```
GLOBAL REAL XI,XX,YY,ZZ, TABLE(5) (5)
                                !Defines real variables and 5x5 matrix.
XX=-20;ZZ=10;XI=0.1!Assigns values to the variable.
!----- Assign values to TABLE matrix -----
TABLE(0) (0)=2; TABLE(0) (1)=22; TABLE(0) (2)=6; TABLE(0) (3)=8; TABLE(0)
(4)=10;
TABLE(1) (0)=12; TABLE(1) (1)=-44; TABLE(1) (2)=16; TABLE(1) (3)=18;
TABLE(1) (4)=20; TABLE(2) (0)=22; TABLE(2) (1)=24; TABLE(2) (2)=26;
TABLE(2) (3)=68; TABLE(2) (4)=30; TABLE(3) (0)=12; TABLE(3) (1)=34;
TABLE(3) (2)=59; TABLE(3) (3)=-38; TABLE(3) (4)=40; TABLE(4) (0)=92;
TABLE(4) (1)=44; TABLE(4) (2)=46; TABLE(4) (3)=10; TABLE(4) (4)=90
WHILE ZZ<70
XX=XX+XI                                !Set conditions
ZZ=ZZ+XI
YY=MAP2 (XX,ZZ, TABLE, -10,5,20,10)!MAP2 command where:
                                !XX and ZZ are variables.
                                !TABLE is a 2x2 matrix that specifies points.
                                !-10 is the first XX point in the matrix.
                                !5 defines the fixed intervals between the XX
                                points.
                                !20 is the first ZZ point in the matrix.
                                !10 defines the fixed intervals between the ZZ
                                !points.
DISP XX,ZZ,YY                        !Displays values of XX, ZZ and YY.
END                                  !Ends MAP2.
STOP                                !Ends program
```

Table 4-13 shows the YY values as a function of XX and ZZ arguments.

Table 4-13. MAP2

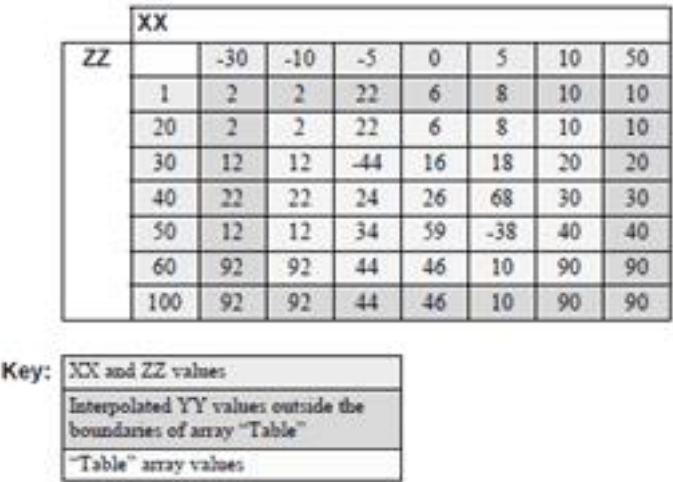


Figure 4-25 illustrates this **MAP2** example on the Scope.

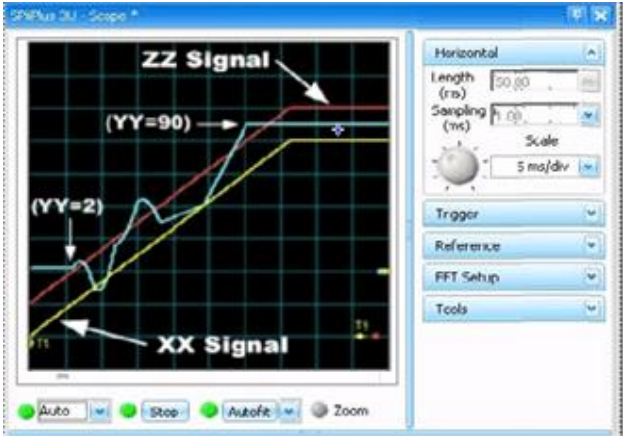


Figure 4-25. MAP2 Example on the Scope

4.9.7.10 MAP2B

Description

MAP2B returns a value from a two dimensional array of points according to the value of two input variables, where the base values of the input variables are fixed and the intervals of the input variables are defined. The return values between the array-points are interpolated according to a third order B-spline. **MAP2B** is useful for dynamic error compensation.

Syntax

MAP2B(XX,ZZ, table, baseX, stepX, baseY, stepY)

Arguments

XX	Real variable name or real expression.
ZZ	Real variable name or real expression.
table	The name of a real two-dimensional array that specifies points
baseX	A real number representing the value of XX that corresponds to the first point in the array.
stepX	A real number representing the value of XX that defines fixed intervals between the array points.
baseY	A real number representing the value of ZZ that corresponds to the first point in the array.
stepY	A real number representing the value of ZZ that defines fixed intervals between the array points.

Return Value

MAP2B returns the third order B-spline interpolated value from the array according to the value of variables **XX** and **ZZ**.

The function detects the following error conditions:

- > Error 3072, Wrong array size, if **table** has only one dimension.
- > Error 3113, The step in the table is zero or negative, when the **stepX** or **stepY** arguments are zero or negative.

Example

```
GLOBAL REAL XI,XX,YY,ZZ,T(5)(5)
                                !Defines real variables and 5x5 matrix.
XX=-20;ZZ=10;XI=0.1!Assigns values to the variables.
!----- Assign values to T matrix -----
T(0)(0)=2;T(0)(1)=22;T(0)(2)=6;T(0)(3)=8;T(0)(4)=10; T(1)(0)=12;
T(1)(1)=-44;T(1)(2)=16;T(1)(3)=18; T(1)(4)=20;T(2)(0)=22;T(2)(1)=24;
T(2)(2)=26; T(2)(3)=68;T(2)(4)=30;T(3)(0)=12;T(3)(1)=34; T(3)(2)=59;
T(3)(3)=-38;T(3)(4)=40;T(4)(0)=92; T(4)(1)=44;T(4)(2)=46;T(4)(3)=10;
T(4)(4)=90
WHILE ZZ<70
XX=XX+XI                                !Set conditions
ZZ=ZZ+XI
YY=MAP2B(XX,ZZ,T,-10,5,20,10!MAP2B command where:
                                !XX and ZZ are variables.
                                !T is a 2x2 matrix that specifies points.
                                !-10 is the first XX point in the matrix.
                                !5 defines the fixed intervals between the XX points.
                                !20 is the first ZZ point in the matrix.
                                !10 defines the fixed intervals between the ZZ
```

```
!points.  
DISP XX,ZZ,YY      !Displays values of XX, ZZ and YY.  
END                !Ends MAP2B.  
STOP               !Ends program
```

Table 4-14 shows the YY values as a function of XX and ZZ arguments.

Table 4-14. MAP2B

ZZ	XX							
		-30	-10	-5	0	5	10	50
1	2	2	22	6	8	10	10	
20	2	2	22	6	8	10	10	
30	12	12	44	16	18	20	20	
40	22	22	24	26	68	30	30	
50	12	12	34	59	-38	40	40	
60	92	92	44	46	10	90	90	
100	92	92	44	46	10	90	90	

Key: XX and ZZ values
Interpolated YY values outside the
boundaries of array "Table"
"Table" array values

Figure 4-26 illustrates this MAP2B example on the Scope.

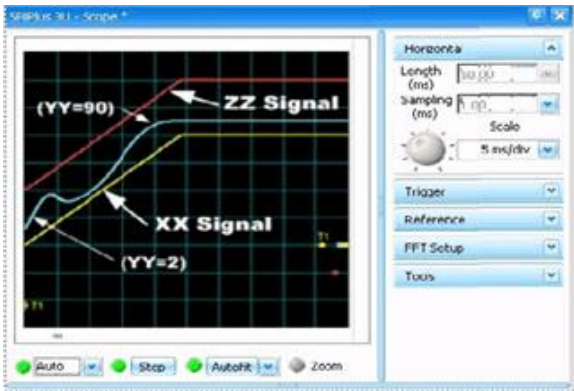


Figure 4-26. MAP2B Example on the Scope

4.9.7.11 MAP2N

Description

MAP2N returns a value from a two dimensional array of points according to the value of two input variables, with look-up arrays for each of the variables. The return values between the array-points are linearly interpolated. **MAP2N** is useful for dynamic error compensation.

Syntax

MAP2N(XX,ZZ, table, X_table, Y_table)

Arguments

XX	Real variable name or real expression.
ZZ	Real variable name or real expression.
table	The name of a real two-dimensional array that specifies points
X_table	A one-dimensional real array that specifies the values of XX that corresponds the point array.
Y_table	A one-dimensional real array that specifies the values of ZZ that corresponds the point array.

Return Value

MAP2N returns the linearly interpolated value from the array according to the value of the variables **XX** and **ZZ**.



- > To obtain a valid return value from the MAP2N function after the Y_table array is changed, the buffer must be recompiled before calling the MAP2N function.
- > If a MAP2N function is called without recompiling the buffer, then the return value is calculated according to the values from the Y_table array that was provided to the function after the last compilation.

Error Conditions

The function detects the following error conditions:

- > Error 3072, Wrong array size, when **table** has only one dimension or **X_table** or **Y_table** have the wrong dimension. **X_table** must contain **M** elements and **Y_table** must contain **N** elements.
- > The values in **X_table** or **Y_table** are not sequenced in ascending order.

Example

```
GLOBAL REAL XI,XX,YY,ZZ,XTABLE(5),YTABLE(5),TABLE(5)(5)
                                !Defines real variables, two arrays and a 5x5 matrix.
XX=-90;ZZ=0;XI=0.1            !Assigns values to the variables.
!----- Assign values to XTABLE array -----
--
XTABLE(0)=13; XTABLE(1)=-8; XTABLE(2)=0; XTABLE(3)=12; XTABLE(4)=51
!----- Assign values to YTABLE array -----
--
YTABLE(0)=16; YTABLE(1)=21; YTABLE(2)=80; YTABLE(3)=82; YTABLE(4)=113
!----- Assign values to TABLE matrix -----
--
TABLE(0)(0)=2; TABLE(0)(1)=22; TABLE(0)(2)=6; TABLE(0)(3)=8; TABLE(0)
(4)=10
TABLE(1)(0)=12; TABLE(1)(1)=-44; TABLE(1)(2)=16; TABLE(1)(3)=18;
```

```
TABLE (1) (4)=20; TABLE (2) (0)=22; TABLE (2) (1)=24; TABLE (2) (2)=26;  
TABLE (2) (3)=68; TABLE (2) (4)=30; TABLE (3) (0)=12; TABLE (3) (1)=34;  
TABLE (3) (2)=59; TABLE (3) (3)=-38; TABLE (3) (4)=40; TABLE (4) (0)=92;  
TABLE (4) (1)=44; TABLE (4) (2)=46; TABLE (4) (3)=10; TABLE (4) (4)=90  
WHILE ZZ<200  
XX=XX+XI!Set conditions  
ZZ=ZZ+XI  
YY=MAP2N(XX,ZZ,TABLE,XTABLE,YTABLE)!MAP2N command where:  
!XX and ZZ are variables.  
!TABLE is a 5x5 matrix specifying mapping points.  
!XTABLE is an array specifying XX points.  
!YTABLE is an array specifying ZZ points.  
!Displays values of XX, ZZ and YY.  
DISP XX,ZZ,YY  
END  
STOP!Ends MAP2N.  
!Ends program
```

Table 4-15 shows the YY values as a function of XX and ZZ arguments.

Table 4-15. MAP2B

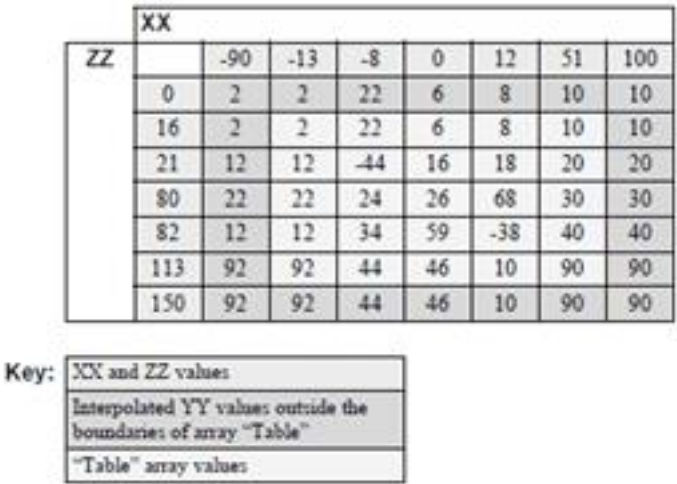


Figure 4-27 illustrates this MAP2N example on the Scope.

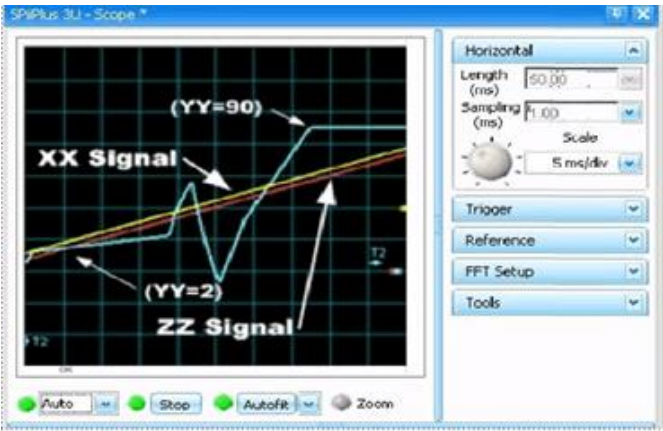


Figure 4-27. MAP2N Example on the Scope

4.9.7.12 MAP2NB

Description

MAP2NB returns a value from a two-dimensional array of points based on the input values of two variables, with a separate look-up array for one variable, and a separate look-up array for the other variable. The return values between the array-points are interpolated according to a third-order B-spline. **MAP2NB** is useful for dynamic error compensation.

Syntax

MAP2NB(*XX,ZZ, Table, X_Table, Y_Table*)

Arguments

<i>XX</i>	Real variable name or real expression.
<i>ZZ</i>	Real variable name or real expression.
<i>table</i>	The name of a real two-dimensional array that specifies points
<i>X_table</i>	A one-dimensional real array that specifies the values of XX that corresponds the point array.
<i>Y_table</i>	A one-dimensional real array that specifies the values of ZZ that corresponds the point array.

Return Value

MAP2NB returns the third-order B-spline interpolated value from the array according to the value of the variables **XX** and **ZZ**.



- > To obtain a valid return value from the MAP2NB function after the Y_table array is changed, the buffer must be recompiled before calling the MAP2NB function.
- > If a MAP2NB function is called without recompiling the buffer, then the return value is calculated according to the values from the Y_table array that was provided to the function after the last compilation.

Error Conditions

The function detects the following error conditions:

- > Error 3072, Wrong array size, when **table** has only one dimension or **X_table** or **Y_table** have the wrong dimension. **X_table** must contain **M** elements and **Y_table** must contain **N** elements.
- > The values in **X_table** or **Y_table** are not sequenced in ascending order.

Example

```

GLOBAL REAL XI,XX,YY,ZZ,XTABLE(5),YTABLE(5),TABLE(5)(5)
                                !Defines real variables, two arrays and a 5x5 matrix.
XX=-90;ZZ=0;XI=0.1            !Assigns values to the variables.
!----- Assign values to XTABLE array -----
--
XTABLE(0)=-13; XTABLE(1)=-8; XTABLE(2)=0; XTABLE(3)=12; XTABLE(4)=51
!----- Assign values to YTABLE array -----
--
YTABLE(0)=16; YTABLE(1)=21; YTABLE(2)=80; YTABLE(3)=82; YTABLE(4)=113
!----- Assign values to TABLE matrix -----
--
TABLE(0)(0)=2; TABLE(0)(1)=22; TABLE(0)(2)=6; TABLE(0)(3)=8; TABLE(0)
(4)=10;
TABLE(1)(0)=12; TABLE(1)(1)=-44; TABLE(1)(2)=16; TABLE(1)(3)=18;
TABLE(1)(4)=20; TABLE(2)(0)=22; TABLE(2)(1)=24; TABLE(2)(2)=26;
TABLE(2)(3)=68; TABLE(2)(4)=30; TABLE(3)(0)=12; TABLE(3)(1)=34;
TABLE(3)(2)=59; TABLE(3)(3)=-38; TABLE(3)(4)=40; TABLE(4)(0)=92;
TABLE(4)(1)=44; TABLE(4)(2)=46; TABLE(4)(3)=10; TABLE(4)(4)=90
WHILE ZZ<200
XX=XX+XI                        !Set conditions
ZZ=ZZ+XI
YY=MAP2NB(XX,ZZ,TABLE,XTABLE,YTABLE)
                                !MAP2NB command where:
                                !XX and ZZ are variables.
                                !TABLE is a 5x5 matrix specifying mapping points.
                                !XTABLE is an array specifying XX points.
                                !YTABLE is an array specifying ZZ points.
                                !Displays values of XX, ZZ and YY.
DISP XX,ZZ,YY
END
                                !Ends MAP2N.
STOP                            !Ends program

```

Table 4-16 shows the YY values as a function of XX and ZZ arguments.

Table 4-16. MAP2NB

XX								
ZZ		-90	-13	-8	0	12	51	100
	0	2	2	22	6	8	10	10
	16	2	2	22	6	8	10	10
	21	12	12	-44	16	18	20	20
	80	22	22	24	26	68	30	30
	82	12	12	34	59	-38	40	40
	113	92	92	44	46	10	90	90
	150	92	92	44	46	10	90	90

Key:	XX and ZZ values
	Interpolated YY values outside the boundaries of array "Table"
	"Table" array values

MAP2B illustrates this MAP2NB example on the Scope.

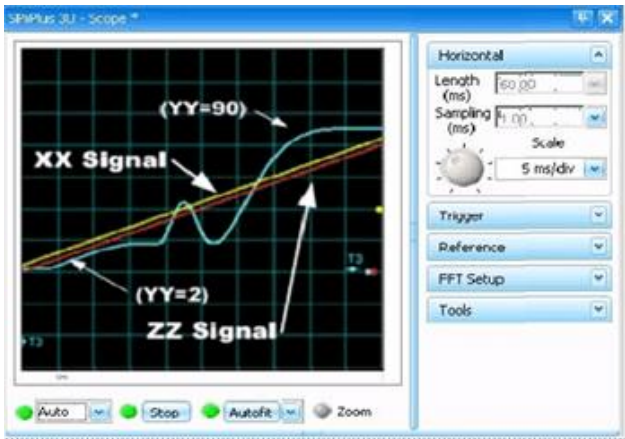


Figure 4-28. MAP2NB Example on the Scope

4.9.7.13 MAP2NS

Description

MAP2NS returns a value from a two-dimensional array of points based on the input values of two variables, with a separate look-up array for one variable, and a separate look-up array for the other variable. The return values between the array-points are interpolated according to a third-order Catmull-Rom spline. The MAP2NS function is useful for dynamic error compensation

Syntax

MAP2NS(XX,ZZ, table, X_table, Y_table)

Arguments

XX	Real variable name or real expression.
ZZ	Real variable name or real expression.

table	The name of a real two-dimensional array that specifies points
X_table	A one-dimensional real array that specifies the values of XX that corresponds the point array.
Y_table	A one-dimensional real array that specifies the values of ZZ that corresponds the point array.

Return Value

MAP2NS returns the third-order Catmull-Rom interpolated value from the array according to the value of the variables **XX** and **ZZ**.



- > To obtain a valid return value from the MAP2NS function after the Y_table array is changed, the buffer must be recompiled before calling the MAP2NS function.
- > If a MAP2NS function is called without recompiling the buffer, then the return value is calculated according to the values from the Y_table array that was provided to the function after the last compilation.

Error Conditions

The function detects the following error conditions:

- > Error 3072, Wrong array size, when **table** has only one dimension or **X_table** or **Y_table** have the wrong dimension. **X_table** must contain **M** elements and **Y_table** must contain **N** elements.
- > The values in **X_table** or **Y_table** are not sequenced in ascending order.

Example

```

GLOBAL REAL XI,XX,YY,ZZ,XTABLE(5),YTABLE(5),TABLE(5)(5)
                                !Defines real variables, two arrays and a 5x5 matrix.
XX=-90;ZZ=0;XI=0.1            !Assigns values to the variables.
!----- Assign values to XTABLE array -----
--
XTABLE(0)=13; XTABLE(1)=-8; XTABLE(2)=0; XTABLE(3)=12; XTABLE(4)=51
!----- Assign values to YTABLE array -----
--
YTABLE(0)=16; YTABLE(1)=21; YTABLE(2)=80; YTABLE(3)=82; YTABLE(4)=113
!----- Assign values to TABLE matrix -----
--
TABLE(0)(0)=2; TABLE(0)(1)=22; TABLE(0)(2)=6; TABLE(0)(3)=8; TABLE(0)
(4)=10;
TABLE(1)(0)=12; TABLE(1)(1)=-44; TABLE(1)(2)=16; TABLE(1)(3)=18;
TABLE(1)(4)=20; TABLE(2)(0)=22; TABLE(2)(1)=24; TABLE(2)(2)=26;
TABLE(2)(3)=68; TABLE(2)(4)=30; TABLE(3)(0)=12; TABLE(3)(1)=34;
TABLE(3)(2)=59; TABLE(3)(3)=-38; TABLE(3)(4)=40; TABLE(4)(0)=92;
TABLE(4)(1)=44; TABLE(4)(2)=46; TABLE(4)(3)=10; TABLE(4)(4)=90
WHILE ZZ<200
XX=XX+XI                      !Set conditions
ZZ=ZZ+XI
YY=MAP2NS(XX,ZZ,TABLE,XTABLE,YTABLE)
                                !MAP2NS command where:
                                !XX and ZZ are variables.
                                !TABLE is a 5x5 matrix specifying mapping points.
                                !XTABLE is an array specifying XX points.
                                !YTABLE is an array specifying ZZ points.
DISP XX,ZZ,YY                !Displays values of XX, ZZ and YY.
END                            !Ends MAP2N.
STOP                          !Ends program

```

Table 4-17 shows the YY values as a function of XX and ZZ arguments.

Table 4-17. MAP2NS

XX								
ZZ		-90	-13	-8	0	12	51	100
	0	2	2	22	6	8	10	10
	16	2	2	22	6	8	10	10
	21	12	12	-44	16	18	20	20
	80	22	22	24	26	68	30	30
	82	12	12	34	59	-38	40	40
	113	92	92	44	46	10	90	90
	150	92	92	44	46	10	90	90

Key:	XX and ZZ values
	Interpolated YY values outside the boundaries of array "Table"
	"Table" array values

Figure 4-29 illustrates this **MAP2NS** example on the Scope.

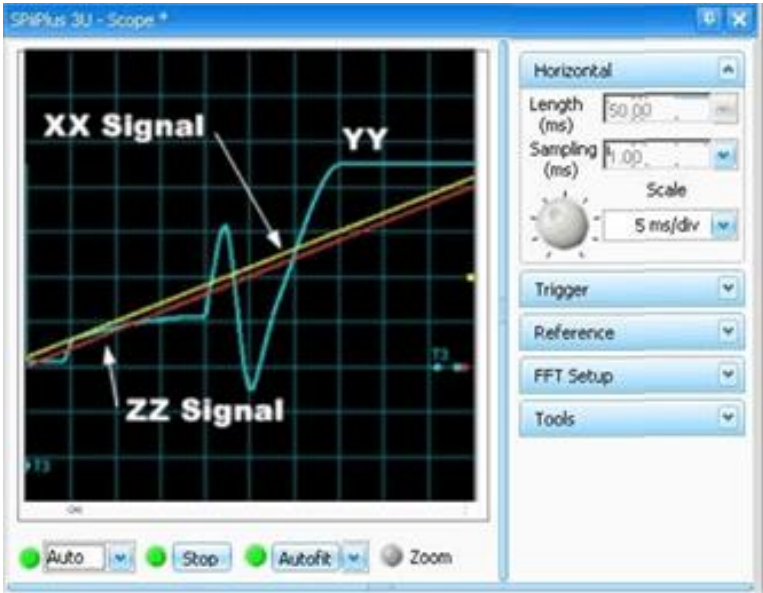


Figure 4-29. MAP2NS Example on the Scope

4.9.7.14 MAP2S

Description

MAP2S returns a value from a two-dimensional array of points based on the input values of two variables, the base values of the two variables, and the fixed defined intervals of the two variables. The return values between the array-points are interpolated according to a third order Catmull-Rom spline. The **MAP2S** function is useful for dynamic error compensation.

Syntax

MAP2S(XX,ZZ, table, baseX, stepX, baseY, stepY)

Arguments

XX	Real variable name or real expression.
ZZ	Real variable name or real expression.
table	The name of a real two-dimensional array that specifies points
baseX	A real number representing the value of XX that corresponds to the first point in the array.
stepX	A real number representing the value of XX that defines the fixed intervals between the array points.
baseY	A real number representing the value of ZZ that corresponds to the first point in the array.
stepY	A real number representing the value of ZZ that defines the fixed intervals between the array points.

Return Value

MAP2S returns the third order Catmull-Rom spline interpolated value from the array according to the value of variables **XX** and **ZZ**.

Error Conditions

The function detects the following error conditions:

- > Error 3072, Wrong array size, when **table** has only one dimension.
- > Error 3113, The step in the table is zero or negative, when the **stepX** or **stepY** arguments are zero or negative.

Example

```
GLOBAL REAL XI,XX,YY,ZZ, TABLE(5)(5)
                                !Defines real variables and 5x5 matrix.
XX=-20;ZZ=10;XI=0.1           !Assigns values to the variable.
!----- Assign values to TABLE matrix -----
TABLE(0)(0)=2; TABLE(0)(1)=22; TABLE(0)(2)=6; TABLE(0)(3)=8; TABLE(0)(4)=10;
TABLE(1)(0)=12; TABLE(1)(1)=-44; TABLE(1)(2)=16; TABLE(1)(3)=18;
TABLE(1)(4)=20; TABLE(2)(0)=22; TABLE(2)(1)=24; TABLE(2)(2)=26;
TABLE(2)(3)=68; TABLE(2)(4)=30; TABLE(3)(0)=12; TABLE(3)(1)=34;
TABLE(3)(2)=59; TABLE(3)(3)=-38; TABLE(3)(4)=40; TABLE(4)(0)=92;
TABLE(4)(1)=44; TABLE(4)(2)=46; TABLE(4)(3)=10; TABLE(4)(4)=90
WHILE ZZ<70
XX=XX+XI                       !Set conditions
ZZ=ZZ+XI
YY=MAP2S(XX,ZZ, TABLE,-10,5,20,10)!MAP2S where:
                                !XX and ZZ are variables.
                                !TABLE is a 2x2 matrix that specifies points.
                                !-10 is the first XX point in the matrix.
                                !5 defines the fixed intervals between the XX points.
                                !20 is the first ZZ point in the matrix.
                                !10 defines the fixed intervals between the ZZ
                                !points.
DISP XX,ZZ,YY                 !Displays values of XX, ZZ and YY.
END                             !Ends MAP2S.
STOP                           !Ends program
```

Table 4-18 shows the YY values as a function of XX and ZZ arguments.

Table 4-18. MAP2S

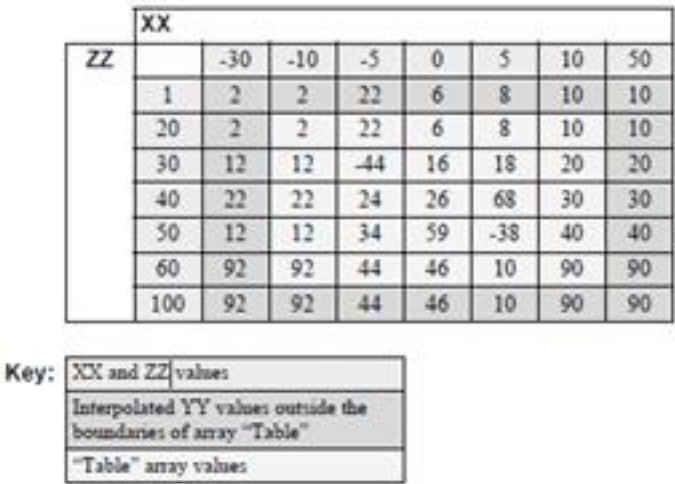


Figure 4-30 illustrates this MAP2S example on the Scope.

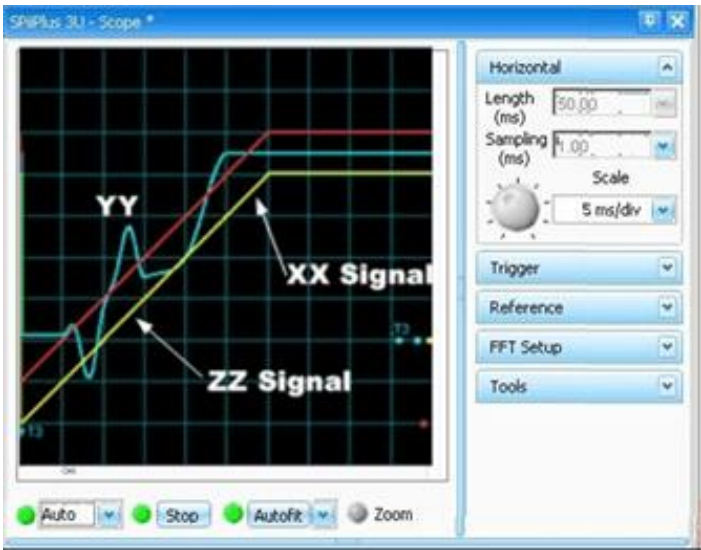


Figure 4-30. MAP2S Example on the Scope

4.9.7.15 MATCH

Description

MATCH calculates axis position (**APOS**) that matches the current reference position (**RPOS**), based on **CONNECT** between **APOS** and **RPOS** of the same axis. If there is no **CONNECT** command, the function returns the value of **RPOS**.

Syntax

real **MATCH** (*axis, from, to*)

Arguments

<i>axis</i>	The axis upon which to apply the MATCH function.
<i>from</i>	The start point to begin searching for matching values of APOS .
<i>to</i>	The end point to stop searching for matching values of APOS .

Comments

The function is useful only in the case of non-default connections.

The connection must be on the same axis.

The function succeeds if the unique root exists in the specified range. If there are several roots in the range, the function returns one of them. If the root does not exist, the function results an error.

Return Value

APOS that matches the current **RPOS** of the same axis.

Error Conditions

Error 3158 - The function cannot find a matching value. The **CONNECT** formula has no root or the root is not single.

Example

```
GLOBAL REAL ARRAY(6), XX
!----- Assign values to ARRAY array -----
ARRAY(0)= 60; ARRAY(1)=40; ARRAY(2)=90; ARRAY(3)=-40; ARRAY(4)=60;
ARRAY(5)=10
VEL(0)=800                !Set VEL(0)
MFLAGS(0).17=1
CONNECT RPOS(0)=APOS(0)+MAP(APOS(0),ARRAY,100,200)
                                !The program executes a CONNECT between RPOS and APOS
                                !for axis X. Where RPOS(0) receives the value of
                                !APOS(0) added to the value of a MAP function.

DEPENDS X,X
SET APOS(0)=0
SET RPOS(0)=MAP(APOS(0),ARRAY,100,200)
PTP X, 1300                !A PTP command sends the motor to 1300 counts,
                                !while the motor is in motion a MATCH function is
                                !calculated and displayed along with RPOS(0) and
                                !APOS(0).

WHILE MST(0).#MOVE
WAIT 300
XX=MATCH(X,0,1300)        !The MATCH function calculates the root of the
                                !current RPOS(0) based on the CONNECT formula.

DISP RPOS(0), APOS(0), XX
END
STOP                        !Ends program
```

The output of the program displays the following **RPOS(0)**, **APOS(0)** values, and the value of **MATCH**. Notice that **MATCH** values are very close to **APOS(0)** values:

RPOS(0)	APOS(0)	MATCH
273.76	228	226.4
554	472.8	471.2
684	717.6	716
1005.6	962.4	960.8
1215.6	1207.2	1205.6
1310	1300	1300

4.9.7.16 RAND

Description

RAND generates a random number

Syntax

real **RAND**(*[min, max, seed]*)

Arguments

<i>min</i>	(optional) Lower boundary of the interval from which randomized number will be selected. Default value is 0.
<i>max</i>	(optional) Upper boundary of the interval from which randomized number will be selected. Default value is 0x7fff.
<i>seed</i>	(optional) Sets the random sequence generator. The number range is limited only by the controller register size - 32bit. If omitted, the controller uses ACSPL+ TIME variable as the seed.

Comments

RAND can generate only one random number per seed number (per interval). For generating a series of random numbers use **TIME** as the seed. In this case the series of random numbers will be in ascending order.

Return Value

The randomized generated number.

Error Conditions

None

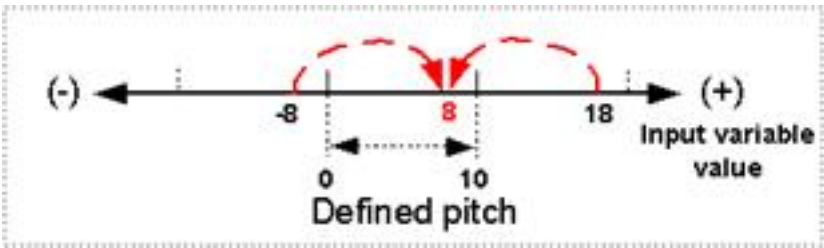
Example

```
RAND (-45, 45, TIME)
```

4.9.7.17 ROLL

Description

The **ROLL** function returns a result rolled-over to within a defined Pitch (range), as illustrated below.



Syntax

ROLL(*X, Pitch*)

Arguments

X	Variable name declared as real or a real expression.
Pitch	A range from 0 to Pitch (positive real value) where the return value will be rolled-over.

Return Value

X - if **X** falls within the range from 0 to **Pitch**.

or

$|X|/Pitch - \text{FLOOR}(|X|/Pitch)*Pitch$, if **X** is less than or greater than **Pitch**.

Error Conditions

None

Example:

<code>YY=ROLL (XX,10)</code>	<code>!Input variable XX is changing between - !20 to 20.</code>
------------------------------	--

Figure 4-31 illustrates this **ROLL** example on the Scope.

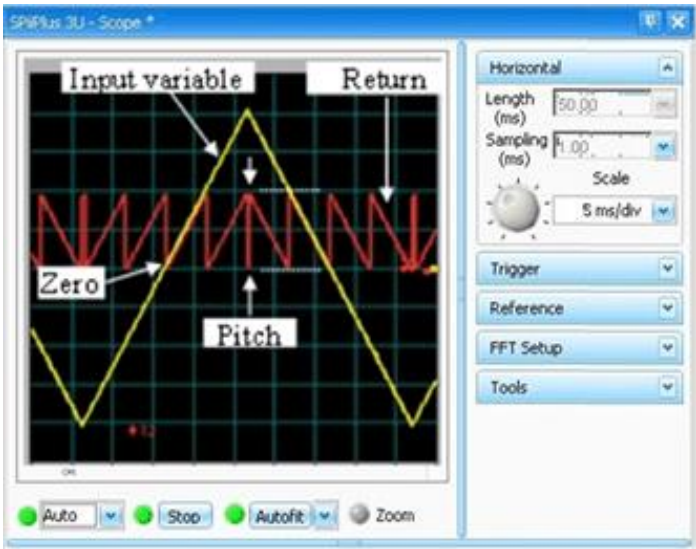


Figure 4-31. ROLL Example on the Scope

4.9.7.18 SAT

Description

SAT returns a result of defined saturation range

Syntax

real **SAT**(*X, Min, Max*)

Arguments

<i>X</i>	Variable name declared as real or a real expression.
<i>Min</i>	Low saturation value. Must be a real number.
<i>Max</i>	High saturation value. Must be a real number.

Return Value

X - if the **X** value falls inside the SAT range.

Min - if **X** value is less than the SAT range.

Max - if **X** value is greater than SAT range.

Error Conditions

None

Example

```
GLOBAL REAL YY,ZZ,XX
EYAL:
ZZ=-20;XX=0.1
WHILE ZZ<=20
    ZZ=ZZ+XX
    YY=SAT (ZZ,-5,15)
END
ZZ=20;XX=0.1
WHILE ZZ>=-20
    ZZ=ZZ-XX
    YY=SAT (ZZ,-5,15)
END
GOTO EYAL
STOP
```

Figure 4-32 illustrates this **SAT** example on the Scope.

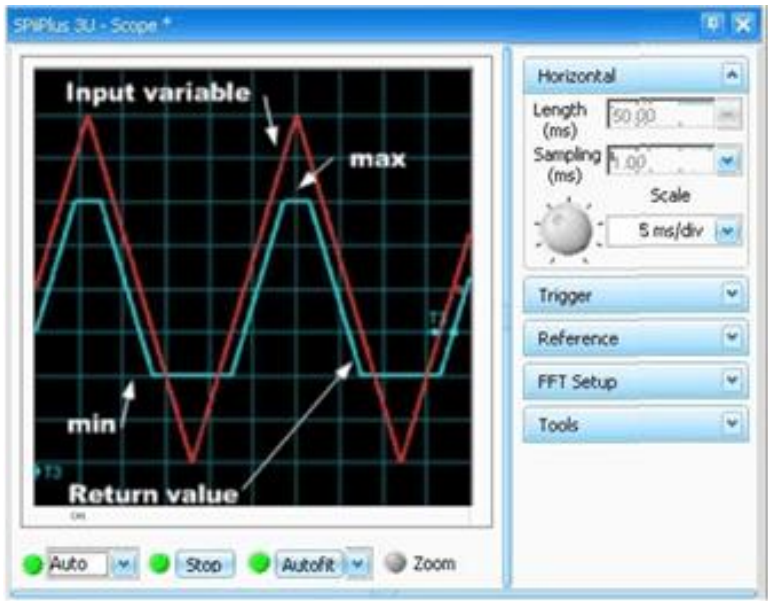


Figure 4-32. SAT Example on the Scope

4.10 Laser Control Functions

The laser control functions are:

Function	Description
LCMODULATION	Initializes Pulse modulation mode and sets initial values for the unit internal registers.
LCFixedDist	Initializes the fixed distance pulse firing mode.
LCFixedInt	Initializes the fixed distance pulse firing mode.
LCRandomDist	Initializes either array based pulse firing mode or gating mode.
LCTickle	Initializes Tickle mode.
LCZone	Sets a laser operation zone area.
LCZoneGet	Returns the limits of the laser operation zone that was previously defined.
LCZoneSet	Changes the minimal and/or maximal laser operation zone limit.

Function	Description
LCStop	Stops any previously initialized laser mode and resets the previously defined mode parameters.
LCSignalSet	Configures the laser control signal (LCS) output conditioning state.
LCSignalGet	Returns the laser control signal (LCS) output conditioning state.
LCOutputSet	Configures the laser physical outputs.
LCOutputGet	Returns the laser outputs configuration.
LCDelaySet	Sets the pulse generation delay in microseconds.
LCDelayGet	Returns the actual currently configured delay in microseconds.
AxListAsMask	Mask for defining axes.

4.10.1 LCMODULATION

The function initializes Pulse modulation mode and sets initial values for the unit internal registers. Which Laser Control unit is initialized is determined by the index argument.

The Laser Control unit can be initialized irrespective if the corresponding axis is enabled or disabled.

Syntax

LCMODULATION (Index, Mode, ModulationFrequency, Width, DutyCycle, AxesUsed, [MinValue, MaxValue, MinVelocity, MaxVelocity])

Arguments

Arguments	Comments
Index	Defines which Laser Control unit is referred. The system allocates 4 indexes (logical axes) for each LCM in the network. The last index of allocated logical axes should be specified. For example, if LCM gets network indexes (logical axes) 4,5,6,7, so index 7 should be used for referring the specific LCM module.

Mode	Operation mode: 1. Mode with fixed frequency 2. Mode with fixed pulse width 3. Mode with fixed duty cycle Modes 1 – 3 are incompatible modes, i.e. setting any of these modes automatically disables the previously set mode. In order to stop any of previously defined modes, 0 value is used.
ModulationFrequency	Pulse modulation frequency in Hz, range from 0.035Hz to 1MHz In <i>mode</i> = 1, the frequency is fixed and is not changed during the process. In <i>mode</i> = 2 or 3, the function only sets the initial frequency. If AxesUsed parameter is defined, during the process the controller will automatically update the frequency as function of vector velocity of the axes referred by AxesUsed. If AxesUsed parameter is zero, the application is responsible to update PFGPAR variable with frequency value.
Width	Pulse Width in milliseconds, range from 6.67nsec to 28.60sec. Only has effect in <i>mode</i> = 2.
DutyCycle	Duty cycle in percentage, range from 0 to 100%. In <i>mode</i> = 1, the function only sets the initial duty cycle. If AxesUsed parameter is defined, during the process the controller will automatically update duty cycle as function of vector velocity of the axes referred by AxesUsed. If AxesUsed parameter is zero, the application is responsible to update PFGPAR variable with duty cycle value, see Duty cycle or frequency update for details. In <i>mode</i> = 2, the argument has no effect, as duty cycle is automatically calculated by the unit as function of frequency. In <i>mode</i> = 3, the duty cycle is fixed and is not changed during the process. In <i>mode</i> = 4, the parameters is ignored.

AxisUsed	Mask that defines which axes are used to calculate vector velocity for controlling duty cycle (mode 1) or frequency (modes 2 and 3). For example, the following mask defines that axes 0 and 2 should be used for vector velocity calculation: 0b0101 (0b – is a prefix for binary value specification) or 0x5 (0x – is a prefix for hexadecimal value specification) To simplify the mask specification, the special ACSPL+ helper-function can be used: <pre>AxesUsed = AxListAsMask(0,2)</pre> Only first 32 axes (0..31) can be referred by AxesUsed. If no mask is specified, the application is responsible to update PFGPAR variable, see Duty cycle or frequency update.
MinValue	[Optional] Minimal value of duty cycle in percentage (if <i>mode</i> = 1) or frequency in Hz (if <i>mode</i> = 2 or <i>mode</i> = 3) if calculated vector velocity equals or below MinVelocity, see Figure 4-33. By default, minimal value of duty cycle is 0% and frequency is 0.035Hz .
MaxValue	[Optional] Maximal value of duty cycle in percentage (if <i>mode</i> = 1) or frequency in Hz (if <i>mode</i> = 2 or <i>mode</i> = 3) if calculated vector velocity equals or above MaxVelocity, see Figure 4-33. By default, maximal value of duty cycle is 100% and frequency is 1MHz.
MinVelocity	[Optional] Minimal velocity for duty cycle or frequency calculation. Below the minimal value, duty cycle or frequency will be limited by MinValue or default, if MinValue is not specified, see Figure 4-33. By default, minimal velocity is zero.
MaxVelocity	[Optional] Maximal velocity for duty cycle or frequency calculation. Above the maximal value, duty cycle or frequency will be limited by MaxValue or default, if MaxValue is not specified, see Figure 4-33. By default, maximal velocity is defined by XVEL parameter.

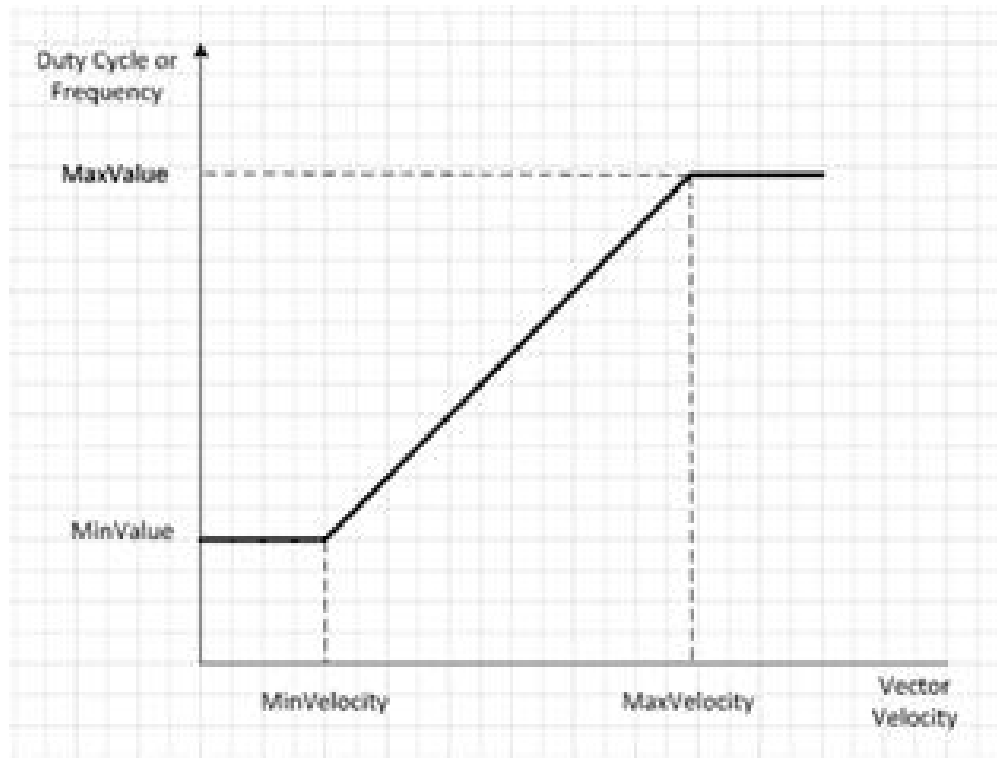


Figure 4-33. Velocity

Status

Modulation active/not active status is represented in AST.#LCMODUL (bit 22).

Example

Add example for regular mode and tickle mode initialization.

```
LCMODULATION (Laser, ! Laser index in the system
              1, ! Mode with fixed frequency
              5000, ! Frequency 5kHz
              0.001, ! Pulse width 1 microsec
              0, ! Initial duty cycle 0%
              0x03, ! Vector velocity of axis 0 and 1 defines duty cycle
              0, ! No Constant Spacing mode
              0, 100, ! Minimal 0% and maximal 100% duty cycle
              10, 1000) ! Minimal 10mm/sec and maximal 1000mm/sec velocity
```

Vector velocity indication (GVEL)

If **AxesUsed** parameter specifies involved axes mask, then the function calculates the vector velocity of all involved axes. The calculated vector velocity is indicated in GVEL parameter. The GVEL parameter index should corresponds to the last (fourth) local logical axis of the specific LCM unit. For example, if LCM gets network indexes (logical axes) 4,5,6,7, so index 7 should be used for monitoring the calculated vector velocity.

4.10.1.1 Duty cycle or frequency update

The **LCMODULATION** function provides ability to automatically calculate vector velocity of the axes involved in a specific motion and update either a laser output duty cycle or frequency accordingly with the actual vector velocity.

In some cases, the user needs to implement more complex user-specific function for duty cycle or frequency update. In this case **AxesUsed** argument should be zero and the user is responsible to update duty cycle or frequency using **PGFPAR** parameter.

4.10.1.2 Duty cycle or Frequency monitoring

The controller provides ability to monitor an actual frequency or duty cycle at real-time by means of **LCFREQ** and **LCDC** controller variables.

LCFREQ and **LCDC** are real type array of 64 elements. Each element corresponds to the specific axis that belongs to LCM unit. The same index that is defined in **LCMODULATION** should be used as index for **LCFREQ** and **LCDC** reference.

	Type	Accessibility	Description
LCFREQ	Real Array 0..63	Read-only	Indicates an actual frequency in [Hz] for modes 2 and 3, or configured frequency value for mode 1.
LCDC	Real Array 0..63	Read-only	Indicates an actual duty cycle in [%] for modes 1, or configured duty cycle value for mode 2 and 3.

4.10.2 LCFixedDist

The function initializes the fixed distance pulse firing mode. This mode is useful if a laser be should be activated with specified fixed intervals between activations along the motion trajectory.

As pulses are generated along an actual motion trajectory. The **AxesUsed** argument defines which axes are used for multi-axis trajectory generation.

The arguments **StartPosition**, **StopPosition** defines start and last points in the user units relatively the user origin, defined by **InitOffset**. The pulse are generated with constant interval, defined by **Interval**.

Additionally, number of extra pulses, specified by **ExtraPulses**, with period, specified by **ExtraPeriod**, can be generated after each pulse at **Interval** position.

The **UserVector** argument provides ability to implement user-defined multi-axes trajectory. The **AxesUsed** argument should be zero in this case.

The **MinPosition** and **MaxPosition** arguments define an operation zone range along the multi-axes trajectory. The laser pulses start outputting after reaching **MinPosition** and stop outputting after reaching **MaxPosition**.

Syntax

```
LCFIXEDDIST (Index, AxesUsed, Width, StartPosition, Interval, StopPosition
[, InitOffset, ExtraPulses, ExtraPeriod, PulseResolution, UserVector,
MinPosition, MaxPosition])
```

Arguments

Arguments	Comments
Index	Defines which Laser Control unit is referred. The system allocates 4 indexes (logical axes) for each LCM in the network. Third index of allocated logical axes should be specified. For example, if LCM gets network indexes (logical axes) 4,5,6,7, so index 6 should be used for referring the specific LCM module.
AxesUsed	Mask that defines the axes, which are used for generating pulses along the multi-axis motion trajectory. For example, the following mask defines that axes 0 and 2 should be used for vector velocity calculation: 0b0101 (0b – is a prefix for binary value specification) or 0x5 (0x – is a prefix for hexadecimal value specification) Only first 32 axes (0..31) can be referred by AxesUsed. If AxesUsed is zero and UserVector is zero, the previously defined fixed distance pulse firing mode is canceled. The AxesUsed should be zero, if UserVector is used for user-defined trajectory calculation.
Width	Pulse Width in milliseconds, from 0.00002664msec (26.64nsec) to 1.745msec
StartPosition	Specifies a start position along the multi-axis motion trajectory. The pulses start generating after reaching this position. The StartPosition is specified relatively to the user origin, defined by InitOffset and in the user units of the axes, defined by AxesUsed.
Interval	Specifies an interval between pulses along the multi-axis motion trajectory. The Interval is specified in the user units of the axes, defined by AxesUsed

StopPosition	Specifies a last position along the multi-axis motion trajectory. The pulses stop generating after reaching this position. The StopPosition is specified relatively to the user origin, defined by InitOffset and the user units of the axes, defined by AxesUsed.
InitOffset	[Optional] Specifies an initial position offset of multi-axis trajectory relatively to the user origin. By default, the function initializes an initial position of multi-axis trajectory as current actual position of the involved axes at the moment the function is called, so InitOffset is zero. For all functions: > For multi-axes-trajectory, the InitOffset should be omitted or set to zero. This causes the trajectory calculation to start from the current position. > For single-axis trajectory, if APOS of the axis, which is specified in AxesUsed, is provided as an argument, then both RPOS of the real axis and the virtual axis will match.
ExtraPulses	[Optional] Number of additional pulses to be generated with ExtraPeriod period after each pulse at each Interval position. By default, no extra pulses are generated.
ExtraPeriod	[Optional] Time period in milliseconds for ExtraPulses additional pulses to be generated after each pulse at each Interval position. By default, no extra pulses are generated.

PulseResolution	[Optional] Pulse Resolution in user units. If the parameter is omitted or specified as zero, then the function automatically calculates the most optimal pulse resolution according with maximal velocity (XVEL) of the axes, defined by AxesUsed, and the internal pulse generator maximal frequency, which is 12.5MHz. An actual pulse resolution is indicated in STEPF parameter. If the user needs to modify the resolution (higher or lower), this parameter allows to do it. The width of internal pulse generator is initialized to its minimal value of 40nsec to allow achieving the maximal pulses frequency.  For ALL functions - if the specified resolution causes the pulse generator frequency to reach its maximal value during the motion, then the Velocity Limit fault is raised and the LCM stops generating pulses.
UserVector	[Optional] The user defined global real scalar variable that contains result of multi-axes trajectory calculation. This option is used if it's required that pulse will be generated along a user-defined multi-axes trajectory, which implements some custom coordinate system transformation. Usually, user defined global scalar variable should be updated each controller cycle, so the calculation should be done in a separate buffer inside an endless LOOP statement. The argument should be zero, if AxesMask is defined.
MinPosition	[Optional] Specifies a minimal position of the operation zone along the motion trajectory. The laser pulses start outputting after reaching this position. The MinPosition is specified relatively to the user origin, defined by InitOffset and in the user units of the axes, defined by AxesUsed.
MaxPosition	[Optional] Specifies a maximal position of the operation zone along the motion trajectory. The laser pulses stop outputting after reaching this position. The MaxPosition is specified relatively to the user origin, defined by InitOffset and in the user units of the axes, defined by AxesUsed.

Example

```
LCFIXEDDIST (Laser, ! Laser index in the system
              0x03, ! Axis 0 and 1 defines multi-axis trajectory for
              generating pulses
```

```
0.001, ! Pulse width 1 microsec
100, ! First pulse position
10, ! Spacing between pulses is user 10 units (µm, mm)
000) ! Last pulse position
```

4.10.3 LCFixedInt

The function is similar to the **LCFixedDist** distance and also initializes the fixed distance pulse firing mode.

Unlike the **LCFixedDist** function, the **LCFixedInt** function does not provide the ability of Extra Pulses generation.

Also the **LCFixedInt** function does not contain the **StartPosition**, **StopPosition**, and **PulseResolution** specifications.

All other capabilities of **LCFixedDist** function are supported.

This mode is useful if a laser is to be activated at specified fixed intervals between activations along the motion trajectory.

As pulses are generated along an actual motion trajectory, the **AxesUsed** argument defines which axes are used for multi-axis trajectory generation.

The user origin is defined by **InitOffset**. The pulse are generated with constant interval, defines by **Interval**.

The **UserVector** argument provides ability to implement user-defined multi-axes trajectory. The **AxesUsed** argument should be zero in this case.

The **MinPosition** and **MaxPosition** arguments define an operation zone range along the multi-axes trajectory. The laser pulses start outputting after reaching **MinPosition** and stop outputting after reaching **MaxPosition**.

Syntax

```
LCFIXEDINT (Index, AxesUsed, Width, Interval, [, InitOffset, UserVector,
MinPosition, MaxPosition])
```

Arguments

Arguments	Comments
Index	Defines which Laser Control unit is referred. The system allocates 4 indexes (logical axes) for each LCM in the network. First index of allocated logical axes should be specified. For example, if LCM gets network indexes (logical axes) 4,5,6,7, so index 6 should be used for referring the specific LCM module.

AxesUsed	Mask that defines the axes, which are used for generating pulses along the multi-axis motion trajectory. For example, the following mask defines that axes 0 and 2 should be used for vector velocity calculation: 0b0101 (0b – is a prefix for binary value specification) or 0x5 (0x – is a prefix for hexadecimal value specification) Only first 32 axes (0..31) can be referred by AxesUsed. If AxesUsed is zero and UserVector is zero, then the previously defined fixed distance pulse firing mode is canceled. The AxesUsed should be zero, if the UserVector is used for user-defined trajectory calculation.
Width	Pulse Width in milliseconds, range from 0.00008ms (80nsec) to 0.08188msec (81.88μsec).
Interval	Specifies an interval between pulses along the multi-axis motion trajectory. The Interval is specified in the user units of the axes, defined by AxesUsed
InitOffset	[Optional] Specifies an initial position offset of multi-axis trajectory relatively to the user origin. By default, the function initializes an initial position of multi-axis trajectory as current actual position of the involved axes at the moment the function is called, so InitOffset is zero.
UserVector	[Optional] The user defined global scalar variable that contains result of multi-axes trajectory calculation. This option is used if it's required that pulse will be generated along a user-defined multi-axes trajectory, which implements some custom coordinate system transformation. Usually, user defined global real scalar variable should be updated each controller cycle, so the calculation should be done in a separate buffer inside an endless LOOP statement. The argument should be zero, if AxesMask is defined.
MinPosition	[Optional] Specifies a minimal position of the operation zone along the motion trajectory. The laser pulses start outputting after reaching this position. The MinPosition is specified relatively to the user origin, defined by InitOffset and in the user units of the axes, defined by AxesUsed.

MaxPosition

[Optional] Specifies a maximal position of the operation zone along the motion trajectory. The laser pulses stop outputting after reaching this position.

The **MaxPosition** is specified relatively to the user origin, defined by **InitOffset** and in the user units of the axes, defined by **AxesUsed**.

Example

```
LCFIXEDINT (Laser, ! Laser index in the system
            0x03, ! Axis 0 and 1 defines multi-axis trajectory for
generating pulses
            0.001, ! Pulse width 1 microsec
            10) ! Spacing between pulses is user 10 units (um, mm)
```

4.10.4 LCRandomDist

The function initializes either array based pulse firing mode or gating mode.

As pulses are generated along an actual motion trajectory, the **AxesUsed** argument defines which axes are used for multi-axis trajectory generation.

The Mode parameter defines either Pulse Firing or Gating mode is initialized.

The arguments **FirstIndex**, **LastIndex** define a first and a last elements in the **Points** array. Only elements from **FirstIndex** to **LastIndex** define points where pulses will be fired. Point coordinates are defined in the user units relatively to the user origin, defined by **InitOffset**.

The **States** argument is used for array based gating mode, where a laser is to be switched on/off at the predefined locations.

Additionally, the number of extra pulses, specified by **ExtraPulses**, with a period, specified by **ExtraPeriod**, can be generated after each pulse.

The **Inversion** argument defines if the pulse and state output need to be inverted.

The **UserVector** argument provides the ability to implement a user-defined multi-axes trajectory. The **AxesUsed** argument should be zero in this case.

The **MinPosition** and **MaxPosition** arguments define an operation zone range along the multi-axes trajectory. The laser pulses start outputting after reaching **MinPosition** and stop outputting after reaching **MaxPosition**.

Syntax

LCRANDOMDIST (**Index**, **Mode**, **AxesUsed**, **Width**, **FirstIndex**, **LastIndex**, **Points**[, **States**, **InitOffset**, **ExtraPulses**, **ExtraPeriod**, **Inversion**, **PulseResolution**, **UserVector**, **MinPosition**, **MaxPosition**])

Arguments

Arguments	Comments
Index	Defines which Laser Control unit is referred. The system allocates 4 indexes (logical axes) for each LCM in the network. Third index of allocated logical axes should be specified.

	For example, if LCM gets network indexes (logical axes) 4,5,6,7, then index 6 is to be used for referring the specific LCM module.
Mode	Defines which mode, Pulse Firing or Gating, is initialized. 1 – Pulse Firing mode 2 – Gating mode For Gating mode, States argument should be specified.
AxesUsed	Mask that defines the axes, which are used for generating pulses along the multi-axis motion trajectory. For example, the following mask defines that axes 0 and 2 should be used for vector velocity calculation: 0b0101 (0b – is a prefix for binary value specification) or 0x5 (0x – is a prefix for hexadecimal value specification) Only first 32 axes (0..31) can be referred by AxesUsed. If AxesUsed is zero and UserVector is zero, then the previously defined fixed distance pulse firing mode is canceled. The AxesUsed should be zero, if the UserVector is used for the user-defined trajectory calculation.
Width	Pulse Width in milliseconds, range from 0.00002664msec (26.64nsec) to 1.745msec
FirstPoint	Specifies a first element in the Points and States arrays. The pulses start generating after reaching the position defined in this element.
LastPoint	Specifies a last element in the Points array. The pulses stop generating after reaching the position defined in this element.
Points	Array of points. Each element of the array defines the point where pulse should be fired. The point coordinates are specified relatively to the user origin, defined by InitOffset in the user units of the axes, defined by AxesUsed.
States	[Optional] Array of states. Each element of the array defines whether the laser should be switched On or OFF. States is only used for the Array based Gating Mode.
InitOffset	[Optional] Specifies an initial position offset of multi-axis trajectory relatively to the user origin. By default, the function initializes an initial position of multi-axis trajectory as current actual position of the involved axes at the moment the function is called, so InitOffset is zero.

ExtraPulses	[Optional] Number of additional pulses to be generated with ExtraPeriod period after each pulse at each Interval position. By default, no extra pulses are generated.
ExtraPeriod	[Optional] Time period in milliseconds for ExtraPulses additional pulses to be generated after each pulse at each Interval position. By default, no extra pulses are generated.
Inversion	[Optional] Inversion of pulse and states output logic 0 (default) – no inversion 1 – inversion
PulseResolution	[Optional] Pulse Resolution in user units. If the parameter is omitted or specified as zero, the function automatically calculates the most optimal pulse resolution according with maximal velocity (XVEL) of the axes, defined by AxesUsed, and the internal pulse generator maximal frequency, which is 12.5MHz. An actual pulse resolution is indicated in STEPFP parameter. If the user needs to modify the resolution (higher or lower), this parameter allows to do it. The width of internal pulse generator is initialized to its minimal value of 40nsec to allow achieving the maximal pulses frequency.
UserVector	[Optional] The user defined global real scalar variable that contains result of multi-axes trajectory calculation. This option is used if it's required that pulse will be generated along a user-defined multi-axes trajectory, which implements some custom coordinate system transformation. Usually, user defined global scalar variable should be updated each controller cycle, so the calculation should be done in a separate buffer inside an endless LOOP statement. The argument should be zero, if AxesMask is defined.
MinPosition	[Optional] Specifies a minimal position of the operation zone along the motion trajectory. The laser pulses start outputting after reaching this position. The MinPosition is specified relatively to the user origin, defined by InitOffset and in the user units of the axes, defined by AxesUsed.

MaxPosition

[Optional] Specifies a maximal position of the operation zone along the motion trajectory. The laser pulses stop outputting after reaching this position.

The **MaxPosition** is specified relatively to the user origin, defined by **InitOffset** and in the user units of the axes, defined by **AxesUsed**.

Example

```
LCFIXEDDIST (Laser, ! Laser index in the system
0x03, ! Axis 0 and 1 defines multi-axis trajectory for generating pulses
0.001, ! Pulse width 1 microsec
100, ! First pulse position
10, ! Spacing between pulses is user 10 units (um, mm)
000) ! Last pulse position
```

4.10.5 LCTickle

The function initializes Tickle mode. In this mode the laser control unit generates a signal at constant frequency and with constant width. Usually, this mode is used for those types of lasers that require gas ionization, during the time period when laser processing is off. By maintaining this mode, the laser will respond faster and more predictably, when the laser processing resumes.

Once this mode is initialized, the laser control unit constantly generates pulses, irrespective to any other operational modes. By default, the tickle mode generate pulses to the same output pin, which is used for the operational modes. In this case, the tickle mode pulses are superimposed on operational mode pulses. If necessary, the tickle mode pulses can be redirected to other available pins, see [Physical outputs configuration](#) for details.

Syntax

LCTICKLE (**Index**, **Frequency**, **Width**[, **Connector**])

Arguments	Comments
Index	Defines which laser control unit is referred. The system allocates 4 indexes (logical axes) for each LCM in the network. The last index of allocated logical axes should be specified. For example, if LCM gets network indexes (logical axes) 4,5,6,7, so index 7 should be used for referring the specific LCM module.
Frequency	Tickle frequency in Hz Minimal frequency is 1149Hz, maximal frequency is 100,000Hz Not any frequency can be configured, so the function automatically rounds the specified frequency to the nearest supported value. Actual value of the configured tickle frequency is updated in LCTFREQ variable. The same index that is defined in LCTICKLE should be used as index for LCTFREQ.

Width	Pulse Width in milliseconds. Minimal width is 0.000107msec (106nsec), maximal width is 0.0273msec (27.3µsec) Not any width can be configured, so the function automatically rounds the specified width to the nearest supported value. Actual value of the configured tickle pulse width is updated in LCTWIDTH variable. The same index that is defined in LCTICKLE should be used as index for LCTWIDTH.
Connector	[Optional] Specifies which connector (J1/J2/J3) to output tickle pulses for: 1 - J1 2 - J2 3 - J3

4.10.6 LCZone

Syntax

LCZONE (**Index**, **AxesUsed**, **MinPosition**, **MaxPosition** [, **InitOffset**, **PulseResolution**])

Arguments	Comments
Index	Defines which Laser Control unit is referred. The system allocates 4 indexes (logical axes) for each LCM in the network. The Index parameter should corresponds to the first logical axis (for Zone 0), second logical axis (for Zone 1) or third logical axis (Zone 2) of the specific LCM unit. For example, if LCM gets network indexes (logical axes) 4,5,6,7, so index 4 should be used for referring Zone 0. It's recommended to refer only Zone 0 or Zone 1, as Zone 2 is used by LCFixedDist or LCFixedInt functions.
AxesUsed	Mask that defines the axes, which are used for generating single-axis or multi-axis trajectory that is used for comparing its actual position with operation zone boundaries. The laser pulses are allowed to be outputted only within operation zone. For example, the following mask defines that axes 0 and 2 should be used for vector velocity calculation: 0b0101 (0b – is a prefix for binary value specification) or 0x5 (0x – is a prefix for hexadecimal value specification) Only first 32 axes (0..31) can be referred by AxesUsed. If AxesUsed is zero, the previously defined zone is canceled.

MinPosition	Specifies a minimal position of the operation zone along the motion trajectory. The laser pulses start outputting after reaching this position. The MinPosition is specified relatively to the user origin, defined by InitOffset and in the user units of the axes, defined by AxesUsed.
MaxPosition	Specifies a maximal position of the operation zone along the motion trajectory. The laser pulses stop outputting after reaching this position. The MaxPosition is specified relatively to the user origin, defined by InitOffset and in the user units of the axes, defined by AxesUsed.
InitOffset	[Optional] Specifies an initial position offset of multi-axis trajectory relatively to the user origin. By default, the function initializes an initial position of multi-axis trajectory as current actual position of the involved axes at the moment the function is called, so InitOffset is zero
PulseResolution	[Optional] Pulse Resolution in user units. If the parameter is omitted or specified as zero, the function automatically calculates the most optimal pulse resolution according with maximal velocity (XVEL) of the axes, defined by AXESUSED, and the internal pulse generator maximal frequency, which is 12.5MHz. An actual pulse resolution is indicated in STEPF parameter. If the user needs to modify the resolution (higher or lower), this parameter allows to do it. The width of internal pulse generator is initialized to its minimal value of 40nsec to allow achieving the maximal pulses frequency.

Example

```
LCZONE (Laser, ! Laser index in the system
        0, ! Zone 0 is used
        0, ! Zone is defined for the axis 0
        1000, 2000) ! Laser zone range is 1000 ... 2000
```

Vector path indication (GPATH)

If the **AxesUsed** parameter specifies involved an axes mask, then the function calculates the single-axis or multi-axis trajectory path. The current value of the path is indicated in the **GPATH** parameter. The **GPATH** parameter index corresponds to the first logical axis (for Zone 0), second logical axis (for Zone 1) or third logical axis (Zone 2) of the specific LCM unit. For example, if LCM gets network indexes (logical axes) 4,5,6,7, then index 5 is to be used for monitoring the current vector path for the Zone 1.

4.10.6.1 LCZoneSet

The minimum and maximum zone limit can be changed using LCZoneSet function.

Syntax

LCZONESET (**Index**, **MinPosition**, **MaxPosition**)

Arguments

Arguments	Comments
Index	Should specify the same zone that was defined by LaserZone function.
MinPosition	Specifies a minimal position of the operation zone along the motion trajectory. The laser pulses start outputting after reaching this position. The MinPosition is specified relatively to the initial position and in the user units, calculated by LaserZone function.
MaxPosition	Specifies a maximal position of the operation zone along the motion trajectory. The laser pulses stop outputting after reaching this position. The MaxPosition is specified relatively to the initial position and in the user units, calculated by LaserZone function.

Example

```
LCZONESET (Zone0, ! Laser index in the system
           1000, 2000) ! Laser zone range is 1000 ... 2000
```

4.10.6.2 LCZoneGet

The function **LCZoneGet** returns the limits of the laser operation zone that was previously defined by **LCZone** function or by **LCZoneSet** function.

Syntax

LCZONEGET (**Index**, **ZoneLimit**)

Arguments

Arguments	Comments
Index	Should specify the same zone that was defined by LaserZone function.
ZoneLimit	0 – to get a minimal zone limit 1 - to get a maximal zone limit

Return Value

The function returns the currently configured limit of the specified laser zone.

Example

```
MinLimit = LCZONEGET (Zone0, ! Laser index in the system  
0) ! Minimal limit of the zone
```

4.10.7 LCStop

The function stops any previously initialized laser mode and resets the previously defined mode parameters. The function is useful in order to ensure that any of the previously defined modes are stopped.

Syntax

LCStop (**Index**)

Arguments

Arguments	Comments
Index	Designates the specific axis to which stopping and resetting the mode will be applied

4.10.8 LCSignalSet

The function configures LCS output conditioning state.

Syntax

LCSignalSet (**Index**, **ConditionMask**)

Arguments

Arguments	Comments
Index	Defines which Laser Control unit is referred. The system allocates 4 axes for each LCM in the network. First of the allocated axes should be specified.

ConditionMask

Bit code that defines the logic conditions for Laser Control Signal (LCS) outputting.

By default, the LCS is generated with no dependance on other signals. Means that only bit 11 (LCS) is set.

Using this function, the user can define additional conditions for LCS outputting. Each bit that is set to 1, adds the dependance of LCS on the specific internal signal.

For example, if necessary that LCS (bit 11) is generated only if InRange 0 signal (bit 4) is "1" and PEG State 0 (bit 12) is "1", then the bit code 0x00001810 should be defined: bits 4, 11 and 12 are set.

Bit	Signal
0	P/D Pulse 0
1	P/D Dir 0
2	PEG Pulse The bit is automatically set by LCFixedDist function or LCRandomDist function in case of the Pulse Firing Mode
3	PEG Active
4	InRange 0 The bit is automatically set by LCZone function
5	P/D Pulse 1
6	P/D Dir 1
7	InRange 1 The bit is automatically set by LCZone function
8	P/D Pulse 2
9	P/D Dir 2
10	InRange 2 The bit is automatically set by LCZone function or LCRandomDist or LCFixedDist or LCFixedInt
11	Laser Control Signal (by default = 1) The bit is automatically set by LCModulation function

Bit	Signal
12	PEG State #0 The bit is automatically set by LCRandomDist function in case of the Gating mode
13	PEG State #1
14	PEG_State #2
15	Tickle output The bit is automatically set by LCTickle function if Connector argument is not specified or omitted
16	Input 0
17	Input 1
18	Input 2
19	Input 3
20	Input 4
21	Input 5
22	Input 6
23	Input 7
24	Output 0
25	Output 1
26	Output 2
27	Output 3
28	Output 4
29	Output 5
30	Output 6
31	Output 7



If it is necessary to set new bits and to preserve the bits that have been previously set, it is recommended to begin by retrieving the current state by using the function **LCSignalGet** and then to set the new bits, see [LCS conditioning example](#).



Inputs 0-7 conditioning is not supported in SPiiPlusCMHP/BA.

4.10.9 LCSignalGet

The function returns LCS output conditioning state.

Syntax

`LCSignalGet (Index)`

Arguments

The function returns the value that represents the outputs configuration state depending on the **Connector** parameter.

Arguments	Comments
Index	Defines which Laser Control unit is referred. The system allocates 4 axes for each LCM in the network. First of the allocated axes should be specified.
	Bit code that defines the logic conditions for Laser Control Signal (LCS) outputting. The meaning if the bits is desribed in LCSignalSet arguments table.

Return value

The function returns bit code that defines the logic conditions for LCS outputting. The meaning if the bits is desribed in **LCSignalSet** arguments table.

4.10.10 LCS conditioning example

The example shows how to configure additional conditions to LCS outputting function.

```
! Configure laser output conditioning
! Bits 4(InRange0), 7(InRange1) are added to LCS conditioning
! Means that LCS is outputted only if both InRange0 and InRange1 signals
are "1"
LCSignalSet (Index, LCSignalGet (Index) | 0x0090)

! Add Tickle output to the LCS
! Add Tickle output (Bit 15) to LCS conditioning
! Means that tickle mode is superimposed on signal generated by PFG
```



```
module (modes 1, 2 or 3)  
LCSignalSet (Index, LCSignalGet (Index) | 0x8000)
```

4.10.11 Physical outputs configuration

The LCM provides the following flexibility in digital outputs configuration: the logical outputs of the internal modules, like P/D pulse, PEG pulse, InRange can be redirected to available pins of J1, J2 and J3 connectors. The configuration is done by [LCOutputSet](#) function, described below. The outputs configuration can be retrieved using [LCOutputGet](#) function.

4.10.11.1 LCOutputSet

The LCOutputSet function configures LCM physical outputs.

Syntax

`LCOutputSet (Index, Connector, Code)`

Arguments	Comments
Index	Defines which Laser Control unit is referred. The system allocates 4 axes for each LCM in the network. First of the allocated axes should be specified.
Connector	Specifies LCM connector: 01 – J1 02 – J2 03 – J3

Arguments	Comments			
Code	Connector value	Return value	#_PULSE output	#_DIR output
	01 or 02 or 03	0x00	None	None
	01	0x01 *See note below.	Counter 0 A signal	Counter 0 B signal
	01	0x02 *See note below.	Pulse 0	Direction 0
	01	0x03	In range 0	"1"
	01 or 02 or 03	0x04	PEG pulse	PEG active
	01 or 02 or 03	0x05	PEG state 0	PEG active
	01 or 02 or 03	0x09	Tickle	"1"
	02	0x01 *See note below.	Counter 1 A signal	Counter 1 B signal
	02	0x02 *See note below.	Pulse 1	Direction 1
	02	0x03	In range 1	"1"
	03	0x01 *See note below.	Counter 2 A signal	Counter 2 B signal
	03	0x02 *See note below.	Pulse 2	Direction 2
	03	0x03	In range 2	"1"



*Only supported by LCM modules with the AqB and P/D output option.

4.10.11.2 LCOutputGet

The **LCOutputGet** function returns LCM outputs configuration.

Syntax

LCOutputGet (**Index**, **Connector**)

Arguments

The function returns the value that represents the outputs configuration state depending on the **Connector** parameter.

Arguments	Comments
Index	Defines which Laser Control unit is referred. The system allocates 4 axes for each LCM in the network. First of the allocated axes should be specified.
Connector	Specifies LCM connector: 01 – J1 02 – J2 03 – J3

Return value

Connector value	Return value	#_PULSE output	#_DIR output
01 or 02 or 03	0x00	None	None
01	0x01	Counter 0 A signal	Counter 0 B signal
01	0x02	Pulse 0	Direction 0
01	0x03	In range 0	"1"
01 or 02 or 03	0x04	PEG pulse	PEG active
01 or 02 or 03	0x05	PEG state 0	PEG active
01 or 02 or 03	0x09	Tickle	"1"
02	0x01	Counter 1 A signal	Counter 1 B signal
02	0x02	Pulse 1	Direction 1

Connector value	Return value	#_PULSE output	#_DIR output
02	0x03	In range 1	"1"
03	0x01	Counter 2 A signal	Counter 2 B signal
03	0x02	Pulse 2	Direction 2
03	0x03	In range 2	"1"

4.10.12 LCDelaySet

The pulse generation delay can also be configured by the user using **LCDelaySet** function. The specified delay will override the default value that is set by default. The currently configured delay can be read using **LCDelayGet** function.

Syntax

LCDelaySet (**Index**, **Delay**)

Arguments

Arguments	Comments
Index	Defines which Laser Control unit is referred. The system allocates 4 axes for each LCM in the network. First of the allocated axes should be specified.
Delay	Delay in microseconds: 0-CTIME*1000(μsec)

4.10.13 LCDelayGet

The function returns the actual currently configured delay in microseconds.

Syntax

LCDelayGet (**Index**)

Arguments

Arguments	Comments
Index	Defines which Laser Control unit is referred. The system allocates 4 axes for each LCM in the network. First of the allocated axes should be specified.

Return Value

The function returns the actual currently configured delay in microseconds.

4.10.14 AxListAsMask

In the functions **LcModulation**, **LcFixedDist**, **LcFixedInt**, **LcRandomDist**, **LcZone** an argument requires a mask to define the axes. The mask specification is defined with **AxListAsMask**.

AxListAsMask (axis_list)

Arguments	Comments
axis_list	List of the axes, separated by comma: 0, 2, 4

The function returns the integer value that represents the axis list as a mask.

```
int AxisMask
AxisMask = AxListAsMask(0,2,4)  ! AxisMask gets value 0x15 (0b00010101)
scripton in the manual.
```

The Dynamic Error Compensation functions are:

Function	Description
ERRORMAP1D	Configures and activates 1D error correction for the mechanical error compensation for the specified zone,
ERRORMAP2D	Configures and activates 2D error correction for the mechanical error compensation of the 'axis0' command for the specified zone
ERRORMAPN1D	Configures and activates 1D error correction for the mechanical error compensation for the specified zone
ERRORMAPN2D	Configures and activates 2D error correction for the mechanical error compensation of the 'axis0' command for the specified zone,
ERRORMAPA1D	Configures and activates 1D error correction for the mechanical error compensation for the specified zone, so that the compensated reference position will be calculated by multiplying the scaling factor by the desired position so that the actual value will be closer to the desired value.
ERRORMAPA2D	Configures and activates 2D error correction for the mechanical error compensation of the specified axis for the specified zone, so that the compensated reference position will be calculated by taking into account the angle for the orthogonality correction so that the actual value will be closer to the desired value.

Function	Description
<code>ERRORMAPOFF</code>	Deactivates error mapping correction for the mechanical error compensation for the specified zone.
<code>ERRORMAPON</code>	Activates error correction for the mechanical error compensation for the specified zone.
<code>#ERRORMAPREP</code>	Generates a report of all activated zones of error mapping for all axes in the system.
<code>ERRORUNMAP</code>	Deactivates error correction for the mechanical error compensation for the specified zone.

4.11.1 `ERRORMAP1D`

Description

The **`ERRORMAP1D`** function configures and activates 1D error correction for the mechanical error compensation for the specified zone, so that the compensated reference position will be calculated by subtracting the linearly (by default) interpolated error from the desired position so that the actual value will be closer to the desired value.

The calculation assumes fixed Intervals between points inside the zone.

Syntax

```
ERRORMAP1D[switches] axis, zone, base, step, correction_map, [referenced_
axis_or_analog_input]
```

Arguments

axis	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.
base	A real number representing the axis command that corresponds to the first point in correction table for mechanical error compensation.
step	A real number representing the fixed interval distance between the two adjacent axis commands.
correction_map	The name of a real one-dimensional array that specifies correction table for mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).

referenced_
axis_or_analog_
input

[Optional] The index of the axis, or the index of the analog input that the mechanical error compensation will be calculated based on its feedback.

Switches

/p	Prevent applying dynamic error compensation on INDEX , MARK , and PEG values
/a	Specifies that the mechanical error compensation will be calculated based on the feedback from the axis specified by the optional parameter.
/i	Specifies that the mechanical error compensation will be calculated based on the feedback from the analog input indicated by the optional parameter.

Error Conditions

The function detects the following error conditions:

- > Error 2044, Index is out of range, when the defined array size is less than the defined number of array points.
- > Error 3113, The step in the table is zero or negative, when the step argument is zero or negative.

Comments

If erroneous parameters are passed to the function, the corresponding runtime error will be generated. The function is intended to be used with arrays only, meaning that an error is generated if a scalar is passed as a parameter.

4.11.2 *ERRORMAPN1D*

Description

The **ERRORMAPN1D** function configures and activates 1D error correction for the mechanical error compensation for the specified zone, so that the compensated reference position will be calculated by subtracting the linearly (by default) interpolated error from the desired position so that the actual value will be closer to the desired value.

The calculation is based on an arbitrary network of points inside the zone.

Syntax

```
ERRORMAPN1D[switches] axis, zone, axis_command, correction_map,  
[referenced_axis_or_analog_input]
```

Arguments

axis	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.
axis_command	The name of a real one-dimensional array that specifies axis command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_map	The name of a real one-dimensional array that specifies correction table for mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
referenced_ axis_or_analog_ input	[Optional] The index of the axis or the index of the analog input to be used for the calculation of the mechanical error compensation.

Switches

/p	Prevent applying dynamic error compensation on INDEX, MARK, and PEG values
/a	Specifies that the mechanical error compensation will be calculated based on the feedback from the axis specified by the optional parameter.
/i	Specifies that the mechanical error compensation will be calculated based on the feedback from the analog input indicated by the optional parameter.

Error Conditions

The function detects the following error conditions:

- > Error 2044, Index is out of range, when the defined array size is less than the defined number of array points.
- > Error 3113, The step in the table is zero or negative, when the step argument is zero or negative.

Comments

In case of erroneous parameters, the relevant runtime error will be generated. The function is intended for use with arrays, meaning that an error is generated if a scalar is given as a parameter.

4.11.3 ERRORMAPA1D**Description**

The **ERRORMAPA1D** function configures and activates 1D error correction for the mechanical error compensation for the specified zone, so that the compensated reference position will be calculated

by multiplying the scaling factor by the desired position so that the actual value will be closer to the desired value.

Syntax

```
ERRORMAP1D[switches] axis, zone, scaling_factor, offset
```

Arguments

axis	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.
scaling_factor	The scaling factor for the linear alignment that will be used for mechanical error compensation. The allowed range for the scaling factor is (0, 2.0).
offset	The offset for the linear alignment that will be used for mechanical error compensation. The offset is actually the mechanical error compensation for the 0-point location.

Switches

/p	Prevent applying dynamic error compensation on INDEX, MARK, and PEG values
----	--

Error Conditions

The function detects the following error conditions:

- > Error 2044, Index is out of range, when the defined array size is less than the defined number of array points.
- > Error 3113, The step in the table is zero or negative, when the step argument is zero or negative.

Comments

In case of erroneous parameters, the corresponding runtime error will be generated. The function is intended to be used for arrays only, meaning that an error is generated if a scalar is given as a parameter.

This command is supported in ADK versions 2.70 and higher.

4.11.4 *ERRORMAP2D*

Description

The **ERRORMAP2D** function configures and activates 2D error correction for the mechanical error compensation of the '**axis0**' or '**axis1**' command (depending on the switch used) for the specified zone, so that the compensated reference position will be calculated by subtracting the linearly (by default) interpolated error from the desired position so that the actual value will be closer to the desired value.

Syntax

```
ERRORMAP2D[switches] (axis0, axis1), zone, base0, step0, base1, step1,
correction_map, [reference_axis_or_analog_input0, referenced_axis_or_
analog_input1]
```

Arguments

axis0	The index of the first axis that the mechanical error compensation will be applied to. Valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis1	The index of the second axis participating in 2D mechanical error compensation. Valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
base0	A real number representing the 'axis0' command that corresponds to the first point in correction table for mechanical error compensation.
step0	A real number representing the fixed interval distance between the two adjacent 'axis0' commands.
base1	A real number representing the 'axis1' command that corresponds to the first point in correction table for mechanical error compensation.
step1	A real number representing the fixed interval distance between the two adjacent 'axis1' commands.
correction_map	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation. The array or MATRIX type should be a GLOBAL REAL STATIC defined in the D-Buffer.
referenced_axis_or_analog_input0	[Optional] The index of the first axis, or the index of the first analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_axis_or_analog_input1	[Optional] The index of the second axis, or the index of the second analog input whose feedback will be used to calculate the mechanical error compensation.

Switches

/0	The mechanical error compensation will be applied to 'axis0' (default)
/1	The mechanical error compensation will be applied to 'axis1'
/a	Specifies that the mechanical error compensation will be calculated based on the feedback from the axis specified by the optional parameter.
/i	Specifies that the mechanical error compensation will be calculated based on the feedback from the analog input indicated by the optional parameter.
/aj	Specifies that the first optional parameter will be treated as an analog input index.
/ak	Specifies that the second optional parameter will be treated as an analog input index.

Error Conditions

The function detects the following error conditions:

- > Error 2044, Index is out of range, when the defined array size is less than the defined number of array points.
- > Error 3113, The step in the table is zero or negative, when the step argument is zero or negative.

Comments

If incorrect parameters are passed to the function the corresponding error will be generated. The function is intended to be used for arrays only, meaning that an error is generated if a scalar is given as a parameter.

4.11.5 ERRORMAPN2D**Description**

The **ERRORMAPN2D** function configures and activates 2D error correction for the mechanical error compensation of the '**axis0**' or '**axis1**' command (depending on the switch used) for the specified zone, so that the compensated reference position will be calculated by subtracting the linearly (by default) interpolated error from the desired position so that the actual value will be closer to the desired value.

Syntax

```
ERRORMAPN2D[switches] (axis0, axis1), zone, axis0_command, axis1_command,
correction_map, [reference_axis_or_analog_input0,
referenced_axis_or_analog_input1]
```

Arguments

axis0	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis1	The index of the second axis participating in 2D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.
axis0_command	The name of a real one-dimensional array that specifies 'axis0' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
axis1_command	The name of a real one-dimensional array that specifies 'axis1' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_map	The name of a real one-dimensional array that specifies correction table for mechanical error compensation. The array should be a GLOBAL REAL STATIC defined in the D-Buffer.
referenced_axis_or_analog_input0	[Optional] The index of the first axis or the index of the first analog input providing the feedback used for calculation of the mechanical error compensation.
referenced_axis_or_analog_input1	[Optional] The index of the second axis or the index of the second analog input providing the feedback used for calculation of the mechanical error compensation.

Switches

/0	The mechanical error compensation will be applied to 'axis0' (default).
/1	The mechanical error compensation will be applied to 'axis1'.
/a	Specifies that the mechanical error compensation will be calculated based on the feedback from the axis specified by the optional parameter.
/i	Specifies that the mechanical error compensation will be calculated based on the feedback from the analog input indicated by the optional parameter.
/aj	Specifies that the first optional parameter will be treated as an analog input index.
/ak	Specifies that the second optional parameter will be treated as an analog input index.

Error Conditions

The function detects the following error conditions:

- > Error 2044, index is out of range, when the defined array size is less than the defined number of array points.
- > Error 3113, the step in the table is zero or negative, when the step argument is zero or negative.

Comments

If incorrect parameters are passed to the function the relevant error will be generated. The function is intended to be used for arrays only, meaning that an error is generated if a scalar is given as a parameter.

4.11.6 *ERRORMAPA2D*

Description

The **ERRORMAPA2D** function configures and activates 2D error correction for the mechanical error compensation of the specified axis for the specified zone, so that the compensated reference position will be calculated by taking into account the angle for the orthogonality correction so that the actual value will be closer to the desired value.

Syntax

```
ERRORMAPA2D[switches] (axis0, axis1), zone, angle
```

Arguments

axis0	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis1	The index of the second axis participating in 2D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
angle	The angle for the orthogonality correction that will be used for mechanical error compensation. The allowed range for the angle is [-45°, 45°].

Switches

/0	The mechanical error compensation will be applied to 'axis0' (default)
/1	The mechanical error compensation will be applied to 'axis1'

Error Conditions

The function detects the following error conditions:

- > Error 2044, Index is out of range, when the defined array size is less than the defined number of array points.

- > Error 3113, The step in the table is zero or negative, when the step argument is zero or negative.

Comments

In case of wrong parameters, the corresponding runtime error will be generated. The function is intended to be used for arrays only, meaning that an error is generated if a scalar is given as a parameter.

This command is supported in ADK versions 2.70 and higher.

4.11.7 *ERRORMAP3DA*

Description

The **ERRORMAP3DA** function configures and activates 3D error correction for the mechanical error compensation of 'axis0', 'axis1', and 'axis2' for the specified zone, so that the compensated reference position will be calculated by adding the interpolated error from the desired position so that the actual value will be closer to the desired value. Interpolation is linear by default, other options are available.

The **ERRORMAP3DA** function receives the indices of three axes, the zone index, the base value of the 'axis0' command, the fixed defined interval of the 'axis0' command, the base value of the 'axis1' command, a fixed defined interval for the 'axis1' command, the base value of the 'axis2' command, the fixed defined interval of the 'axis2' command, and 10 2D correction tables correlated to the specified 'axis2' coordinates for mechanical error compensation.

Syntax

```
ERRORMAP3DA[command options] (axis0, axis1, axis2), zone, base1, step1,
base2, step2, base3, step3, correction_map0, correction_map1, correction_
map2, correction_map3 , correction_map4, correction_map5, correction_
map6, correction_map7, correction_map8, correction_map9[, reference_axis_
or_analog_input0, referenced _axis_or_analog_input1, referenced _axis_or_
analog_input2]
```

Arguments

axis0	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis1	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis2	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.

base0	A real number representing the 'axis0' command that corresponds to the first point in correction table for mechanical error compensation.
step0	A real number representing the fixed interval distance between the two adjacent 'axis0' commands.
base1	A real number representing the 'axis1' command that corresponds to the first point in correction table for mechanical error compensation.
step1	A real number representing the fixed interval distance between the two adjacent 'axis1' commands.
base2	A real number representing the 'axis2' command that corresponds to the first point in correction table for mechanical error compensation.
step2	A real number representing the fixed interval distance between the two adjacent 'axis2' commands.
correction_ map0	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '0' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map1	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '1' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map2	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '2' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map3	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '3' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map4	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '4' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map5	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '5' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).

correction_ map6	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '6' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map7	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '7' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map8	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '8' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map9	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '9' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
referenced_ axis_or_analog_ input0	[Optional] The index of the first axis, or the index of the first analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input1	[Optional] The index of the second axis, or the index of the second analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input2	[Optional] The index of the second axis, or the index of the third analog input whose feedback will be used to calculate the mechanical error compensation.

Switches

/0	The mechanical error compensation will be applied to 'axis0' (default)
/1	The mechanical error compensation will be applied to 'axis1'
/2	The mechanical error compensation will be applied to 'axis2'
/a	Specified that the mechanical error compensation will be calculated be on the feedback from the axis specified by the optional parameter.
/aj	Specifies that the first optional parameter to be treated as an analog input index.
/ak	Specifies that the second optional parameter to be treated as an analog input index.
/al	Specifies that the third optional parameter to be treated as an analog input index.

Error Conditions

The function detects the following error conditions.

- > **2044** - Index is out of range, when the defined array size is less than the defined number of array points.
- > **3113** - The step in the table is zero or negative, when the step argument is zero or negative.
- > **3384** - The specified suffix combination is invalid. Please see the documentation for more details.
- > **3413** - The supplied correction maps (2-dimensional arrays or MATRIX types) should have the same dimensions. Please see the documentation for more details.
- > **3414** - The Specified referenced axes/analog inputs are invalid. Please see the documentation for more details.

Comments

If incorrect parameters are passed to the function the corresponding runtime error is generated. The function is intended to be used only with arrays; thus, an error is generated if a scalar is given as a parameter.

4.11.8 *ERRORMAP3D2*

Description

The **ERRORMAP3D2** function configures and activates 3D error correction for the mechanical error compensation of the 'axis0' command, 'axis1' command, and 'axis2' command for the specified zone, so that the compensated reference position will be calculated by adding the interpolated error from the desired position so that the actual value will be closer to the desired value. Interpolation is linear by default; other options are available.

The **ERRORMAP3D2** function receives indexes of three axes, zone index, base value of the 'axis0' command, fixed defined interval of the 'axis0' command, base value of the 'axis1' command, fixed defined interval of the 'axis1' command, base value of the 'axis2' command, fixed defined interval of the 'axis2' command, and two 2D correction tables (in correlation to the specified 'axis2' coordinates) for mechanical error compensation.

Syntax

```
ERRORMAP3D2[command options] (axis0, axis1, axis2), zone, base0, step0,
base1, step1, base2, step2, correction_map0, correction_map1[, reference_
axis_or_analog_input0, referenced _axis_or_analog_input1, referenced _
axis_or_analog_input2]
```

Arguments

axis0	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis1	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.

axis2	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.
base0	A real number representing the 'axis0' command that corresponds to the first point in correction table for mechanical error compensation.
step0	A real number representing the fixed interval distance between the two adjacent 'axis0' commands.
base1	A real number representing the 'axis1' command that corresponds to the first point in correction table for mechanical error compensation.
step1	A real number representing the fixed interval distance between the two adjacent 'axis1' commands.
base2	A real number representing the 'axis2' command that corresponds to the first point in correction table for mechanical error compensation.
step2	A real number representing the fixed interval distance between the two adjacent 'axis2' commands.
correction_ map0	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '0' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map1	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '1' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
referenced_ axis_or_analog_ input0	[Optional] The index of the first axis, or the index of the first analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input1	[Optional] The index of the second axis, or the index of the second analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input2	[Optional] The index of the second axis, or the index of the third analog input whose feedback will be used to calculate the mechanical error compensation.

Switches

/0	The mechanical error compensation will be applied to 'axis0' (default)
/1	The mechanical error compensation will be applied to 'axis1'
/2	The mechanical error compensation will be applied to 'axis2'
/a	Specified that the mechanical error compensation will be calculated be on the feedback from the axis specified by the optional parameter.
/aj	Specifies that the first optional parameter will be regarded as an analog input index.
/ak	Specifies that the second optional parameter will be regarded as an analog input index.
/al	Specifies that the third optional parameter will be regarded as an analog input index.
/e	Specifies that the extrapolation will be used for Z values that are beyond the original observation range. That is, we are assuming that existing trends will continue (by using the nearest correction table to estimate the correction). This method is subject to greater uncertainty and a higher risk of producing meaningless results than interpolation.

Comments

- > When incorrect parameters are specified, the relevant compile-time or run-time error is generated. See [ACSPL+ Runtime Errors](#) for explanation of error 2044 and [ACSPL+ Compilation Errors](#) for explanations of errors 3113, 3384, 3413, 3414. The function is intended for use with arrays only; an error is generated if a scalar is given as a parameter.

Examples**Example 1**

This example uses two 2-dimensional arrays to create 2-dimensional correction maps that are used for the 3D Dynamic error compensation. Each map represents a different value (height) of the Z axis. An XYZ cuboid (rectangular prism) zone starts at coordinate -100 for X, 30 for Y, and 30 for Z. It has a fixed interval of 10 mm for each axis. The user units are represented in mm.

```
D-Buffer:
global real static X_Correction1(3) (10)
global real static X_Correction2(3) (10)

Buffer:
local int X_axis
local int Y_axis
local int Z_axis
local int zone
local int X_base
local int X_step
local int Y_base
local int Y_step
```

```

local int Z_base local int Z_step

X_axis=0      ! First moving axis
Y_axis=1      ! Second moving axis
Z_axis=2      ! Third moving axis
zone =0       ! Index of error mapping zone in use
X_base=-100   ! X starting coordinate of error mapping zone
Y_base=30     ! Y starting coordinate of error mapping zone
X_step=10     ! X interval in error map table
Y_step=10     ! Y interval in error map table
Z_base=30     ! Z starting coordinate of error mapping zone
Z_step=10     ! Z interval in error map table
              ! Correction table for Error Mapping for X axis
              ! of the first Z height - declared in D-Buffer
X_Correction1(0)(0)=0; X_Correction1(0)(1)=0.3; X_Correction1(0)(2)=-0.4;
X_Correction1(0)(3)=-0.7; X_Correction1(0)(4)=-0.1;
X_Correction1(0)(5)=0.22; X_Correction1(0)(6)=0.55; X_Correction1(0)(7)=0.2;
X_Correction1(0)(8)=-0.3;
X_Correction1(0)(9)=0;
X_Correction1(1)(0)=0; X_Correction1(1)(1)=0.4; X_Correction1(1)(2)=-0.4;
X_Correction1(1)(3)=-0.5; X_Correction1(1)(4)=-0.2;
X_Correction1(1)(5)=0.24; X_Correction1(1)(6)=0.3; X_Correction1(1)(7)=0.32;
X_Correction1(1)(8)=-0.13;
X_Correction1(1)(9)=0;
X_Correction1(2)(0)=0; X_Correction1(2)(1)=0.1; X_Correction1(2)(2)=-0.2;
X_Correction1(2)(3)=-0.4; X_Correction1(2)(5)=0.3;
X_Correction1(2)(6)=0.4; X_Correction1(2)(7)=0.33; X_Correction1(2)(8)=-0.1;
X_Correction1(2)(9)=0
! Correction table for Error Mapping for X axis of the second Z height
! - declared in D-Buffer
X_Correction2(0)(0)=0; X_Correction2(0)(1)=0.23; X_Correction2(0)(2)=-0.12;
X_Correction2(0)(3)=-0.3; X_Correction2(0)(4)=-0.2
X_Correction2(0)(5)=0.26; X_Correction2(0)(6)=0.15; X_Correction2(0)(7)=0.02;
X_Correction2(0)(8)=-0.15;
X_Correction2(0)(9)=0
X_Correction2(1)(0)=0; X_Correction2(1)(1)=0.24; X_Correction2(1)(2)=-0.14;
X_Correction2(1)(3)=-0.34;
X_Correction2(1)(4)=-0.2
X_Correction2(1)(5)=0.14; X_Correction2(1)(6)=-0.23; X_Correction2(1)(7)=-0.32; X_
Correction2(1)(8)=-0.13; X_Correction2(1)(9)=0
X_Correction2(2)(0)=0; X_Correction2(2)(1)=-0.1; X_Correction2(2)(2)=-0.32;
X_Correction2(2)(3)=-0.4; X_Correction2(2)(4)=0.15
X_Correction2(2)(5)=0.36; X_Correction2(2)(6)=0.44; X_Correction2(2)(7)=0.23; X_Correction2
(2)(8)=-0.11;
X_Correction2(2)(9)=0
! X axis error mapping function configuration
ERRORMAP3D2/0 (X_axis, Y_axis, Z_axis), zone, X_base, X_step, Y_base, Y_step, Z_base, Z_
step, X_Correction1, X_Correction2
! Enable error mapping
ERRORMAPON X_axis, zone
STOP

```

Example 2

This example uses two 2-dimensional arrays to create 2-dimensional correction maps, that are used for the 3D Dynamic error compensation. Each map represents a different analog input value. An XYZ cuboid (rectangular prism) zone starts at coordinate -100 for X, 30 for Y, and 30% for the Z specified analog input. It has a fixed interval of 10 mm for each axis, and 10 percent for the Z analog input. The user units are represented in mm.

```

D-Buffer:
global real static X_Correction1(3)(10)
global real static X_Correction2(3)(10)
Buffer:
local int X_axis
local int Y_axis
local int Z_axis
local int zone
local int X_base
local int X_step
local int Y_base
local int Y_step

local int Z_base
local int Z_step
local int Z_referenced_analog_input

X_axis=0      ! First moving axis
Y_axis=1      ! Second moving axis
Z_axis=2      ! Third moving axis
zone =0       ! Index of error mapping zone in use
X_base=-100   ! X staring coordinate of error mapping zone
Y_base=30     ! Y staring coordinate of error mapping zone
X_step=10     ! X interval in error map table
Y_step=10     ! Y interval in error map table
Z_base=30     ! Z staring percentage of analog input
Z_step=10     ! Z interval
Z_referenced_analog_input = 1 ! AIN(1)!Correction table for Error Mapping
for X axis of the first Z value - declared in D-Buffer
X_Correction1(0)(0)=0; X_Correction1(0)(1)=0.3; X_Correction1(0)(2)=-0.4;
X_Correction1(0)(3)=-0.7; X_Correction1(0)(4)=-0.1;
X_Correction1(0)(5)=0.22; X_Correction1(0)(6)=0.55;
X_Correction1(0)(7)=0.2; X_Correction1(0)(8)=-0.3;
X_Correction1(0)(9)=0;
X_Correction1(1)(0)=0; X_Correction1(1)(1)=0.4; X_Correction1(1)(2)=-0.4;
X_Correction1(1)(3)=-0.5; X_Correction1(1)(4)=-0.2;
X_Correction1(1)(5)=0.24; X_Correction1(1)(6)=0.3; X_Correction1(1)
(7)=0.32; X_Correction1(1)(8)=-0.13;
X_Correction1(1)(9)=0;
X_Correction1(2)(0)=0; X_Correction1(2)(1)=0.1; X_Correction1(2)(2)=-0.2;
X_Correction1(2)(3)=-0.4; X_Correction1(2)(5)=0.3;
X_Correction1(2)(6)=0.4; X_Correction1(2)(7)=0.33; X_Correction1(2)(8)=-
0.1; X_Correction1(2)(9)=0
! Correction table for Error Mapping for X axis of the second Z value -
declared in D-Buffer
X_Correction2(0)(0)=0; X_Correction2(0)(1)=0.23;
X_Correction2(0)(2)=-0.12; X_Correction2(0)(3)=-0.3;
X_Correction2(0)(4)=-0.2
X_Correction2(0)(5)=0.26; X_Correction2(0)(6)=0.15;

```

```

X_Correction2(0)(7)=0.02; X_Correction2(0)(8)=-0.15;
X_Correction2(0)(9)=0
X_Correction2(1)(0)=0; X_Correction2(1)(1)=0.24;
X_Correction2(1)(2)=-0.14; X_Correction2(1)(3)=-0.34;
X_Correction2(1)(4)=-0.2
X_Correction2(1)(5)=0.14;
X_Correction2(1)(6)=-0.23; X_Correction2(1)(7)=-0.32;
X_Correction2(1)(8)=-0.13; X_Correction2(1)(9)=0
X_Correction2(2)(0)=0; X_Correction2(2)(1)=-0.1;
X_Correction2(2)(2)=-0.32; X_Correction2(2)(3)=-0.4;
X_Correction2(2)(4)=0.15
X_Correction2(2)(5)=0.36; X_Correction2(2)(6)=0.44;
X_Correction2(2)(7)=0.23; X_Correction2(2)(8)=-0.11;
X_Correction2(2)(9)=0
! X axis error mapping function configuration
ERRORMAP3D2/0a1 (X_axis, Y_axis, Z_axis), zone, X_base, X_step, Y_base,
Y_step, Z_base, Z_step, X_Correction1, X_Correction2, X_axis, Y_axis, Z_
referenced_analog_input
! Enable error mapping
ERRORMAPON X_axis, zone
STOP

```

4.11.9 ERRORMAP3D3

Description

The **ERRORMAP3D3** function configures and activates 3D error correction for the mechanical error compensation of the 'axis0' command, 'axis1' command, and 'axis2' command for the specified zone, so that the compensated reference position will be calculated by adding the interpolated error from the desired position so that the actual value will be closer to the desired value. Interpolation is linear by default; other options are available.

ERRORMAP3D3 function receives indexes of three axes, zone index, base value of the 'axis0' command, fixed defined interval of the 'axis0' command, base value of the 'axis1' command, fixed defined interval of the 'axis1' command, base value of the 'axis2' command, fixed defined interval of the 'axis2' command, and three 2D correction tables (in correlation to the specified 'axis2' coordinates) for mechanical error compensation.

Syntax

```

ERRORMAP3D3[command options] (axis0, axis1, axis2), zone, base0, step0,
base1, step1, base2, step2, correction_map0, correction_map1, correction_
map2[, reference_axis_or_analog_input0, referenced_axis_or_analog_
input1, referenced_axis_or_analog_input2]

```

Arguments

axis0	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis1	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis2	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.
base0	A real number representing the 'axis0' command that corresponds to the first point in correction table for mechanical error compensation.
step0	A real number representing the fixed interval distance between the two adjacent 'axis0' commands.
base1	A real number representing the 'axis1' command that corresponds to the first point in correction table for mechanical error compensation.
step1	A real number representing the fixed interval distance between the two adjacent 'axis1' commands.
base2	A real number representing the 'axis2' command that corresponds to the first point in correction table for mechanical error compensation.
step2	A real number representing the fixed interval distance between the two adjacent 'axis2' commands.
correction_ map0	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '0' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map1	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '1' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).

correction_ map2	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '2' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
referenced_ axis_or_analog_ input0	[Optional] The index of the first axis, or the index of the first analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input1	[Optional] The index of the second axis, or the index of the second analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input2	[Optional] The index of the second axis, or the index of the third analog input whose feedback will be used to calculate the mechanical error compensation.

Switches

/0	The mechanical error compensation will be applied to 'axis0' (default)
/1	The mechanical error compensation will be applied to 'axis1'
/2	The mechanical error compensation will be applied to 'axis2'
/a	Specified that the mechanical error compensation will be calculated be on the feedback from the axis specified by the optional parameter.
/aj	Specifies that the first optional parameter will be regarded as an analog input index.
/ak	Specifies that the second optional parameter will be regarded as an analog input index.
/al	Specifies that the third optional parameter will be regarded as an analog input index.
/e	Specifies that the extrapolation will be used for Z values that are beyond the original observation range. That is, we are assuming that existing trends will continue (by using the nearest correction table to estimate the correction). This method is subject to greater uncertainty and a higher risk of producing meaningless results than interpolation.

Comments

- > When incorrect parameters are specified, the relevant compile-time or run-time error is generated. See [ACSPL+ Runtime Errors](#) for explanation of error 2044 and [ACSPL+](#)

[Compilation Errors](#) for explanations of errors 3113, 3384, 3413, 3414. The function is intended for use with arrays only; an error is generated if a scalar is given as a parameter.

Example

This example uses three 2-dimensional arrays to create 2-dimensional correction maps, that are used for the 3D Dynamic error compensation. Each map represents a different value(height) of the Z axis. The zone starts at coordinate -100 for X, 30 for Y, and 30 for Z. It has a fixed interval of 10 mm for each axis. The user units are represented in mm.

```
D-Buffer:
global real static X_Correction1(3)(10)
global real static X_Correction2(3)(10)
global real static X_Correction3(3)(10)

Buffer:
local int X_axis
local int Y_axis
local int Z_axis
local int zone
local int X_base
local int X_step
local int Y_base
local int Y_step

local int Z_base
local int Z_step

X_axis=0      ! First moving axis
Y_axis=1      ! Second moving axis
Z_axis=2      ! Third moving axis
zone = 0      ! Index of error mapping zone in use
X_base=-100   ! X starting coordinate of error mapping zone
Y_base=30     ! Y starting coordinate of error mapping zone
X_step=10     ! X interval in error map table
Y_step=10     ! Y interval in error map table
Z_base=30     ! Z starting coordinate of error mapping zone
Z_step=10     ! Z interval in error map table!Correction table for Error
Mapping for X axis of the first Z height - declared in D-Buffer
X_Correction1(0)(0)=0; X_Correction1(0)(1)=0.3; X_Correction1(0)(2)=-0.4;
X_Correction1(0)(3)=-0.7; X_Correction1(0)(4)=-0.1;
X_Correction1(0)(5)=0.22; X_Correction1(0)(6)=0.55;
X_Correction1(0)(7)=0.2; X_Correction1(0)(8)=-0.3;
X_Correction1(0)(9)=0;
X_Correction1(1)(0)=0; X_Correction1(1)(1)=0.4; X_Correction1(1)(2)=-0.4;
X_Correction1(1)(3)=-0.5; X_Correction1(1)(4)=-0.2;
X_Correction1(1)(5)=0.24; X_Correction1(1)(6)=0.3;
X_Correction1(1)(7)=0.32; X_Correction1(1)(8)=-0.13;
X_Correction1(1)(9)=0;
X_Correction1(2)(0)=0; X_Correction1(2)(1)=0.1; X_Correction1(2)(2)=-0.2;
X_Correction1(2)(3)=-0.4; X_Correction1(2)(5)=0.3;
```

```

X_Correction1(2)(6)=0.4; X_Correction1(2)(7)=0.33;
X_Correction1(2)(8)=-0.1; X_Correction1(2)(9)=0
! Correction table for Error Mapping for X axis of the second Z height -
declared in D-Buffer
X_Correction2(0)(0)=0; X_Correction2(0)(1)=0.23;
X_Correction2(0)(2)=-0.12; X_Correction2(0)(3)=-0.3;
X_Correction2(0)(4)=-0.2
X_Correction2(0)(5)=0.26; X_Correction2(0)(6)=0.15;
X_Correction2(0)(7)=0.02; X_Correction2(0)(8)=-0.15;
X_Correction2(0)(9)=0
X_Correction2(1)(0)=0; X_Correction2(1)(1)=0.24;
X_Correction2(1)(2)=-0.14; X_Correction2(1)(3)=-0.34;
X_Correction2(1)(4)=-0.2
X_Correction2(1)(5)=0.14; X_Correction2(1)(6)=-0.23;
X_Correction2(1)(7)=-0.32; X_Correction2(1)(8)=-0.13;
X_Correction2(1)(9)=0
X_Correction2(2)(0)=0; X_Correction2(2)(1)=-0.1;
X_Correction2(2)(2)=-0.32; X_Correction2(2)(3)=-0.4;
X_Correction2(2)(4)=0.15
X_Correction2(2)(5)=0.36; X_Correction2(2)(6)=0.44;
X_Correction2(2)(7)=0.23; X_Correction2(2)(8)=-0.11;
X_Correction2(2)(9)=0
! Correction table for Error Mapping for X axis of the third Z height -
declared in D-Buffer
X_Correction3(0)(0)=0; X_Correction3(0)(1)=0.23;
X_Correction3(0)(2)=-0.12; X_Correction3(0)(3)=-0.3;
X_Correction3(0)(4)=-0.2
X_Correction3(0)(5)=0.26; X_Correction3(0)(6)=0.15;
X_Correction3(0)(7)=0.02; X_Correction3(0)(8)=-0.15;
X_Correction3(0)(9)=0
X_Correction3(1)(0)=0; X_Correction3(1)(1)=0.24;
X_Correction3(1)(2)=-0.14; X_Correction3(1)(3)=-0.34;
X_Correction3(1)(4)=-0.2
X_Correction3(1)(5)=0.14; X_Correction3(1)(6)=-0.23;
X_Correction3(1)(7)=-0.32; X_Correction3(1)(8)=-0.13;
X_Correction3(1)(9)=0
X_Correction3(2)(0)=0; X_Correction3(2)(1)=-0.1;
X_Correction3(2)(2)=-0.32; X_Correction3(2)(3)=-0.4;
X_Correction3(2)(4)=0.15
X_Correction3(2)(5)=0.36; X_Correction3(2)(6)=0.44;
X_Correction3(2)(7)=0.23; X_Correction3(2)(8)=-0.11;
X_Correction3(2)(9)=0

! X axis error mapping function configuration
ERRORMAP3D3/0 (X_axis, Y_axis, Z_axis), zone, X_base, X_step, Y_base, Y_
step, Z_base, Z_step, X_Correction1, X_Correction2, X_Correction3
! Enable error mapping
ERRORMAPON X_axis, zone

```

4.11.10 *ERRORMAPN3D2*

Description

The **ERRORMAPN3D2** function configures and activates 3D error correction for the mechanical error compensation of the 'axis0' command, 'axis1' command, and 'axis2' command for the specified zone, so that the compensated reference position will be calculated by adding the linearly (by default) interpolated error from the desired position so that the actual value will be closer to the desired value.

ERRORMAPN3D2 function receives indexes of three axes, zone index, 'axis0' command table, 'axis1' command table, 'axis2' command table, and two 2D correction tables (in correlation to the specified 'axis2' coordinates) for mechanical error compensation.

Syntax

```
ERRORMAPN3D2[command options] (axis0, axis1, axis2), zone, axis0_command,
axis1_command, axis2_command, correction_map0, correction_map1[,
reference_axis_or_analog_input0, referenced_axis_or_analog_input1,
referenced_axis_or_analog_input2]
```

Arguments

axis0	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis1	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis2	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.
axis0_command	The name of a real one-dimensional array that specifies 'axis0' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
axis1_command	The name of a real one-dimensional array that specifies 'axis1' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
axis2_command	The name of a real one-dimensional array that specifies 'axis2' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_	The name of a real two-dimensional array or MATRIX type that specifies a

map0	correction table for mechanical error compensation in relation to axis 2's first specified coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map1	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to axis 2's second specified coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
referenced_ axis_or_analog_ input0	[Optional] The index of the first axis, or the index of the first analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input1	[Optional] The index of the second axis, or the index of the second analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input2	[Optional] The index of the second axis, or the index of the third analog input whose feedback will be used to calculate the mechanical error compensation.

Switches

/0	The mechanical error compensation will be applied to 'axis0' (default)
/1	The mechanical error compensation will be applied to 'axis1'
/2	The mechanical error compensation will be applied to 'axis2'
/a	Specified that the mechanical error compensation will be calculated be on the feedback from the axis specified by the optional parameter.
/aj	Specifies that the first optional parameter will be regarded as an analog input index.
/ak	Specifies that the second optional parameter will be regarded as an analog input index.
/al	Specifies that the third optional parameter will be regarded as an analog input index.
/e	Specifies that the extrapolation will be used for Z values that are beyond the original observation range. That is, we are assuming that existing trends will continue (by using the nearest correction table to estimate the correction). This method is subject to greater uncertainty and a higher risk of producing meaningless results than interpolation.

Comments

- > When incorrect parameters are specified, the relevant compile-time or run-time error is generated. See [ACSPL+ Runtime Errors](#) for explanation of error 2044 and [ACSPL+ Compilation Errors](#) for explanations of errors 3113, 3384, 3413, 3414. The function is intended for use with arrays only; an error is generated if a scalar is given as a parameter.

Example

```
D-Buffer:
global real static X_Correction1(3)(10)
global real static X_Correction2(3)(10)
global real static X_Axis_Coordinates(10)
global real static Y_Axis_Coordinates(3)
global real static Z_Axis_Coordinates(3)

Buffer:
local int X_axis
local int Y_axis
local int Z_axis
int Zone

X_axis = 0      ! First moving axis
Y_axis = 1      ! Second moving axis
Z_axis = 2      ! third moving axis
Zone=0         ! Index of error mapping zone in use
               ! Error map zone definition for X axis
X_Axis_Coordinates(0)=-100; X_Axis_Coordinates(1)=-90;
X_Axis_Coordinates(2)=-80;
X_Axis_Coordinates(3)=-70; X_Axis_Coordinates(4)=-60;
X_Axis_Coordinates(5)=-50;
X_Axis_Coordinates(6)=-40; X_Axis_Coordinates(7)=-30;
X_Axis_Coordinates(8)=-20;
X_Axis_Coordinates(9)=-10
! Error map zone definition for Y axis
Y_Axis_Coordinates(0)=30; Y_Axis_Coordinates(1)=40;
Y_Axis_Coordinates(2)=50
! Error map zone definition for Z axis
Z_Axis_Coordinates(0)=30; Z_Axis_Coordinates(1)=32;
! Correction table for Error Mapping for X axis of the first Z height -
declared in D-Buffer
X_Correction1(0)(0)=0; X_Correction1(0)(1)=0.3;
X_Correction1(0)(2)=-0.4; X_Correction1(0)(3)=-0.7;
X_Correction1(0)(4)=-0.1; X_Correction1(0)(5)=0.22;
X_Correction1(0)(6)=0.55; X_Correction1(0)(7)=0.2;
X_Correction1(0)(8)=-0.3; X_Correction1(0)(9)=0
X_Correction1(1)(0)=0; X_Correction1(1)(1)=0.4;
X_Correction1(1)(2)=-0.4; X_Correction1(1)(3)=-0.5;
X_Correction1(1)(4)=-0.2; X_Correction1(1)(5)=0.24;
X_Correction1(1)(6)=0.3; X_Correction1(1)(7)=0.32;
X_Correction1(1)(8)=-0.13; X_Correction1(1)(9)=0
```

```

X_Correction1(2)(0)=0; X_Correction1(2)(1)=0.1;
X_Correction1(2)(2)=-0.2; X_Correction1(2)(3)=-0.4;
X_Correction1(2)(4)=0.1; X_Correction1(2)(5)=0.3;
X_Correction1(2)(6)=0.4; X_Correction1(2)(7)=0.33;
X_Correction1(2)(8)=-0.1; X_Correction1(2)(9)=0
! Correction table for Error Mapping for X axis of the second Z height -
declared in D-Buffer
X_Correction2(0)(0)=0; X_Correction2(0)(1)=0.23;
X_Correction2(0)(2)=-0.12; X_Correction2(0)(3)=-0.3;
X_Correction2(0)(4)=-0.2; X_Correction2(0)(5)=0.26;
X_Correction2(0)(6)=0.15; X_Correction2(0)(7)=0.02;
X_Correction2(0)(8)=-0.15; X_Correction2(0)(9)=0
X_Correction2(1)(0)=0; X_Correction2(1)(1)=0.24;
X_Correction2(1)(2)=-0.14; X_Correction2(1)(3)=-0.34;
X_Correction2(1)(4)=-0.2; X_Correction2(1)(5)=0.14;
X_Correction2(1)(6)=-0.23; X_Correction2(1)(7)=-0.32;
X_Correction2(1)(8)=-0.13; X_Correction2(1)(9)=0
X_Correction2(2)(0)=0; X_Correction2(2)(1)=-0.1;
X_Correction2(2)(2)=-0.32; X_Correction2(2)(3)=-0.4;
X_Correction2(2)(4)=0.15; X_Correction2(2)(5)=0.36;
X_Correction2(2)(6)=0.44; X_Correction2(2)(7)=0.23;
X_Correction2(2)(8)=-0.11; X_Correction2(2)(9)=0

! X axis error mapping function configuration
ERRORMAPN3D2/0 (X_axis, Y_axis, Z_axis), Zone, X_Axis_Coordinates, Y_
Axis_Coordinates, Z_Axis_Coordinates, X_Correction1, X_Correction2

! Enable Error Mapping
ERRORMAPON X_axis, Zone
STOP

```

4.11.11 ERRORMAPN3D3

Description

The **ERRORMAPN3D3** function configures and activates 3D error correction for the mechanical error compensation of the 'axis0' command, 'axis1' command, and 'axis2' command for the specified zone, so that the compensated reference position will be calculated by adding the linearly (by default) interpolated error from the desired position so that the actual value will be closer to the desired value.

ERRORMAPN3D3 function receives indexes of three axes, zone index, 'axis0' command table, 'axis1' command table, 'axis2' command table, and three 2D correction tables (in correlation to the specified 'axis2' coordinates) for mechanical error compensation.

Syntax

```

ERRORMAPN3D3[command options] (axis0, axis1, axis2), zone, axis0_command,
axis1_command, axis2_command, correction_map0, correction_map1,
correction_map2[, reference_axis_or_analog_input0, referenced _axis_or_
analog_input1, referenced _axis_or_analog_input2]

```

Arguments

axis0	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis1	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis2	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.
axis0_ command	The name of a real one-dimensional array that specifies 'axis0' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
axis1_ command	The name of a real one-dimensional array that specifies 'axis1' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
axis2_ command	The name of a real one-dimensional array that specifies 'axis2' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map0	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to axis 2's first specified coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map1	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to axis 2's second specified coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map2	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to axis 2's third specified coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
referenced_ axis_or_analog_ input0	[Optional] The index of the first axis, or the index of the first analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ 	[Optional] The index of the second axis, or the index of the second analog

axis_or_analog_input1	input whose feedback will be used to calculate the mechanical error compensation.
referenced_axis_or_analog_input2	[Optional] The index of the second axis, or the index of the third analog input whose feedback will be used to calculate the mechanical error compensation.

Switches

/0	The mechanical error compensation will be applied to 'axis0' (default)
/1	The mechanical error compensation will be applied to 'axis1'
/2	The mechanical error compensation will be applied to 'axis2'
/a	Specified that the mechanical error compensation will be calculated be on the feedback from the axis specified by the optional parameter.
/aj	Specifies that the first optional parameter will be regarded as an analog input index.
/ak	Specifies that the second optional parameter will be regarded as an analog input index.
/al	Specifies that the third optional parameter will be regarded as an analog input index.
/e	Specifies that the extrapolation will be used for Z values that are beyond the original observation range. That is, we are assuming that existing trends will continue (by using the nearest correction table to estimate the correction). This method is subject to greater uncertainty and a higher risk of producing meaningless results than interpolation.

Comments

- > When incorrect parameters are specified, the relevant compile-time or run-time error is generated. See [ACSPL+ Runtime Errors](#) for explanation of error 2044 and [ACSPL+ Compilation Errors](#) for explanations of errors 3113, 3384, 3413, 3414. The function is intended for use with arrays only; an error is generated if a scalar is given as a parameter.

Example

```
D-Buffer:
global real static X_Correction1(3)(10)
global real static X_Correction2(3)(10)
global real static X_Correction3(3)(10)
global real static X_Axis_Coordinates(10)
```



```

global real static Y_Axis_Coordinates(3)
global real static Z_Axis_Coordinates(3)

Buffer:
local int X_axis
local int Y_axis
local int Z_axis
int Zone

X_axis = 0      ! First moving axis
Y_axis = 1      ! Second moving axis
Z_axis = 2      ! third moving axis
Zone=0          ! Index of error mapping zone in use! Error map zone definition
for X axis
X_Axis_Coordinates(0)=-100; X_Axis_Coordinates(1)=-90; X_Axis_Coordinates
(2)=-80;
X_Axis_Coordinates(3)=-70; X_Axis_Coordinates(4)=-60; X_Axis_Coordinates
(5)=-50;
X_Axis_Coordinates(6)=-40; X_Axis_Coordinates(7)=-30; X_Axis_Coordinates
(8)=-20;
X_Axis_Coordinates(9)=-10
! Error map zone definition for Y axis
Y_Axis_Coordinates(0)=30; Y_Axis_Coordinates(1)=40; Y_Axis_Coordinates
(2)=50
! Error map zone definition for Z axis
Z_Axis_Coordinates(0)=30; Z_Axis_Coordinates(1)=32; Z_Axis_Coordinates
(1)=34;
! Correction table for Error Mapping for X axis of the first Z height -
declared in D-Buffer
X_Correction1(0)(0)=0; X_Correction1(0)(1)=0.3; X_Correction1(0)(2)=-0.4;
X_Correction1(0)(3)=-0.7;
X_Correction1(0)(4)=-0.1; X_Correction1(0)(5)=0.22; X_Correction1(0)
(6)=0.55; X_Correction1(0)(7)=0.2;
X_Correction1(0)(8)=-0.3; X_Correction1(0)(9)=0
X_Correction1(1)(0)=0; X_Correction1(1)(1)=0.4; X_Correction1(1)(2)=-0.4;
X_Correction1(1)(3)=-0.5;
X_Correction1(1)(4)=-0.2; X_Correction1(1)(5)=0.24; X_Correction1(1)
(6)=0.3; X_Correction1(1)(7)=0.32;
X_Correction1(1)(8)=-0.13; X_Correction1(1)(9)=0
X_Correction1(2)(0)=0; X_Correction1(2)(1)=0.1; X_Correction1(2)(2)=-0.2;
X_Correction1(2)(3)=-0.4;
X_Correction1(2)(4)=0.1; X_Correction1(2)(5)=0.3; X_Correction1(2)
(6)=0.4; X_Correction1(2)(7)=0.33;
X_Correction1(2)(8)=-0.1; X_Correction1(2)(9)=0
! Correction table for Error Mapping for X axis of the second Z height -
declared in D-Buffer
X_Correction2(0)(0)=0; X_Correction2(0)(1)=0.23; X_Correction2(0)(2)=-
0.12; X_Correction2(0)(3)=-0.3;
X_Correction2(0)(4)=-0.2; X_Correction2(0)(5)=0.26; X_Correction2(0)

```

```

(6)=0.15; X_Correction2(0)(7)=0.02;
X_Correction2(0)(8)=-0.15; X_Correction2(0)(9)=0
X_Correction2(1)(0)=0; X_Correction2(1)(1)=0.24; X_Correction2(1)(2)=-
0.14; X_Correction2(1)(3)=-0.34;
X_Correction2(1)(4)=-0.2; X_Correction2(1)(5)=0.14; X_Correction2(1)(6)=-
0.23; X_Correction2(1)(7)=-0.32;
X_Correction2(1)(8)=-0.13; X_Correction2(1)(9)=0
X_Correction2(2)(0)=0; X_Correction2(2)(1)=-0.1; X_Correction2(2)(2)=-
0.32; X_Correction2(2)(3)=-0.4;
X_Correction2(2)(4)=0.15; X_Correction2(2)(5)=0.36; X_Correction2(2)
(6)=0.44; X_Correction2(2)(7)=0.23;
X_Correction2(2)(8)=-0.11; X_Correction2(2)(9)=0
! Correction table for Error Mapping for X axis of the third Z height -
declared in D-Buffer
X_Correction3(0)(0)=0; X_Correction3(0)(1)=0.23; X_Correction3(0)(2)=-
0.12; X_Correction3(0)(3)=-0.3; X_Correction3(0)(4)=-0.2
X_Correction3(0)(5)=0.26; X_Correction3(0)(6)=0.15; X_Correction3(0)
(7)=0.02; X_Correction3(0)(8)=-0.15;
X_Correction3(0)(9)=0
X_Correction3(1)(0)=0; X_Correction3(1)(1)=0.24; X_Correction3(1)(2)=-
0.14; X_Correction3(1)(3)=-0.34;
X_Correction3(1)(4)=-0.2
X_Correction3(1)(5)=0.14; X_Correction3(1)(6)=-0.23; X_Correction3(1)
(7)=-0.32; X_Correction3(1)(8)=-0.13; X_Correction3(1)(9)=0
X_Correction3(2)(0)=0; X_Correction3(2)(1)=-0.1; X_Correction3(2)(2)=-
0.32; X_Correction3(2)(3)=-0.4; X_Correction3(2)(4)=0.15
X_Correction3(2)(5)=0.36; X_Correction3(2)(6)=0.44; X_Correction3(2)
(7)=0.23; X_Correction3(2)(8)=-0.11;
X_Correction3(2)(9)=0

! X axis error mapping function configuration
ERRORMAPN3D3/0 (X_axis, Y_axis, Z_axis), Zone, X_Axis_Coordinates, Y_
Axis_Coordinates, Z_Axis_Coordinates, X_Correction1, X_Correction2, X_
Correction3

! Enable Error Mapping
ERRORMAPON X_axis, Zone
STOP

```

4.11.12 ERRORMAP3D5

Description

The **ERRORMAP3D5** function configures and activates 3D error correction for the mechanical error compensation of the 'axis0' command, 'axis1' command, and 'axis2' command for the specified zone, so that the compensated reference position will be calculated by adding the interpolated error from the desired position so that the actual value will be closer to the desired value. Interpolation is linear by default; other options are available.

The **ERRORMAP3D5** function receives indices of three axes, zone index, the base value of the 'axis0' command, a fixed defined interval of the 'axis0' command, the base value of the 'axis1' command, a

fixed defined interval of the 'axis1' command, the base value of the 'axis2' command, a fixed defined interval of the 'axis2' command, and five 2D correction tables (in correlation to the specified 'axis2' coordinates) for mechanical error compensation.

Syntax

```
ERRORMAP3D5[command options] (axis0, axis1, axis2), zone, base1, step1,
base2, step2, base3, step3, correction_map0, correction_map1, correction_
map2, correction_map3, correction_map4[, reference_axis_or_analog_input0,
referenced _axis_or_analog_input1, referenced _axis_or_analog_input2]
```

Arguments

axis0	The index of the axis to which the mechanical error compensation will be applied, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis1	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis2	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.
base0	A real number representing the 'axis0' command that corresponds to the first point in correction table for mechanical error compensation.
step0	A real number representing the fixed interval distance between the two adjacent 'axis0' commands.
base1	A real number representing the 'axis1' command that corresponds to the first point in correction table for mechanical error compensation.
step1	A real number representing the fixed interval distance between the two adjacent 'axis1' commands.
base2	A real number representing the 'axis2' command that corresponds to the first point in correction table for mechanical error compensation.
step2	A real number representing the fixed interval distance between the two adjacent 'axis2' commands.

correction_ map0	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '0' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map1	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '1' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map2	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '2' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map3	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '3' coordinate. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map4	The name of a real two-dimensional array or MATRIX type that specifies correction table for mechanical error compensation in relation to axis 2 step '4' coordinate. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
referenced_ axis_or_analog_ input0	[Optional] The index of the first axis, or the index of the first analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input1	[Optional] The index of the second axis, or the index of the second analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input2	[Optional] The index of the second axis, or the index of the third analog input whose feedback will be used to calculate the mechanical error compensation.

Switches

/0	The mechanical error compensation will be applied to 'axis0' (default)
/1	The mechanical error compensation will be applied to 'axis1'
/2	The mechanical error compensation will be applied to 'axis2'
/a	Specified that the mechanical error compensation will be calculated be on the feedback from the axis specified by the optional parameter.

/aj	Specifies that the first optional parameter to be treated as an analog input index.
/ak	Specifies that the second optional parameter to be treated as an analog input index.
/al	Specifies that the third optional parameter to be treated as an analog input index.

Error Conditions

The function detects the following error conditions.

- > **2044** - Index is out of range, when the defined array size is less than the defined number of array points.
- > **3113** - The step in the table is zero or negative, when the step argument is zero or negative.
- > **3384** - The specified suffix combination is invalid. Please see the documentation for more details.
- > **3413** - The supplied correction maps (2-dimensional arrays or MATRIX types) should have the same dimensions. Please see the documentation for more details.
- > **3414** - The Specified referenced axes/analog inputs are invalid. Please see the documentation for more details.

Comments

If incorrect parameters are passed to the function the corresponding runtime error is generated. The function is intended to be used only with arrays; thus, an error is generated if a scalar is given as a parameter.

4.11.13 *ERRORMAPN3D5*

Description

The **ERRORMAPN3D5** function configures and activates 3D error correction for the mechanical error compensation of the 'axis0' command, 'axis1' command, and 'axis2' command for the specified zone, so that the compensated reference position will be calculated by adding the interpolated error from the desired position so that the actual value will be closer to the desired value. Interpolation is linear by default; other options are available.

ERRORMAPN3D5 function receives indices of three axes, zone index, 'axis0' command table, 'axis1' command table, 'axis2' command table, and 5 2D correction tables correlated to the specified 'axis2' coordinates for mechanical error compensation.

Syntax

```
ERRORMAPN3D5[command options] (axis0, axis1, axis2), zone, axis0_command,
axis1_command, axis2_command, correction_map0, correction_map1,
correction_map2, correction_map3, correction_map4[, reference_axis_or_
analog_input0, referenced _axis_or_analog_input1, referenced _axis_or_
analog_input2]
```

Arguments

axis0	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis1	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis2	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.
axis0_ command	The name of a real one-dimensional array that specifies 'axis0' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
axis1_ command	The name of a real one-dimensional array that specifies 'axis1' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
axis2_ command	The name of a real one-dimensional array that specifies 'axis2' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map0	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the first specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map1	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the second specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map2	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the third specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map3	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the fourth specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).

correction_ map4	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the fifth specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
referenced_ axis_or_analog_ input0	[Optional] The index of the first axis, or the index of the first analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input1	[Optional] The index of the second axis, or the index of the second analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input2	[Optional] The index of the second axis, or the index of the third analog input whose feedback will be used to calculate the mechanical error compensation.

Switches

/0	The mechanical error compensation will be applied to 'axis0' (default)
/1	The mechanical error compensation will be applied to 'axis1'
/2	The mechanical error compensation will be applied to 'axis2'
/a	Specified that the mechanical error compensation will be calculated be on the feedback from the axis specified by the optional parameter.
/aj	Specifies that the first optional parameter will be regarded as an analog input index.
/ak	Specifies that the second optional parameter will be regarded as an analog input index.
/al	Specifies that the third optional parameter will be regarded as an analog input index.

Error Conditions

The function detects the following error conditions.

- > **2044** - Index is out of range, when the defined array size is less than the defined number of array points.
- > **3384** - The specified suffix combination is invalid. Please see the documentation for more details.
- > **3413** - The supplied correction maps (2-dimensional arrays or MATRIX types) should have the same dimensions. Please see the documentation for more details.
- > **3414** - The Specified referenced axes/analog inputs are invalid. Please see the documentation for more details.

Comments

If incorrect parameters are passed to the function the corresponding runtime error is generated. The function is intended to be used only with arrays; thus, an error is generated if a scalar is given as a parameter.

4.11.14 *ERRORMAPN3DA*

Description

The **ERRORMAPN3DA** function configures and activates 3D error correction for the mechanical error compensation of the 'axis0' command, 'axis1' command, and 'axis2' command for the specified zone, so that the compensated reference position will be calculated by adding the interpolated error from the desired position so that the actual value will be closer to the desired value. Interpolation is linear by default; other options are available.

ERRORMAPN3DA function receives indexes of three axes, zone index, 'axis0' command table, 'axis1' command table, 'axis2' command table, and 10 2D correction tables (correlated to the specified 'axis2' coordinates) for mechanical error compensation.

Syntax

```
ERRORMAPN3DA[command options] (axis0, axis1, axis2), zone, axis0_command,
axis1_command, axis2_command, correction_map0, correction_map1,
correction_map2, correction_map3, correction_map4, correction_map5,
correction_map6, correction_map7, correction_map8, correction_map9[,
reference_axis_or_analog_input0, referenced_axis_or_analog_input1,
referenced_axis_or_analog_input2]
```

Arguments

axis0	The index of the axis that the mechanical error compensation will be applied to, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis1	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
axis2	The index of the second axis participating in 3D mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index of the mechanical error compensation, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1.
axis0_command	The name of a real one-dimensional array that specifies 'axis0' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
axis1_command	The name of a real one-dimensional array that specifies 'axis1' command values used for correction table of mechanical error compensation. The

	array type should be GLOBAL REAL STATIC (defined in D-Buffer).
axis2_ command	The name of a real one-dimensional array that specifies 'axis2' command values used for correction table of mechanical error compensation. The array type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map0	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the first specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map1	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the second specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map2	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the third specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map3	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the fourth specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map4	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the fifth specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map5	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the sixth specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map6	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the seventh specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map7	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the eighth specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map8	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the

	ninth specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
correction_ map9	The name of a real two-dimensional array or MATRIX type that specifies a correction table for mechanical error compensation in relation to the tenth specified coordinate of the Z-axis. The array or MATRIX type should be GLOBAL REAL STATIC (defined in D-Buffer).
referenced_ axis_or_analog_ input0	[Optional] The index of the first axis, or the index of the first analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input1	[Optional] The index of the second axis, or the index of the second analog input whose feedback will be used to calculate the mechanical error compensation.
referenced_ axis_or_analog_ input2	[Optional] The index of the second axis, or the index of the third analog input whose feedback will be used to calculate the mechanical error compensation.

Switches

/0	The mechanical error compensation will be applied to 'axis0' (default)
/1	The mechanical error compensation will be applied to 'axis1'
/2	The mechanical error compensation will be applied to 'axis2'
/a	Specified that the mechanical error compensation will be calculated be on the feedback from the axis specified by the optional parameter.
/aj	Specifies that the first optional parameter will be regarded as an analog input index.
/ak	Specifies that the second optional parameter will be regarded as an analog input index.
/al	Specifies that the third optional parameter will be regarded as an analog input index.

Error Conditions

The function detects the following error conditions.

- > **2044** - Index is out of range, when the defined array size is less than the defined number of array points.
- > **3384** - The specified suffix combination is invalid. Please see the documentation for more details.
- > **3413** - The supplied correction maps (2-dimensional arrays or MATRIX) should have the same dimensions. Please see the documentation for more details.

- > **3414** - The Specified referenced axes/analog inputs are invalid. Please see the documentation for more details.

Comments

If incorrect parameters are passed to the function the corresponding runtime error is generated. The function is intended to be used only with arrays; thus, an error is generated if a scalar is given as a parameter.

4.11.15 *ERRORMAPOFF*

Description

ERRORMAPOFF function receives axis index and zone index. The **ERRORMAPOFF** function deactivates error mapping correction for the mechanical error compensation for the specified zone.

Syntax

```
ERRORMAPOFF axis, zone
```

Arguments

axis	The axis index, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1. If '-1' is specified, all zones of specified axis will be affected.

4.11.16 *ERRORMAPON*

ERRORMAPON function receives axis index and zone index. The **ERRORMAPON** function activates error correction for the mechanical error compensation for the specified zone.

Syntax

```
ERRORMAPON axis, zone
```

Arguments

axis	The axis index, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1. If '-1' is specified, all zones of specified axis will be affected.

4.11.17 *#ERRORMAPREP*

Description

#ERRORMAPREP function generates a report of all activated zones of error mapping for all axes in the system.

Syntax

```
#ERRORMAPREP
```

4.11.18 ERRORUNMAP

Description

ERRORUNMAP function receives axis index and zone index. The **ERRORUNMAP** function deactivates error correction for the mechanical error compensation for the specified zone.

Syntax

```
ERRORUNMAP axis, zone
```

Arguments

axis	The axis index, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
zone	The zone index, valid numbers are: 0, 1, 2, ... up to the maximum number of zones minus 1. If '-1' is specified, all zones of specified axis will be affected.

5. ACSPL+ Standard Structures

5.1 LCI Standard Structure

The LCI structure is supported in versions 3.10 and higher.

5.1.1 LCI Functions

5.1.1.1 PowerPWMOut

Description

This function initializes Pulse Modulation mode .

Syntax

```
PowerPWMOut (Mode, Freq, Width, DutyCycle, [MinValue, MaxValue,
MinVelocity, MaxVelocity])
```

Arguments

Mode	Modulation mode: 0 – Fixed parameters mode 1 - Fixed Frequency 2 - Fixed Pulse Width 3 - Fixed duty cycle Modes 0 – 3 are incompatible modes, i.e. setting any of these modes automatically disables the previously set mode.
Freq	Pulse modulation frequency in Hz, range from 0.035Hz to 1MHz. In mode = 1, the frequency is fixed and is not changed during the process. In mode = 2 or 3, the function only sets the initial frequency.
Width	Pulse Width in milliseconds, range for 6.67 nsec to 28.60 sec.
DutyCycle	Duty cycle in percentage, range from 0% to 100%. When mode = 0, DutyCycle value is not applicable When mode = 1, the function only sets the initial duty cycle. When mode = 2, the argument has no effect, as duty cycle is automatically calculated by the unit as function of frequency. When mode = 3, the duty cycle is fixed and is not changed during the process
MinValue	[Optional] Minimum value of volatile parameter (depends on mode) , if calculated vector velocity is less than or equal to MinVel

MaxValue	[Optional] Maximum value of volatile parameter (depends on mode) , if calculated vector velocity is greater than or equal toMaxVel
MinVel	[Optional] Minimum velocity for parameter calculation. Below the minimum value, the parameter will be limited by MinValue or default, if MinValue is not specified By default, minimum velocity is zero.`
MaxVel	[Optional] Maximum velocity for parameter calculation. Above the maximum value, the parameter will be limited by MaxValue or default, if MaxValue is not specified. By default, maximum velocity is defined by XVEL parameter

Example**5.1.1.2 PowerAnalogOut****Description**

The function defines the range of Analog Output value depending on the actual velocity

Syntax

```
PowerAnalogOut (AnalogOutInd, [MinValue, MaxValue, MinVelocity,
MaxVelocity] )
```

Arguments

AnalogOutInd	Analog output index (0 or 1)
MinValue	[Optional] Minimum value of volatile parameter if calculated vector velocity equals or below MinVel, default is 0.
MaxValue	[Optional] Maximum of volatile parameter if calculated vector velocity equals or above MaxVel, default is 100 (%)
MinVel	[Optional] Minimum velocity for parameter calculation. Below the minimum value, parameter will be limited by MinValue or default, if MinValue is not specified By default, Minimum velocity is zero.

MaxVel

[Optional] Maximum velocity for parameter calculation.

Above the maximum value, parameter will be limited by MaxValue or default, if MaxValue is not specified.

By default, Maximum velocity is defined by **XVEL** parameter

```
!D-buffer declaration
LCI lc

!Program buffer
lc.MotionAxes = AxListAsMask(X,Y)
!Define laser power control by analog output
! Output value is 0 Volt for velocity < 10 and 10 Volt for velocity > VEL
(X)*2
! Use Analog output - 1
lc.PowerAnalogOut(1, 0, 10, 10, VEL(X)*2)

lc.LaserEnable()
```

5.1.1.3 PowerDigitalOut

Description

The function defines the range of Digital Output value depending on the actual velocity.

SYNTAX

```
PowerDigitalOut (SynchPulseWidth, [MinValue, MaxValue, MinVelocity,
MaxVelocity] )
```

Arguments

SynchPulseWidth	The width of the synchronization pulse in msec. The pulse fires every time when all Digital Outputs are updated and user can read the Power value
MinValue	[Optional] Minimum value of volatile parameter (depends on mode) , if calculated vector velocity equals or below MinVel. Default is 0
MaxValue	[Optional] Maximum of volatile parameter (depends on mode) , if calculated vector velocity equals or above MaxVel. Default is 255
MinVel	[Optional] Minimum velocity for parameter calculation. Below the minimum value, parameter will be limited by MinValue or by the default value if MinValue is not specified. By default, minimum velocity is zero.

MaxVel	[Optional] Maximum velocity for parameter calculation. Above the maximum value, the parameter will be limited by MaxValue or by the default value if MaxValue is not specified. By default, maximum velocity is defined by the XVEL parameter.
--------	---

5.1.1.4 FixedDistPulse

Description

The function initializes the fixed distance pulse firing mode. This mode is useful if a laser beam should be activated at specified fixed intervals between activations along an actual motion trajectory. The **AxesUsed** field defines which axes are used for multi-axis trajectory generation.

Syntax

```
int FixedDistPulse (PulseWidth, Interval[, StartPos, EndPos, InitialPos,
MinValue, MaxValue, MinVelocity, MaxVelocity])
```

Return Type

Occupied channel index as integer value

Arguments

PulseWidth	Pulse Width in milliseconds (40ns – 161ms)
Interval	Specifies an interval in user units between pulses along the multi-axis motion trajectory.
StartPos	[Optional] Specifies a start position in user units along the multi-axis motion trajectory. The pulses start generating after reaching this position. The value is 0 by default.
EndPos	[Optional] Specifies a last position in user units along the multi-axis motion trajectory. The pulses stop generating after reaching this position. If the parameter is omitted, then there is no limitation.
InitialPos	[Optional] Specifies an initial position offset relative to the user origin. By default InitialPos is zero. For a multi-axes-trajectory, the InitialPos should be omitted or set to zero.
MinValue	[Optional] Minimum value of volatile parameter (depends on mode), if calculated vector velocity is greater than or equal to MinVel
MaxValue	[Optional] Maximum of volatile parameter (depends on mode), if calculated vector velocity less than or equal to MaxVel
MinVelocity	[Optional] Minimum velocity for parameter calculation. Below the minimum value, parameter will be limited by MinValue or default, if MinValue is not specified

	By default, minimum velocity is zero.`
MaxVelocity	[Optional] Maximum velocity for parameter calculation. Above the maximum value, parameter will be limited by MaxValue or default, if MaxValue is not specified. By default, maximum velocity is defined by XVEL parameter

Comments

The ExtraPulses and ExtraPeriod fields define the extra pulses generation. If at least one of parameters is 0, no extra pulses are generated.

The PiercePulsesNum and PiercePulseWidth fields define the pierce pulse generation. If at least one of parameters is 0, no pierce pulses are generated.

The MotionAxes axes mask field is used for trajectory calculation.

The PulseResolution field is used for operation pulse resolution. If the default PulseResolution is 0, the pulse resolution should be calculated according to the **XVEL** parameter and Maximum LCI Frequency.

When the motion occurs in the negative direction, it is not sufficient to specify a negative interval; StartPos and EndPos must also be specified. See example below.

Example

```
int AxX = 0 ! Actual moving X axis

real PulseWidth = 0.1 ! Generated pulse width 0.1 mS

real Interval = -1 ! interval between pulses = -1mm
                  ! to move in negative direction

lc.SetMotionAxes(AxX) ! Axes 0,1 used for vector velocity calculation
! optional Parameters StartPos = 0 and EndPos = -30 need to be specified
lc.FixedDistPulse(PulseWidth, Interval, 0, -30)

! Initialization of Fixed Interval function

lc.LaserEnable() ! LCI ready to generate pulses in motion

ptp/e (AxX), -30! motion command

stop
```

5.1.1.5 DistanceArrPulse

Description

The function initializes either array-based pulse firing mode.

Syntax

```
int DistanceArrPulse (arrPos, arWidth, ArraySize)
```

Return Type

Occupied channel index as integer value

Arguments

arPos	Array of points. Each element in the array defines the point where a pulse should be fired.
arWidth	Pulse Width value or Pulse Width array in milliseconds. Each width in the array corresponds to a point in the Position array. The size of array should be equals to size of Position array. If the parameter is a real value, pulse width is applied for all elements in the Position array.
ArraySize	The size of Points and Width arrays.

Comments

The ExtraPulses and ExtraPeriod fields define the extra pulses generation. If at least one of parameters is 0, no extra pulses are generated.

The MotionAxes axes mask field is used for trajectory calculation.

The PulseResolution field is used for operation pulse resolution. In case of the default PulseResolution is 0, the pulse resolution should be calculated according to the Comments

The ExtraPulses and ExtraPeriod fields define the extra pulses generating. If at least one of parameters is 0, no extra pulses are generated.

The MotionAxes axes mask field is used for trajectory calculation.

The PulseResolution field is used for operation pulse resolution. In case of the default PulseResolution is 0, the pulse resolution should calculate according to **XVEL** parameter and Maximum LCI Frequency.

Command requires "Array and Segment Based Modes" LCI ordering option to use.

Example

5.1.1.6 CoordinateArrPulse

Description

The function initializes either array-based pulse firing mode based on multiple axis position.

Syntax

```
int CoordinateArrPulse (ArraySize, WidthArr[, XPosArr, YPosArr , ZPosArr, APosArr, BPosArr, CPosArr])
```

Return Type

Occupied first channel index as integer value

Arguments

ArraySize	The size of Points and Width arrays.
WidthArr	Pulse Width value or Pulse Width array in milliseconds. Each width in the array corresponds to a point in the Positions arrays. If the parameter is a real value, pulse width is applied for all items of Positions arrays.
XPosArr	(optional) Array of X axis positions. Each element of the array defines the point where a pulse should be fired.
YPosArr	(optional) Array of Y axis positions. Each element of the array defines the point where a pulse should be fired.
ZPosArr	(optional) Array of Z axis positions. Each element of the array defines the point where a pulse should be fired.
APosArr	(optional) Array of A axis positions. Each element of the array defines the point where a pulse should be fired.
BPosArr	(optional) Array of B axis positions. Each element of the array defines the point where a pulse should be fired.
CPosArr	(optional) Array of C axis positions. Each element of the array defines the point where a pulse should be fired.

Comments

Function occupies several channels, depending on the MotionAxes field.

If the parameter is 0, no extra pulses are generated.

The MotionAxes axes mask field is used for trajectory calculation.

The PulseResolution field is used for operation pulse resolution. In case of the default PulseResolution is 0, the pulse resolution should calculate according to **XVEL** parameter and Maximum LCI Frequency.

Command requires "Array and Segment Based Modes" LCI ordering option to use.

Example*5.1.1.7 Tickle***Description**

The function initializes the Tickle mode. In this mode the laser control unit generates a signal at constant frequency and with constant width. Usually this mode is used for those types of lasers that require gas ionization during the time period when laser processing is off. By setting this mode the laser will respond faster and more predictably when laser processing resumes. Once this mode is

initialized, the laser control unit constantly generates pulses without regard to any other operational modes.

Syntax

```
Tickle (Freq, Width)
```

Return Type

None

Arguments

Freq	Tickle frequency in Hz. Minimum frequency is 1149Hz, maximum frequency is 100,000Hz Not all frequencies can be configured, so the function automatically rounds the specified frequency to the nearest supported value.
Width	Pulse Width in milliseconds Minimal width is 0.000107msec (107nsec), maximal width is 0.0273msec (27.3µsec) Not all width values are supported, so the function automatically rounds the specified width to the nearest supported value.

5.1.1.8 LaserEnable

Description

The function enables laser pulse generation.

Syntax

```
LaserEnable ()
```

5.1.1.9 LaserDisable

Description

The function disables laser pulse generation.

Syntax

```
LaserDisable ()
```

5.1.1.10 DistanceArrGate

Description

The function adds the gating mode to the system. In this mode laser is to be switched on or off at a predefined position. The position is defined by distance position.

Syntax

```
int DistanceArrGate (arPos, arStates,Size)
```

Return Type

None

Arguments

arPos	Array of points. Each element of the array defines the point where a state signal should be changed.
arStates	Array of states. Each element of the array defines the state value in the corresponding point from arPos array. The array size should be equal to points array (arPos)
Size	Size of the two arrays used for gating definition.

Comments

The MotionAxes axes mask field is used for trajectory calculation. The PulseResolution field is used for operation pulse resolution. If the default PulseResolution is 0, the pulse resolution should be calculated according to the **XVEL** parameter and Maximum LCI Frequency.

Command requires "Array and Segment Based Modes" LCI ordering option to use.

5.1.1.11 CoordinateArrGate

Description

The function adds the gating mode to the system. In this mode laser is to be switched on or off at a predefined position. The position defined by axes coordinates.

Syntax

```
int CoordinateArrGate (PointsNum, StatesArr[, XPosArr, YPosArr, ZPosArr,  
APosArr, BPosArr, CPosArr ])
```

Return Type

First occupied channel index as integer value

Arguments

PointsNum	The actual points number
StatesArr	States array of integers
XPosArr	(optional)Array of X axis positions. Each element of the array defines the point where a pulse should be fired.
YPosArr	(optional)Array of Y axis positions.Each element of the array defines the point where a pulse should be fired.

ZPosArr	(optional)Array of Z axis positions.Each element of the array defines the point where a pulse should be fired.
APosArr	(optional)Array of A axis positions.Each element of the array defines the point where a pulse should be fired.
BPosArr	(optional)Array of B axis positions. Each element of the array defines the point where a pulse should be fired.
CPosArr	(optional)Array of C axis positions.Each element of the array defines the point where a pulse should be fired.

Comments

The trajectory calculation is built according to defined Position arrays.

The PulseResolution field is used for operation pulse resolution. If PulseResolution is 0, the pulse resolution should be calculated according to the **XVEL** parameter and Maximum LCI Frequency.

This command is available when the "Array and Segment Based Modes" LCI ordering option is purchased.

5.1.1.12 AddZone

Description

Add the laser activation zone. If the zone is defined, the laser can be activated only in the specified zone.

Syntax

```
int AddZone (MinPos, MaxPos[, MotionAxes] )
```

Return Type

Occupied channel index as integer value

Arguments

MinPos	Specifies a minimum position of the operation zone along the motion trajectory. The laser pulses start outputting after reaching this position
MaxPos	Specifies a maximum position of the operation zone along the motion trajectory. The laser pulses stop outputting after reaching this position.
MotionAxes	[Optional] Mask that defines the axes, which are used for generating pulses along the multi-axis motion trajectory. If parameter is 0 or omitted, the default axes mask is taken from MotionAxes.

Example

5.1.1.13 SetZone

Description

Change the laser activation zone for specified channel. If zone is defined, the laser can be activated only in the specified zone

Syntax

```
int SetZone (Channel, MinPos, MaxPos)
```

Return Type

None

Arguments

Channel	Channel or Operation ID where the zone was defined
MinPos	Specifies a minimal position of the operation zone along the motion trajectory. The laser pulses start outputting after reaching this position
MaxPos	Specifies a maximal position of the operation zone along the motion trajectory. The laser pulses stop outputting after reaching this position.

Example

5.1.1.14 SetCondition

Description

Define laser activation condition in one of the 3 condition registers by a bitwise mask.

Syntax

```
lc.SetCondition (ConditionReg, SetBitMask [,ClearBitMask])
```

Return Type

None

Arguments

ConditionReg	Register index (0,1,2)
SetBitMask	Bitwise condition mask to set
ClearBitMask	(optional) Bitwise condition mask to clear

Comments

Table 4-19. Condition Mask for Register 0

Bit #	Signal
0	PWM
1	Tickle
2-17	For internal usage
18	In Range 0
19	In Range 1
20	In Range 2
21	In Range 3
22	In Range 4
23	In Range 5
24	In Range 6
25	In Range 7

Table 4-20. Condition Mask for Register 1

Bit #	Signal
0	PEG Pulse 0
1	PEG Pulse 1
2	PEG Pulse 2
3	PEG Pulse 3
4	PEG Pulse 4
5	PEG Pulse 5
6	PEG Active 0
7	PEG Active 1
8	PEG Active 2

Bit #	Signal
9	PEG Active 3
10	PEG Active 4
11	PEG Active 5
12	PEG State of PEG 0
13	PEG State of PEG 1
14	PEG State of PEG 2
15	PEG State of PEG 3
16	PEG State of PEG 4
17	PEG State of PEG 5

Table 4-21. Condition Mask for Register 2

Bit #	
0	General Purpose Input 0
1	General Purpose Input 1
2	General Purpose Input 2
3	General Purpose Input 3
4	General Purpose Input 4
5	General Purpose Input 5
6	General Purpose Input 6
7	General Purpose Input 7
8	General Purpose Output 0
9	General Purpose Output 1
10	General Purpose Output 2
11	General Purpose Output 3
12	General Purpose Output 4

Bit #	
13	General Purpose Output 5
14	General Purpose Output 6
15	General Purpose Output 7

Example

```
!Set bit 0 and clear bit 1 in register 1
SetCondition (1, 0x1 , 0x2)
```

5.1.1.15 *GetCondition*

Description

Return the current laser activation condition mask from the specified register.

Syntax

```
INT GetCondition (ConditionReg)
```

Return Type

Condition mask. See tables in [SetCondition](#) description.

Arguments

ConditionReg	Register index (0,1,2)
--------------	------------------------

Comments

Example

5.1.1.16 *SegmentGate*

Description

Activate automatic gating for segment motion.

Syntax

```
int SegmentGate ([InitialState])
```

Arguments

InitialState	(optional)The gating initial state: 1 or 0 (default). The gating state is set to requested value immediately after function call
--------------	--

Return Type

Channel ID allocated for this operation. Return -1 if failed.

5.1.1.17 *SegmentPulse*

Description

Activate automatic pulse firing for segment motion.

Syntax

```
int SegmentPulse (Mode, PulseWidth)
```

Return Type

Channel ID allocated for this operation. Return -1 if failed.

Arguments

Mode	This parameter defines the type and occurrence of the synchronization signal: 2 – Pulse on demand. The LINE/ARC1/ARC2 special switch (/p) defines a segment at the beginning of which the pulse occurs. 3 – Pulse is generated automatically at the beginning of every segment (starting from the second wegment)
PulseWidth	Pulse width (msec)

5.1.1.18 *SetExtClockSync*

Description

Enable or Disable External Laser Clock synchronization.

Syntax

```
SetExtClockSync(Enable [,Polarity, Delay] )
```

Enable	Enable or Disable External Laser Clock synchronization mode (1 – Enable, 0 - Disable)
Polarity	Optional 1 or 0 1 – Synchronization on rising edge(default) 0 – Synchronization on falling edge
Delay	Optional Delay relative to External Laser Clock edge in µsec. The default is 0.

5.1.1.19 *PowerPWMBurst*

Description

The function initializes pulse modulation mode with specified number of pulses

Syntax

```
PowerPWMBurst (Freq, Width, PulseNum, [SynchFlag])
```

Arguments

Freq	Pulse modulation frequency in Hz, range from 0.035Hz to 1MHz.
Width	Pulse Width in milliseconds. Range from 6.67nsec to 28.60sec
PulseNum	The number of outgoing PWM pulses
SynchFlag	(Optional). If parameter is non-zero, the function works synchronously i.e., waits till the pulse sequence finished, otherwise initializes the pulse sequence without waiting. The default is 1 (works synchronously)

Comments

The **LaserEnable** function should be called ahead beforehand to enable the sequence of pulses. If the function works asynchronously, the program should check the `PWMBurstReady` field to ensure that the pulse sequence is finished.

Example

```
lc.LaserEnable()

!Initialize the 10 pulses with frequency 1 kHz
lc.PowerPWMBurst(1000, 0.5, 10, 0)

!wait till pulse sequence finished
till lc.PowerBurstReady = 1
```

5.1.1.20 SetSafetyMasks

Description

Enable or disable Safety and Fault inputs.

Syntax

```
SetSafetyMasks (SafetyInput, FaultInput )
```

Arguments

SafetyInput	The 1 value masks the Safety Input and the system ignores Safety errors
FaultInput	The 1 value masks the Fault Input and the system ignores the Faults

5.1.1.21 Stop

Description

Cancel laser mode or state for specified channel.

Syntax

```
Stop (Channel)
```

Return Type

None

Arguments

Channel	Channel or Operation ID. If omitted or negative, cancel all active operations.
---------	--

5.1.1.22 *SetMechPlatformAxes*

Description

Define the axes configuration for the mechanical platform.

Syntax

```
SetMechPlatformAxes (X_Ax, Y_Ax, Z_Ax , A_Ax, B_Ax, C_Ax)
```

Return Type

None

Arguments

X_Ax	Axis index corresponds to X axis of mechanical platform
Y_Ax	Axis index corresponds to Y axis of mechanical platform
Z_Ax	Axis index corresponds to Z axis of mechanical platform
A_Ax	Axis index corresponds to A axis of mechanical platform
B_Ax	Axis index corresponds to B axis of mechanical platform
C_Ax	Axis index corresponds to C axis of mechanical platform

Comments

If parameter omitted or -1, axis is not available in the current mechanical platform configuration.

5.1.1.23 *SetMotionAxes*

Description

Define the axes used for subsequent laser operations.

Syntax

```
SetMotionAxes (Axes_list)
```

Arguments

Axes_list	Single axis or axis group, valid numbers are: 0...5
-----------	---

Return Value

None

Comments

The function defines the mechanical platform to use. Valid values for axis designation are as follows:

Axis	Code
X	0
Y	1
Z	2
A	3
B	4
C	5

Example

```
!Set platform axes X and Y for subsequent operations.  
lcUnit.SetMotionAxes (0,1)
```

5.1.1.24 SetSystemDelay

Description

Change the internal pulse generation delay.

Syntax

```
SetSystemDelay (Channel, TimeDelay[,Relative] )
```

Arguments

Channel	Channel ID
TimeDelay	A new system delay in μ sec
Relative	If 1, the TimeDelay value is added to the current delay, otherwise the TimeDelay value is used as a new value for system delay

Return Value

None

5.1.1.25 *GetSystemDelay*

Description

Return the current system delay in μsec for the specified channel.

Syntax

```
GetSystemDelay (Channel)
```

Arguments

Channel	Channel ID. If omitted or negative cancel defined operation for all channels
---------	--

Return Value

None

5.1.1.26 *SetConfigOut*

Description

Configure the Digital Output signal

Syntax

```
SetConfigOut (OutInd, Channel, Code)
```

Arguments

OutInd	Index of Configurable Output (0...7) or 10 for LPC output
Channel	Channel index (0...7) whose signal routed to output
Code	Enumerator defines the signal routed to the output. See table below.

Return Value

None

Comments

Code	Signal Description
0	Cancel Routing
2	P/D Pulse
3	P/D Direction
4	A signal of AqB
5	B signal of AqB
6	InRange

Code	Signal Description
7	PEG Pulse
8	PEG State
9	PEG Active

If output index equals 10 (LPC output), channel parameter is irrelevant and the code parameter can accept the following values:

Code	Signal Description
0	No signal via LPC
1	Output PWM via LPC
2	LPC output use as synchronization pulse signal for 8 bit digital port
3	Reserved

Example

```
! Output 2 is configured as the PEG State of channel 1:
lc.SetConfigOut(2, 1, 8)
```

5.1.1.27 *AssignChannels*

Description

Dedicate channel for a specific operation.

Syntax

```
AssignChannels (OperationsArray)
```

Arguments

OperationsArray	Operation enumeration array, each member contains the operation type dedicated for specific channel
-----------------	---

Return Value

None

Comments

Operation Enumeration

100	Fixed Distance Pulse
101	Distance Array Pulse

102	Coordinate Array pulse
103	Distance Array Gating
104	Segment-based
105	Coordinate Array Gating
0x10000	Zone (In Range). InRange code can be combined using an OR operator with another operation code

Example*5.1.1.28 SetCustomPosCalc***Description**

Define the user function intended for position calculation. The firmware calls this function every controller cycle. The function returns the calculated value as a real variable.

Syntax

```
SetCustomPosCalc (Channel, FuncRef)
```

Arguments

Channel	Channel or Operation ID
FuncRef	Reference to the function, which calculates the new position value. The function must match this pattern: <code>real fastcall FuncName()</code>

Return Value

None

*5.1.1.29 SetCustomVelCalc***Description**

Define the user function intended for custom velocity-based calculation. The firmware calls this function every controller cycle. The function returns the calculated value as a real variable.

Syntax

```
SetCustomVelCalc (Operation, FuncRef)
```

Arguments

Operation	Operation ID (Modulation, Analog Power etc.)
FuncRef	Reference to the function, which calculates the new value. The function must match this pattern: <code>real fastcall FuncName()</code>

5.1.1.30 SetCustomVelVar

Description

Define the user custom position calculation variable. Every controller cycle the firmware takes the variable as a new calculated value.

Syntax

```
SetCustomVelVar (Operation, VarRef)
```

Arguments

Operation	Operation ID (Modulation, Analog Power etc.)
VarRef	Reference to the global ACSPL+ variable

Return Value

None

5.1.2 LCI Structure Fields

Field Name	Type	Accessibility	Description
MotionAxes	int	R/W	Default axes mask used by LCI functions
PosResolution	real	R/W	Default Pulse Resolution used by LCI functions
Internal PosResolution (8)	real	R/W	Contains the actual Pulse Resolution used by LCI functions, one element per channel
PWMDutyCycle	real	R	Current laser duty cycle as percent
PWMFrequency	real	R	Current laser frequency in Hz
PWMPulseWidth	real	R	PWM pulse width in ms
TickleFrequency	real	R	Tickle frequency in Hz
TicklePulseWidth	real	R	
PWMActive	in	T	Boolean, 1 if modulation mode is active
TickleActive	int	R	Boolean, 1 if Tickle mode is active

Field Name	Type	Accessibility	Description
InRange	int	R	Boolean 1/0
LaserEnabled	int	R	Boolean 1/0
OperationMode(8)	int	R	Operation mode specified for channel
Positions(8)	int	R	FPOS for specific channel
UserPos	real	R	Current Channel Position in user units
MultiAxWinSize	real	R/W	Comparison window width for coordinate-based operations
ExtraPulsesQty	int	R/W	The number of additional pulses generated with ExtraPulsesPeriod after each pulse at each firing position. The parameter is applicable for pulses generated by the following functions: FixedDistPulse, DistanceArrPulse, CoordinateArrPulse.
ExtraPulsesPeriod	real	R/W	The period in milliseconds for ExtraPulses additional pulses to be generated after each pulse at each firing position. The parameter is applicable for pulses generated by the following functions: FixedDistPulse, DistanceArrPulse, CoordinateArrPulse
PiercePulsesNum	int	R/W	Number of pierce pulses at the beginning
PiercePulsesWidth	real	R/W	ms
GateOnDelay	real	R/W	Stat on delay in ms in gating mode

Field Name	Type	Accessibility	Description
GateOffDelay	real	R/W	State off delay in ms in gating mode
PulseDelay	real	R	Laser firing pulse delay in Distance Array Pulse modes
PowerAOutVal	real	R	Analog Output value in %
PowerDigOutVal	int	R	Digital Output value

5.1.2.1 MotionAxes

Description

MotionAxes is a R/W integer field and contains the default axes mask used by LCI functions.

Syntax

```
lcUnit.MotionAxes = AxListAsMask(X,
```

5.1.2.2 PosResolution

Description

PosResolution is a R/W real field and contains the default Pulse Resolution used by LCI functions. If the default Pulse Resolution is 0, the pulse resolution should be calculated according to **XVEL** parameter and Maximum LCI Frequency.

5.1.2.3 InternalPosResolution

Description

InternalPosResolution is a Read-only real array with one element for each LCI channel and containing the actual Pulse Resolution used by LCI functions.

5.1.2.4 PWMDutyCycle

Description

PWMDutyCycle is a Read-Only real field. It presents the current laser duty cycle (%)

Syntax

```
DISP lcUnit.PWMDutyCycle
```

5.1.2.5 PWMFrequency

Description

PWMFrequency is a Read-Only real field. It presents the current laser frequency (Hz)

Syntax

```
DISP lcUnit.PWMFrequency
```

*5.1.2.6 PWMPulseWidth***Description**

PWMPulseWidth is a Read-Only real field. It presents the current PWM pulse width (ms)

Syntax

```
DISP lcUnit.PWMPulseWidth
```

*5.1.2.7 TickleFrequency***Description**

TickleFrequency is a Read-Only real field. It presents the current Tickle frequency (Hz)

Syntax

```
DISP lcUnit.TickleFrequency
```

*5.1.2.8 TicklePulseWidth***Description**

TicklePulseWidth is a Read-Only real field. It returns the current Tickle pulse width in ms.

Syntax

```
DISP lcUnit.TicklePulseWidth
```

*5.1.2.9 PWMActive***Description**

PWMActive is a Read-Only integer field. It returns a Boolean value: 1 if modulation mode is switched on, otherwise 0.

*5.1.2.10 TickleActive***Description**

TickeActive is a Read-Only integer field. It returns a Boolean value: 1 if Tickle mode is switched on, otherwise 0.

*5.1.2.11 InRange***Description**

InRange is a Read-Only integer field. It returns a Boolean value: 1 if the unit is inside of defined range, otherwise 0.

5.1.2.12 *LaserEnabled*

Description

Laser Enabled is a Read-Only integer field. It returns a Boolean value: 1 if the laser is enabled, otherwise 0.

5.1.2.13 *OperationMode*

Description

OperationMode is an integer array with one element for each LCI channel. It returns the operation mode defined for a specific channel.

5.1.2.14 *Positions*

Description

Positions is an integer array with one element for each LCI channel. It returns the pulse counter value for a specific channel.

5.1.2.15 *UserPos*

Description

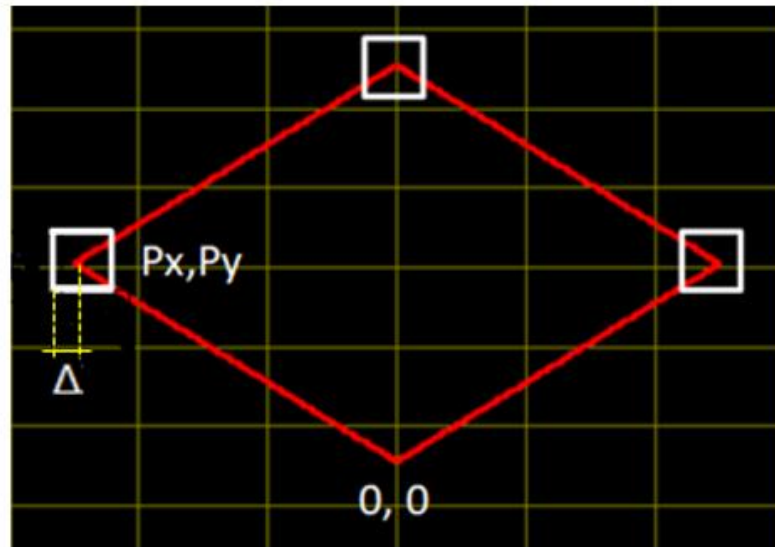
UserPos is an array of real, with one element for each LCI channel. Each element returns the current channel position in user units. UserPos is calculated as follows:

$$\text{UserPos} = \text{Positions} * \text{InternalPosResolution}$$

5.1.2.16 *MultiAxWinSize*

Description

MultiAxWinSize returns a real value defining the window size in coordinate based mode. The value defines the range (Δ) inside which the pulse and gate signals should be fired. The field is applicable for the **CoordinateArrPulse** and **CoordinateArrGate** functions. The parameter is in user units. The default is 1 resolution count. Δ indicates the window in the following diagram.



5.1.2.17 *ExtraPulsesQty*

Description

ExtraPulsesQty returns the number of additional pulses generated with ExtraPulsesPeriod after the initial pulse at each firing position. The parameter is applicable for pulses generated by the following functions: **FixedDistPulse**, **DistanceArrPulse**, **CoordinateArrPulse**. The value 0 means no extra pulses are generated (default).

5.1.2.18 *ExtraPulsesPeriod*

Description

The period in milliseconds for additional pulses defined in ExtraPulses to be generated after each pulse at each firing position. The parameter is applicable for pulses generated by the following functions: **FixedDistPulse**, **DistanceArrPulse**, **CoordinateArrPulse**. The value 0 means no extra pulses are generated (default).

5.1.2.19 *PiercePulsesNum*

Description

The number of pierce pulses generated during a pierce pulse cycle. The parameter is applicable for pulses generated by **FixedDistPulse** function. The value 0 means no pierce pulses are generated (default).

5.1.2.20 *PiercePulsesWidth*

Description

The width of pierce pulses in milliseconds. The parameter is applicable for pulses generated by **FixedDistPulse** function. The 0-value means, no pierce pulses are generated (default).

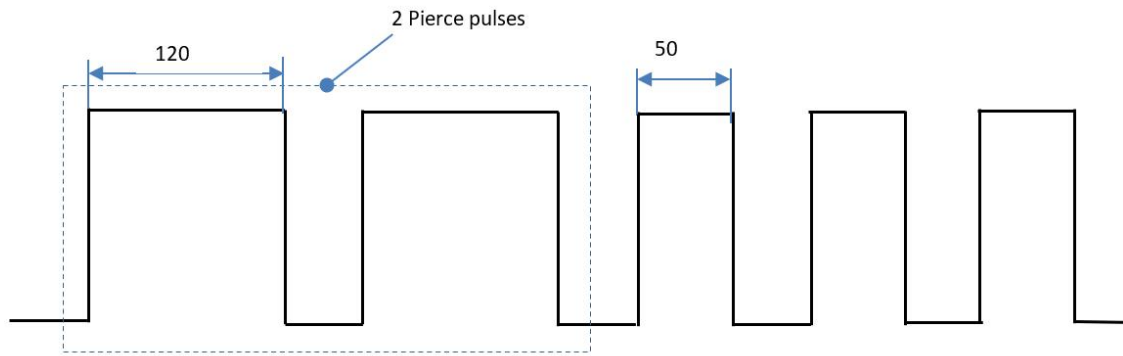


Figure 4-34. 2 Pulses with Pulse Width 120 μ sec.

5.1.2.21 *GateOnDelay*

Description

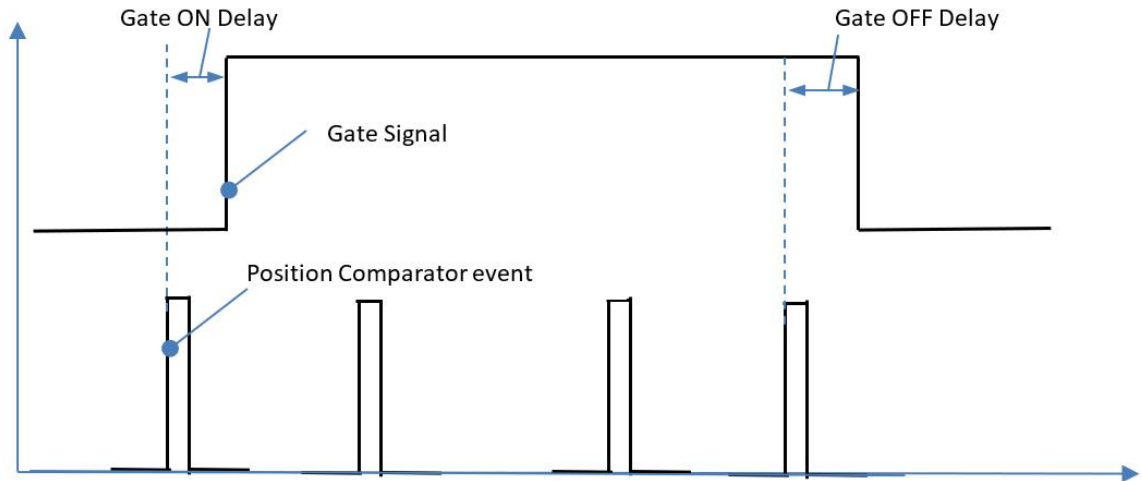
Returns state on delay in milliseconds when in Gating Mode. Default is 0. See [Figure 4-35](#)

5.1.2.22 *GateOffDelay*

Description

Returns state off delay in milliseconds when in Gating Mode. Default is 0.

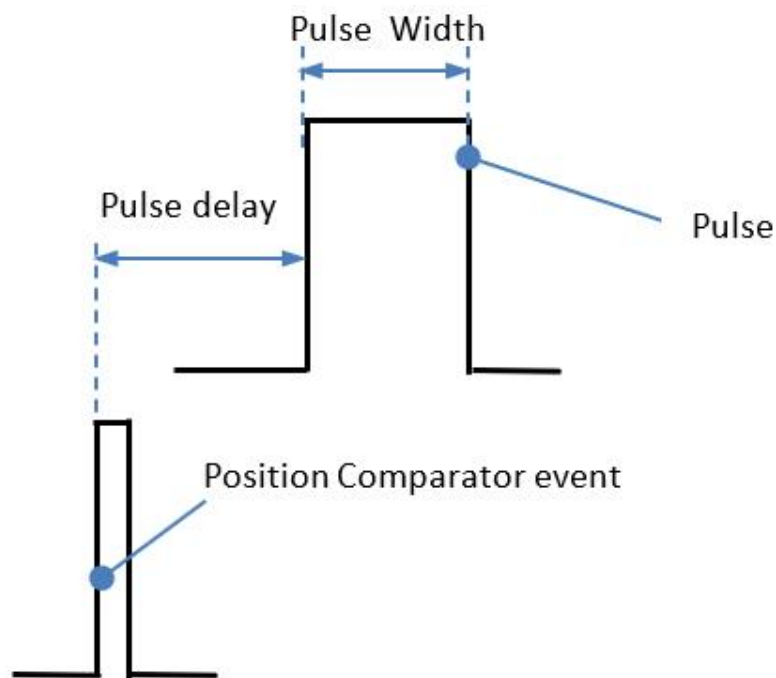
Figure 4-35. Delays in Gating Mode



5.1.2.23 PulseDelay

Description

Laser firing pulse delay in Distance Array Pulse modes. The default is 0.



5.1.2.24 PowerAOutVal

Description

PowerAOutVal is a Real Read-Only real field. It returns the current analog output value. The range is 0 to 100%. The field is updated if the Power Control via Analog Output mode is activated.

5.1.2.25 Faults

Description

Faults is a Read-only Integer Field. It contains a set of bits representing the current LCI error state, according to the following table:

Bit #	Description
0	Laser Fault, fault input is On
1	Laser safety, safety input is Off
2	Velocity limit fault, motion axes velocity exceeds the maximum frequency supported by LCI

5.1.2.26 PWMBurstReady

Description

PWMBurstReady is a read-only Integer field. It is interpreted as a boolean flag: 1 means that the system is ready to perform the next PWM burst.

5.2 Diagnostics and Preventive Maintenance (DPM)

The DPM related STRUCTs and functions answer the need to monitor system performance degradation. The user can implement the required measurements and diagnostics with a minimal effort.

The DPM feature allows the user to identify:

1. Performance degradation
 - > Increase of following error
 - > Increase in settling time
 - > Increase in velocity ripple during constant velocity
2. System degradation (changes that indicate some mechanical or electrical deterioration)
 - > Increase of current consumption
 - > Increase of temperature
3. Part failure (root cause of deterioration)
 - > Coupling break
 - > Cable break / partial break
 - > Motor deterioration due to over temperature
4. Maintenance based on smart measurement
 - > Bearing life measurement for preventive maintenance (like the warning that the car generates about the time for periodic maintenance.)
 - > Moving cable cycles measurement for preventive maintenance
 - > Drive life shortening due to repeated short circuits

This object is supported in version 3.10 and higher.

5.2.1 DPM_Measurement

Description

DPM_Measurement is an ACSPL+ composite data type(STRUCT) with different fields of data types and functions that are used to configure and store the results of the MeasureProcess()/MeasurePeriodically() functions. It is part of the DPM(Diagnostics and Preventive Maintenance) feature. To declare an instance of the DPM_Measurement STRUCT the user will use the reserved word DPM_Measurement followed by the desired struct name.

Syntax

```
DPM_Measurement Struct_name
```

See the *ACSPL+ Commands & Variables Reference Guide* for details about specific commands, functions, and variables.

5.2.1.1 DPM_Measurement Fields

Variable Type and Name	Description
real variable_value	It holds the raw instant measured value of the variable specified by the user. This variable is updated every MPU cycle once the measurement is initiated. Default = 0. Types of variables that can be monitored: > Any ACSPL+ standard variable. > User variables that are defined in the D-Buffer as global and static.(Including Structs variables if the Struct is defined as global and static)
real variable_abs_value	It holds the absolute value of the variable_value . This variable is updated every MPU cycle once the measurement is initiated.
real measured_abs_value	Each MPU cycle this variable is set (or updated) to variable_abs_value * when_to_measure This variable holds the absolute value of the variable_value when when_to_measure is 1, and 0 otherwise.
real sampled_value	Holds the latest sample of measured_abs_value. See definition of a sample under sampling_type

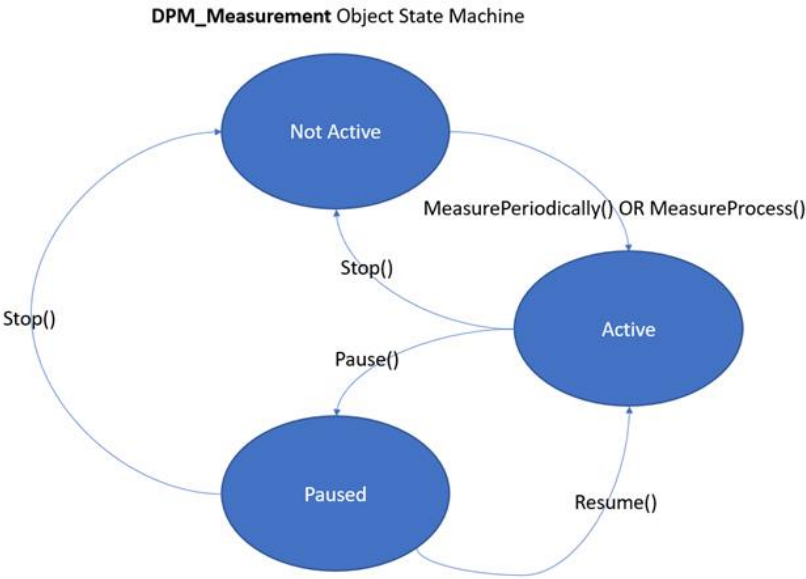
Variable Type and Name	Description
int measurement_status	Holds the current status of the DPM_measurement structure 0 – measurement is NOT ACTIVE 1 – measurement is PAUSED 2 – measurement is ACTIVE
int measurement_type	Specifies the current measurement type. 0 – None 1 – Process measurement 2 – Periodical measurement
int sample_set_size	Specifies the size of the set of adjacent samples to be used for the calculation of the moving_average_value, std_dev_value, and the rms_value. Values: Ranges between 1 and 512.
int when_to_measure	when_to_measure defines the event slots during which the variable_abs_value is measured and assigned to measured_abs_value. Values : 0 or 1. For example: Motor Current during the acceleration phase of a moving axis. For measurements related to motion, the DPM_Motion_Status struct can be used. Types of variables that can be used: > Any ACSPL+ integer standard variable. > Any user variable that is defined in the D-Buffer as integer global and static (Including Struct variables if the Struct is defined as global and static).
real sampling_time	sampling_time defines the sampling period in milliseconds when using the MeasurePeriodically() function. Minimum value: MPU cycle time. Maximum value: 100,000,000 msec (>24h). MeasurePeriodically() sample will be taken first on the positive edge of when_to_measure, and then every sample_time period (as long as when_to_measure is 1(True)). The resolution of sampling_time is the MPU cycle time.
int sampling_type	The sampling_type is used by the MeasureProcess() function, and defines the nature of a sample:

Variable Type and Name	Description
	> When to sample > What is a sample and how it is calculated 0 – The measured_abs_value is sampled upon the positive edge of when_to_measure, and the sampled_value is set to this value. As an example, it may be used to measure the settling time of moves. 1 – The measured_abs_values are summed up every MPU cycle, and the sampled_value is set to the average value of the summation upon the negative edge of when_to_measure. For example, it may be used to measure current during acceleration. sampled_value presents the average of the absolute current during the acceleration period. This variable is ignored when using MeasurePeriodically(). In a periodic measurement, the sampling_time variable is used. See sampling_time definition for more information.
real sample_counter	Specifies the number of samples that have been collected since the beginning of the measurement process. See definition of a sample at sampling_type
real peak_value	Holds the highest value of a sample that has been taken since the beginning of the measurement. It can be set to 0 by the user by calling ClearPeak().
real moving_average_value	Holds the average value of the samples in the set. The number of samples in the set is specified by sample_set_size. See definition of a sample at sampling_type
real std_dev_value	Holds the standard deviation of the samples in the set. The number of samples in the set is specified by sample_set_size.
real rms_value	Holds the RMS (Root mean square) of the samples in the set. The number of samples in the set is specified by the sample_set_size.

5.2.1.2 DPM_Measurement Functions

Function Name	Arguments	Description
MeasureProcess (monitored_variable , when_to_measure , sample_set_size , sampling_type)	real/Int monitored_variable – A variable to be monitored. (see monitored_variable field definition for more details) int when_to_measure – A variable that acts as a flag. when the value of the variable is different than 0, a measurement will take place. (see when_to_measure field definition for more details) int sample_set_size – Defines the samples set size. (see sample_set_size field definition for more details) int sampling_type – Defines the sampling behavior(see sampling_type definition for more details)	This function is used to initiate monitoring and measurements of variables that are related to processes and especially motion processes that are generated by atomic motion commands (PTP, BPTP, JOG). Example: The average current during the acceleration phase of a moving axis.
MeasurePeriodically (monitored_variable , when_to_measure , sample_set_size , sampling_time)	real/Int monitored_variable – A variable to be monitored. (see monitored_variable field definition for more details) int when_to_measure – A variable that acts as a flag. When the value of the variable is different than 0, a measurement will take place. (see when_to_measure field definition for more details) int sample_set_size – Defines the samples set size. (see sample_set_size field definition for more details)	This function is used to initiate the monitoring and measurements of the supplied variable over time. Example 1: measurement of temperature that is monitored by a sensor that is connected to an analog input Example 2: 2D position error during one “long” arbitrary XY move

Function Name	Arguments	Description
	real sampling_time – Defines the period between samples(see sampling_time definition for more details)	
Stop()	None	Stops the measurement if it is active or paused. It does not affect the measurement if it is not active.
Pause()	None	Pauses the measurement if it is active. It does not affect the measurement if it is not active.
Resume()	None	Resume previously paused measurement. Does not affect the measurement if it is not paused.
ClearAll()	None	Sets variable_value , variable_abs_value , measured_abs_value , sampled_value , sample_counter , peak_value , moving_average_value , std_dev_value , rms_value and when_to_measure to 0.
ClearPeak()	None	Sets peak_value to 0.



5.2.2 DPM_Motion_Status

Description

DPM_Motion_Status is an ACSPL+ composite data type (STRUCT), with different fields of data types and functions that are used to configure and to provide an indication on the phase of the motion for a user-selected axis. It is part of the DPM (Diagnostics and Preventive Maintenance) feature. To declare an instance of the DPM_Motion_Status STRUCT the user uses the reserved word DPM_Motion_Status followed by the desired struct name.

Syntax

```
DPM_Motion_Status struct_name
```

5.2.2.1 DPM_Motion_Status Fields

Variable Name	Description
int during_motion	The variable is set to true (=1) once the motion profile starts and is set automatically to false (=0) once the motion profile is completed.
int during_accel	The variable is set to true (=1) once the motion profile reaches the acceleration phase(can be constant acceleration or during jerk) and is set automatically to false (=0) once the acceleration phase is completed.

Variable Name	Description
int during_decel	The variable is set to true (=1) once the motion profile reaches the deceleration(can be constant deceleration or during jerk) and is set automatically to false (=0) once the deceleration phase is completed.
int during_cv	The variable is set to true (=1) once the motion profile reaches the Constant Velocity (CV) phase and is set to false (=0) once deceleration has started.
int selected_axis	The variable specifies the selected axis for which its motion phases are monitored.
int on_off	When 1, the motion phases are monitored. When 0, the phases are not monitored.

5.2.2.2 DPM_Motion_Status Functions

Function	Arguments	Description
SelectAxis (Axis)	Axis - Valid values are: 0, 1, 2, ... up to the number of axes in the system minus 1.	Axis - Specifies the axis is monitored once MonitorOn() is called. *Can only be used if the on_off variable is OFF (=0)(monitoring is not active).
MonitorOn()	None	Provides the user the ability to activate monitoring of the selected_axis. Once the MonitorOn() has been called, the motion phase and settling status of move commands related to the Selected_Axis will be indicated by the STRUCT variables.

Function	Arguments	Description
MonitorOff()	None	Provides the user the ability to deactivate monitoring of the selected_axis. Move commands that are issued after calling MonitorOff() will not be monitored.
ClearAll()	None	Sets during_motion , during_accel , during_decel , and during_cv to 0

5.2.3 DPM Example - Adding current measurement during acceleration phase to an existing application

This example demonstrates how measurements can be added to existing applications without modifying existing code. In this example, the current during acceleration of every move of the Y-axis is measured.

To implement the measurement for an existing application:

1. Add code to an AUTOEXEC routine that initiates the measurement.
2. Add a condition monitoring (ON) routine that raises a flag when a threshold is exceeded.

The setting and resetting of **motion_status_Y.MonitorOn()** / **MonitorOff()** provides a mechanism to control which moves are monitored and measured.

```

D-Buffer:
axisdef Y=1
Global static DPM_Measurement actual_current_during_accel_Y
Global static DPM_Motion_Status motion_status_Y
Global static REAL Y_Accel_Peak_Current_Threshold, Y_Accel_Moving_
Average_Current_Threshold, CURRENT_Y_AMP
Global INT Samplesample_set_size_Y, DRIVE_PEAK_CURRENT_AMP, ADC_RANGE
Global static INT measure_continuously

Buffer 0:
AUTOEXEC:
Y_Accel_Peak_Current_Threshold = 5
Y_Accel_Moving_Average_Current_Threshold = 2.5
measure_continuously = 1
sample_set_size_Y = 20
DRIVE_PEAK_CURRENT_AMP = 10
ADC_RANGE = 32767 ! Current is sampled with a 16 bit ADC

motion_status_Y.SelectAxis(1) ! Set axis 1 as the axis that provides the
motion phases to the the motion_status_Y STRUCT once the motion_status_
Y.MonitorOn() function is called
motion_status_Y.MonitorOn() !Monitoring is activated. The STRUCT will be
updated for each move that follows, and measurement will be made
accordingly

```

```

actual_current_during_accel_Y.Stop() !Stops the measurement if it is
active
!Measurement activation
actual_current_during_accel_Y.MeasureProcess(CURRENT_Y_AMP, motion_
status_Y.during_accel, sample_set_size_Y, measure_continuously)

WHILE(1) !Run every MPU cycle
CURRENT_Y_AMP = DOUT(1) * (DRIVE_PEAK_CURRENT_AMP /ADC_RANGE) !converts
DOUT units to Amperes
END
STOP

Buffer 1:
ENABLE 1
LOOP 10
PTP 1,1000
BPTP 1,3000
PTP 1,4000
PTP 1,0
.
.
.
END
STOP
!acceleration phase peak threshold has been exceeded
ON actual_current_during_accel_Y.peak_value > Y_Accel_Peak_Current_
Threshold
DISP "Y peak current threshold is exceeded"
actual_current_during_accel_Y.ClearPeak()
RET
!acceleration phase moving average threshold has been exceeded
ON actual_current_during_accel_Y.moving_average_value > Y_Accel_Moving_
Average_Current_Threshold
DISP "Y moving average current threshold is exceeded"
RET

```

5.3 Motion Duration

5.3.1 MotionDuration Struct

The MotionDuration struct can be used to calculate motion duration according to motion type and motion parameters.

The MotionDuration struct calculation functions calculate motion duration in seconds. Functions of the struct can be executed in buffer only, and it will wait till the execution is completed.

Struct Fields

Field Name	Type	Accessibility	Range	Comments
Distance	real	R/W		user units
Velocity	real	R/W	from - 1.79769e+30 8 to 1.79769e+30 8	Default = 10000
Acceleration	real	R/W	from 2.22507e-308 to 1.79769e+30 8	Default = 100000
Deceleration	real	R/W	from 2.22507e-308 to 1.79769e+30 8	Default = 100000
Jerk	real	R/W	from 2.22507e-308 to 1.79769e+30 8	Default = 2E7
Snap	real	R/W	from 2.22507e-308 to 1.79769e+30 8	Deafault = 100E7
InitialVelocity	real	R/W		Default = 0
InitialAcceleration	real	R/W		Default = 0
FinalVelocity	real	R/W		Default = 0
FinalAcceleration	real	R/W		Default = 0
EncoderFactor	real	R/W	from 1e-15 to 1e+15	Default = 1
T_MotionOverallTime	real	R		

Field Name	Type	Accessibility	Range	Comments
T31_ JerkBuildupAccelerationBuild up	real	R		
T1_ ConstantJerkAccelerationBuil dup	real	R		
T32_ JerkFinishAccelerationBuildu p	real	R		
T2_ConstantAcceleration	real	R		
T33_ JerkBuildupAccelerationFinis h	real	R		
T3_ ConstantJerkAccelerationFinish	real	R		
T34_ JerkFinishAccelerationFinish	real	R		
T4_ConstantVelocity	real	R		
T35_ JerkBuildupDecelerationBuil dup	real	R		
T5_ ConstantJerkDecelerationBui ldup	real	R		
T36_ JerkFinishDecelerationBuildu p	real	R		
T6_ConstantDeceleration	real	R		
T37_ JerkBuildupDecelerationFinis h	real	R		
T7_ ConstantJerkDecelerationFin	real	R		

Field Name	Type	Accessibility	Range	Comments
ish				
T38_JerkFinishDecelerationFinish	real	R		

Struct Functions

ReadFrom(int AxisNum)	Read Velocity, Acceleration, Deceleration, Jerk, Snap and EncoderFactor from specified axis and write them to appropriate fields in struct.
CalculatePTPDuration()	Calculates PTP (3rd order motion profile) motion duration and duration of each motion phase using struct parameters Relevant output members: T_MotionOverallTime, T1_ConstantJerkAccelerationBuildup, T2_ConstantAcceleration, T3_ConstantJerkAccelerationFinish, T4_ConstantVelocity, T5_ConstantJerkDecelerationBuildup, T6_ConstantDeceleration, T7_ConstantJerkDecelerationFinish
CalculateBPTPDuration()	Calculates BPTP motion duration using struct parameters Relevant output member: T_MotionOverallTime
CalculateSPTPDuration()	Calculates SPTP (4th order motion profile) motion duration and duration of each motion phase using struct parameters Output members: T_MotionOverallTime, T31_JerkBuildupAccelerationBuildup, T1_ConstantJerkAccelerationBuildup, T32_JerkFinishAccelerationBuildup, T2_ConstantAcceleration, T33_JerkBuildupAccelerationFinish, T3_ConstantJerkAccelerationFinish, T34_JerkFinishAccelerationFinish,

```

T4_ConstantVelocity,
T35_JerkBuildupDecelerationBuildup,
T5_ConstantJerkDecelerationBuildup,
T36_JerkFinishDecelerationBuildup,
T6_ConstantDeceleration,
T37_JerkBuildupDecelerationFinish,
T7_ConstantJerkDecelerationFinish,
T38_JerkFinishDecelerationFinish

```

Comments

- > Two options exist for initialization of the editable fields of the MotionDuration struct:
 - a. Initialize each writeable field manually, one by one.
 - b. Use the **ReadFrom** function to read the Velocity, Acceleration, Deceleration, Jerk, Snap and EncoderFactor fields from a specified axis. The Distance field will still need to be initialized manually.

Struct members: InitialVelocity, InitialAcceleration, FinalVelocity and FinalAcceleration will not be affected by calling the **ReadFrom** function.

- > Zero value in fields Velocity, Acceleration, Deceleration, Jerk, Snap and EncoderFactor will implicitly be replaced by the default value.
- > There may be calculated results that are not relevant for all types of motion; they will be initialized to their default value.

Example

```

#D-Buffer:
global static MotionDuration motion_duration

```

```

#Buffer 0:
motion_duration.ReadFrom(0)
motion_duration.Distance=1
motion_duration.InitialVelocity=0
motion_duration.InitialAcceleration=0
motion_duration.FinalVelocity=0
motion_duration.FinalAcceleration=0
motion_duration.CalculatePTPDuration()

```

Terminal output:

```

?motion_duration
Struct type : MotionDuration
real Distance
          1

```

```

real Velocity
    10
real Acceleration
    1000
real Deceleration
    1000
real Jerk
    100000
real Snap
    1E+007
real InitialVelocity
    0
real InitialAcceleration
    0
real FinalVelocity
    0
real FinalAcceleration
    0
real EncoderFactor
    1
READONLY real T_MotionOverallTime
    0.12
READONLY real T31_JerkBuildupAccelerationBuildup
    0
READONLY real T1_ConstantJerkAccelerationBuildup
    0.01
READONLY real T32_JerkFinishAccelerationBuildup
    0
READONLY real T2_ConstantAcceleration
    0
READONLY real T33_JerkBuildupAccelerationFinish
    0
READONLY real T3_ConstantJerkAccelerationFinish
    0.01
READONLY real T34_JerkFinishAccelerationFinish
    0
READONLY real T4_ConstantVelocity
    0.08
READONLY real T35_JerkBuildupDecelerationBuildup
    0
READONLY real T5_ConstantJerkDecelerationBuildup
    0.01
READONLY real T36_JerkFinishDecelerationBuildup
    0
READONLY real T6_ConstantDeceleration
    0
READONLY real T37_JerkBuildupDecelerationFinish
    0
READONLY real T7_ConstantJerkDecelerationFinish

```



```
0.01  
READONLY real T38_JerkFinishDecelerationFinish  
0
```

5.4 Safety Zones

The SafetyZone structure allows the user to restrict movement to a defined range ("No Exit") or prevent movement into a defined range or set of ranges ("No Entry"). These ranges may be defined in one, two, or three dimensions.

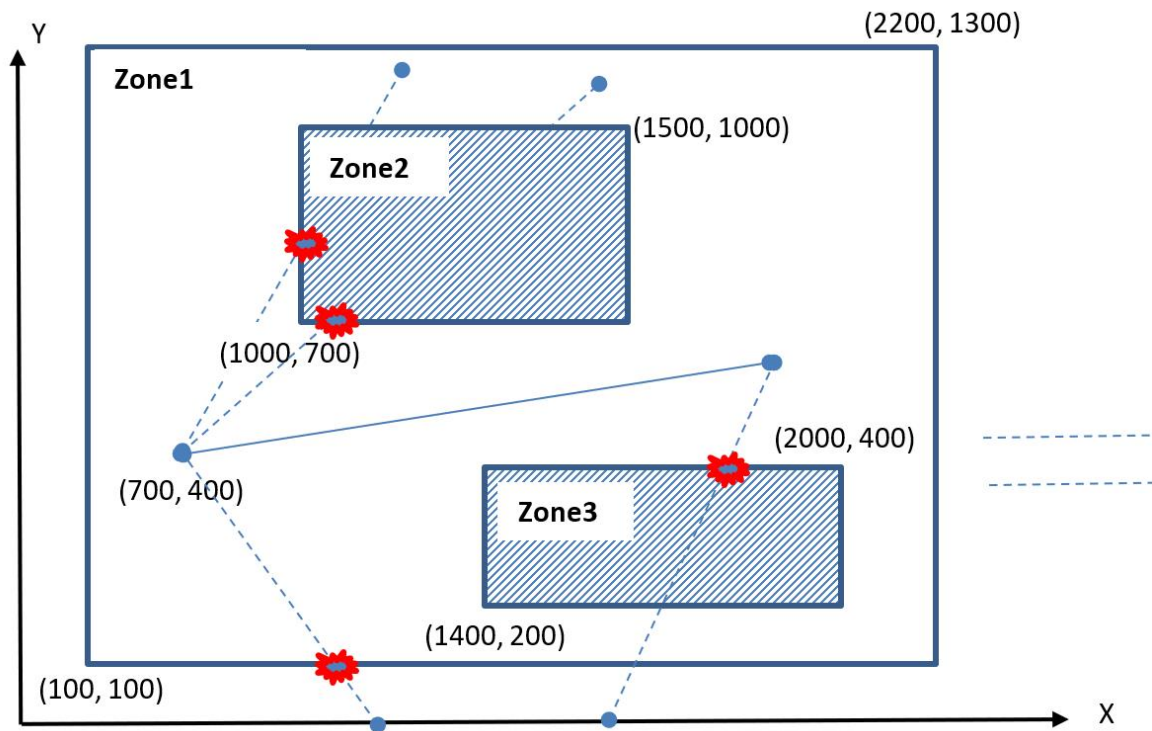


Table 4-22. Example of Two-dimensional Safety Zone Definition

In this example, the "No Exit" zone is defined as the rectangle from (100, 100) to (2200, 1300). Any motion bringing the load outside of this zone will cause an exception and the system will execute the Kill command, stopping all motion. Two "No Enter" Zones are also defined; any attempt to enter these zones will also cause an exception and execution of the Kill command. Each zone is defined by a separate SafetyZone structure.

5.4.1 SafetyZone Structure and ACSPL+ Global Functions

The SafetyZone structure includes the following fields.

Name	Description
ConditionType	0 - No Enter Zone 1 - No Exit Zone
Enabled	0 - Zone not enabled

Name	Description
	1 - Zone Enabled
Faults	The Read Only integer flag. A non-zero value means that the motion has been stopped by a safety zone condition.

The SafetyZone structure includes the following functions.

Name	Description
SetAxes	Defines the Safety Zone physical axes indexes
Enable	Enables the Safety Zone
Disable	Disables the Safety Zone
SetZone1D	Defines a one-dimensional Safety Zone
SetZone2D	Defines a two-dimensional Safety Zone
SetZone3D	Defines a three-dimensional Safety Zone
SetZoneDualCarriage	Defines a dual carriage Safety Zone
SetMargins	Sets Safety Zone margins dynamically
AddOffset	Adds offsets to Safety Zones dynamically
SetCustomPosValidFunc	Defines a custom user function to validate positions related to Safety Zone.

5.4.1.1 *SZonesEn*

Description

The function enables all previously disabled Safety Zones.

Syntax

```
SZonesEn
```

5.4.1.2 *SZonesDis*

Description

The function disables all previously enabled Safety Zones and removes them from system list.

Syntax

```
SZonesDis
```

5.4.2 SetAxes

Description

The function defines the Safety Zone physical axes indexes.

Syntax

```
SetAxes (X_Index, [Y_Index, [Z_Index]])
```

Arguments

X_Index	Controller axis index used to implement Safety Zone X axis
Y_Index	Controller axis index used to implement Safety Zone Y axis
Z_Index	Controller axis index used to implement Safety Zone Z axis

Comments

For 1D Safety zone only X_Index parameter is relevant, for 2D Zone only X_Index and Y_Index are relevant. The default system settings are:

X_Index=0, Y_Index=1, Z_Index=2

Example

```
global SafetyZone sz1, sz2
!Define axis 20 as X axis of sz1 Safety Zone, 21 as Y and 22 as Z
sz1.SetAxes(20,21,22)
!Define axis 20 as X axis of Safety Zone sz2, 22 as Y
sz2.SetAxes(20,22)
```

5.4.3 Enable

Description

This function enables a Safety Zone.

Syntax

```
Enable ()
```

Comments

The function enables the Safety Zone and adds it to available zones list. The Safety Zone should be defined previously by one of these functions: **SetZone1D**, **SetZone2D**, **SetZone3D**, **SetCustomZone**, or **SetZoneDualCarriage**.

5.4.4 Disable

Description

This function disables a Safety Zone.

Syntax

```
Disable()
```

5.4.5 SetZone1D

Description

The function defines a one-dimensional Safety Zone.

Syntax

```
SetZone1D (LeftLimit, RightLimit)
```

Arguments

LeftLimit	Safety Zone left limit
RightLimit	Safety Zone right limit

Comments



Figure 4-36. Illustration of one-dimensional Safety Zone

5.4.6 SetZone2D

Description

This function defines the parameters of a two-dimensional Safety Zone.

Syntax

```
SetZone2D (LeftLimitX, RightLimitX, LeftLimitY, RightLimitY)
```

Arguments

LeftLimitX	Safety Zone left limit on X Axis
RightLimitX	Safety Zone right limit on X Axis
LeftLimitY	Safety Zone left limit on Y Axis
RightLimitY	Safety Zone right limit on Y Axis

Comments

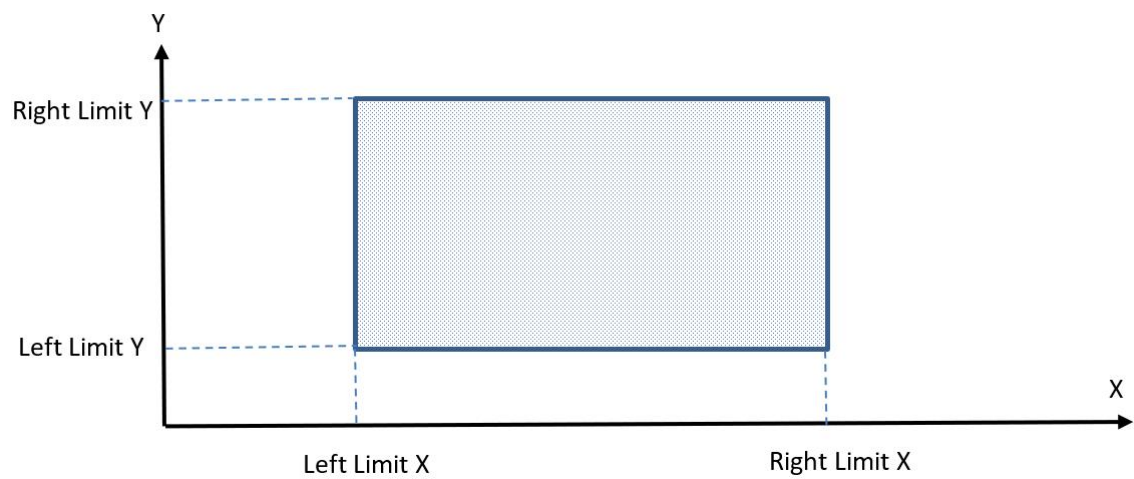


Figure 4-37. Illustration of two-dimensional safety zone

5.4.7 SetZone3D

Description

The function defines the parameters of a three-dimensional Safety Zone .

Syntax

```
SetZone3D (LeftLimitX, RightLimitX, LeftLimitY, RightLimitY,LeftLimitZ,
RightLimitZ)
```

Arguments

LeftLimitX	Safety Zone left limit on X Axis
RightLimitX	Safety Zone right limit on X Axis
LeftLimitY	Safety Zone left limit on Y Axis
RightLimitY	Safety Zone right limit on Y Axis
LeftLimitZ	Safety Zone left limit on Z Axis
RightLimitZ	Safety Zone right limit on Z Axis

Comments

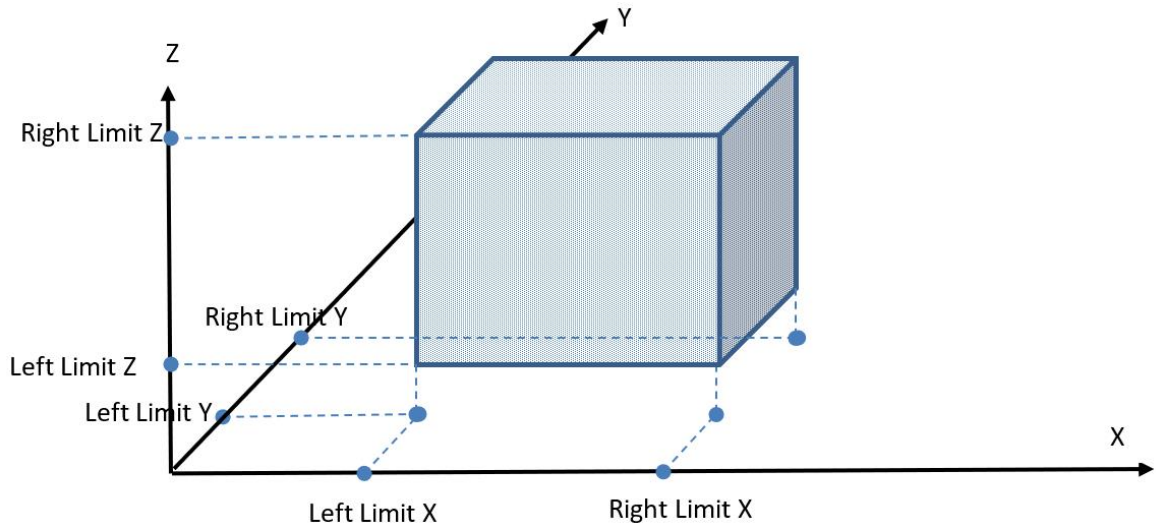


Figure 4-38. Illustration of three-dimensional Safety Zone

5.4.8 SetCustomZone

Description

The function defines the custom Safety Zone. The user ACSPL+ fastcall function is using for motion validation .

Syntax

```
SetCustomZone(ValidationFunc)
```

Arguments

ValidationFunc	The name of a FASTCALL ACSPL+ function. The function format is: int FASTCALL FuncName()
----------------	--

Comments

Define the user function intended for position validation calculation. The firmware calls this function every controller cycle. If the function returns zero value, Safety Zone exception occurred. The motion limits are determined solely by the User Custom Function.

Example

See TBD Example

5.4.9 SetZoneDualCarriage

Description

The function defines the gap between the two elements of a dual carriage configuration. In a dual carriage configuration, two loads move independently along a single axis. The purpose of this Safety Zone is to ensure a minimum distance between the two loads , independent of their absolute positions.

Syntax

```
SetZoneDualCarriage (Gap)
```

Comments

The Dual Carriage system has two linear stages on one track (see picture below). Two stages have coaxial and co-directional axes. The goal of Safety Zone is to prevent the two stages from colliding. The minimal possible distance between the stages defined by Gap parameter. When the distance is going to be less than Gap, kill motion process starts.

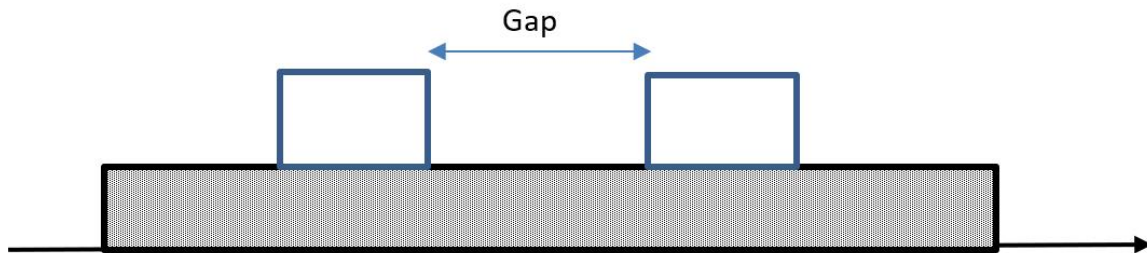


Figure 4-39. Illustration of Dual Carriage Safety Zone Gap

Example

```
global  SafetyZone sz1

!Program buffer
int Ax1 = 0
int Ax2 = 1
SET FPOS (Ax1)=0
SET FPOS (Ax2)=200

sz1.SetAxes (Ax1,Ax2)
sz1.SetZoneDualCarriage (50)
sz1.Enable ()

enable (Ax1,Ax2)
ptp (Ax1), 200
ptp (Ax2), 0

stop
```

5.4.10 SetMargins

Description

This function sets Safety Zone margins. It may be called dynamically, changing Safety Zone dimensions during motion.

Syntax

```
SetMargins (X_Margin[, Y_Margin][, Z_Margin])
```

Arguments

X_Margin	Margin on X Axis
Y_Margin	(optional) Margin on Y Axis
Z_Margin	(optional) Margin on Z Axis

Comments

The margins enlarge the Safety Zone if the value is positive. If the value is negative the Safety Zone becomes smaller. Margins change the Zone validation area, but don't change the Zone limits. For a 1D Safety zone only the `x_Margin` parameter is relevant, for a 2D Zone only `x_Margin` and `y_Margin` are relevant.

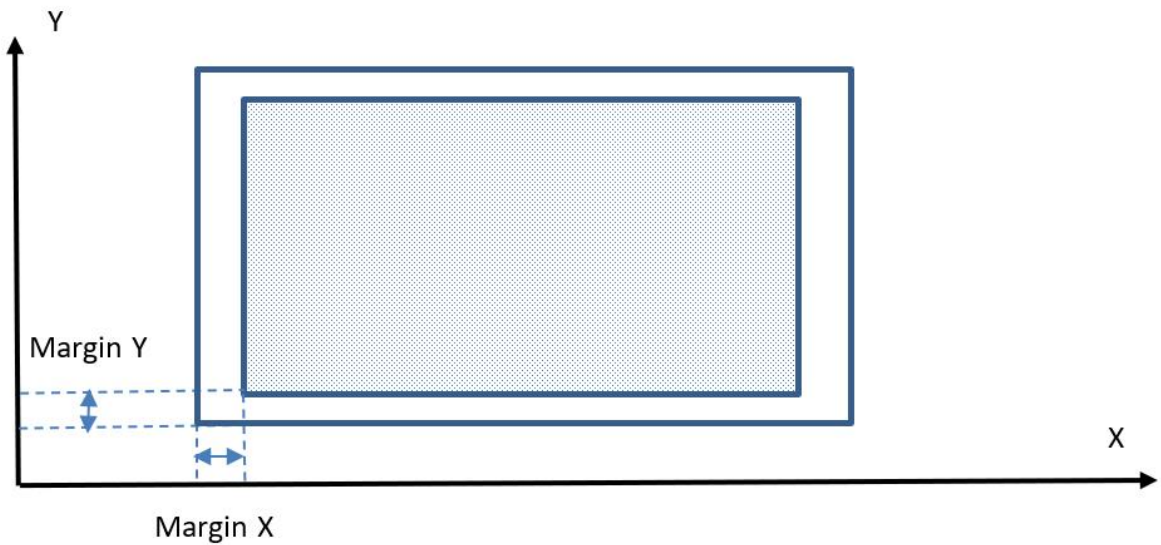


Figure 4-40. SetMargins Enlarging Safety Zone

5.4.11 AddOffset

Description

This function sets Safety Zone Offsets. The offset or offsets are relative to the current offset values and may be changed dynamically.

Syntax

```
AddOffset (X_Offset[, Y_Offset][, Z_Offset])
```


Arguments

X_Offset	Offset on X Axis
Y_Offset	(Optional) Offset on Y Axis
Z_Offset	(Optional) Offset on Z Axis

Comments

The zone moves additively, according to the specified values.

Example

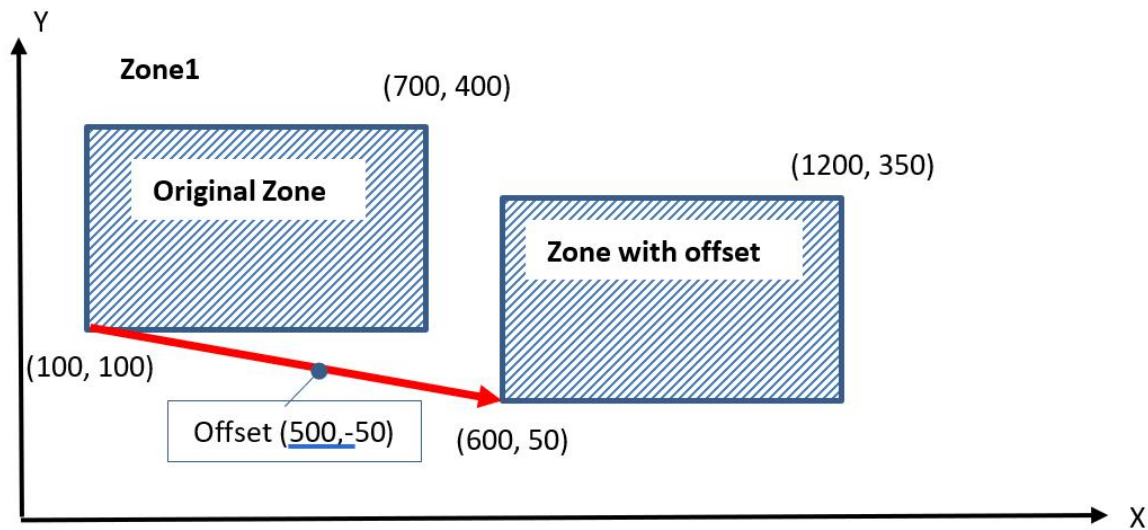


Figure 4-41. Changing Safety Zone Offsets

```
!D-buffer declaration
global SafetyZone sz1
global int const NO_ENTER_ZONE = 0
global int const NO_EXIT_ZONE = 1

!Program buffer
sz1.SetZone2D(100, 700, 100, 400)
sz1.ConditionType = NO_ENTER_ZONE
sz1.Enable()
sz1.AddOffset(500, -50)
```

5.4.12 GetLeftLimit

Description

The function returns the left limit value for the specified axis .

Syntax

```
real GetLeftLimit (Axis)
```

Arguments

Axis	Queried axis: X - 0, Y - 1, Z - 2
------	-----------------------------------

5.4.13 *GetRightLimit*

Description

The function returns the right limit value for the specified axis .

Syntax

```
real GetRightLimit (Axis)
```

Arguments

Axis	Queried axis: X - 0, Y - 1, Z - 2
------	-----------------------------------

5.4.14 *GetMargin*

Description

The function returns the Margin value for the specified axis.

Syntax

```
real GetMargin (Axis)
```

Arguments

Axis	Queried axis: X - 0, Y - 1, Z - 2
------	-----------------------------------

5.4.15 *Customized Safety Zone Definition*

In this example the code declares a customized function to trigger a Safety Zone exception, and installs it after calling SetZone2D.

```
!D-buffer declaration
global SafetyZone sz1
global int const NO_ENTER_ZONE = 0
global int const NO_EXIT_ZONE = 1

int fastcall IsValidPos()
{!Enable motion while Analog Input is less than 20%
    ret (AIN(0) < 20)
}
!Program buffer

sz1.SetCustomZone(IsValidPos)
sz1.Enable()
```

5.5 Autofocus Mode

Autofocus is a mode in which the standard control operation of the Servo Processor is switched from maintaining the desired position and velocity as specified by the controller, to executing a control algorithm that receives up to 2 local input signals (directly by the Servo Processor) and maintains a desired value for this input.

The inputs used in this mode are

- > Control input – the input used for the control algorithm
- > Quality input – input used to determine if the signal quality is sufficient to execute the control algorithm

The source of the inputs can be signals received from a distance sensor, light sensors, pressure sensor, and the like.

Signals may be digital or analog and more specific data is necessary to determine how to retrieve the input information from these signals.

The new feature relies on a standard implementation of Servo Processor and supports flexible configuration for the signals source and how to retrieve the control inputs from them.

This feature is available for the IDMxx/ECMxx/UDMxx products.

Usage

1. The user defines (with an **AutoFocusDefinition** struct) signals A,B in the Servo Processor. This is done by setting the parameters in the signalA, SignalB fields of the struct. For instance, signalsA is defined as an SPI signal and signalB refers to a Servo Processor variable.
2. Based on the defined signalA and signalB, the user now defines the quality and control signals.
3. The user assigns these definitions to the desired axis.
4. The user activates the feature for the axis using the activation function; this sets **ASTX**(Axis).**#AFACTIVE**.
5. When the quality signal, as defined by the user passes a threshold, as defined in the quality signal parameter, the **ASTX**(Axis).**#AF_RANGE** is set and the feature engages.
6. When focus is reached (focus signal < threshold) the **ASTX**(Axis).**#AF_FOCUS** is set.
7. If the focus signal reaches saturation (focus signal > saturation_threshold), the focus signal is no longer valid, and both **ASTX**(Axis).**#AF_RANGE** & **ASTX**(Axis).**#AF_FOCUS** bits are reset.
8. To deactivate the feature at any stage, the user executes the deactivation function.

5.5.1 Autofocus Signal Source Struct

Description

The **AutoFocusSource** struct cannot be independently instantiated and is only used as a field of the **AutoFocusDefinition** struct. The struct represents a signal source and allows the user to define the two sources, A and B, that are used by the Autofocus feature. The A signal can be retrieved from both the SPI or A variable.

Struct Fields

Field Name	Type	Accessibility	Default Value	Range	Comments
SPINumofWord	Int	Read-only	0		Only relevant for SPI signal
SPIOffset	Int	Read-only	0	0-7	Only relevant for SPI signal
SPIValidOffset	Int	Read-only	0 (invalid)	1-4	Only relevant for SPI signal
SPIValidMask	Int	R/W	0		Only relevant for SPI signal
SPIValidCmp	Int	R/W	0		Only relevant for SPI signal
SPIIsMsb	Int	R/W	0		Only relevant for SPI signal
Mask	Int	R/W	0		Relevant for both signals
XorMask	Int	R/W	0		Relevant for both signals
ShiftRight	Int	R/W	0		Relevant for both signals
ShiftLeft	Int	R/W	0		Relevant for both signals
DspVarAddress	Int	R/W	0 (Illegal)		Relevant for Variable only

Struct Functions

Function Signature	Description
IsValid()	Checks that the signal has been set to SPI/ Address. Also checks for validity of field values
IsAddress()	Was the signal set to Address?
IsSPI()	Was the signal set to SPI?
SetAddress(INT address)	Set parameters for Direct address signal – the user must set the address directly (retrieved using the GETSPA function) Parameter address – the address
SetSPI(INT SPINumOfWord, INT SPIOffset, INT SPIValidOffset)	Set parameters for SPI signal spiNumOfWord – number of SPI words spiOffset – SPI offset spiValidOffset – SPI validation daa offset

5.5.2 Autofocus Definition Struct

Description

The **AutoFocusDefinition** struct defines the Servo Processor parameters needed by the Autofocus feature. The struct is only used for definition and does not reflect real-time execution information.

The struct uses the 2 Signals, A and B, that are defined as two of its fields to define the quality/control signals. Signals A and B are **AutoFocusSource** struct types themselves and can only be accessed by referring to the fields.

Struct Fields

Field Name	Type	Accessibility	Default Value	Range	Comments
AppliedAxis	Int	Read-only	-1 (Illegal)	> 0	Is updated after application
QualitySignalSourceOption	Int	Read-only	0 (Illegal)	1-4	How to use A,B to derive the signal. One of several allowed options: > 0 – NONE > 1 – A+B > 2 – A-B > 3 – A > 4 – B > 5 – $\frac{A-B}{A+B}$ > 6-31 reserved
QualitySignalFactor	Real	Read-only	0		Source factor

Field Name	Type	Accessibility	Default Value	Range	Comments
QualitySignalOffset	Real	Read-only	0		Source offset
QualitySignalThreshold	Real	Read-only	0		Source threshold
ControlSignalSourceOption	Int	Read-only	0 (Illegal)	1-4	How to use A and B to derive the signal. One of several allowed options: > 0 – NONE > 1 – A+B > 2 – A-B > 3 – A > 4 – B > 5 – $\frac{A-B}{A+B}$ > 6-31 reserved
ControlSignalFactor	Real	Read-only	0		Source factor
ControlSignalOffset	Real	Read-only	0		Source offset
ControlSignalThreshold	Real	Read-only	0		Source threshold
signalA	AutoFocusSource	R/W	Default values		

Field Name	Type	Accessibility	Default Value	Range	Comments
signalB	AutoFocusSource	R/W	Default values		

Struct Functions

Function Signature	Description
IsValid()	Are the parameters set valid and ready to be applied to an axis?
ApplyToAxis(Int Axis)	Apply Autofocus parameters to a specified axis. This does not activate the feature (see Activation function). If the operation fails, an error will be presented. If the operation succeeds, the AppliedAxis field is updated
DisableForAxis(Int Axis)	Disable the feature by setting the Autofocus parameters for the specified axis to NULL.
SetQualityParameters(Int SrcOption, Real Factor, Real offset, Real threshold)	Set parameters for quality signal. If source is invalid, an error is presented and the operation fails Parameter SrcOption – quality signal source Parameter Factor – quality signal source Parameter offset – quality signal offset Parameter threshold – quality signal threshold
SetControlParameters (Int SrcOption, Real Factor, Real offset, Real threshold)	Set parameters for Control signal, if source is Illegal, an error is presented, and the operation fails > SrcOption – Control signal source > Factor – Control signal source > offset – Control signal offset > threshold – Control signal threshold

5.5.3 ACTIVATEAF - Activation Function

Description

Even after setting Autofocus parameters to an axis, the Autofocus feature is not active. To activate it the user must use this ACSPL activation method.

The direct effect of the function is setting **ASTX**(Axis).#**AFACTIVE** bit to 1.

The function is license protected and will fail if there is no license for the feature.

Syntax

```
Void activateaf(Int Axis)
```

Arguments

Axis	Axis index, valid numbers are: 0,1,2,... Up to the number of the axes in the system minus 1.
------	--

Example

```
ACTIVATEAF(1) !Activate AF feature for axis 1
STOP
```

5.5.4 DEACTIVATEAF - Deactivation Function

Description

Deactivation of the Autofocus feature for a specific axis.

The direct effect of the function is reset the **ASTX**(Axis).#**AFACTIVE** bit to 0.

The function is license protected and will fail if there is no license for the feature.

Syntax

```
Void DEACTIVATEAF(Int Axis)
```

Arguments

```
DEACTIVATEAF(1) !Deactivate AF feature for axis 1
STOP
```

5.5.5 ASTX - Execution Status

Description

This variable communicates the Autofocus status to the user.

See **ASTX** for bit specification.

Autofocus Example

```
AutoFocusDefinition autoFocusDef

!set Signal A to SPI
autoFocusDef.signalA.SpiIsMsb=0;
autoFocusDef.signalA.Mask= 0xffff;
autoFocusDef.signalA.SetSPI(2,2,3);
autoFocusDef.signalA.SpiValidCmp= 0xc0;
autoFocusDef.signalA.SpiValidMask= 0xe0;

!set signal B to be a DSP variable with a specific address
ptrA=getspa(0,"axes[2].sin");
autoFocusDef.signalB.SetAddress(ptrA)

autoFocusDef.SetControlParameters(2, 0.2 ,0 ,150,4500); !option=A-B
autoFocusDef.SetQualityParameters(1, 1, 0, 4000);

!set AXIS
autoFocusDef.ApplyToAxis(1)
activateaf(1) !Feature activation

TILL ASTX(1).#AF_RANGE
DISP "IN REANGE"

TILL ASTX(1).#AF_FOCUS
DISP "INFOCUS"
STOP
```

Response to KILL

As a response to a **KILL** command, the **ASTX(axis).#AFACTIVE** bit is reset, and the feature needs to be activated again.

5.5.6 FE

Description

FE is a real array, with one element for each axis in the system, and is used for displaying the error in Autofocus mode.

Tag

436

Accessibility

Read-Only

.NET Library Methods

ReadVariable

C Library Functions

acsc_ReadReal

5.5.7 *ERRF*

Description

ERRF is a real array, with one element for each axis in the system, and is used for defining the non-critical Position Error criterion for Autofocus mode.

Syntax

```
ERRF(axis_index) = value
```

Arguments

Axis_index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	Value of each member ranges between 2.22507e-308 and 1.79769e+308, Default = 100.

Tag

434

Comments

ERRF defines the non-critical position error fault (**FAULT**(axis_index).#PE) criterion when the motor is in focus mode.

As a configuration variable, the **ERRF** value is normally defined by SPiiPlus MMI Application Studio -> Toolbox -> Setup -> Adjuster Wizard during the setup procedure of the system.

Accessibility

Read-Write

Related ACSPL+ Variables

FAULT(axis_index).#PE

FE, CERRF

.NET Library Methods

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

5.5.8 *CERRF*

Description

CERRF is a real array with one element for each axis in the system, and is used for defining the critical Position Error criterion for Autofocus mode.

Syntax

```
CERRF(axis_index) = value
```

Arguments

Axis_ index	Designates the specific axis, valid numbers are: 0, 1, 2, ... up to the number of axes in the system minus 1.
value	value of each member ranges between 2.22507e-308 and 1.79769e+308, Default = 1000.

Tag

435

Comments

CERRF defines critical position error fault (**FAULT**(axis_index).#CPE) criterion when the motor is in focus mode.

As a configuration variable, the **CERRF** value is normally defined by SPiiPlus MMI Application Studio - >Toolbox -> Setup -> Adjuster Wizard during the setup procedure of the system.

Accessibility

Read-Write

Related ACSPL+ Variables**FAULT**(axis_index).#CPE**FE, ERRF****.NET Library Methods**

ReadVariable, WriteVariable

C Library Functions

acsc_ReadReal, acsc_WriteReal

5.6 Gantry Support

5.6.1 CrossCouplingCompensation Struct

Description

The **CrossCouplingCompensation** struct activates the cross axis compensation algorithm. The application controls initiation and stopping of the compensation.

Struct Fields

CrossAxis in the fields' name refers to the load or loads carried on the cross beam. These values define the coordinate system relative to the cross beam.

Field Name	Type	Accessibility	Default Value	Range	Comments
YawAxisIndex	Int	R/W	The Index of the first valid Yaw Axis in the system	0 to number of axes in the system-1	The index of the yaw axis
NumberOfCrossAxes	Int	R/W	1	1 to minimum(10, number of axes -2)	Number of elements in Cross Axis related arrays
CrossAxisIndex*	Int array	R/W	0	0 to number of axes in the system	The indices of the cross axes (cannot be Yaw or Longitudinal indexes)
CrossAxisPolarity*	Int array	R/W	-1	-1 or 1	Determines the motion direction of the CrossAxis.

Field Name	Type	Accessibility	Default Value	Range	Comments
					1: The motion direction is from the yaw axis index towards the primary axis index. -1: The motion direction is from the primary axis index towards the yaw axis index .
CrossAxisMaxPosition*	Real array	R/W	0	-1e302:1e302	Represents the maximum value on the cross beam in the coordinate system of the CrossAxis.
CrossAxisMinPosition*	Real array	R/W	0	-1e302:1e302	Represents the minimum value on the cross beam in the coordinate system of the CrossAxis.

Field Name	Type	Accessibility	Default Value	Range	Comments
CrossAxisMassRatio*	Real array	R/W	0	0:1	Mass ratio of the coming and non-moving parts of the cross axis Mass ratio of the CrossAxis to the total mass, including cross beam and the mass of the CrossAxis. If there is more than one CrossAxis, all the CrossAxis masses are added to the dominator. If, for example, there is a single CrossAxis, the CrossAxisMassRatio is $\frac{m_{CrossAxis(i)}}{m_{Crossbeam} + m_{CrossAxis}}$ In case of several CrossAxis, CrossAxisMassRatio of cross (i) is $\frac{m_{CrossAxis(i)}}{m_{Crossbeam} + \sum m_{CrossAxis}}$
CrossAxisForceRatio	Real	R/W	0	-100:100	Represents the force ratio compensation. Can be automatically calculated (if IsUserDefined is 0), or user-defined (if IsUserDefined is 1).
IsActive	INT	R	0(FALSE)	TRUE or FALSE (0 or 1)	Represents whether the compensation is active or not

Field Name	Type	Accessibility	Default Value	Range	Comments
IsUserDefined	INT	R/W	0(FALSE)	TRUE or FALSE (0 or 1)	0: The CrossAxisForceRatio is automatically calculated based on the CrossAxis parameters. 1: The algorithm is controlled by user. The user should define the CrossAxisForceRatio value.

Struct Functions

Name	Description
StartCompensation()	Starts the calculation of the CrossAxisForceRatio. Once it is called, the IsActive flag is set to 1
StopCompensation()	Stops the compensation; IsActive flag is set to 0

Comments

- > The cross-axis compensation algorithm is supported in the following products:
 - > CMxa
 - > ECMsm / UDMsm / IDMsm
 - > ECMma / UDMma / IDMma
 - > ECMdx / UDMdx / IDMdx
- > To start the compensation process, the following conditions must be met:
 - > Gantry axes are enabled, commutated, and homed.
 - > The cross axes are homed.

- > **DCOM**(YawAxisIndex) is 0.
- > Learning Boost is not active for the specified YawAxisIndex.
- > To prevent any damage to the system, the cross axes must be located symmetrically around the center of mass.
- > The following conditions force the compensation process to stop:
 - > **StopCompensation()** function is called.
 - > Gantry axes are disabled.
- > If the cross axes are not homed the compensation process cannot begin. An error message will appear if the user attempts to start the process while the axes are not homed.
- > The struct parameters cannot be changed during an active compensation process. Any assignment of new values to these parameters will take effect after stopping and restarting the compensation process.
- > During the compensation process, it is not possible to change **DCOM**(YawAxisIndex) value.
- > All the cross axes parameters should be expressed in the same user-units.
- > CrossAxisMaxPosition and CrossAxisMinPosition represent the maximum and minimum values on the cross beam in the coordinate system of the CrossAxis (see [Figure 5-1](#)). Therefore, the difference between the CrossAxisMaxPosition and the CrossAxisMinPosition of all the cross axes should be identical.

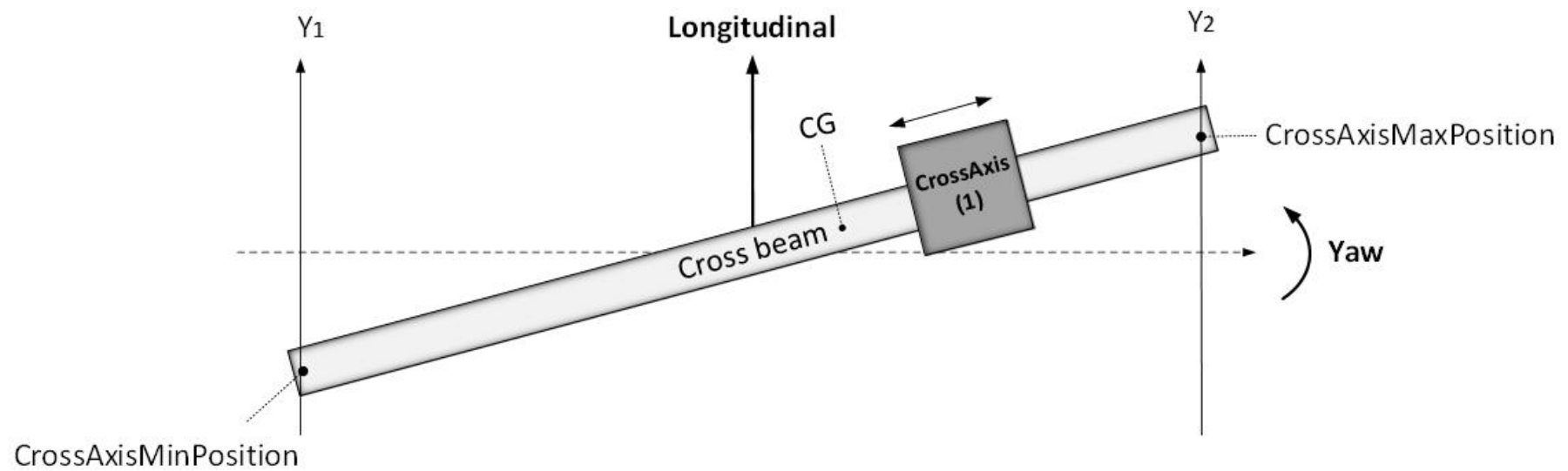


Figure 5-1. Cross Coupled System Example

Example

The following example shows a gantry system with 2 cross axes, where the cross masses are $m_{\text{cross } 1} = 3 \text{ kg}$, $m_{\text{cross } 2} = 2 \text{ kg}$, and the non-moving gantry mass is $M = 15 \text{ kg}$. The gantry motors are connected to axes 0,2. CrossAxis(1) is connected to axis 1, and CrossAxis(2) is connected to axis 3. The length of the cross beam is 100 mm. Each cross axis has its own coordinate system:

- > CrossAxis(1): Position "0" is located 20 mm from axis 2. The motion direction is from axis 0 to axis 2.
- > CrossAxis(2): Position "0" is located 90 mm from axis 2. The motion direction is from axis 2 to axis 0.

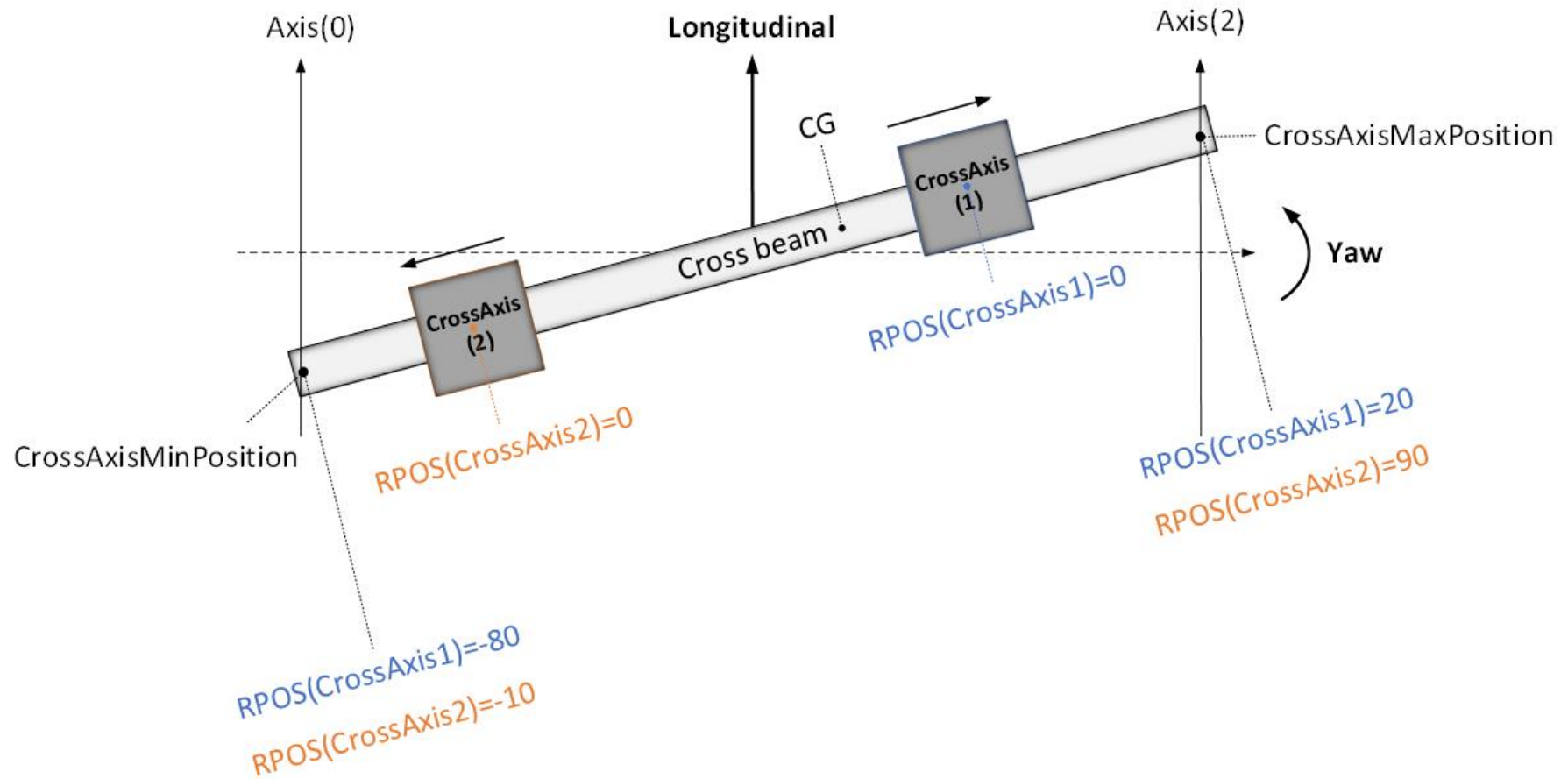


Figure 5-2. CrossCoupling System with Example Position Values

```

D-Buffer:
global static CrossCouplingCompensation Cross

#-Buffer:
real m_cross_beam = 15; ! Non-moving mass (kg)
real m_cross1 = 3;      ! CrossAxis(1) mass (kg)
real m_cross2 = 2;      ! CrossAxis(2) mass (kg)

Cross.YawAxisIndex = 2;      ! YAW axis index is 2
Cross.NumberOfCrossAxes = 2; ! 2 Cross axes
Cross.IsUserDefined = 0;     ! The CrossAxisForceRatio value is automatically calculated
                             ! based on the CrossAxis parameters.

! CrossAxis(1) parameters
Cross.CrossAxisIndex(0) = 1;
Cross.CrossAxisPolarity(0) = -1;      ! Motion direction: 0->2
Cross.CrossAxisMinPosition(0) = -80;
Cross.CrossAxisMaxPosition(0) = 20;
Cross.CrossAxisMassRatio(0) = m_cross1/( m_cross_beam + m_cross1 + m_cross2);

! CrossAxis(2) parameters
Cross.CrossAxisIndex(1) = 3;
Cross.CrossAxisPolarity(1) = 1;      ! Motion direction: 2->0
Cross.CrossAxisMinPosition(1) = -10;
Cross.CrossAxisMaxPosition(1) = 90;
Cross.CrossAxisMassRatio(1) = m_cross2/(m_cross_beam + m_cross1 + m_cross2);

Cross.StartCompensation();
STOP

```

6. Terminal Commands

Terminal commands are those commands that are specific to the SPiiPlus MMI Application Studio Communication Terminal utility and are not part of the ACSPL+, nor can they be incorporated into ACSPL+ programming. As soon as the command is received through one of the communication channels, it is executed.

This chapter covers all of the available Terminal commands.

6.1 Entering Terminal Commands

Terminal commands are entered in the **Communication Terminal** (the general structure of which is shown in [Figure 6-1](#)).



Figure 6-1. Communication Terminal Window

The **Communication Terminal** window is described in the *SPiiPlus MMI Application Studio User Guide*.



Terminal commands are case sensitive.

Terminal commands are divided into:

- > [Query Commands](#)
- > [Program Management Commands](#)
- > [System Commands](#)

6.2 Query Commands

Query commands are designated by the question mark (?) and are entered through the **Communication Terminal**.

Command & Syntax	Description	Example
?[ACSPL+ variable]	Returns the current value of a standard variable .	?TCP/IP – Returns the TCP/IP for the Ethernet port N1.
?[ACSPL+ variable ACSPL+ variable]...	Returns the current value of listed standard variables.	?TCP/IP, TCP/PORT – Returns the TCP/IP for the Ethernet port N1 and the TCP port number.
?[buffer number]: [local user-defined variable]	Returns the current values of a local user variable or array defined in a program buffer.	?1:MY_VAR – Returns the values of a local user-defined variable named MY_VAR.
?[array_ variable (index)]	Returns the current value of a specific element in the given array. The brackets enclosing the index are optional.	?FPOS(0) – Returns the feedback position of 0 axis. ?FPOS0: Returns the feedback position of 0 axis.
?[global user array_ variable [(index)]]	Returns the current values of a global user variable or array. Or the value of an element in the array if index is included.	?ARRAY1 – Returns the values of a global user-defined array variable named ARRAY1. ?ARRAY1(3) – Returns the value of the fourth element of a global user-defined array variable named ARRAY1.
?[matrix_ variable (row_ index) (col_ index)]	Returns the current value(s) of an element in a two dimensional matrix. The indices may be entered as a range, e.g., (0,4) is the first through fifth, inclusively.	?MATRIX – Returns all of the values (in tabular format) of the two-dimensional array named MATRIX. ?MATRIX(1)(0) – Returns the value of the second element in the first column of MATRIX. ?MATRIX(0) – Returns all the values of the first row of MATRIX. ?MATRIX(0,2)(0,1) – Returns a range of values, those of the first through third rows in the first and second columns of MATRIX.
?[buffer number]	Returns the current status and information about a program buffer	?0 – Returns the current status and information about program buffer 0.

Command & Syntax	Description	Example
?#	Returns the current status of all program buffers.	
?\$[axis number]	Returns the current status of the motor for the specified axis.	?\$16 – Returns the current status of the axis 16 motor.
?\$	Returns the current status of all motors.	
?VR	Returns the firmware version	
?SN	Returns the controller serial number and hardware version - indicated by a letter.	
??[error_code]	Returns the error code number. If the error code is included in the query, returns error description.	??3260 : Returns “Motor cannot start because the motor is disabled”
??[variable_name]	Returns a brief description of the variable.	??CERRA - Returns “Critical Position Error In Accelerating”

6.2.1 Default Query Formats

All the queries described above produce a variable report in a default format depending on the queried variable. In some cases the default format produces unsatisfactory results. For very large or small real values the output may appear misleading because very large or small values may require more positions than allocated by the default.

The default format for all real variables is similar to the C “%10G” format, where each real value is represented by 10 digits. The controller automatically chooses the position of decimal point and the number of digits right to the decimal point. If required, the controller uses an exponential format, like 3.14E-13.

The default format for all user integer variables and for all standard integer variables, except State flags, and I/O variables, is similar to the C “%10i” format that specifies 10 digits for each integer number.

State and Flag variables are reported in special format. Each State and Flag variable is a collection of bits. Each bit within the variable has its own function. The variable is reported in a format that displays the state of each bit with a short explanation.

The standard I/O arrays IN and OUT are reported in binary format.

6.2.2 Predefined Query Output Formats

The controller provides several predefined formats that can be used instead of the default format for querying variables:

Format	Description
?D/	Decimal format. This format is identical to the default format for integer variables. When applied to a state variable, the format displays the decimal presentation of the variable. C-equivalent: %10i.
?X/	Hexadecimal format. When applied to an integer variable, this format displays the hexadecimal presentation of the variable. C-equivalent: %08X
?B/	Binary format. This format is identical to the default format for the IN and OUT variables. When applied to an integer variable, the format displays the binary presentation of the variable.
?E/	Extended format. This format is useful for very large or small real values, when the default format produces ambiguous results because the default does not provide enough positions to display very large or very small numbers. When applied to a real variable, the format displays each value in 20 positions. C-equivalent: %20G

Examples

Display the motor state in decimal format:

```
?D/ MST
3      15      15      3      0      0      0      0
X/Y_MST
0      0      0      0      0      0      0      0
```

Display the state of motor x in binary format

```
?B/ MST
00000000,00000000,00000000,00000011
```

Display the status of variable UserReal in extended format

```
?E/UserReal
1.0000000000000001
```


6.2.3 User-Defined Query Output Format

If the default format or any other predefined format is not suitable for the user needs, the user may specify a format using C notation. The specification is placed just before the variable name in curled brackets. The specification applies only to the value of the name that is specified in the command. If an array is queried, the specification applies to each element of the array.

C notation provides an unlimited number of possible formats.

Examples

Format	Description
?{%12.3f}FPOS	The motor feedback values for all axes are displayed in 12 digits, fixed decimal point, 3 digits after the point. The same format applies to all 8 values.
?{%8.0f}X_FPOS	8 digits, no decimal point, no fraction digits.
?{XFPOS = %8.0f}X_FPOS	The response will look like XFPOS = 1234.
?{%08X} X_MST	8 digits, hexadecimal format with leading zeros

6.3 Program Management Commands

Program Management commands are used for:

- > Controlling program execution
- > Viewing and editing program content

The Program Management commands are designated by the pound (#) character.

6.3.1 Program Management Command Arguments

Program Management commands can take two types of arguments:

Buffer Designation

Buffer designation can be specified in two forms:

- > An integer number, between 0 and 16, that addresses a specific buffer (16 addresses the D-buffer)
- > The pound character, #, that addresses all program buffers in one command

Only the following commands can address all buffers:

- > L – List
- > C – Compile
- > S, SR – Stop, Stop and Reset
- > P – Pause
- > F, FI – Find
- > BR – Reset Breakpoints

All other commands operate with one buffer only, and must specify the buffer number.

Line Designation

A line designation can appear in one of three forms, as shown in the following table:

Table 6-1. Line Designation

Line Qualifier Type	Description
Single number	Specifies only one line.
Two numbers separated by comma	Specifies a range of lines. If the second number is larger than a total number of lines in the program, the list range spans the last program line.
Label preceded by a slash (/) character	Specifies the line by a designated label (the text following the slash).
Label preceded by a slash (/) character, then a comma and number	Specifies a range of lines starting from the line with a designated label.

Examples

The following are examples of using the **L** command.

List line 4 in buffer 3.

#3L4

```
4: till ^MST(0).#MOVE
```

List lines from 1 to 3 inclusively in buffer 5.

#5L1,3

```
1: movePTP:
2: VEL(0) = 20000
3: ptp 0, 4000
```

List the line that contains the label **MovePTP** in buffer 5.

#5L/MovePTP

```
1: movePTP:
```

List 3 lines, starting from the label **movePTP** in buffer 5.

#5L/MovePTP, 3

```
1: movePTP:
2: VEL(0) = 20000
3: ptp 0, 4000
```

6.3.2 Program Buffer Commands

6.3.2.1 Open/Close Buffer (#)

Description

The # (Open/Close Buffer) command is used to open and close a buffer for the purpose of entering code.

Syntax

#buffer_number[I][line_number]

Arguments

<i>buffer_number</i>	buffer_number qualifier in the command specifies the buffer, a number between 0 and 16.
<i>I</i>	Optional, if included, opens the buffer for insertion of code.
<i>line_number</i>	Optional, if included, the command opens the buffer specified by buffer_number and sets the insert line before the line specified by line_number . line_number can be specified in any of the three forms listed in.

Comments

The # command opens the buffer and sets the insert line as follows:

- > If the buffer is empty, the insert line is 1.
- > If the buffer already contains a program, the insert line is set after the last line of the program.

Only one buffer can be opened at a time.

To close the buffer, # is entered without the buffer number (since only one buffer is open, the controller knows which buffer to close).

Examples

The following are examples of opening and closing buffers.

```
#0          Open buffer 0
0:00001>    Buffer 0 is empty
#3          Open buffer 3
3:00006>    This indicates that buffer 3 contains 5 lines. The insert
            line is set to 6.
#3          Open buffer 3 for insertion
3:00006>    The command does not specifies a line qualifier. It is
            identical to the command #3.
#3I2       Open buffer 3 and set insertion point prior to line 2.
3:00002>    Insert line is set before line 2
#          Close the buffer
:          The prompt indicates that all buffers are closed
```

When a program buffer is open, ACSPL+ commands are stored in the buffer (and are not executed immediately). The controller checks the syntax of inserted ACSPL+ lines and immediately reports any errors detected.

The following is an example of an editing session:

```
#0                                Open buffer 0
0:00001>                          Buffer 0 is empty
>VEL(0) = 20000                   Enter the program lines sequentially. After each
                                  line, the insert line number is increased by 1.

0:00002>
ptp X, 4000
0:00003>
till ^MST(0).#MOVE
0:00004>

stop
0:00005>

#0L                                List the program
1: VEL(0) = 20000
2: ptp  0, 4000
3: till ^MST
(0).#MOVE
4: stop
0:0005>                          The buffer remains open
#0I1                              Change the insert line number to 1
0:0001>                          Insert line is set before the first line
MovePTP:                          Insert label
0:0002>
#0L                                List the program
1: MovePTP:
2: VEL(0) = 20000
3: ptp  0, 4000
4: till ^MST(0).#MOVE
5: stop
0:0002>                          The buffer remains open
#                                Close the buffer
```

6.3.2.2 D

Description

The **D** (Delete) command deletes the specified lines in the buffer.

Syntax

#buffer_numberDline_number[,line_number]

Arguments

<i>buffer_number</i>	buffer_number qualifier in the command specifies the buffer, a number between 0 and 16.
<i>line_number</i>	line_number in the D command is obligatory. line_number can be: > A single number, specifying one specific line, or > Two numbers separated by comma that specify a range of lines. If the second number is larger than a total number of lines in the program, the delete range includes the last program line.

Comments

If a buffer is open, the **D** command that addresses the buffer shifts the insert line to before the first undeleted line.

Example

#0L	Open buffer 0
0:00001>	Buffer 0 is empty
#3	Open buffer 3
3:00006>	Buffer 3 contains 5 lines. The insert line is set to 6.
#3I	
3:00006>	The command does not specifies a line qualifier. It is identical to the command #3
#3I2	
3:00002>	Insert line is set before line 2
#0D3	Delete line 3 in buffer 0
#	Close the buffer
:	The prompt indicates that all buffers are closed

6.3.2.3 F/IF

Description

The **F/IF** (Find/Find Case-sensitive) commands are used to search for a specific text in a specified buffer or in all buffers.

Syntax

#**buffer_number**{**F/IF**}/**search_string** [,**line_number**]

Arguments

<i>buffer_number</i>	buffer_number qualifier in the command specifies the buffer, a number between 0 and 16.
<i>search_string</i>	The text being sought.
<i>line_number</i>	Optional, if included, line_number defines the start line for the search. Otherwise, the search starts from the first line.

Comments

search_string must be specified as a label, that is, it must be preceded by a slash (/), or as a label and number separated by comma. **search_string** can be any text, such as, a variable name, ACSPL+ command, constant, label or keyword.

The search terminates when the first entry of the specified text is found, or the buffer end is reached. The command reports the line that contains the text, or an error message if the text was not found.

To find the next entry, the user must execute the command again, specifying the new start line number of the reported line plus one.

If the # character is specified instead of the buffer number, the search command addresses all buffers. In this case the command finds the first entry in each buffer.

Examples

The following are examples of using the **F** command.

#0F/X	Find "X" in buffer 0
0002	Response: found entry in line 2
#0F/X,3	Find the next entry, starting from line 3
0003	Response: found entry in line 3
#0F/X,5	Find the next entry, starting from line 5
?1078	Response: No more entries
##F/stop	Find stop in all buffers
Buffer #0: no matches	
Buffer #1: 4: stop	Entry is found in line 4 of Buffer 1
Buffer #2: no matches	
Buffer #3: 4: stop	Entry is found in line 4 of Buffer 3
Buffer #4: 1: stop	Entry is found in line 1 of Buffer 4
Buffer #5: no matches	
Buffer #6: no matches	
Buffer #7: no matches	Entry is found in line 238 of Buffer 8
Buffer #8: 238: stop	
Buffer #9: no matches	
Buffer #10: no matches	

6.3.2.4 L

Description

The **L** (List) command is used for displaying a program listing.

Syntax

#buffer_numberL[line_number]

Arguments

<i>buffer_ number</i>	buffer_number specifies the buffer, a number between 0 and 16; or you can use the pound (#) to designate all buffers.
<i>line_ number</i>	Optional, if included, line_number defines a specific line to be listed. Otherwise, the search starts from the first line. line_number can be specified in any of the three forms listed in Line Designation .

Comments

The listing contains all program lines preceded by line numbers. Each line appears exactly as it was inserted. No automatic formatting is provided.

To address all buffers the # character is used instead of the buffer number, for example, the command **##L** provides a listing of all programs in all buffers. If **line_number** is included and a buffer does not contain the specified line number, only the buffer number is listed.

If the buffer is empty, the list includes the buffer designation followed by the first line (0) which is blank.



It is recommended that you place a remark with a short program description in the first line of each program. This enables using the command **##L1** to get quick information about all loaded programs.

Examples

The following are examples of the **L** command.

Example 1:

Provide a program listing for buffer #3:

#3L

```
1: MovePTP:
2: VEL(0) = 20000
3: ptp X, 4000
4: till ^MST(0).#MOVE
5: stop
```

Example 2:

Provide a program listing for the first line in all buffers:

```
##L1           List the contents of line 1 in all buffers.
Buffer 0       Response
0:
Buffer 1
1: ! Homing of all axes
Buffer 2
1: ! Registration motion
Buffer 3
```

```

1: MovePTP:
Buffer 4
1: ! PLC program
Buffer 5                               Buffer 5 is empty
0:
Buffer 6                               Buffer 6 is empty
0:
Buffer 7                               Buffer 7 is empty
0:
Buffer 8                               Buffer 8 is empty
0:
Buffer 9                               Buffer 9 is empty
0:

```

6.3.3 RESET

Description

The **RESET** command is used to reset the controller to factory default state.

Syntax

#RESET

Comments

The **RESET** command can be issued even if the application is in the Protected mode in which case the password, if included, is not needed.

6.3.4 Listing Program Variables

There are three types of variables:

- > ACSPL+ Variables – variables contained in the ACSPL+ language set
- > SP Variables – variables incorporated in the controller
- > User-Defined Variables – variables that have been declared by the user

For each type of variable there is a Communication Terminal command for listing them.

6.3.4.1 VGR

Description

The **VGR** command lists the categories within which the ACSPL+ variables are grouped. The categories of the ACSPL+ variables are:

- > Axis_State
- > Monitoring
- > Motion
- > Safety_Control
- > Inputs_Outputs
- > Program_Execution_Control
- > System_Configuration

- > Axis_Configuration
- > Communication
- > Commutation
- > Data_Collection
- > Servo_Loop
- > Miscellaneous
- > Obsolete

Syntax

#VGR [group_name]

Arguments

group_name	One of the ACSPL+ variable categories.
-------------------	--

Comments

If **group_name** is omitted, the command lists only the categories. If **group_name** is included, the command lists the ACSPL+ sub-categories within the category.



The category must be entered in exactly the same format as given in the list above.

6.3.4.2 VSD

Description

The **VSD** command lists all ACSPL+ variables with a short description.

Syntax

#VSD [group_name]

Arguments

group_name	One of the ACSPL+ variable categories (see VGR).
-------------------	---

Comments

When **group_name** is included in the VSD command, the ACSPL+ variables within the specified category and a brief description of each variable is listed.

6.3.4.3 VS/VSG

Description

The **VS/VSG** commands are used to list the variables that are incorporated in the ACSPL+ language set.

Syntax

#VS

#VSG [group_name]**Arguments****group_name**One of the ACSPL+ variable categories (see [VGR](#)).**Comments**

When **group_name** is included in the **VSG** command, the names of the ACSPL+ variables within the specified category are listed.

Example

The following is an example of the **VS** command.

```
#VS          List ACSPL+ variable names
Response
ACC
AFLAGS
AIN
AOUT
APOS
AST
BAUD
BOFFTIME
BONTIME
MASK
FPOS
FVEL
FVFIL
GACC
GETIME
GJERK
GMOT
GMQU
GMTYPE
```



For brevity, only a portion of the response is shown here.

6.3.4.4 VSF/VSGF**Description**

Both the **VSF** and **VSGF** commands, in addition to the global variable names, display the variable type, the number of elements (for arrays only), address of the variable in the controller memory and the step between array elements (for arrays only).

Syntax**#VSF**

#VSGF [group_name]

Arguments

<i>group_name</i>	One of the ACSPL+ variable categories (see VGR).
-------------------	---

Comments

When **group_name** is included in the **VSGF** command, the names of the ACSPL+ standard variables within the specified category and their details are listed.

6.3.4.5 VG/VGF

Description

The **VG** command lists all global variable names in the system.

The **VGF** command, in addition to the global variable names, lists the variable type, the number of elements (for arrays only), address of the variable in the controller memory and the step between array elements (for arrays only).

Syntax

#[buffer_no]VG

#[buffer_no]VGF [variable_name]

Arguments

<i>buffer_no</i>	A number ranging from 0 to 16, representing a specific buffer.
<i>variable_name</i>	A specific ACSPL+ variable.

Comments

If **buffer_no** is included, **VG** and **VGF** list all the global variables in the specified buffer.

If **variable_name** specifying an ACSPL+ variable is included, **VGF** lists the details just for the specified variable

6.3.4.6 VL/VLF

Description

The **VL** command lists all local variable names in the system.

The **VLF** command, in addition to the local variable names, lists the variable type, the number of elements (for arrays only), address of the variable in the controller memory and the step between array elements (for arrays only).

Syntax

#[buffer_no]VL

#[buffer_no]VLF [variable_name]

Arguments

<i>buffer_no</i>	A number ranging from 0 to 16, representing a specific buffer.
<i>variable_name</i>	A specific ACSPL+ variable.

Comments

If **buffer_no** is included, **VL** and **VLF** list all the local variables in the specified buffer.

If **variable_name** specifying an ACSPL+ variable is included, **VGF** lists the details just for the specified variable.

6.3.4.7 V/VF

Description

The **V/VF** (List User-Defined Variable Names only/List User-Defined Variables with Description) commands are used to list the user-defined variables that are found in compiled programs.

Syntax

#[buffer_number]V

#[buffer_number]VF

Arguments

<i>buffer_no</i>	A number ranging from 0 to 16, representing a specific buffer.
------------------	--

Comments

If **buffer_no** is not specified, the list includes the user-defined variables in all compiled buffers. The **V** command only displays the names of the user-defined variables. The **VF**, on the other hand, in addition to the variable names, it displays the variable type, the number of elements (for arrays only), address of the variable in the controller memory and the step between array elements (for arrays only).

The list can be saved to a file by clicking **Save** in the **Communication Terminal** window.

Examples

The following are examples of the **V** and **VF** commands.

#9V	Provide a list of user variables in buffer 9
	Response
ITIME Global TS_AMP1 Local TS_AMP0 Global #9VF	Provide a list of user variables in buffer 9 with additional information
	Response
ITIME Global real@00DA0C50	

```
TS_AMP1 Local    int@00DA1A30
TS_AMP0          Global  int@00DA0C80
```

6.3.4.8 VSP

Description

The **VSP** (List Servo Processor Variables) command provides a list of the SP variables that are defined in the program in the specified SP.

Syntax

#VSPservo_number

Arguments

servo_number

servo_number is number of the Servo Processor.

Comments

Each variable name in the list is accompanied by an SP address of the variable.

The list can be saved to a file by clicking **Save** in the Terminal window.

Example:

```
#VSP0                                List the variables in SP 0
                                     Response
_RMS_SUM @088
A_RMS_SUM @089
X_NOTCH_OUT_PR_H @08A
X_NOTCH_OUT_PR2_H @08B
X_NOTCH_IN_PR @08C
X_NOTCH_IN_PR2 @08D
X_NOTCH_FRC @08E
X_NOTCH_FRC_L @08F
A_NOTCH_OUT_PR_H @091
A_NOTCH_IN_PR @092
```

6.3.4.9 VST/VSGT

Description

Both the **VST** and **VSGT** commands display a list of ACSPL+ variables to which **PROTECT** can be applied.

Syntax

#VST

#VSGT [group_name]

Arguments

group_name

One of the ACSPL+ variable categories (see [VGR](#)).

Comments

When **group_name** is included with the **VSGT** command, the ACSPL+ variables within the specified category are listed.

6.3.4.10 VSTF/VSGTF/VSDT

Description

The **VSTF**, **VSGTF**, and **VSDT** commands all list the variable names, the variable type, the number of elements (for arrays only), address of the variable in the controller memory and the step between array elements (for arrays only) of those ACSPL+ variables to which protection can be applied.

Syntax

#VSTF

#VSGTF [group_name]

#VSDT [group_name]

Arguments

group_name

One of the ACSPL+ variable categories (see [VGR](#)).

Comments

When **group_name** is included with the **VSGTF** or **VSDT** command, the ACSPL+ variables within the specified category are listed.

6.3.4.11 VGV

Description

The **#VGV** command is used to remove global variables that have been set via **Communication Terminal** or **STATIC** variables defined in the D-Buffer.

Syntax

#VGV [global_var]

Arguments

global_var

Name of a global variable

Comments

If the **global_var** parameter is included, **VGV** removes only this variable; otherwise it removes all global variables.

The **#VGV** command deletes **STATIC** variables but does not free their memory. If, after using the command, the user encounter memory shortage issues, a controller reboot is needed to reset the memory.

6.3.4.12 VGS/VGSF

Description

The **VGS** command is used to list all **STATIC** variables currently defined.

The **VGSF** command, in addition to **STATIC** variable names, also lists the type, number of elements (arrays only) and address of the variable in the controller memory.

Syntax

```
#VGS [Static_Var]
#VGSF [Static_Var]
```

Arguments

Static_Var

Optional parameter, check if *Static_Var* is a **STATIC** variable that is currently defined.
Leave empty to list all variables.

6.3.5 Program Handling Commands

These commands are used for compiling, executing, pausing and halting the program.

The commands in this set are:

- > **C** - Compile Program
- > **X** - Execute Program
- > **S/SR** - Stop/Stop & Reset Program
- > **P** - Pause Program

The following diagram shows the program buffer states and the Communication Terminal commands that affect the buffer states:

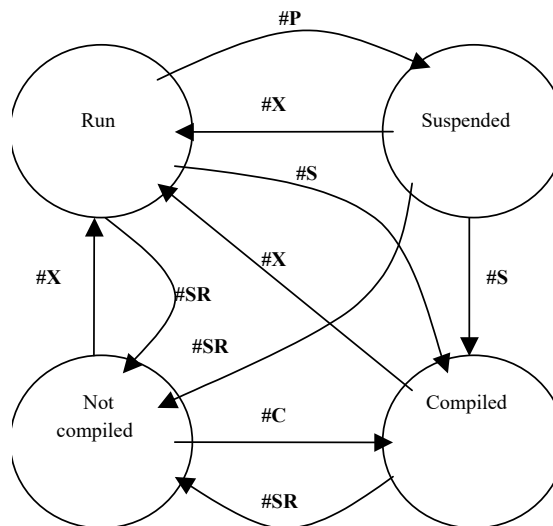


Figure 6-2. Interaction of Program Buffer States

The program buffer enters the Not Compiled state:

- > After any change in the program text.
- > Following execution of the **#S** (Stop) or **#R** (Reset) command.

The program buffer enters the **Compiled** state:

- > Following execution of the **#C** (Compile) command the program buffer is transferred from the **Not Compiled** to the **Compiled** state.
- > The **#S** (Stop) command transfers the program buffer from the Run or Suspended state to Compiled state.
- > Following program termination with an ACSPL+ **STOP/STOPALL** command (or **RET** if an autoroutine was executed).
- > When the program fails due to an error.

The program buffer enters the **Run** state:

- > When the **#X** (Execute) command is issued.
- > When another program executes a **STOP/STOPALL** command, or autoroutine condition is satisfied.

6.3.5.1 C

Description

The **C** (Compile) command compiles a program in the buffer or all programs in all buffers, depending on the buffer qualifier.

The **C** command must not include a line qualifier and is prohibited when the buffer is in the Run or Suspended states.

Syntax

#buffer_numberC

Arguments

*buffer_
number*

buffer_number specifies the buffer, a number between 0 and 16; or you can use the pound (**#**) to designate all buffers.

Comments

The **C** command is not obligatory in order to execute a program. When the **X** (Execute) command is issued, the controller automatically compiles the program if it was not previously compiled. However, a separate compilation step is required in the following cases:

- > To check the program correctness without executing it.
- > The program is not intended for direct starting, but contains autoroutines. The autoroutines are ready for execution only after compilation.
- > The program is intended for starting from another program by the **START** command. The program started by the **START** command must be compiled before the **START** command can be executed.

If the program is successfully compiled, the controller prints a short report of how many lines were compiled. If an error was encountered, the controller reports the error code and the line number in which the error was found.

Examples

The following are examples of the **C** command.

#0C	Compile the program in buffer 0
5 lines compiled	Response, the program was compiled successfully
#9C	Compile the program in buffer 9
?2026 in line 16	Response, Error 2026 was found in line 16
??2026	Explain error 2026.
Undefined variable name	Response
##C	Compile the programs in all buffers
	Response
Buffer 0: 5 lines compiled	The program was compiled successfully
Buffer 1: 18 lines compiled	The program was compiled successfully
Buffer 2: empty	The buffer is empty
Buffer 3: 5 lines compiled	The program was compiled successfully
Buffer 4: empty	The buffer is empty
Buffer 5: empty	The buffer is empty
Buffer 6: empty	The buffer is empty
Buffer 7: empty	The buffer is empty
Buffer 8: empty	The buffer is empty
Buffer 9: ?2026 in line 16	Error 2026 was found in line 16

6.3.5.2 X

Description

The **X** (Execute) command starts a program in a specific buffer, and can be executed in any program state except the **Run** state.

Syntax

#buffer_numberX[line_number]

Arguments

<i>buffer_number</i>	buffer_number qualifier in the command specifies the buffer, a number between 0 and 16.
<i>line_number</i>	Optional, line_number can be a line number or a label. Execution starts from the specified line. If line_number is omitted, the program starts from the first line.

Comments

buffer_number must specify one buffer only.

If the state of the program is **Not Compiled**, the controller first compiles the program and then starts it. If an error is encountered during compilation, the program does not start.

If the state of the program is **Suspended**, the X command resumes the program execution. In this case the command must not contain **line_number** because upon execution the program resumes from the point where the it was suspended.

Example

#1X	Execute the program in buffer 1
?1	Query status of buffer 1

Buffer 1: 18 lines, running at line 7 Response

6.3.5.3 S/SR

Description

The **S/SR** Commands are used for terminating program execution:

- > **S** - Stop
The **S** command terminates program execution in a buffer or the execution of all programs in all buffers.
- > **SR** - Stop and Reset
The **SR** (Stop and Reset) command terminates program execution in a buffer or the execution of all programs in all buffers, and resets the buffer or all buffers to the **Not Compiled** state. The command provides the de-compile function, which is useful if the program contains autoroutines that are ready to start when the buffer is in the **Compiled** state.

Syntax

#buffer_number{S|SR}

Arguments

<i>buffer_number</i>	buffer_number specifies the buffer, a number between 0 and 16; or you can use the pound (#) to designate all buffers.
----------------------	--

Comments

If **buffer_number** is omitted, this will stop, or stop and reset all programs in all buffers, or you can use the # character as the **buffer_number**, for example, **##S**, which will do the same.

Program termination commands must not include **line_numbers**.

The **S** command can be issued in any program state.



The issuance of the **SR** command effectively prevents the activation of autoroutines.

Example

#1S	Terminate the program in buffer 1
?1	Query status of buffer 1
Buffer 1: 18 lines, terminated in line 7	Response
#1SR	Reset the program in buffer 1
?1	Query status of buffer 1
Buffer 1: 18 lines, not compiled	Response
##SR	Reset all programs in all buffers

6.3.5.4 P

Description

The **P** (Pause) command suspends program execution in a buffer.

Syntax

#buffer_numberP

Arguments

<i>buffer_number</i>	buffer_number specifies the buffer, a number between 0 and 16; or you can use the pound character, #, to designate all buffers.
----------------------	--

Comments

Generally **buffer_number** refers to one buffer only. The # character may be used instead of a buffer number, for example, **##P**, in which case the execution of all programs in all buffers is suspended.

Pause commands must not include **line_numbers**.

Pause commands are allowed in any program state, but in all states other than **Run** the command has no effect.

If the program is in the **Suspended** state, the **X** command resumes execution. The **S/SR** command transfers the buffer to the **Compiled** state. The **S/SR** command transfers the buffer to the **Not Compiled** state.

Example

#1P	Suspend the program in buffer 1
?1	Query status of buffer 1
Buffer 1: 18 lines, suspended in line 7	Response
#1SR	Reset the program in buffer 1
?1	Query status of buffer 1
Buffer 1: 18 lines, not compiled	Response
##P	Suspend all programs in all buffers

6.3.6 Debug Commands

The following debug commands are supported:

- > **XS** - Execute one program line
- > **XD** - Execute program in debug mode
- > **BS** - Set breakpoint at specified line
- > **BR** - Reset breakpoint

6.3.6.1 XS

Description

The **XS** (Execute one step) command executes one program line.

Syntax

#buffer_numberXSline_number

Arguments

<i>buffer_number</i>	buffer_number specifies the buffer, a number between 0 and 16.
<i>line_number</i>	line_number gives the line to be executed, it can be a line number or a label.

Comments

The **buffer_number** qualifier in the command must specify one buffer only.

After executing the specified **line_number**, the buffer automatically enters the **Suspended** state.

6.3.6.2 XD

Description

The **XD** (Execute in Debug mode) command executes the program up to the next breakpoint (see [BS](#)).

Syntax

#buffer_numberXD

Arguments

<i>buffer_number</i>	buffer_number specifies the buffer, a number between 0 and 16.
----------------------	---

Comments

The **buffer_number** qualifier in the command must specify one buffer only.

The command is similar to the **X** command. The difference is that the **X** command ignores breakpoints in the program. If the program is started by the **XD** command, it will stop when it reaches a breakpoint. At the breakpoint the program transfers to the **Suspended** state and can be started again by the [X](#), [XS](#), or **XD** commands.

6.3.6.3 BS

Description

The **BS** (Set Breakpoint) command sets a breakpoint at the specified line.

Syntax

#buffer_numberBSline_number

Arguments

<i>buffer_number</i>	buffer_number specifies the buffer, a number between 0 and 16.
<i>line_number</i>	line_number specifies the line at which to set the breakpoint, it can be a line number or a label.

Comments

The **buffer_number** qualifier in the command must specify one buffer only.

Any number of breakpoints can be set in a program. For breakpoints to be active, the program must be started with the **XD** command.



In a program listing, the lines with breakpoints are indicated by an asterisk.

6.3.6.4 BR

Description

The **BR** (Reset Breakpoint) command resets the breakpoint at the specified line or all breakpoints.

Syntax

#buffer_numberBR[line_number]

Arguments

<i>buffer_number</i>	buffer_number specifies the buffer, a number between 0 and 16; or you can use the pound (#) to designate all buffers.
<i>line_number</i>	Optional, if included, the command resets one breakpoint at this line. line_number can be a line number or a label.

Comments

The **buffer_number** qualifier in the command must specify one buffer only.

If **line_number** is omitted, the command resets all breakpoints in the buffer.

If the buffer qualifier is specified as #, for example, **##BR**, and **line_number** is omitted, the command resets all breakpoints in all buffers.

Example

```
#0L                                     List buffer 0
1: MovePTP:                             Resonse
2: VEL(0) = 20000
3: ptp X, 4000
4: till ^MST(0).#MOVE
5: stop
#0BS3                                   Set breakpoint at line 3
#0L                                     List buffer 0 (note the
                                       asterisk indicating the
                                       breakpoint)
1: MovePTP:                             Response
2: VEL(0) = 20000
3: *ptp X, 4000
4: till ^MST(0).#MOVE
5: stop
#0XD                                   Execute the program in
                                       debug mode in buffer 0
?0                                     Query buffer 0 state
```

Buffer 0: 5 lines, suspended in line 3	Response
#0XS	Execute line 3 of buffer 0
?0	Query buffer 0 state
Buffer 0: 5 lines, suspended in line 4	Response
#0XD	Execute the rest of the program in buffer 0
?0	Query buffer 0 state
Buffer 0: 5 lines, terminated in line 5	Response

6.4 System Commands

System commands provide you with information contained in the system.

6.4.1 SI

Description

The **SI** (System Information) command returns System Information about the SPiiPlus controller including serial number, firmware version, configuration, name and SP programs.

Syntax

#SI

Arguments

None

Example

```
#SI
Network System Name:      140
Controller Firmware Version: 1.95.00.00
Controller Serial Number:  NTM000000A
Controller Part Number:   SP+NTM-080000001NNN
Hardware:
  MPU board:              Nexcom EBC220 500MHz
  MPU board ID:           5
  MPU number:             DOM4F00010462
  EtherCAT Master:        N/A
  Master Shift:           Enabled
  Ethernet Adapter:       RealTek RTL8139
    ID:                   3
    IP Address:           10.0.0.140
    MAC Address:          00 10 F3 0D B2 23
  EtherCAT Adapter:       RealTek RTL8139
    ID:                   3
    MAC Address:          00 50 C2 88 91 4A
Axes:
  Dummy:                  none
  DC Brush:               0,1,2,3,4,5,6,7
  DC Brushless:           0,1,2,3,4,5,6,7
  P/D Stepper:            8,9,10,11,12,13,14,15
  Linear drives:          none
```

```

PWM drives:                4,5,6,7
Digital Current Loop:      4,5,6,7
Integrated drives:         4,5,6,7
    Axis (4):              4.0A continuous/5.0A peak
    Axis (5):              4.0A continuous/5.0A peak
    Axis (6):              4.0A continuous/5.0A peak
    Axis (7):              4.0A continuous/5.0A peak
Remote HSSI drives:        0,1,2,3,4,5,6,7
Dual loop:                 0,1,2,3,4,5,6,7
Position Event Generation (PEG):
    PEG pulse:             0,1,2,4,5,6
    PEG states:            0,1,2,4,5,6
Options:
    Total Number of Axes:  0
    SIN-COS Encoders:      0
    Input Shaping:         No
    SPiiPlus PLC:          No
    Axes with Customized Servo Algorithms: 0
    Customized Servo Algorithms Mask: 0x0000
    Non-ACS Servo Axes:    0
    Non-ACS Stepper Axes:  0
    Non-ACS I/O Nodes:     0
Network Unit 0:
    ID:                    0
    DIP:                   0
    Part Number:           NT-LT-8
    Vendor ID:             0x00000540
    Product ID:            0x01020000
    Revision:              1
    Serial Number:         16
    HW ID:                 0x00000064
    FPGA version:          0x0000001A
Options:
    SIN-COS Encoders:      0
    Motor Type Limitations: None
Axes Assignment:           0,1,2,3,4,5,6,7
Inputs/Outputs Assignment:
    Digital inputs (IN):   0.0,0.1,0.2,0.3,0.4,0.5,0.6,0.7
    Digital outputs (OUT): 0.0,0.1,0.2,0.3,0.4,0.5,0.6,0.7
    Analog inputs (AIN):   0,1,2,3
    Analog outputs (AOUT): 0,1,2,3
    HSSI channels:         4
    Ext. inputs (EXTIN):   0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15
    Ext. outputs (EXTOUT): 0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15
Integrated Component "SPiiPlus DC-LT-8":
    Type:                  Controller (11)
    Address:               0x007
    Subsystems:            2
    Production date:       04/04/11
    HW revision:           A

```

```

S/N:                      3N000016
Integrated Component "PSM3U-48V-0.7kW":
  Type:                   Power supply (7)
  Address:                0x005
  Subsystems:            1
  Voltage:               48V - 48V
  Power:                 700W
  Production date:      08/12/10
  HW revision:          6
  S/N:                  66
Integrated Component "DDM3U-4-60V-4/5A":
  Type:                   PWM drive (5)
  Address:                0x002
  Subsystems:            4
  Axes:                  4,5,6,7
    Drive 0:  Axis 4
    Drive 2:  Axis 5
    Drive 1:  Axis 6
    Drive 3:  Axis 7
  Voltage:               60V - 60V
  Nominal current:      4.000000A
  Peak current:         5.000000A
  RMS protection Time:  3476.000000
  Production date:      27/10/10
  HW revision:          10
  S/N:                  27
Network Unit 1:
  ID:                    2
  DIP:                   63
  Part Number:           PDMnt-4-08-08-00-00
  Vendor ID:             0x00000540
  Product ID:            0x02040000
  Revision:              0
  Serial Number:         0
  Options:
    SIN-COS Encoders:    0
    Motor Type Limitations: None
  Axes Assignment:       8,9,10,11
  Inputs/Outputs Assignment:
    Digital inputs (IN):  1.0,1.1,1.2,1.3,1.4,1.5,1.6,1.7
    Digital outputs (OUT): 1.0,1.1,1.2,1.3,1.4,1.5,1.6,1.7
    Analog inputs (AIN):  none
    Analog outputs (AOUT): none
Integrated Component "PDM-4-8-8":
  Type:                   Single-Slot Unit (14)
  Address:                0x207
  Subsystems:            1
  Axes:                  8,9,10,11
    Drive 0:  Axis 8
    Drive 1:  Axis 9

```



```

        Drive 2:  Axis 10
        Drive 3:  Axis 11
    Production date:      01/01/10
    HW revision:          0
    S/N:                  0
Network Unit 2:
    ID:                   3
    DIP:                   7
    Vendor ID:             0x00000540
    Product ID:            0x02040000
    Revision:              0
    Serial Number:         0
    Options:
        SIN-COS Encoders:  0
        Motor Type Limitations: None
    Axes Assignment:      12,13,14,15
    Inputs/Outputs Assignment:
        Digital inputs (IN):  2.0,2.1,2.2,2.3,2.4,2.5,2.6,2.7
        Digital outputs (OUT): 2.0,2.1,2.2,2.3,2.4,2.5,2.6,2.7
        Analog inputs (AIN):  none
        Analog outputs (AOUT): none
    Integrated Component "PDM-4-8-8":
        Type:                Single-Slot Unit (14)
        Address:              0x307
        Subsystems:           1
        Axes:                 12,13,14,15
            Drive 0:  Axis 12
            Drive 1:  Axis 13
            Drive 2:  Axis 14
            Drive 3:  Axis 15
        Production date:      01/01/10
        HW revision:          0
        S/N:                  0
SP0 Program Info:
    Monitor version:1
    Creation Date: Sun Apr 03 08:26:19 2011
    Saving Tool: SPiiPlus NT Servo Application File Generator v.6.83.07.00
    SPiiPlus NT Servo Processor Program.
    Date=      June 14th 2010
    Version=   1.0
    Firmware=  1.0
    ACS Motion Control Ltd.,
    Control and Applications Development,
    Copyright (c) 2010. All Rights Reserved.
SP1 Program Info:
    Monitor version:1
    Creation Date: Sun Apr 03 08:26:19 2011
    Saving Tool: SPiiPlus NT Servo Application File Generator v.6.83.07.00
    SPiiPlus NT Servo Processor Program.

```

```

Date=      June 14th 2010
Version=   1.0
Firmware=  1.0
ACS Motion Control Ltd.,
Control and Applications Development,
Copyright (c) 2010. All Rights Reserved.
SP2 Program Info:
  Monitor version:ffffffff
  Default Servo Processor Info.
SP3 Program Info:
  Monitor version:ffffffff
  Default Servo Processor Info.

```

6.4.2 SIR

Description

The **SIR** (System Information Report) command provides information about the controller.

Syntax

#SIR/Section|ALL/Key|ALL/

Arguments

There are, as a minimum, six Sections:

- > **Hardware**
This section contains information about the controller's hardware. It has the following keys:
 - > **Model**
The Model ID for the controller card (hardware prefix). It is a three digit number that can be:
 - 001 SPiiPlus DDM-4
 - 020 SPiiPlus CM
 - 030 SPiiPlus SA
 - 040 SPiiPlus 3U-4
 - 041 SPiiPlus 3U-8
 - 042 SPiiPlus 3U-DDM
 - 043 SPiiPlus M
 - 044 SPiiPlus M(A)
 - 050 SPiiPlus-LF
 - 060 SPiiPlus NT
 - > **FM**
The Firmware version number.
 - > **Platform**
The controller card type ID, (first two numbers of hardware prefix).
 - > **SN**
The controller serial number.
 - > **PN**
The controller part number.

- > **MPU**
A number identifying the controller MPU, which can be:
 - 0) Unknown
 - 1) RTD 686GX-233MHz
 - 2) Sensoray 301-133MHz
 - 3) Netcom CM589/CM585-300MHz
 - 4) Kontron MOPS6-266MHz
 - 5) Nexcom EBC220-500MHz
- > **MPUN**
The MPU serial number.
- > **PAL**
The controller PAL version.
- > **Controller_Version**
Card version for the controller.
- > **SP**
Number of Servo-Processor units in the controller.
- > **Master_Shift**
Enabled - master shift is disabled
Disabled - master shift is enabled
- > **Options**
This section contains information about the controller's options. It has the following keys:
 - > **Total NumberOf Axes**
Maximum number of allowed axes.
 - > **Sin Cos Encoders**
Maximum number of allowed SIN-COS encoders.
 - > **Input Shaping**
Indicates if Input Shaping is allowed or not:
Yes - Input Shaping is allowed
No - Input Shaping is not allowed
 - > **Sin Cos Encoders**
Maximum number of allowed SIN-COS encoders.
 - > **Axes With Customized Servo Algorithms**
Maximum number of allowed axes to be used with customized servo algorithms.
 - > **Customized Servo Algorithms Mask**
A 4 hexadecimal digits (starts with 0x) that serves as a mask of allowed customized servo algorithms.
 - > **Non ACS Servo Axes**
Maximum number of allowed Non-ACS Servo axes that can be used.
 - > **Non ACS Stepper Axes**
Maximum number of allowed Non-ACS Stepper axes that can be used.
 - > **Non ACS IO Nodes**
Maximum number of allowed Non-ACS I/O Nodes that can be used.

> **Network**

This section contains information about the controller's Ethernet channels. It contains the following keys:

> **NIC1**

Code for the type of the first network adapter (the same codes are used for the second NIC if one exists, see NIC2 below) which can be:

000> Not present

001> NE2000 compatible Ethernet card

002> Intel 82559 PCI Ethernet card

003> RealTek RTL8139 PCI Ethernet card

> **NIC1_IP**

Number for the first NIC IP address.

> **NIC1_MAC**

12 hexadecimal digits providing the MAC address for the first NIC



This number is fictitious in the Simulator.

> **NIC2**

Code for the type of the second network adapter if it exists. Values are the same as those for **NIC1**, otherwise it is zero.

> **NIC2_IP**

Number for the second NIC IP address, if the second network adapter exists, otherwise it is zero.

> **NIC2_MAC**

12 hexadecimal digits providing the MAC address for the second NIC if it exists, otherwise it is zero. As for **NIC1_MAC**, this information is fictitious in Simulator.

> **Axes_support**

This section contains information about the features that each axis has. It has the following keys:

> **Dummy**

A list of dummy axes numbers, separated by commas.

> **DC_Brush**

All axes that support DC brush motors.

> **DC_Brushless**

All axes that support DC Brushless motors.

> **PD_Stepper**

All axes that support P/D Stepper motors.

> **LDM3U**

All axes that are controlled by an internal LDM3U drive.

> **Digital_Current_Loop**

All axes that support a drive with digital current loop.

- > **PWM**
All axes controlled by an internal PWM drive.
- > **Integrated**
All axes controlled by an Integrated drive (PWM or LDM).
- > **HSSI_Drive**
All axes that support an HSSI drive.
- > **Dual_Loop**
All axes that support Dual Loop control.
- > **PEG_Pulse**
All axes that support the PEG Pulse feature.
- > **PEG_State**
All axes that support the PEG State feature.
- > **UNIT#**
There is a UNIT section for each unit in the system, they are numbered from 0 up to the number of units minus 1, for example, **UNIT0** is the first unit in the system. Each UNIT section contains information about the unit. The keys are as follows:
 - > **ID**
The ID of the unit.
 - > **DIP**
The DIP switch of the unit.
 - > **Network Axes**
The axes indices, separated by commas, of all axes allocated to the unit.
 - > **Digital Inputs**
All digital input variable indices, separated by commas, that are allocated to the unit.
 - > **Digital Outputs**
All digital output variable indices, separated by commas, that are allocated to the unit.
 - > **Analog Inputs**
All analog input variable indices, separated by commas, that are allocated to the unit.
 - > **AnalogOutputs**
All analog output variable indices, separated by commas, that are allocated to the unit.
 - > **HSSIVariables**
All HSSI variable indices (input/output pairs), separated by commas, that are allocated to the unit.
 - > **HSSI Channels**
All HSSI channel indices, separated by commas, for the HSSI variables that are allocated for the unit.



Element N in this list corresponds to element N in the **HSSIRegisters** list.

> **HSSI Registers**

List of HSSI registers for the HSSI variables, separated by commas, that are allocated for the unit.

> **AXIS#**

There is an AXIS section for each axis in the system, they are numbered from 0 up to the total number of axes in the system minus 1, for example, **AXIS0** is the first axis in the system. Each AXIS section contains information about the axis. The keys are as follows:

> **Current Nominal**

The nominal current, in amperes, of the axis.

> **Current Peak**

The peak current, in amperes, of the axis.

> **XRM Smax**

Maximum nominal current, in % of peak, of the axis.

> **XRMST max**

Maximum time constant for RMS protection.

> **Drive Interface.**

Interface numbers for the drive are:

1. PWM
2. External ± 10
3. LDM3U
4. ED2
5. Network
6. Digital LDM3U

> **Max Command Current**

Maximum command for the drive.

> **Serial Number**

Serial number for the axis drive. If the drive has no serial number, nothing is displayed.

Example 1

```
#SIR/Options/LearningBoost/
[Options]
LearningBoost = Yes
```

Example 2

```
#SIR/Hardware/FW/
[Hardware]
FW = 3.11.64.00
```

Example 3

```
#SIR/ALL/ALL/  
[Hardware]  
Model = 60  
FW = 1.95.03.00  
SN = 3N000021A  
PN = SP+NT  
MPU = 5  
MPUN = DOMA400088768  
SP = 3  
Master_Shift = Disabled  
[Network]  
NIC1 = 3  
NIC1_IP = -2097151990  
NIC1_MAC = 0010F31A09E3  
NIC2 = 0  
NIC2_IP = 0  
NIC2_MAC = 000000000000  
[Options]  
TotalNumberOfAxes = 32  
SinCosEncoders = 32  
InputShaping = Yes  
AxesWithCustomizedServoAlgorithms = 0  
CustomizedServoAlgorithmsMask = 0x0000  
NonACSServoAxes = 32  
NonACSStepperAxes = 32  
NonACSIONodes = 32  
[Axes_support]  
Dummy = 0  
DC_Brush = 18446744069414584575  
DC_Brushless = 18446744069414584575  
PD_Stepper = 18446744069414584320  
LDM3U = 0  
Digital_Current_Loop = 0  
PWM = 0  
Integrated = 0  
HSSI_Drive = 18446744069414584575  
Dual_Loop = 18446744069414584575  
PEG_Pulse = 18446744069414584439  
PEG_State = 18446744069414584439  
[UNIT0]  
ID = 0  
DIP = 0  
NetworkAxes = 0,1,2,3,4,5,6,7  
DigitalInputs = 0  
DigitalOutputs = 0  
AnalogInputs = 0,1,2,3  
AnalogOutputs = 0,1,2,3
```

```
HSSIVariables = 0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15
HSSIChannels = 0,0,0,0,1,1,1,1,2,2,2,2,3,3,3,3
HSSIRegisters = 0,1,2,3,0,1,2,3,0,1,2,3,0,1,2,3
[UNIT1]
ID = 2
DIP = 0
NetworkAxes = none
DigitalInputs = none
DigitalOutputs = none
AnalogInputs = none
AnalogOutputs = none
HSSIVariables = none
HSSIChannels = none
HSSIRegisters = none
[AXIS0]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS1]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS2]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS3]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
```



```
SerialNumber =  
[AXIS4]  
CurrentNominal = 0.000000  
CurrentPeak = 0.000000  
XRMSmax = 50.000000  
XRMSTmax = 3230.000000  
Voltage = 0  
DriveInterface = 2  
MaxCommandCurrent = 5242  
SerialNumber =  
[AXIS5]  
CurrentNominal = 0.000000  
CurrentPeak = 0.000000  
XRMSmax = 50.000000  
XRMSTmax = 3230.000000  
Voltage = 0  
DriveInterface = 2  
MaxCommandCurrent = 5242  
SerialNumber =  
[AXIS6]  
CurrentNominal = 0.000000  
CurrentPeak = 0.000000  
XRMSmax = 50.000000  
XRMSTmax = 3230.000000  
Voltage = 0  
DriveInterface = 2  
MaxCommandCurrent = 5242  
SerialNumber =  
[AXIS7]  
CurrentNominal = 0.000000  
CurrentPeak = 0.000000  
XRMSmax = 50.000000  
XRMSTmax = 3230.000000  
Voltage = 0  
DriveInterface = 2  
MaxCommandCurrent = 5242  
SerialNumber =  
[AXIS8]  
CurrentNominal = 0.000000  
CurrentPeak = 0.000000  
XRMSmax = 50.000000  
XRMSTmax = 3230.000000  
Voltage = 0  
DriveInterface = 2  
MaxCommandCurrent = 5242  
SerialNumber =  
[AXIS9]  
CurrentNominal = 0.000000  
CurrentPeak = 0.000000
```

```

XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS10]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS11]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS12]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS13]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS14]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2

```

```

MaxCommandCurrent = 5242
SerialNumber =
[AXIS15]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS16]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS17]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS18]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS19]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS20]
CurrentNominal = 0.000000

```

```

CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS21]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS22]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS23]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS24]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS25]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0

```

```
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS26]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS27]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS28]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS29]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS30]
CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
[AXIS31]
```

```

CurrentNominal = 0.000000
CurrentPeak = 0.000000
XRMSmax = 50.000000
XRMSTmax = 3230.000000
Voltage = 0
DriveInterface = 2
MaxCommandCurrent = 5242
SerialNumber =
:

```

6.4.3 MEMORY

Description

Retrieves RAM availability.

Syntax

#MEMORY

Arguments

None.

Return Value

Available RAM.

Example

Up to Version 3.11

```

#MEMORY
There is      15 percent(s) of memory in use.
There are 114.83 total MBytes of physical memory.
There are  96.91 free  MBytes of physical memory.

```

From Version 3.12

```

#MEMORY
Memory Allocation for buffers:
for all buffers code: 14832 kB
for all buffers source: 6416 kB
for all buffers local variables: 8704 kB
for global variables: 1024 kB

```

6.4.4 IR

Description

The **IR** (Integrity Report) command activates integrity validation and provides a report of current integrity state. The report displays a list of files. Each list entry displays a file name, expected file size and checksum of the file and actual file size and checksum.



It takes some time to compile the Integrity Report, in order to avoid a Timeout error:

1. Right-click the controller in the Workspace Tree and select **Properties**.
2. Increase the **Connection Timeout** to 10,000 ms.
3. Click **Connect**.

Syntax

#IR

Arguments

None

Comments

If any integrity problem is detected, the command raises fault bit **S_FAULT.#INTGR**.

Example

```
#IR
```

	Size		Checksum	
	Registered	Actual	Registered	Actual
c:\				
SB1218PC.frm	001E2870	001E2870	03672ED6	03672ED6
SB1218PC.bin	0000812A	0000812A	DDC2E529	DDC2E529
c:\sb4\dsp\				
dsp.###	0004FF18	0004FF18	61F6BA11	61F6BA11
dsp.##1	0003E11B	0003E11B	783AE65B	783AE65B
adj0.\$\$\$	0000122B	0000122B	69003B3B	69003B3B
adj1.\$\$\$	00001A99	00001A99	055D4E11	055D4E11
adj2.\$\$\$	000019DA	000019DA	BC928A33	BC928A33
adj3.\$\$\$	00001A9B	00001A9B	4403628F	4403628F
adj4.\$\$\$	0000197E	0000197E	DE12BDD7	DE12BDD7
adj5.\$\$\$	00001229	00001229	A974E209	A974E209
adj6.\$\$\$	00001229	00001229	A977E20C	A977E20C
adj7.\$\$\$	0000122B	0000122B	61EE382F	61EE382F
c:\sb4\startup\				
par.\$\$\$	000001C9	000001C9	D8A0220A	D8A0220A
par0.\$\$\$	00000A16	00000A16	B2783961	B2783961
par1.\$\$\$	00000A13	00000A13	5E2EAD92	5E2EAD92
par2.\$\$\$	00000A14	00000A14	5A972E6D	5A972E6D
par3.\$\$\$	00000A13	00000A13	B25EDAD0	B25EDAD0
par4.\$\$\$	00000A13	00000A13	CD6E01E2	CD6E01E2
par5.\$\$\$	00000A13	00000A13	F6873205	F6873205
par6.\$\$\$	00000A15	00000A15	39E27520	39E27520
par7.\$\$\$	00000A15	00000A15	77CA503D	77CA503D
par8.\$\$\$	00000A13	00000A13	43A67043	43A67043
par9.\$\$\$	00000A13	00000A13	67BC915E	67BC915E
par10.\$\$\$	00000A89	00000A89	454F04F9	454F04F9
par11.\$\$\$	00000A89	00000A89	5A67281F	5A67281F

```

par12.$$$      00000A89      00000A89      6F7F4B45      6F7F4B45
par13.$$$      00000A89      00000A89      84976E6B      84976E6B
par14.$$$      00000A89      00000A89      99AF9191      99AF9191
par15.$$$      00000A89      00000A89      AEC7B4B7      AEC7B4B7
par16.$$$      00000A89      00000A89      C3DFD7DD      C3DFD7DD
par17.$$$      00000A89      00000A89      D8F7FB03      D8F7FB03
par18.$$$      00000A89      00000A89      EE101E29      EE101E29
par19.$$$      00000A89      00000A89      0328414F      0328414F
par20.$$$      00000A89      00000A89      6B641D1C      6B641D1C
par21.$$$      00000A89      00000A89      807C4042      807C4042
par22.$$$      00000A89      00000A89      95946368      95946368
par23.$$$      00000A89      00000A89      AAAC868E      AAAC868E
par24.$$$      00000DF1      00000DF1      26E26061      26E26061
par25.$$$      00000DF1      00000DF1      4B01777C      4B01777C
par26.$$$      00000DF1      00000DF1      6F208E97      6F208E97
par27.$$$      00000DF1      00000DF1      933FA5B2      933FA5B2
par28.$$$      00000DF1      00000DF1      B75EBCCD      B75EBCCD
par29.$$$      00000DF1      00000DF1      DB7DD3E8      DB7DD3E8
par30.$$$      00000DF1      00000DF1      B18A230C      B18A230C
par31.$$$      00000DF1      00000DF1      D5A93A27      D5A93A27
c:\sb4\user\
I              000001A0      000001A0      414453BB      414453BB
V              00000330      00000330      414453BC      414453BC
X_FILE         00000070      00000070      82665362      82665362
PMAP           00000340      00000340      F29BADD0      F29BADD0
ONE            00000FC0      00000FC0      4144573F      4144573F
TWO            00002F00      00002F00      414453A7      414453A7
X_ERR          00000040      00000040      42A2135C      42A2135C
System Integrity is OK

```

6.4.5 U

Description

The **U** (Usage) command is used for monitoring MPU usage. It returns the maximum, average, and minimum values as a percent.

Syntax

#U

Arguments

None

Comments

The controller continuously measures the time taken by real-time tasks. When the **U** command is received, the controller analyzes the measured times during the last 50 controller cycles and calculates minimal, maximal and average time. The results are reported in percents.

6.4.6 TD

Description

The **TD** command returns the names of all user-defined variables and arrays that are in the controller flash memory.

Syntax

#TD

Arguments

None

6.4.7 SC

Description

The **SC** (Safety Control) command reports the current safety system configuration.

The controller response includes the following:

- > active safety groups
- > the configuration of each fault for each motor

Syntax

#SC

Arguments

None

Example

#SC																		
Bit	Name	Fault Description	0	1	2	4	5	6	8	9	10	11	12	13	14	15		
0	#RL	Hardware Right Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K		
1	#LL	Hardware Left Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K		
2	#NT	Network Error	D	D	D	-	D	D	D	D	D	D	D	D	D	D		
4	#HOT	Motor Overheat	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
5	#SRL	Software Right Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K		
6	#SLL	Software Left Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K		
7	#ENCNC	Encoder Not Connected	D	D	D	D	D	D	D	D	D	D	D	D	D	D		
8	#ENC2NC	Encoder 2 Not Connected	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
9	#DRIVE	Drive Fault	D	D	D	D	D	D	D	D	D	D	D	D	D	D		
10	#ENC	Encoder Error	D	D	D	D	D	D	D	D	D	D	D	D	D	D		
11	#ENC2	Encoder 2 Error	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
12	#PE	Position Error																
13	#CPE	Critical Position Error	D	D	D	D	D	D	D	D	D	D	D	D	D	D		
14	#VL	Velocity Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K		
15	#AL	Acceleration Limit	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
16	#CL	Overcurrent	D	D	D	D	D	D	D	D	D	D	D	D	D	D		
17	#SP	Servo Processor Alarm	D	D	D	D	D	D	D	D	D	D	D	D	D	D		
20	#HSSINC	HSSI Not Connected	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
25	#PROG	Program Error	K	K	K	K	K	K	K	K	K	K	K	K	K	K		
26	#MEM	Memory Overflow	K	K	K	K	K	K	K	K	K	K	K	K	K	K		
27	#TIME	MPU Overuse				-												
28	#ES	Hardware Emergency Stop	D	D	D	-	D	D	D	D	D	D	D	D	D	D		
29	#INT	Servo Interrupt	D	D	D	D	D	D	D	D	D	D	D	D	D	D		
30	#INTGR	File Integrity																
31	#FAILURE	Component Failure	D	D	D	D	D	D	D	D	D	D	D	D	D	D		
Legend:																		
-		Fault Detection Disabled																
Blank		No Default Response																
K		Kill Motion Response																
D		Disable Axis Response																
KD		Kill Motion Followed by Disable Axis Response																
+		Generalized Fault																
Bit	Name	Fault Description	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31

0	#RL	Hardware Right Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
1	#LL	Hardware Left Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
2	#NT	Network Error	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
4	#HOT	Motor Overheat	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
5	#SRL	Software Right Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
6	#SLL	Software Left Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
7	#ENCNC	Encoder Not Connected	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
8	#ENC2NC	Encoder 2 Not Connected	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
9	#DRIVE	Drive Fault	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
10	#ENC	Encoder Error	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
11	#ENC2	Encoder 2 Error	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12	#PE	Position Error																
13	#CPE	Critical Position Error	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
14	#VL	Velocity Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
15	#AL	Acceleration Limit	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
16	#CL	Overcurrent	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
17	#SP	Servo Processor Alarm	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
20	#HSSINC	HSSI Not Connected	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
25	#PROG	Program Error	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
26	#MEM	Memory Overflow	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
27	#TIME	MPU Overuse																
28	#ES	Hardware Emergency Stop	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
29	#INT	Servo Interrupt	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
30	#INTGR	File Integrity																
31	#FAILURE	Component Failure	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D

Legend:

- Fault Detection Disabled
- Blank No Default Response
- K Kill Motion Response
- D Disable Axis Response
- KD Kill Motion Followed by Disable Axis Response
- + Generalized Fault

Bit	Name	Fault Description	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
0	#RL	Hardware Right Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
1	#LL	Hardware Left Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
2	#NT	Network Error	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
4	#HOT	Motor Overheat	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
5	#SRL	Software Right Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
6	#SLL	Software Left Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
7	#ENCNC	Encoder Not Connected	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
8	#ENC2NC	Encoder 2 Not Connected	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
9	#DRIVE	Drive Fault	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
10	#ENC	Encoder Error	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
11	#ENC2	Encoder 2 Error	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12	#PE	Position Error																
13	#CPE	Critical Position Error	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
14	#VL	Velocity Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
15	#AL	Acceleration Limit	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
16	#CL	Overcurrent	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
17	#SP	Servo Processor Alarm	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
20	#HSSINC	HSSI Not Connected	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
25	#PROG	Program Error	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
26	#MEM	Memory Overflow	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
27	#TIME	MPU Overuse																
28	#ES	Hardware Emergency Stop	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
29	#INT	Servo Interrupt	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
30	#INTGR	File Integrity																
31	#FAILURE	Component Failure	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D

Legend:

- Fault Detection Disabled
- Blank No Default Response
- K Kill Motion Response
- D Disable Axis Response
- KD Kill Motion Followed by Disable Axis Response
- + Generalized Fault

Bit	Name	Fault Description	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63
0	#RL	Hardware Right Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
1	#LL	Hardware Left Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
2	#NT	Network Error	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
4	#HOT	Motor Overheat	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
5	#SRL	Software Right Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
6	#SLL	Software Left Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
7	#ENCNC	Encoder Not Connected	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
8	#ENC2NC	Encoder 2 Not Connected	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
9	#DRIVE	Drive Fault	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
10	#ENC	Encoder Error	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
11	#ENC2	Encoder 2 Error	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
12	#PE	Position Error																

13	#CPE	Critical Position Error	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
14	#VL	Velocity Limit	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
15	#AL	Acceleration Limit	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
16	#CL	Overcurrent	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
17	#SP	Servo Processor Alarm	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
20	#HSSINC	HSSI Not Connected	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
25	#PROG	Program Error	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
26	#MEM	Memory Overflow	K	K	K	K	K	K	K	K	K	K	K	K	K	K	K
27	#TIME	MPU Overuse															
28	#ES	Hardware Emergency Stop	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
29	#INT	Servo Interrupt	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
30	#INTGR	File Integrity															
31	#FAILURE	Component Failure	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D
Legend:																	
-	Fault Detection Disabled																
Blank	No Default Response																
K	Kill Motion Response																
D	Disable Axis Response																
KD	Kill Motion Followed by Disable Axis Response																
+	Generalized Fault																

6.4.8 ETHERCAT

Description

The **ETHERCAT** command is used for obtaining complete information about the connected EtherCAT slaves.

The information it displays is:

- > Slave number
- > Vendor ID
- > Product ID
- > Revision
- > Serial number
- > EtherCAT physical address
- > DC support
- > Mailbox support
- > PdoIndex

Afterwards the list of network variables is listed. Each variable is described with:

- > Name (as in XML)
- > Offset inside the telegram (magic number that is used for mapping)
- > IN or OUT description
- > Data size

Syntax

#ETHERCAT

Arguments

None

Example 1

```
#ETHERCAT
EtherCAT bus scan found 4 nodes:
Node 0:
=====
Name:          Device 1 (SPiiPlus NT-LT-8-New)
Vendor ID:     0x00000540    Product ID:    0x01020000
PHYS ADDR:     0x03E9    Alias:    0x0000
PD IN:         Offset    26.0 Size    118
PD OUT:        Offset    26.0 Size    136
STATE:         OP
Node 1:
=====
Name:          Device 2 (SPiiPlus NT-LT-8-New)
Vendor ID:     0x00000540    Product ID:    0x01020000
PHYS ADDR:     0x03EA    Alias:    0x0000
PD IN:         Offset    162.0 Size    118
PD OUT:        Offset    162.0 Size    136
STATE:         OP
Node 2:
=====
Name:          Device 3 (SPiiPlus PDMnt-4-08-08-00-00-0)
Vendor ID:     0x00000540    Product ID:    0x02040000
PHYS ADDR:     0x03EB    Alias:    0x0000
PD IN:         Offset    298.0 Size     5
PD OUT:        Offset    298.0 Size    56
STATE:         OP
Node 3:
=====
Name:          Device 4 (SPiiPlus PDMnt-4-08-08-00-00-0)
Vendor ID:     0x00000540    Product ID:    0x02040000
PHYS ADDR:     0x03EC    Alias:    0x0000
PD IN:         Offset    354.0 Size     5
PD OUT:        Offset    354.0 Size    56
STATE:         OP

Network variables:
=====
Offset  Size  Dir  Name
  26     32  In   Command response1
  30     32  In   Command response2
  34     32  In   Command response3
  38     32  In   Command response4
  42     32  In   Command response5
  46     32  In   Command response6
  50     32  In   Command response7
  54     32  In   Command response8
  58     32  In   Feedback Position1
  62     32  In   Reference Position1
  66     32  In   Drive status1
```

70	32	In	GP data 1
74	32	In	Feedback Position2
78	32	In	Reference Position2
82	32	In	Drive status2
86	32	In	GP data 2
90	32	In	Feedback Position3
94	32	In	Reference Position3
98	32	In	Drive status3
102	32	In	GP data 3
106	32	In	Feedback Position4
110	32	In	Reference Position4
114	32	In	Drive status4
118	32	In	GP data 4
122	16	In	Digital inputs
124	16	In	FPGA status
126	16	In	HSSI 1
128	16	In	HSSI 2
130	16	In	HSSI 3
132	16	In	HSSI 4
134	16	In	HSSI 5
136	16	In	HSSI 6
138	16	In	HSSI 7
140	16	In	HSSI 8
142	16	In	Sync Counter
162	32	In	Command response1
166	32	In	Command response2
170	32	In	Command response3
174	32	In	Command response4
178	32	In	Command response5
182	32	In	Command response6
186	32	In	Command response7
190	32	In	Command response8
194	32	In	Feedback Position1
198	32	In	Reference Position1
202	32	In	Drive status1
206	32	In	GP data 1
210	32	In	Feedback Position2
214	32	In	Reference Position2
218	32	In	Drive status2
222	32	In	GP data 2
226	32	In	Feedback Position3
230	32	In	Reference Position3
234	32	In	Drive status3
238	32	In	GP data 3
242	32	In	Feedback Position4
246	32	In	Reference Position4
250	32	In	Drive status4
254	32	In	GP data 4
258	16	In	Digital inputs
260	16	In	FPGA status

262	16	In	HSSI 1
264	16	In	HSSI 2
266	16	In	HSSI 3
268	16	In	HSSI 4
270	16	In	HSSI 5
272	16	In	HSSI 6
274	16	In	HSSI 7
276	16	In	HSSI 8
278	16	In	Sync Counter
298	8	In	LIMITS
299	8	In	DIGITAL_INPUTS
300	8	In	FAULTS
301	8	In	NODE_NUM
302	8	In	WD_COUNTER
354	8	In	LIMITS
355	8	In	DIGITAL_INPUTS
356	8	In	FAULTS
357	8	In	NODE_NUM
358	8	In	WD_COUNTER
422	32	In	DC1
426	32	In	DC2
430	32	In	DC3
434	32	In	DC4
438	32	In	DC5
442	32	In	DC6
446	32	In	DC7
450	32	In	DC8
454	32	In	DC9
458	32	In	DC10
462	32	In	DC11
466	32	In	DC12
470	32	In	DC13
474	32	In	DC14
478	32	In	DC15
482	32	In	DC16
486	32	In	DC17
490	32	In	DC18
494	32	In	DC19
498	32	In	DC20
502	32	In	DC21
506	32	In	DC22
510	32	In	DC23
514	32	In	DC24
518	32	In	DC25
522	32	In	DC26
526	32	In	DC27
530	32	In	DC28
534	32	In	DC29
538	32	In	DC30
542	32	In	DC31

546	32	In	DC32
550	32	In	DC33
554	32	In	DC34
558	32	In	DC35
562	32	In	DC36
566	32	In	DC37
570	32	In	DC38
574	32	In	DC39
578	32	In	DC40
594	32	In	DC1
598	32	In	DC2
602	32	In	DC3
606	32	In	DC4
610	32	In	DC5
614	32	In	DC6
618	32	In	DC7
622	32	In	DC8
626	32	In	DC9
630	32	In	DC10
634	32	In	DC11
638	32	In	DC12
642	32	In	DC13
646	32	In	DC14
650	32	In	DC15
654	32	In	DC16
658	32	In	DC17
662	32	In	DC18
666	32	In	DC19
670	32	In	DC20
674	32	In	DC21
678	32	In	DC22
682	32	In	DC23
686	32	In	DC24
690	32	In	DC25
694	32	In	DC26
698	32	In	DC27
702	32	In	DC28
706	32	In	DC29
710	32	In	DC30
714	32	In	DC31
718	32	In	DC32
722	32	In	DC33
726	32	In	DC34
730	32	In	DC35
734	32	In	DC36
738	32	In	DC37
742	32	In	DC38
746	32	In	DC39
750	32	In	DC40
26	16	Out	Command1

28	16	Out	Command2
30	16	Out	Command3
32	16	Out	Command4
34	16	Out	Command5
36	16	Out	Command6
38	16	Out	Command7
40	16	Out	Command8
42	32	Out	Command Arg1
46	32	Out	Command Arg2
50	32	Out	Command Arg3
54	32	Out	Command Arg4
58	32	Out	Command Arg5
62	32	Out	Command Arg6
66	32	Out	Command Arg7
70	32	Out	Command Arg8
74	32	Out	Direct Command1
78	32	Out	Reference Acceleration1
82	32	Out	Reference Velocity1
86	32	Out	Reference Position1
90	32	Out	Controller status1
94	32	Out	Direct Command2
98	32	Out	Reference Acceleration2
102	32	Out	Reference Velocity2
106	32	Out	Reference Position2
110	32	Out	Controller status2
114	32	Out	Direct Command3
118	32	Out	Reference Acceleration3
122	32	Out	Reference Velocity3
126	32	Out	Reference Position3
130	32	Out	Controller status3
134	32	Out	Direct Command4
138	32	Out	Reference Acceleration4
142	32	Out	Reference Velocity4
146	32	Out	Reference Position4
150	32	Out	Controller status4
154	16	Out	analog output1
156	16	Out	analog output2
158	16	Out	digital output
160	16	Out	Sync Counter
162	16	Out	Command1
164	16	Out	Command2
166	16	Out	Command3
168	16	Out	Command4
170	16	Out	Command5
172	16	Out	Command6
174	16	Out	Command7
176	16	Out	Command8
178	32	Out	Command Arg1
182	32	Out	Command Arg2
186	32	Out	Command Arg3

190	32	Out	Command Arg4
194	32	Out	Command Arg5
198	32	Out	Command Arg6
202	32	Out	Command Arg7
206	32	Out	Command Arg8
210	32	Out	Direct Command1
214	32	Out	Reference Acceleration1
218	32	Out	Reference Velocity1
222	32	Out	Reference Position1
226	32	Out	Controller status1
230	32	Out	Direct Command2
234	32	Out	Reference Acceleration2
238	32	Out	Reference Velocity2
242	32	Out	Reference Position2
246	32	Out	Controller status2
250	32	Out	Direct Command3
254	32	Out	Reference Acceleration3
258	32	Out	Reference Velocity3
262	32	Out	Reference Position3
266	32	Out	Controller status3
270	32	Out	Direct Command4
274	32	Out	Reference Acceleration4
278	32	Out	Reference Velocity4
282	32	Out	Reference Position4
286	32	Out	Controller status4
290	16	Out	analog output1
292	16	Out	analog output2
294	16	Out	digital output
296	16	Out	Sync Counter
298	16	Out	PULSE_WIDTH
300	16	Out	INTERVAL1_1
302	16	Out	INTERVAL1_2
304	16	Out	INTERVAL1_3
306	16	Out	INTERVAL2_1
308	16	Out	INTERVAL2_2
310	16	Out	INTERVAL2_3
312	16	Out	INTERVAL3_1
314	16	Out	INTERVAL3_2
316	16	Out	INTERVAL3_3
318	16	Out	INTERVAL4_1
320	16	Out	INTERVAL4_2
322	16	Out	INTERVAL4_3
324	16	Out	PULSE_QTY1_1
326	16	Out	PULSE_QTY1_2
328	16	Out	PULSE_QTY1_3
330	16	Out	PULSE_QTY2_1
332	16	Out	PULSE_QTY2_2
334	16	Out	PULSE_QTY2_3
336	16	Out	PULSE_QTY3_1
338	16	Out	PULSE_QTY3_2

340	16	Out	PULSE_QTY3_3
342	16	Out	PULSE_QTY4_1
344	16	Out	PULSE_QTY4_2
346	16	Out	PULSE_QTY4_3
348	8	Out	ENABLE
349	8	Out	DIGITAL_OUTPUT
350	8	Out	WD_COUNTER
351	16	Out	SEVEN_SEG
353	8	Out	SPARE
354	16	Out	PULSE_WIDTH
356	16	Out	INTERVAL1_1
358	16	Out	INTERVAL1_2
360	16	Out	INTERVAL1_3
362	16	Out	INTERVAL2_1
364	16	Out	INTERVAL2_2
366	16	Out	INTERVAL2_3
368	16	Out	INTERVAL3_1
370	16	Out	INTERVAL3_2
372	16	Out	INTERVAL3_3
374	16	Out	INTERVAL4_1
376	16	Out	INTERVAL4_2
378	16	Out	INTERVAL4_3
380	16	Out	PULSE_QTY1_1
382	16	Out	PULSE_QTY1_2
384	16	Out	PULSE_QTY1_3
386	16	Out	PULSE_QTY2_1
388	16	Out	PULSE_QTY2_2
390	16	Out	PULSE_QTY2_3
392	16	Out	PULSE_QTY3_1
394	16	Out	PULSE_QTY3_2
396	16	Out	PULSE_QTY3_3
398	16	Out	PULSE_QTY4_1
400	16	Out	PULSE_QTY4_2
402	16	Out	PULSE_QTY4_3
404	8	Out	ENABLE
405	8	Out	DIGITAL_OUTPUT
406	8	Out	WD_COUNTER
407	16	Out	SEVEN_SEG
409	8	Out	SPARE
422	32	Out	REV_DC1
426	32	Out	REV_DC2
430	32	Out	REV_DC4
434	32	Out	REV_DC4
438	32	Out	REV_DC5
442	32	Out	REV_DC6
446	32	Out	REV_DC7
450	32	Out	REV_DC8
454	32	Out	REV_DC9
458	32	Out	REV_DC10
462	32	Out	REV_DC11

```

466      32      Out      REV_DC12
470      32      Out      REV_DC13
474      32      Out      REV_DC14
478      32      Out      REV_DC15
482      32      Out      REV_DC16
486      32      Out      REV_DC17
490      32      Out      REV_DC18
494      32      Out      REV_DC19
498      32      Out      REV_DC20
502      32      Out      REV_DC21
506      32      Out      REV_DC22
510      32      Out      REV_DC23
514      32      Out      REV_DC24
518      32      Out      REV_DC25
522      32      Out      REV_DC26
526      32      Out      REV_DC27
530      32      Out      REV_DC28
534      32      Out      REV_DC29
538      32      Out      REV_DC30
542      32      Out      REV_DC31
546      32      Out      REV_DC32
550      32      Out      REV_DC33
554      32      Out      REV_DC34
558      32      Out      REV_DC35
562      32      Out      REV_DC36
566      32      Out      REV_DC37
570      32      Out      REV_DC38
574      32      Out      REV_DC39
578      32      Out      REV_DC40
594      32      Out      REV_DC1
598      32      Out      REV_DC2
602      32      Out      REV_DC4
606      32      Out      REV_DC4
610      32      Out      REV_DC5
614      32      Out      REV_DC6
618      32      Out      REV_DC7
622      32      Out      REV_DC8
626      32      Out      REV_DC9
630      32      Out      REV_DC10
634      32      Out      REV_DC11
638      32      Out      REV_DC12
642      32      Out      REV_DC13
646      32      Out      REV_DC14
650      32      Out      REV_DC15
654      32      Out      REV_DC16
658      32      Out      REV_DC17
662      32      Out      REV_DC18
666      32      Out      REV_DC19
670      32      Out      REV_DC20
674      32      Out      REV_DC21

```

```

678      32    Out    REV_DC22
682      32    Out    REV_DC23
686      32    Out    REV_DC24
690      32    Out    REV_DC25
694      32    Out    REV_DC26
698      32    Out    REV_DC27
702      32    Out    REV_DC28
706      32    Out    REV_DC29
710      32    Out    REV_DC30
714      32    Out    REV_DC31
718      32    Out    REV_DC32
722      32    Out    REV_DC33
726      32    Out    REV_DC34
730      32    Out    REV_DC35
734      32    Out    REV_DC36
738      32    Out    REV_DC37
742      32    Out    REV_DC38
746      32    Out    REV_DC39
750      32    Out    REV_DC40

```

Example 2 from a SPiiPlusES

```

Network variables:
=====
Offset  Size  Dir  PdoIndex  Name
72      32   In   0x1600    Command response1
76      32   In   0x1600    Command response2
80      32   In   0x1600    Command response3
84      32   In   0x1600    Command response4
88      32   In   0x1600    Feedback Position1
92      32   In   0x1600    Reference Position1
96      32   In   0x1600    Drive status1
100     32   In   0x1600    GP data 1A
104     32   In   0x1600    Feedback Position2
108     32   In   0x1600    Reference Position2
112     32   In   0x1600    Drive status2
116     32   In   0x1600    GP data 2A
120     16   In   0x1600    Drive Output1
122     16   In   0x1600    Drive Output2
124     16   In   0x1600    Digital inputs
126     16   In   0x1600    FPGA status

```

6.4.9 #ETHERCAT2

The #ETHERCAT2 command can be used for gaining complete information about the connected EtherCAT nodes on the Secondary EtherCAT network.

Example:

```
#ETHERCAT2
EtherCAT Master is in OP state.
EtherCAT Bus Scan Found 1 Nodes:
Node 0:
=====
Name:          Device 0   (IOMnt-32-32-00-00-01)
Vendor ID:     0x00000540   Product ID:   0x05040000   Revision:
0x00000001
ESC Type:      ET1100      ESC Revision:   0       ESC Build:    2

Phys. Addr:    0x03E9      Alias:        0x0001
Ports:         A(MII/RMII) - ON      B(MII/RMII) - OFF
Type:          Mandatory
PD IN:         Offset    72.0 Size      7.0
PD OUT:        Offset    72.0 Size      6.0
State:         OP

Network variables:
=====
Offset  Size  Dir  ObjectIndex  SubIndex  PdoIndex  Name
72.0    8    In   0x6000      0x01      0x1600    Digital Inputs 0
73.0    8    In   0x6000      0x02      0x1600    Digital Inputs 1
74.0    8    In   0x6000      0x03      0x1600    Digital Inputs 2
75.0    8    In   0x6000      0x04      0x1600    Digital Inputs 3
76.0    8    In   0x6000      0x05      0x1600    WD_COUNTER
77.0    8    In   0x6000      0x06      0x1600    SYSTEM
72.0    8    Out  0x7000      0x01      0x1A00    Digital Outputs 0
73.0    8    Out  0x7000      0x02      0x1A00    Digital Outputs 1
74.0    8    Out  0x7000      0x03      0x1A00    Digital Outputs 2
75.0    8    Out  0x7000      0x04      0x1A00    Digital Outputs 3
76.0    8    Out  0x7000      0x05      0x1A00    WD_COUNTER
77.0    8    Out  0x7000      0x06      0x1A00    System
```

6.4.10 *ECMAPREP*

Description

The **ECMAPREP** command displays a report of all variables mapped using the **ECIN** , **ECOUT**, **ECEXTIN**, and **ECEXTOUT** functions. Bit notation is available for all variables..

Syntax

#ECMAPREP [/0 /1]

/0	First EtherCAT network. Equivalent to no switch
/1	Second EtherCAT network

Example

```
Example
#ECMAPREP
EtherCAT Network 0:
=====
Input 1:
=====
Variable I0, at 0x856E0A24 (integer)
Array Length 0
Data Length 2
EC Offset 204.0
Limits 0
Read Only = false

EtherCAT Network 1:
=====
Input 2:
=====
Variable I2, at 0x856E0A44 (integer)
Array Length 0
Data Length 1
EC Offset 210.0
Limits 0
Read Only = false
```

6.4.11 CC

Description

The **CC** command returns data on the existing communication channels.

Syntax

#CC

Example

```
#CC
Channel Type      Mode
1      Serial    Command Rate:115200 0N 1
2      Serial    Command Rate:115200 0N 1
6      TCP/IP    ( 701) Peer:None
7      TCP/IP    ( 701) Peer:10.0.0.52
8      TCP/IP    ( 701) Peer:10.0.0.65
9      TCP/IP    ( 701) Peer:10.0.0.58
10     UDP       ( 700) Peer:N/A
36     TCP/IP    ( 701) Peer:None
37     TCP/IP    ( 701) Peer:None
38     TCP/IP    ( 701) Peer:None
39     TCP/IP    ( 701) Peer:None
12     PCI bus   Command
16     TCP/IP    MODBUS Slave Connection:network Peer:None
```

6.4.12 PLC**Description**

This command is valid only if SpiiPlus PLC is running in the system.

The **PLC** command provides some very important information about the SpiiPlus PLC co-existence inside the SpiiPlus firmware.

The information it displays is:

- > PLC cycle (in ms):
PLC cycle means what is the frequency that PLC program is executed is. If Maximum is equal to CTIME, the PLC cycle is always identical to Motion Controller realtime tick. The data that is shown is:
 - > Avg:
 - > Min:
 - > Max:
 - > PLC Usage consumption (when active):
How much of the controller usage does the PLC execution take when it is running arranged by:
 - > Avg:
 - > Min:
 - > Max:
- Where the values shown are percentages.
- > PLC program: (Program status)
The status can be:

- > Running
- > Stopped
- > Not valid

Syntax**#PLC****6.4.13 LOG****Description**

The controller supplies a log of important events to the user.

- > The log can keep the last 500 events in memory.
- > There is a command to set the time stamp for the log (setting current time).
- > There is a command to display the log entries (similar to the Communication Terminal #SI command), the user can use a host program to save these.

Initially the log includes:

- > Motor errors
- > Change in an axis **FAULT** variable
- > System Errors.
- > Program errors (ACSPL+ buffer termination error).
- > Reported I2C drive errors.
- > Reported I2C component errors.
- > EtherCAT errors

The LOG report has the following format

Error Type	Format
Motor Errors	TIME axis error, MERR(AXIS) = ERROR,
Change in an axis FAULT variable	TIME FAULTS(AXIS): PREVIOUS_VALUE --> CURRENT_VALUE
System Errors	TIME system error, S_ERR = ERROR
Program errors (ACSPL+ buffer termination error)	TIME program error, PERR(BUFFER) = ERROR, PERL(BUFFER) = LINE
Reported I2C drive errors	TIME driver alarm extended error, address:ADDRESS, axis:AXIS, error:ERROR
Reported I2C component errors	TIME component fault extended error, address:ADDRESS, error:ERROR
EtherCAT errors	TIME network error, ECERR = ERROR

Syntax**#LOG****Example**

```
[2011/03/28] [11:15:17.140] FAULTS(57): 0x0 --> 0x20000
[2011/03/28] [11:15:17.140] FAULTS(58): 0x0 --> 0x20000
[2011/03/28] [11:15:17.140] FAULTS(59): 0x0 --> 0x20000
[2011/03/28] [11:15:17.140] FAULTS(60): 0x0 --> 0x20000
[2011/03/28] [11:15:17.140] FAULTS(61): 0x0 --> 0x20000
[2011/03/28] [11:15:17.140] FAULTS(62): 0x0 --> 0x20000
[2011/03/28] [11:15:17.140] FAULTS(63): 0x0 --> 0x20000
[2011/03/28] [11:15:17.403] system error, S_ERR = 5151
[2011/03/28] [11:15:25.502] FAULTS(0): 0x20000 --> 0x20080
[2011/03/28] [11:15:26.526] FAULTS(1): 0x20000 --> 0x20080
[2011/03/28] [11:15:26.527] FAULTS(2): 0x20000 --> 0x20080
[2011/03/28] [11:15:26.527] FAULTS(3): 0x20000 --> 0x20080
[2011/03/28] [11:15:26.527] FAULTS(4): 0x20000 --> 0x20080
[2011/03/28] [11:15:26.527] FAULTS(5): 0x20000 --> 0x20080
[2011/03/28] [11:15:26.527] FAULTS(6): 0x20000 --> 0x20080
[2011/03/28] [11:15:26.527] FAULTS(7): 0x20000 --> 0x20080
[2011/03/28] [11:15:34.352] FAULTS(0): 0x20080 --> 0x80
[2011/03/28] [11:15:34.352] FAULTS(1): 0x20080 --> 0x80
[2011/03/28] [11:15:34.352] FAULTS(2): 0x20080 --> 0x80
[2011/03/28] [11:15:34.352] FAULTS(3): 0x20080 --> 0x80
[2011/03/28] [11:15:34.352] FAULTS(4): 0x20080 --> 0x80
[2011/03/28] [11:15:34.352] FAULTS(5): 0x20080 --> 0x80
[2011/03/28] [11:15:34.352] FAULTS(6): 0x20080 --> 0x80
[2011/03/28] [11:15:34.352] FAULTS(7): 0x20080 --> 0x80
[2011/03/28] [11:17:09.270] program error, PERR(33) = 3232, PERL(33) = 0
[2011/03/28] [11:17:48.000] program error, PERR(33) = 3232, PERL(33) = 0
[2011/03/28] [11:17:57.958] axis error, MERR(3) = 5017
```

6.4.14 LOG_HOST_TICKS**Description**

LOG_HOST_TICKS adds a time offset to the controller's system time when reporting events, so that the time of reported events will match the host millisecond counter.

The controller adds the difference between HOST_TICKS (in milliseconds) and controller's time to event time when reporting events. The controller considers its own power-up time as: 1970/1/1 00:00:00.

If the HOST_TICKS supplied is the correct number of milliseconds since that date, the controller displays the correct date and time for the GMT time-zone. A time before 1970/1/1 00:00:00 (controller power-up) is displayed as negative milliseconds before this date.

Syntax**#LOG_HOST_TICKS**

Example

```

const unsigned __int64 SEC_IN_MIN    = 60;
const unsigned __int64 MSEC_IN_SEC  = 1000;
char SETLOGTIME[100] = "#LOG ";
int SETLOGTIMELENGTH = (int)strlen(SETLOGTIME);
struct __timeb64 timebuffer;
double CurrentTime;
// Set time zone from TZ environment variable. If TZ is not set,
// the operating system is queried to obtain the default value
// for the variable.
_tzset();
// GET CURRENT LOCAL TIME
_ftime64_s(&timebuffer);
CurrentTime = (((double)timebuffer.time)
              - (((double)timebuffer.timezone) * SEC_IN_MIN)) * MSEC_IN_SEC
              + timebuffer.millitm;
SETLOGTIMELENGTH = sprintf_s(SETLOGTIME, 100, "#LOG %f\r", CurrentTime);
if (!acsc_Command(( Handle, // communication handle
                  SETLOGTIME, // pointer to the buffer that contains
                           // executed controller's command
                  SETLOGTIMELENGTH, // size of this buffer
                  NULL// waiting call
                  ))
{
    printf("transaction error: %d\n", acsc_GetLastError());
}

```

6.4.15 LOGP

Description

Presents G-code run-time errors detected during running the program in simulation mode (using **START/s** command).

Syntax

```
#LOGP buffer_number
```

Arguments

buffer_number	The number of buffer that was ran in simulation mode using START/s command.
---------------	---

Comments

- > If the program contains calls to sub-routines residing in the D-Buffer, then all run-time errors occurred inside sub-routine will be addressed by the sub-routine calling line number.
- > Buffer recompilation or rerun clears the list of run-time errors detected during the last running process.



Motion generator related run-time errors are not detected, since the simulation takes place only on G-code level.

7. SPiiPlus Error Codes

This chapter contains explanations of the Error Codes that may appear.

The chapter contains only those Error Codes returned by the controller and does not include errors associated with the C Library. Errors that are detected by the C Library are never returned by the controller. A host application that calls a C Library function can receive these codes if a function call failed. For explanations of the C Library errors see the *SPiiPlus C Library Reference Guide*.

The ACSPL+ Error Codes range 1000 to 7099 and are assigned as follows:

ACSPL+ Syntax Errors – numbers 1000 to 1999

ACSPL+ Compilation Errors – numbers 2000 to 2999

ACSPL+ Runtime Errors – numbers 3000 to 3999

Errors – numbers 5000 to 5150

System Errors - numbers 5151 to 5999

EtherCAT Errors - numbers 6000 to 6999

EtherCAT Slave Errors - numbers 7001-7099

G-Code Error Codes - numbers 2000 to 4000, not related to ACSPL+ error codes



Error code values of 7100 through 8999 are reserved for future use and are currently not used.
Error code values of 9000 and above are reserved for user-defined error codes.

7.1 ACSPL+ Syntax Errors

These appear in response to syntax errors that the controller detects when a program is entered into the buffer and are reported immediately in the prompt that is displayed in response to the command.

The error code values range between 1000 and 1999.

Table 7-1. ACSPL+ Syntax Errors

Error Code	Error Message	Remarks
1001	The program is suspended	The program is being run in the Step mode, and has been suspended at the current step.
1002	The program was terminated by user	The program has reached a user-set breakpoint.
1020	Illegal subcommand	A subroutine command that is not recognized has been entered.

Error Code	Error Message	Remarks
1021	SP command requires axis specification	The required axis designation is missing.
1022	Illegal command	A program command that is not recognized has been entered.
1023	Read-only variable cannot be assigned	The command specifies assignment to a read-only variable.
1024	Set variable cannot be reported	
1025	Time interval between two characters in command is more than 2 seconds	The compiler recognizes that a follow-on command has not arrived within 2 seconds. The error is relevant when an application is using a call to a SPiiPlus C Library routine, and usually indicates a communication problem.
1026	Serial Number, Part Number, or Software Options were already specified	An attempt was made to respecify the controller's serial number, part number, or software options.
1027	Variable requires axis specification	An ACSPL+ variable missing the required axis specification was used.
1028	Scalar variable cannot accept axis specification	A scalar variable was used with an axis specification.
1029	Extra characters after the command	A command has been entered with extraneous characters or parameters that are not recognized.
1030	Too many parameters	The entry contains too many parameters.
1031	Illegal array in the array command	The array in the command either does not exist, or its size does not match the requirements of the command.
1032	Illegal data in array	The array contains data in a format that does not match requirements.

Error Code	Error Message	Remarks
1033	Illegal edit command	The edit command that has been entered cannot be executed.
1034	Illegal index value	The command includes numerical index specification, but the specified index is not a number.
1035	Index is out of range	The error is caused by one the following: > The specified index value is more than the number of elements in the array > The specified index value is negative > The specified index values are incompatible (first value in the range greater than last).
1036	Internal error	An internal error has been detected..
1037	Illegal variable name	The command requires specification of an ACSPL+ variable name, but the specified name is not that of an ACSPL+ variable.
1038	Wrong checksum in the command	The command contains checksum, and the checksum value is wrong. The error is relevant when an application includes SPiiPlus C Library routines, and indicates communication problems.
1039	Only one motion per axis can be planned in advance	An attempt was made to setup more than one motion for a specified axis.
1040	Unable to open file	The command specifies an internal file in the flash memory that does not exist.

Error Code	Error Message	Remarks
1041	Assigned value is out of range	The command attempts to assign an ACSPL+ variable with a value that is out of the range allowed for this variable.
1042	Operation failed because of exception	
1043	Program cannot start because the buffer was not compiled	The command attempts to start an un-compiled ACSPL+ program. To compile a program, in the SPiiPlus MMI Application Studio: > In the Program Manager, right-click the buffer and select Compile Buffer, or > Use the #nC command in the Communication Terminal, where n is the buffer number.
1044	Command cannot be executed while the program is running	The command attempts to affect a running ACSPL+ program. Stop the program before executing the command. To stop a program, in the SPiiPlus MMI Application Studio: > In the Program Manager, right-click the buffer and select Stop Buffer, or > Use the #nS command in the Communication Terminal, where n is the buffer number.
1045	Numerical error in standard function	The command includes an ACSPL+ function that caused a numerical error. Check if the arguments of the function fall into the allowed range.
1046	Write file error	The command has caused writing to the flash memory that failed. Possible reason is a flash memory overflow because of too many stored user files.

Error Code	Error Message	Remarks
1047	Read file error	The command has caused reading from the flash memory that failed. A reoccurring error points to a serious problem in the controller hardware or firmware.
1048	More axes than were defined in the motion	The POINT command specifies more axes than were specified in the motion that the command refers to.
1049	Axis already belongs to a group	An attempt was made to assign an axis to a group when it has already been assigned to another group.
1050	Conflict with user-defined axis group	The command is incompatible with a user-defined axis group that was defined before. The axes specified in the command must either all belong to one user-defined axis group, or not to intersect with any user-defined axis group.
1051	Line number is out of range	The command specifies a line number that does not exist in the specified program buffer.
1052	Buffer number is out of range	The command specifies illegal buffer number. The controller contains 16 program buffers numbered from 0 to 15, plus the D buffer.
1053	Wrong type	The command addresses an ACSPL+ variable and the type of the variable is different from the type specified in the command. The error is relevant when an application includes a SPiiPlus C Library routine. The error indicates problems in communication.
1054	This type of motion is valid for single axis only	An attempt was made to send a single-axis motion to more than one axis.

Error Code	Error Message	Remarks
1055	Command requires line number specification	The command is missing a required line number specification.
1056	Parameter defining Master has illegal value	The parameter specifying the master has an illegal value (see MASTER).
1057	Previous superimposed motion is in progress	
1058	Slave is not synchronized to master	The slave controller has not been synchronized with the master controller.
1059	Command PTP/V must specify velocity value	The PTP/v command was issued without a value for velocity.
1060	Illegal memory command	The memory management command is improperly formatted.
1061	)' wasn't found	The command contains a non-paired left bracket.
1062	Command is too long	Command exceeds maximum permitted length
1063	Variable is not defined in the buffer	The command addresses a variable that is not declared in the specified buffer, or the specified buffer is not compiled.
1064	Undefined global variable	The command addresses a global variable that is not declared in any buffer, or the buffer that contains the declaration is not compiled. Variable name must be in upper case.
1065	The command cannot be executed while the current motion is in progress	The command is in conflict with one or more of the currently executed motions. To kill a motion use KILL/KILLALL or KILLALL .
1066	Attempt to compile or start empty buffer	

Error Code	Error Message	Remarks
1067	GO command failed	The controller has no motions waiting for the GO command.
1068	Referenced axis does not execute a motion (motion was terminated?)	The command refers to a motion, but the specified axis is not involved in any motion.
1069	This command can only be used with MPTP , PATH or PVSPLINE motion	Commands POINT and MPOINT are compatible only with the MPTP...ENDS , PATH...ENDS , or PVSPLINE...ENDS motions.
1070	Attempt to add segment after ENDS command	The command attempts to add a segment or a point to the motion that was closed before with an ENDS of the MPTP...ENDS , PATH...ENDS , or PVSPLINE...ENDS commands.
1071	File name was expected	The command must specify a name of internal file in the flash memory.
1072	Wrong array size	The command specifies an array, but the motion to which the command refers, or other command arguments require an array of another size.
1073	Text for search is not specified	The command must specify a text for search operation.
1074	Only standard or SP variable is allowed in the command	The command requires an ACSPL+ or SP variable name to be specified.
1075	Name is not a standard or user variable	The specified variable name is not an ACSPL+ or user-defined variable name.
1076	Undefined label	The command requires a label specification. The program that contains the label specified in the command must be compiled in a buffer.

Error Code	Error Message	Remarks
1077	Protection violation	The command attempts to assign a protected variable when the controller is in Protected mode. The controller must be put in the Not Protected mode before protected variables can be assigned. To do this, use the SPiiPlus MMI Application Studio Protection Wizard in Development Tools .
1078	Variable can be changed only while the motor is disabled	The command attempts to assign a value to a variable that can be changed only if the motor is disabled. A DISABLE/DISABLEALL command must be put in before the value can be assigned to the variable.
1079	Motion cannot start because one or more motors are disabled	Motor(s) are in motion and have to be disabled before the motion can be initiated. A DISABLE/DISABLEALL command must be put in prior to initiating motion.
1080	Default Connection flag is set for the motor	
1081	Incompatible suffixes	The command includes command options that cannot be used together.
1082	Commands BEGIN, END, KILL, GO require axis specification	A GO or KILL/KILLALL command has been entered without an axis specification.
1083	Array requires axis specification	An array requiring an axis specification has been entered without an axis specification.
1084	Illegal array command	Either the array does not exist, or it is not in a compiled buffer.
1085	Extra number after the command	The command specifies a superfluous numerical argument.

Error Code	Error Message	Remarks
1086	Variable name must be specified	The command requires a variable name specification.
1087	Command cannot be executed while the axis is moving	The command specifies one or more axes that are involved in a motion. The command can be executed only after the motion finishes.
1088	Variable can be queried only in compiled buffer	The command attempts to query a variable in a buffer that was not compiled.
1089	Label can be referenced only in compiled buffer	The command attempts to reference a label in a buffer that was not compiled.
1090	This type of motion is not allowed for grouped axis	The slave or track command specifies an axis that was included in a user-defined group. Only a single axis can be specified in the slave or track command.
1091	Less arguments than required	The motion command specifies fewer coordinate values than required by the axis specification.
1092	More arguments than required	The motion command specifies more coordinate values than required by the axis specification.
1093	Bit selector value is greater than 31 or less than 0	The expression specified in the bit selector yields a value greater than 31 or less than 0.
1094	Empty line range is specified	The range of lines specified refers to empty lines.
1095	No master-slave motion is in progress	The command addresses master-slave motion that has not yet started.
1096	'}' was not found	The command includes a non-paired left curly bracket.

Error Code	Error Message	Remarks
1097	Previous data collection is in progress	The command attempts to start a data collection while the previous data collection for the same axis is still in progress.
1098	Stalled motion requires limits specification	The motion command, which includes a command option of stalled motion, requires specification of the motion limits.
1099	Extra numbers after the command	The motion command includes superfluous numerical arguments.
1100	Received command has no message ID	The command does not contain the ID of the message to which it refers.
1101	The program is suspended, the start line number is not allowed	The command attempts to start a program from a specified line when the program is in suspended state. A program in suspended state can be only resumed from the next line. The line specification is not allowed. The only way to start a suspended program from arbitrary line is to stop the program, and to start it again from the desired line.
1102	Zero divide	The an illegal divide by zero is attempted by the command.
1103	Invalid reference	The command contains an invalid reference.
1104	No ACSPL program is waiting for input	The command is attempting to send input, but no program is open for receipt.
1105	Format error	The command is incorrectly formatted.

Error Code	Error Message	Remarks
1106	SP function failed	The function that accesses SP memory cannot be executed. Check the address specified in the function call. The address must fall into the range of 0 to 511.
1107	Current empty space in the dynamic buffer is not enough for the command	The D-Buffer is full and no further items can be entered.
1108	Invalid real number	The command includes specification of a real number that cannot be interpreted as a valid real.
1109	The command is not allowed in SAFE mode	Safe communication mode is a host communication format based on block-by-block transfer of data with delivery confirmation.
1110	At least one variable must be specified for data collection	Data collection was attempted without specifying at least one variable.
1111	Too long reply requested	The time of the reply request parameter is too long.
1112	No matches were found	The search command did not find any match.
1113	The step in the table is zero or negative	The command reference to a table element is either zero or negative.
1114	The program finished without a STOP command	The program that runs in a buffer executed the last command and this was not a STOP/STOPALL command.
1115	Stack underflow (RET without CALL)	The program that runs in a buffer caused stack underflow. This occurs if the program executes the RET command that without the paired CALL command.

Error Code	Error Message	Remarks
1116	Stack overflow (too many nested CALLs)	The program that runs in a buffer caused stack overflow. This occurs if the program executes too many nested CALL commands. Check that the program does not specify infinite recursion (subroutine calls itself infinitely).
1117	Attempt to enter autoroutine directly	The program that runs in a buffer comes to the ON command (see ON...RET). ON must never be executed in the normal program flow.
1118	Illegal axis number	The command specifies an axis by number or expression, and the resulting value is not a legal axis number. Valid axis numbers range between 0 and the number of axes in the system minus one.
1119	Integer overflow	The integer value is too large.
1120	The motion must involve the first two axes from the group	The segmented motion must always involve two axes. If a user-defined group contains more than two axes, the segmented motion must involve the first two axes in the group.
1121	Unknown #-constant	The command specifies an unknown symbolic constant.
1122	Bit selection is not applicable	Bit selector cannot be applied to real value.
1123	Illegal bit selector	Value of bit selector must fall in the range of 0 to 31.

Error Code	Error Message	Remarks
1124	Attempt to enable motor failed	The enable command failed. Additional information can be obtained from the corresponding element of the MERR variable (motor disable code).
1125	Error in SP program	The command caused unsuccessful loading of an SP program. The file with the SP program contains an error.
1126	Illegal SP number	The command specifies an illegal SP number. Valid SP numbers are 0, 1, 2, and 3.
1127	Editing of this buffer is disabled	The command attempts to open the buffer for editing or to change the program in a Protected buffer. Editing a buffer is disabled if the controller is in the Protected mode and bit 1 of the corresponding element of PFLAGS is set to 1.
1128	Configuration changed. Commands SAVE and HWRES must be executed.	The program has changed the configuration. The configuration has to be saved to the controller flash and the controller restarted. Select the controller in the Workspace Tree and click the Save to Flash button. Then right-click the controller and select Reboot .
1129	In binary command the name specification must end with /	The command syntax requires a name to be followed by the / (slash) character.
1130	Segment sequence for the previous motion was not terminated with ENDS command	Commands MPTP...ENDS , MSEG...ENDS , PATH...ENDS , PVSPLINE...ENDS are followed by a sequence of points or segments. The sequence must terminate with ENDS .

Error Code	Error Message	Remarks
1131	SP program is incompatible with one or more products	SP program interface is unrecognized.
1132	The file is not a legal user data file	The file designator is not valid.
1133	Discrepancy in types of the saved and restored arrays	Error detected in array types when comparing the saved and restored versions.
1134	Discrepancy in sizes of the saved and restored arrays	Error detected in array size when comparing the saved and restored versions.
1135	Operation failed because of communication overload	Error occurs when there is either a fault in the communication, or too much communication is being conducted.
1136	Wrong relation between first point, last point and interval	Usually caused by the first point being greater than the last point.
1137	Illegal analog output number	The command contains an illegal analog output number.
1138	Incompatible SP and analog output	Conflict between analog output value in the command and that contained in the SP.
1139	Illegal input	The controller tries to interpret the inserted string as a response to the executed input command, but the string does not follow the required format.
1140	The function is not supported	An attempt was made to use a function that is not supported in SPiiPlus.
1141	Timeout	Communication timeout has occurred. This can be corrected by right-clicking the controller in the Workspace Tree, selecting Properties , and increasing the Connection Timeout value.

Error Code	Error Message	Remarks
1142	Arguments are inconsistent	Conflict with the command arguments and their values.
1143	Memory overflow	Usually caused by infinite loops.
1144	Simulator does not support this command	The command cannot be executed by Simulator.
1145	The specified DPR address is less than 128 or exceeds the DPR size	SPiiPlus does not contain a dual-port RAM; therefore this is not relevant.
1146	Collision with other variable in DPR	SPiiPlus does not contain a dual-port RAM; therefore this is not relevant.
1147	Incomplete command (intrusion of other process?)	The command is not correctly formulated.
1148	Requested more SINCOS encoder multipliers than installed	The user tried to select more Sin-Cos multipliers than allowed.
1149	Illegal SP address	Valid SP numbers are 0, 1, 2, and 3.
1150	Only even numbers are allowed as DPR address	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).
1151	This is a DEMO version of Simulator. 5 minutes of work remains.	The demo version of the Simulator has a limited session length. The Simulator is going to stop after 5 minutes.
1152	The DEMO version of Simulator has terminated	The demo version of the Simulator has a limited session length. The session time has elapsed.
1153	Illegal query command	The controller cannot recognize the query command.

Error Code	Error Message	Remarks
1154	The command can be only used with MSEG motion	The command was incorrectly used - it can only be used with MSEG...ENDS .
1155	Motion cannot start because the previous motion failed. Use FCLEAR command.	To start the motion enter the FCLEAR command.
1156	Profile key must be specified as /SECTION/KEY/	
1157	Illegal configuration string. Use only characters K,D,+,-.	Illegal characters were used in the configuration string.
1158	Cannot find matching value. The formula has no root or the root is not single.	
1159	Axis number is specified more than once	The axis designation parameter is repeated in the command.
1160	The axis cannot be used in a group (see AFLAGS)	The specified axis has been flagged as not to be grouped using AFLAGS .
1161	Illegal communication channel	An invalid communication channel has been specified.
1162	Illegal tag value	The command Tag parameter does not match the intended command.
1163	Illegal configuration parameters	Invalid system configuration parameters have been entered.
1164	Illegal password	The password that has been entered does not match the required password.
1165	Attempt to execute optional function which is not installed	
1166	Flash file operation failed (overlapped file operations?)	
1167	Wrong start position is specified	

Error Code	Error Message	Remarks
1168	File not opened	
1169	D-Buffer cannot be changed while any other buffer is compiled	An attempt was made to make changes to the D-Buffer while other buffers were being compiled.
1171	The number of axes in coordinated motion is restricted to 4	
1172	Illegal category name	
1173	EtherCAT offset is out of range	
1174	Can't find EtherCAT network variable	
1175	EtherCAT Master is not ready	
1176	Data size should be specified	
1177	This slave has no mailbox support	
1178	Invalid CoE SDO parameter	
1179	EtherCAT slave is in invalid state for this operation	
1180	General EtherCAT error	
1181	EtherCAT Timeout error	
1182	Can't split non-existing axes group	An axes group that was not previously defined was specified for splitting.
1183	Attempt to split group with active motion on	Cannot split an axis group while in motion.
1184	Mapped variable must be defined in D-Buffer	
1185	Number of connected EtherCAT devices is not covered by SW options	
1186	Can't initialize SP injection	

Error Code	Error Message	Remarks
1187	There is an active injection on this SP	
1188	Can't enable motor, DIP switch settings are incorrect.	
1189	The command can only be used with XSEG motion	
1190	PLC option on this device is not enabled	
1191	PLC is already running	
1192	General PLC error	
1193	Wrong array size or type	
1194	Not allowed Encoder Type	
1195	Requested more absolute encoder multipliers than installed	
1196	Requested absolute encoder is not supported	
1197	The slave is incompatible with current controller configuration	
1198	Invalid encoder type	See ACSPL+ variable E_TYPE (axis) in the Command and Variable Reference Guide
1199	Invalid absolute encoder parameter type	See ACSPL+ variable E_TYPE (axis) in the Command and Variable Reference Guide
1200	JIT and Dynamic modes are not allowed for D-Buffer	
1201	End-of-Sequence is illegal for this motion	
1202	JIT and Dynamic modes are not allowed at the same time	

Error Code	Error Message	Remarks
1203	DEBUG mode is not allowed for JIT and Dynamic buffers	
1204	JIT and Dynamic modes require the buffer to be empty	
1205	Illegal zone number	
1206	Cannot change a constant variable	
1207	One or more arguments are out of range	
1208	Function ended without return statement	
1209	Entered a function not by calling, STOP statement might be missing	
1210	Incompatible matrix size	
1211	Too many segments	See MSEG...ENDS .
1212	ARC arguments are inconsistent	See MSEG...ENDS , ARC1 and ARC2 .
1213	Stopper is prohibited for master-slave motion	See STOPPER .
1214	Adjacent stoppers are prohibited	See STOPPER .
1215	In cyclic path the first and the last points must coincide	
1216	Velocity is specified, but the motion command had no V command option	Velocity argument can be specified in POINT , LINE , ARC1 or ARC2 only if the corresponding motion command specifies /v .
1217	Segment of zero length	
1218	ARC radius is too small	
1219	Specified motion delay is out of range	
1220	Command is incompatible with connect	

Error Code	Error Message	Remarks
1221	20 KHz motion generation incompatible with CTIME or number of axes	See SPiiPlus ADK Suite Release Notes section "Controller Cycle Time (CTIME) Support" for Controller CTIME Limitations
1222	20 KHz motion limitation reached	
1223	Time specified is not possible under current motion safety limitations	
1224	Offset doesn't exist	
1225	Microblaze monitor is not ready	
1226	Address couldn't be found	
1227	Maximum number of Registers was already added	
1228	Firmware is not properly installed	
1229	SLABITS does not match Single Turn and Multi Turn number of bits	
1230	Non Linear Control Feature is not supported	
1231	Illegal key	An invalid Key argument has been entered in either the GETCONF or SETCONF function.
1232	Illegal index	An invalid Index argument has been entered in either the GETCONF or SETCONF function.
1233	Illegal value	The Value argument of the SETCONF command specifies a value that is not compatible with the Key argument.
1234	Value in SETCONF must be 0 or 1	See SETCONF .
1235	SETCONF function is disabled	

Error Code	Error Message	Remarks
1236	SETCONF cannot be executed while the motor is enabled	
1238	The operation can be executed only when the HSSI channel is in command mode	
1239	HSSI channel cannot switch to command mode because it affects one or more remote drivers. Reset corresponding MFLAGS.#HSSI bits before.	See MFLAGS .
1240	Operation is temporarily disabled	
1241	Operation failed	
1242	The feature is not supported	
1243	Max number of 20 KHz moving axes reached	
1244	Illegal number of coordinates specified	
1245	The specified channel is already being used. Please use the MBCLOSE function to close the connection	
1251	Operation is temporarily disabled	
1252	Motor cannot be enabled while a motion is in termination process	
1253	Unable to work with Dummy motor	The command cannot be executed for a dummy motor. Check flag MFLAGS.#DUMMY (see MFLAGS).
1254	The operation requires the motor to be enabled	The ENABLE/ENABLE ALL command has to be run.
1255	The operation requires the motor to be disabled	The DISABLE/DISABLEALL command has to be run.

Error Code	Error Message	Remarks
1256	The operation is valid only for brushless motor without Hall sensor	
1257	The operation failed because the brushless motor is not commutated	Refer to the chapter on the Adjuster Wizard in the <i>SPiiPlus MMI Application Studio User Guide</i> for details on commutation.
1258	The operation failed because the motor is in open-loop mode	
1259	Motion cannot start because the motor is defined as dummy	
1260	Motion cannot start because the motor is disabled	The ENABLE/ENABLE ALL command has to be run.
1261	Motion cannot start because the brushless motor is not commutated	Refer to the chapter on the Adjuster Wizard in the <i>SPiiPlus MMI Application Studio User Guide</i> for details on commutation.
1262	Motion cannot start because the motor is in open-loop mode	
1263	Motion cannot start because the previous motion failed. Use FCLEAR command.	To start the motion enter the FCLEAR command.
1265	SP program does not support this operation	
1266	Invalid PEG pulse width	Pulse width set in or is incorrect.
1267	Maximal number of time-based PEG pulses is 65,535	Not relevant to PEG operations.
1268	Invalid period of time-based PEG pulses	Not relevant to PEG operations.
1269	PEG pulse width must be less than time-based pulse period	Not relevant to operations.
1270	PEG is not supported for the specified axis	See .

Error Code	Error Message	Remarks
1271	Step does not agree with start/end points	
1272	Incremental PEG step must correspond to encoder counts within the $(-2^{31}, 2^{31}-1)$ range, zero is excluded	The PEG step in encoder counts must be in the range of -2^{31} to $2^{31}-1$.
1273	The specified axis does not exist in this controller model	Refers to controllers that support 4 axes only, and the specified axis is above this.
1274	The specified axis is defined as dummy only	
1275	Only stepper motor is supported for the specified axis	
1276	The brushless motor is not supported for the specified axis	
1277	Dual loop control is not supported for the specified axis	
1278	Remote HSSI driver is not supported for the specified axis	
1279	PEG states are not supported for the specified axis	
1280	The stepper motor is not supported for the specified axis	
1281	No Hall support for 2-Phase motors	
1282	Value out of range	
1283	Number of points for Random PEG exceeds the limit	
1284	Axis is not connected to PEG engine	
1285	NanoMotion Piezoceramic Motor is not supported for the specified axis	

Error Code	Error Message	Remarks
1286	Fast Sin-Cos Encoder is not supported for the specified axis	
1287	Laser Modes are not supported for the specified axis	
1288	Time-based PEG is not supported for the specified axis	
1289	This function is not supported for the specified axis	
1290	SPINJECT/SPRT and Fast GPRT cannot be used in parallel	
1291	P/D interface is required for this function	
1292	Secondary Feedback is not supported	
1293	SLABITS does not match Single Turn and Multi Turn number of bits	
1294	Motion cannot start because the Gantry Complementary brushless motor is not commutated	
1295	Commut is not allowed when Compensation is in process	
1300	Non Linear Control License is required	
1301	Another EtherCAT port is already closed	
1302	Not possible to save network topology configuration when more than one line break exists	
1303	ERROR SDO: Object does not exist in the object dictionary.	
1304	The value is too high (check E_PAR_D value)	

Error Code	Error Message	Remarks
1305	The network variable is already mapped to ACSPL+ variable	
1306	Invalid axis list	
1307	FoE Protocol is not supported by Slave	
1308	The Disk is full or the file is too big	
1309	Function supported only by CiA402 Drive	
1310	CiA402 Drive is not in OP State. PDO is not enabled.	
1311	Cannot convert REAL type to INT	
1312	More than one hold command is not allowed	
1313	Continue command is already in process	
1314	Requested Homing Method is not supported	
1316	Homing in Gantry mode requires additional parameters	
1317	FoE Error: Access denied	
1318	Matrix is not invertible	
1319	PI AME interface is not initialized	
1320	PI AME interface is busy	
1321	Axis is not commutated	
1322	Homing is required	
1323	DCOM is not 0	
1324	Parameters are not consistent	

Error Code	Error Message	Remarks
1325	Main FPGA file downloading is in progress	
1326	Operation aborted, PEG is in process	
1327	FPGA versions do not match	
1328	FPGA version is the same. FPGA Upgrade aborted.	
1329	Invalid value, digital input index should range between 0-99 and bit index should range between 0-31	
1330	Invalid value, digital output index should range between 0-99 and bit index should range between 0-31	
1331	This output is already mapped as a mechanical brake to a different axis	
1332	Hardware limit swapping (MFLAGSX.#HLIMSWAP) and limit routing are mutually exclusive	
1333	File name MAX length is 100 chars	
1334	Operation failed because Learning Boost is active	
1335	Illegal SPATH specification	
1336	Nurbs motion was not initiated	
1337	Illegal NURBS/SPATH parameter value	
1338	Profile has ended or last knots already added	
1339	Invalid number of axes	
1340	Dummy point must also specify a Knot value	

Error Code	Error Message	Remarks
1341	Illegal dummy point, two points are allowed before coordinates and two after	
1342	The measurement you are trying to activate is already active. (see DPM_Measurement Stop() function for more details)	
1343	An invalid sample type has been specified. Type can either be 0 or 1. (see DPM_Measurement sample type for more details)	
1344	The sampling time specified is invalid, the minimum value is the controller cycle time(?CTIME) and the maximum value is: 100000000msec	
1345	The sample-set size specified is invalid. Minimum size: 1. Maximum size: 512.	
1346	The specified "when_to_measure" flag is invalid. (see DPM_Measurement documentation for more details)	
1347	The measurement you are trying to activate is paused. (see DPM_Measurement Stop()/Resume() functions for more details)	
1348	Cannot change the selected axis while monitoring is ON. (see DPM_Motion_Status SelectAxis()/MonitorOff() functions for more details)	
1349	The specified "monitored_variable" is invalid. (see DPM_Measurement documentation for more details)	
1350	Illegal attempt to copy STRING	

Error Code	Error Message	Remarks
1351	Illegal String operation - value too long	
1352	DCOM value cannot be changed during Gantry Cross Axis Compensation	
1353	Cross Axis Force Ratio parameter is out of range	
1354	Number Of Cross Axes parameter is out of range	
1355	Cross Axis Index parameter is invalid	
1356	Cross Axis Polarity parameter is out of range	
1357	Cross Axis Mass Ratio parameter is out of range	
1358	Cross Axis Max Position and Cross Axis Min Position are not consistent	
1359	Yaw Axis Index parameter is invalid	
1360	LCI unit initialization failed	
1361	Unable to start the LCI operation	
1362	Trajectory axes are not defined	
1363	LCI Safety input is off	
1364	LCI laser fault input is on	
1365	LCI supported velocity limit exceeded	
1366	LCI unit not found	
1367	LCI. Unable to allocate channel for specified operation	
1368	LCI function timeout	

Error Code	Error Message	Remarks
1371	Enable of Gantry axes is required for Compensation to start	
1372	Gantry Mode is required	
1373	Compensation for the specified axis is already in process	
1374	The force ratio is too high.Place the cross masses symmetrically around the center of gravity	
1375	The specified IP address is invalid. Please see MBOPEN documentation for more details.	
1376	The specified server ID is invalid (valid values are between [1 - 247]). Please see MBOPEN documentation for more details.	
1377	The specified word-order is invalid. (valid values are 0 for big-endian and 1 for little-endian) Please see MBOPEN documentation for more details.	
1378	Unable to open more than 3 Modbus connections at any given time. Please see MBOPEN documentation for more details.	
1379	The specified communication handle is invalid.	
1380	A connection with the specified IP was not found.	
1381	The specified address is invalid.	
1382	The specified request frequency is invalid. The minimum value is 5 milliseconds. Please see the documentation for more details.	

Error Code	Error Message	Remarks
1383	The specified variable length and the specified number of elements do not match. Please see the documentation for more details.	
1384	The specified suffix combination is invalid. Please see the documentation for more details.	
1385	Read-Only variables cannot be written.	
1386	The specified request ID is invalid.	
1387	The specified variable(or part of it) is already being written to by another Modbus request. Use # MBMAPREP for more details.	
1388	The specified server address is already being written to by another Modbus request. Use # MBMAPREP for more details.	
1389	The specified number of elements is greater than the specified variable length. Please see the documentation for more details.	
1390	The maximum number of integer values that can be "mapped" in a single request is 16. Please see the documentation for more details.	
1391	The maximum number of short values that can be "mapped" in a single request is 32. Please see the documentation for more details.	
1392	The maximum number of single-precision floating-point values that can be "mapped" in a single request is 16. Please see the documentation for more details.	

Error Code	Error Message	Remarks
1393	The maximum number of double-precision floating-point values that can be "mapped" in a single request is 8. Please see the documentation for more details.	
1394	The maximum number of coil values that can be "mapped" in a single request is 32. Please see the documentation for more details.	
1395	The maximum number of discrete input values that can be "mapped" in a single request is 32. Please see the documentation for more details.	
1396	Failed to establish a connection with the specified server device.	
1397	The maximum number of mapping requests per server is 32. Please see the documentation for more details.	
1398	A connection with the specified IP is already open and with different configuration parameters. To change the connection parameters, close and open a new connection.	
1399	The specified Modbus channel operates in sequential mode. This mode does not support user-defined request frequency. Each request is sent after a response for the previous one has arrived, but no less than 10 milliseconds from the last time it has been sent. Please see the documentation for more details.	
1410	The minimum value can not be greater than the maximum value. Please see the documentation for more details.	

Error Code	Error Message	Remarks
1411	G-code: Run-time errors were detected during the program execution in simulation mode. Use the #LOGP <buffer number> command to present the detected errors.	
1412	This motion type is not supported	
1413	The supplied correction maps (2-dimensional arrays) should have the same dimensions. Please see the documentation for more details.	
1414	The Specified referenced axes/analog inputs are invalid. Please see the documentation for more details.	
1415	Internal Error	
1416	Stage model dll is not loaded	
1417	Stage model dll's name does not match the stage model implemented by the dll	
1418	Invalid number of virtual axes	
1419	Invalid number of physical axes	
1420	Virtual axis index is out of range	
1421	Physical axis index is out of range	
1422	Virtual and machine axes conflict	
1423	Tool length index is out of range	
1424	Fixture offset index is out of range	
1425	Sync positions array must be set when feedback transform is not supported	
1426	Custom virtual axes must be set in this FW version	

Error Code	Error Message	Remarks
1427	Custom virtual axes must be set for 128 axes systems	
1428	Not implemented feature	

7.2 ACSPL+ Compilation Errors

The error codes in ACSPL+ compilation errors range between 2000 and 2999.

ACSPL+ compilation errors are reported either immediately when the erroneous line is inserted, or subsequently, when the controller attempts to compile an ACSPL+ program. If a program in a buffer undergoes compilation and an error is detected, the error code is stored in the corresponding element of the **PERL** array and the erroneous line number is stored in the corresponding element of the **PERL** array.

Table 7-2. ACSPL+ Compilation Errors

Error Code	Error Message	Remarks
2002	Unrecognized command	The program line contains a sequence that is not recognized as a legal command. If the line starts from a standard variable name, check the following: > All letters are uppercase > Spelling is correct If the line starts from a user variable name, check the following: > The variable is defined before use > The letter case matches the definition > Spelling is correct
2003	Unexpected delimiter, incomplete command	A command contained in the line is incomplete.
2006	Unexpected END	The END command is specified without a paired IF , LOOP or WHILE command.

Error Code	Error Message	Remarks
2007	Two adjacent operators	Two binary operator are specified in the expression without an operand between.
2008	Left bracket was expected	Left bracket was not found in a position where it is expected. Possible reasons for this error are array name without index or function call without arguments.
2009	Right bracket was expected	The expression or sub-expression has an unpaired left bracket.

Error Code	Error Message	Remarks
2010	Comma was expected	Comma character was not found in a position where it is expected. Possible reasons for this error are: > Missing comma between the arguments of the command or function > The command or function call has less arguments than required.
2011	Equals sign was expected	Equal sign in assignment is missing.
2012	Direct command was expected	
2013	DO or LOOP without END	
2015	Integer positive constant is expected	The declaration of array contains a size definition other than positive integer constant.

Error Code	Error Message	Remarks
2016	Only label or line number is allowed as start point	The start command can contain the start line definition. The start line definition can appear only as a label or a positive line number.
2018	Scalar variable cannot be indexed or used with axis specification	The command contains a scalar variable followed by index specification or prefixed by axis specification.
2019	Write-only variable cannot be read	
2020	Read-only variable cannot be assigned	The command attempts to assign a read-only variable.
2021	Label was expected	GOTO or CALL must specify a label name.
2022	Array name was expected	The command or function requires an array name as one of its arguments. The array name must not be indexed.
2023	Variable name was expected	The declaration specifies an illegal variable name. A variable name must begin with a letter or underscore and can include letters, digits and underscores.

Error Code	Error Message	Remarks
		A variable name must not include any of the following: > Keywords (IF, WAIT, SIN, etc.) or standard variable names (FPOS, MST, etc.) > Standard label (AUTOEXEC).
2024	Undefined label	GOTO or CALL specify a label name that was not defined in the program.
2025	Duplicated label	The program contains two identical labels.
2026	Undefined variable name	The command contains a variable name that was not declared in the program.
2027	Duplicated variable name	The program declares two variables with identical names.

Error Code	Error Message	Remarks
2030	Number user array elements can not exceed the XARRSIZE parameter	The value of the XARRSIZE parameter is limited to a maximum of 30,000 elements. The limitation applies also to two-dimensional arrays. The total number of elements in a two-dimensional array is equal to the product of its sizes by each dimension. The total number of elements must not exceed 30,000.
2031	Global of different type/dimension was defined in other buffer	The program declares a global variable with the name that was already declared in other buffer with different type or dimension.
2033	Mandatory argument is omitted	The command or function call contains fewer arguments than required.
2034	More arguments than required	The command or function call contains more arguments than required.
2035	Wrong argument type	The command or function call contains an argument of incorrect type.

Error Code	Error Message	Remarks
2036	Function that does not return a value cannot be used in expression	The expression cannot include a function that has no return value.
2037	Axis specification was expected	This error refers to a logical check. The command must specify an axis. Axis specification can appear: as an integer designation, like 0, 1, 32, or as an expression that evaluates to an integer within the range of 0 and the number of axes in the system minus 1.
2038	Axis specification was expected	This error refers to a physical check. The command must specify an axis that physically exists. Axis specification can appear: as an integer designation, like 0, 1, 32, or as an expression that evaluates to an integer within the range of 0 and the number of axes in the system minus 1.
2042	Array index must be a non-negative integer constant	
2043	Internal error	

Error Code	Error Message	Remarks
2044	Index is out of range	The constant index value is greater than the array size minus one, or is negative.
2045	Illegal axis value	The axis value specified by a numerical constant is greater than the number of axes in system minus one, or is negative.
2048	Argument must be specified as + or - sign	The command requires a direction specification that must be presented by character + or -.
2049	Illegal suffix for this command	The command specifies a command option, which is illegal for this command.
2050	Name of standard variable was expected	The command requires an argument that specifies one of the ACSPL+ variables.
2051	Only APOS, RPOS, FPOS and F2POS are allowed in SET command	The SET command must specify assignment to one element of one of the following variables: APOS, RPOS, FPOS, F2POS. The variables must be in upper case.

Error Code	Error Message	Remarks
2052	Variable name was expected	The command requires an argument that specifies a scalar variable or an element of array. A general expression or a constant is not allowed for this argument.
2053	Constant argument was expected	The command requires an argument that specifies a numerical constant.
2054	Illegal buffer number	The command must specify a constant buffer number. The constant must fall in the range of from 0 to 15.
2055	Assigned value is out of range	The command attempts to assign a variable with a value that falls out of the range allowed for the variable.
2056	Zero divide	The command attempts to divide by zero.
2057	Only VEL,ACC,DEC,JERK,KDEC are allowed in IMM command	IMM must specify assignment to one element of one of the following variables: VEL , ACC , DEC , JERK , KDEC .

Error Code	Error Message	Remarks
2058	Bit selection cannot be applied to real variable	The bit selector is specified for a real variable.
2059	ELSE without IF	ELSE is specified without corresponding IF (see IF , ELSEIF , ELSE...END).
2060	ELSEIF without IF	ELSEIF is specified without corresponding IF (see IF , ELSEIF , ELSE...END).
2061	LOOP without END	LOOP has no corresponding END (see LOOP...END).
2062	DO without END	
2063	IF without END	IF has no corresponding END (see IF , ELSEIF , ELSE...END).
2064	Memory overflow	The application requires too much memory. Reduce the number and size of the local and global variables used in the application, or use a more powerful model of the controller.

Error Code	Error Message	Remarks
2065	Axis constant or axis expression in brackets expected	The command must include an axis specification in parentheses, for example: FMASK (4).16 , where 4 is the axis specification.
2066	Too many axis specifiers	The axis specification includes too many axis specifiers.
2067	An axis is specified more than once	The axis specification includes two identical axis specifiers.

Error Code	Error Message	Remarks
2068	Sign constant or sign expression in brackets expected	The command must include a sign specification. Each sign in the specification defines a direction of one involved axis. Sign specification can appear: > As a sequence of sign characters, like +, ++, -+- > As a list of expressions enclosed in brackets where (0, 1, 0) is equivalent to ++.
2069	Too many sign specifiers	The sign specification includes too many sign specifiers.
2070	Unknown #-constant	The command includes an unknown symbolic constant.
2071	Local variable is not allowed in this command	The command does not allow the use of local variables. Only ACSPL+ variables and user global variables that have been declared can be used.

Error Code	Error Message	Remarks
2072	WHILE without END	WHILE has no corresponding END (see WHILE...END).
2073	WAIT,TILL,GOTO,CALL,RET,LOOP,WHILE,ON,INPUT,AXISDEF, ENCREAD are not allowed in immediate command	The specified commands cannot be entered through the SPiiPlus MMI Application Studio Communication Terminal .
2074	Only RPOS variable is allowed after CONNECT	CONNECT must specify assignment to one element of one of the RPOS variables.
2075	Only MPOS variable is allowed after MASTER	MASTER must specify assignment to one element of one of the MPOS variables.
2076	Constant bit selector is greater than 31 or less than 0	The command includes a constant bit selector. The value of bit selector must fall into the range from 0 to 31.
2077	Array name requires indexing	The array name must be followed by an index specification. For a two-dimensional array, two index specifiers are required.

Error Code	Error Message	Remarks
2078	Current empty space in the dynamic buffer is not enough for the command	The dynamic buffer contains too many commands queued for execution. The application must wait for the controller to execute and remove one or more commands from the buffer.
2079	GOTO,CALL,RET,LOOP,WHILE,IF,ON are not allowed in a dynamic or JIT buffer	Commands GOTO, CALL, LOOP...END, WAIT, IF, ELSEIF, ELSE...END, ON...RET cannot be used in the dynamic buffer.
2080	Local variable declaration is not allowed in immediate command	Local variables cannot be used in a SPiiPlus MMI Application Studio Communication Terminal command.
2081	Variable declaration is not allowed in a dynamic or JIT buffer	Variable declaration cannot be executed in a dynamic buffer.
2082	Illegal string argument	The command requires file name specification.
2083	Integer overflow	The result of constant integer expression is more than 2147483647 or less than -2147483648.

Error Code	Error Message	Remarks
		Consider using real constants instead of integer. To change constant type to real, add decimal point to the end of the constant.
2084	Integer constants are allowed in the range from -2147483648 to +2147483647	Consider using real constants instead of integer. To change constant type to real, add decimal point to the end of the constant.
2085	Protection violation	The controller is in the Protected mode and the command violates one of the protection rules.
2086	Protection attribute cannot be changed for this parameter	
2087	Only constant 0 or 1 is allowed at the right side	The command allows only 0 or 1 to the right of the equals sign.
2088	No dual-port RAM in this controller	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).

Error Code	Error Message	Remarks
2089	Bit selection is not available for DPR variable	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).
2090	Only global variables can be defined in DPR	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).
2091	DPR address must be specified	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).

Error Code	Error Message	Remarks
2092	Only even numbers from 128 to 504 are allowed as DPR address	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).
2093	Collision with other variable in DPR	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).
2094	DPR variable is not allowed in this command	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).
2095	Illegal line number	The specified line number must be a positive non-zero integer.

Error Code	Error Message	Remarks
2096	Only even numbers from 128 to 1016 are allowed as DPR address	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).
2097	Illegal SP number	Valid SP numbers are 0, 1, 2, and 3.
2098	Illegal SP variable specification	Invalid SP variable was entered.
2099	Undefined SP variable	Non-SP variable was entered.
2100	User-defined array was expected	
2101	Illegal character constant	
2102	Illegal tag	
2103	Tag can be specified only for global variables	
2104	BLOCK cannot be nested	See BLOCK...END .
2105	BLOCK without END	See BLOCK...END .
2106	Illegal declaration	
2107	Axis redefinition	
2108	Assignment to constant	Value has to be assigned to a variable.
2109	ALL cannot be used in this command	
2110	Ambiguous axis specification	

Error Code	Error Message	Remarks
2111	Illegal format of real constant	
2112	This function cannot be used as argument of CONNECT, TRIGGER functions, or in Autoroutine condition	See CONNECT , TRIGGER or ON...RET (for the autoroutine structure).
2113	Shared memory variables must be global	
2114	Shared memory variables can be declared only in the D-Buffer	
2115	Bit selector must be integer scalar or constant	
2116	No shared memory support in this controller	
2117	G-code programs execution requires option	
2118	CALL,CONNECT,MASTER are not allowed in D-buffer	
2119	Static/Constant variables can only be declared in the D-Buffer	
2120	Static variable must be global	
2121	Static variable of a different type/dimension was already defined. Use #VGV command to deallocate.	
2122	A Static variable of the same name is already defined	
2123	A Static variable was expected	
2124	Dimension mismatch on array initialization.	
2125	Array Initialization is possible in D-Buffer only	
2126	Axes X,Y,Z cannot be defined as Rotary axes	
2127	Closing curly bracket not expected	
2128	Curly bracket expected	
2129	It is illegal to define a function inside another function	
2130	Function parameter definition expected	

Error Code	Error Message	Remarks
2131	Wrong function arguments	
2132	Action not supported	
2133	Function signature does not conform to previous declaration	
2134	A parameter by the same name already exists	
2135	A function by that name already exists	
2136	A Function was defined but never implemented	
2137	A void/Composite type function is not applicable	
2138	A Function should not be defined as Static/Local/Global/Const	
2139	Array dimensions mismatch	
2140	Array dimensions must be positive and known at compilation time	
2141	A variable by that name already exists	
2142	Function call is not allowed in a DISP statement	
2143	Only user defined primitive variables can be passed by reference	
2144	Constant Parameters are not allowed	
2145	First array dimension should not be specified	
2146	A function was defined without a return statement	
2147	Functions variable type expected	
2148	Structs can only be defined in D-Buffer	
2149	Struct name was expected	
2150	Structs can only contain fields and functions	
2151	Structs can only be defined as global in D_buffer or local otherwise	

Error Code	Error Message	Remarks
2152	Field is not recognised	
2153	Field selection expected	
2154	Global/Static not allowed	
2155	Struct Functions can only accept primitive types	
2156	Struct already has a property by that name	
2157	Void type cannot be defined	
2158	Illegal definition of struct inside struct	
2159	All struct fields must be defined before functions	
2160	Illegal array type	
2161	Name Is Taken	
2162	Illegal Field/Function definition	
2163	Illegal variable qualification	
2164	A Struct by that name already exists	
2165	A return value was expected	
2166	The Type selected was not properly defined	
2167	Maximum number of structs reached	
2168	Function name was expected	
2169	The function has already been implemented	
2170	Fastcall functions do not allow DISP/TILL/WAIT/Function Call/G-Code operations	
2171	It is illegal for a name to start with {N/n}{0-9}	
2172	Only a D-Buffer FASTCALL function can be passed as parameter, all parameters must be INT/REAL	
2173	Trying to assign to read-only field	

Error Code	Error Message	Remarks
2174	Trying to send a read-only field by reference	
2175	The function does not conform	
2176	Default Value for function parameter missing	
2177	Array default parameter must be 0	
2178	Default values can only be defined for INT/REAL/MATRIX/Arrays	
2179	Default parameters must be specified in declaration only	
2180	The increment operator can only be applied to an Integer type variable	
2181	Dimension mismatch on matrix initialization.	
2182	Illegal matrix operation.	
2183	The argument matrix must be square.	
2184	The second argument matrix must be square.	
2185	The matrices are not of the same size.	
2186	The first argument matrix columns number does not match the second argument matrix rows number.	
2187	The result matrix size mismatch.	
2188	Matrix was expected.	
2189	Operators concatenation is not allowed.	
2190	Matrix type requires 2 dimensions.	
2191	Only ACSPL+ Standard Structs can be defined as static	
2192	This Struct type must only define static variables	
2193	Struct fields cannot be initialized	
2194	Some ACSPL commands cannot be used as function name i.e. "DISP", "CONNECT" ...	
2195	Connect/Trigger expressions do not support user defined	

Error Code	Error Message	Remarks
	functions	
2196	Some parameters must have a fixed memory - Static user variable or standard ACSPL variable	
2197	Real time C function library requires option	
2198	Real time C function library can only be declared in D-Buffer	
2199	Real time C function library declarations not allowed in a struct	
2200	Failed to load library	
2201	Real time C function library must be FASTCALL	
2202	Real time C function arguments cannot be structs	
2203	Real time C function cannot have ACSPL+ body	
2204	Failed to load function	
2205	Only FASTCALL functions are applicable in this context	
2206	Number of real time C function libraries exceeds the limit	
2207	No return value is expected	
2208	Illegal SWITCH syntax used	
2209	Illegal String syntax used	
2210	Illegal String operation	
2211	Illegal String definition	
2212	Function variables definition inside a syntactical strucutre is not allowed	
2213	Non Default CTor is not avaiable	
2214	Illegal String operation - value too long	
2215	Creating an instance is not allowed	
2216	Instances of this type must be defined as GLOBAL	

7.3 ACSPL+ Runtime Errors

The ACSPL+ runtime error code values range between 3000 and 3999.



Codes from 3000 to 3019, however, do not indicate an error. For example, code 3001 reports that the program is suspended and code 3002 reports that the user has terminated the program.
Codes from 3020 to 3999 indicate run-time errors.

If an error occurs in immediate execution of ACSPL+ command, the error is indicated immediately in the prompt. If an error occurs when an ACSPL+ program is executed in a buffer, no immediate indication is provided. Instead, the error code and the line number are stored in the corresponding elements of the **PERL** and **PERL** arrays.

Table 7-3. ACSPL+ Runtime Errors

Error Code	Error Message	Remarks
3020	Illegal subcommand	
3021	SP command requires axis specification	
3022	Illegal command	
3023	Read-only variable cannot be assigned	A command specifies an assignment to a read-only variable.
3024	Set variable cannot be reported	
3025	Time interval between two characters in command is more than 2 seconds	
3026	Serial Number, Part Number, or Software Options were already specified	
3027	Variable requires axis specification	
3028	Scalar variable cannot accept axis specification	A scalar variable cannot be prefixed with an axis specification.
3029	Extra characters after the command	
3030	Too many parameters	

Error Code	Error Message	Remarks
3031	Illegal array in the array command	
3032	Illegal data in array	
3033	Illegal edit command	
3034	Illegal index value	The command specifies an assignment to a read-only variable.
3035	Index is out of range	The reason of the error is one the following: > The specified index value is more or equal to the number of elements in the array > The specified index value is negative > The specified index values are incompatible (first value in the range greater than last).
3036	Internal error	
3037	Illegal variable name	The command requires specification of a standard variable name, but the specified name is not a name of an ACSPL+ variable.
3038	Wrong checksum in the command	
3039	Only one motion per axis can be planned in advance	
3040	Unable to open file	The command specifies a file in the flash memory that does not exist.

Error Code	Error Message	Remarks
3041	Assigned value is out of range	The command attempts to assign the standard variable with a value that is out of the range allowed for this variable.
3042	Operation failed because of exception	
3043	Program cannot start because the buffer was not compiled	The command attempts to start an ACSPL+ program that has not been compiled. To compile a program, in the SPiiPlus MMI Application Studio: > In the Program Manager, right-click the buffer and select Compile Buffer, or > Use the #nC command in the Communication Terminal, where n is the buffer number.
3044	Command cannot be executed while the program is running	The command attempts to affect a running ACSPL+ program. Stop the program before executing the command. To stop a program, in the SPiiPlus MMI Application Studio: > In the Program Manager, right-click the buffer and select Stop Buffer, or > Use the #nS command in the Communication Terminal, where n is the buffer number.

Error Code	Error Message	Remarks
3045	Numerical error in standard function	The command includes an ACSPL+ function that caused a numerical error. Check if the arguments of the function fall into the allowed range.
3046	Write file error	The command caused a failed write to flash memory. A re-occurring error of this type indicates a serious problem in the controller hardware or firmware.
3047	Read file error	The command caused a failed read from flash memory. A re-occurring error of this type points to a serious problem in the controller hardware or firmware.
3048	More axes than were defined in the motion	The POINT command specifies more axes than were specified in the motion that the command refers to.
3049	Axis already belongs to a group	
3050	Conflict with user-defined axis group	The command is incompatible with a previously defined user-defined axis group. The axes specified in the command may either belong to one user-defined axis group, or may not intersect with a user-defined axis group.
3051	Line number is out of range	The command specifies a line number that does not exist in the specified program buffer.

Error Code	Error Message	Remarks
3052	Buffer number is out of range	The command specifies an illegal buffer number. The controller has program buffers numbered 0 to 15, where 15 is the D Buffer..
3053	Wrong type	The command addresses a standard variable of a different type from the variable specified in the command. This error never occurs when the user communicates with the controller through a communication terminal. The error occurs only when an application communicates with the controller using the SPiiPlus C Library. The error indicates a communication problem.
3054	This type of motion is valid for single axis only	
3055	Command requires line number specification	The command must contain a line number specification.
3056	Parameter defining Master has illegal value	
3057	Previous superimposed motion is in progress	
3058	Slave is not synchronized to master	
3059	Command PTP/V must specify velocity value	
3060	Illegal memory command	The memory management command is improperly formatted.
3061	'}' wasn't found	The command contains a non-paired left bracket.

Error Code	Error Message	Remarks
3062	Command is too long	
3063	Variable is not defined in the buffer	The command addresses a variable that is not declared in the specified buffer, or the specified buffer is not compiled.
3064	Undefined global variable	The command addresses a global variable that is not declared in any buffer, or the buffer that contains the declaration is not compiled.
3065	The command cannot be executed while the current motion is in progress	The command is in conflict with one or more of the currently executed motions. To kill a motion use KILL/KILLALL .
3066	Attempt to compile or start empty buffer	
3067	GO command failed	The controller has no motions waiting for GO .
3068	Referenced axis does not execute a motion (motion was terminated?)	The command specifies an axis, but no motion was specified for this axis.
3069	This command can only be used with MPTP, PATH or PVSPLINE motion	The command specifies an axis, but the motion specified for the axis is incompatible with the command.
3070	Attempt to add segment after ENDS command	The command attempts to add a segment or a point to the motion that has already been closed by ENDS .
3071	File name was expected	The command must specify the name of an internal file in the flash memory.

Error Code	Error Message	Remarks
3072	Wrong array size	The command specifies an array, but the motion that the command refers to, or other command arguments require an array of another size.
3073	Text for search is not specified	The command must specify a text for search operation.
3074	Only standard or SP variable is allowed in the command	The command requires an ACSPL+ or SP variable name to be specified.
3075	Name is not a standard or user variable	
3076	Undefined label	The command requires a label specification. The program that contains the label specified in the command must be compiled in a buffer.
3077	Protection violation	The command attempts to assign a protected variable when the controller is in protected mode. The controller must be in configuration mode before protected variables can be assigned.
3078	Variable can be changed only while the motor is disabled	The command attempts to assign a variable that can be changed only if the motor is disabled: > If the variable is axis-related, disable the corresponding motor before assigning the variable. > If the variable is not axis-related, disable all motors before assigning the variable.

Error Code	Error Message	Remarks
3079	Motion cannot start because one or more motors are disabled	The motion command involves one or more motors that are disabled.
3080	Default Connection flag is set for the motor	The command cannot be executed because the default connection flag is set for the motor. The default connection flag is bit 17, i.e., #DEFCON, of the MFLAGS variable (see MFLAGS).
3081	Incompatible suffixes	The command includes switches that cannot be used together.
3082	Commands BEGIN, END, KILL, GO require axis specification	
3083	Array requires axis specification	
3084	Illegal array command	
3085	Extra number after the command	The command specifies a superfluous numerical argument.
3086	Variable name must be specified	The command requires a variable name specification.
3087	Command cannot be executed while the axis is moving	The command specifies one or more axes that are involved in a motion. The command can be executed only after the motion finishes.
3088	Variable can be queried only in compiled buffer	The command attempts to query a variable in a buffer that was not compiled.
3089	Label can be referenced only in compiled buffer	The command attempts to reference a label in a buffer that was not compiled.

Error Code	Error Message	Remarks
3090	This type of motion is not allowed for grouped axis	The slave or track command specifies an axis that is included in a user-defined group. Only a single axis can be specified by SLAVE or TRACK .
3091	Less arguments than required	The motion command specifies less coordinate values than required by the axis specification.
3092	More arguments than required	The motion command specifies more coordinate values than required by the axis specification.
3093	Bit selector value is greater than 31 or less than 0	The expression specified in the bit selector yields a value greater than 31 or less than 0.
3094	Empty line range is specified	
3095	No master-slave motion is in progress	
3096	'}' was not found	The command includes a non-paired left curly bracket.
3097	Previous data collection is in progress	The command attempts to start a data collection while the previous data collection for the same axis is still in progress.
3098	Stalled motion requires limits specification	The motion command, which includes a command option of stalled motion, requires specification of the motion limits.
3099	Extra numbers after the command	The motion command includes superfluous numerical arguments.
3100	Received command has no message ID	

Error Code	Error Message	Remarks
3101	The program is suspended, the start line number is not allowed	The command attempts to start a program from a specified line when the program is in suspended state. A program in suspended state can be only resumed from the next line. The line specification is not allowed. The only way to start a suspended program from a specific line is to stop the program, and to start it again from the desired line.
3102	Zero divide	
3103	Invalid reference	
3104	No ACSPL program is waiting for input	
3105	Format error	The command is incorrectly formatted.
3106	SP function failed	The function that accesses SP memory cannot be executed. Check the address specified in the function call. The address must fall in the range of 0 to 511.
3107	Current empty space in the dynamic buffer is not enough for the command	
3108	Invalid real number	The command includes specification of a real number that cannot be interpreted as a valid real number.
3109	The command is not allowed in SAFE mode	
3110	At least one variable must be specified for data collection	
3111	Too long reply requested	

Error Code	Error Message	Remarks
3112	No matches were found	The search command did not find any matches.
3113	The step in the table is zero or negative	
3114	The program finished without a STOP command	The program that is running in a buffer executed the last command, and it was not a STOP/STOPALL command.
3115	Stack underflow (RET without CALL)	The program that runs in a buffer caused a stack underflow. Program execution requires CALL .
3116	Stack overflow (too many nested CALLs)	The program that is running in a buffer caused stack overflow. This occurs if the program executes too many nested calls. Check that the program does not specify infinite recursion (subroutine is in an infinite loop).
3117	Attempt to enter autoroutine directly	The program that is running in a buffer comes to the ON command (see ON...RET). ON must never be executed in the normal program flow.
3118	Illegal axis number	The command specifies an axis by number or expression, and the resulting value is not a legal axis number. Valid axis numbers range between 0 to the number of axes in the system minus one.
3119	Integer overflow	

Error Code	Error Message	Remarks
3120	The motion must involve the first two axes from the group	The segmented motion must always involve two axes. If a user-defined group contains more than two axes, the segmented motion must involve the first two axes in the group.
3121	Unknown #-constant	The command specifies an unknown symbolic constant.
3122	Bit selection is not applicable	Bit selector cannot be applied to a real value.
3123	Illegal bit selector	Value of bit selector must fall into the range 0..32.
3124	Attempt to enable motor failed	ENABLE/ENABLE ALL failed. Additional information can be obtained from the corresponding element of the MERR variable (motor disable code).
3125	Error in SP program	The command caused unsuccessful loading of an SP program. The file with the SP program contains an error.
3126	Illegal SP number	The command specifies an illegal SP number. Legal SP numbers are from 0 to 3.
3127	Editing of this buffer is disabled	The command attempts to open the buffer for editing or to change the program in the buffer. Editing of a buffer is disabled if bit 1 of the corresponding element of PFLAGS = 1.
3128	Configuration changed. Commands SAVE and HWRES must be executed.	

Error Code	Error Message	Remarks
3129	In binary command the name specification must end with /	The command syntax requires a name to be followed by the / (slash) character.
3130	Segment sequence for the previous motion was not terminated with ENDS command	MPTP , MSEG , PATH are followed by a number of the point or segment commands - segment definition sequence. The segment definition sequence must be closed with ENDS. See MPTP...ENDS , MSEG...ENDS , and PATH...ENDS .
3131	SP program is incompatible with one or more products	
3132	The file is not a legal user data file	READ specifies a file in the flash memory that is not a legal user data file. Only files created with WRITE are legal user data files.
3133	Discrepancy in types of the saved and restored arrays	The array and the user data file specified by READ contain data of different types. READ must specify an array of the same type and dimension as the array that was specified by the WRITE command that created the file. For example, the error occurs if a real array was saved in the file by WRITE , but READ tries to load the file into an integer array.

Error Code	Error Message	Remarks
3134	Discrepancy in sizes of the saved and restored arrays	The array and the user data file specified by READ are different sizes. READ must specify an array of the same type and dimension as the array that was specified by the WRITE command that created the file.
3135	Operation failed because of communication overload	
3136	Wrong relation between first point, last point and interval	The interval value specified by or does not correspond to the start and final points. For example, the error occurs if the final point is less than the start point, but the step is positive.
3137	Illegal analog output number	The command specifies a number of analog outputs which is not available in the controller.
3138	Incompatible SP and analog output	The command specifies an analog output number that cannot be accessed in the specified SP.
3139	Illegal input	The controller tries to interpret the string as a response to the executed input command but the string does not follow the required format.
3140	The function is not supported	
3141	Timeout	
3142	Arguments are inconsistent	Check arguments against the specifications for the or commands.

Error Code	Error Message	Remarks
3143	Memory overflow	
3144	Simulator does not support this command	The command cannot be executed by the simulator.
3145	The specified DPR address is less than 128 or exceeds the DPR size	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).
3146	Collision with other variable in DPR	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).
3147	Incomplete command (intrusion of other process?)	
3148	Requested more SINCOS encoder multipliers than installed	
3149	Illegal SP address	The SP address must fall in the range of from 0 to 512.
3150	Only even numbers are allowed as DPR address	SPiiPlus does not contain a dual-port RAM. If the program has been imported from an older non-NT version and this error appears, delete any DPR command (such as COPYFROMDPR or COPYTODPR).

Error Code	Error Message	Remarks
3151	This is a DEMO version of Simulator. 5 minutes of work remains.	The demo version of the Simulator has a limited session time. The simulator is going to stop after five minutes.
3152	The DEMO version of Simulator has terminated	The demo version of the Simulator has a limited session time. The session time has elapsed.
3153	Illegal query command	
3154	The command can be only used with MSEG motion	Only PROJECTION , LINE , ARC1 , ARC2 , and STOPPER commands apply to MSEG...ENDS .
3155	Motion cannot start because the previous motion failed. Use FCLEAR command.	In the Strict mode (S_FLAGS .#FCLEAR = 1) the motion cannot start after a fault has occurred. Use the command FCLEAR to clear the result of the fault.
3156	Profile key must be specified as /SECTION/KEY/	
3157	Illegal configuration string. Use only characters K,D,+,-.	
3158	Cannot find matching value. The formula has no root or the root is not single.	
3159	Axis number is specified more than once	
3160	The axis cannot be used in a group (see AFLAGS)	
3161	Illegal communication channel	
3162	Illegal tag value	
3163	Illegal configuration parameters	
3164	Illegal password	

Error Code	Error Message	Remarks
3165	Attempt to execute optional function which is not installed	
3166	Flash file operation failed (overlapped file operations?)	
3167	Wrong start position is specified	
3168	File not opened	
3169	D-Buffer cannot be changed while any other buffer is compiled	
3171	The number of axes in coordinated motion is restricted to 4	
3172	Illegal category name	
3173	EtherCAT offset is out of range	
3174	Can't find EtherCAT network variable	
3175	EtherCAT Master is not ready	
3176	Data size should be specified	
3177	This slave has no mailbox support	
3178	Invalid CoE SDO parameter	
3179	EtherCAT slave is in invalid state for this operation	
3180	General EtherCAT error	
3181	EtherCAT Timeout error	
3182	Can't split non-existing axes group	
3183	Attempt to split group with active motion on	
3184	Mapped variable must be defined in D-Buffer	

Error Code	Error Message	Remarks
3185	Number of connected EtherCAT devices is not covered by SW options	
3186	Can't initialize SP injection	
3187	There is an active injection on this SP	
3188	Can't enable motor, DIP switch settings are incorrect.	
3189	The command can only be used with XSEG motion	
3190	PLC option on this device is not enabled	
3191	PLC is already running	
3192	General PLC error	
3193	Wrong array size or type	
3194	Not allowed Encoder Type	
3195	Requested more absolute encoder multipliers than installed	
3196	Requested absolute encoder is not supported	
3197	The slave is incompatible with current controller configuration	
3198	Invalid encoder type	
3199	Invalid absolute encoder parameter type	
3200	JIT and Dynamic modes are not allowed for D-buffer	JIT and Dynamic modes are not allowed for D-buffer
3201	End-of-Sequence is illegal for this motion	The ends command cannot be specified for this type of motion.
3202	JIT and Dynamic modes are not allowed at the same time	

Error Code	Error Message	Remarks
3203	DEBUG mode is not allowed for JIT and Dynamic buffers	
3204	JIT and Dynamic modes require the buffer to be empty	
3205	Illegal zone number	
3206	Cannot change a constant variable	
3207	One or more arguments are out of range	
3208	Function ended without return statement	
3209	Entered a function not by calling, STOP statement might be missing	
3210	Incompatible matrix size	
3211	Too many segments	
3212	ARC arguments are inconsistent	ARC1 or ARC2 specify inconsistent arguments. The desired arc cannot be calculated.
3213	Stopper is prohibited for master-slave motion	STOPPER cannot be used with segmented MASTER - SLAVE motion.
3214	Adjacent stoppers are prohibited	Two adjacent STOPPER commands are not allowed.
3215	In cyclic path the first and the last points must coincide	In a cyclic segmented motion, the first point of the first segment must coincide with the last point of the last segment.
3216	Velocity is specified, but the motion command had no V command option	Velocity argument can be specified in POINT , LINE , ARC1 or ARC2 only if the corresponding motion command specifies /v .

Error Code	Error Message	Remarks
3217	Segment of zero length	Segments of zero length are not allowed.
3218	ARC radius is too small	
3219	Specified motion delay is out of range	
3220	Command is incompatible with connect	
3221	20 KHz motion generation incompatible with CTIME or number of axes	
3222	20 KHz motion limitation reached	
3223	Time specified is not possible under current motion safety limitations	
3224	Offset doesn't exist	
3225	Microblaze monitor is not ready	
3226	Address couldn't be found	
3227	Maximum number of Registers was already added	
3228	Firmware is not properly installed	
3229	SLABITS does not match Single Turn and Multi Turn number of bits	
3230	Non Linear Control Feature is not supported	
3231	Illegal key	The Key argument of the GETCONF or SETCONF command specifies a value that is not supported in this controller.
3232	Illegal index	The Index argument of the GETCONF or SETCONF command specifies a value that is not supported in this controller.

Error Code	Error Message	Remarks
3233	Illegal value	The Value argument of the SETCONF command specifies a value that is not compatible with the Key argument.
3234	Value in SETCONF must be 0 or 1	For the specified Key only 0 or 1 are legal in the Value argument. of SETCONF
3235	SETCONF function is disabled	The specified SETCONF function is disabled in the current controller mode.
3236	SETCONF cannot be executed while the motor is enabled	The specified SETCONF function cannot be executed while the motor is enabled.
3238	The operation can be executed only when the HSSI channel is in command mode	
3239	HSSI channel cannot switch to command mode because it affects one or more remote drivers. Reset corresponding MFLAGS.#HSSI bits before.	
3240	Operation is temporarily disabled	
3241	Operation failed	
3242	The feature is not supported	
3243	Max number of 20 KHz moving axes reached	
3244	Illegal number of coordinates specified	
3245	The specified channel is already being used. Please use the MBCLOSE function to close the connection	
3251	Operation is temporarily disabled	
3252	Motor cannot be enabled while a motion is in termination process	

Error Code	Error Message	Remarks
3253	Unable to work with Dummy motor	The command cannot be executed for a dummy motor. The addressed motor is configured as a dummy, (see the #DUMMY bit of MFLAGS).
3254	The operation requires the motor to be enabled	The command can be executed only while the motor is enabled.
3255	The operation requires the motor to be disabled	The command can be executed only while the motor is disabled.
3256	The operation is valid only for brushless motor without Hall sensor	The command cannot be executed for any motor type other than brushless, and only for brushless motors that do not have a Hall sensor.
3257	The operation failed because the brushless motor is not commutated	The command cannot be executed because the defined brushless motor is not commutated. The #BRUSHOK bit of MFLAGS reflects the state of the brushless commutation. To commutate the motor, execute COMMUT or a specific initialization program.
3258	The operation failed because the motor is in open-loop mode	The command cannot be executed because the motor is in open-loop state. The open-loop mode may be set using the #OPEN bit of MFLAGS.

Error Code	Error Message	Remarks
3259	Motion cannot start because the motor is defined as dummy	The motion command cannot be executed because one or more of the motors involved are dummy. Motor dummy state is defined by the #DUMMY bit of MFLAGS.
3260	Motion cannot start because the motor is disabled	The motion command cannot be executed because one or more of the motors involved are disabled.
3261	Motion cannot start because the brushless motor is not commutated	The command cannot be executed because the defined brushless motor is not commutated. The #BRUSHOK bit of MFLAGS reflects the state of the brushless commutation. To commutate the motor, execute COMMUT or a specific initialization program.
3262	Motion cannot start because the motor is in open-loop mode	The motion command cannot be executed because one or more of the motors involved are in open-loop state. Open-loop state is defined by bit MST.#OPEN. The open-loop mode may be set using the #OPEN bit of MFLAGS.

Error Code	Error Message	Remarks
3263	Motion cannot start because the previous motion failed. Use FCLEAR command.	The motion command cannot execute because the previous motion for one or more motors failed and the controller is set to the Strict mode. The Strict mode is defined by bit <code>S_FLAGS.#FCLEAR</code>. Use <code>FCLEAR</code> to clear the result of the fault.
3265	SP program does not support this operation	The command is not supported by the current version of the controller. Specifically, the SP program does not support the operation. Check if the SP program has been changed.
3266	Invalid PEG pulse width	
3267	Maximal number of time-based PEG pulses is 65,535	
3268	Invalid period of time-based PEG pulses	
3269	PEG pulse width must be less than time-based pulse period	
3270	PEG is not supported for the specified axis	
3271	Step does not agree with start/end points	
3272	Incremental PEG step must correspond to encoder counts within the $(-2^{31}, 2^{31}-1)$ range, zero is excluded	
3273	The specified axis does not exist in this controller model	
3274	The specified axis is defined as dummy only	
3275	Only stepper motor is supported for the specified axis	

Error Code	Error Message	Remarks
3276	The brushless motor is not supported for the specified axis	
3277	Dual loop control is not supported for the specified axis	
3278	Remote HSSI driver is not supported for the specified axis	
3279	PEG states are not supported for the specified axis	
3280	The stepper motor is not supported for the specified axis	
3281	No Hall support for 2-Phase motors	
3282	Value out of range	
3283	Number of points for Random PEG exceeds the limit	
3284	Axis is not connected to PEG engine	
3285	NanoMotion Piezoceramic Motor is not supported for the specified axis	
3286	Fast Sin-Cos Encoder is not supported for the specified axis	
3287	Laser Modes are not supported for the specified axis	
3288	Time-based PEG is not supported for the specified axis	
3289	This function is not supported for the specified axis	
3290	SPINJECT/SPRT and Fast GPRT cannot be used in parallel	
3291	P/D interface is required for this function	
3292	Secondary Feedback is not supported	

Error Code	Error Message	Remarks
3293	SLABITS does not match Single Turn and Multi Turn number of bits	
3294	Motion cannot start because the Gantry Complementary brushless motor is not commutated	
3295	Commut is not allowed when Compensation is in process	
3300	Non Linear Control License is required	
3301	Another EtherCAT port is already closed	
3302	Not possible to save network topology configuration when more than one line break exists	
3303	ERROR SDO: Object does not exist in the object dictionary.	
3304	The value is too high (check E_PAR_D value)	
3305	The network variable is already mapped to ACSPL+ variable	
3306	Invalid axis list	
3307	FoE Protocol is not supported by Slave	
3308	The Disk is full or the file is too big	
3309	Function supported only by CiA402 Drive	
3310	CiA402 Drive is not in OP State. PDO is not enabled.	
3311	Cannot convert REAL type to INT	
3312	More than one hold command is not allowed	
3313	Continue command is already in process	

Error Code	Error Message	Remarks
3314	Requested Homing Method is not supported	
3316	Homing in Gantry mode requires additional parameters	
3317	FoE Error: Access denied	
3318	Matrix is not invertible	
3319	PI AME interface is not initialized	
3320	PI AME interface is busy	
3321	Axis is not commutated	
3322	Homing is required	
3323	DCOM is not 0	
3324	Parameters are not consistent	
3325	Main FPGA file downloading is in progress	
3326	Operation aborted, PEG is in process	
3327	FPGA versions do not match	
3328	FPGA version is the same. FPGA Upgrade aborted.	
3329	Invalid value, digital input index should range between 0-99 and bit index should range between 0-31	
3330	Invalid value, digital output index should range between 0-99 and bit index should range between 0-31	
3331	This output is already mapped as a mechanical brake to a different axis	
3332	Hardware limit swapping (MFLAGSX.#HLIMSWAP) and limit routing are mutually exclusive	

Error Code	Error Message	Remarks
3333	File name MAX length is 100 chars	
3334	Operation failed because Learning Boost is active	
3335	Illegal SPATH specification	
3336	Nurbs motion was not initiated	
3337	Illegal NURBS/SPATH parameter value	
3338	Profile has ended or last knots already added	
3339	Invalid number of axes	
3340	Dummy point must also specify a Knot value	
3341	Illegal dummy point, two points are allowed before coordinates and two after	
3342	The measurement you are trying to activate is already active. (see DPM_Measurement Stop() function for more details)	
3343	An invalid sample type has been specified. Type can either be 0 or 1. (see DPM_Measurement sample type for more details)	
3344	The sampling time specified is invalid, the minimum value is the controller cycle time (?CTIME) and the maximum value is: 100000000msec	
3345	The sample-set size specified is invalid. Minimum size: 1. Maximum size: 512.	
3346	The specified "when_to_measure" flag is invalid. (see DPM_Measurement documentation for more details)	

Error Code	Error Message	Remarks
3347	The measurement you are trying to activate is paused. (see DPM_Measurement Stop()/Resume() functions for more details)	
3348	Cannot change the selected axis while monitoring is ON. (see DPM_Motion_Status SelectAxis() / MonitorOff() functions for more details)	
3349	The specified " monitored_variable " is invalid. (see DPM_Measurement documentation for more details)	
3350	Illegal attempt to copy STRING	
3351	Illegal String operation - value too long	
3352	DCOM value cannot be changed during Gantry Cross Axis Compensation	
3353	Cross Axis Force Ratio parameter is out of range	
3354	Number Of Cross Axes parameter is out of range	
3355	Cross Axis Index parameter is invalid	
3356	Cross Axis Polarity parameter is out of range	
3357	Cross Axis Mass Ratio parameter is out of range, should be [0..1]	
3358	Cross Axis Max Position and Cross Axis Min Position are not consistent	
3359	Yaw Axis Index parameter is invalid	
3360	LCI unit initialization failed	
3361	Unable to start the LCI operation	
3362	Trajectory axes are not defined	

Error Code	Error Message	Remarks
3363	LCI Safety input is off	
3364	LCI laser fault input is on	
3365	LCI supported velocity limit exceeded	
3366	LCI unit not found	
3367	LCI. Unable to allocate channel for specified operation	
3368	LCI function timeout	
3371	Enable of Gantry axes is required for Compensation to start	
3372	Gantry Mode is required	
3373	Compensation for the specified axis is already in process	
3374	The force ratio is too high.Place the cross masses symmetrically around the center of gravity	
3375	The specified IP address is invalid. Please see MBOPEN documentation for more details.	
3376	The specified server ID is invalid (valid values are between [1 - 247]). Please see MBOPEN documentation for more details.	
3377	The specified word-order is invalid. (valid values are 0 for big-endian and 1 for little-endian) Please see MBOPEN documentation for more details.	
3378	Unable to open more than 3 Modbus connections at any given time. Please see MBOPEN documentation for more details.	
3379	The specified communication handle is invalid.	
3380	A connection with the specified IP was not	

Error Code	Error Message	Remarks
	found.	
3381	The specified address is invalid	
3382	The specified request frequency is invalid. the minimum value is 5 milliseconds. Please see the documentation for more details.	
3383	The specified variable length and the specified number of elements do not match. Please see the documentation for more details.	
3384	The specified suffix combination is invalid. Please see the documentation for more details.	
3385	Read-Only variables cannot be written.	
3386	The specified request ID is invalid.	
3387	The specified variable(or part of it) is already being written to by another Modbus request. Use # MBMAPREP for more details.	
3388	The specified server address is already being written to by another Modbus request. Use # MBMAPREP for more details.	
3389	The specified number of elements is greater than the specified variable length. Please see the documentation for more details.	
3390	The maximum number of integer values that can be "mapped" in a single request is 16. Please see the documentation for more details.	
3391	The maximum number of short values that can be "mapped" in a single request is 32. Please see the documentation for more details.	

Error Code	Error Message	Remarks
3392	The maximum number of single-precision floating-point values that can be "mapped" in a single request is 16. Please see the documentation for more details.	
3393	The maximum number of double-precision floating-point values that can be "mapped" in a single request is 8. Please see the documentation for more details.	
3394	The maximum number of coil values that can be "mapped" in a single request is 32. Please see the documentation for more details.	
3395	The maximum number of discrete input values that can be "mapped" in a single request is 32. Please see the documentation for more details.	
3396	Failed to establish a connection with the specified server device.	
3397	The maximum number of mapping requests per server is 32. Please see the documentation for more details.	
3398	A connection with the specified IP is already open and with different configuration parameters. To change the connection parameters, close and open a new connection.	
3399	The specified Modbus channel operates in sequential mode. This mode does not support user-defined request frequency. Each request is sent after a response for the previous one has arrived, but no less than 10 milliseconds from the last time it has been sent. Please see the documentation for more details.	
3410	The minimum value can not be greater than the maximum value. Please see the documentation for more details.	

Error Code	Error Message	Remarks
3411	G-code: Run-time errors were detected during the program execution in simulation mode. Use the #LOGP <buffer number> command to present the detected errors.	
3412	This motion type is not supported	
3413	The supplied correction maps (2-dimensional arrays) should have the same dimensions. Please see the documentation for more details.	
3414	The Specified referenced axes/analog inputs are invalid. Please see the documentation for more details.	
3415	Internal error	
3416	Stage model dll is not loaded	
3417	Stage model dll's name does not match the stage model implemented by the dll	
3418	Invalid number of virtual axes	
3419	Invalid number of physical axes	
3420	Virtual axis index is out of range	
3421	Physical axis index is out of range	
3422	Virtual and machine axes conflict	
3423	Tool length index is out of range	
3424	Fixture offset index is out of range	
3425	Sync positions array must be set when feedback transform is not supported	
3426	Custom virtual axes must be set in this FW version	
3427	Custom virtual axes must be set for 128 axes systems	

Error Code	Error Message	Remarks
3428	Not implemented feature	

7.4 Errors

Motion Termination error code values range between 5000 and 5150.



Codes from 5000 to 5008, however, do not indicate an error. They report normal motion termination.

Codes from 5009 and higher appear when a motion is terminated or a motor is disabled due to a fault detected by the controller.

When a motion terminates abnormally, or a motor is disabled, the error code is stored in the [MERR](#) variable. If there is an initialization fault during startup, the error code is stored in the [S_ERR](#) variable.

Table 7-4. ACSPL+ Motion Termination Errors

Error Code	Error Message	Remarks
5000	Motion has not finished	Motion was terminated before reaching final point.
5001	Motion generation finished	The motion came to the final point and was successfully completed.
5002	Motion was killed by user	
5003	Motion was terminated by user	The motion was terminated by HALT before the final point was reached
5004	Motor was disabled by user	The motion was disabled by DISABLE/DISABLEALL .
5005	Motion was terminated because a motor was disabled	The motion was disabled by DISABLE/DISABLEALL .
5006	Motion was killed	The motion was terminated by KILL/KILLALL before the final point was reached
5007	Motor was disabled due to another motor becoming disabled	If a fault occurs in an axis disabling the motor thereby disabling other motors, code 5007 is stored in the MERR variable for all affected axes.

Error Code	Error Message	Remarks
5008	Motion was killed due to another motion being killed	If a fault occurs in an axis killing the motion of a motor thereby killing the motion other motors, code 5008 is stored in the MERR variable for all affected axes.
5009	Common motion failed	
5010	Hardware Right Limit	The motion was killed because of a right limit fault.
5011	Hardware Left Limit	The motion was killed because of a left limit fault.
5012	Network Error	Check EtherCAT Cables for continuity, check ECERR variable, make sure no loose connections, Run System Viewer and Diagnostics and check Error Registers for possible CRC errors on specific network nodes to identify failure point.
5014	Motor Overheat	The motor was disabled or the motion failed because of an overheat fault.
5015	Software Right Limit	The motion is killed because of a software right limit fault.
5016	Software Left Limit	The motion is killed because of a software left limit fault..
5017	Encoder 1 Not Connected	The motor was disabled because of an encoder not connected fault. Note how error is caught in hardware. Difference between A,A- or B,B- differential signal. If A=A- Error is thrown. Check continuity of cable pinouts.
5018	Encoder 2 Not Connected	The motor was disabled because of an encoder 2 not connected fault. Note how error is caught in hardware. Difference between A,A- or B,B- differential signal. If A=A- Error is thrown. Check continuity of cable pinouts.
5019	Drive Fault	The motor was disabled because of a drive alarm fault.
5020	Encoder 1 Error	The motor was disabled or the motion failed because of an encoder error fault. For Quadrature and SinCos encoders the signal monitors the logical

Error Code	Error Message	Remarks
		state transition between A,B signals. For contiuous motion the transition should be {[1,0],[1,1],[0,1],[0,0]}. If the transition ever skips states it indicates count loss. For Absolute encoders check configuration parameters. Problem could stem from signal contiuity or induced noise. Ensure head alignment and tape is clean.
5021	Encoder 2 Error	The motor was disabled or the motion failed because of an encoder 2 error fault. For Quadrature and SinCos encoders the signal monitors the logical state transition between A,B signals. For contiuous motion the transition should be {[1,0],[1,1],[0,1],[0,0]}. If the transition ever skips states it indicates count loss. For Absolute encoders check configuration parameters. Problem could stem from signal contiuity or induced noise. Ensure head alignment and tape is clean.
5022	Non-Critical Pos. Error	The motor was disabled or the motion failed because of a position error fault.
5023	Critical Position Error	The motor was disabled or the motion failed because of a critical position error fault.
5024	Velocity Limit	The motor was disabled or the motion failed because of a velocity limit fault. See ACSPL+ variable XVEL(axis)
5025	Acceleration Limit	The motor was disabled or the motion failed because of an acceleration limit fault. See ACSPL+ variable XACC(axis)
5026	Drive/Motor Overcurrent	The motor was disabled or the motion failed because of an overcurrent fault. Ensure stability of system. User can monitor servo processor variables: SP#(dsp#) Current Command axis axis _ number, or SP#(dsp#) Phase q current axis_number in the scope to establish when current ramps up. Error is triggered exceeding XCURI, XCURV, XRMSM, XRMSD values.

Error Code	Error Message	Remarks
5027	Servo Processor Alarm	The motor was disabled or the motion failed because of a servo processor alarm fault.  Activated when an axes does not have a physical drive associated to it.
5028	Safe Torque Off	The motor was disabled because STO not connected.
5030	HSSI Not Connected	
5031	Broken Wire	
5032	Attempt of motion while a fault is active	
5033	Attempt of motion in disabled direction	The motion is killed because of an attempt to move to left direction when the Left Limit or Left Software Limit fault is On, or move to right direction when the Right Limit or Right Software Limit fault is On. User can verify proper encoder count direction in "Verification" page in the Adjuster Wizard.
5034	MPU Overheat	See GETCONF (76,index) in Command and Variable Reference Guide
5035	Program Error	The motor was disabled or the motion failed because of a program error fault.
5036	Memory Overflow	The motor was disabled or the motion failed because of a memory overuse fault.
5037	MPU Overuse	The motor was disabled or the motion failed because of a time overuse fault. For case of XSEG use user should ensure each segment does not execute in less than minimum time given in the SpiiPlus Release notes section: "Controller Cycle Time (CTIME) Support" for the given controller. User can also monitor Usage related variables given in the SpiiPlus Command and Variable reference guide.
5038	Hardware Emergency Stop	The motor was disabled because of an emergency stop fault.

Error Code	Error Message	Remarks
5039	Servo Interrupt	The motor was disabled or the motion failed because of a servo interrupt fault.
5041	Component Failure	The motor was disabled or the motion failed because of power supply failure or not supplied bus voltage
5042	Motor was disabled due to hall error	Motor was disabled due to hall error. User can monitor ACSPL variable SLSTHALL or use the hall monitor in "Verification" portion of the Adjuster Wizard to ensure all hall states are read. Check continuity of cable.
5043	Motion was killed due to external profile error	
5044	External Network Error	External network error
5045	Motor was disabled due to SYNC loss	Motor was disabled due to SYNC loss
5046	Motion was disabled due to GPRT overflow	Motor was disabled due to GPRT overflow
5047	The axis remains in HOLD state.	
5048	Motor Overcurrent	Ensure stability of system. User can monitor servo processor variables SP# (dsp#) Current Command axis axis_number, or SP# (dsp#) Phase q current axis_number in the scope to establish when current ramps up. Error is triggered exceeding XCURI , XCURV , XRMSM , XRMSD values.
5049	Drive Overcurrent	User can monitor servo processor variables: SP# (dsp#) Current Command axis axis_number, or SP# (dsp#) Phase q current axis_number in the scope to establish when current ramps up. Error is triggered exceeding XRMSD value.
5050	Safe Torque Off	STO Error
5051	Safe Torque Off: STO_1 24V	STO1 24V source disconnected

Error Code	Error Message	Remarks
5052	Safe Torque Off: STO_2 24V	STO2 24V source disconnected
5053	Safe Torque Off: STO_1 24V and STO_2 24V	STO1 and STO2 24V sources disconnected
5054	Safe Torque Off: SS1_1 5V	SS1-1 5V source disconnected
5055	Safe Torque Off: SS1_2 5V	SS1-2 5V source disconnected
5056	Safe Torque Off: SS1_1 5V and SS1_2 5V	SS1-1 and SS1-2 5V sources disconnected
5057	Safe Torque Off: SS1 Timing Error	See Timing Error section in STO Application Note for more details.
5060	Driver Alarm	Drive alarm: No fault
5061	Driver Alarm: Short circuit	The motor was disabled or the motion failed because of a short circuit condition in the drive. Measure resistance of motor phases to ensure no physical short in motor or cable. Error is triggered by rapid increase in SP#(dsp#) Phase q current axis_axis_number
5062	Driver Alarm: External protection activated	The motor was disabled or the motion failed because of an external protection condition in the drive.
5063	Driver Alarm: Power supply too low	The motor was disabled or the motion failed because of a power supply too low condition in the drive.
5064	Driver Alarm: Power supply too high	The motor was disabled or the motion failed because of a power supply too high condition in the drive.
5065	Driver Alarm: Temperature too high	The motor was disabled or the motion failed because of a power supply too high condition in the drive.
5066	Driver Alarm: Power supply 24VF1	The motor was disabled or the motion failed because of a power supply 24VF1 condition in the drive.

Error Code	Error Message	Remarks
5067	Driver Alarm: Power supply 24VF2	The motor was disabled or the motion failed because of a power supply 24VF2 condition in the drive.
5068	Driver Alarm: Emergency stop	The motor was disabled or the motion failed because of an emergency stop condition in the drive.
5069	Driver Alarm: Power Down	The motor was disabled or the motion failed because of a power-down condition in the drive.
5070	Driver Alarm: Phase Loss	
5071	Driver Alarm: Drive not ready	
5072	Driver Alarm: Overcurrent	
5073	Driver Alarm: Reserved	
5074	Driver Alarm: Dumper fault	
5075	Driver Alarm: Digital Drive Interface not connected	The motor was disabled or the motion failed because of a Digital Drive Interface Not Connected condition in the drive. Check the cable connections to the drive.
5076	Driver Alarm: Drive Saturation	
5077	Driver Alarm: Regeneration module fault	
5080	Component Failure	
5081	Component Failure: Power supply 0 fault	
5082	Component Failure: Power supply 1 fault	

Error Code	Error Message	Remarks
5083	Component Failure: Regeneration module fault	
5084	Component Failure: Power supply too high	The motor was disabled due to excessive high voltage of the Power Supply. To correct this, either reduce the motor speed. Or, in the case of PSM3U 8 Kw or 10 Kw being installed, connect an External Regeneration (see <i>MC4U Hardware Guide</i>).
5085	Component Failure: Temperature too high	The motor was disabled, the Power Supply temperature is too high. Check that the unit cooling fans are working, if not, replace the cooling fans. Make sure that the air flow around the unit is clear of any obstructions.
5086	Component Failure: Unknown error	
5087	Component Failure: Unknown error	
5088	Component Failure: Unknown error	
5089	Component Failure: Power down	Indicates missing 3 phase 230Vac servo supply voltage. Check that the AC input voltage is connected correctly.
5090	Component Failure: Drive supply phase lost	This applies only to 3-phase AC input power. In the event that one of the AC input supply phases is lost or one of the AC input fuses is blown all axis drivers which are supplied by this power supply are disabled. Make sure that the JP6 jumper is installed in the PSM3U-320V-XXkW board if a three- phase input supply is used.
5091	Component Failure: Power supply not ready. Try to enable axis again within 10 seconds	This is caused by the inrush power protection detector. It temporarily suspends the power input. Wait for a few seconds for the power to normalize before starting.

Error Code	Error Message	Remarks
5092	Component Failure: Unknown error	
5093	Component Failure: Unknown error	
5094	Component Failure: Power supply damper not OK	There is a failure in the damper circuit in the Power Supply. There may be a short in the External Regeneration, or Internal Regeneration.
5095	Component Failure: Unknown error	
5100	Current bias measured is out of range	At the beginning of the enable process the controller measures the current and calculates an average bias (offset). If the result is greater than the predefined maximum value (2% of maximum output), the controller generates the error and axis remains disabled. 1. The first thing to check is if the drive has STO. If 24V is not connected to STO inputs, the drive will always measure offsets of 100% (in case of our high voltage units). This is because the bootstrap mechanism needed to charge the current loop measurement circuits doesn't work (it requires transistors to be ON). This is the most common case of such fault. 2. If that's not the case - Check if motor was moving during the enable process Try to disconnect the motor and see if the results are different 3. Check the value of SLBIASA, SLBIASB. If non of the above, and it's more than 5% it usually indicates a problem. Usually it's much less. If you see very big values (>50%) and surely indicates a problem with the drive. If offsets are smaller than 10%, there is still a way to increase the threshold of the protection. SETCONF(207, axis, threshold in %) will allow you to increase the threshold of the "current bias out of range protection". Not recommended to use if offsets are too big, 4. Don't disable the automatic measurement in this

Error Code	Error Message	Remarks
		case. A very big offset indicates that something is wrong with the drive and better not try to work like that.
5101	Autocommutation failure (phase error)	Error occurs if COMMUT fails. The error indicates wrong phase sequencing, or incorrect commutation parameters. To avoid this error it is recommended that the initial commutation be performed from the Adjuster. The Adjuster commutation program verifies the phase sequence and commutation parameters. This must be done before COMMUT is executed.
5102	Autocommutation failure (position error)	Error occurs if COMMUT fails. The error indicates bad or marginal servo tuning, or incorrect commutation parameters. To prevent this error make sure that the axis is properly tuned - use the Adjuster Wizard or FRF Analyzer .
5103	ENABLE is prohibited while Constant Current is ON	
5104	ENABLE failed because the motor is moving	ENABLE cannot be executed whenever the motor is in motion (see ENABLE/ENABLE ALL).
5105	E_TYPE does not match HSSI-HES DIP switch settings	Check the HSSI-HES DIP switch settings.
5106	EtherCAT node failure	
5107	The axis uses a licensed servo feature that is not allowed	
5108	Number of axes that use licensed servo feature exceeds the number allowed	
5109	Current bias measurements process was not completed	

Error Code	Error Message	Remarks
5110	Command failed due to transaction timeout	
5111	ENABLE failed because the system need to be reconfigured	
5121	Encoder Error: CRC Error	
5122	Encoder Error: Busy	
5123	Encoder Error: Encoder Not Ready	
5124	Encoder Error: Timeout	
5125	Encoder Error: FPGA File not found	
5126	Encoder Error: Error Flag	
5127	Encoder Error: An internal incompatibility was detected. Full system upgrade is required.	
5128	Encoder Error: Watchdog	
5130	Safety Zone Fault	

7.5 Encoder Errors

Table 7-5. Encoder Errors

Error Code	Error Message	Remarks
5121	Encoder Error: CRC Error	Result of CRC check of the received value
5122	Encoder Error: Busy	Indicates that encoder is busy - specifics depend on encoder brand

Error Code	Error Message	Remarks
5123	Encoder Error: Encoder Not Ready	Endat: The status register is not completely updated. Not All checks have been performed. Data transmission is not yet completed.
5124	Encoder Error: Timeout	
5125	Encoder Error: FPGA File not found	FPGA file is missing, therefore, the Absolute Encoder for the required axis cannot be operated.
5126	Encoder Error: Error Flag	Error
5127	Encoder Error: An internal incompatibility was detected. Full system upgrade is required.	
5128	Encoder error: Watchdog	

7.6 System Errors

System Errors error code values range between 5151 and 5999. They are generally caused by problems with the firmware of the servo processor (SP) program.

Table 7-6. ACSPL+ System Errors

Error Code	Error Message
5151	Integrity of the firmware or user application is broken
5152	Initialization problem: Main Interrupt is not detected
5154	Initialization problem: Main Interrupt period is wrong
5156	Initialization problem: Main Interrupt Event cannot be created
5158	Initialization problem: SP initialization failed
5160	Initialization problem: SP program activation failed
5162	Initialization problem: SP and MP are not synchronous
5163	Initialization problem: EthernetIP initialization failed.
5164	Initialization problem: Improper controller card is detected
5165	Initialization problem: Improper configuration file is detected
5166	Initialization problem: SP is not detected by hardware
5167	Initialization problem: Number of drive modules does not meet the configuration
5168	Initialization problem: Drive Interface is not connected
5169	Initialization problem: One or more parameters are out of range for the current controller configuration (See #LOG for details). Default values are assigned
5170	Initialization problem: One or more EEPROM parameters are different from the current controller configuration. Values from EEPROM are assigned
5171	Initialization problem: One axis attached to two drives in configuration file
5172	Initialization problem: Nominal current parameter in configuration file is different from the current controller configuration. Value from EEPROM is assigned

Error Code	Error Message
5173	Initialization problem: Peak current parameter in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5174	Initialization problem: Power parameter in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5175	Initialization problem: Voltage (min or max) parameter in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5176	Initialization problem: Type parameter in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5177	Initialization problem: Number of subsystems parameter in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5178	Initialization problem: Component was detected by I2C but was not found in configuration file
5179	Initialization problem: Slave address parameter in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5180	Initialization problem: Drive number parameter in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5181	Initialization problem: RMS protection time constant parameter in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5183	Initialization problem: Number of axes parameter in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5184	Initialization problem: One of digital inputs parameters in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5185	Initialization problem: One of digital outputs parameters in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5186	Initialization problem: One of analog inputs parameters in configuration file is different from the current controller configuration. Value from EEPROM is assigned

Error Code	Error Message
5187	Initialization problem: One of analog outputs parameters in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5188	Initialization problem: Number of HSSI channels parameter in configuration file is different from the current controller configuration. Value from EEPROM is assigned
5189	Initialization problem: Unit ID is wrong. The unit definition is ignored
5190	Initialization problem: Number of nodes does not meet the configuration
5191	Initialization problem: Number of axes does not meet controller options
5192	Initialization problem: Incorrect DIP switch settings
5193	Initialization problem: Firmware is not properly installed
5194	Initialization problem: One or more EtherCAT nodes are incompatible with current controller configuration
5195	Initialization problem: One or more EtherCAT nodes are incompatible with current controller cycle time (CTIME)
5196	Initialization problem: One or more EtherCAT nodes are incompatible with Ring Topology
5197	Initialization problem: Cycle time of one or more EtherCAT nodes is different from the controller cycle time (CTIME). System should be reconfigured
5198	Initialization problem: One or more EtherCAT nodes are incompatible with ENI. System should be reconfigured
5199	Initialization problem: One or more Gantry pairs were changed. Controller reboot is required
5201	Initialization problem: Wrong range of input parameters. Servo Processor program activation failed
5202	Initialization problem: File not found. Servo Processor program activation failed
5203	Initialization problem: Read file error. Servo Processor program activation failed
5204	Initialization problem: Memory allocation error. Servo Processor program activation failed

Error Code	Error Message
5205	Initialization problem: Checksum error. Servo Processor program activation failed
5206	Initialization problem: EtherCAT error. Servo Processor program activation failed
5207	Initialization problem: EtherCAT timeout. Servo Processor program activation failed
5208	Initialization problem: Servo Processor program is incompatible with one or more products

7.7 EtherCAT Errors

EtherCAT errors range from 6000 to 6999 and are latched in the [ECERR](#) variable.

Table 7-7. ACSPL+ EtherCAT Errors

Error Code	Error Message	Remarks
6000	General EtherCAT Error. Check # LOG output for more details.	General EtherCAT Error. Check #LOG output for more details.
6001	EtherCAT cable not connected	Check that the EtherCAT connections are firmly seated.
6002	EtherCAT master is in incorrect state	On start up all slaves did not succeed to initialize to full OP state. Can be caused by wrong configuration or a problem in a Slave
6003	Not all EtherCAT frames can be processed	The Master has detected that at least one frame that was sent has not returned. This implies a hardware problem in the cables or Slaves.
6004	EtherCAT Slave Error	Slave did not behave according to EtherCAT state machine – internal Slave failure.
6005	EtherCAT initialization failure	The EtherCAT-related hardware in the Master could not be initialized. Check the EtherCAT hardware.

Error Code	Error Message	Remarks
6006	EtherCAT cannot complete the operation	The bus scan could not be completed. This implies hardware level problems in the EtherCAT network.
6007	EtherCAT work count error	Every Slave increments the working counter in the telegram. If this error is triggered, it means that a Slave has failed. Possible root cause: cable interruption, Slave reset, Slave hardware failure,
6008	Not all EtherCAT slaves are in OP state	One or more of the Slaves has changed its state to other than OP, or it may be due to a Slave restart or internal fault that internally forces the Slave to go to PREOP or SAFEOP.
6009	EtherCAT protocol timeout	The Master has detected that the Slave does not behave as expected for too long, and reports timeout. Implies a Slave hardware problem. Try power down, and system restart.
6010	Slave initialization failed	The Master cannot initialize a Slave by the configuration file. It can be caused by either wrong configuration, or a hardware problem in the Slave.
6011	Bus configuration mismatch	The bus topology differs from that in configuration file.
6012	CoE emergency	A Slave with CoE support has reported an emergency message.
6013	EtherCAT Master won't enter INIT state	Hardware fault, for example DHD with broken (logic) supply.
6014	EtherCAT ring topology requires network reconfiguration	System should be reconfigured to use the NetworkBoost feature.
6015	One or more EtherCAT cables are not connected	One or more EtherCAT cables are not connected; can happen only when using the NetworkBoost feature.

Error Code	Error Message	Remarks
6016	The actual network configuration doesn't match the last approved configuration.	
6017	Change in configuration was detected. Optional Group was added. It should be approved.	
6018	EtherCAT Master won't enter PREOP state	EtherCAT Master is in INIT state and won't enter the PREOP state.
6019	EtherCAT Master won't enter SAFEOP state	EtherCAT Master is in PREOP state and won't enter the SAFEOP state.
6020	EtherCAT Master won't enter OP state	EtherCAT Master is in SAFEOP state and won't enter the OP state.
6021	Group ID mismatch between configuration file and actual configuration	
6022	Change in configuration was detected. Optional Group was removed. It should be approved.	
6023	EtherCAT cable(s) are crossed. Check proper cables connection from OUT port to IN port.	
6024	One or more EtherCAT slaves causes frames to be lost. Check proper cables connection from OUT port to IN port.	

7.8 EtherCAT Slave Errors

EtherCAT Slave errors range from 7000 to 7999 and are latched in the [ECALERR](#) variable.

The error codes are defined according to AL Status Code (ETG 1020).

Table 7-8. EtherCAT Slave Errors

ACS Error Code	Error Message	Remarks
7001	General Error	General Error which is not defined in the following list
7002	Mailbox No Memory	Local Application runs out of memory (e.g. dynamic memory allocation for emergency messages)
7003	Invalid device setup	
7004	Unknown error	If the slave checks if the SII/EEPROM content matches the firmware, e.g. process data description or revision number, and detects a mismatch
7006	Firmware and EEPROM do not match. Slave needs BOOT-INIT transition	
7007	Firmware update not successful. Old firmware still running	Error occurred during firmware update
7014	License Error (HW/SW license invalid or evaluation period expired)	HW/SW license invalid or evaluation period expired
7017	Invalid requested state change	Requested state change is invalid
7018	Unknown requested state	Requested state change is unknown
7019	Bootstrap state is not supported by the slave	
7020	No valid firmware	
7021	Invalid Mailbox Configuration in Boot	The mailbox SyncManager configuration is not valid in Bootstrap state
7022	Invalid Mailbox Configuration in PREOP	The mailbox SyncManager configuration is not valid in PREOP state

ACS Error Code	Error Message	Remarks
7023	Invalid Sync Manager Configuration	Invalid sync manager configuration. Possible reason can be invalid PDO size configuration
7024	No valid inputs available	Slave application cannot provide valid inputs
7025	No valid outputs available	Slave application cannot provide valid outputs
7026	Synchronization error	Multiple synchronization errors. Device is not synchronized any more
7027	Sync Manager Watchdog	No process data received yet or not received within a specified timeout value
7028	Invalid Sync Manager Types	
7029	Invalid Sync Manager Output Configuration	SyncManager configuration for output process data is invalid (check ENI)
7030	Invalid Sync Manager Input Configuration	SyncManager configuration for input process data is invalid (check ENI)
7031	Invalid Watchdog	Watchdog Settings are invalid (e.g. SyncManger watchdog trigger is enabled but no watchdog timeout is defined)
7032	Slave needs cold start	Slave device requires a power off – power on reset
7033	Slave needs INIT	Slave application requests INIT state
7034	Slave needs PREOP	Slave application requests PREOP state
7035	Slave needs SAFEOP	Slave application requests SAFEOP state
7036	Invalid Input Mapping	Input process data mapping do not match to expected mapping
7037	Invalid Output Mapping	Output process data mapping do not match to expected mapping
7038	Inconsistent Settings	General settings mismatch

ACS Error Code	Error Message	Remarks
7039	Free Run Mode is Not Supported	Slave doesn't support Free Run Mode
7040	SyncMode Not Supported	Slave doesn't support SYNC mode
7041	Free Run Needs 3-Buffer Mode	FreeRun Mode, sync manager has to run in 3Buffer Mode
7042	Background watchdog	
7043	No Valid Inputs and Outputs	
7044	Fatal Sync Error	Fatal Sync Error: Sync0 or Sync1 are not received any more
7045	No Sync Error	Sync not received: In SAFEOP the slave waits for the first Sync0/Sync1 events before switching to OP, if these events were not received during the SAFEOP to OP-Timeout time the slave should refuse the state transition to OP with this AL Status Code (SystemTimeOffset too big, no DC event received)
7046	EtherCAT cycle time smaller than the minimum cycle time supported by the slave	
7048	Invalid DC SYNC Configuration	Distributed Clocks configuration is invalid due to application requirements
7049	Invalid DC Latch Configuration	DC Latch configuration is invalid due to application requirements
7050	DC PLL Sync Error	Master not synchronized, at least one DC event received
7051	Invalid DC IO Error	Multiple synchronization errors. IO is not synchronized any more
7052	Invalid DC Timeout Error	Multiple synchronization errors. Too much SM Events missed

ACS Error Code	Error Message	Remarks
7053	DC Invalid Sync Cycle Time	
7054	DC Sync0 Cycle Time	DC Sync0 Cycle time does not fit to application requirements
7055	DC Sync1 Cycle Time	DC Sync1 Cycle time does not fit to application requirements
7066	Mailbox EoE	
7067	Mailbox CoE	
7068	Mailbox FoE	
7080	EEPROM No Access	EEPROM not assigned to PDI
7081	EEPROM Error	EEPROM access error
7200	FoE Error: Vendor specific FoE error	
7201	FoE Error: Not found	
7202	FoE Error: Access denied	
7203	FoE Error: Disk full	
7204	FoE Error: Illegal access	
7205	FoE Error: Wrong packet number	
7206	FoE Error: Already exists	
7207	FoE Error: No user	
7208	FoE Error: Bootstrap state only	
7209	FoE Error: Not valid in Bootstrap state	

ACS Error Code	Error Message	Remarks
7210	FoE Error: No rights	
7211	FoE Error: Program error	
7212	FoE Error: Wrong checksum	
7213	FoE Error: Firmware is incompatible with Hardware	
7214	FoE Error: No file to read	
7215	FoE Error: File header does not exist	
7216	FoE Error: Flash problem	
7217	FoE Error: File incompatible	
7218	FoE Error: slave does not support FoE in Bootstrap	
7219	FoE Error: File is bigger than max file size	
8001	Function code received in the query is not recognized or allowed by server.	
8002	Data address of some or all the required entities are not allowed or do not exist in server.	

ACS Error Code	Error Message	Remarks
8003	A value contained in the query data field is not an allowable value for the server.	
8004	An unrecoverable error occurred while the server was attempting to perform the requested action.	
8005	Server has accepted request and is processing it, but a long duration of time is required.	
8006	Server is engaged in processing a long-duration command. Client should retry later.	
8012	The received response transaction identifier did not match the expected value.	
8013	The received response length did not match the expected length.	
8014	The received response value is invalid. This may be due to the value being out of the allowed range, an attempt to write to a protected variable, etc.	
8015	No response has been received from the server device for the	

ACS Error Code	Error Message	Remarks
	timeout period(5 seconds). The connection with the server device has been terminated.	
8016	The connection with the server device has been closed.	

7.9 MODBUS Errors

Table 7-9. Modbus Errors

Error Code	Error Message	Remarks
8001	Function code received in the query is not recognized or allowed by server.	
8002	Data address of some or all the required entities are not allowed or do not exist in server.	
8003	A value contained in the query data field is not an allowable value for the server.	
8004	An unrecoverable error occurred while the server was attempting to perform the requested action.	
8005	Server has accepted request and is processing it, but a long duration of time is required.	
8006	Server is engaged in processing a long-duration command. Client should retry later.	
8012	The received response transaction identifier did not match the expected value.	
8013	The received response length did not match the expected length.	
8014	The received response value is invalid. This may be due to the value being out of the allowed range, an attempt to write to a protected variable, etc.	
8015	No response has been received from the server device for	

Error Code	Error Message	Remarks
	the timeout period(5 seconds). The connection with the server device has been terminated.	
8016	The connection with the server device has been closed.	

8. G-Code Error Codes

This chapter contains explanations of the Error Codes that may appear.

- > [G-Code Syntax Errors](#)
- > [G-Code Compilation Errors](#)
- > [G-Code Runtime Errors](#)

8.1 G-Code Syntax Errors

Code	Meaning
1500	G-code: General G-Code runtime error
1501	G-code: Incompatible G commands in one statement
1502	G-code: The value should be positive
1503	G-code: G2/G3 can be specified in the main plane only
1504	G-code: An illegal arc with the current radius compensation parameters
1505	G-code: Segments of radius compensation should be separated by G0 or G1 segment
1506	G-code: Impossible case in radius compensation
1507	G-code: Too many S or R addresses
1508	G-code: Wrong digital output specification in S or R address
1509	G-code: Unsupported G function
1510	G-code: Unsupported M function
1511	G-code: I,J,K, or R specified outside arc definition -or- R specified outside cylindrical interpolation definition(G207)
1512	G-code: I,J,K and R cannot be specified together
1513	G-code: Wrong parameters in arc definition

Code	Meaning
1514	G-code: Internal error
1515	G-code: Wrong index in gParamAddr/gParamValue
1516	G-code: gParamAddr/gParamValue used out of subroutine
1517	G-code: XSEG address specified outside XSEG Motion Parameterization (No G200)
1518	G-code: XSEG address specified outside XSEG Segment Parameterization (No G01, G02 or G03)
1519	G-code: XSEG address specified outside XSEG Motion/Segment Parameterization (No G200/No G01, G02 or G03)
1520	G-code: XSEG addresses - Cannot use (,Y) together with addresses (,J), (,A) & (,D)
1521	G-code: XSEG comma-addresses (,g), (,u) and (,h) cannot be specified together
1522	G-code: Cannot use XSEG motion parameters command G200 with any of G01, G02 or G03
1523	G-code: XSEG addresses - Cannot use M61/M62 with addresses (,01) to (,04)
1524	G-code: XSEG addresses - At least both (,01) and (,02) addresses must be used on the same line
1525	G-code: XSEG addresses - Cannot use (,M) together with (,F) or (F)
1526	G-code: XSEG addresses - Cannot use (,M) together with (,L)
1527	G-code: Incompatible G with custom G208 in one statement

Code	Meaning
1528	G-code: G208 should define at least two primary axes, according to active trajectory plane (G17/G18/G19)
1529	G-code: G208 should define minimum of 2 XSEG axes and maximum of 6 XSEG axes
1530	G-code: Segment processing time (T) option should be specified only with in-plane G01/G02/G03
1531	G-code: Cannot use (T) option - Segment processing time, together with (,F) or (F)
1532	G-code: The value should be above a minimum value (see GSP Reference Guide)
1533	G-code: Incompatible G with custom G207/G205 in one statement
1534	G-code: G207 should define one linear axis followed by one rotational axis, and cylinder radius, in one statement
1535	G-code: G54 should define a WCS number between 1 to 12
1536	G-code: GUFAC is out of range (1e-15, 1e15) or zero
1537	G-code: Radius should be positive
1538	G-code: Blended motion should not specify feedrate and segment time together
1539	G-code: Illegal parameter for Blended motion
1540	G-code: Mandatory parameter for BSEG not specified
1541	G-code: Bseg parameter must be greater than 0
1542	G-code: G84 only allows rotation angle and center parameters to be specified

Code	Meaning
1543	G-code: Wrong M-code comand use
1544	G-code: Unmapped axis name used
1545	G-code: Tool height compensation is not allowed as second segment of Radius Compensation
1546	G-code: Incompatible mode to activate Local Coordinate System (G68)
1547	G-code: Too many G68 parameters
1548	G-code: Incompatible G code when Local Coordinate System is active
1549	G-code: Illegal case of segment synchronization
1550	G-code: Illegal NURBS based motion specification
1551	G-code: An incompatible address with NURBS/Path_Smoothing was specified
1552	G-code: An Illegal number of NURBS/Path_Smoothing main axes was specified
1553	G-code: Path Smoothing motion was not initiated
1554	G-code: Illegal Address for Path Smoothing
1555	G-code: Buffer is in Path Smoothing mode, please switch mode to use G06 command
1556	G-code: The address must contain a value
1557	G-code: The address must not contain a value
1558	G-code: SPath arc specification missing
1559	G-code: G54 and G54.4 are not allowed in the same line

8.2 G-Code Compilation Errors

Code	Meaning
2500	G-code: General G-Code compilation error
2501	G-code: Unknown G-Code command
2502	G-code: Value was expected
2503	G-code: Illegal syntax used
2504	G-code: Illegal value
2505	G-code: Address should not have a value
2506	G-code: Address value should be global or standard variable
2507	G-code: The output range specified is too big
2508	G-code: Wrong digital output specification in S or R address
2509	G-code: Comment lacks right parenthesis
2510	G-code: Illegal G specified
2511	G-code: Illegal M specified
2512	G-code: ']' expected
2513	G-code: '[' is not expected
2514	G-code: Address value should be a global variable
2515	G-code: All comments must start with '!'
2516	G-code: Addresses that are not allowed together were specified
2517	G-code: Illegal Synchronized code specification

8.3 G-Code Runtime Errors

3500	G-code: General G-Code runtime error
3501	G-code: Incompatible G commands in one statement
3502	G-code: The value should be positive

3503	G-code: G2/G3 can be specified in the main plane only
3504	G-code: An illegal arc with the current radius compensation parameters
3505	G-code: Segments of radius compensation should be separated by G0 or G1 segment
3506	G-code: Impossible case in radius compensation
3507	G-code: Too many S or R addresses
3508	G-code: Wrong digital output specification in S or R address
3509	G-code: Unsupported G function
3510	G-code: Unsupported M function
3511	G-code: I,J,K, or R specified outside arc definition -or- R specified outside cylindrical interpolation definition(G207)
3512	G-code: I,J,K and R cannot be specified together
3513	G-code: Wrong parameters in arc definition
3514	G-code: Internal error
3515	G-code: Wrong index in gParamAddr/gParamValue
3516	G-code: gParamAddr/gParamValue used out of subroutine
3517	G-code: XSEG address specified outside XSEG Motion Parameterization (No G200)
3518	G-code: XSEG address specified outside XSEG Segment Parameterization (No G01, G02 or G03)
3519	G-code: XSEG address specified outside XSEG Motion/Segment Parameterization (No G200/No G01, G02 or G03)
3520	G-code: XSEG addresses - Cannot use (,Y) together with addresses (,J), (,A) & (,D)
3521	G-code: XSEG comma-addresses (,g), (,u) and (,h) cannot be specified together
3522	G-code: Cannot use XSEG motion parameters command G200 with any of G01, G02 or G03

3523	G-code: XSEG addresses - Cannot use M61/M62 with addresses (,01) to (,04)
3524	G-code: XSEG addresses - At least both (,01) and (,02) addresses must be used on the same line
3525	G-code: XSEG addresses - Cannot use (,M) together with (,F) or (F)
3526	G-code: XSEG addresses - Cannot use (,M) together with (,L)
3527	G-code: Incompatible G with custom G208 in one statement
3528	G-code: G208 should define at least two primary axes, according to active trajectory plane (G17/G18/G19)
3529	G-code: G208 should define minimum of 2 XSEG axes and maximum of 6 XSEG axes
3530	G-code: Segment processing time (T) option should be specified only with in-plane G01/G02/G03
3531	G-code: Cannot use (T) option - Segment processing time, together with (,F) or (F)
3532	G-code: The value should be above a minimum value (see GSP Reference Guide)
3533	G-code: Incompatible G with custom G207/G205 in one statement
3534	G-code: G207 should define one linear axis followed by one rotational axis, and cylinder radius, in one statement
3535	G-code: G54 should define a WCS number between 1 to 12
3536	G-code: GUFAC is out of range (1e-15, 1e15) or zero
3537	G-code: Radius should be positive
3538	G-code: Blended motion should not specify feedrate and segment time together
3539	G-code: Illegal parameter for Blended motion
3540	G-code: Mandatory parameter for BSEG not specified
3541	G-code: Bseg parameter must be greater than 0
3542	G-code: G84 only allows rotation angle and center parameters to be specified

3543	G-code: Wrong M-code comand use
3544	G-code: Unmapped axis name used
3545	G-code: Tool height compensation is not allowed as second segment of Radius Compensation
3546	G-code: Incompatible mode to activate Local Coordinate System (G68)
3547	G-code: Too many G68 parameters
3548	G-code: Incompatible G code when Local Coordinate System is active
3549	G-code: Illegal case of segment synchronization
3550	G-code: Illegal NURBS based motion specification
3551	G-code: An incompatible address with NURBS/Path_Smoothing was specified
3552	G-code: An Illegal number of NURBS/Path_Smoothing main axes was specified
3553	G-code: Path Smoothing motion was not initiated
3554	G-code: Illegal Address for Path Smoothing
3555	G-code: Buffer is in Path Smoothing mode, please switch mode to use G06 command
3556	G-code: The address must contain a value
3557	G-code: The address must not contain a value
3558	G-code: SPath arc specification missing
3559	G-code: G54 and G54.4 are not allowed in the same line

Appendix A. PEG And MARK Mapping Tables

A.1 ASSIGNPEG Mapping

Mapping PEG engines to encoders

Table A-1. Mapping PEG Engines to Encoders (Servo Processor 0) for SPiiPlusNT/DC-LT/HP/LD

Bit Code	Encoder 0(X)	Encoder 1(Y)	Encoder 2(A)	Encoder 3(B)
000 (default)	PEG0	PEG1	PEG2	no
001	PEG0	PEG1	no	PEG2
010	PEG0 PEG2	PEG1	no	no
011	PEG0	PEG1 PEG2	no	no
100	PEG0 PEG1 PEG2	no	no	no
101	no	PEG0 PEG1 PEG2	no	no

Table A-2. Mapping PEG Engines to Encoders (Servo Processor 1) for SPiiPlusNT/DC-LT/HP/LD

Bit Code	Encoder 4(Z)	Encoder 5(T)	Encoder 6(C)	Encoder 7(D)
000 (default)	PEG4	PEG5	PEG6	no
001	PEG4	PEG5	no	PEG6
010	PEG4 PEG6	PEG5	no	no
011	PEG4	PEG5 PEG6	no	no
100	PEG4 PEG5 PEG6	no	no	no
101	no	PEG4 PEG5 PEG6	no	no

**Table A-3. Mapping PEG Engines to Encoders (Servo Processor 0) for SPiiPlus
CMnt/CMhv/CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa/UDMhv/UDMnt/UDMpa/UDMpm/UDMpc/UDMcb**

Bit Code	Encoder 0(X)	Encoder 1(Y)	Encoder 2(A)	Encoder 3(B)
000 (default)	PEGO	PEG1	PEG2 ²	
001	PEGO	PEG1	no	PEG2 ^{1, 2}
010	PEGO PEG2 ²	PEG1	no	
011	PEGO	PEG1 PEG2 ²	no	
100	PEGO PEG1 PEG2 ²	no	no	
101	no	PEGO PEG1 PEG2 ²	no	
110			PEGO ^{1, 2} PEG1 ^{1, 2} PEG2 ^{1, 2}	

Bit Code	Encoder 0(X)	Encoder 1(Y)	Encoder 2(A)	Encoder 3(B)
111			PEGO ^{1, 2} PEG2 ^{1, 2}	PEG1 ^{1, 2}

¹ These combinations are **not** supported by UDMpc-x.

² These combinations are **not** supported by UDMnt-x.

Table A-4. Mapping PEG Engines to Encoders (Servo Processor 0) for UDMlc/UDlIt/UDlhp/UDMmc/PDlcl

Bit Code	Encoder 0	Encoder 1	Encoder 2	Encoder 3
000 (default)	PEGO	no	no	no
001	no	PEGO	no	no
010	no	no	PEGO	no
011	no	no	no	PEGO ¹
100	no	no	no	no
101	no	no	no	no
110	no	no	no	no

¹ These combinations are **not** supported by LCM-x.

Table A-5. Mapping PEG Engines to Encoders (Servo Processor 0) for NPMpm/NPMpc-

Bit Code	Encoder 0	Encoder 1	Encoder 2	Encoder 3
000 (default)	PEG0	PEG1	no	no
001	PEG1	PEG0	no	no
010	PEG0 PEG1	no	no	no
011	no	PEG0 PEG1	no	no
100	no	PEG1	PEG0	no
101	no	PEG1	no	PEG0
110	PEG0	no	PEG1	no
111	PEG0	no	no	PEG1

Bit Code assignment example

Assume the system includes two Servo Processors:

- > Servo Processor (node) 0: CMba with 4 encoders (0,1,2,3) and 3 PEG engines (0,1,2).
- > Servo Processor (node) 1: NPMpm with 4 encoders (4,5,6) and 2 PEG engines (4,5).

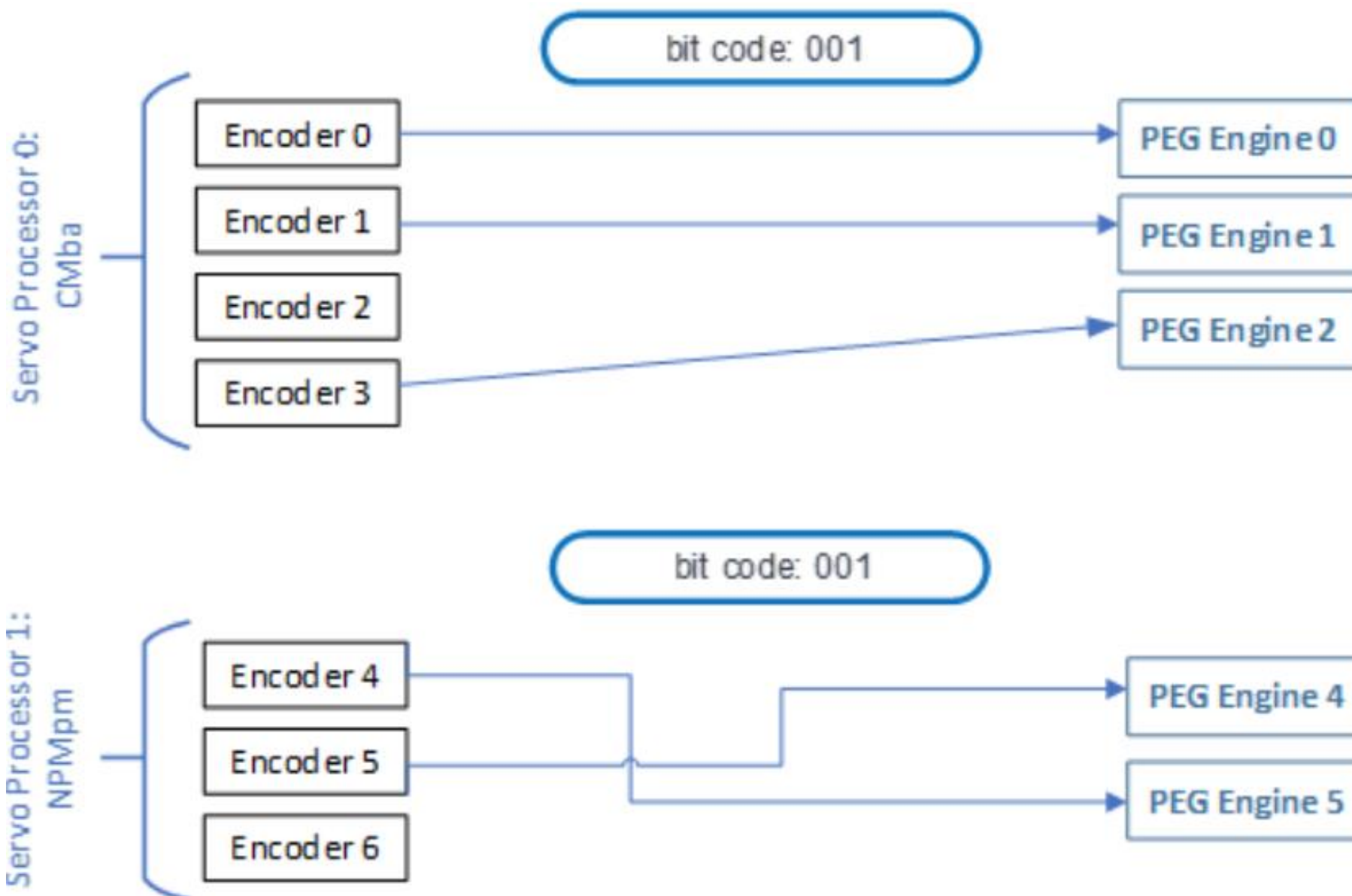
The bit code 001 for an axis associated with Servo Processor 0 (CMba) performs the following assignment:

- > PEG 0 engine is triggered by Encoder 0.

- > PEG 1 engine is triggered by Encoder 1.
- > PEG 2 engine is triggered by Encoder 3.

The bit code 001 for an axis associated with Servo Processor 1 (NPMpm) performs the following assignment:

- > PEG 4 engine is triggered by Encoder 5.
- > PEG 5 engine is triggered by Encoder 4.



General purpose outputs assignment to use as PEG pulse outputs

Table A-6. General Purpose Outputs Assignment for Use as PEG Pulse Outputs (Servo Processor 0) for SPiiPlusNT/DC-LT/HP/LD

Bit Code	GP Out 0	GP Out 1	GP Out 2	GP Out 3
0000 (default)	GP Out 0	GP Out 1	GP Out 2	GP Out 3
0001	PEG0_PULSE	GP Out 1	GP Out 2	GP Out 3
0010	GP Out 0	PEG2_PULSE	GP Out 2	GP Out 3
0011	GP Out 0	GP Out 1	PEG1_PULSE	GP Out 3
0100	GP Out 0	GP Out 1	GP Out 2	Reserved
0101	GP Out 0	PEG2_PULSE	GP Out 2	Reserved
0110	PEG0_PULSE	GP Out 1	PEG1_PULSE	GP Out 3
0111	PEG0_PULSE	PEG2_PULSE	PEG1_PULSE	Reserved
1000 - 1111	Reserved	Reserved	Reserved	Reserved

Table A-7. General Purpose Outputs Assignment for Use as PEG Pulse Outputs (Servo Processor 1) for SPiiPlusNT/DC-LT/HP/LD

Bit Code	GP Out 4	GP Out 5	GP Out 6	GP Out 7
0000 (default)	GP Out 4	GP Out 5	GP Out 6	GP Out 7
0001	PEG4_PULSE	GP Out 5	GP Out 6	GP Out 7

Bit Code	GP Out 4	GP Out 5	GP Out 6	GP Out 7
0010	GP Out 4	PEG6_PULSE	GP Out 6	GP Out 7
0011	GP Out 4	GP Out 5	PEG5_PULSE	GP Out 7
0100	GP Out 4	GP Out 5	GP Out 6	Reserved
0101	GP Out 4	PEG6_PULSE	GP Out 6	Reserved
0110	PEG4_PULSE	GP Out 5	PEG5_PULSE	GP Out 7
0111	PEG4_PULSE	PEG6_PULSE	PEG5_PULSE	Reserved
1000 - 1111	Reserved	Reserved	Reserved	Reserved

Table A-8. General Purpose Outputs Assignment for Use as PEG Pulse Outputs (Servo Processor 0) for SPiiPlus CMnt/UDMpm/CMhv/UDMhv-

Bit Code	GP Out 0	GP Out 1	GP Out 2	GP Out 3
0000 (default)	GP Out 0	GP Out 1	GP Out 2	GP Out 3
0001	PEG0_PULSE	GP Out 1	GP Out 2	GP Out 3
0010	GP Out 0	PEG1_PULSE	GP Out 2	GP Out 3
0011	GP Out 0	GP Out 1	PEG2_PULSE	GP Out 3
0100	GP Out 0	GP Out 1	GP Out 2	GP Out 3

Bit Code	GP Out 0	GP Out 1	GP Out 2	GP Out 3
0101	GP Out 0	PEG1_PULSE	GP Out 2	GP Out 3
0110	PEGO_PULSE	GP Out 1	PEG2_PULSE	GP Out 3
0111	PEGO_PULSE	PEG1_PULSE	PEG2_PULSE	GP Out 3
1000 - 1111	Reserved	Reserved	Reserved	Reserved

Table A-9. General Purpose Outputs Assignment for Use as PEG Pulse Outputs (Servo Processor 0) for UDMnt/UDMpa/UDMcb

Bit Code	GP Out 0	GP Out 1
0000 (default)	GP Out 0	GP Out 1
0001	PEGO_PULSE	GP Out 1
0010	GP Out 0	PEG1_PULSE
0011	PEG1_PULSE	GP Out 1
0100	GP Out 0	PEGO_PULSE
0101	PEGO_PULSE	PEG1_PULSE
0110 - 1111	Reserved	Reserved

Bit Code assignment example

For example, for an axis associated with Servo Processor 0 (SPiiPlusNT / DC-LT / HP / LD), **0110** switches **GP Out 0** to **PEG0_PULSE** and **GP Out 2** to **PEG1_PULSE**.

The same **Bit Code** applied to an axis associated with Servo Processor 1 switches **GP Out 4** to **PEG4_PULSE** and **GP Out 6** to **PEG5_PULSE**.

All other **GP Out** assignments are unchanged.

Table A-10. Engine to Encoder Assignment for IDMxx, ECMxx, and UDMxx

PEG Engine	Bits	HEX Code	Encoder 0	Encoder 1	Encoder 2	Encoder 3
0	0..7	00 (default)	PEG0			
		01		PEG0		
		02			PEG0	
		03				PEG0
1 ¹	8..15	00	PEG1			
		01 (default)		PEG1		
		02			PEG1	
		03				PEG1

PEG Engine	Bits	HEX Code	Encoder 0	Encoder 1	Encoder 2	Encoder 3
2 ¹	16..23	00	PEG2			
		01		PEG2		
		02 (default)			PEG2	
		03				PEG2
3 ¹	24..31	00	PEG3			
		01		PEG3		
		02			PEG3	
		03				PEG3
1 This row applies to XXMsm, XXMma, and XXMdx						
2These rows apply only to XXMsm and XXMma						

Instructions: the above table is used to build a hexadecimal value for the engines_to_encoders_code argument. Byte x determines that PEG engine x will be triggered by a specific encoder.

Example for IDMSm/ECMSm

```
ASSIGNPEG 0, 0x03020100
```

This code configures the default mapping:

- > PEG engine 0 is triggered by Encoder 0.

- > PEG1 engine 1 is triggered by Encoder 1.
- > PEG engine 2 is triggered by Encoder 2.
- > PEG engine 3 is triggered by Encoder 3.

```
ASSIGNPEG 0x02010203
```

This code configures the following mapping:

- > PEG engine 0 is triggered by Encoder 3.
- > PEG engine 1 is triggered by Encoder 2.
- > PEG engine 2 is triggered by Encoder 1.
- > PEG engine 3 is triggered by Encoder 2.

A.2 ASSIGNPOUTS Mapping

Mapping of PEG engine outputs to physical outputs

Table A-11. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0) for SPiiPlusNT/DC-LT/HP/LD

PEG OUTPUT PIN NAME Bit Code	0 X_PEG	1 Y_PEG	2 Z_PEG	3 T_PEG	4 H_DO_1 (HSSI)
000 (default)	PEGO_PULSE	PEG1_PULSE	PEG4_PULSE	PEG5_PULSE	HSSI1_DO
001	PEG4_STATE0	PEG1_STATE0	PEG2_PULSE	PEGO_STATE0	PEGO_STATE1
010	Reserved	Reserved	PEG1_STATE1	PEG2_STATE0	PEG2_STATE1
011	Reserved	Reserved	Reserved	Reserved	Reserved
100	Reserved	Reserved	Reserved	Reserved	Reserved
111	FGP_OUT0	FGP_OUT1	FGP_OUT2	FGP_OUT3	Reserved

Table A-12. SPiiPlusNT/DC-LT/HP/LD Mapping of Engine Outputs to Physical Outputs (Servo Processor 1)

PEG OUTPUT PIN NAME Bit Code	5 X_STATE0	6 X_STATE1	7 X_STATE2	8 H_DO_0 (HSSI)	9 H_CON_0 (HSSI)
000 (default)	PEG0_STATE0	PEG0_STATE1	PEG0_STATE2	HSSIO_DO	HSSIO_CON
001	PEG1_STATE0	PEG1_STATE1	PEG2_STATE0	PEG0_STATE2	PEG0_STATE0
010	PEG4_PULSE	PEG5_PULSE	PEG6_PULSE	PEG5_STATE0	PEG2_STATE1
011	Reserved	PEG4_STATE0	PEG4_STATE1	Reserved	PEG6_STATE1
100	Reserved	Reserved	Reserved	Reserved	Reserved
111	FGP_OUT4	FGP_OUT5	FGP_OUT6	Reserved	Reserved

Table A-13. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0) for CMnt/UDMpm/UDMpc/CMhv/UDMhv

Bit Code	PEG OUTPUT PIN NAME	0 PEG0	1 PEG1	5 STATE0	6 STATE1
000 (default)		PEG0_PULSE	PEG1_PULSE	PEG0_OUT0	PEG0_OUT1
001		Encoder X Phase A	Encoder X Phase B	PEG1_OUT0	PEG1_OUT1
010		Encoder Y Phase A	Encoder Y Phase B	PEG2_OUT0	PEG2_OUT1
011		Reserved	Reserved	Encoder X Phase A	Encoder X Phase B
100		Reserved	Reserved	Encoder Y Phase A	Encoder Y Phase B
101		Reserved	Reserved	Encoder X INDEX	Encoder X INDEX
110		Reserved	Reserved	Reserved	Encoder Y INDEX
111		FGP_OUT0	FGP_OUT1	Reserved	Reserved

Table A-14. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, OUT 0-4) for CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa

PEG OUTPUT PIN NAME Bit Code	0 (0)_PEG_PULSE	1 (1)_PEG_PULSE	2 (1)_STATE0	3 (1)_STATE1	4 (1)_STATE2
000 (default)	PEG0_PULSE	PEG1_PULSE	PEG1_STATE0	PEG1_STATE1	PEG1_STATE2
001	Encoder X Phase A	Encoder X Phase B	PEG0_STATE0	PEG0_STATE1	PEG0_STATE2
010	Encoder Y Phase A	Encoder Y Phase B	PEG2_STATE0	PEG2_STATE1	PEG2_STATE2
011	Encoder A Phase A	Encoder A Phase B	Encoder A Phase A	Encoder A Phase B	Encoder A INDEX
100	Reserved	PEG2_PULSE	Reserved	Reserved	Reserved
101	PEG0_PULSE or PEG2_PULSE	PEG2_PULSE or PEG1_PULSE	Reserved	Reserved	Reserved
110	PEG0_PULSE or PEG1_PULSE or PEG2_PULSE	PEG0_PULSE or PEG1_PULSE or PEG2_PULSE	Reserved	Reserved	Reserved
111	FGP_OUT0	FGP_OUT1	FGP_OUT2	Reserved	Reserved

Table A-15. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, OUT_5-9) for CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa

PEG OUTPUT PIN NAME Bit Code	5 (0)_STATE0	6 (0)_STATE1	7 (0)_STATE2	8 (0)_STATE3	9 (1)_STATE3
000 (default)	PEGO_STATE0	PEGO_STATE1	PEGO_STATE2	PEGO_STATE3	PEG1_STATE3
001	PEG1_STATE0	PEG1_STATE1	PEG1_STATE2	PEG1_STATE3	PEGO_STATE3
010	PEG2_STATE0	PEG2_STATE1	PEG2_STATE2	PEG2_STATE3	PEG2_STATE3
011	Encoder X Phase A	Encoder X Phase B	Encoder Y Phase A	Encoder Y Phase B	PEG2_STATE0
100	Encoder Y Phase A	EncoderY Phase B	Reserved	Encoder Y INDEX	PEG2_STATE1
101	Encoder X INDEX	Encoder X INDEX	Reserved	Reserved	Reserved
110	PEG2_PULSE	Encoder Y INDEX	PEG2_PULSE	PEG2_PULSE	Reserved
111	Reserved	Reserved	Reserved	Reserved	Reserved

Table A-16. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0) for UDMnt/UDMpa/UDMcb

PEG OUTPUT PIN NAME Bit Code	0 PEG0	1 PEG1
000 (default)	PEG0_PULSE	PEG1_PULSE
001	Encoder X Phase A	Encoder X Phase B
010	Encoder Y Phase A	Encoder Y Phase B
011	PEG1_STATE0	PEG0_STATE0
100	PEG0_STATE0	PEG1_STATE0
101	Reserved	Reserved
110	Reserved	Reserved
111	FGP_OUT0	FGP_OUT1

Table A-17. Mapping of Engine Outputs to Physical Outputs (Servo Processor 0) for UDMlc/UDMmc/UDIlt/UDIhp/PDIcl

PEG OUTPUT PIN NAME Bit Code	0 PEGO
000 (default)	PEGO_PULSE
001	Reserved
010	Reserved
011	Reserved
100	Reserved
101	Reserved
110	Reserved
111	FGP_OUT0

Table A-18. NPMpm/NPMpc Mapping of Engine Outputs to Physical Outputs (Servo Processor 0)

PEG OUTPUT PIN NAME Bit Code	0 PEGO	1 PEG1
000 (default)	PEGO_PULSE	PEG1_PULSE
001	PEGO_STATE0	PEG1_STATE0
010	PEG1_STATE0	PEGO_STATE0
011	PEGO_STATE1	PEG1_STATE1
100	PEG1_STATE1	PEGO_STATE1
101	Reserved	Reserved
110	Reserved	Reserved
111	FGP_OUT0	FGP_OUT1

Bit Code: 111

The **Bit Code: 111**, both for Servo Processor 0 and Servo Processor 1, is used for switching the physical output pins to Fast General Purpose Outputs: **FGP_OUT0** to **FGP_OUT6**. The state of the Fast General Purpose Outputs can be read or changed using the ACSPL+ **OUT(x)** variable. The Fast General Purpose Outputs are mapped as follows:

FGP_OUT0 is mapped to bit 16 of the ACSPL+ **OUT(x)** variable

FGP_OUT1 is mapped to bit 17 of the ACSPL+ **OUT(x)** variable

FGP_OUT2 is mapped to bit 18 of the ACSPL+ **OUT(x)** variable

FGP_OUT3 is mapped to bit 19 of the ACSPL+ **OUT(x)** variable

FGP_OUT4 is mapped to bit 20 of the ACSPL+ **OUT(x)** variable

FGP_OUT5 is mapped to bit 21 of the ACSPL+ **OUT(x)** variable

FGP_OUT6 is mapped to bit 22 of the ACSPL+ **OUT(x)** variable

Table A-19. IDMSm/ECMSm/UDMSm Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 0-5)

PEG OUTPUT PIN NAME Bit Code	0 OUT_CNFG_0	1 OUT_CNFG_1	2 OUT_CNFG_2	3 OUT_CNFG_3	4 OUT_CNFG_4	5 OUT_CNFG_5
Connectors/pinouts	J11-I/O: 36(+),37(-)	J11-I/O: 38(+),39(-)	J11-I/O: 40 (+),41(-)	J11-I/O: 25 (+),10(-)	J10-LIMITS: 9 (+),22(-)	J10-LIMITS: 10 (+),23 (-)
0000	PEG0_State0	PEG0_State1	PEG0_State2	PEG0_State3	Reserved	Reserved
0001	PEG1_State0	PEG1_State1	PEG1_State2	PEG1_State3	Reserved	Reserved
0010	PEG2_State0	PEG2_State1	PEG2_State2	PEG2_State3	Reserved	Reserved
0011	PEG3_State0	PEG3_State1	PEG3_State2	PEG3_State3	Reserved	Reserved
0100	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
0101	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
0110	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
0111	GP_OUT0 (Default)	GP_OUT1 (Default)	GP_OUT2 (Default)	GP_OUT3 (Default)	GP_OUT4 (Default)	GP_OUT5 (Default)
1000	PEG0_Pulse	PEG0_Pulse	PEG0_Pulse	PEG0_Pulse	Encoder 0 Phase A	Encoder 0 Phase A

PEG OUTPUT PIN NAME Bit Code	0 OUT_CNFG_0	1 OUT_CNFG_1	2 OUT_CNFG_2	3 OUT_CNFG_3	4 OUT_CNFG_4	5 OUT_CNFG_5
Connectors/pinouts	J11-I/O: 36(+),37(-)	J11-I/O: 38(+),39(-)	J11-I/O: 40 (+),41(-)	J11-I/O: 25 (+),10(-)	J10-LIMITS: 9 (+),22(-)	J10-LIMITS: 10 (+),23 (-)
1001	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse	Encoder 1 Phase A	Encoder 1 Phase A
1010	PEG2_Pulse	PEG2_Pulse	PEG2_Pulse	PEG2_Pulse	Encoder 2 Phase A	Encoder 2 Phase A
1011	PEG3_Pulse	PEG3_Pulse	PEG3_Pulse	PEG3_Pulse	Encoder 3 Phase A	Encoder 3 Phase A
1100	Reserved	Reserved	Reserved	Reserved	Encoder 0 Phase B	Encoder 0 Phase B
1101	Reserved	Reserved	Reserved	Reserved	Encoder 1 Phase B	Encoder 1 Phase B
1110	Reserved	Reserved	Reserved	Reserved	Encoder 2 Phase B	Encoder 2 Phase B
1111	Reserved	Reserved	Reserved	Reserved	Encoder 3 Phase B	Encoder 3 Phase B

Table A-20. IDMSm/ECMSm/UDMSm Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 6-11)

PEG OUTPUT PIN NAME Bit Code	6 OUT_CNFG_6	7 OUT_CNFG_7	8 PEG0	9 PEG1	10 PEG2	11 PEG3
Connectors/pinouts	J10-LIMITS: 11 (+),24 (-)	J10-LIMITS: 12 (+),25 (-)	J11-I/O: 21(+),6 (-)	J11-I/O: 22(+),7 (-)	J11-I/O: 23 (+),8(-)	J11-I/O: 24 (+),9(-)
0000	Reserved	Reserved	PEG0_Pulse (Default)	PEG1_Pulse (Default)	PEG2_Pulse (Default)	PEG3_Pulse (Default)
0001	Reserved	Reserved	PEG0_State0	PEG0_State1	PEG0_State2	PEG0_State3
0010	Reserved	Reserved	PEG1_State0	PEG1_State1	PEG1_State2	PEG1_State3
0011	Reserved	Reserved	PEG2_State0	PEG2_State1	PEG2_State2	PEG2_State3
0100	Reserved	Reserved	PEG3_State0	PEG3_State1	PEG3_State2	PEG3_State3
0101	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
0110	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
0111	GP_OUT6 (Default)	GP_OUT7 (Default)	GP_OUT8	GP_OUT9	GP_OUT10	GP_OUT11
1000	Encoder 0 Phase A	Encoder 0 Phase A	PEG0_Pulse	PEG0_Pulse	PEG0_Pulse	PEG0_Pulse

PEG OUTPUT PIN NAME Bit Code	6 OUT_CNFG_6	7 OUT_CNFG_7	8 PEGO	9 PEG1	10 PEG2	11 PEG3
Connectors/pinouts	J10-LIMITS: 11 (+),24 (-)	J10-LIMITS: 12 (+),25 (-)	J11-I/O: 21(+),6 (-)	J11-I/O: 22(+),7 (-)	J11-I/O: 23 (+),8(-)	J11-I/O: 24 (+),9(-)
1001	Encoder 1 Phase A	Encoder 1 Phase A	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse
1010	Encoder 2 Phase A	Encoder 2 Phase A	PEG2_Pulse	PEG2_Pulse	PEG2_Pulse	PEG2_Pulse
1011	Encoder 3 Phase A	Encoder 3 Phase A	PEG3_Pulse	PEG3_Pulse	PEG3_Pulse	PEG3_Pulse
1100	Encoder 0 Phase B	Encoder 0 Phase B	Reserved	Reserved	Reserved	Reserved
1101	Encoder 1 Phase B	Encoder 1 Phase B	Reserved	Reserved	Reserved	Reserved
1110	Encoder 2 Phase B	Encoder 2 Phase B	Reserved	Reserved	Reserved	Reserved
1111	Encoder 3 Phase B	Encoder 3 Phase B	Reserved	Reserved	Reserved	Reserved

Table A-21. ECMsa/IDMsa/UDMsa Mapping of Engine Outputs to Physical Outputs (Servo Processor 0)

PEG OUTPUT PIN NAME	8 PEG0	9 PEG1
Bit Code		
Connectors/pinouts	J7-I/O: 1(+), 9(-)	J7-I/O: 2(+),10(-)
0000	PEG0_Pulse(Default)	PEG1_Pulse(Default)
0001	PEG0_State0	PEG0_State1
0010	PEG1_State0	PEG1_State1
0011	PEG2_State0	PEG2_State1
0100	PEG3_State0	PEG3_State1
0101	Reserved	Reserved
0110	Reserved	Reserved
0111	GP_OUT8	GP_OUT9
1000	PEG0_Pulse	PEG0_Pulse
1001	PEG1_Pulse	PEG1_Pulse

PEG OUTPUT PIN NAME	8 PEG0	9 PEG1
Bit Code		
Connectors/pinouts	J7-I/O: 1(+), 9(-)	J7-I/O: 2(+),10(-)
1010	PEG2_Pulse	PEG2_Pulse
1011	PEG3_Pulse	PEG3_Pulse
1100	Reserved	Reserved
1101	Reserved	Reserved
1110	Reserved	Reserved
1111	Reserved	Reserved

Table A-22. ECMma / IDMma / UDMma Mapping of Engine Outputs to Physical Outputs (Servo Processor 0)

PEG OUTPUT PIN NAME Bit Code	0 OUT0	1 OUT1	2 OUT2	3 OUT3	8 PEG0	9 PEG1	10 PEG2	11 PEG3
Connectors/ pinouts	J11-I/O: 36	J11-I/O: 37	J11-I/O: 38	J11-I/O: 39	J9-I/O: 21	J9-I/O: 22(+),7(-)	J9-I/O: 23(+),8(-)	J9-I/O: 24(+),9(-)
0000	PEG0_ State0	PEG0_ State1	PEG0_ State2	PEG0_ State3	PEG0_ Pulse (Default)	PEG1_Pulse (Default)	PEG2_ Pulse (Default)	PEG3_Pulse (Default)
0001	PEG1_ State0	PEG1_ State1	PEG1_ State2	PEG1_ State3	PEG0_ State0	PEG0_ State1	PEG0_ State2	PEG0_ State3
0010	PEG2_ State0	PEG2_ State1	PEG2_ State2	PEG2_ State3	PEG1_ State0	PEG1_ State1	PEG1_ State2	PEG1_ State3
0011	PEG3_ State0	PEG3_ State1	PEG3_ State2	PEG3_ State3	PEG2_ State0	PEG2_ State1	PEG2_ State2	PEG2_ State3
0100	Reserved	Reserved	Reserved	Reserved	PEG3_ State0	PEG3_ State1	PEG3_ State2	PEG3_ State3
0101	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

PEG OUTPUT PIN NAME	0 OUT0	1 OUT1	2 OUT2	3 OUT3	8 PEG0	9 PEG1	10 PEG2	11 PEG3
Bit Code								
Connectors/ pinouts	J11-I/O: 36	J11-I/O: 37	J11-I/O: 38	J11-I/O: 39	J9-I/O: 21	J9-I/O: 22(+),7(-)	J9-I/O: 23(+),8(-)	J9-I/O: 24(+),9(-)
0110	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
0111	GP_OUT0 (Default)	GP_OUT1 (Default)	GP_OUT2 (Default)	GP_OUT3 (Default)	GP_OUT4	GP_OUT5	GP_OUT6	GP_OUT7
1000	PEG0_ Pulse	PEG0_ Pulse	PEG0_ Pulse	PEG0_ Pulse	PEG0_ Pulse	PEG0_ Pulse	PEG0_ Pulse	PEG0_ Pulse
1001	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse
1010	PEG2_ Pulse	PEG2_ Pulse	PEG2_ Pulse	PEG2_ Pulse	PEG2_ Pulse	PEG2_ Pulse	PEG2_ Pulse	PEG2_Pulse
1011	PEG3_ Pulse	PEG3_ Pulse	PEG3_ Pulse	PEG3_ Pulse	PEG3_ Pulse	PEG3_ Pulse	PEG3_ Pulse	PEG3_Pulse
1100	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1101	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

PEG OUTPUT PIN NAME Bit Code	0 OUT0	1 OUT1	2 OUT2	3 OUT3	8 PEG0	9 PEG1	10 PEG2	11 PEG3
Connectors/ pinouts	J11-I/O: 36	J11-I/O: 37	J11-I/O: 38	J11-I/O: 39	J9-I/O: 21	J9-I/O: 22(+),7(-)	J9-I/O: 23(+),8(-)	J9-I/O: 24(+),9(-)
1110	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
1111	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

Table A-23. ECMdx / IDMdx / UDMdx Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 0-5)

PEG OUTPUT PIN NAME Bit Code	0 OUT0	1 OUT1	2 OUT2	3 OUT3	4 OUT4	5 OUT5
Connectors/pinouts	J9-I/O: 33	J9-I/O: 34	J9-I/O: 35	J9-I/O: 3	J9-I/O: 36	J9-I/O: 37
0000	PEG0_State0	PEG0_State1	PEG0_State2	PEG0_State3	Reserved	Reserved
0001	PEG1_State0	PEG1_State1	PEG1_State2	PEG1_State3	Reserved	Reserved
0010	PEG2_State0	PEG2_State1	PEG2_State2	PEG2_State3	Reserved	Reserved
0011	PEG3_State0	PEG3_State1	PEG3_State2	PEG3_State3	Reserved	Reserved
0100	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
0101	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
0110	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
0111	GP_OUT0 (Default)	GP_OUT1 (Default)	GP_OUT2 (Default)	GP_OUT3 (Default)	GP_OUT4 (Default)	GP_OUT5 (Default)
1000	PEG0_Pulse	PEG0_Pulse	PEG0_Pulse	PEG0_Pulse	Encoder 0 Phase A	Encoder 0 Phase A

PEG OUTPUT PIN NAME	0 OUT0	1 OUT1	2 OUT2	3 OUT3	4 OUT4	5 OUT5
Bit Code						
Connectors/pinouts	J9-I/O: 33	J9-I/O: 34	J9-I/O: 35	J9-I/O: 3	J9-I/O: 36	J9-I/O: 37
1001	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse	Encoder 1 Phase A	Encoder 1 Phase A
1010	PEG2_Pulse	PEG2_Pulse	PEG2_Pulse	PEG2_Pulse	Encoder 2 Phase A	Encoder 2 Phase A
1011	PEG3_Pulse	PEG3_Pulse	PEG3_Pulse	PEG3_Pulse	Encoder 3 Phase A	Encoder 3 Phase A
1100	Reserved	Reserved	Reserved	Reserved	Encoder 0 Phase B	Encoder 0 Phase B
1101	Reserved	Reserved	Reserved	Reserved	Encoder 1 Phase B	Encoder 1 Phase B
1110	Reserved	Reserved	Reserved	Reserved	Encoder 2 Phase B	Encoder 2 Phase B
1111	Reserved	Reserved	Reserved	Reserved	Encoder 3 Phase B	Encoder 3 Phase B

Table A-24. ECMdx / IDMdx / UDMdx Mapping of Engine Outputs to Physical Outputs (Servo Processor 0, Outputs 6-11)

PEG OUTPUT PIN NAME Bit Code	6 OUT6	7 OUT7	8 PEG0	9 PEG1	10 PEG2	11 PEG3
Connectors/pinouts	J9-I/O: 38	J9-I/O: 4	J9-I/O: 21(+), 6(-)	J9-I/O: 22(+),7(-)	J9-I/O: 23(+), 8(-)	J9-I/O: 24(+),9(-)
0000	Reserved	Reserved	PEG0_Pulse (Default)	PEG1_Pulse (Default)	PEG2_Pulse (Default)	PEG3_Pulse (Default)
0001	Reserved	Reserved	PEG0_State0	PEG0_State1	PEG0_State2	PEG0_State3
0010	Reserved	Reserved	PEG1_State0	PEG1_State1	PEG1_State2	PEG1_State3
0011	Reserved	Reserved	PEG2_State0	PEG2_State1	PEG2_State2	PEG2_State3
0100	Reserved	Reserved	PEG3_State0	PEG3_State1	PEG3_State2	PEG3_State3
0101	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
0110	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
0111	GP_OUT6	GP_OUT7	GP_OUT8	GP_OUT9	GP_OUT10	GP_OUT11
1000	Encoder 0 Phase A	Encoder 0 Phase A	PEG0_Pulse	PEG0_Pulse	PEG0_Pulse	PEG0_Pulse

PEG OUTPUT PIN NAME	6 OUT6	7 OUT7	8 PEG0	9 PEG1	10 PEG2	11 PEG3
Bit Code						
Connectors/pinouts	J9-I/O: 38	J9-I/O: 4	J9-I/O: 21(+), 6(-)	J9-I/O: 22(+),7(-)	J9-I/O: 23(+), 8(-)	J9-I/O: 24(+),9(-)
1001	Encoder 1 Phase A	Encoder 1 Phase A	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse	PEG1_Pulse
1010	Encoder 2 Phase A	Encoder 2 Phase A	PEG2_Pulse	PEG2_Pulse	PEG2_Pulse	PEG2_Pulse
1011	Encoder 3 Phase A	Encoder 3 Phase A	PEG3_Pulse	PEG3_Pulse	PEG3_Pulse	PEG3_Pulse
1100	Encoder 0 Phase B	Encoder 0 Phase B	Reserved	Reserved	Reserved	Reserved
1101	Encoder 1 Phase B	Encoder 1 Phase B	Reserved	Reserved	Reserved	Reserved
1110	Encoder 2 Phase B	Encoder 2 Phase B	Reserved	Reserved	Reserved	Reserved
1111	Encoder 3 Phase B	Encoder 3 Phase B	Reserved	Reserved	Reserved	Reserved

A.3 ASSIGNMARK Mapping

Mark Inputs to Encoders Mapping

Table A-25. Mark-1 Inputs to Encoders Mapping for SPiiPlusNT/DC-LT/HP/LD

Bit code	Latching of MARK Encoder 0(X)	Latching of MARK Encoder 1(Y)	Latching of MARK Encoder 4(Z)	Latching of MARK Encoder 5(T)	Latching of MARK Encoder 2(A)	Latching of MARK Encoder 3(B)	Latching of MARK Encoder 6(C)	Latching of MARK Encoder 7(D)
00000 (default)	X_MARK1	Y_MARK1	Z_MARK1	T_MARK1	-	-	-	-
00001	Y_MARK1	Z_MARK1	T_MARK1	X_MARK1	-	-	-	-
00010	Z_MARK1	T_MARK1	X_MARK1	Y_MARK1	-	-	-	-
00011	T_MARK1	X_MARK1	Y_MARK1	Z_MARK1	-	-	-	-
00100	-	Y_MARK1	Z_MARK1	T_MARK1	X_MARK1	-	-	-
00101	X_MARK1	-	Z_MARK1	T_MARK1	-	Y_MARK1	-	-
00110	X_MARK1	Y_MARK1	-	T_MARK1	-	-	Z_MARK1	-
00111	X_MARK1	Y_MARK1	Z_MARK1	-	-	-	-	T_MARK1
01000	-	-	Z_MARK1	T_MARK1	X_MARK1	Y_MARK1	-	-
01001	X_MARK1	Y_MARK1	-	-	-	-	Z_MARK1	T_MARK1

Bit code	Latching of MARK Encoder 0(X)	Latching of MARK Encoder 1(Y)	Latching of MARK Encoder 4(Z)	Latching of MARK Encoder 5(T)	Latching of MARK Encoder 2(A)	Latching of MARK Encoder 3(B)	Latching of MARK Encoder 6(C)	Latching of MARK Encoder 7(D)
01010	X_MARK1	-	-	-	-	Y_MARK1	Z_MARK1	T_MARK1
01011	-	Y_MARK1	-	-	X_MARK1	-	Z_MARK1	Z_MARK1
01100	-	-	Z_MARK1	-	X_MARK1	Y_MARK1	-	T_MARK1
01101	-	-	-	T_MARK1	X_MARK1	Y_MARK1	Z_MARK1 pin	-
01110	-	-	-	-	X_MARK1	Y_MARK1	Z_MARK1	T_MARK1
01111	-	-	-	-	Y_MARK1	Z_MARK1	T_MARK1	X_MARK1
10000	-	-	-	-	Z_MARK1	T_MARK1	X_MARK1	Y_MARK1
10001	-	-	-	-	T_MARK1	X_MARK1	Y_MARK1	Z_MARK1

Example

```
ASSIGNMARK 1, 1, 0x0b00010
```

By using SPiiPlusNT as the first node, entering the command performs the following assignments for these inputs:

- > Latching of Encoder 0(X) occurs once Z_MARK1 physical pin gets an input.
- > Latching of Encoder 1(Y) occurs once T_MARK1 physical pin gets an input.
- > Latching of Encoder 4(Z) occurs once X_MARK1 physical pin gets an input.

- > Latching of Encoder 5(T) occurs once Y_sMARK1 physical pin gets an input.

Table A-26. Mark-2 Inputs to Encoders Mapping for SPiiPlusNT/DC-LT/HP/LD

Bit code	Latching of M2ARK Encoder 0(X)	Latching of M2ARK Encoder 1(Y)	Latching of M2ARK Encoder 4(Z)	Latching of M2ARKEncoder 5(T)	Latching of M2ARKEncoder 2(A)	Latching of M2ARKEncoder 3(B)	Latching of M2ARKEncoder 6(C)	Latching of M2ARK Encoder 7(D)
00000 (default)	GP IN6	GP IN7	GP IN4	GP IN5	-	-	-	-
00001	GP IN7	GP IN4	GP IN5	GP IN6	-	-	-	-
00010	GP IN4	GP IN5	GP IN6	GP IN7	-	-	-	-
00011	GP IN5	GP IN6	GP IN7	GP IN4	-	-	-	-
00100	-	GP IN7	GP IN4	GP IN5	GP IN6	-	-	-
00101	GP IN6	-	GP IN4	GP IN5	-	GP IN7	-	-
00110	GP IN6	GP IN7	-	GP IN5	-	-	GP IN4	-
00111	GP IN6	GP IN7	GP IN4	-	-	-	-	GP IN5
01000	-	-	GP IN4	GP IN5	GP IN6	GP IN7	-	-
01001	GP IN6	GP IN7	-	-	-	-	GP IN4	GP IN5
01010	GP IN6	-	-	-	-	GP IN7	GP IN4	GP IN5

Bit code	Latching of M2ARK Encoder 0(X)	Latching of M2ARK Encoder 1(Y)	Latching of M2ARK Encoder 4(Z)	Latching of M2ARKEncoder 5(T)	Latching of M2ARKEncoder 2(A)	Latching of M2ARKEncoder 3(B)	Latching of M2ARKEncoder 6(C)	Latching of M2ARK Encoder 7(D)
01011	-	GP IN7	-	-	GP IN6	-	GP IN4	GP IN5
01100	-	-	GP IN4	-	GP IN6	GP IN7	-	GP IN5
01101	-	-	-	GP IN5	GP IN6	GP IN7	GP IN4	-
01110	-	-	-	-	GP IN6	GP IN7	GP IN4	GP IN5
01111	-	-	-	-	GP IN7	GP IN4	GP IN5	GP IN6
10000	-	-	-	-	GP IN4	GP IN5	GP IN6	GP IN7
10001	-	-	-	-	GP IN5	GP IN6	GP IN7	GP IN4

Example

```
ASSIGNMARK 1, 2, 0x0b00010
```

By using SPiiPlusNT as the first node, entering the command performs the following assignments for these inputs:

- > Latching of M2ARK of Encoder 0(X) occurs once GP IN4 physical pin gets an input.
- > Latching of M2ARK of of Encoder 1(Y) occurs once GP IN5 physical pin gets an input.
- > Latching of M2ARK of of Encoder 4(Z) occurs once GP IN6 physical pin gets an input.
- > Latching of M2ARK of Encoder 5(T) occurs once GP IN7 physical pin gets an input.

Table A-27. Mark-1 Inputs to Encoders Mapping for with SPiiPlus CMnt/UDMpm-x/UDMpc/CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa/CMhv/UDMhv

Bit code	Latching of Mark1 Encoder 0(X)	Latching of Mark1 Encoder 1(Y)	Latching of Mark1 Encoder 2(A)	Latching of Mark1 Encoder 3(B)
000 (default)	Mark1 of encoder 0(X) pin	Mark1 of encoder 1(Y) pin	Mark1 of encoder 0(X) pin	Mark1 of encoder 1(Y) pin
001	GP IN6	Mark1 of encoder 0(X) pin	Mark2 of encoder 0(X) pin	Mark1 of encoder 0(X) pin
010	-	GP IN4	GP IN6	GP IN6
011	-	GP IN6	-	-

Example

```
ASSIGNMARK 1, 1, 0x0b0001
```

By using CMhp as the first node, entering the command above performs the following assignments for these inputs:

- > Latching of Encoder 0 occurs once IN6 pin (pin 5, J9 connector at CMhp) gets an input.
- > Latching of Encoder 1 occurs once X(0)_MARK1+ physical pin (pin 12, J9 connector at CMhp) gets an input.
- > Latching of Encoder 2 occurs once X(0)_MARK2+ physical pin (pin 13, J9 connector at CMhp) gets an input.
- > Latching of Encoder 3 occurs once X(0)_MARK1+ physical pin (pin 12, J9 connector at CMhp) gets an input.

Table A-28. Mark-2 Inputs to Encoders Mapping for with SPiiPlus CMnt/UDMpm/UDMpc/CMba/CMhp/CMxa/UDMba/UDMhp/UDMxa/CMhv/UDMhv

Bit code	Latching of Mark2 Encoder 0(X)	Latching of Mark2 Encoder 1(Y)	Latching of Mark2 Encoder 2(A)	Latching of Mark2 Encoder 3(B)
000 (default)	Mark2 of axis 0(X) pin	Mark2 of axis 1(Y) pin	GP IN6	GP IN7
001	Mark1 of axis 1(Y) pin	Mark1 of axis 1(Y) pin	Mark1 of axis 1(Y) pin	Mark1 of axis 1(Y) pin
010	Mark2 of axis 1(Y) pin	GP IN5	Mark2 of axis 1(Y) pin	-
011	GP IN7	GP IN7	GP IN7	-

Example

```
ASSIGNMARK 0x0b010
```

By using CMhp as the first node, entering the command above performs the following assignments for these inputs:

- > Latching of Encoder 0 occurs once Y(1)_MARK2+ physical pin (pin 15, J9 connector at CMhp) gets an input.
- > Latching of Encoder 1 occurs once IN5 pin (pin 23, J9 connector at CMhp) gets an input.
- > Latching of Encoder 2 occurs once Y(1)_MARK2+ physical pin (pin 15, J9 connector at CMhp) gets an input.

Table A-29. IDMxx/ECMxx/UDMsm/UDMsA/UDMma Encoder Mapping

Byte	3	2	1	0
	Latching of Encoder 3(B)	Latching of Encoder 2(A)	Latching of Encoder 1(Y)	Latching of Encoder 0(Y)
MARK0 input pin	0x00	0x00	0x00	0x00 (default)
MARK1 input pin	0x01	0x01	0x01(default)	0x01
MARK2 input pin	0x02	0x02(default)	0x02	0x02
MARK3 input pin	0x03(default)	0x03	0x03	0x03

The table above allows the user to build a hexadecimal value for the inputs_to_encoder_bit_code argument:

ASSIGNMARK axis, type, 0XAABBCCDD

Where AA is the code for encoder 3, BB is the code for encoder 2, CC is the code for encoder 1, and DD is the code for encoder 0.

Example 1 (default case)

```
ASSIGNMARK 1, 1, 0x03020100
```

By using UDMsm as the first node, entering the command above performs the following assignments for these inputs:

- > Latching of Encoder 0 occurs once MARK0 physical pin (pin 16, J11 connector) gets an input.
- > Latching of Encoder 1 occurs once MARK1 physical pin (pin 17, J11 connector) gets an input.
- > Latching of Encoder 2 occurs once MARK2 physical pin (pin 18, J11 connector) gets an input.
- > Latching of Encoder 3 occurs once MARK3 physical pin (pin 19, J11 connector) gets an input.

Example 2

```
ASSIGNMARK 1, 1, 0x03010102
```

By using UDMsm as the first node, entering the command above performs the following assignments for these inputs:

- > Latching of Encoder 0 occurs once MARK2 physical pin (pin 18, J11 connector) gets an input.
- > Latching of Encoder 1 occurs once MARK1 physical pin (pin 17, J11 connector) gets an input.
- > Latching of Encoder 2 occurs once MARK1 physical pin (pin 17, J11 connector) gets an input.
- > Latching of Encoder 3 occurs once MARK3 physical pin (pin 19, J11 connector) gets an input.

Smarter



Motion

2008 2009 2010 2011 2012
www.acsmotioncontrol.com

5 HaTnufa St.
Yokneam Illit, 2066717
Israel
Tel: (+972) (4) 654 6440 Fax: (+972) (4) 654 6443

Contact us: sales@acsmotioncontrol.com | www.acsmotioncontrol.com

ACS
MOTION CONTROL